

HOL-2535-01-VCF-L

Virtualization 101

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Lab Overview - HOL-2535-01-VCF-L - Virtualization 101

Lab Description

[2]

This introductory lab demonstrates the core features and functions of vSphere and vCenter. This is an excellent place to begin your Virtualization 101 experience.

This lab will walk you through the core features of vSphere and vCenter, including storage and networking.

The lab is broken into 3 Modules and the Modules can be taken in any order.

Lab Guidance

[3]

Note: It may take more than 90 minutes to complete this lab. You may only finish 2-3 of the modules during your time. However, you may take this lab as many times as you want. The modules are independent of each other so you can start at the beginning of any module and proceed from there. Use the Table of Contents to access any module in the lab. The Table of Contents can be accessed in the upper right-hand corner of the Lab Manual.

This introductory lab demonstrates the core features and functions of vSphere and vCenter. This is an excellent place to begin your Virtualization 101 experience.

This lab will walk you through the core features of vSphere and vCenter, including storage and networking.

The lab is broken into 3 Modules and the Modules can be taken in any order.

Lab Module List:

- **Module 1 - An Introduction to Management with vCenter Server (60 Minutes)(Basic)** This module will start with an interactive simulation of an ESXi installation. ESXi is the foundation of vSphere and is sometimes referred to as the host. After the installation, the ESXi Host Client will be reviewed. It is a web-based management tool that allows you to manage a single ESXi host at a time.
- **Module 2 - An Introduction to vSphere Networking and Security (60 Minutes)(Basic)** Learn how to add, configure, and work with vSphere Standard and Distributed Switches for most types of in and out network traffic of the ESXi host through vCenter, as well as the advanced networking features it provides.
- **Module 3 - An Introduction to vSphere Storage (60 Minutes)(Basic)** This module will provide you with an overview of the different types of storage available in vSphere. You will go through different scenarios such as creation and configuration of vSphere Datastores, management of storage vMotion, among others.

Each Module will take approximately 60-90 minutes to complete, but based on your experience this could take more or less time.

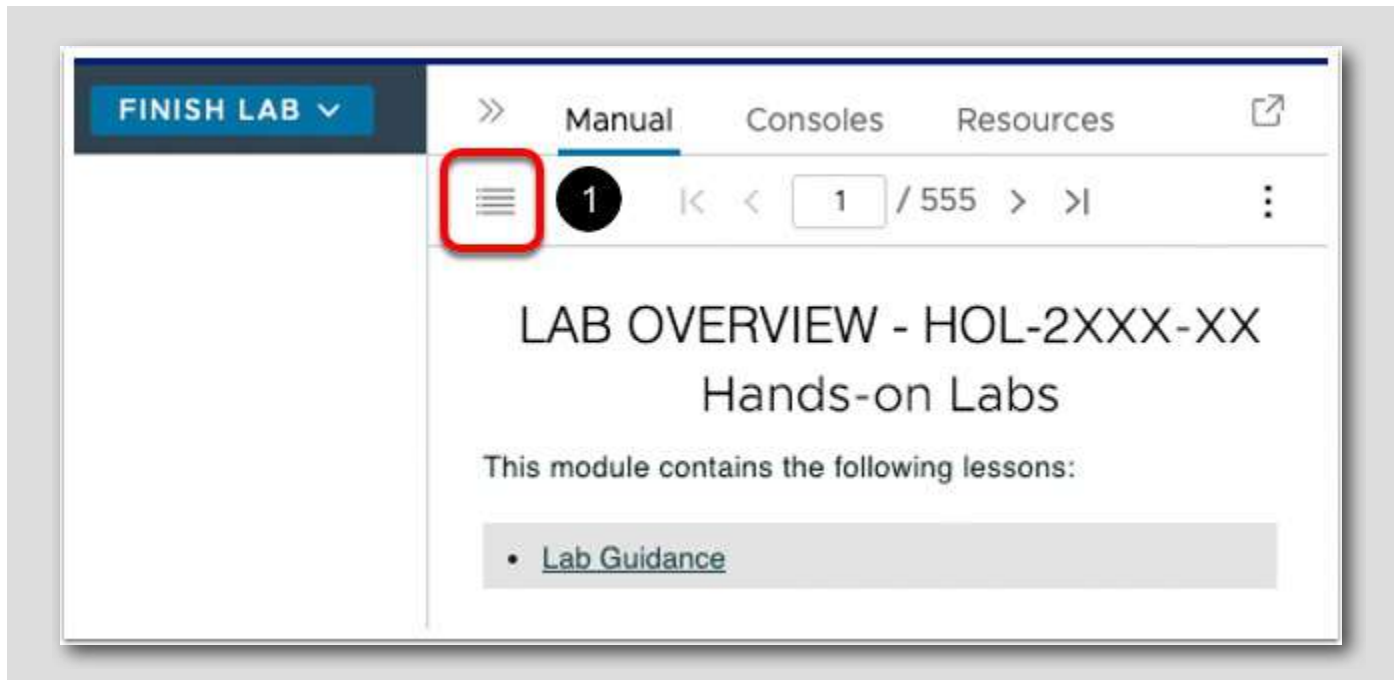
We have included videos throughout the modules. To get the most out of these videos, it is recommended that you have headphones to hear the audio. The timing of each video is noted next to the title. In some cases, videos are included for tasks we are unable to show in a lab environment, while others are there to provide additional information. Some of these videos may contain an earlier edition of vSphere, however, the steps and concepts are primarily the same.

Lab Captains:

- Doug Baer, Lab Godfather, USA
- Dave Cook, Product Marketing Engineer, USA
- Matthew Meyer, Client Services Consultant, USA
- Milena Chen, In Spirit, Costa Rica

First time using Hands-on Labs?

[4]

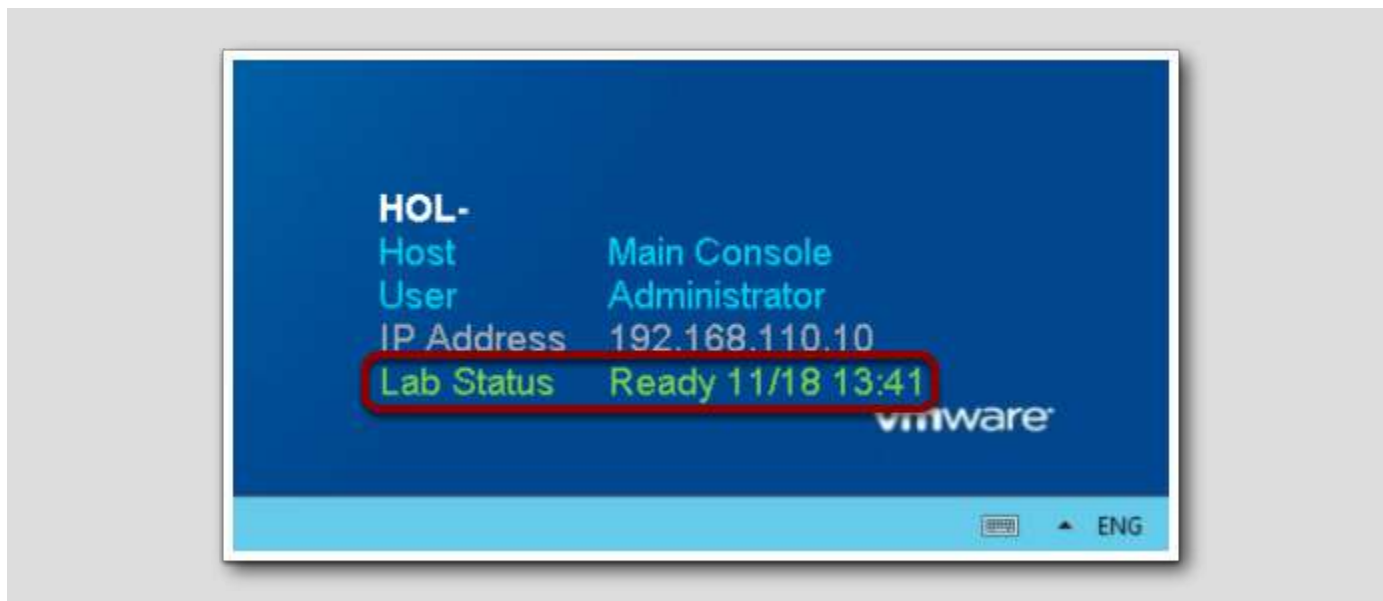


Welcome!

1. If this is your first time taking a lab navigate to the **Appendix** in the Table of Contents to review the interface and features before proceeding. For returning users, feel free to start your lab by clicking next in the manual.

You are ready....is your lab?

[5]



Please verify that your lab has finished all the startup routines and is ready for you to start. If you see anything other than "Ready", please wait a few minutes. If after 5 minutes your lab has not changed to "Ready", please ask for assistance.

Virtualization 101

[6]

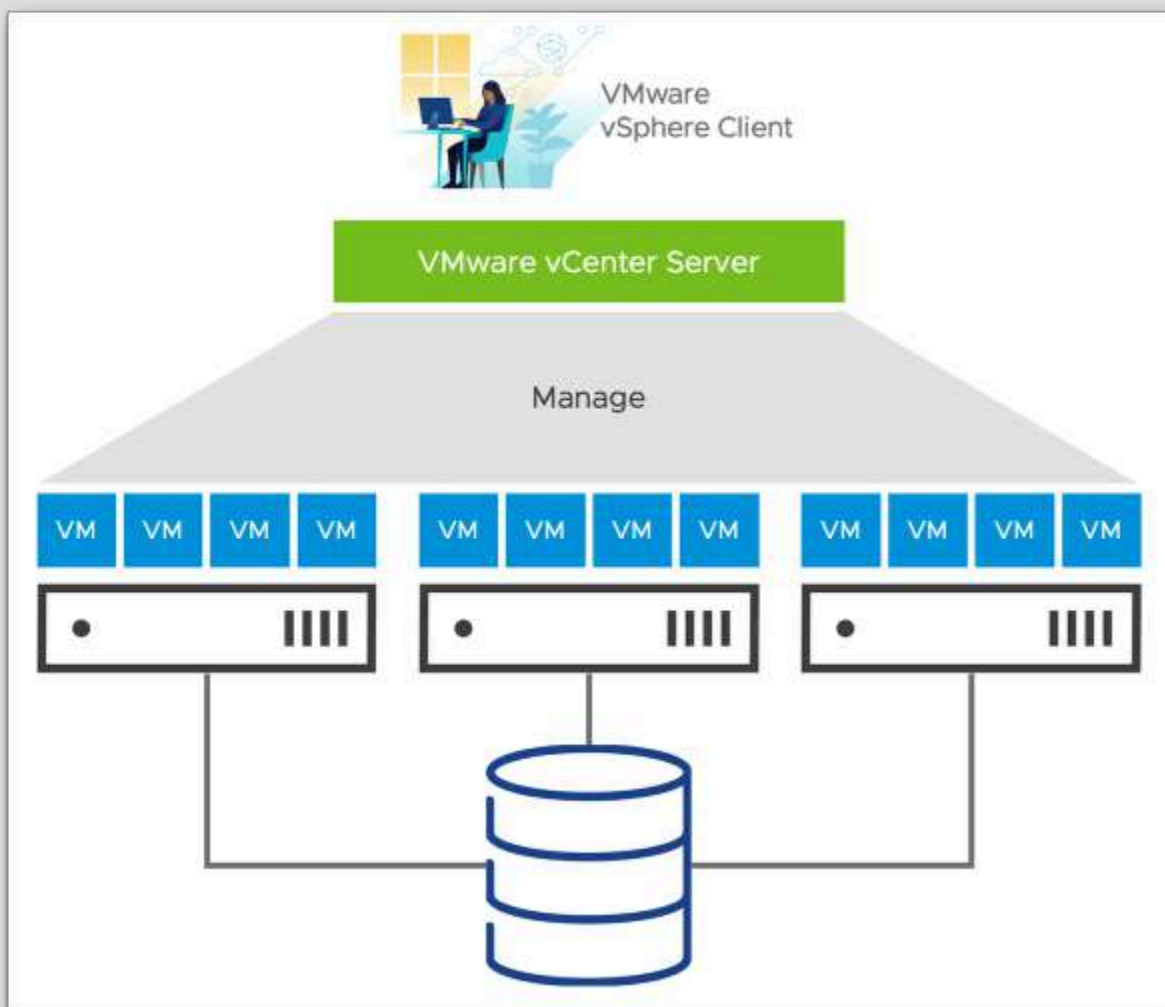
If you are not familiar with Virtualization, this lesson will give you an introduction to it.

If you are familiar with virtualization or have taken this lab previously, you can jump ahead to [Module 1 - Introduction to management with vCenter Server](#).

What is Virtualization:

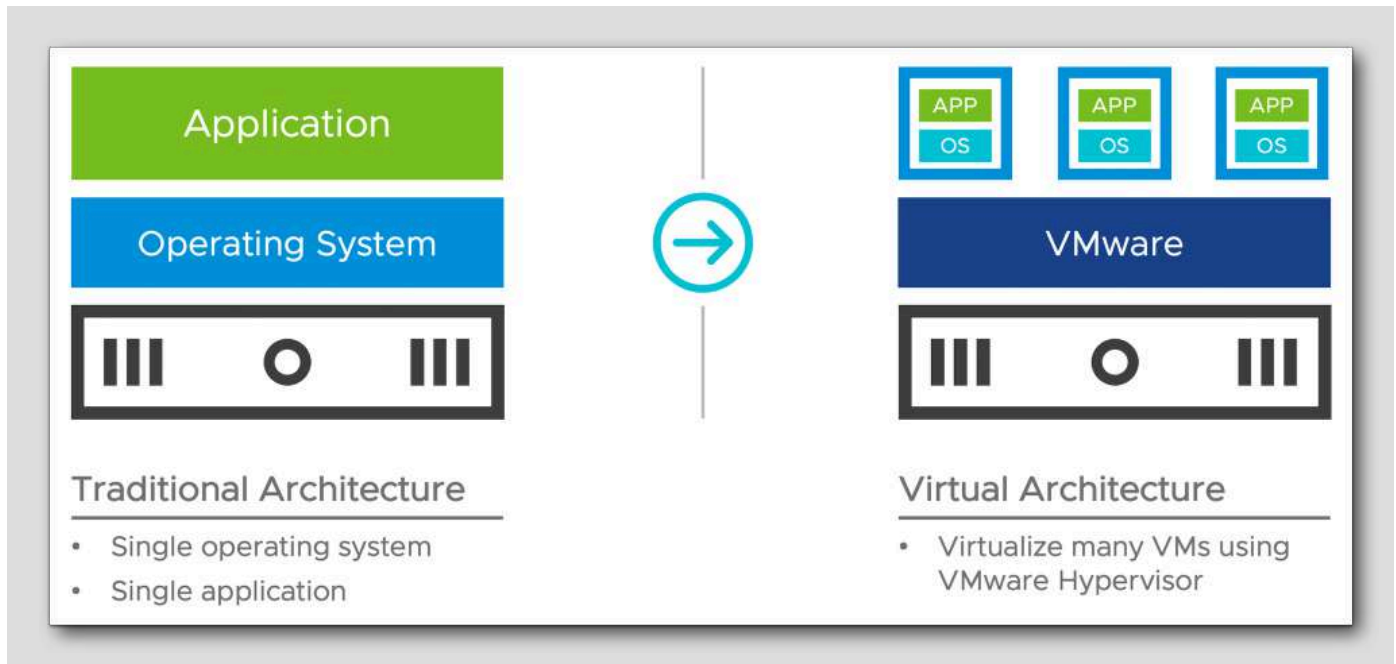
[7]

Today's x86 computer hardware was designed to run a single operating system and a single application, leaving most machines vastly underutilized. Virtualization lets you run multiple virtual machines on a single physical machine, with each virtual machine sharing the resources of that one physical computer across multiple environments. Different virtual machines can run different operating systems and multiple applications on the same physical computer.



Virtualization Defined

[8]



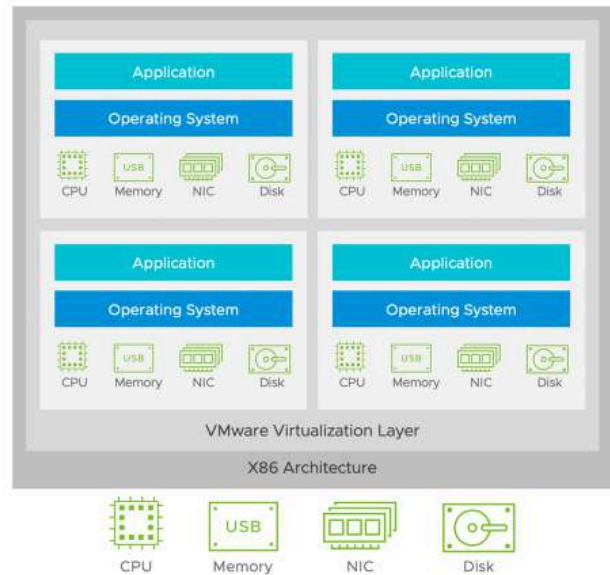
Virtualization is placing an additional layer of software called a hypervisor on top of your physical server. The hypervisor enables you to install multiple operating systems and applications on a single server.

Separation

[9]

Server virtualization is separating the OS from the Hardware ...

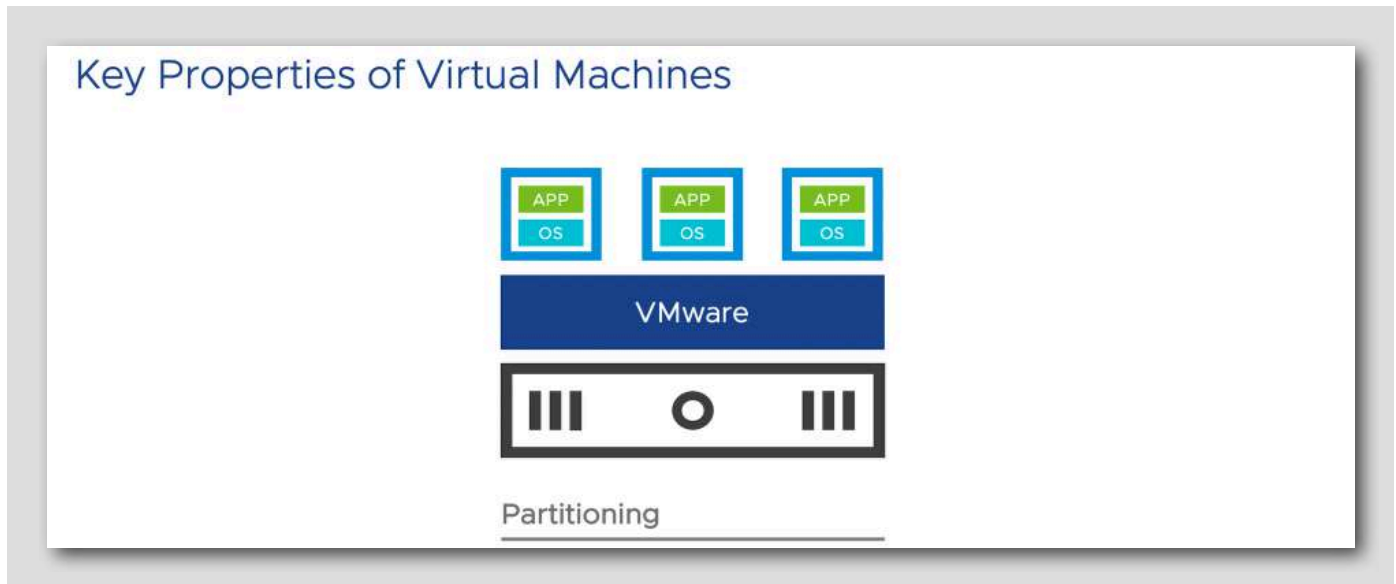
... by presenting a complete x86 platform to the OS



By isolating the operating system from the hardware, you can create a virtualization-based x86 platform. VMware's hypervisor-based virtualization products and solutions provide you the fundamental technology for x86 virtualization.

Partitioning

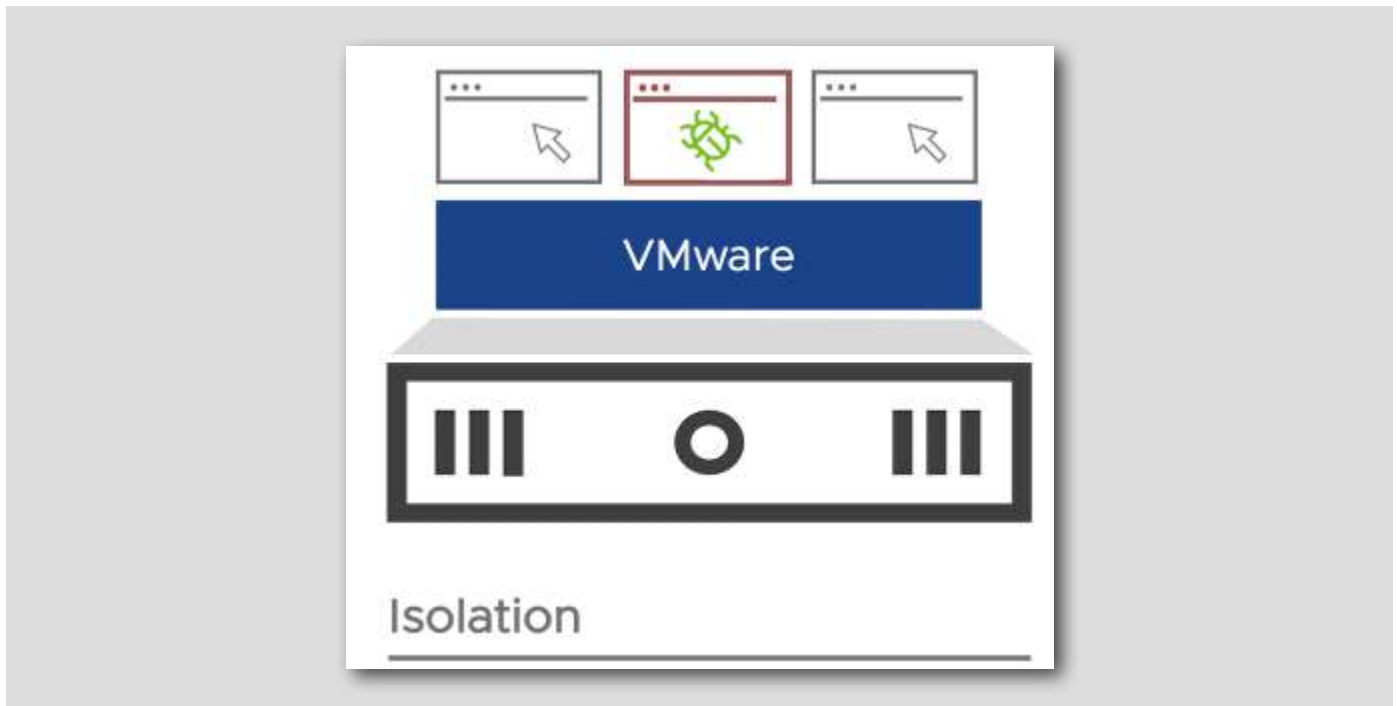
[10]



Partitioning helps improve utilization by running multiple operating systems on one physical machine while also divides the system resources between the virtual machines.

Isolation

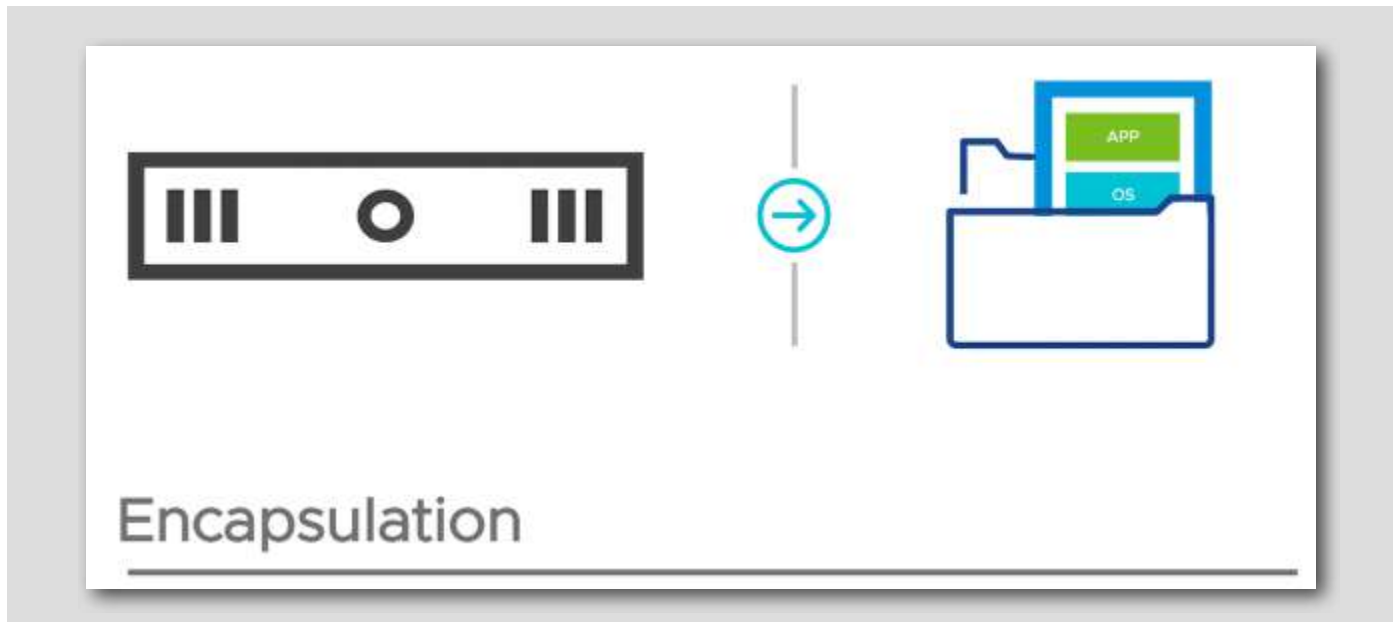
[11]



You can isolate a VM to find and fix bugs and faults without affecting other VMs and operating systems. Fault and security isolation can be performed at a hardware level. Once fixed, an entire VM Restore can be performed in minutes through advanced resource controls.

Encapsulation

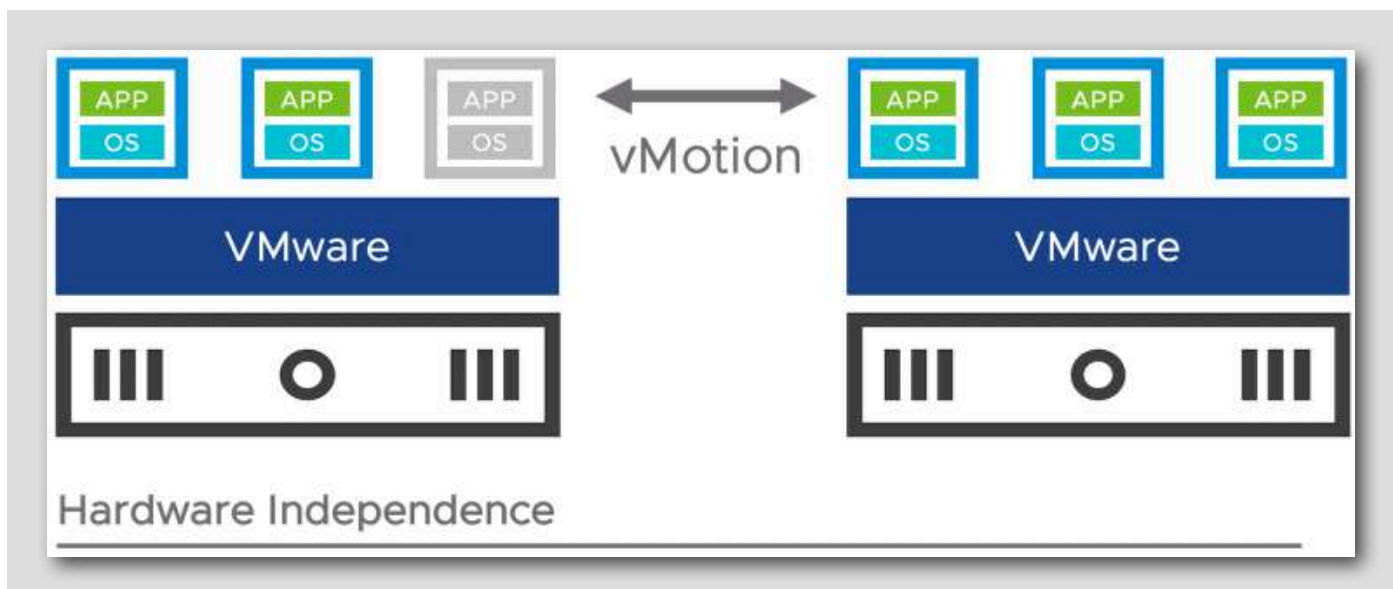
[12]



Encapsulation simplifies management by helping you copy, move and restore VMs by treating entire VMs as files.

Hardware Independence

[13]



VMs are not dependent on any physical hardware or vendor, making your IT more flexible and scalable. You can provision and migrate any VM to any physical server.

Benefits

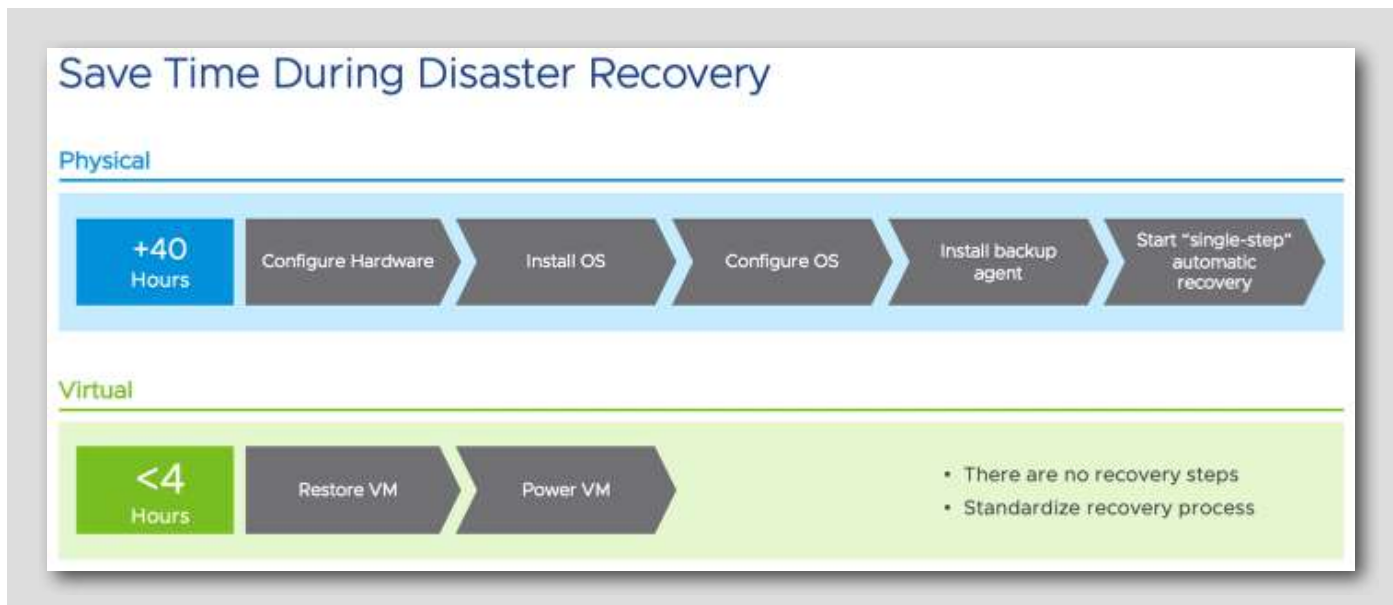
[14]



Virtualization enables you to consolidate servers and contain applications, resulting in high availability and scalability of critical applications.

Simplify Recovery

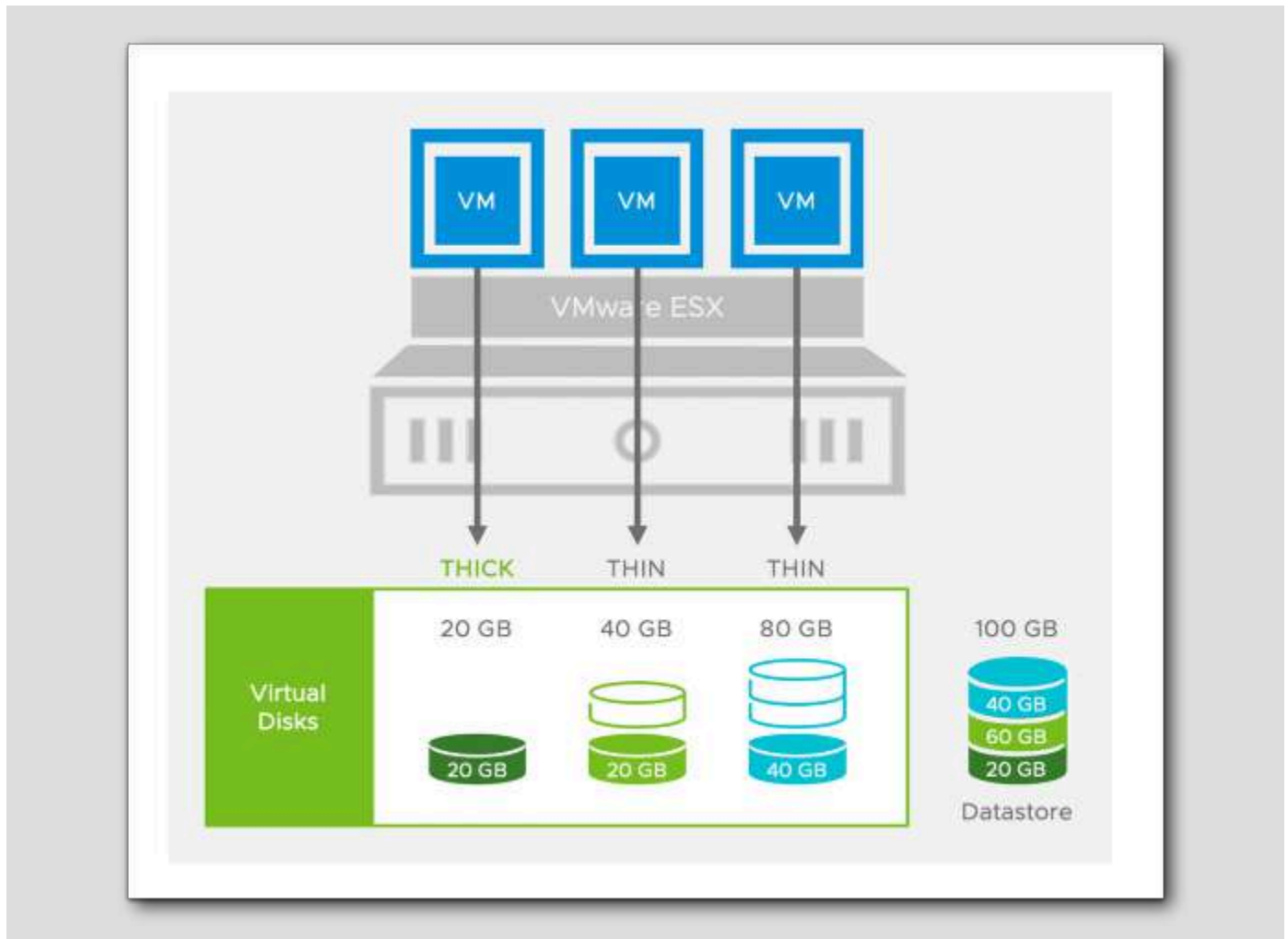
[15]



Virtualization eliminates the need for any hardware configuration, OS reinstallation and configuration, or backup agents. A simple restore can recover an entire VM.

Reduce Storage Costs and Optimize Storage Utilization and Efficiency

[16]



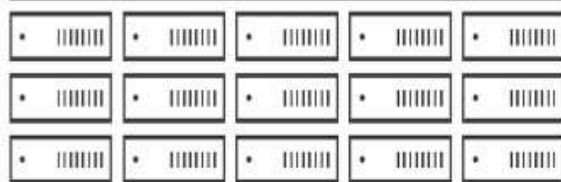
A technology called thin provisioning helps you optimize space utilization and reduce storage costs. It provides storage to VMs when it's needed, and shares space with other VMs.

Cost Avoidance

[17]

The CapEx Story: Better Use of Existing Infrastructure

Before VMware

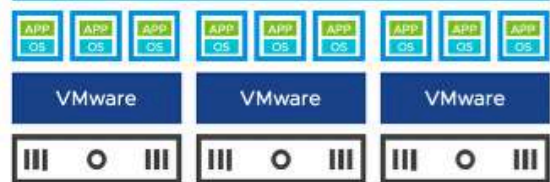


Servers	10
Utilization	8%
Annual cost per server	\$4,000
Total Cost	\$40,000

More apps per machine =
less machines



After VMware



Servers	3
Utilization	80%
Annual cost per server	\$4,000
Total Cost	\$12,000

\$28,000 in cost avoidance

SOURCE: IT Business Edge, "The Business Value of Server Virtualization" -cost average a 2x CPU server in three-year amortized hardware purchase, and annual support and maintenance contract costs 9/02

vmware

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Module 1 - Introduction to Management with vCenter Server (60 minutes)

Introduction

[19]

This module will start with an interactive simulation of an ESXi installation. ESXi is the foundation of vSphere and is sometimes referred to as the host. After the installation, the ESXi Host Client will be reviewed. It is a web-based management tool that allows you to manage a single ESXi host at a time.

The remainder of the module will focus on using the vSphere Client to access vCenter Server and manage your entire virtual infrastructure using one interface. Virtual machines will be created, with more details covered on how to manage and monitor the environment. Lastly, you will migrate some workloads with VMware vMotion and learn how to use the performance charts and graphs in the vSphere Client.

ESXi Host Client

[20]

The VMware Host Client is an HTML5-based client that is used to connect to and manage single ESXi hosts.

You can use the VMware Host Client to perform administrative and basic troubleshooting tasks, as well as advanced administrative tasks on your target ESXi host. You can also use the VMware Host Client to conduct emergency management when vCenter Server is not available.

It is important to know that the VMware Host Client is different from the vSphere Web Client, regardless of their similar user interfaces. You use the vSphere Web Client to connect to vCenter Server and manage multiple ESXi hosts, whereas you use the VMware Host Client to manage a single ESXi host.

For additional details on the VMware Host Client, please see this PDF (<https://docs.vmware.com/en/VMware-vSphere/8.0/vsphere-esxi-host-client-212-guide.pdf>)

This lesson will walk through some of the most frequently used features in the ESXi Host Client.

Open the Lab Main Console

[21]

Click the button below to switch back to the Main Console

[vlp:switch-console|Main Console|Switch Back to the Main Console]

Launch Firefox

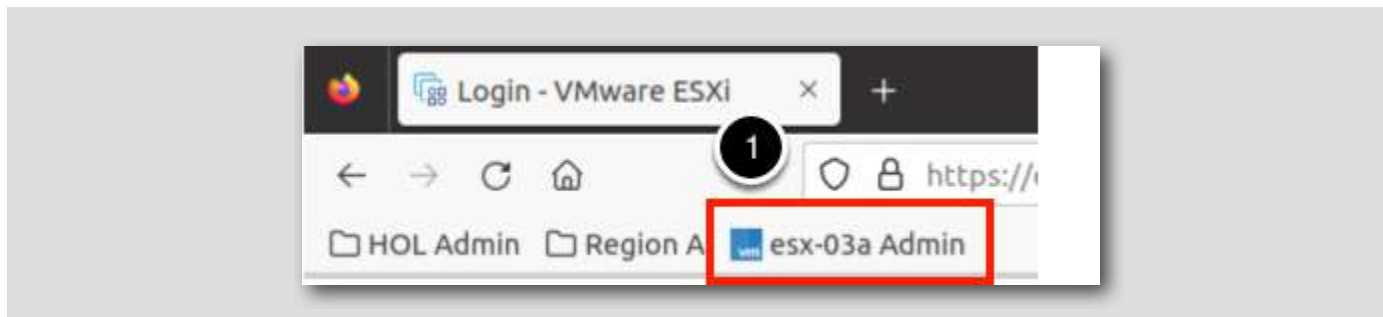
[22]



1. Click on the **Firefox** Icon on the Linux Task Bar

Select esx-03a

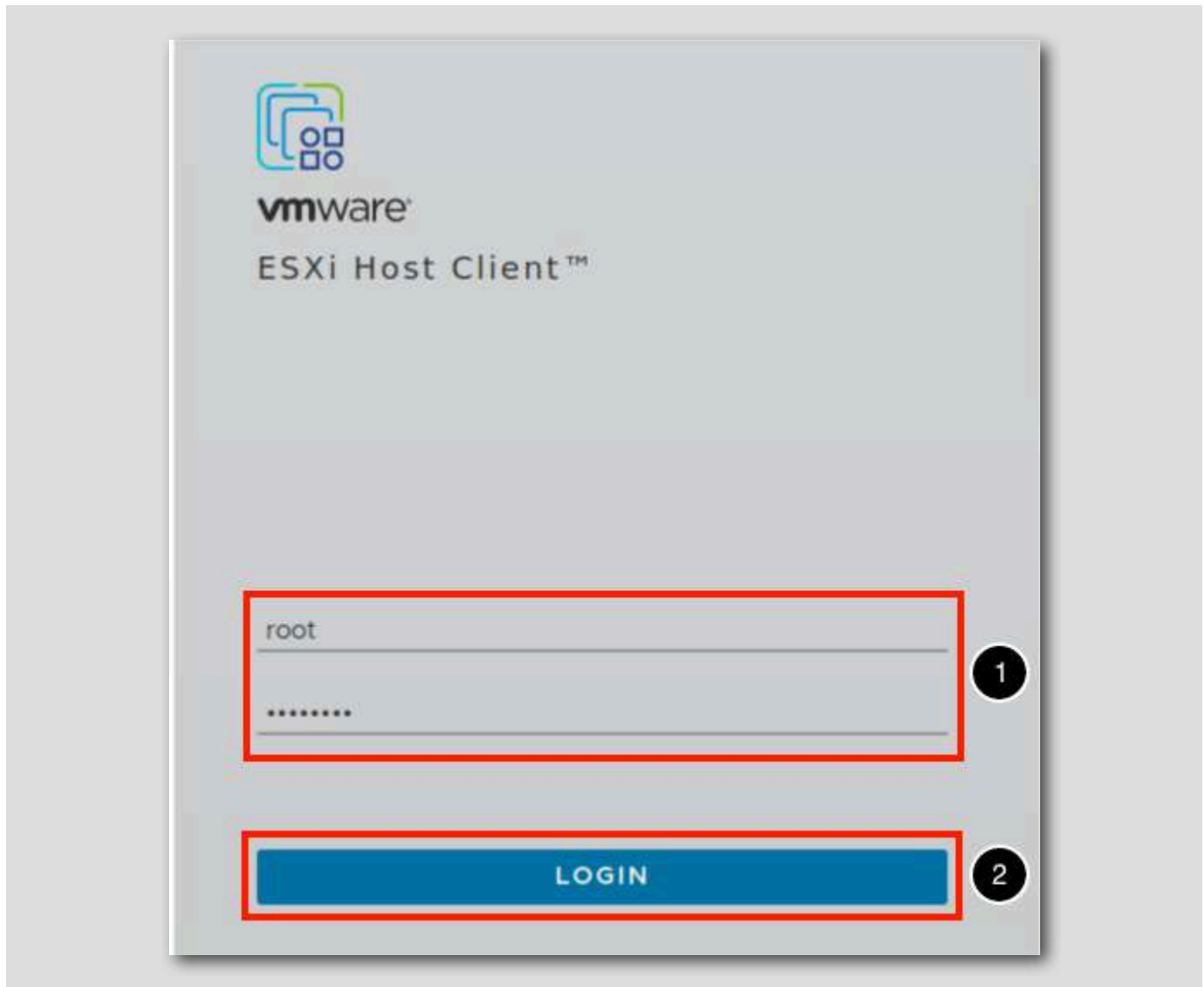
[23]



1. From the Bookmarks bar, select esx-03a Admin

Login

[24]

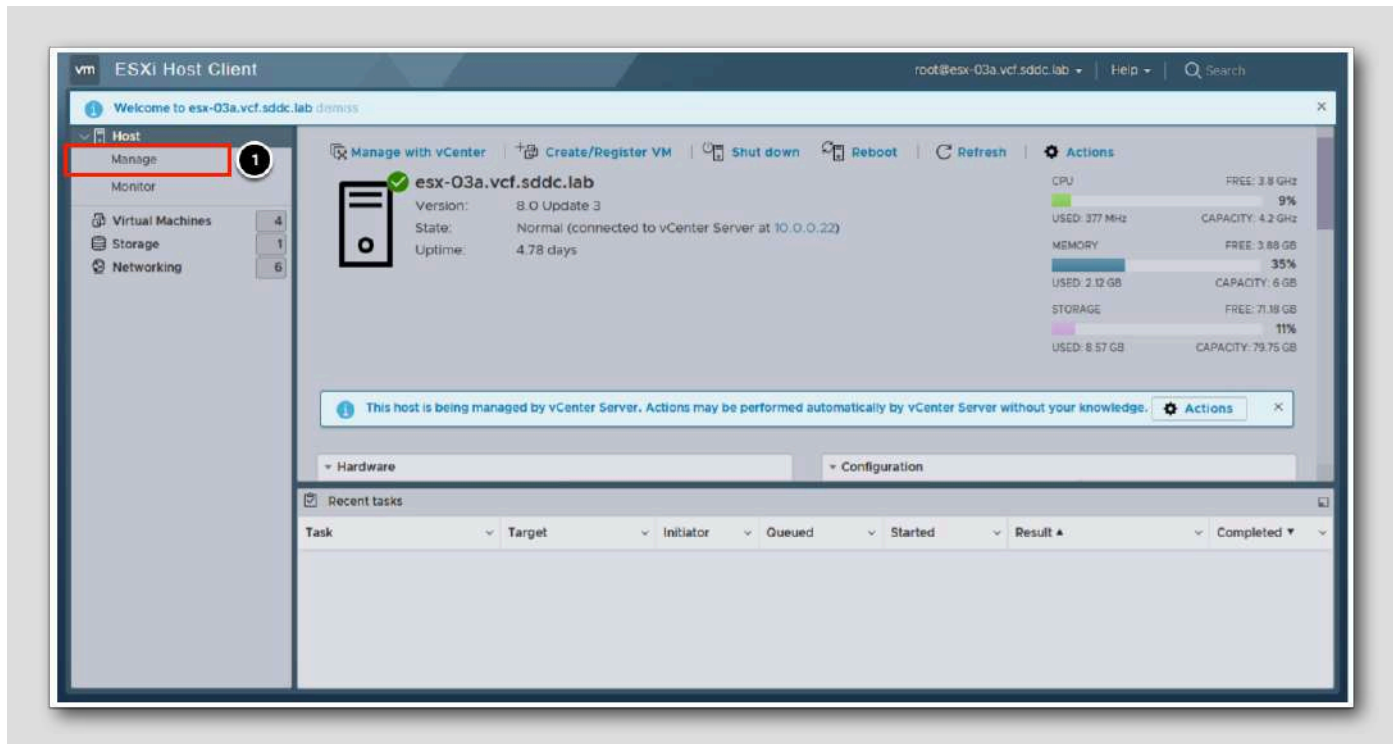


1. Login with the following credentials:

- User name: root
- Password: VMware123!

2. Click the Login button

ESXi Host Client



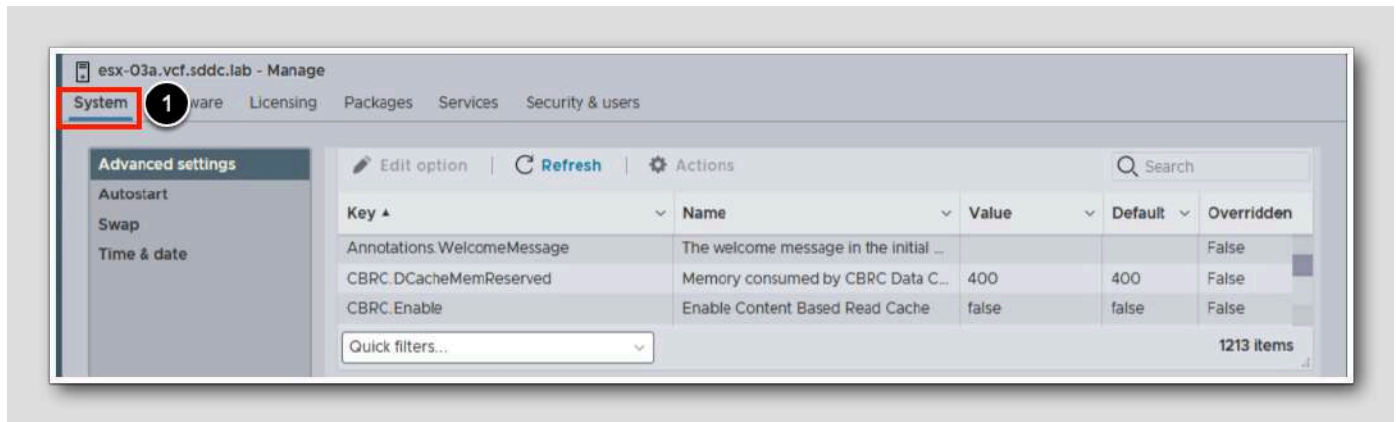
The ESXi Host, in this case, **esx-03a**, can now be directly managed. This can be useful in test/dev environments where a vCenter Server is not present or in a production environment where the vCenter Server is not reachable.

The initial screen shows high-level details and recent tasks. There are also various power options for the host and an Actions menu for the most common tasks. Note that the server is currently in Maintenance Mode, which will be discussed in a future lesson.

1. Click on **Manage**

System

[26]

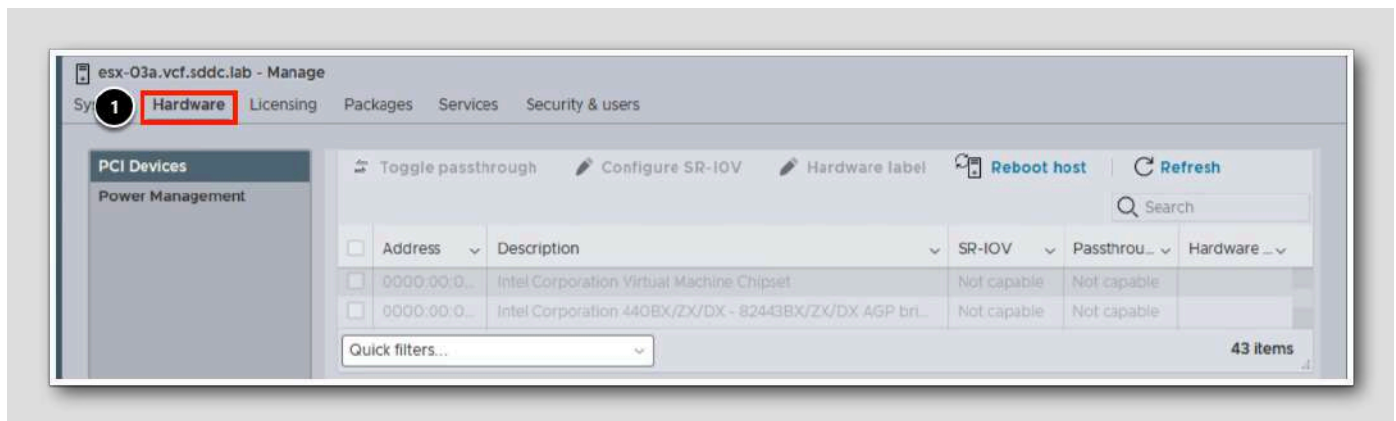


1. Click on the **System** tab.

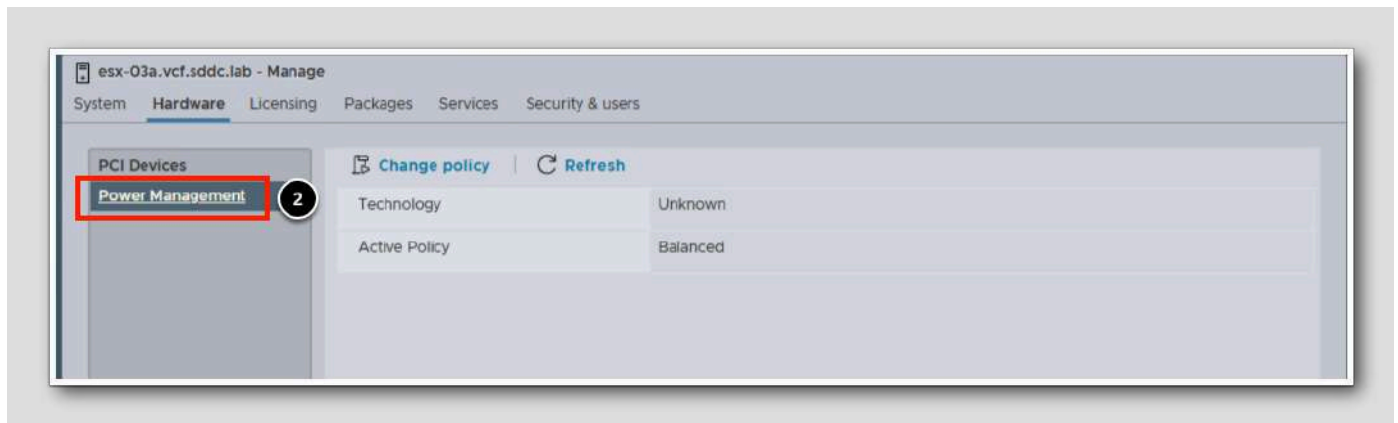
You can find that the most common options set are the date and time for the host. It can be set and synchronized with an NTP server or set manually. In addition, Autostart settings for the host can be configured here as well.

Hardware

[27]



1. Click on the **Hardware** tab



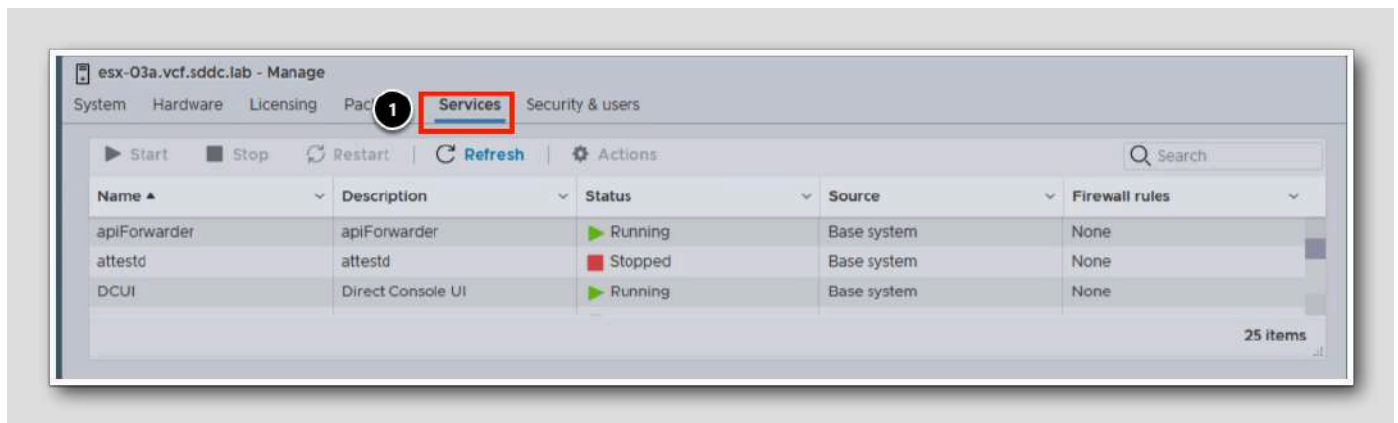
Note: Due to this being a nested virtual environment, seeing items such as "Unknown" under 'Technology' is expected behavior. You can safely ignore this and move on.

2. Click Power Management

This is where power management policies can be set for the host.

Services

[28]

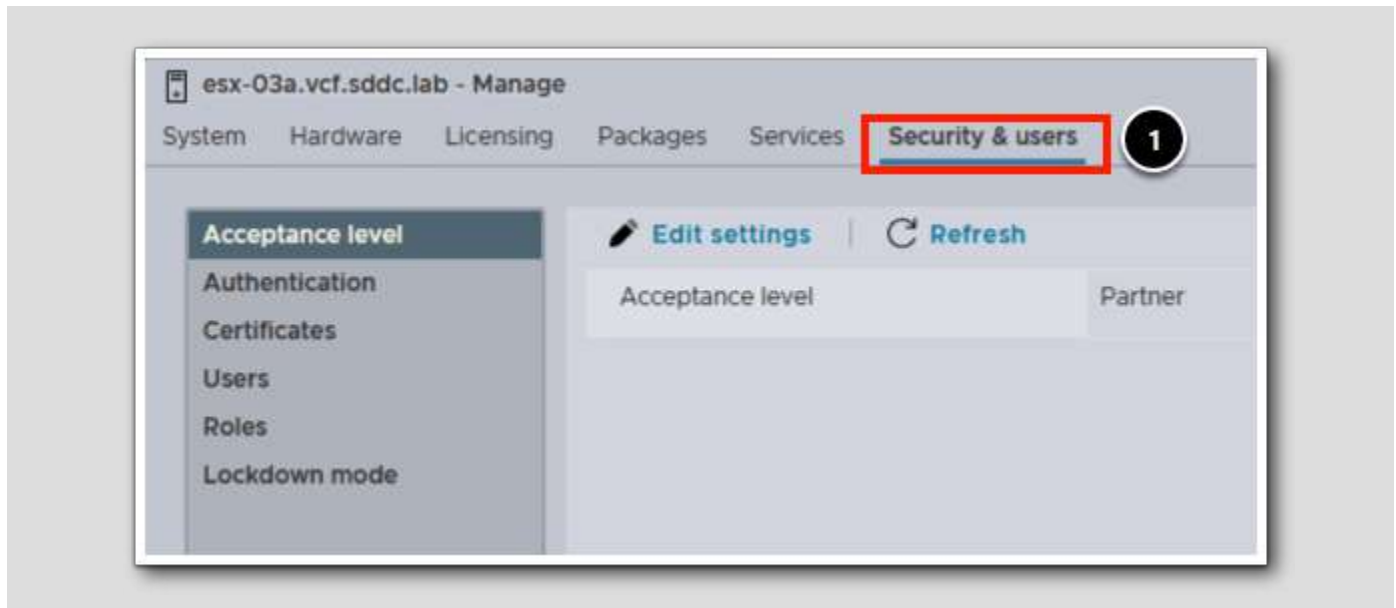


1. Click the Services tab

Services like SSH access and the Direct Console UI can be stopped and started from this screen.

Security and Users

[29]

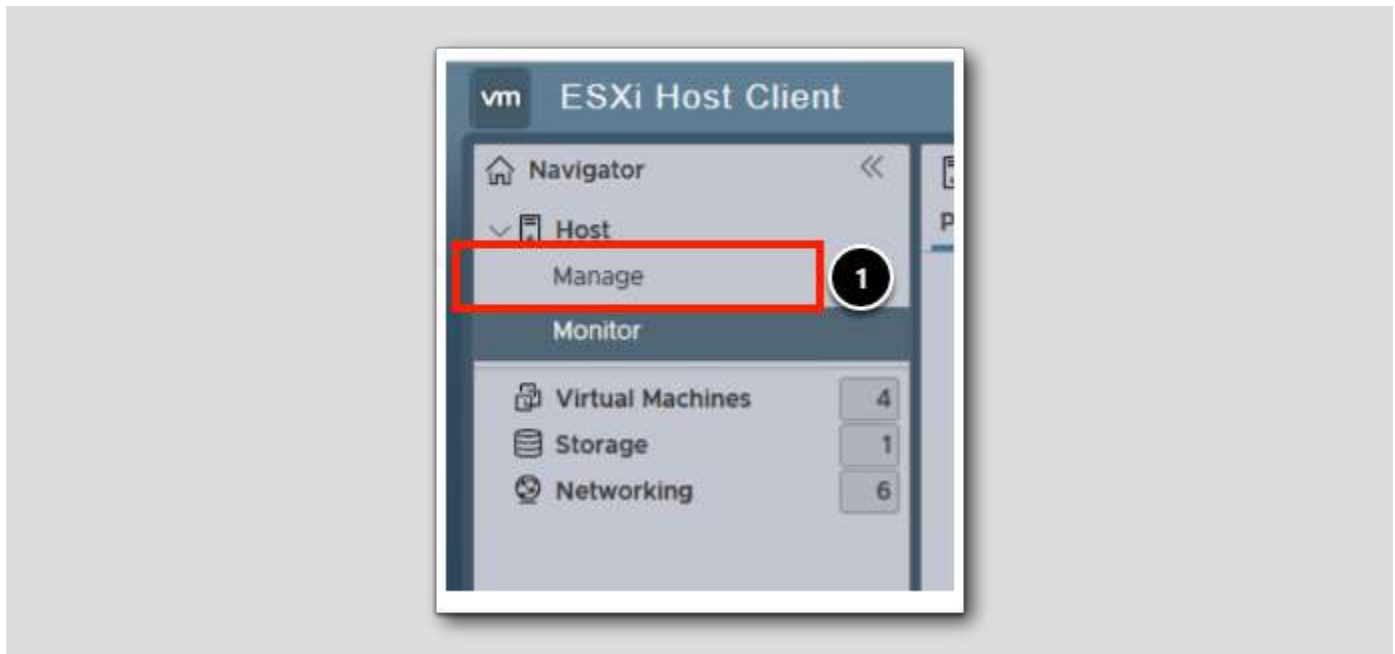


On the Security & Users tab, security options such as authentication to Active Directory and Certificates can be set here. There is also the ability to create additional roles and user accounts for the host itself. This option uses accounts that are local only to the host and not shared with any other hosts or vCenter Server. vCenter Server is set up to use single sign-on which makes account management much easier. This will be reviewed in the lessons that follow.

1. Click on **Security & users**

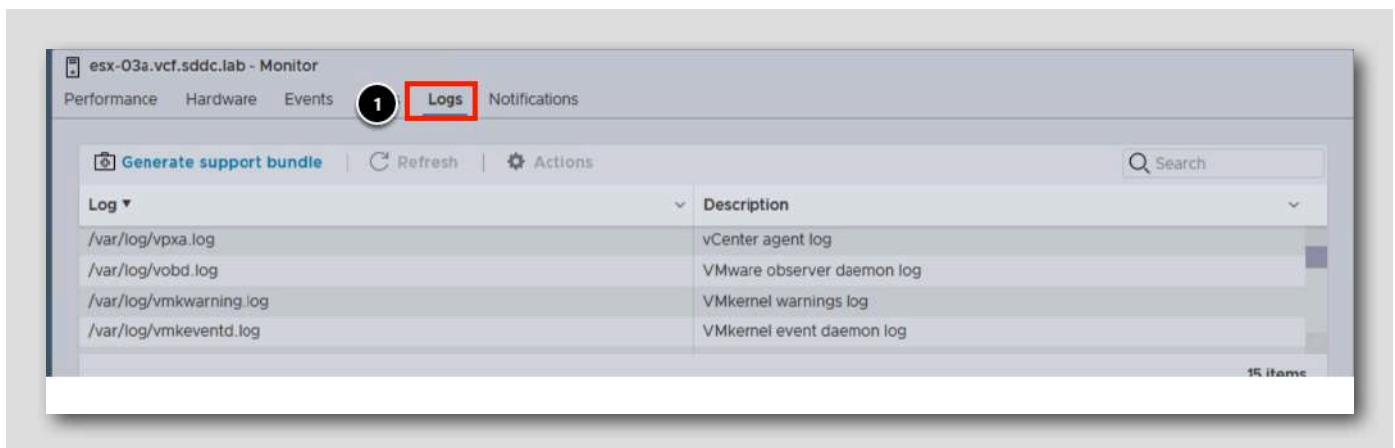
Monitor

[30]



The Monitor section includes Performance Charts, Hardware monitoring, an event log and other useful monitoring information.

1. Click on **Monitor**

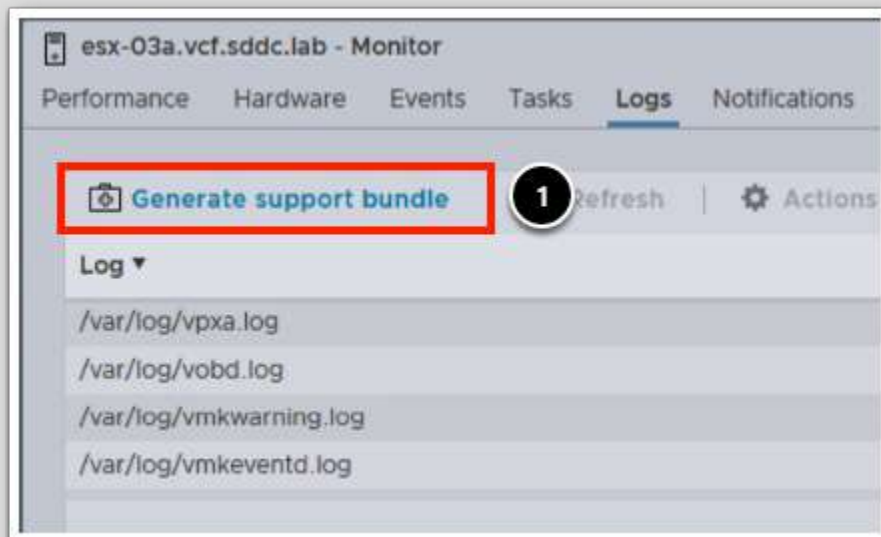


1. Click the **Logs** tab

On the Logs tab, a support bundle can be created that includes log files and system information that can be helpful in troubleshooting issues.

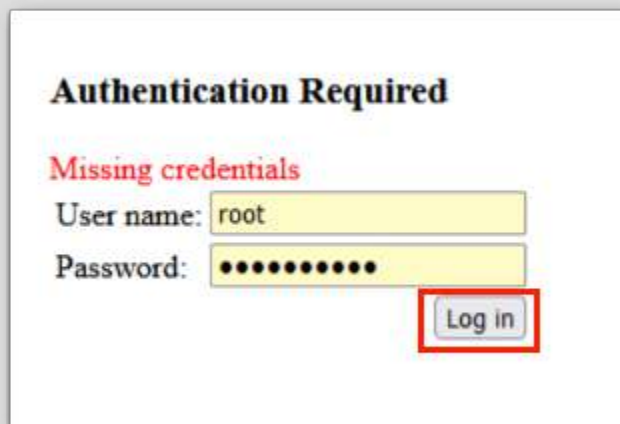
Generate Support Bundle

[31]



1. Click the Generate support bundle button

This operation will automatically download the support file. It will take a couple of minutes.



You may be asked to provide credentials. Use the same information you used to log in:

- Username: root
- Password: VMware123!

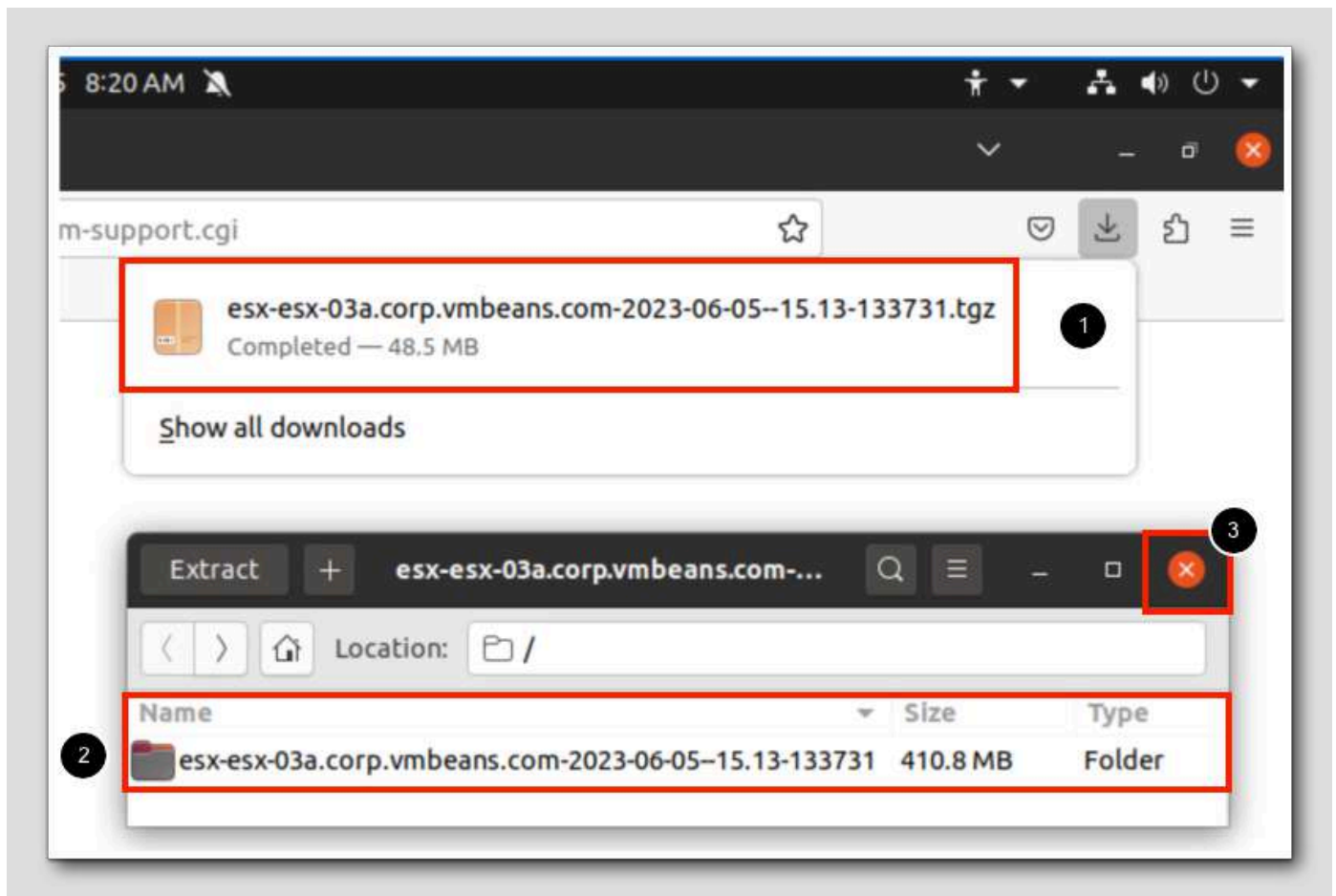
1. Click on **Log in**

Save the Support Bundle

[32]



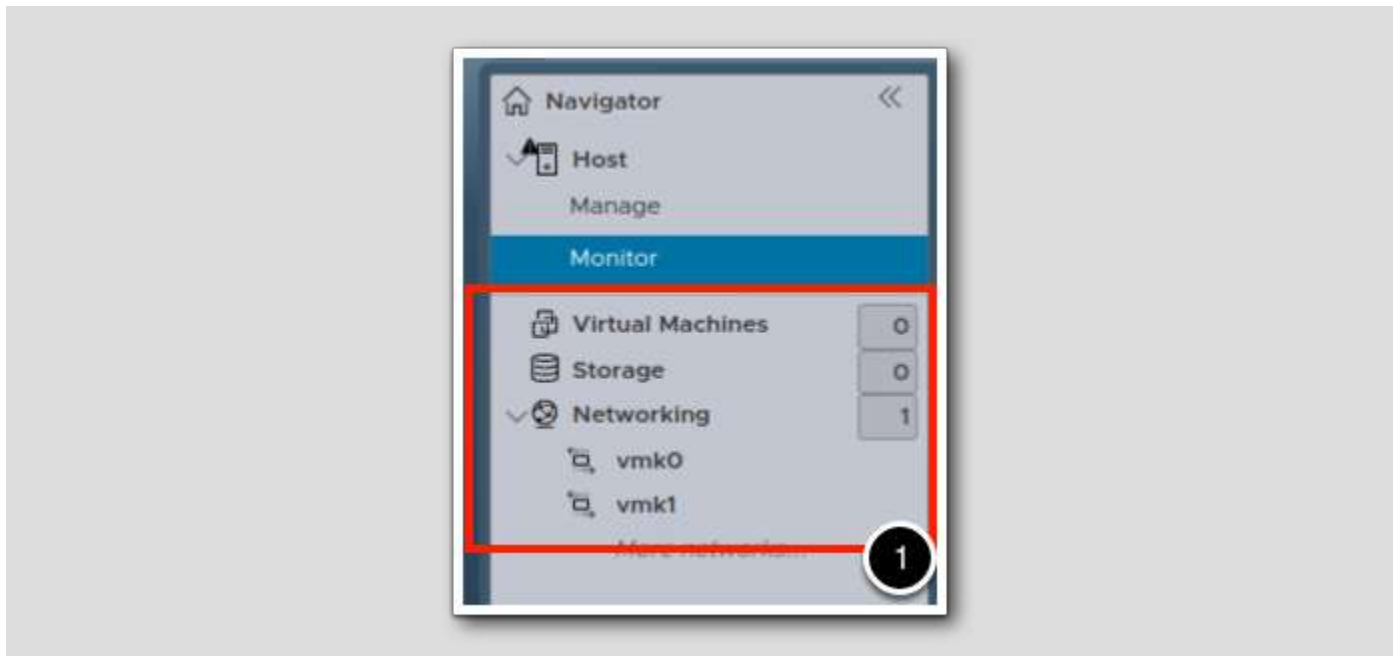
1. Click on **Save** if the pop up shows up.



1. Click on the **downloaded file**
2. A **pop-up window** will appear with the downloaded support file. Review file if needed.
3. **Close window** when finished.

VMs, Storage and Networking

[33]



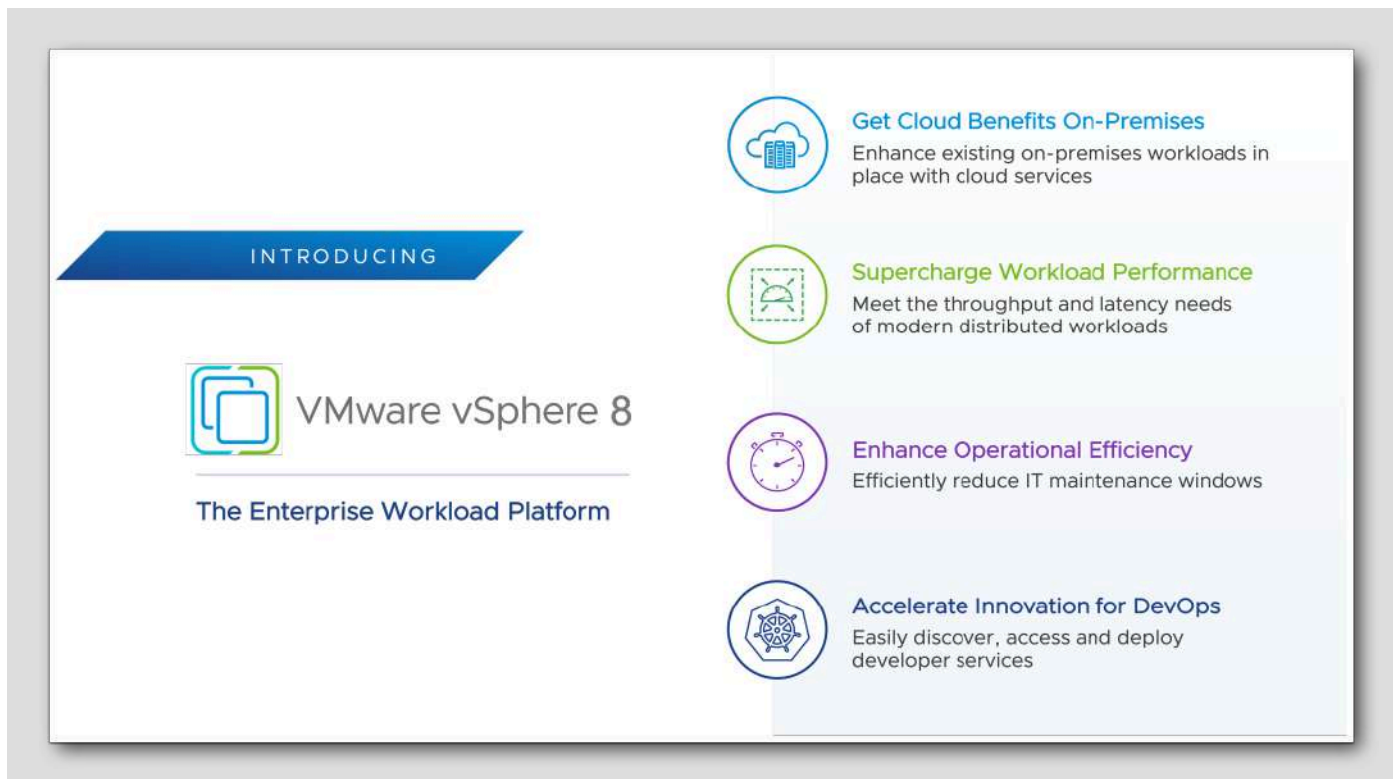
1. In addition to managing and monitoring the host, **Virtual Machines** can be created, **Storage** and **Networking** can be configured at the host level.

Since these features will be covered throughout the lab and the actions performed are identical, just at the vCenter Server level, we will not be reviewing them here.

The ESXi Host Client can be very useful in situations where a vCenter Server is not present to manage the host. However, when a vCenter Server is present, it is the preferred option and provides better tools to manage your infrastructure as a whole.

vCenter 8 Overview

[34]



vCenter Server unifies resources from individual hosts so that those resources can be shared among virtual machines in the entire datacenter. It accomplishes this by managing the assignment of virtual machines to the hosts and the assignment of resources to the virtual machines within a given host based on the policies that the system administrator sets.

VMware vSphere 8 is the enterprise workload platform that brings the benefits of cloud to on-premises workloads. It supercharges performance with DPU and GPU based acceleration, enhances operational efficiency through the VMware Cloud Console, seamlessly integrates with add-on hybrid cloud services, and accelerates innovation with an enterprise-ready integrated Kubernetes runtime that runs containers alongside VMs.

INTRODUCING



VMware vSphere 8

The Enterprise Workload Platform



Get Cloud Benefits On-Premises

Enhance existing on-premises workloads in place with cloud services



Supercharge Workload Performance

Meet the throughput and latency needs of modern distributed workloads



Enhance Operational Efficiency

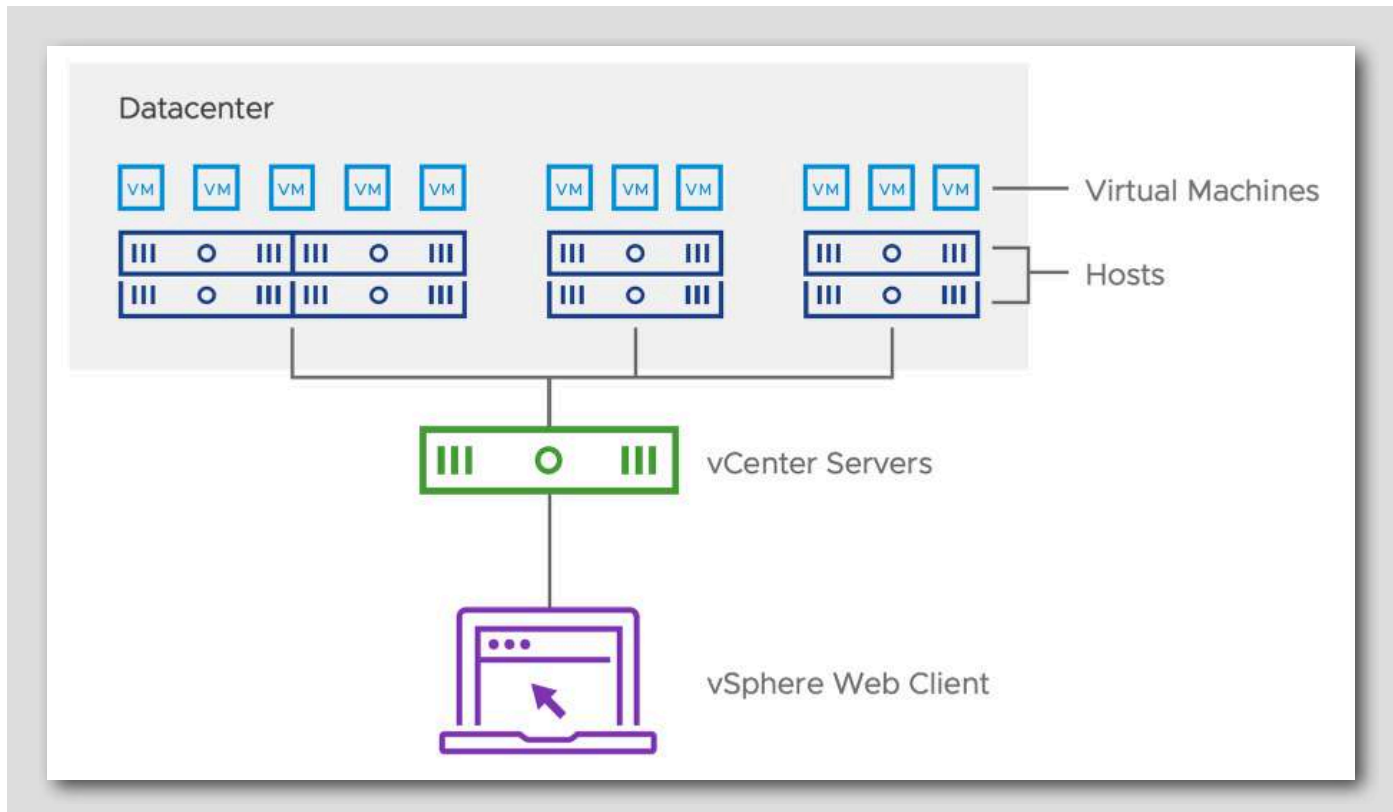
Efficiently reduce IT maintenance windows



Accelerate Innovation for DevOps

Easily discover, access and deploy developer services

vSphere Components

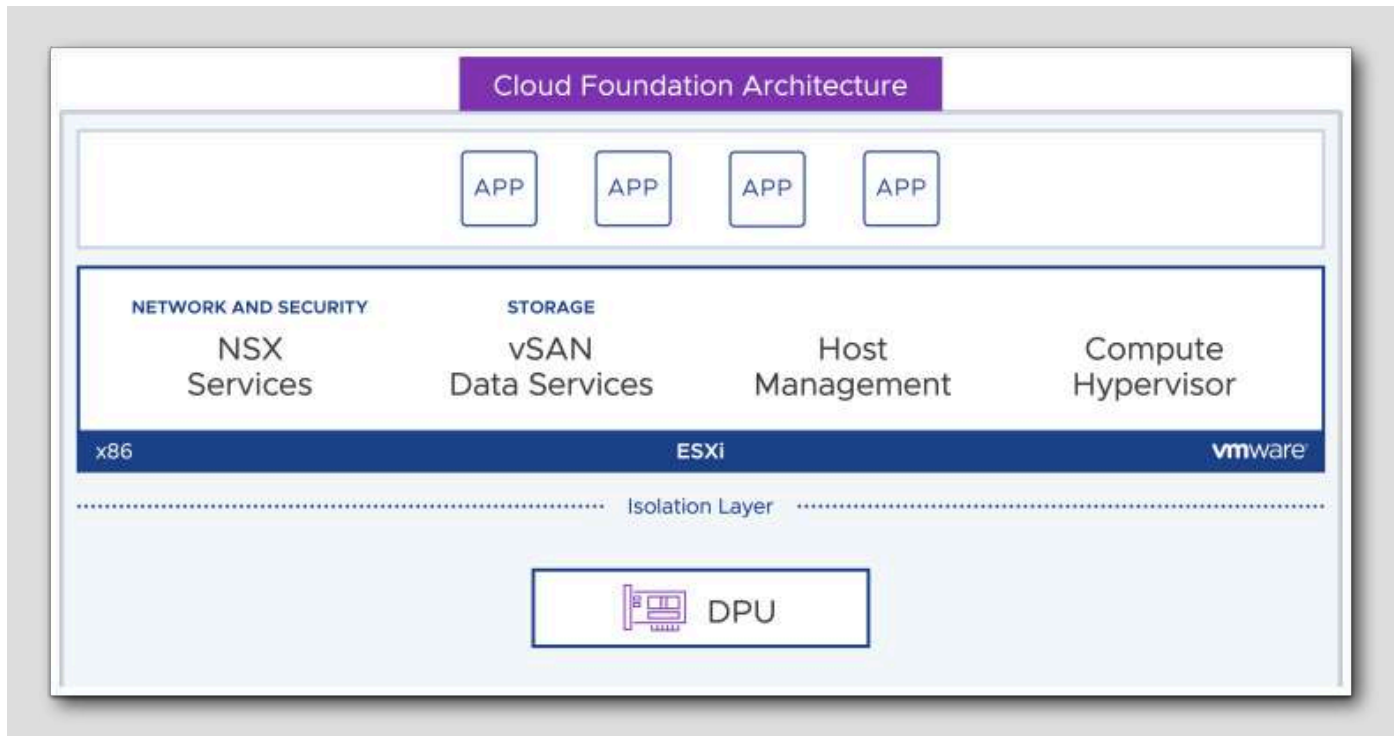


The above diagram shows how vCenter fits in the vSphere stack. With vCenter installed, you have a central point of management. vCenter Server allows the use of advanced vSphere features such as vSphere Distributed Resource Scheduler (DRS), vSphere High Availability (HA), vSphere vMotion, and vSphere Storage vMotion.

The other component is the vSphere Web Client. The vSphere Web Client is the interface to vCenter Server and multi-host environments. It also provides console access to virtual machines. The vSphere Web Client lets you perform all administrative tasks by using an in-browser interface.

Data Processing Units

[36]

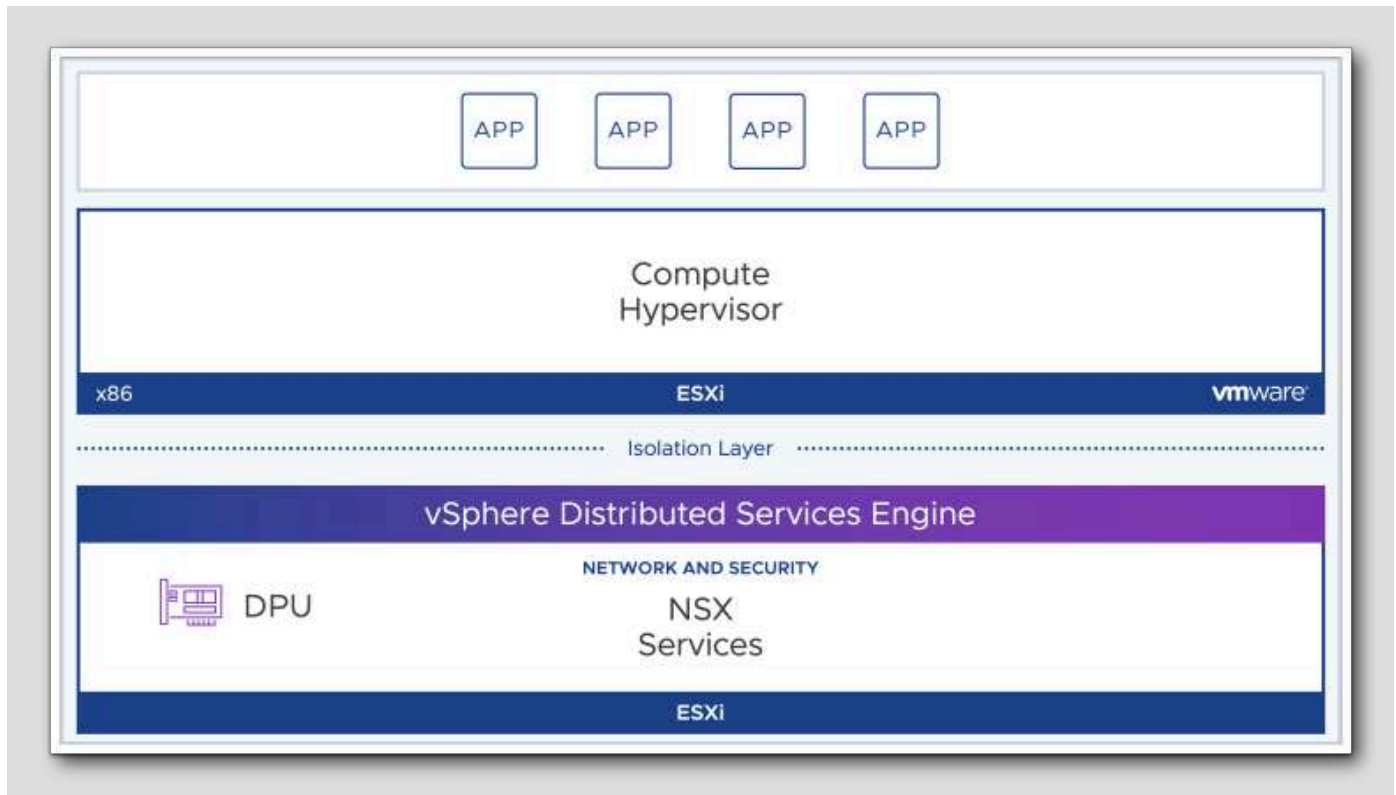


vSphere Distributed Services Engine unlocks the power of Data Processing Units (DPUs) for hardware accelerated data processing to improve infrastructure performance, boost infrastructure security and simplify DPU lifecycle management. vSphere 8 makes using DPUs easy for workloads to take advantage of these benefits.

Data Processing Units (DPU) exist today and live in the hardware layer, similar to a PCIe device like a NIC or GPU. Today networking, storage and host management services run in the instance of ESXi virtualizing the x86 compute layer.

vSphere Distributed Services Engine

[37]



In vSphere 8, an additional instance of ESXi is installed directly on the Data Processing Unit. This allows ESXi services to be offloaded to the DPU for increased performance.

In vSphere 8 GA, we support greenfield installations with support for network offloading with NSX. vSphere Distributed Services Engine is life-cycle managed using vSphere Lifecycle Manager. When remediating a host that contains a DPU ESXi installation, the DPU ESXi version is always remediated with the parent host and kept in version lock-step.

Video: Getting Started with vSphere Distributed Services Engine (9:55)

[38]

https://www.youtube.com/watch?v=7_vl4_K-Ht8



Discover how VMware is leveraging new hardware technology with Data Processing Units; to boost performance, security, and consolidation ratios among other advantages.

vCenter Single Sign On

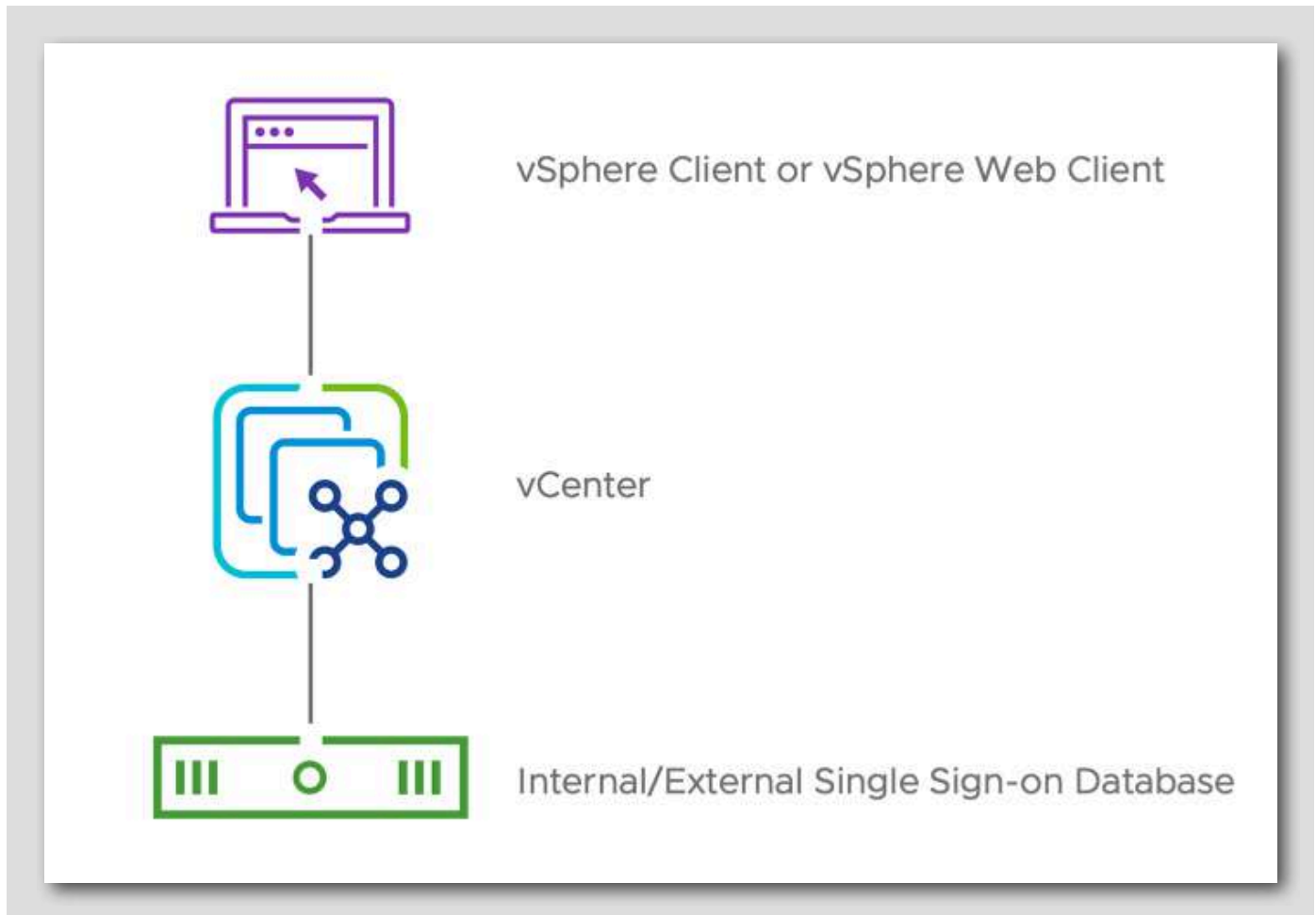
[39]

vSphere 5.1 introduced vCenter Single Sign On (SSO) as part of the vCenter Server management infrastructure. This change affects the vCenter Server installation, upgrading, and operation. Authentication by vCenter Single Sign On makes the VMware cloud infrastructure platform more secure by allowing the vSphere software components to communicate with each other through a secure token exchange mechanism, instead of requiring each component to authenticate a user separately with a directory service like Active Directory.

Beginning in vSphere 7.0, deploying a new vCenter Server or upgrading to vCenter Server 7.0 requires the use of the vCenter Server appliance, a preconfigured virtual machine optimized for running vCenter Server. The new vCenter Server contains all Platform Services Controller services, preserving the functionality and workflows, including authentication, certificate management, tags, and licensing. It is no longer necessary nor possible to deploy and use an external Platform Services Controller. All Platform Services Controller services are consolidated into vCenter Server, and deployment and administration are simplified.

vCenter Single Sign On - Typical Deployment

[40]



Starting with version 5.1, vSphere includes a vCenter Single Sign-On service as part of the vCenter Server management infrastructure.

Authentication with vCenter Single Sign-On makes vSphere more secure because the vSphere software components communicate with each other by using a secure token exchange mechanism, and all other users also authenticate with vCenter Single Sign-On.

When a user logs in to a vSphere component, or when vCenter Server solution user accesses another vCenter Server service, vCenter Single Sign-On performs authentication. Users must be authenticated with vCenter Single Sign-On and have the necessary privileges for interacting with vSphere objects.

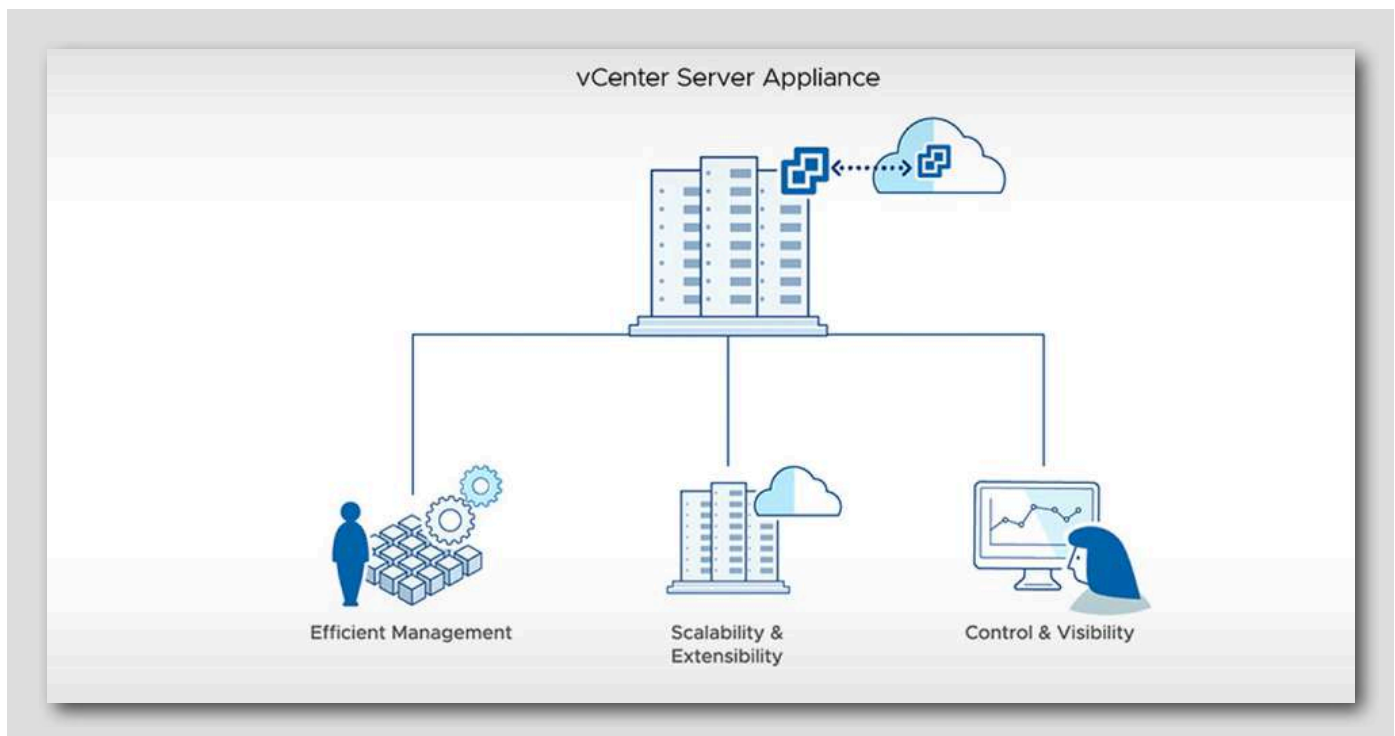
vCenter Single Sign-On authenticates both solution users and other users.

- Solution users represent a set of services in your vSphere environment. During installation, VMCA assigns a certificate to each solution user by default. The solution user uses that certificate to authenticate to vCenter Single Sign-On. vCenter Single Sign-On gives the solution user a SAML token, and the solution user can then interact with other services in the environment.
- When other users log in to the environment, for example, from the vSphere Client, vCenter Single Sign-On prompts for a user name and password. If vCenter Single Sign-On finds a user with those credentials in the corresponding identity source, it assigns the user a SAML token. The user can now access other services in the environment without being prompted to authenticate again.

Which objects the user can view, and what a user can do, is determined by vCenter Server permission settings.

Overview of the vCenter Server Appliance

[41]



The vCenter Server appliance is a preconfigured virtual machine that is optimized for running vCenter Server and the associated services. The vCenter Server appliance package contains the following software:

- Photon OS 3.0
- The vSphere authentication services
- PostgreSQL
- VMware vSphere Lifecycle Manager Extension
- VMware vCenter Lifecycle Manager

Version 8.0 of vCenter Server is deployed with virtual hardware version 10, which supports 64 virtual CPUs per virtual machine in ESXi. During the deployment, you can choose the vCenter Server appliance size for your vSphere environment size and the storage size for your database requirements.

For more information regarding vCenter installation options and capabilities, please see the [vCenter Server Installation and Setup](#) guide.

More Information on Single Sign On

[42]

The second Module in this lab, Introduction to vSphere Networking and Security covers SSO in more detail.

However, you can also refer to the [vSphere Authentication Guide](#) for more in-depth requirements and considerations for SSO architecture in vCenter 7.

vCenter Server and Creating a Virtual Machine

[43]

The previous lesson reviewed the ESXi Host Client, which can be used to manage one ESXi host at a time. This lesson will introduce the vSphere Client which is used to connect to vCenter Server to manage your collective infrastructure as a whole. In addition, the process of creating a virtual machine will also be covered.

The vSphere Client is the primary method for system administrators and end-users to interact with the virtual data center environment created by VMware vSphere. vSphere manages a collection of objects that make up the virtual data center, including hosts, clusters, virtual machines, data storage, and networking resources.

The vSphere Client is a Web browser-based application that you can use to manage, monitor, and administer the objects that make up your virtualized data center. You can use the vSphere Client to observe and modify the vSphere environment in the following ways.

- Viewing health, status, and performance information on vSphere objects
- Issuing management and administration commands to vSphere objects
- Creating, configuring, provisioning, or deleting vSphere objects

You can extend vSphere in different ways to create a solution for your unique IT infrastructure. You can extend the vSphere Client with additional GUI features to support these new capabilities, with which you can manage and monitor your unique vSphere environment.

Launch Firefox

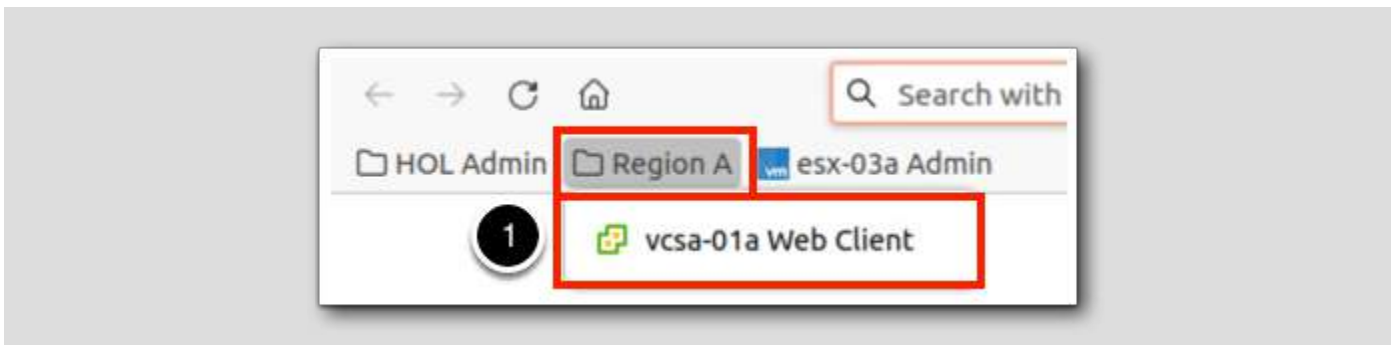
[44]



If you are not already in **Firefox**, click on Firefox on your desktop. If you are already in Firefox, **open a new tab**.

Select vSphere Client

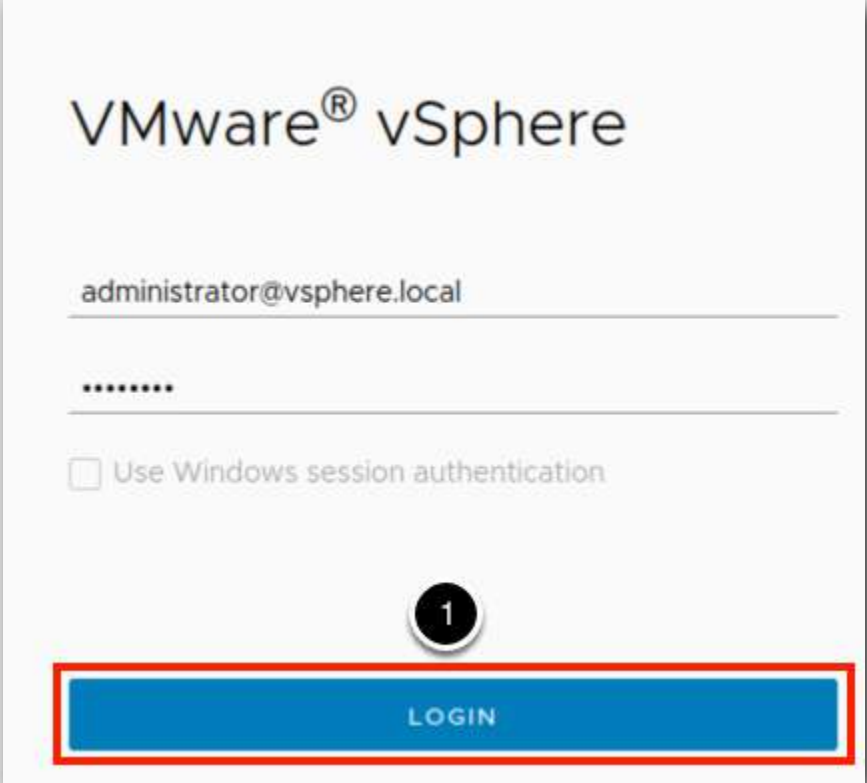
[45]



1. Under the Region A Bookmark's Folder, click the vSphere Web Client bookmark.

Login to vCenter

[46]



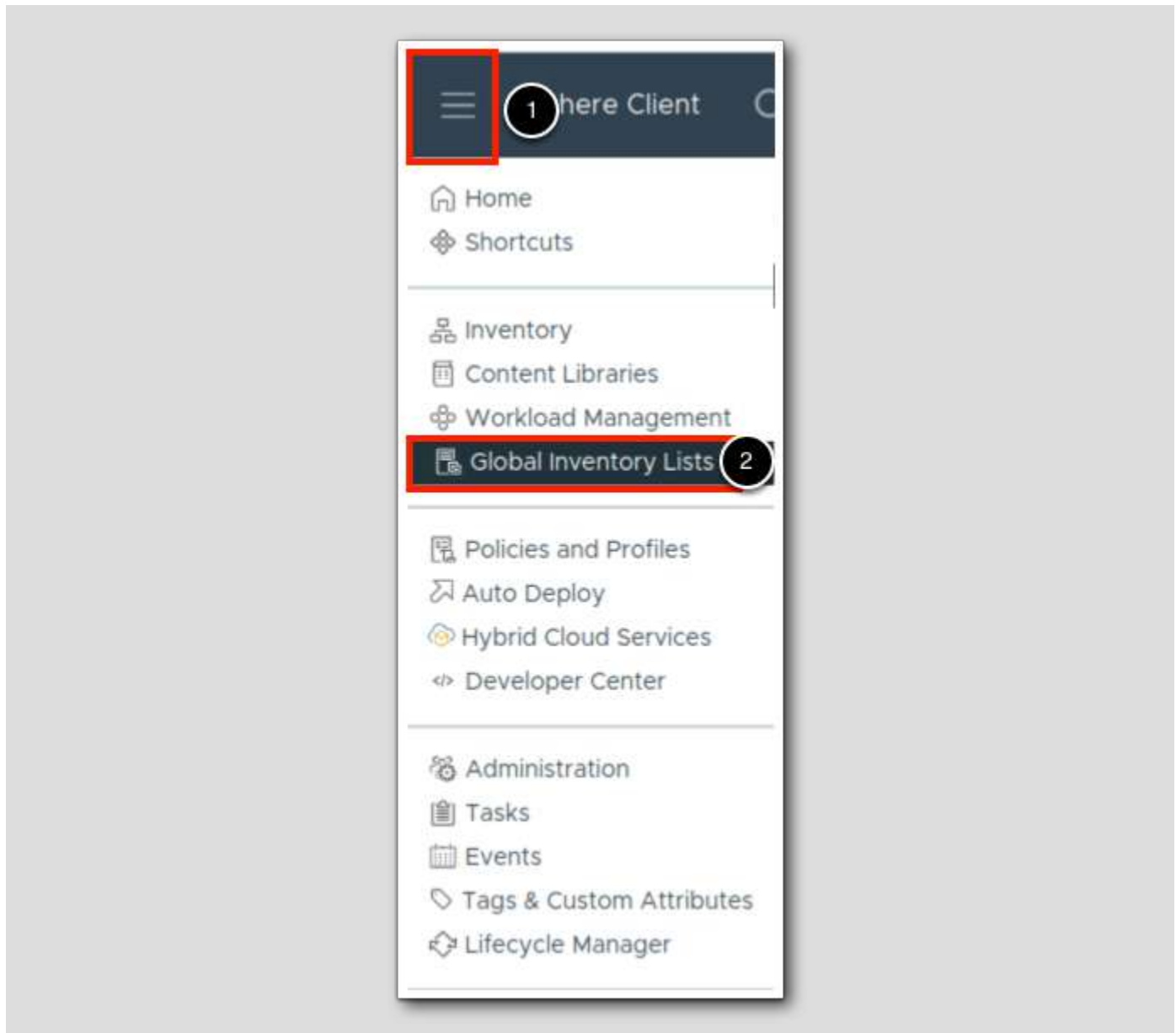
The image shows the VMware vSphere login interface. At the top, the text "VMware® vSphere" is displayed. Below this, there are two input fields: the first contains the username "administrator@vsphere.local" and the second contains a masked password "*****". Below the password field is a checkbox labeled "Use Windows session authentication". At the bottom of the form is a blue "LOGIN" button, which is highlighted with a red rectangular border. A small black circle with the number "1" is positioned above the "LOGIN" button.

If your 'LOGIN' button is grey or cannot be click, you can click on the username and pasword fields, this should enable your 'LOGIN' button.

1. Click on **Login** to proceed - The credentials should already be pre-populated

vCenter Inventory

[47]



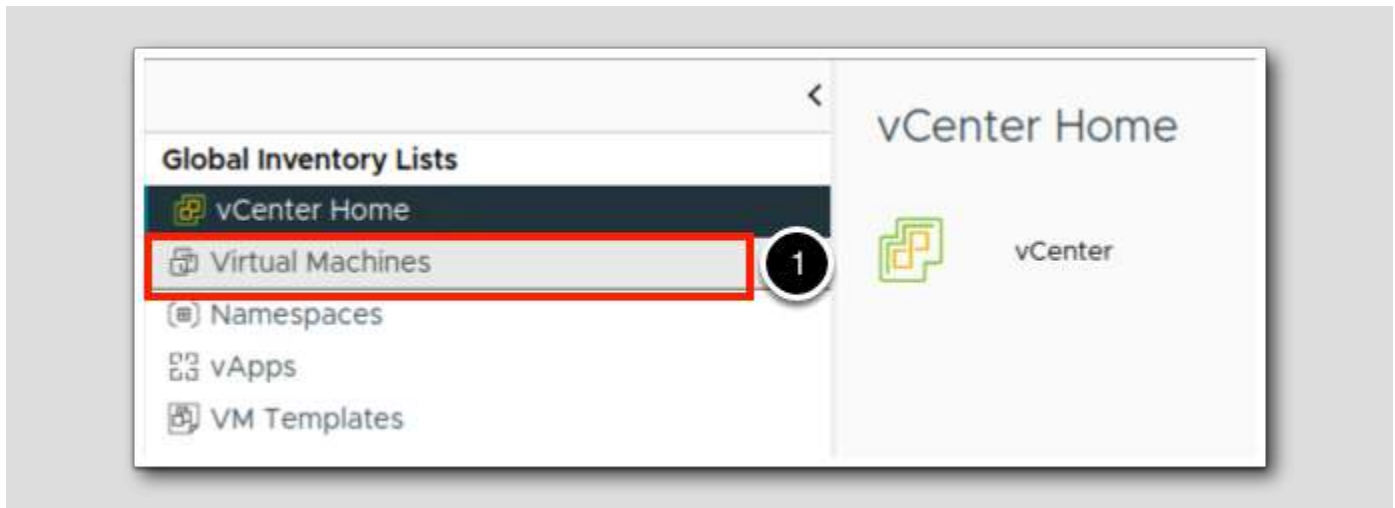
By default, you are brought to a view that shows the Hosts and Clusters attached to vCenter. Get a more complete look by viewing the Global Inventory Lists.

1. Click on the **Menu** hamburger icon on the top left and select **Global Inventory Lists**.

Clicking Global Inventory Lists will take you to the inventory page where you find all the objects associated with vCenter Server systems such as data centers, hosts, clusters, networking, storage, and virtual machines.

Child objects, Data Centers, and Hosts

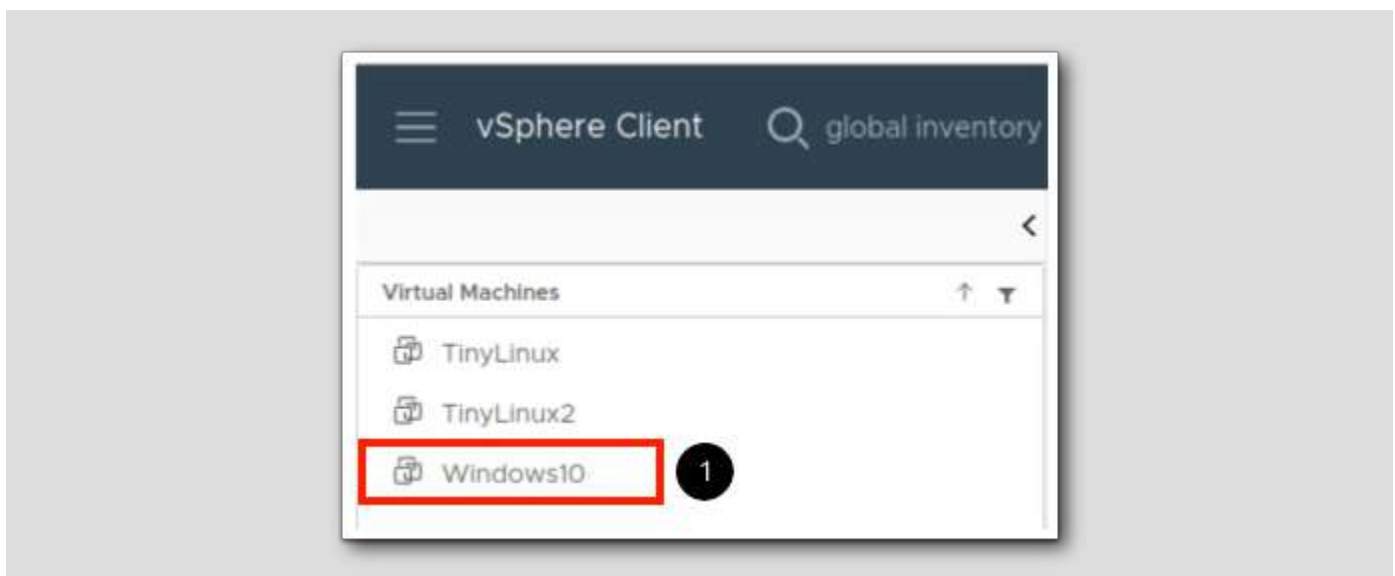
[48]



1. Click the "Virtual Machines" inventory item. By selecting this inventory item, you are presented with a list of the VMs which are located in this environment.

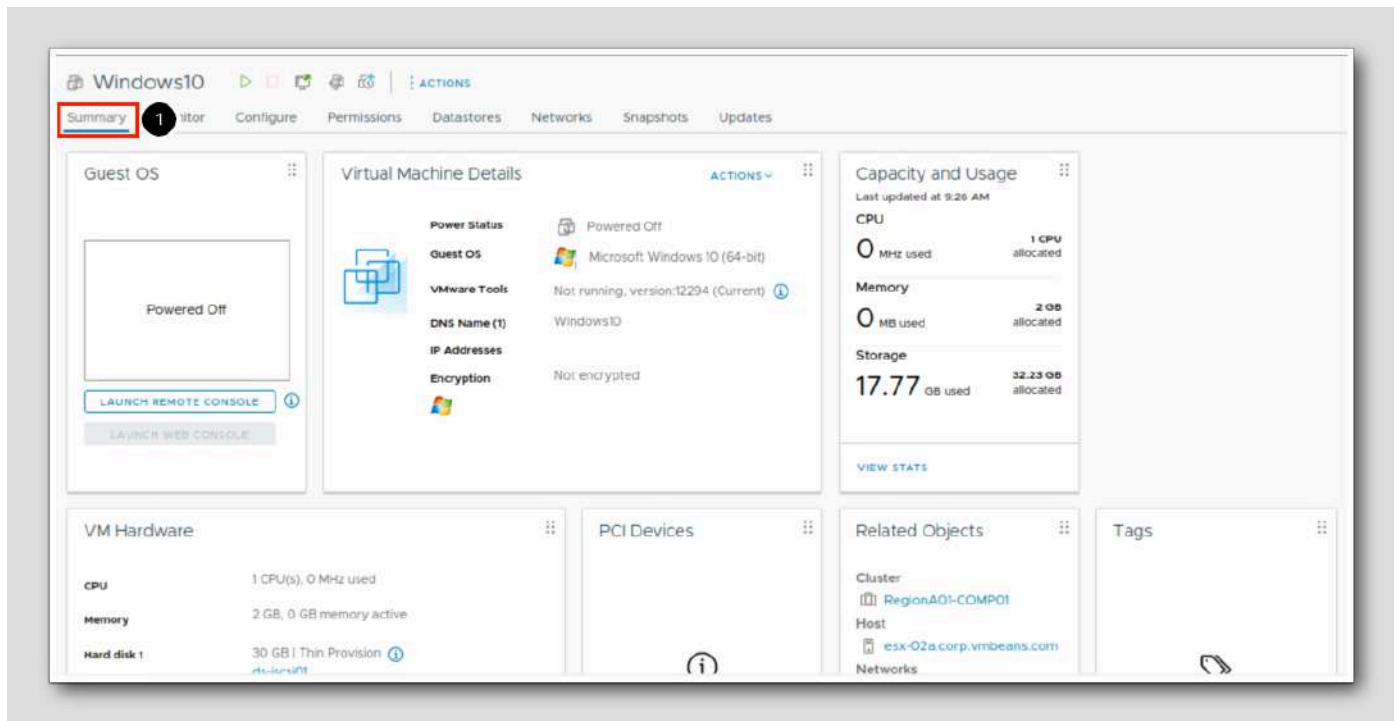
Virtual Machine Summary

[49]



Here are all the virtual machines associated with this vCenter instance.

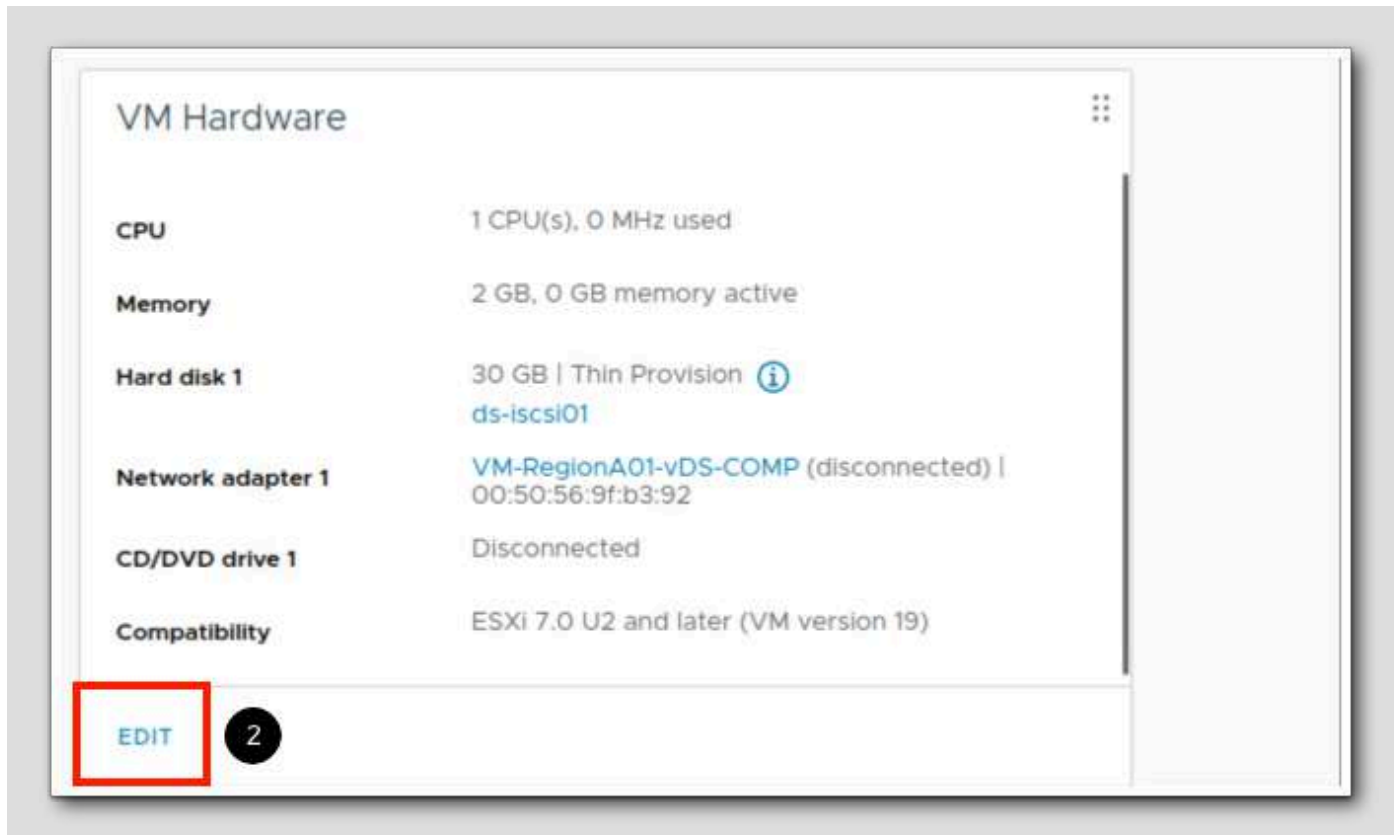
1. Click the "Windows10" virtual machine.



1. You will land in the "Summary" Tab for that virtual machine. In this page, you are able to see all the details regarding the virtual machine.

Edit the Settings for the Virtual Machine

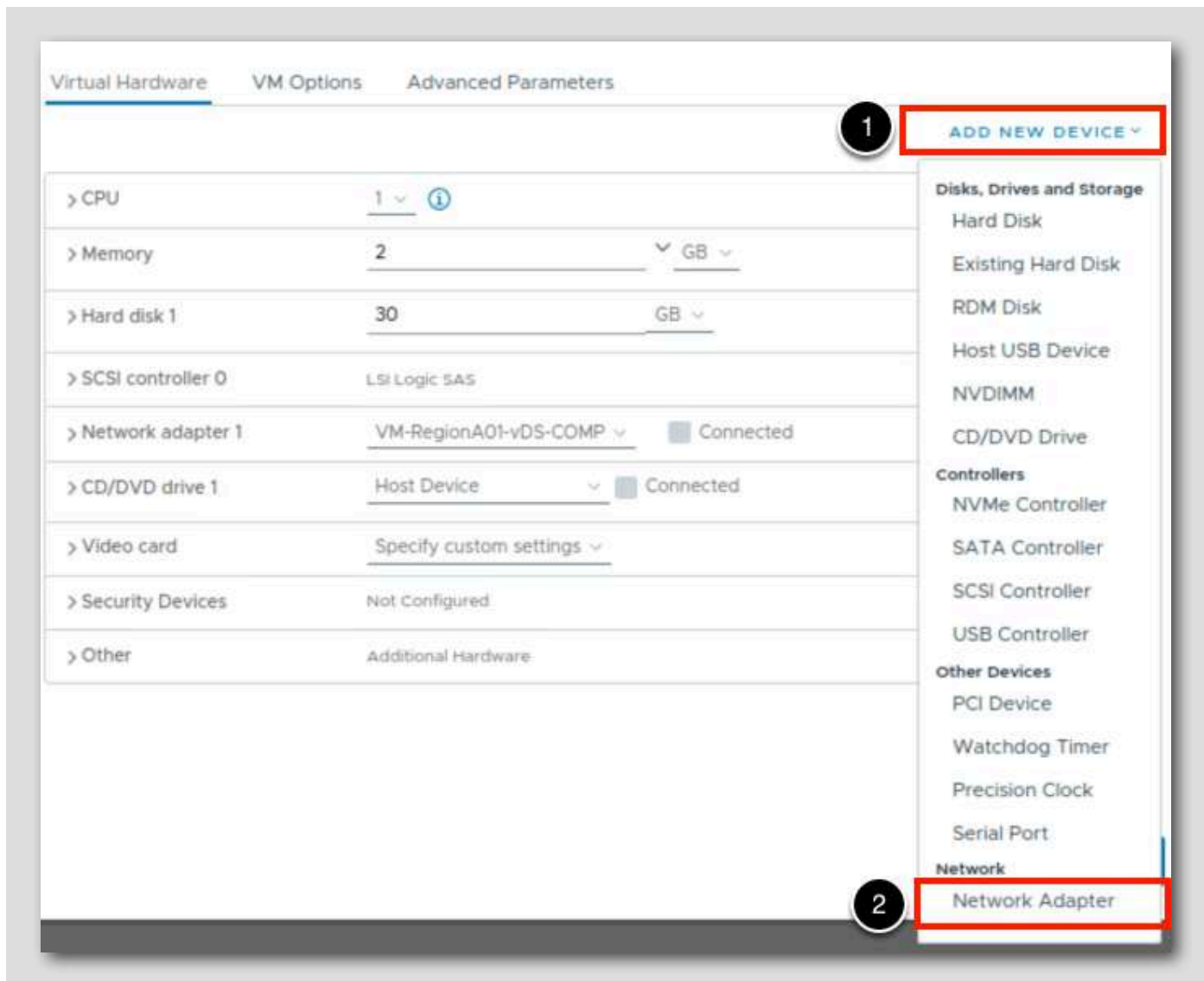
[50]



1. Review the VM Hardware details for the windows10 virtual machine. Note that there is currently only one network adapter.
2. Click on Edit so a second network adapter can be added to the virtual machine. You may need to scroll down to see this option.

Add a Second Network Adapter

[51]

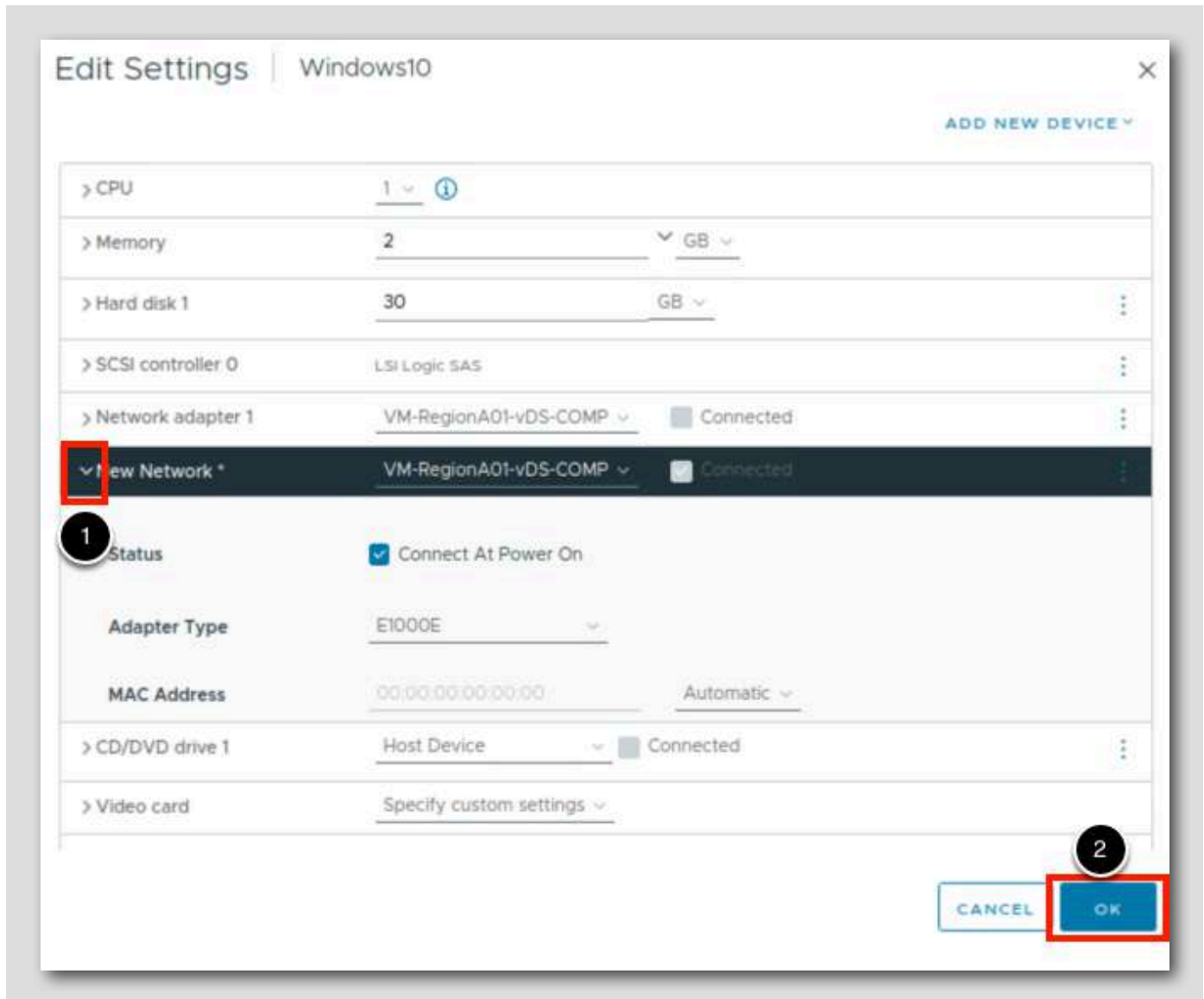


Add another network adapter to the windows10 machine.

1. In the Edit settings window, click the **Add New Device** drop-down menu.
2. Select **Network Adapter** from the list. You may need to scroll down to see the option.

Configure the Second Network Card

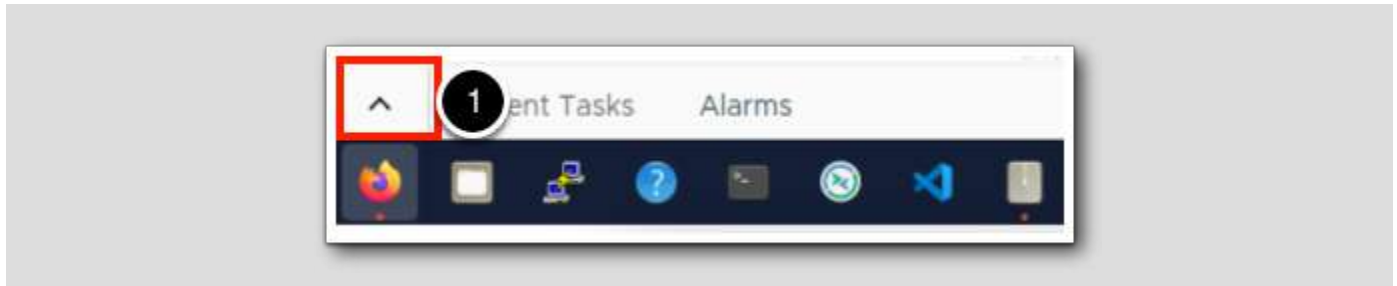
[52]



1. Click the arrow next to the **New Network** to expand and view its settings. Notice that the MAC address is blank at this point. A new MAC address will be generated once this NIC is added or we are able to specify (with some rules) our own MAC address.
2. Click on **OK** to add the device to the VM. When you select **OK** a new task will be created.

Recent Tasks List (1/2)

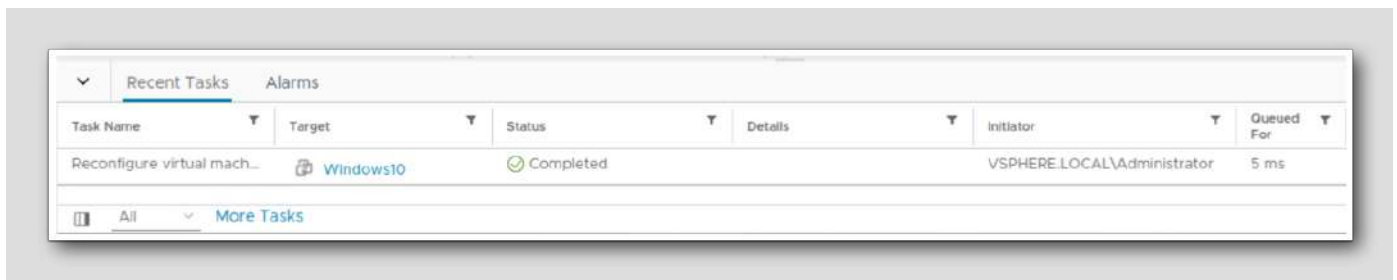
[53]



1. Click on on the **Arrow** to view the **Recent Tasks** to watch the task's progress.

Recent Tasks List (2/2)

[54]

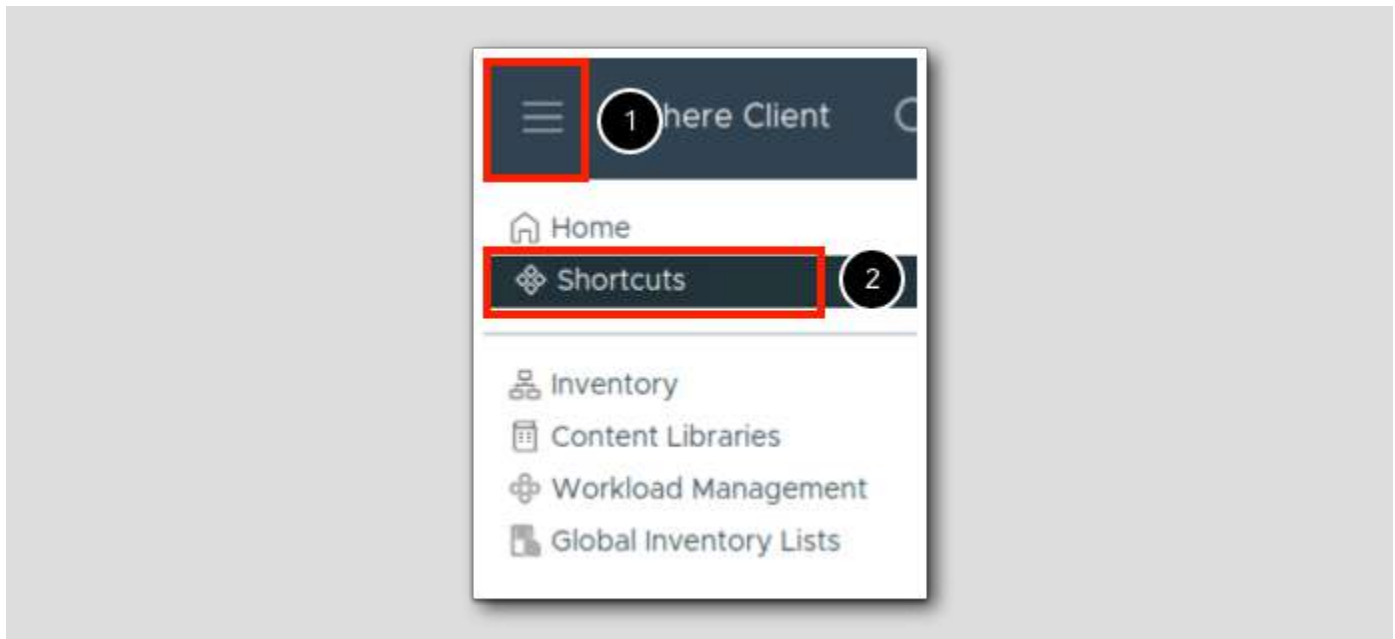


Review the "Recent Tasks" list. Once the task is complete, a second Network Adapter should be shown in the "VM Hardware" section. Note the networks are in a disconnected state because the VM is powered off.

Once you are done viewing the Recent Tasks list, click the down-arrows to minimize it.

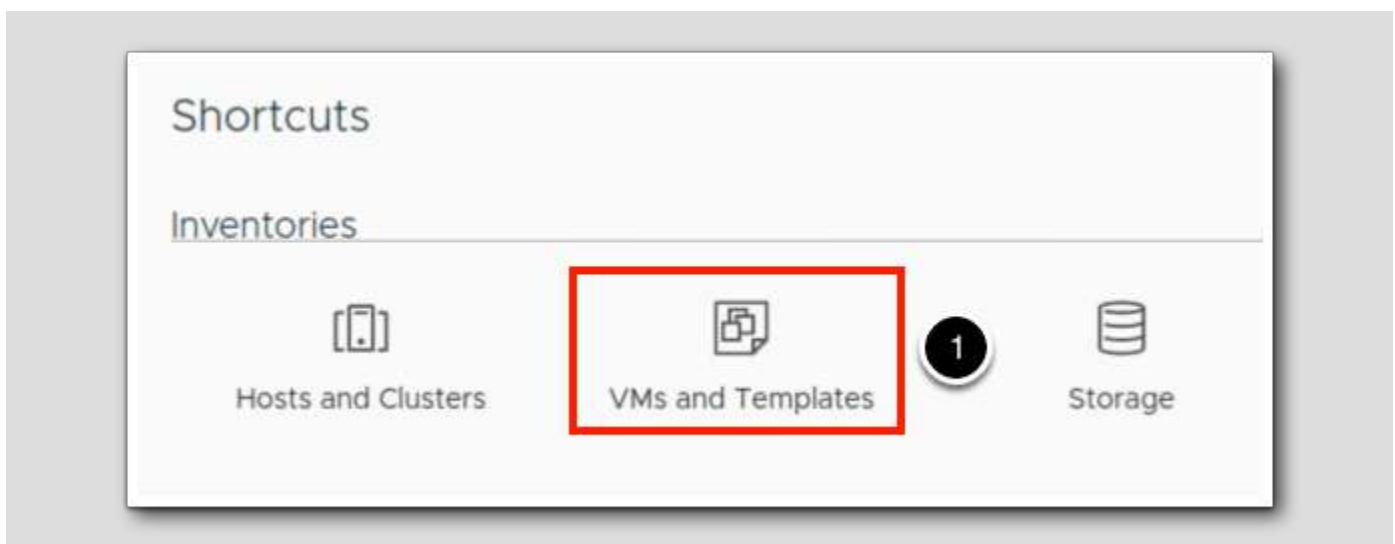
Create a Virtual Machine

[55]



In the next steps, we will create a virtual machine and then, install an operating system.

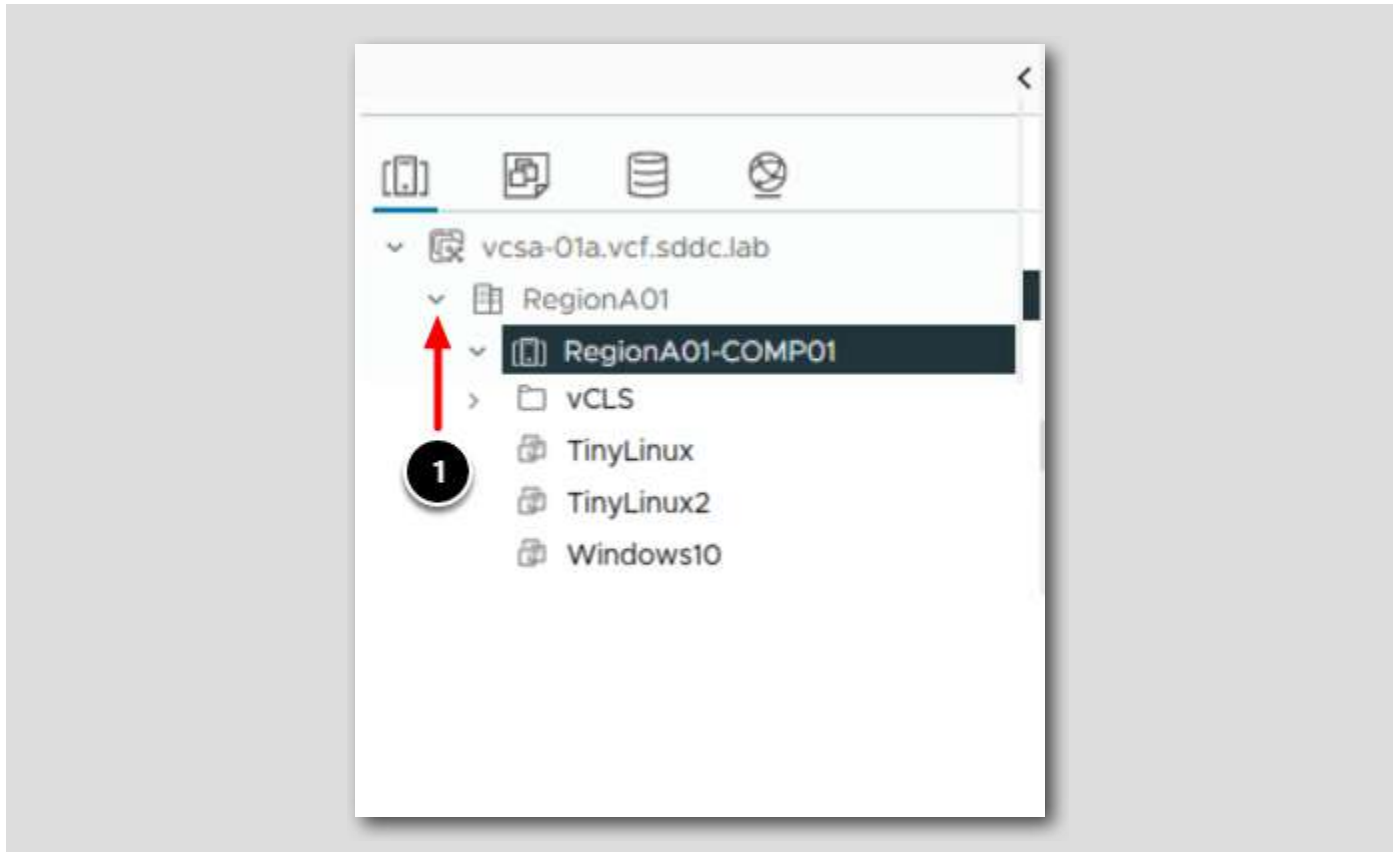
1. To return to the VMs and Templates view, click on **Menu**.
2. Click into **Shortcuts**



1. Select VMs and Templates.

Select and Expand Datacenter

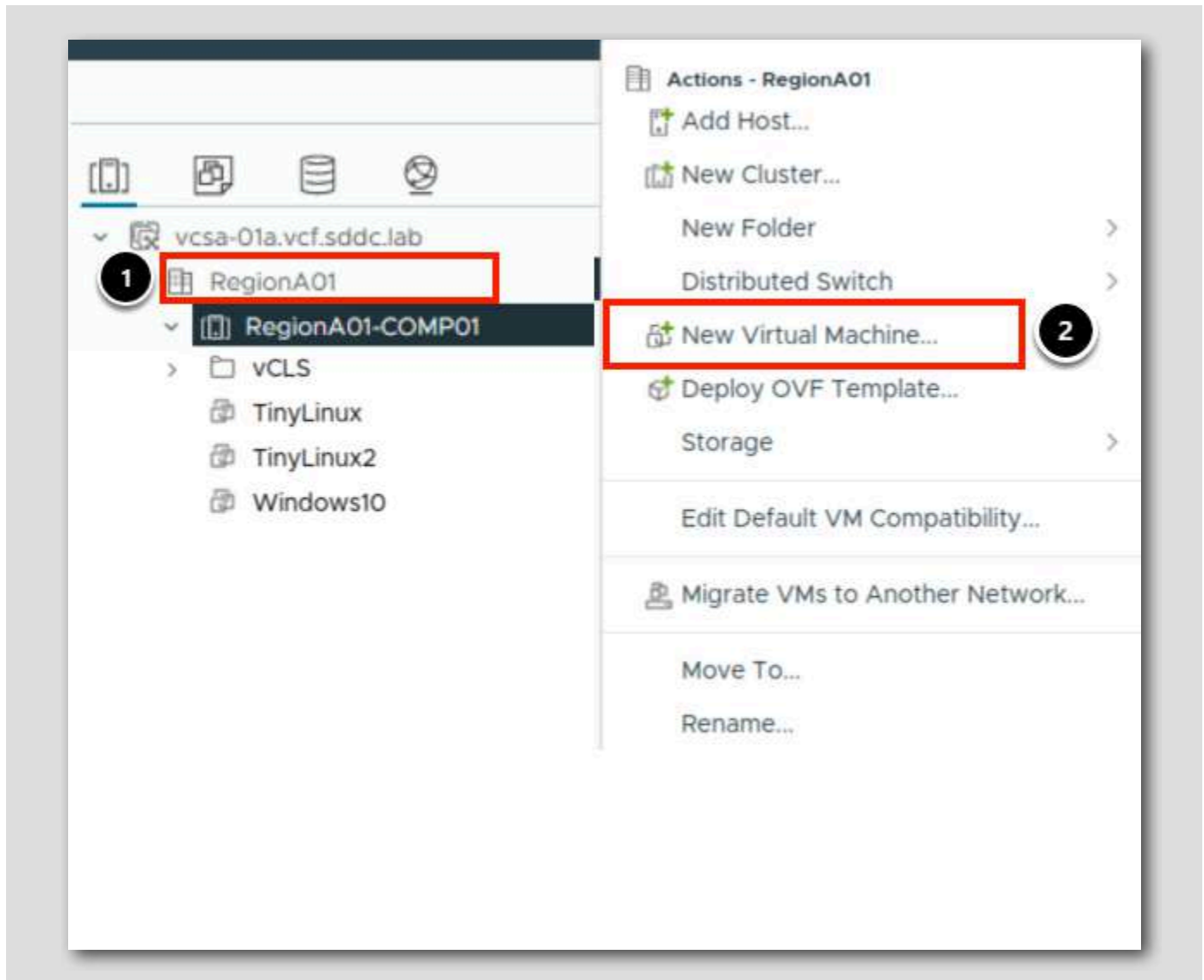
[56]



1. Expand **RegionA01** Datacenter if you cannot see the virtual machines under it can be seen.

Start the New Virtual Machine Wizard

[57]

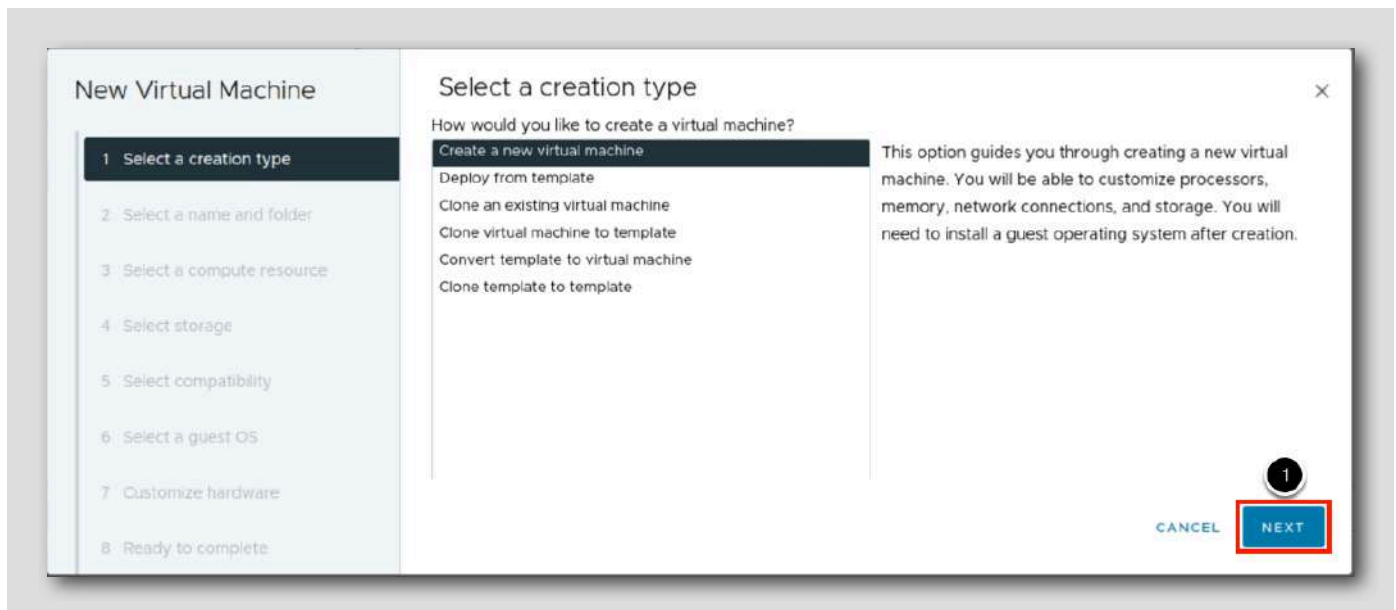


1. Right-click on **RegionA01** Datacenter.
2. Click **New Virtual Machine** to start the new virtual machine wizard.

This wizard is used to create a new Virtual Machine and place it in the vSphere inventory.

Virtual Machine Wizard

[58]



1. Since the **Create a new virtual machine** wizard is highlighted, just click **Next**.

Name the Virtual Machine

[59]

Select a name and folder

Specify a unique name and target location

Virtual machine name:

Select a location for the virtual machine.

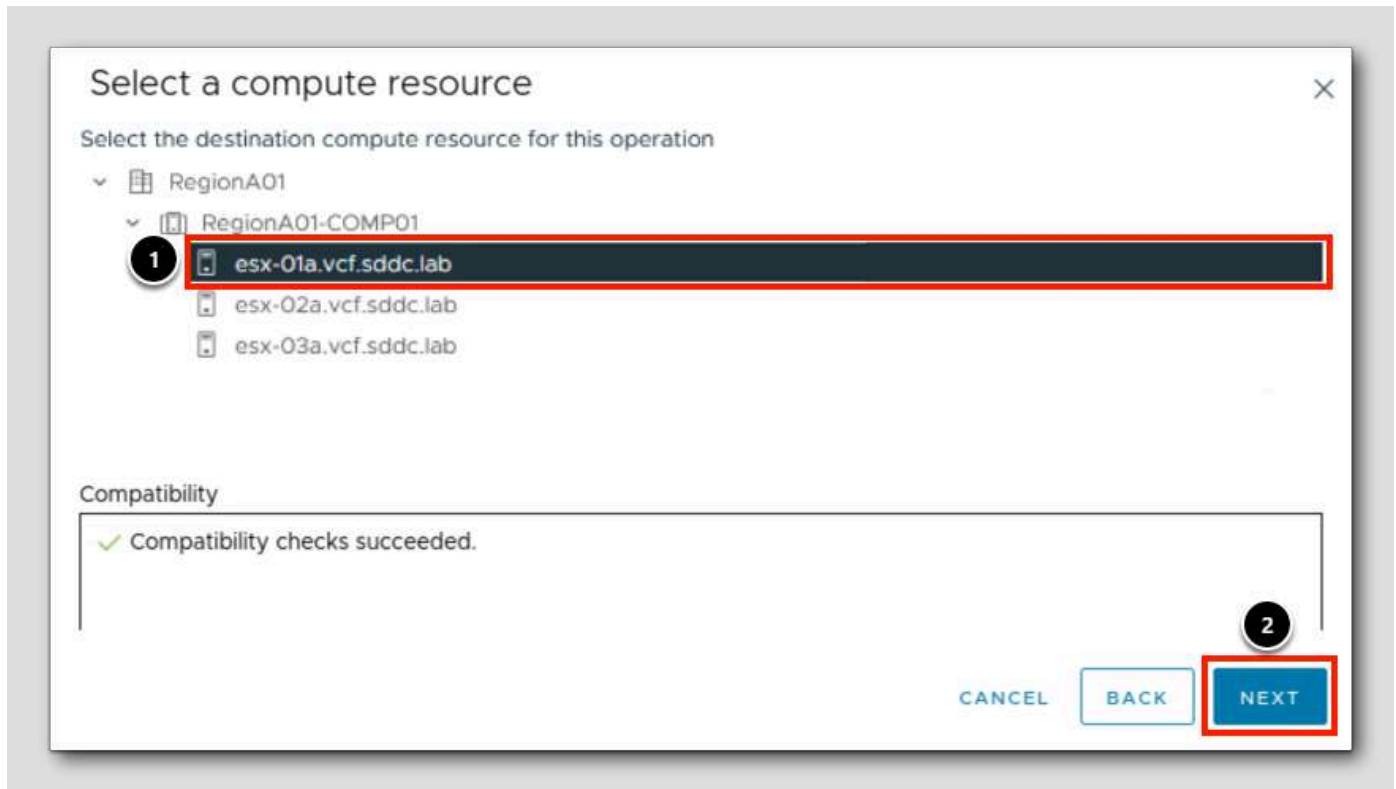
- ▼ vcsa-01a.vcf.sddc.lab
 - > RegionA01

CANCEL BACK NEXT

1. Enter **web-serv01** for the name of the new virtual machine.
2. Click **Next**.

Virtual Machine Placement

[60]



Because Distributed Resource Scheduler (DRS) is not enabled, you just have to select a host to use for the VM. More details on DRS will be covered later in this module.

1. Expand **RegionA01** and **RegionA01-COMP01**, and select **esx-01a.corp.vmbeans.com**
2. Click **Next**.

Select Storage

[61]

Select storage

Select the storage for the configuration and disk files

☐ Encrypt this virtual machine (Requires Key Management Server)

VM Storage Policy Datastore Default

☐ Disable Storage DRS for this virtual machine

	Name	Storage Compatibility	Capacity	Provisioned	Free	Type	Cluster
1	ds-iscsi01	--	84.75 GB	39.29 GB	61.28 GB	VMFS 6	
	ds-nfs01	--	5.78 GB	88 KB	5.78 GB	NFS v3	

Compatibility

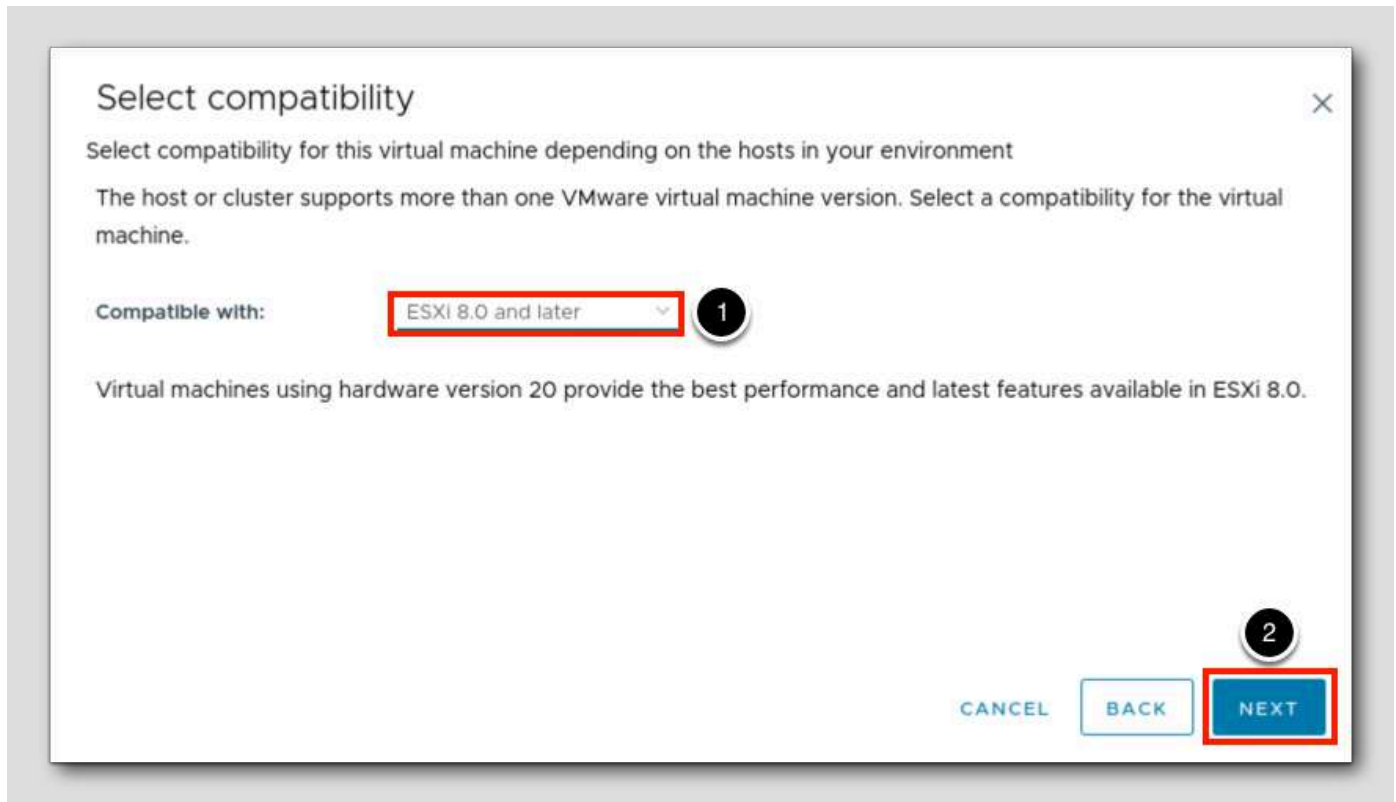
✓ Compatibility checks succeeded.

CANCEL BACK 2 NEXT

1. Select ds-iscsi01 datastore as the storage option.
2. Click Next.

Compatibility

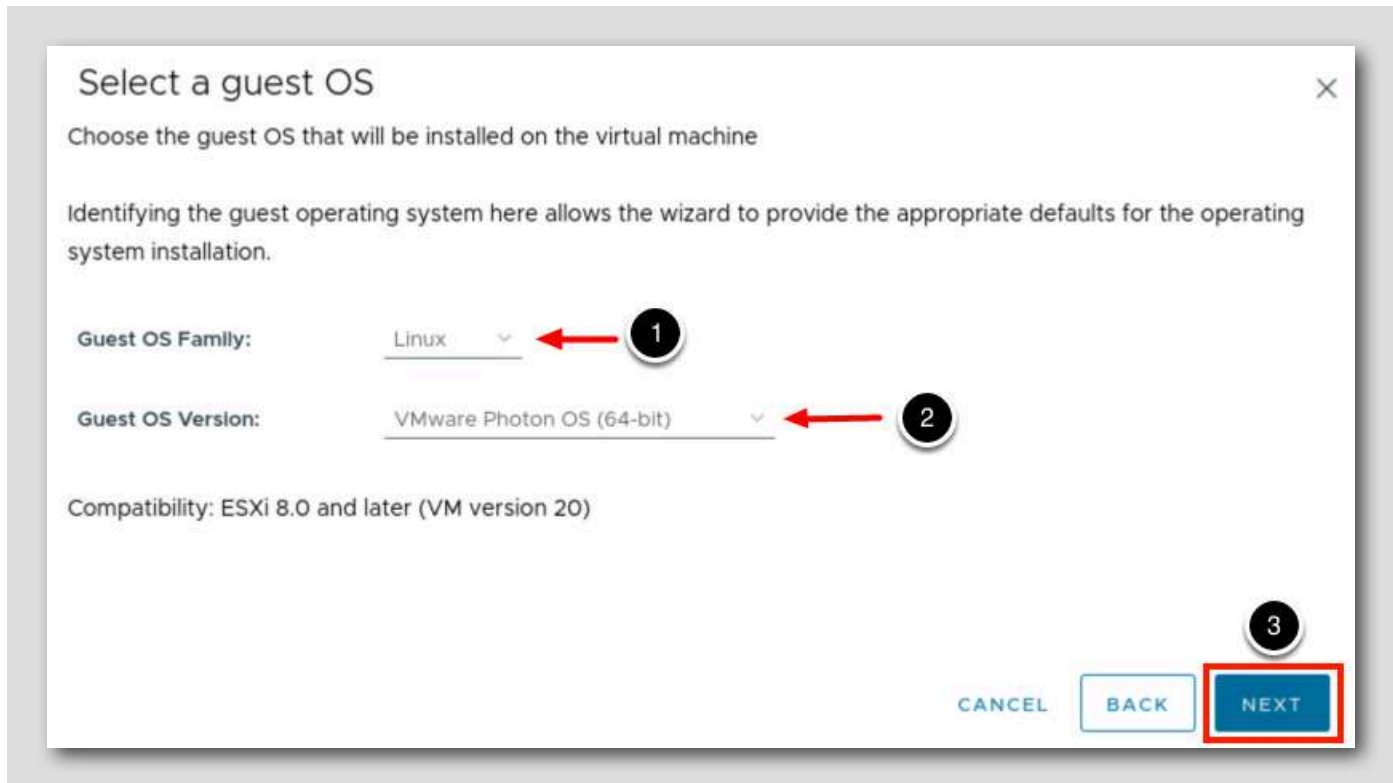
[62]



1. Select ESXi 8.0 and later.
2. Click **Next** to accept.

Guest OS

[63]



In this step, we will be selecting what operating system we will be installing. When we select the operating system, the supported virtual hardware and recommended configuration is used to create the virtual machine. Keep in mind this does not create a virtual machine with the operating system installed, but rather creates a virtual machine that is tuned appropriately for the operating system you have selected.

1. For the **Guest OS Family**, select Linux from the drop-down menu.
2. For the **Guest OS Version**, select VMware Photon OS (64-bit).
3. Click **Next** to continue.

Customize Hardware

The screenshot shows the 'Customize hardware' dialog box. At the top right is a close button (X) and a link 'ADD NEW DEVICE'. The hardware list includes:

- CPU: 1 (with an info icon)
- Memory: 2 GB
- New Hard disk *: 16 GB (highlighted with a red box and a '1' in a circle)

Below the hardware list, the following settings are shown:

- Maximum Size: 64.97 GB
- VM storage policy: Datastore Default
- Location: Store with the virtual machine
- Disk Provisioning: Thin Provision (highlighted with a red box and a '2' in a circle)

At the bottom right, there are three buttons: 'CANCEL', 'BACK', and 'NEXT' (highlighted with a red box and a '3' in a circle).

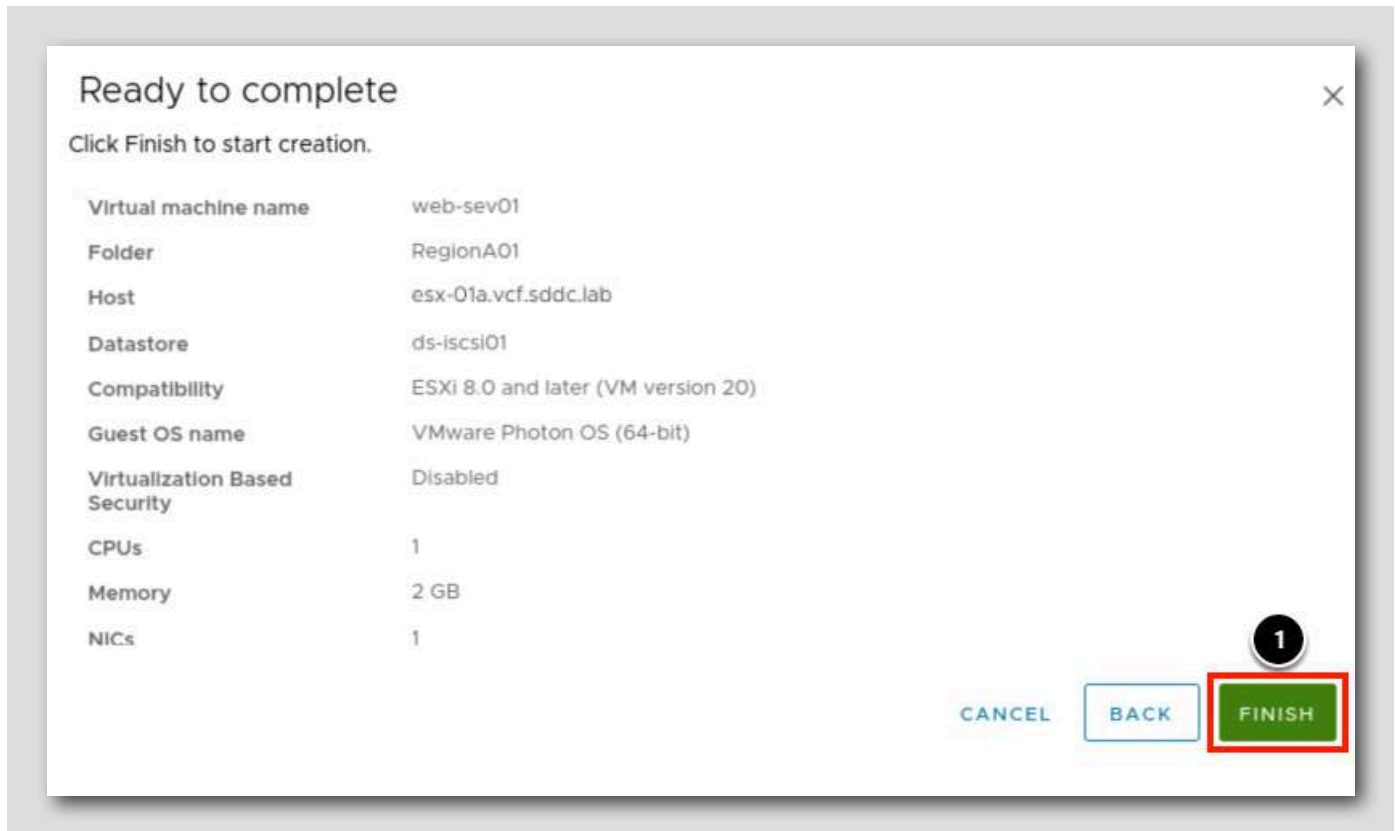
The recommended virtual hardware settings are shown as the default. These can be modified if needed.

1. Scroll down and expand the **New Hard disk*** section.
2. For **Disk Provisioning**, select **Thin Provision**.
3. Click **Next**.

Note: if you miss choosing "Thin Provision", you will run out of disk space for this lab and your Virtual Machine may not power on. Disk space is limited in this lab environment.

Ready to Complete

[65]

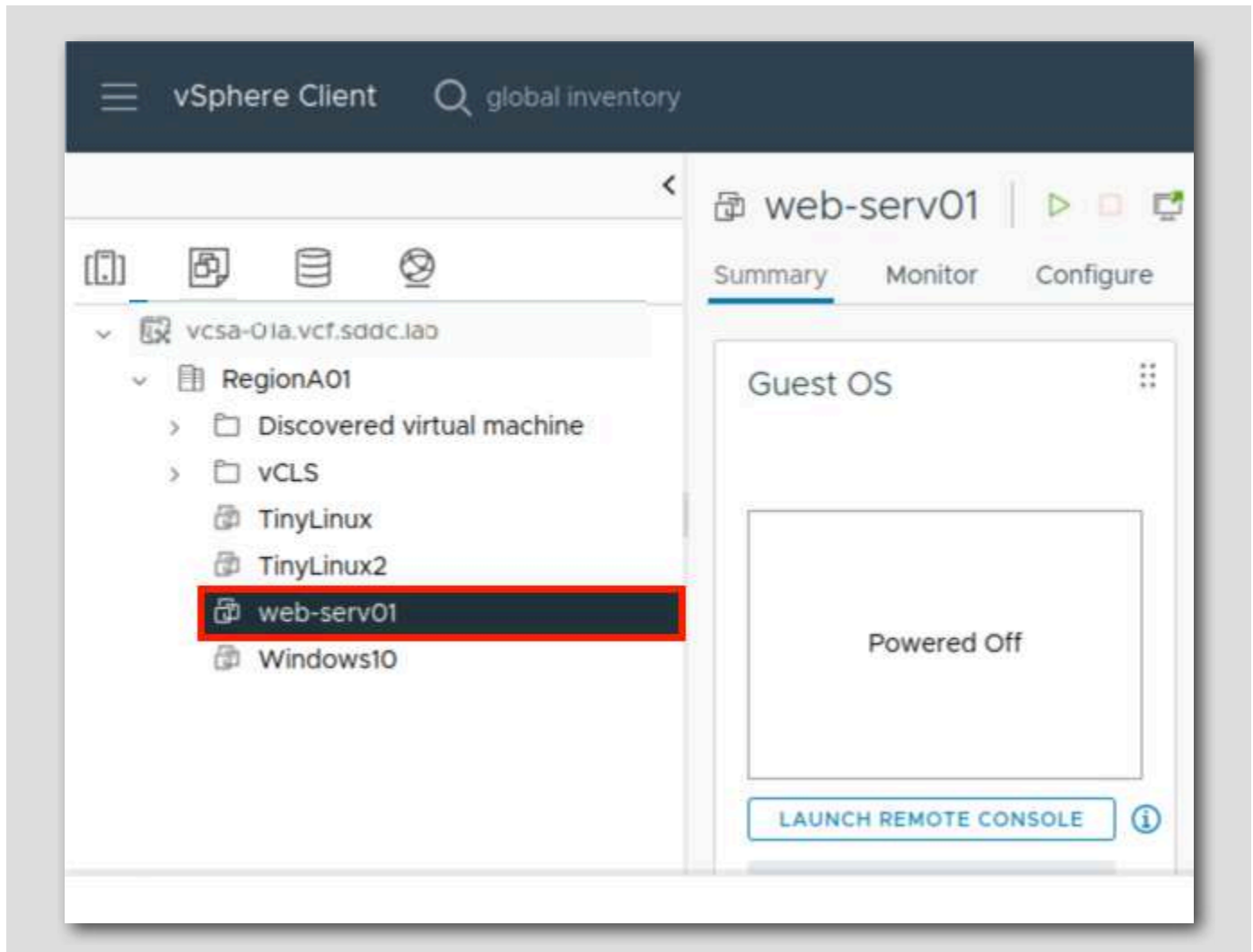


The settings for the virtual machine can be verified prior to it being created.

1. Click **Finish** to create the virtual machine.

Newly Created Virtual Machine

[66]

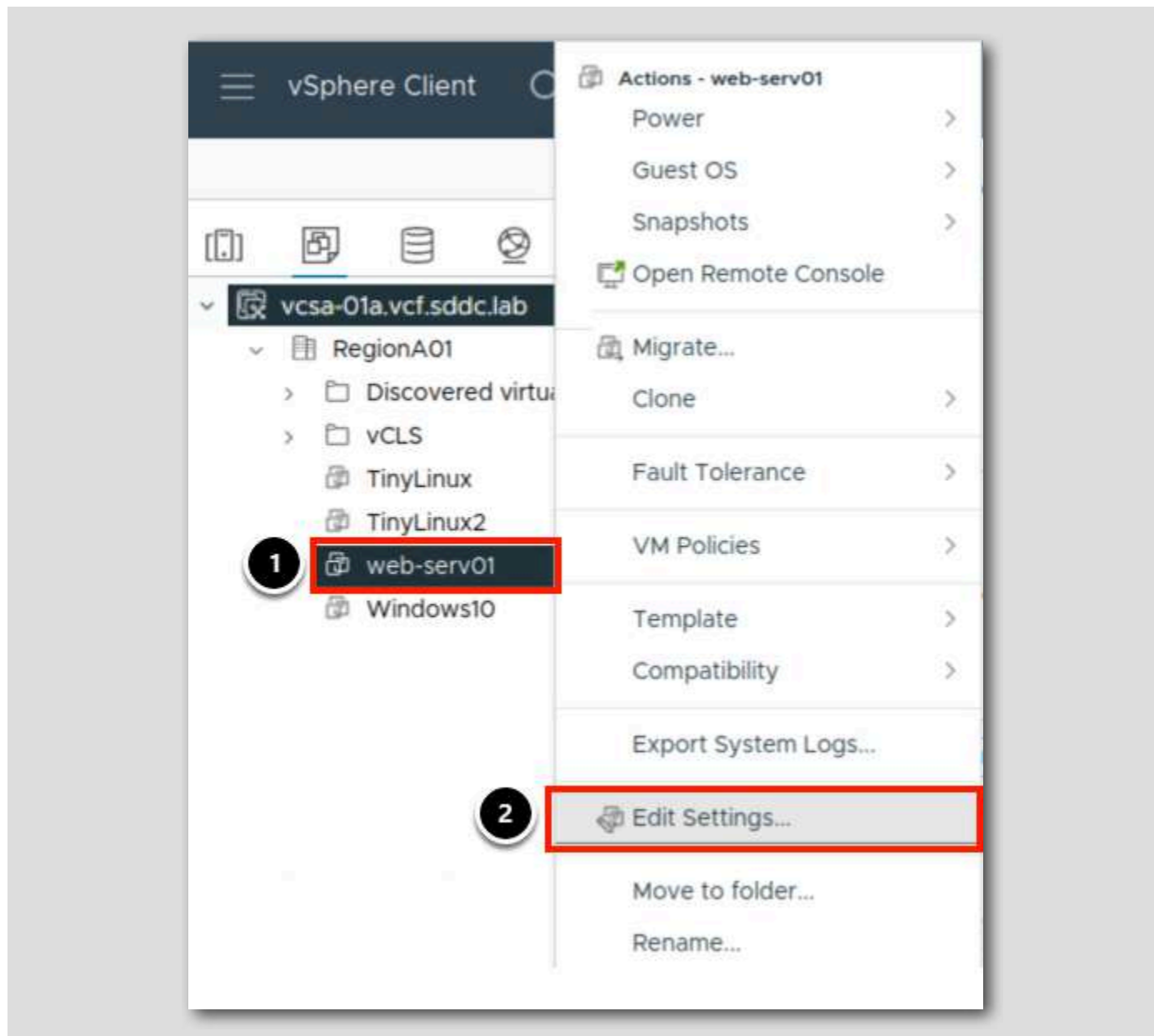


Congratulations on creating your first virtual machine **web-serv01**!

In the next steps, Photon OS will be installed on the virtual machine.

Attaching an ISO to a Virtual Machine

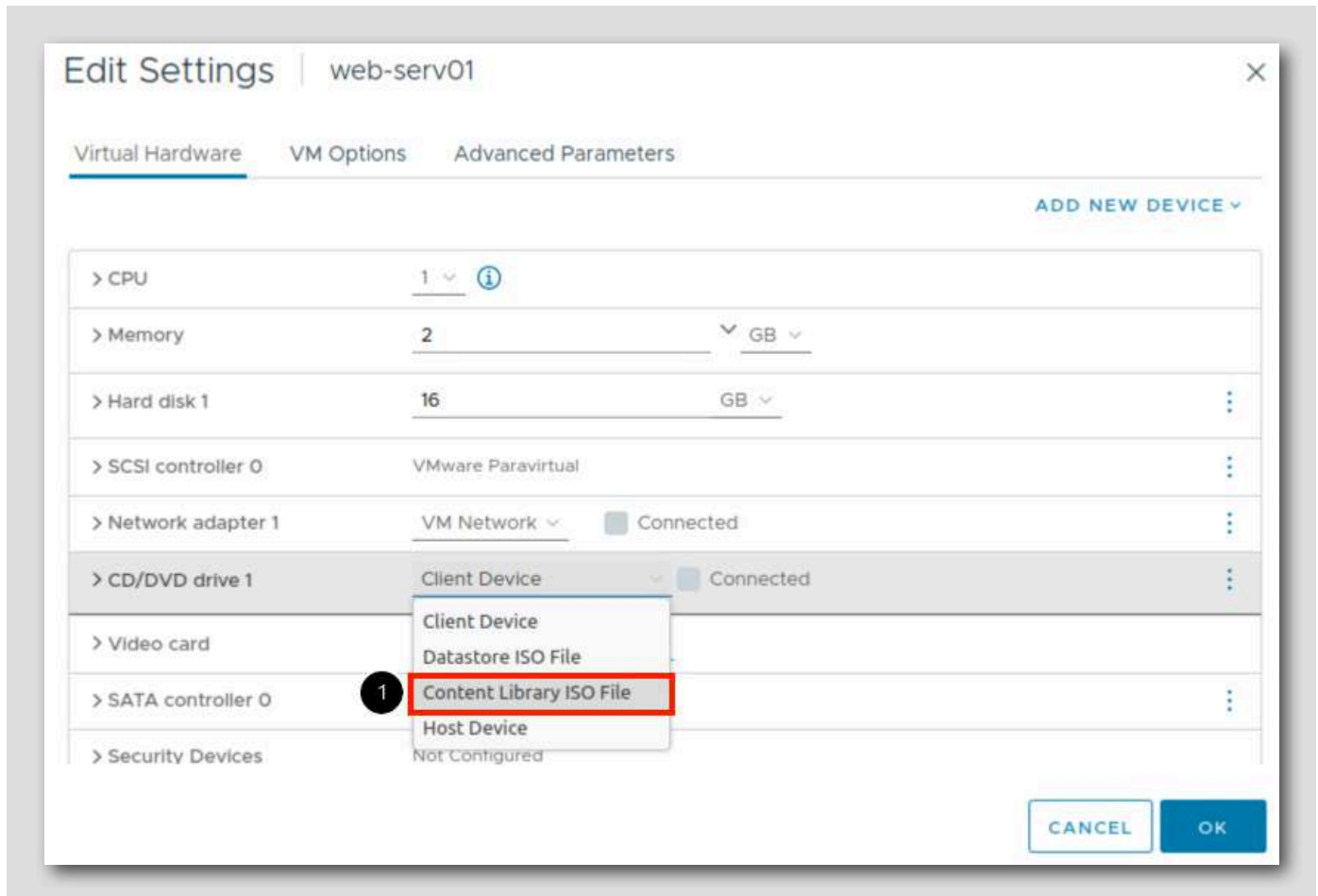
[67]



To make it easier to install operating systems on virtual machines, ISO images can be used. These can be kept in the same storage used for virtual machines. In addition, vCenter offers a Content Library as a repository. Content Libraries can then be synchronized to ensure every location is using the same versions.

1. To attach an ISO image to the virtual machine we just created, make sure **web-serv01** is selected.
2. Right-click on **web-serv01** and select **Edit Settings...**

Content Library ISO File

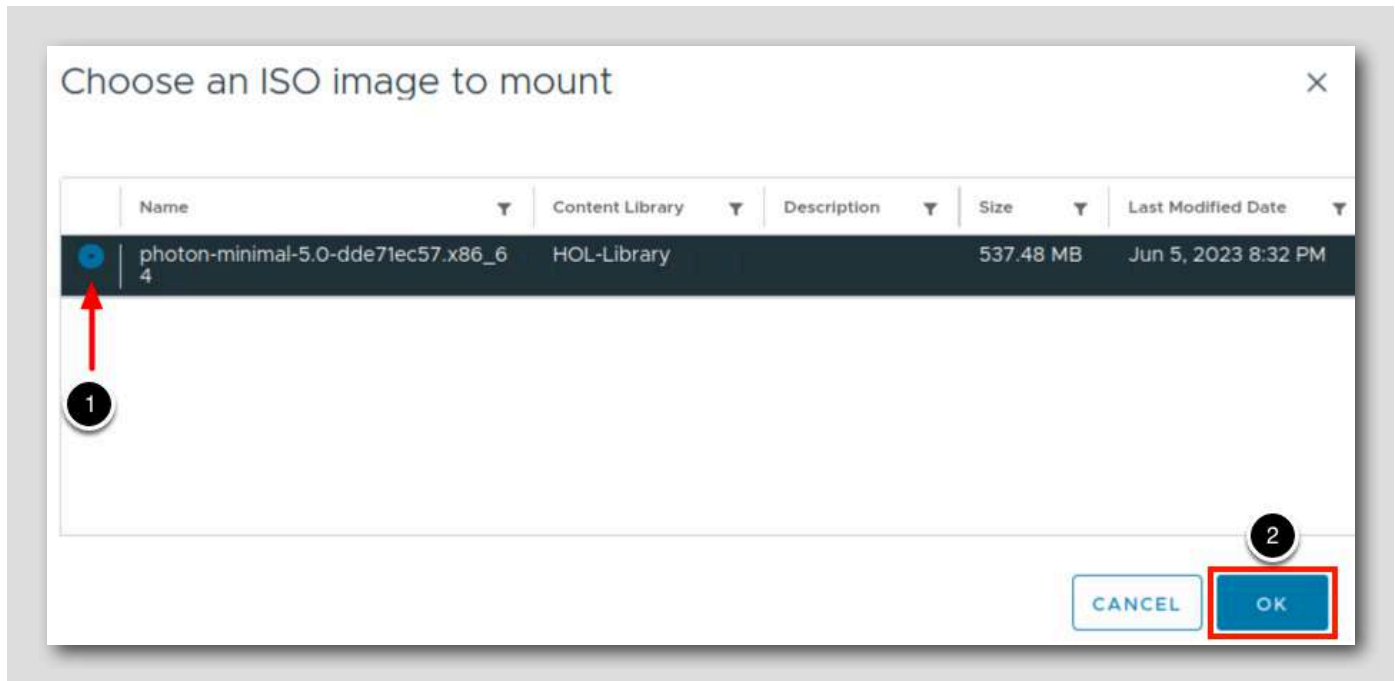


1. From the CD/DVD drive 1 drop-down menu, select Content Library ISO File.

This will open a file explorer to select that file.

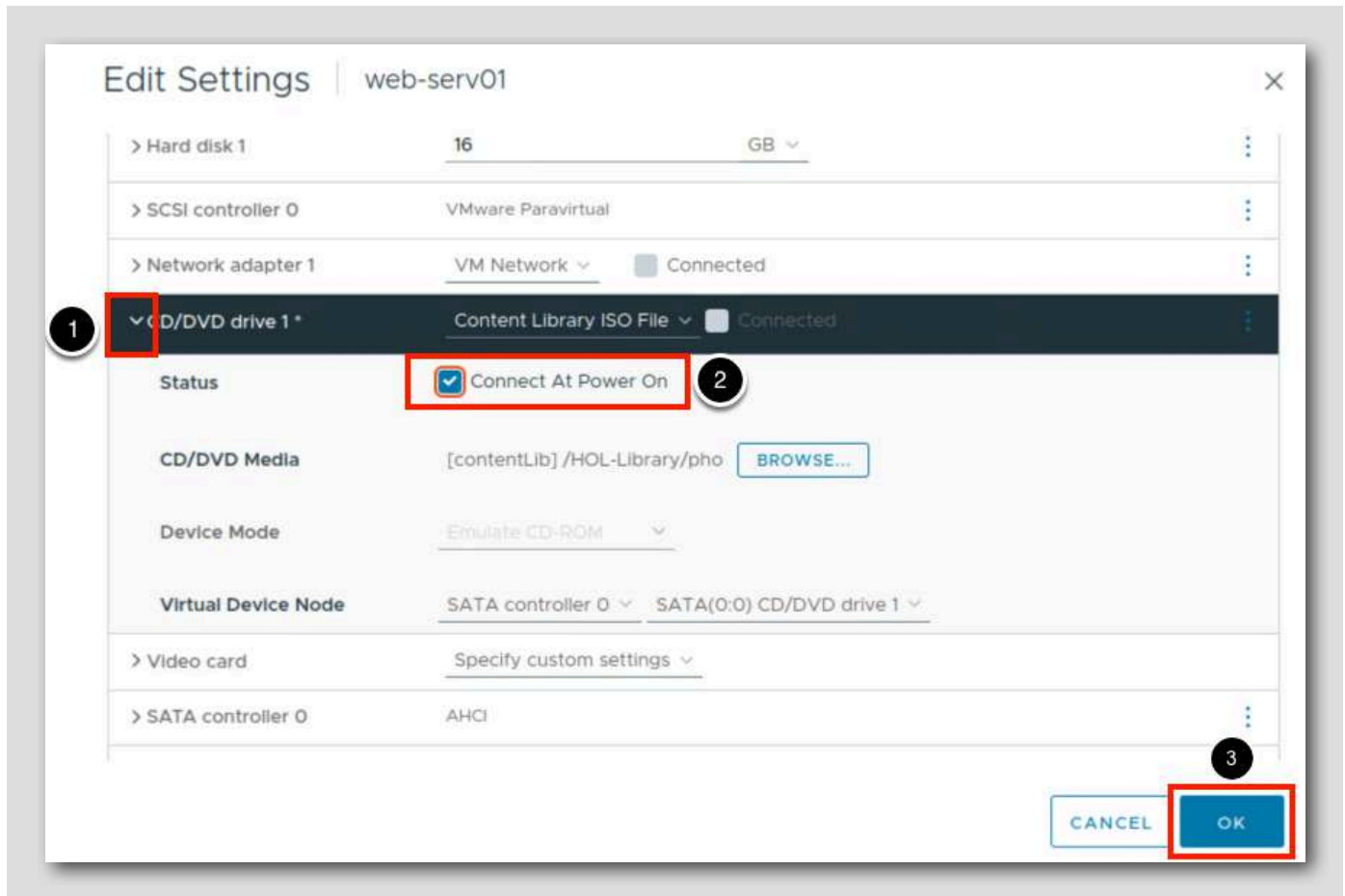
Select Photon

[69]



1. Click the radio button next to **photon-minimal-5.0-dde71ec57.x86_64**
2. Click OK.

Connect the Drive



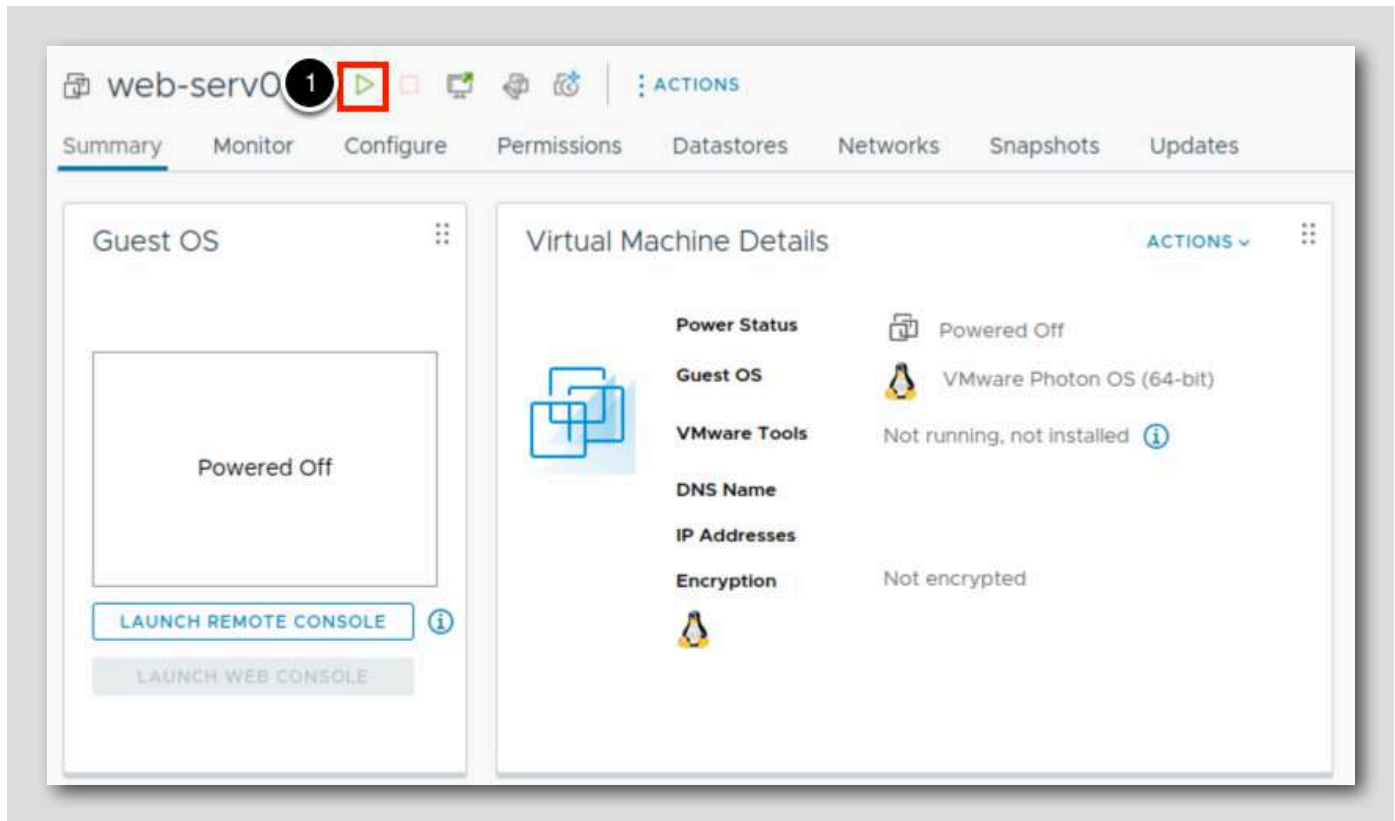
Finally, we want to attach or connect the ISO image to the virtual machine.

1. Expand the CD/DVD drive 1 menu
2. Click the **Connect At Power On** check box next to **Status**
3. Click **OK**.

Note: The virtual machine's BIOS is set to boot from CDROM only on first boot. If you miss this step you will need to manually adjust the BIOS to boot from CDROM just like a physical machine.

Power on web-serv01

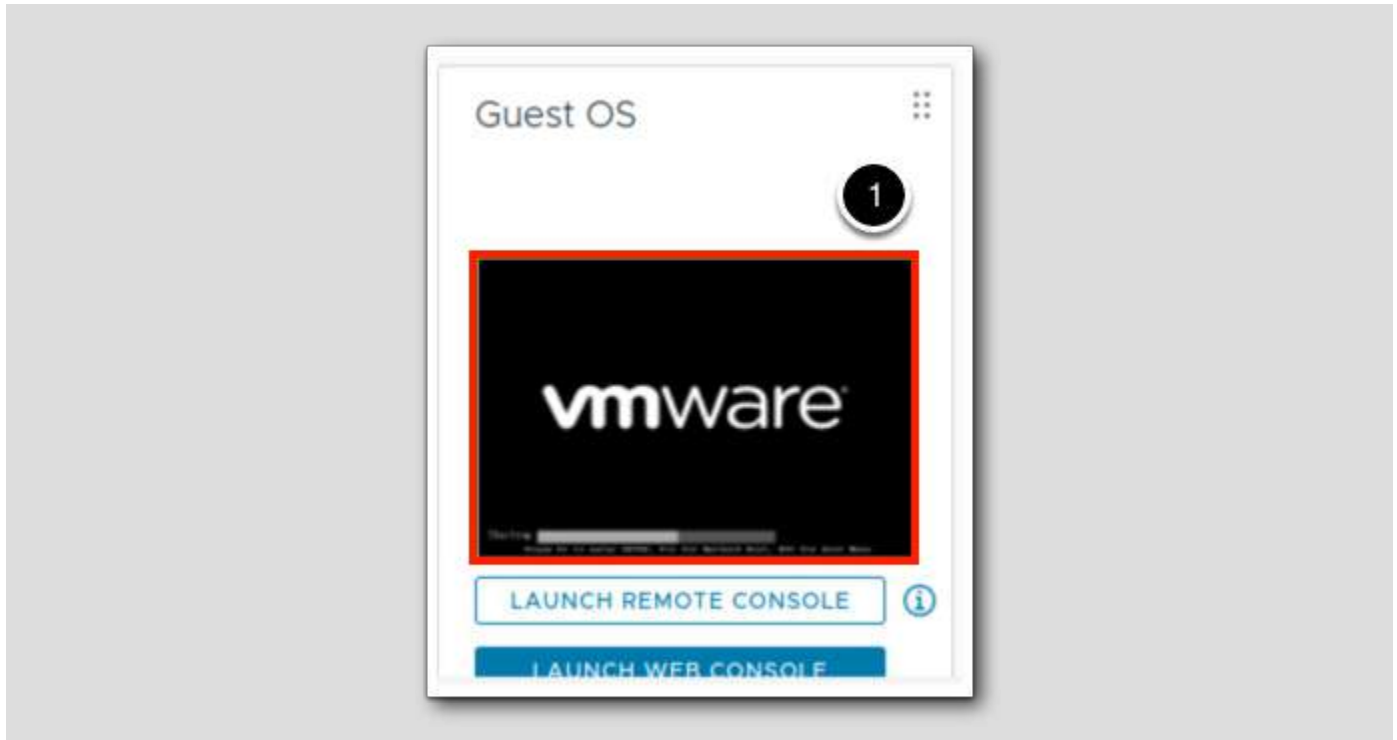
[71]



1. Click the green play button to power on the virtual machine and start the installation.

Launch Console

[72]

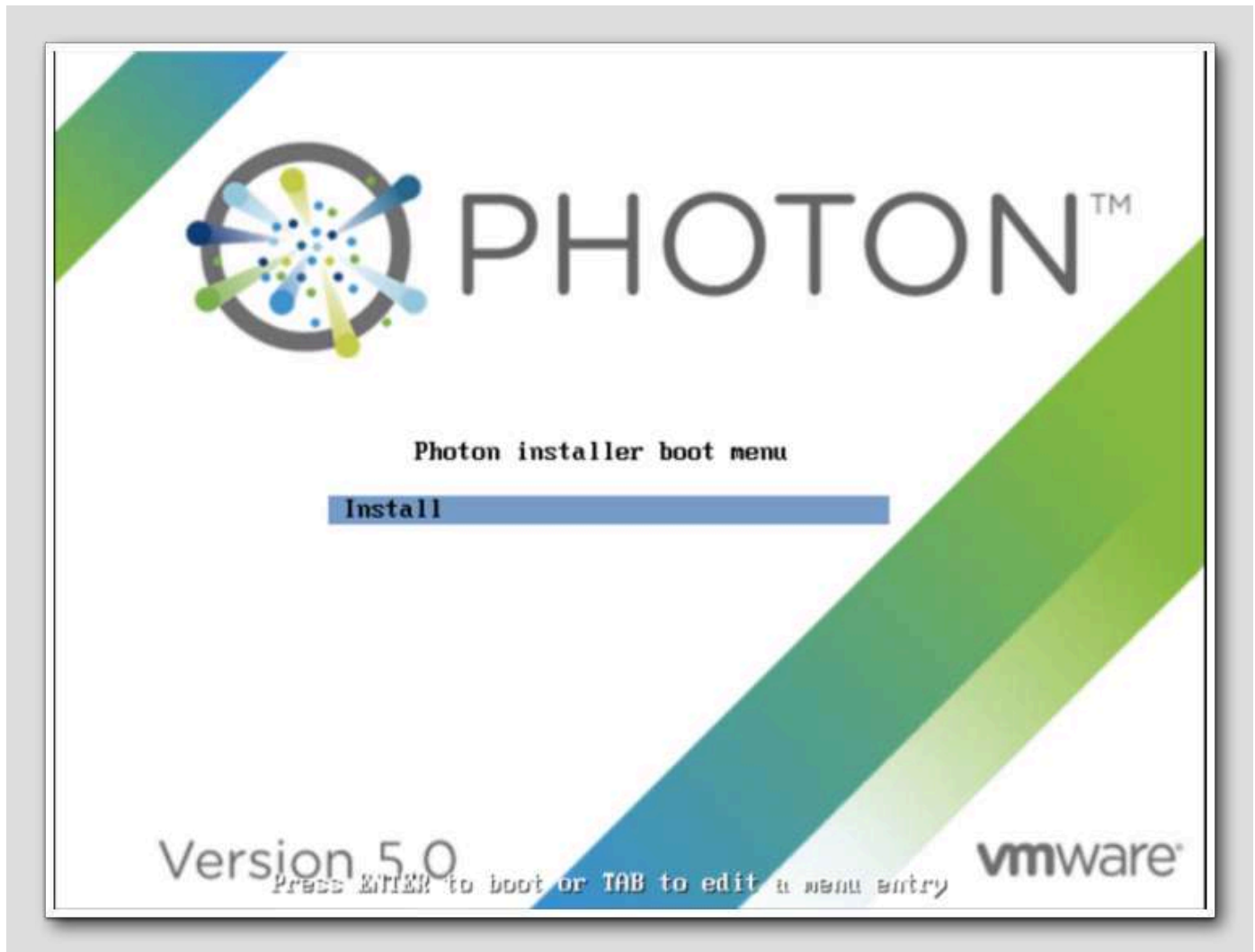


1. To launch the console window, click anywhere in the console window screen. Alternatively, you can click on **Launch Remote Console**

Note you also have the option of using the VMware Remote Console (VMRC). This console is a separate application that needs to be installed on your local device as opposed to the Web Console which will launch in a new browser tab. The VMRC can be useful in certain situations when you need more capabilities, like attaching devices or power cycling options.

Photon Boot Screen

[73]

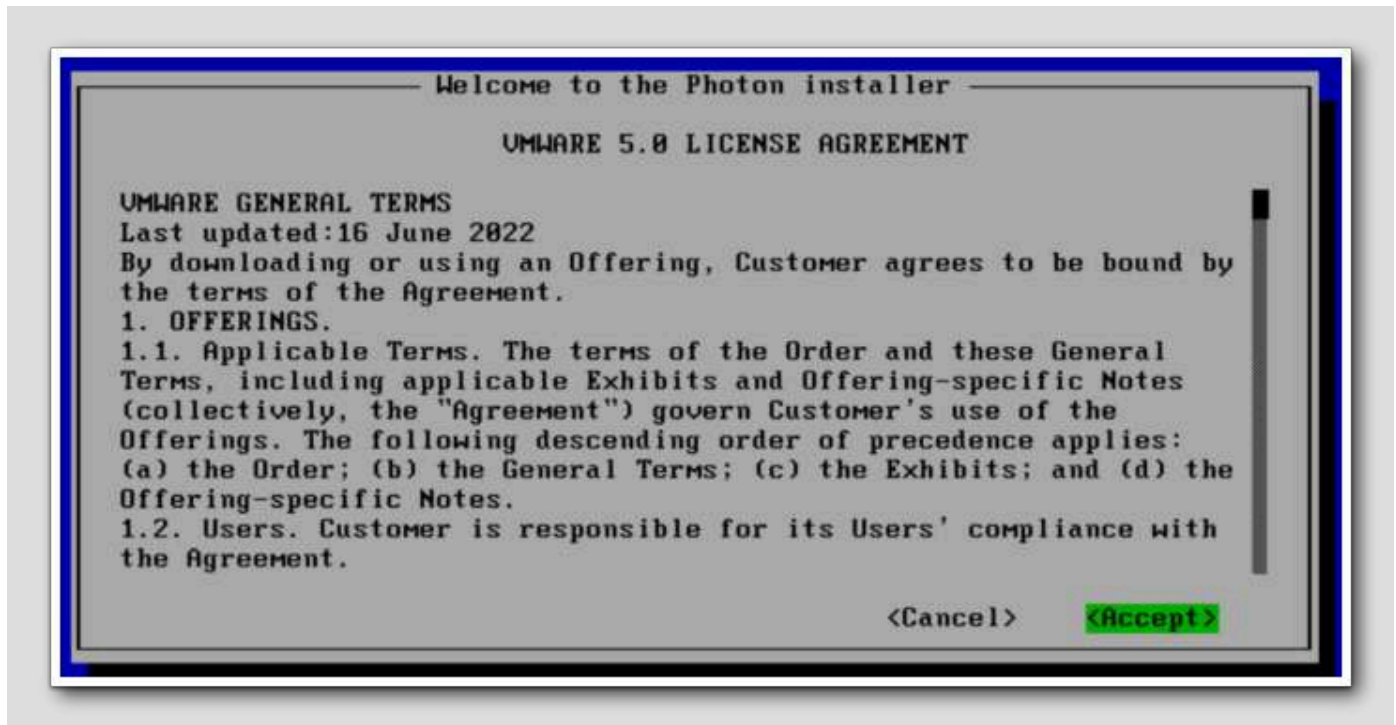


A new tab will open and you will be presented with the Photon OS boot screen.

1. Press the **Enter** key to start the installation process.

License Agreement

[74]



After the boot process is complete, you will be presented with a license agreement.

1. Press Enter to accept.

Select Disk

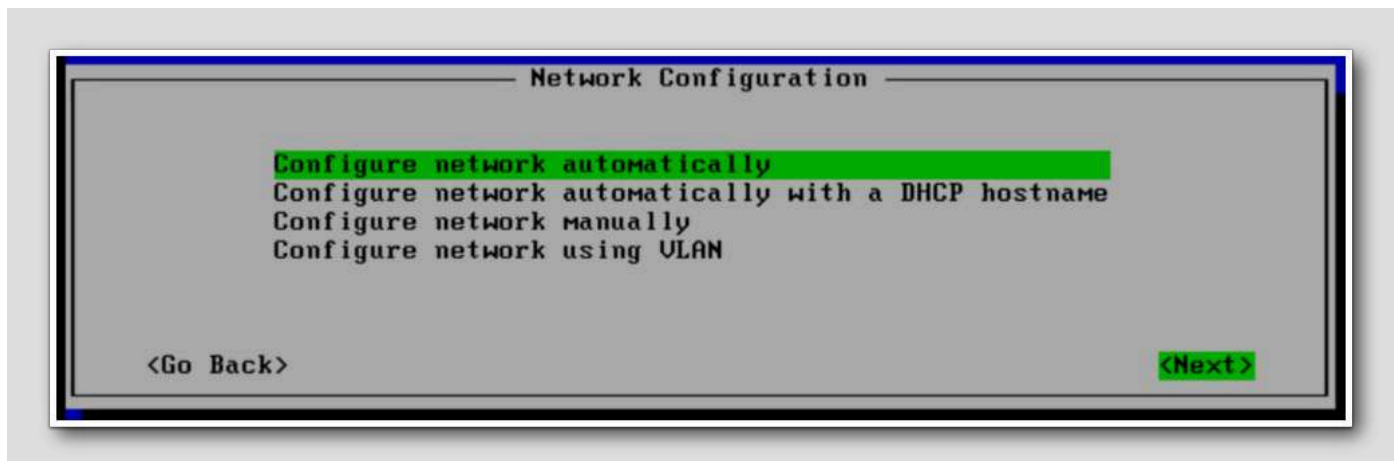
[75]



1. Press **Enter** to accept the selected disk and use the auto partitioning option.

Select your Network Configuration

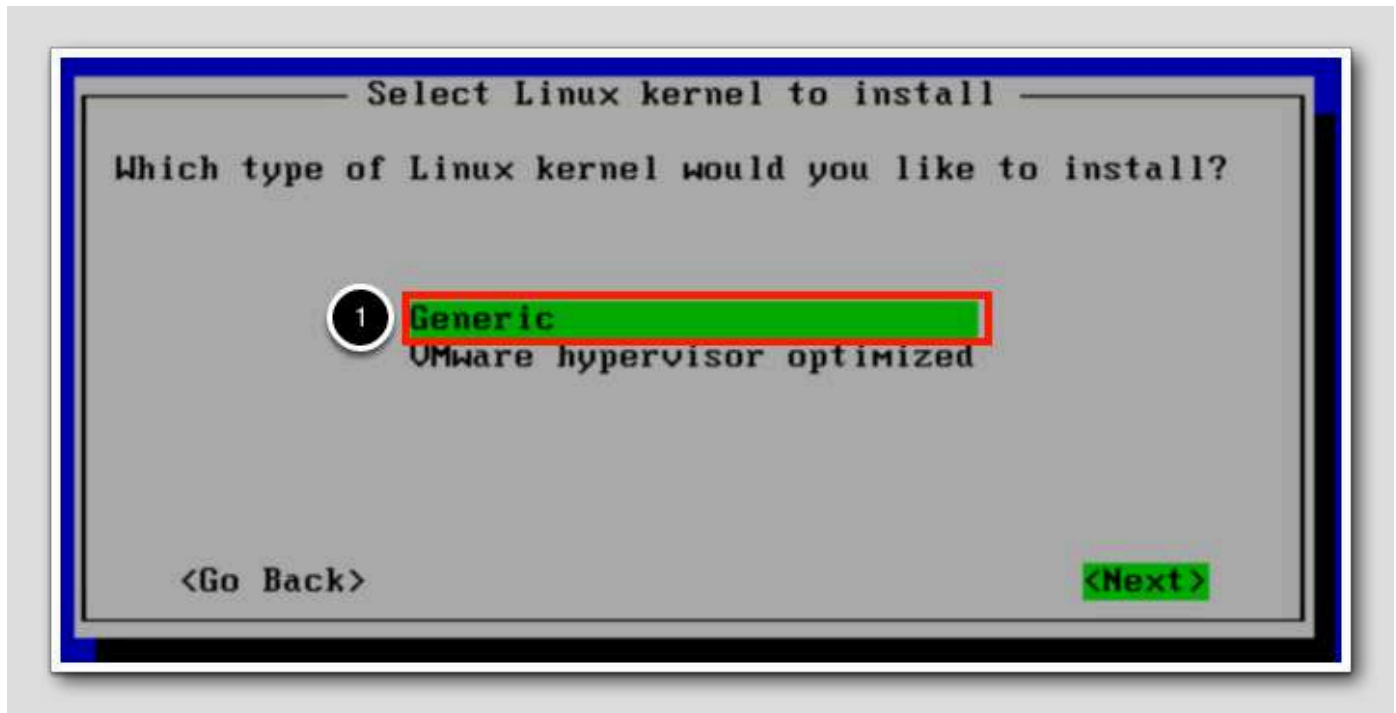
[76]



1. Select **Configure network automatically** and press **Enter**

Linux Kernel

[77]

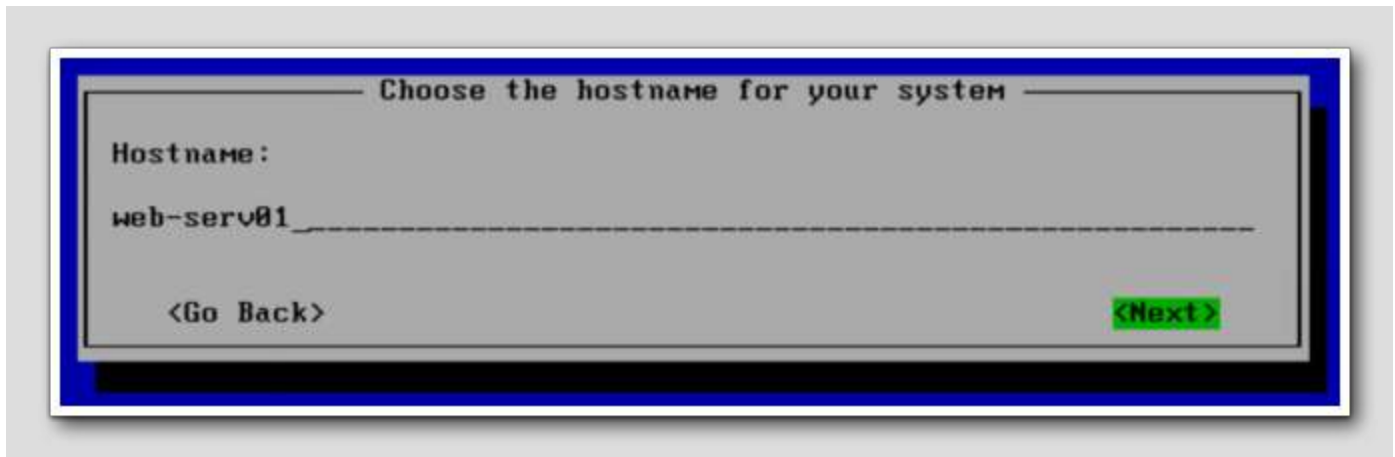


1. Leave the Generic option, and press Enter.

NOTE: If Hypervisor optimized is selected, the virtual machine will not boot. This is due to the unique environment the Hands-on Labs are running in.

Rename Host

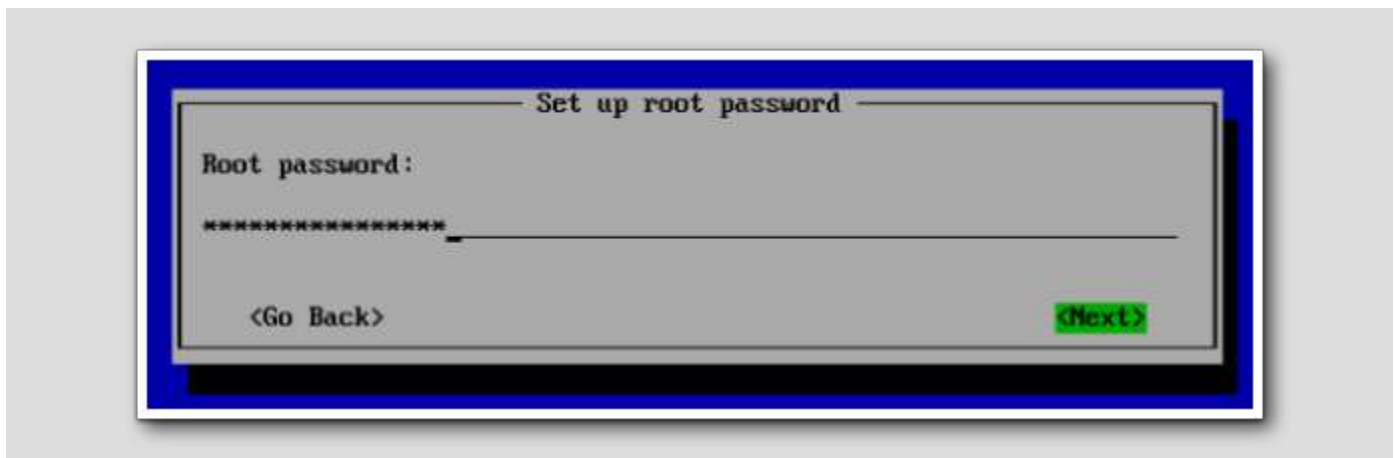
[78]



1. Use the Backspace key to remove the default hostname.
2. Type web-serv01.
3. Press the Enter key.

Password

[79]

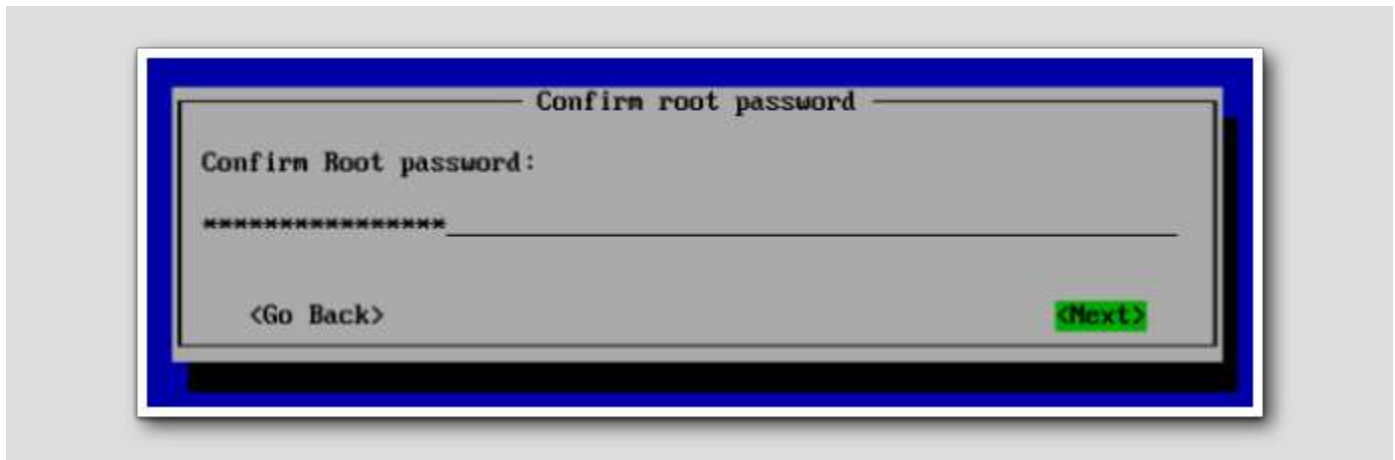


1. For the password, use VMware1!VMware1!

Note that Photon requires a complex, non-dictionary password, which is why the typical password is being repeated.

Confirm Password

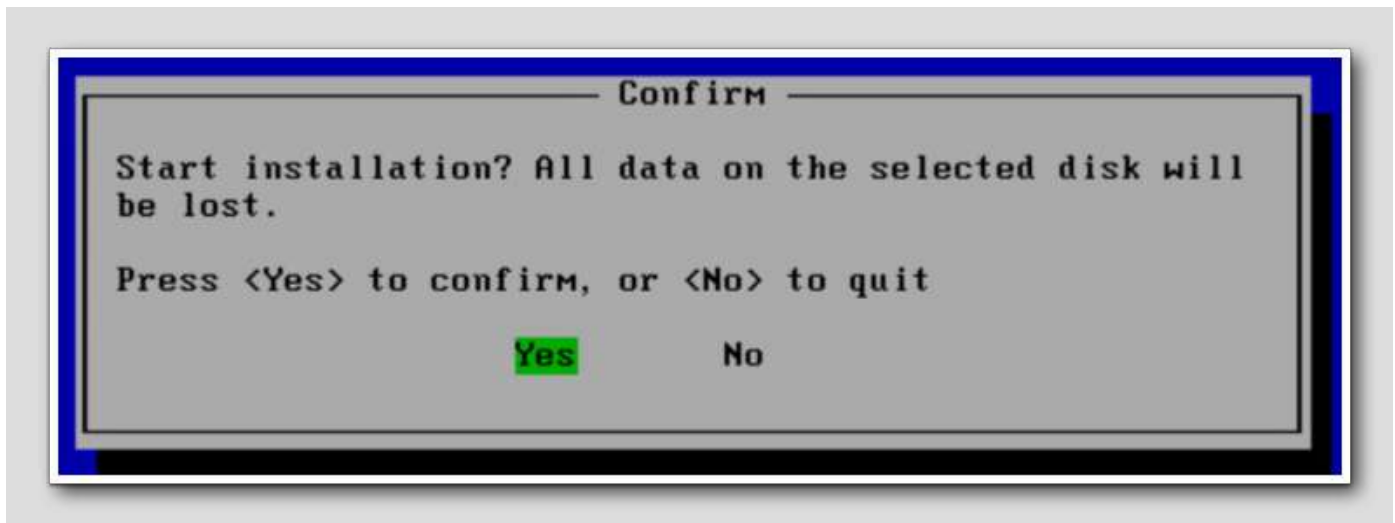
[80]



1. Type VMware1!VMware1! again to confirm the password.
2. Press the Enter key.

Confirm the Installation of Photon

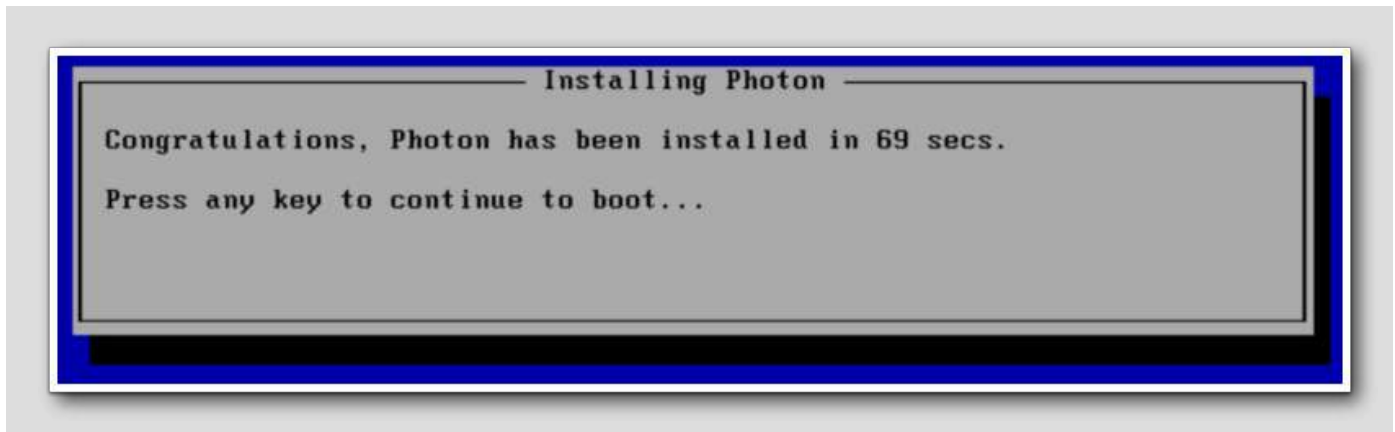
[81]



1. Use your keyboard arrows to select Yes, and press Enter

Installation Complete

[82]

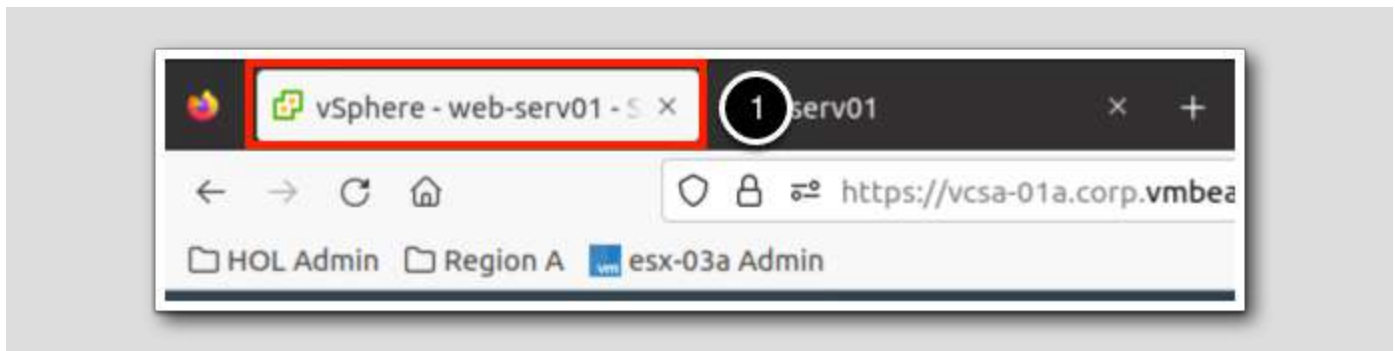


After a minute or two, the installation will be complete.

1. Press a key to reboot the virtual machine. After a minute or two, the system should boot the login prompt.

vSphere Tab

[83]

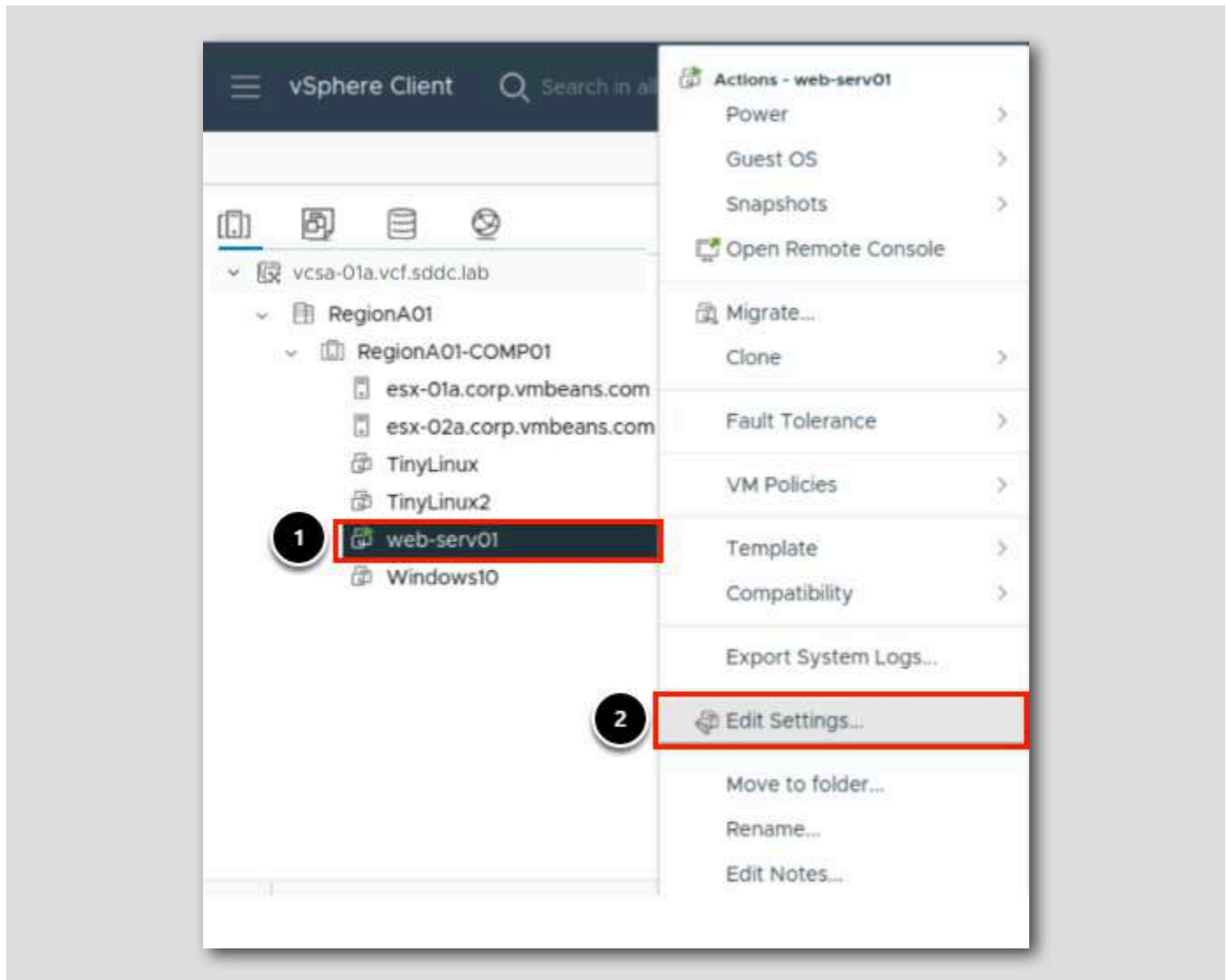


Now that the operating system has been installed and is up and running, the ISO image needs to be disconnected from the virtual machine.

1. Select the vSphere- web-serv01 tab.

Edit Settings

[84]

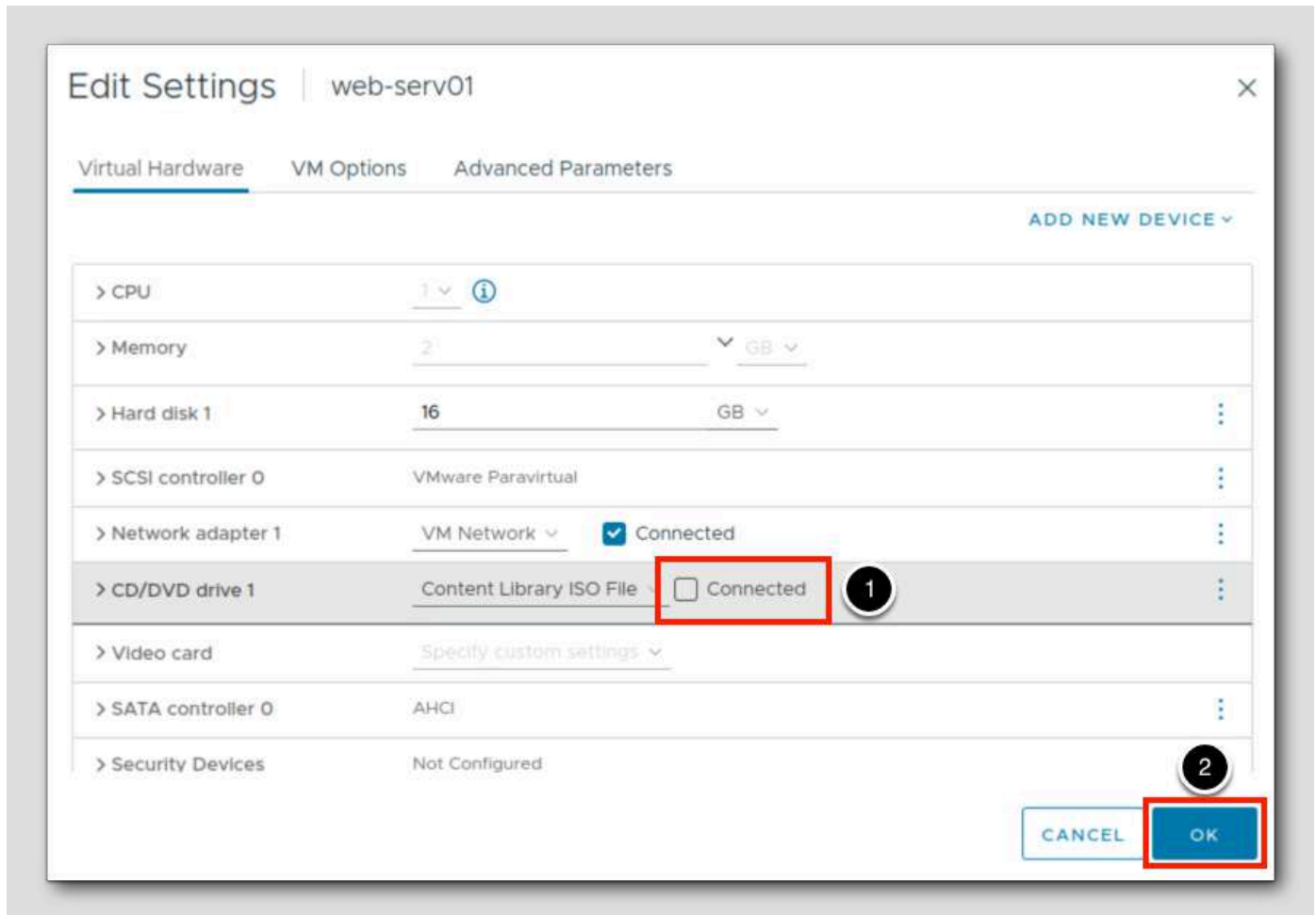


Make sure **web-serv01** is still highlighted.

1. Right-click on **web-serv01**.
2. Select **Edit Settings...**

Disconnect CD/DVD

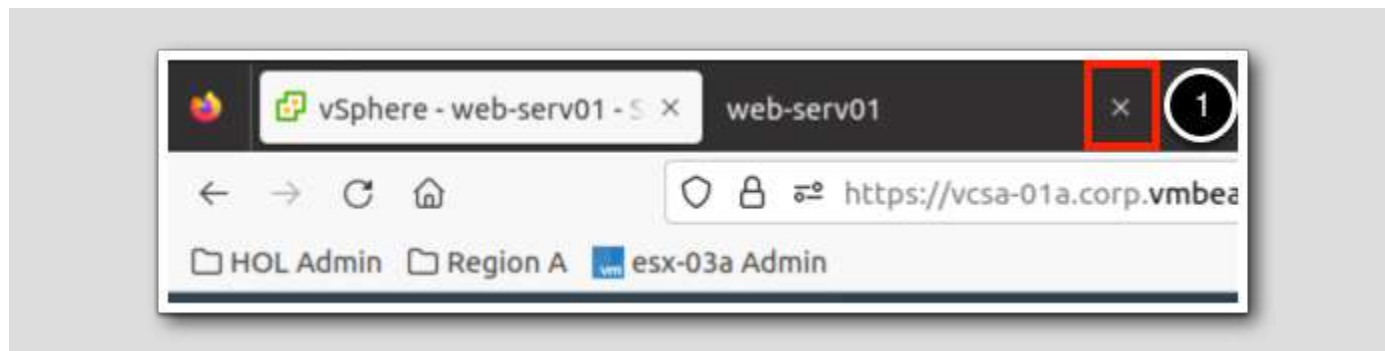
[85]



1. Verify that the Connected checkbox in the CD/DVD drive 1* section is unchecked
2. Click OK

web-serv01 Console

[86]



1. Click the 'X' to close the console window for web-serv01.

Cloning Virtual Machines and Using Templates

[87]

VMware provides several ways to provision vSphere virtual machines. In the last lesson, you saw how to create a virtual machine and manually install the operating system.

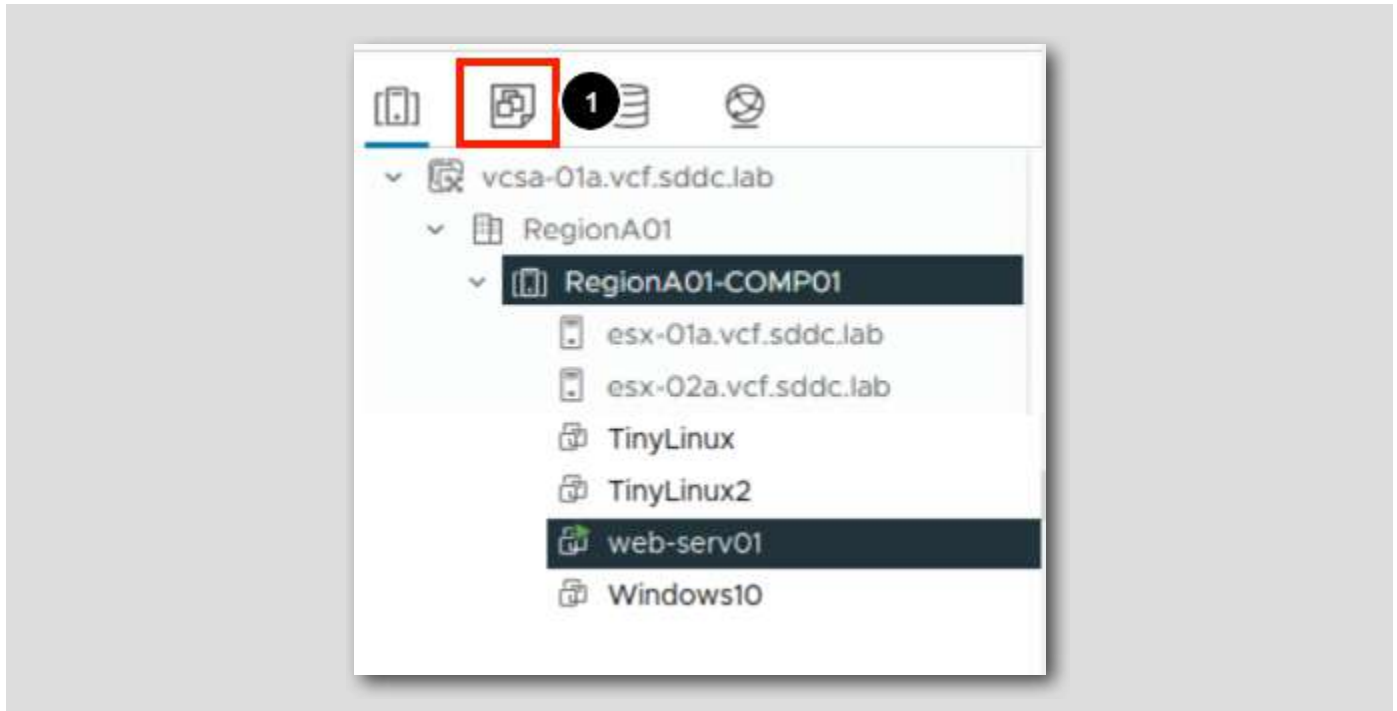
The virtual machine that was created can then be used as a base image from which to clone other virtual machines. Cloning a virtual machine can save time if you are deploying many similar virtual machines. You can create, configure, and install software on a single virtual machine. You can clone it multiple times, rather than creating and configuring each virtual machine individually.

Another provisioning method is to clone a virtual machine to a template. A template is a master copy of a virtual machine that you can use to create and provision virtual machines. Creating a template can be useful when you need to deploy multiple virtual machines from a single baseline but want to customize each system independently of the next. A common value point for using templates is to save time. If you have a virtual machine that you will clone frequently, make that virtual machine a template, and deploy your virtual machines from that template.

In this lesson, you will clone an existing Virtual Machine to a Template and deploy a new Virtual Machine from that Template.

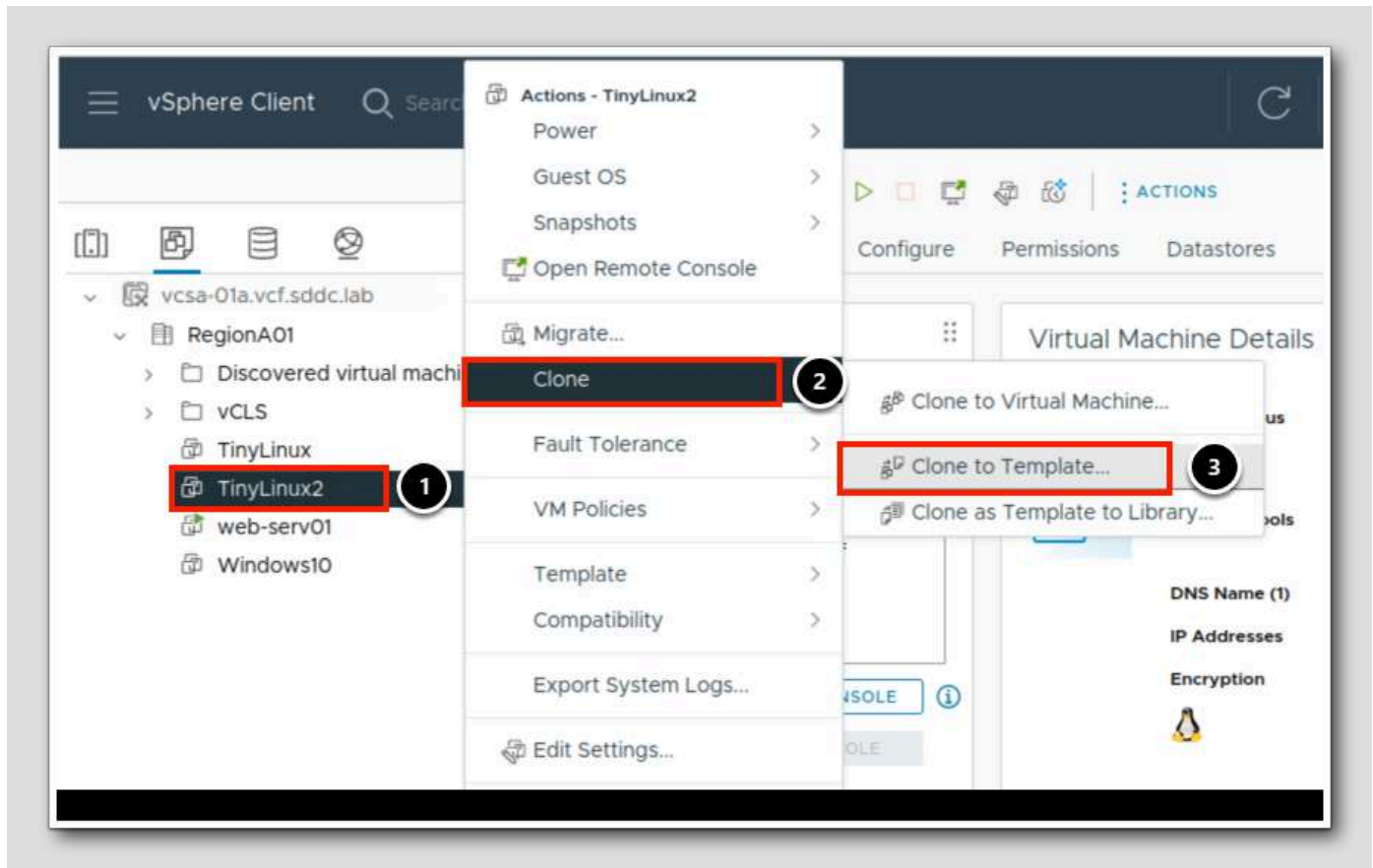
Navigate to the VMs and Templates Management Pane

[88]



1. Click on the **VMs and Templates** icon on the left hand side menu.

Launch the Clone Virtual Machine to Template wizard



1. Right-click the Virtual Machine TinyLinux2.
2. Select Clone.
3. Select Clone to Template...

Select a Name and Folder

Select a name and folder

Specify a unique name and target location

VM template name:

Select a location for the template.

- ▼ vcsa-01a.corp.vmbeans.com
 - > RegionA01

CANCEL NEXT

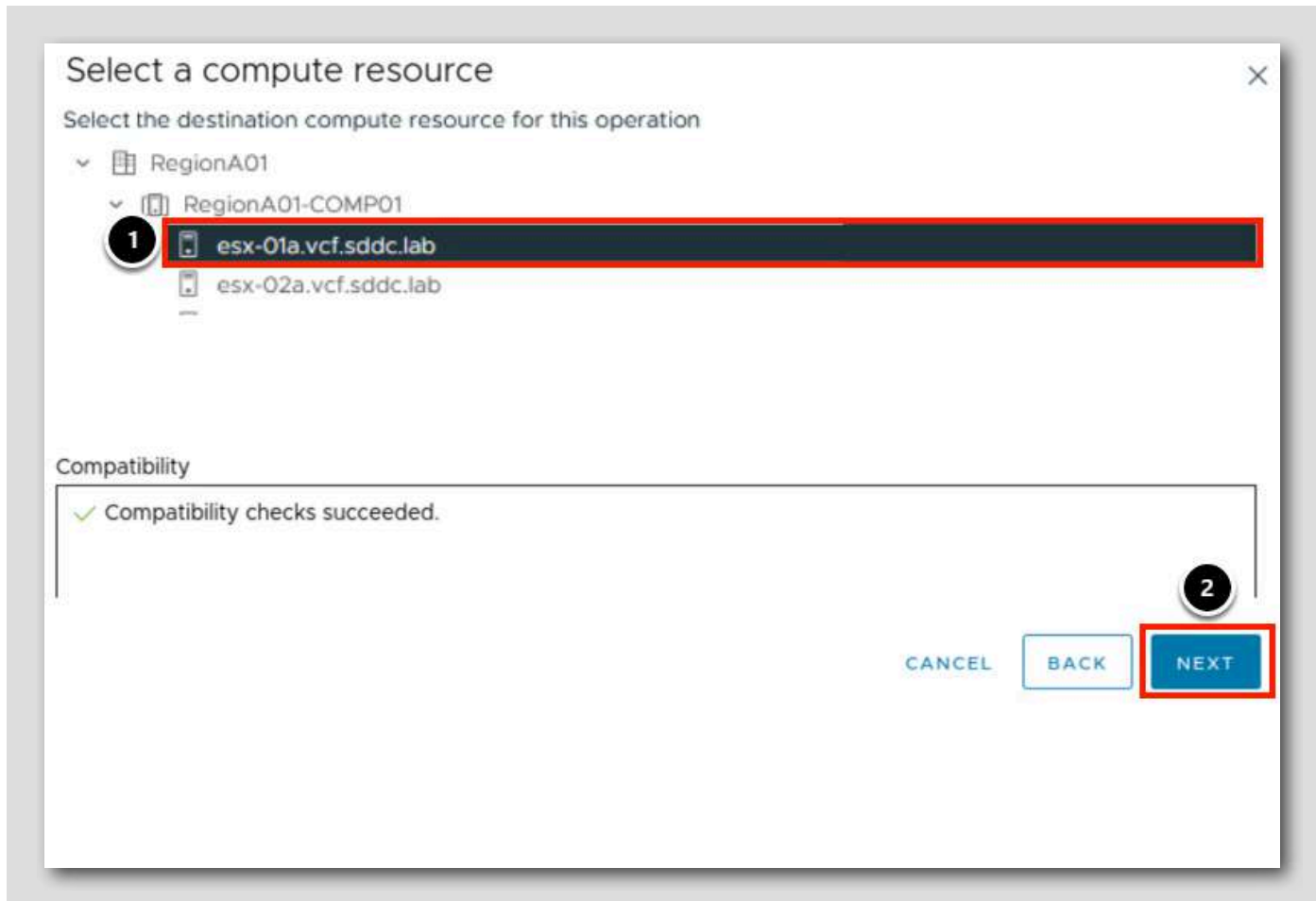
1. In the Clone Virtual Machine to Template wizard, provide a name for the Template - **TinyLinux2 Template**

Please leave the location as **RegionA01** for this lab.

2. Click **Next**

Select Compute Resource

[91]



Select a compute resource:

1. Expand **RegionA01-COMP01** and choose **esx-01a.vcf.sddc.lab**
2. Click **Next**.

Select Storage

Select storage

BATCH CONFIGURE **CONFIGURE PER DISK**

☐ Encrypt this virtual machine (Requires Key Management Server)

Select virtual disk format: Thin Provision

VM Storage Policy: Keep existing VM storage policies

☐ Disable Storage DRS for this virtual machine

	Name	Storage Compatibility	Capacity	Provisioned	Free	Type	Cluster
☐	ds-iscsi...	--	84.75 GB	54.56 GB	62.02 GB	VMFS 6	
☒	ds-nfs01	--	78 GB	88 KB	5.78 GB	NFS v3	

items per page: 10 2 items

Compatibility

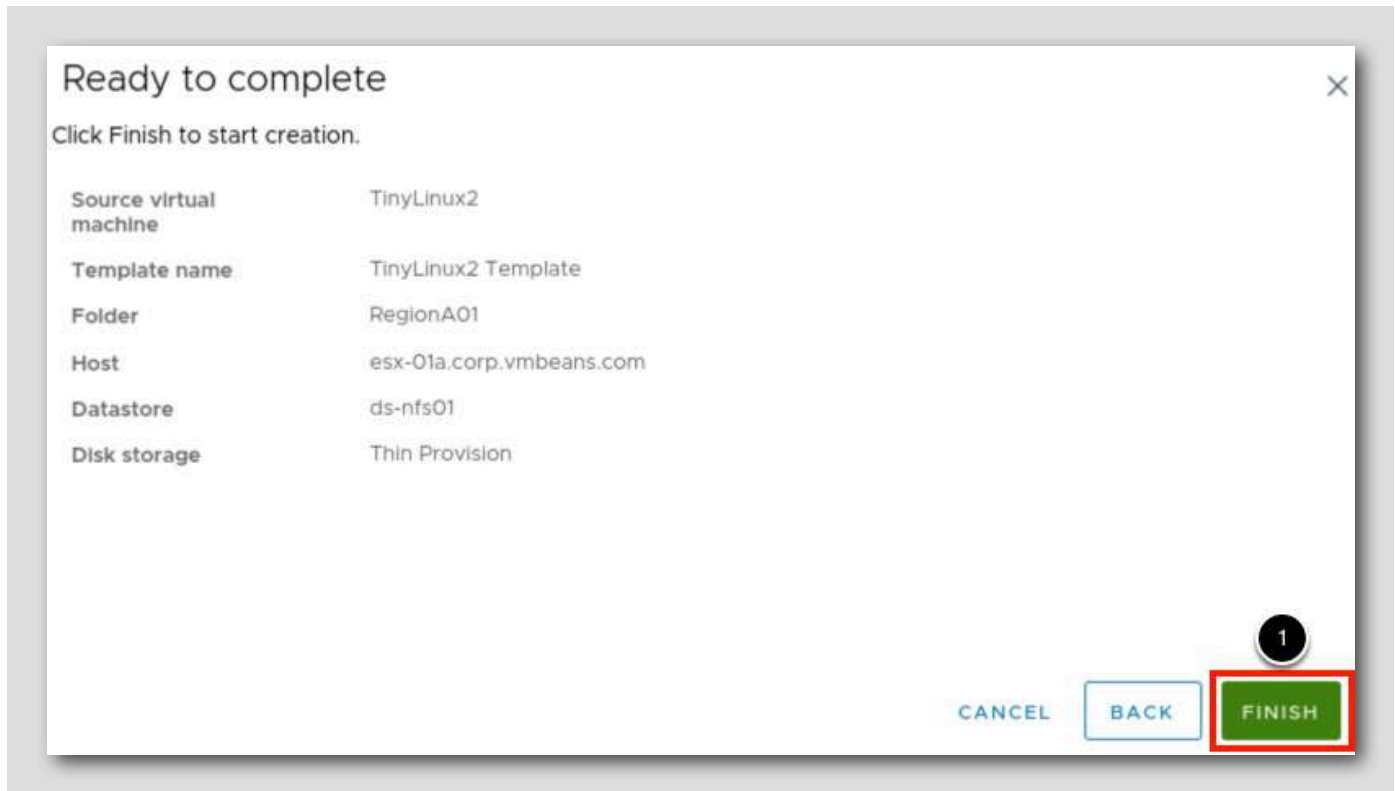
✓ Compatibility checks succeeded.

CANCEL **BACK** **NEXT**

1. Use the scroll bar on the right to view the datastores and select `ds-nfs01` as the datastore.
2. Press the **Next** button.

Review the VM Template Settings

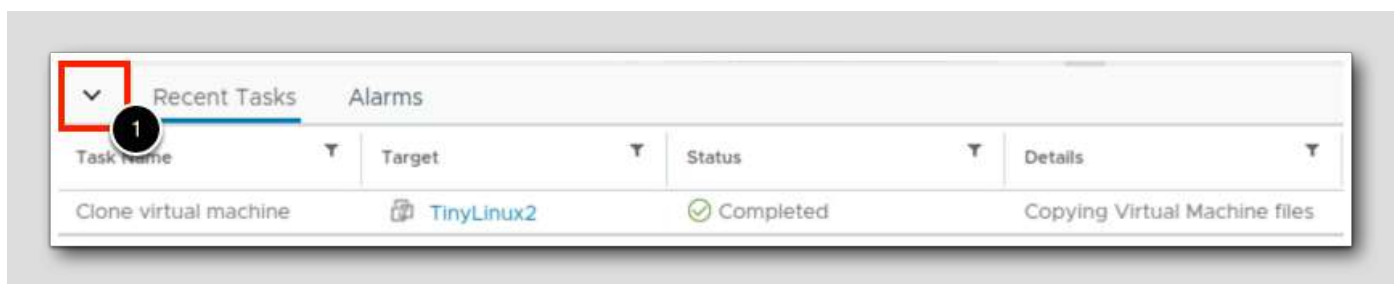
[93]



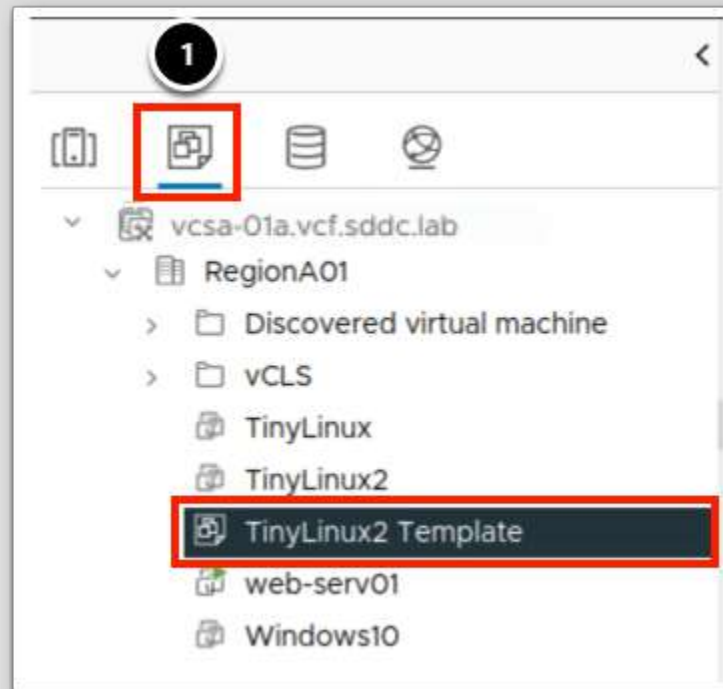
1. Review the VM Template settings and press the Finish button.

Monitor Task Progress

[94]



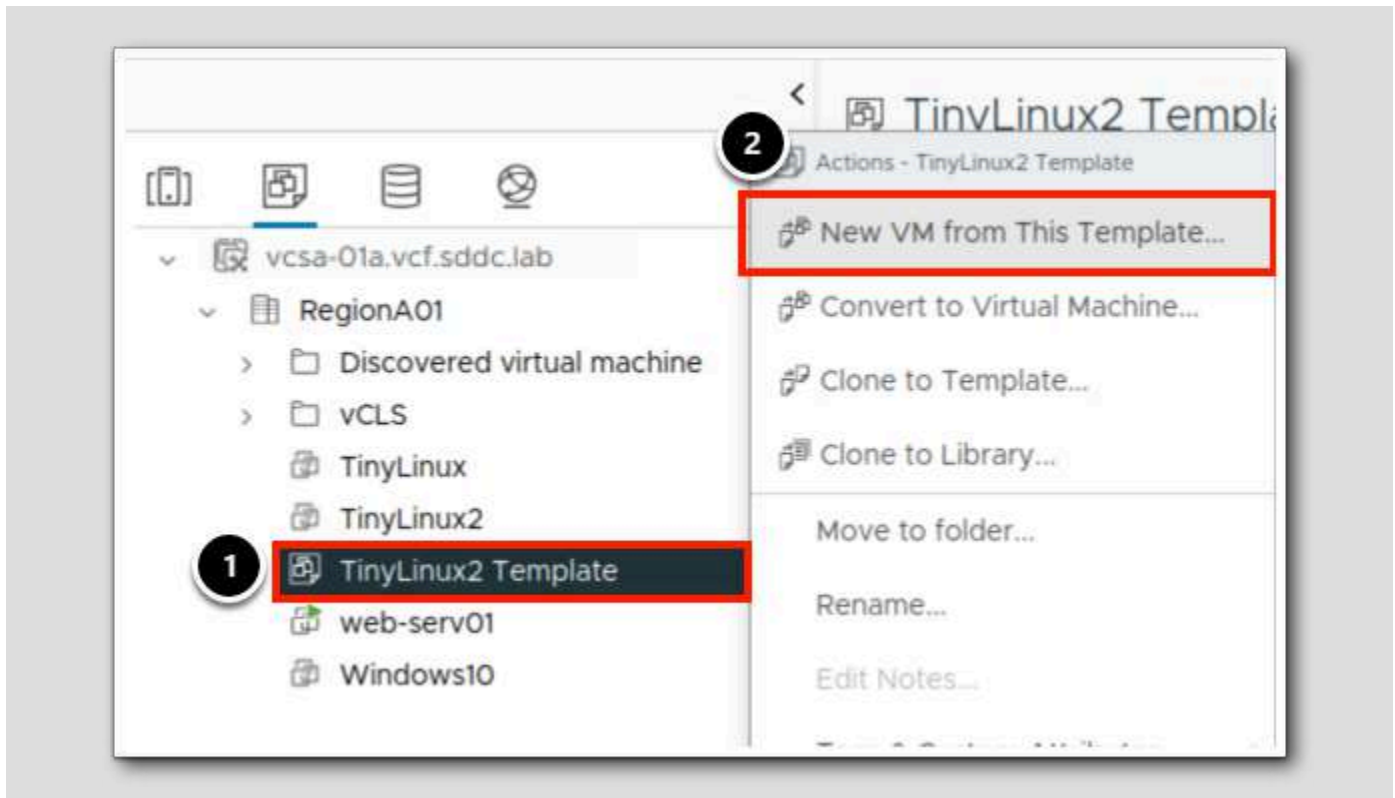
1. You can monitor the progress in the recent task window.



1. Once the task has been completed, click on the VM and Templates icon. TinyLinux 2 Template object should be on the inventory pane.

Launch the Deploy From Template wizard

[95]



1. Select the Template, TinyLinux2 Template
2. Right click on TinyLinux2 Template and select New VM from This Template.

Select a Name and Folder

Select a name and folder

Specify a unique name and target location

Virtual machine name:

Select a location for the virtual machine.

vcsa-01a.corp.vmbeans.com

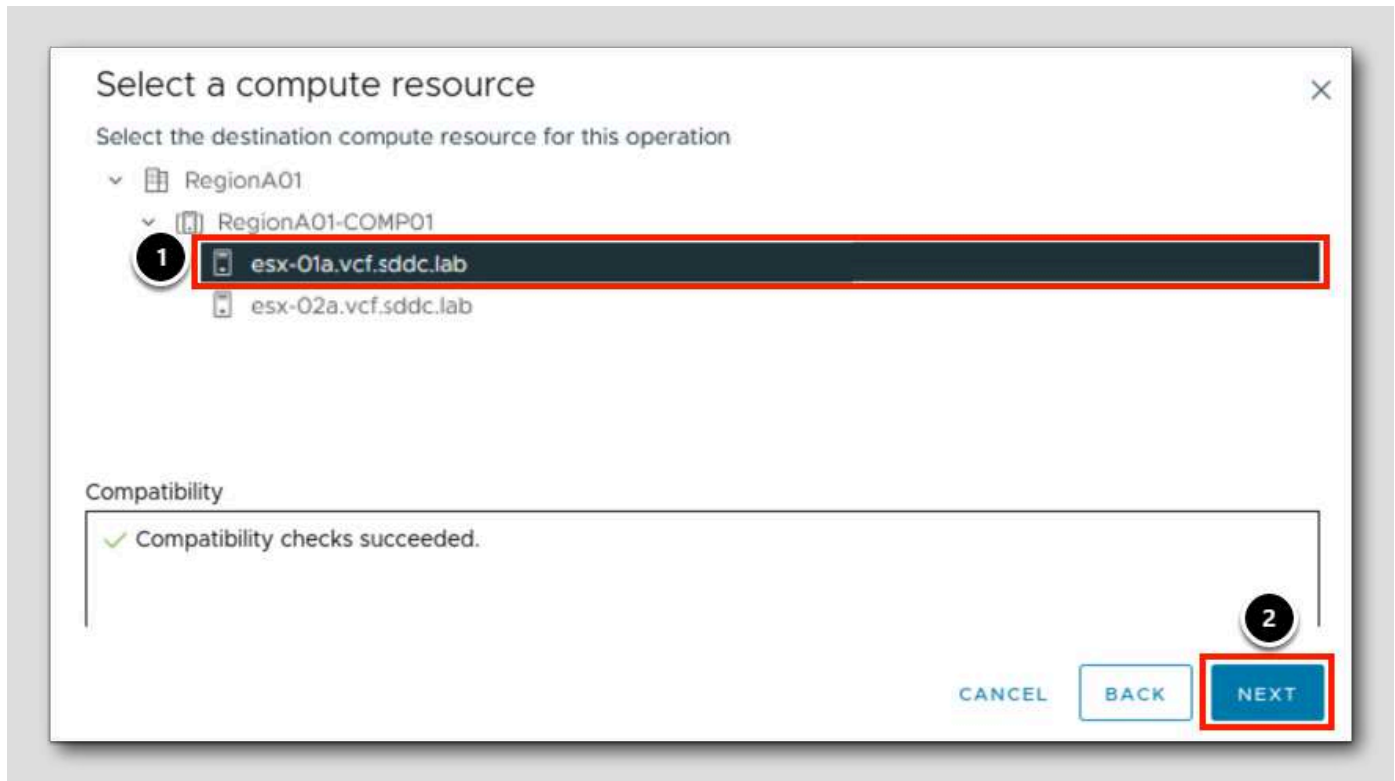
RegionA01

CANCEL NEXT

1. Enter **app-serv01** for the name of the new virtual machine.
2. Leave the default location of **RegionA01** Datacenter.
3. Click the **Next** button.

Select Compute Resource

[97]



1. Expand RegionA01-COMP01 and select esx-01a.vcf.sddc.lab
2. Click Next.

Select Storage

Select storage

Select the storage for the configuration and disk files

BATCH CONFIGURE **CONFIGURE PER DISK**

☐ Encrypt this virtual machine (Requires Key Management Server)

Select virtual disk format: Same format as source

VM Storage Policy: keep existing VM storage policies

☐ Disable Storage DRS for this virtual machine

	Name	Storage Compatibility	Capacity	Provisioned	Free	Type	Cluster
1	ds-iscsi01	--	84.75 GB	54.56 GB	62.02 GB	VMFS 6	
	ds-nfs01	--	5.78 GB	732.91 MB	5.78 GB	NFS v3	

Items per page: 10 2 items

Compatibility

✓ Compatibility checks succeeded.

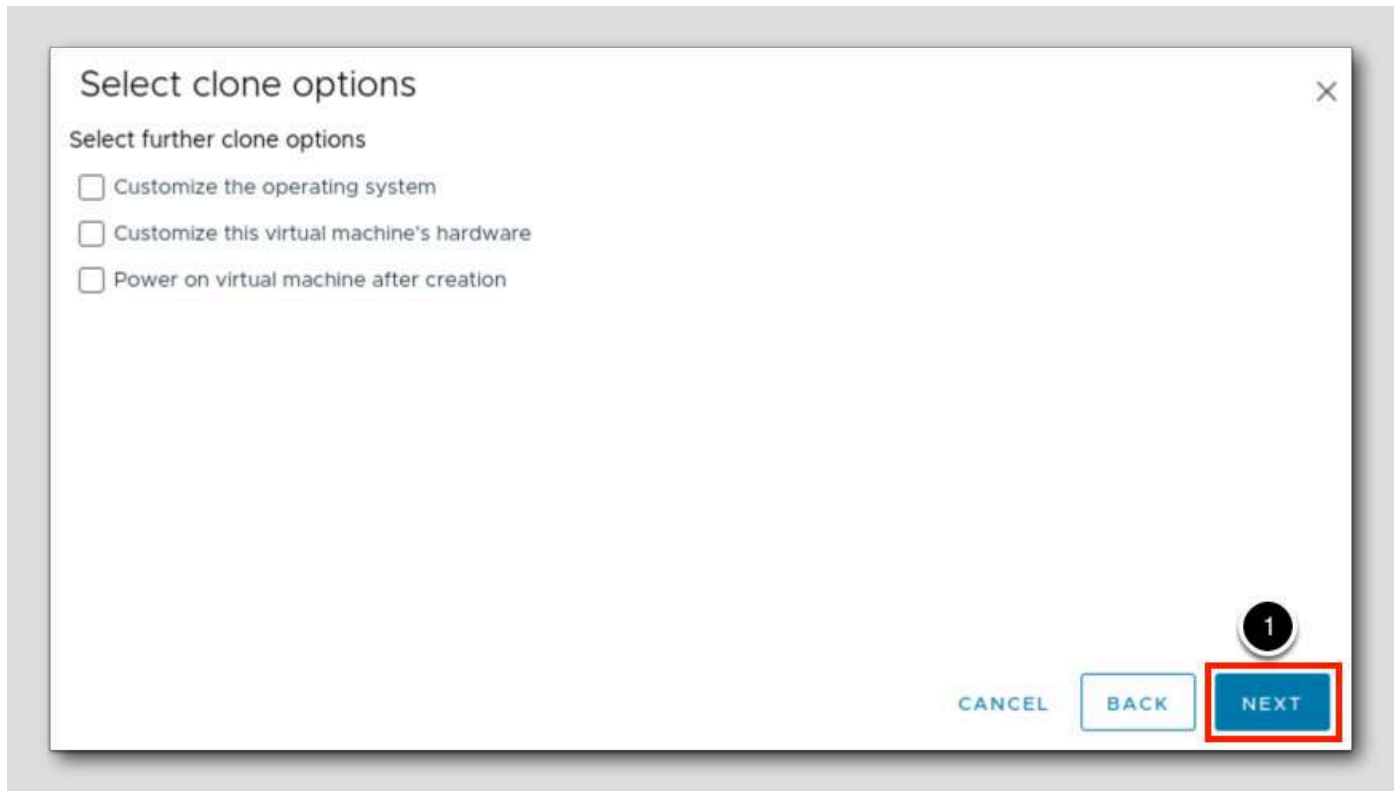
2

CANCEL BACK **NEXT**

1. Select ds-iscsi01 as your storage option
2. Click Next.

Select Clone Options

[99]

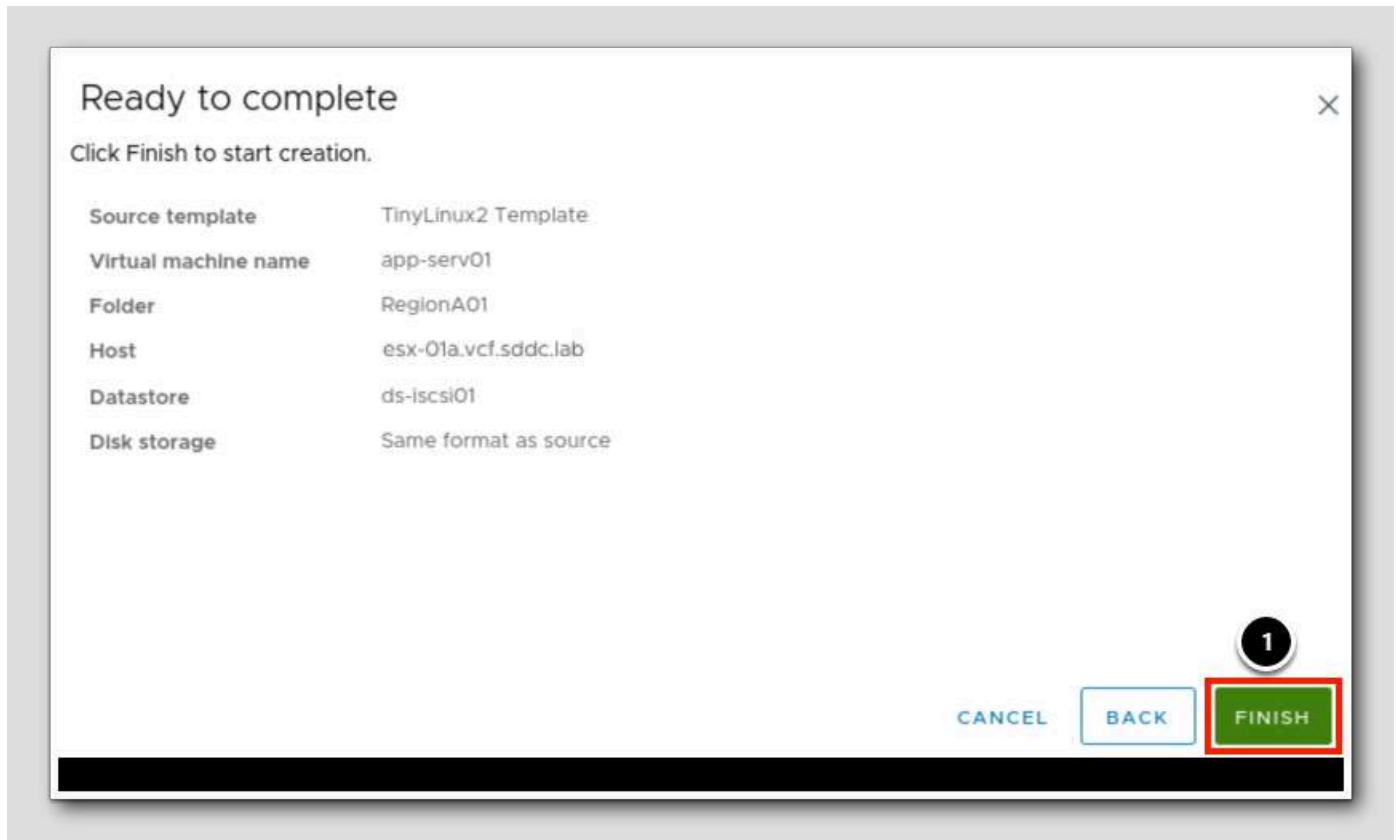


When cloning a virtual machine from a template, the guest operating system and virtual hardware can be modified. For this example, we will not customize the operating system or hardware.

1. Click **Next**.

Ready to Complete

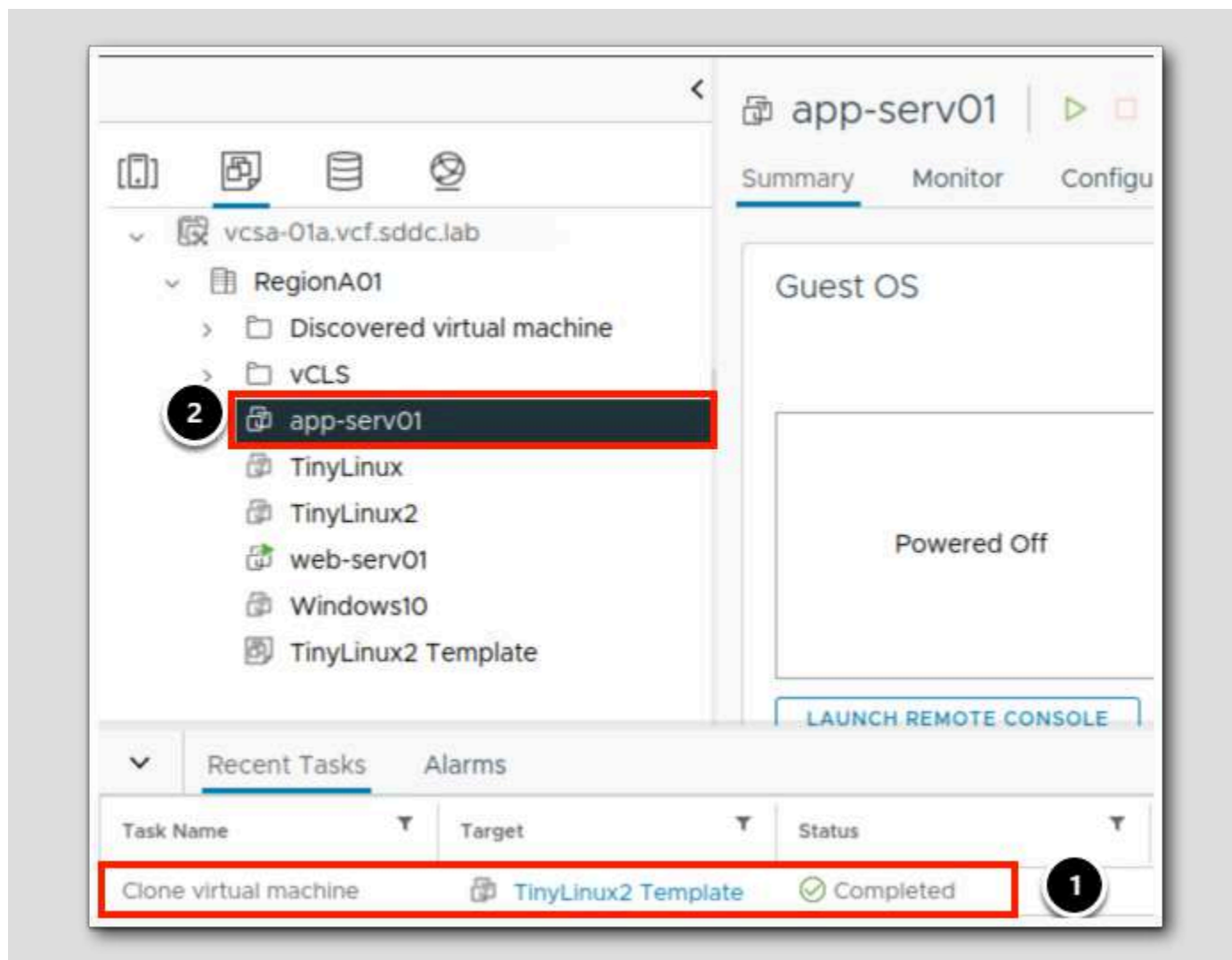
[100]



1. Review the deployment options and then click **Finish**.

Monitor Task Progress

[101]



1. You can view the **Recent Tasks** window to monitor the virtual machine being created from the template.
2. When the task is complete, you will see the **app-serv01** virtual machine in the inventory pane.

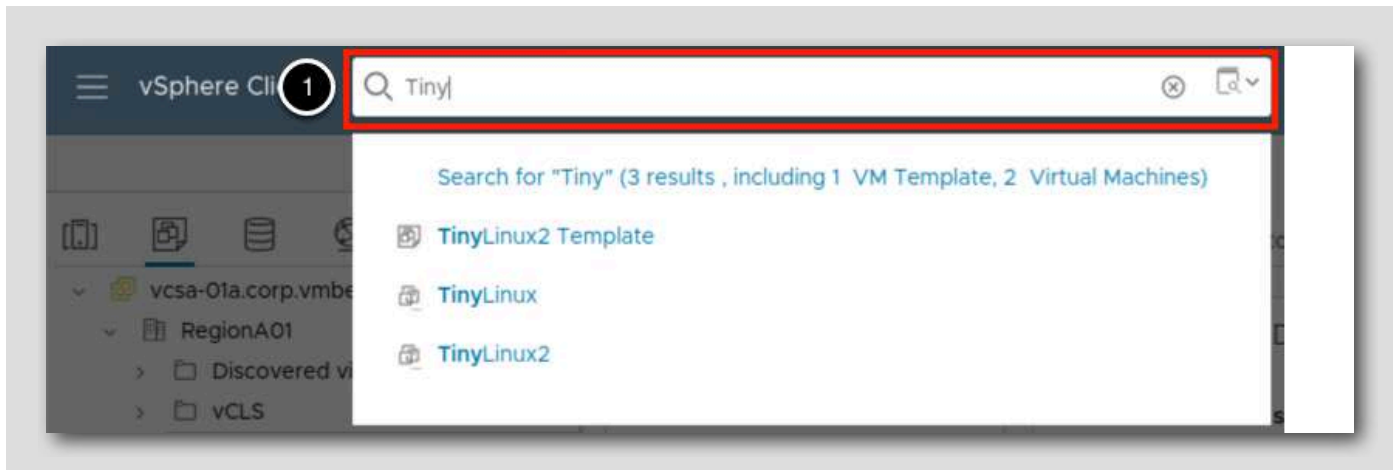
Using Tagging and Search to Find Objects Quickly

[102]

The vSphere Client provides some powerful search options. This lesson will guide you through the different search options to find the inventory of interest quickly. Also, the vCenter Inventory Service enables users to create custom defined tags that can be categorized and added to any inventory objects in the environment. These tags are searchable metadata and reduce the time to find inventory object information. This lab will cover how to create tags and use the tags for a search.

Search for Virtual Machines

[103]



At the top of the vSphere Client is a search bar that can be used to find objects quickly. This can be an object's name, like app-serv01 or an ESXi host. Tags can also be attached to objects and the search feature can be used to find them as well.

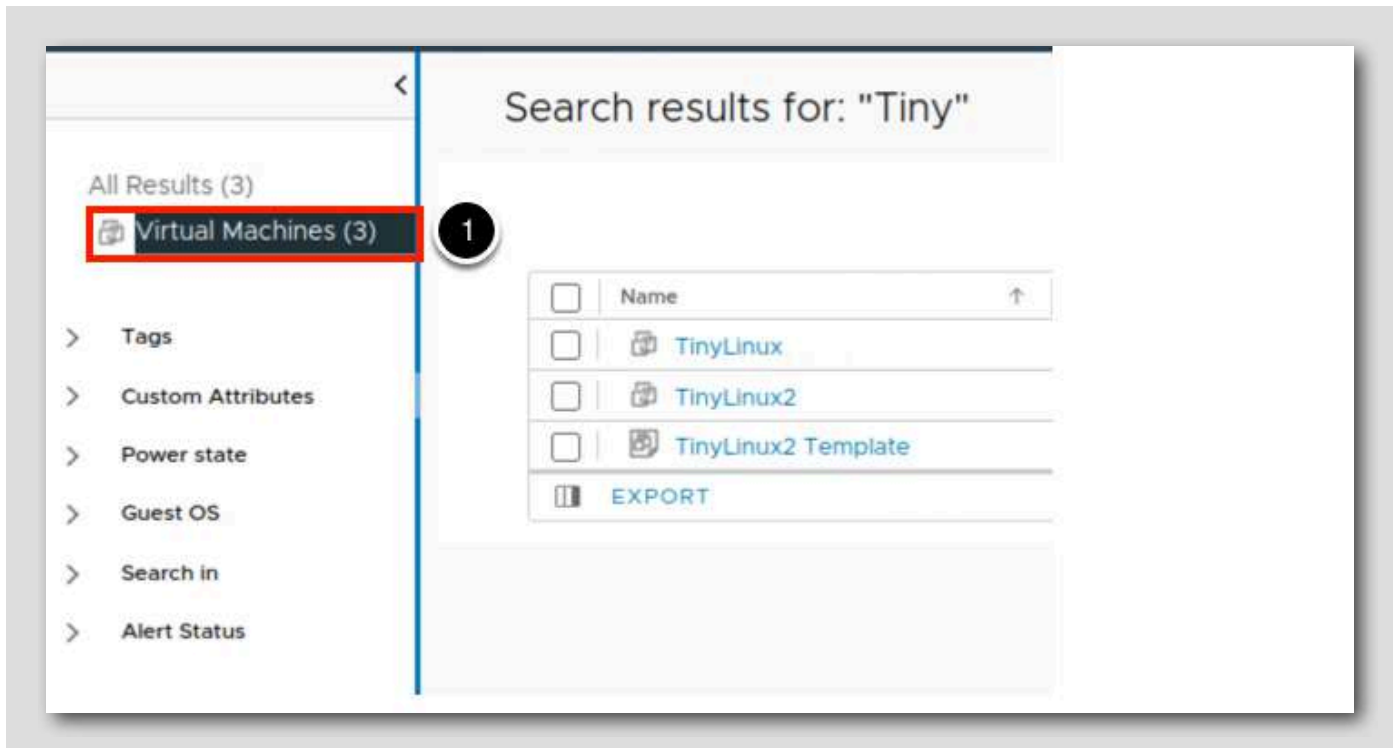
1. Click on the search bar at the top of the screen and type Tiny.

You can see all of the objects that contain the word tiny.

2. Press the Enter key.

Search Results

[104]

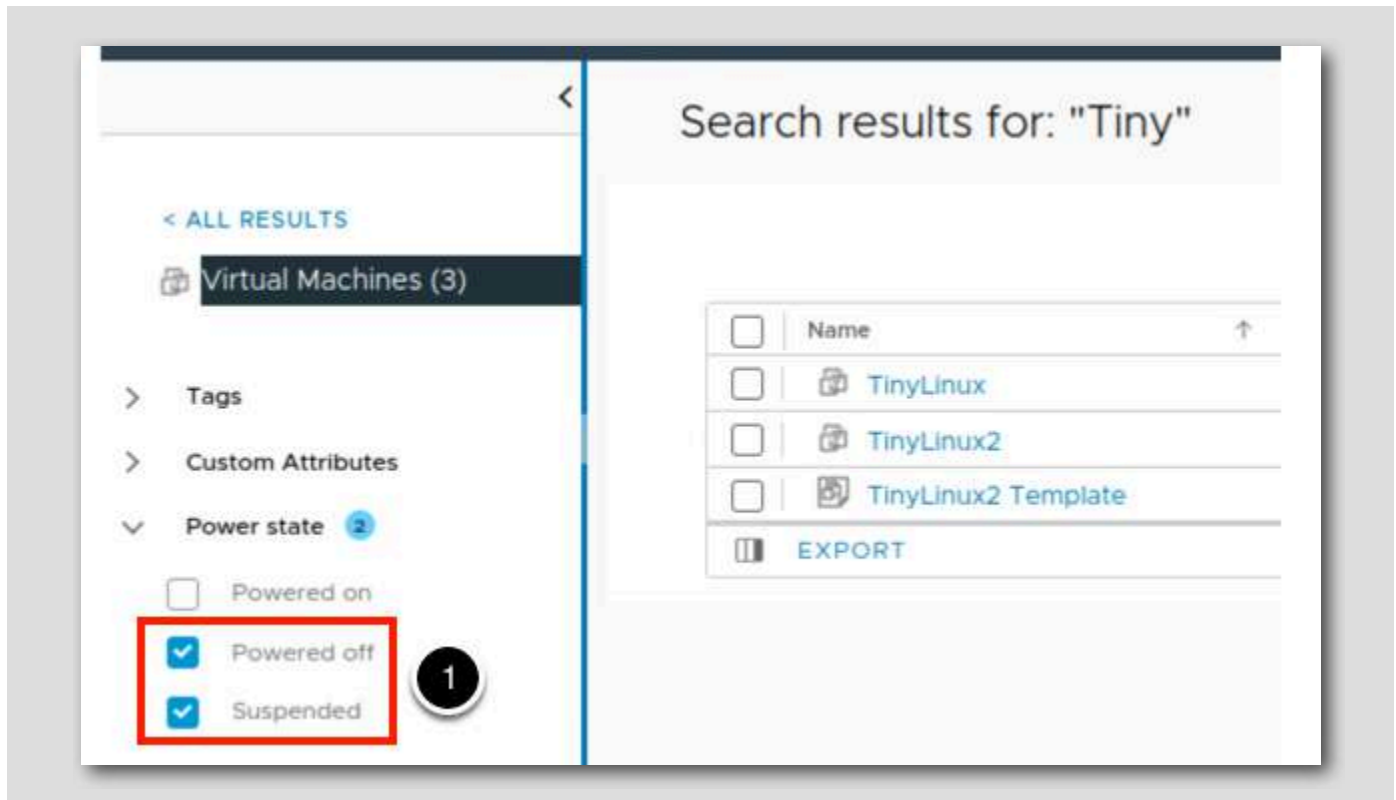


On this page, you can see all the results for objects that contain the word **tiny**. If you have a large inventory, the results can be narrowed down further by selecting the object type you are looking for. Tags or Custom Attributes could be used to narrow the search results down. Selecting the object type can help you quickly find the object you are looking for.

1. Click on **Virtual Machines**.

Filter Results

[105]



You can then filter the results down even further by specifying:

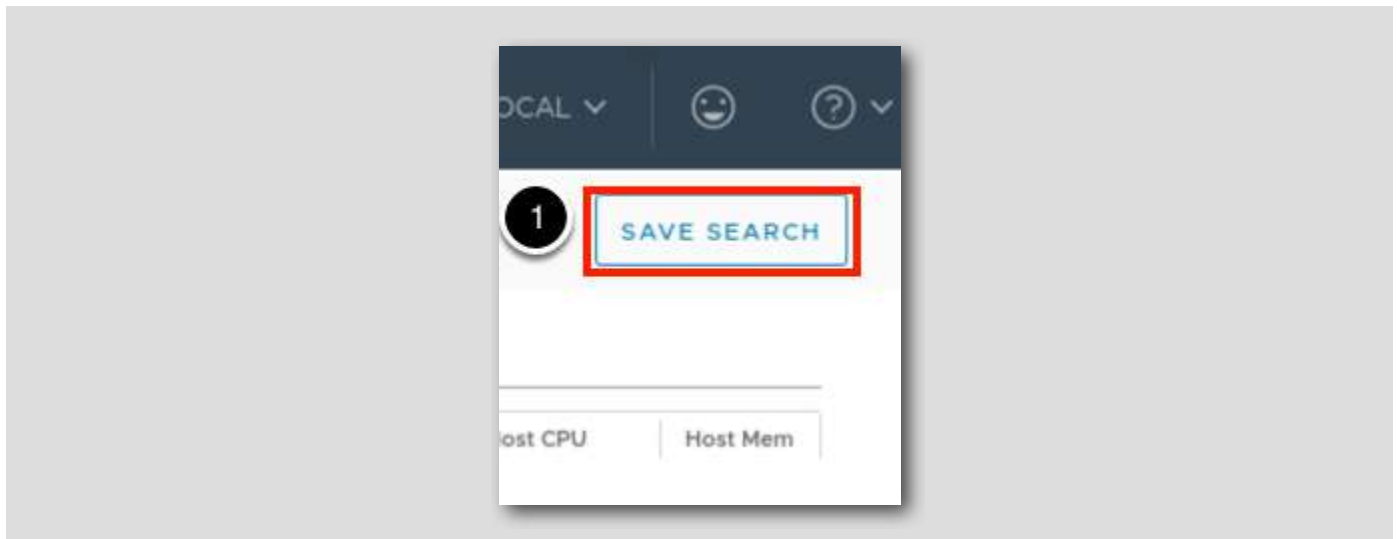
- The Power state of the virtual machine
- What operating system is running in the virtual machine
- What Host, Cluster or Datacenter to search in

1. Tick the box next to **Powered Off** and **Suspended**.

The search field is updated with the results.

Save the Search

[106]

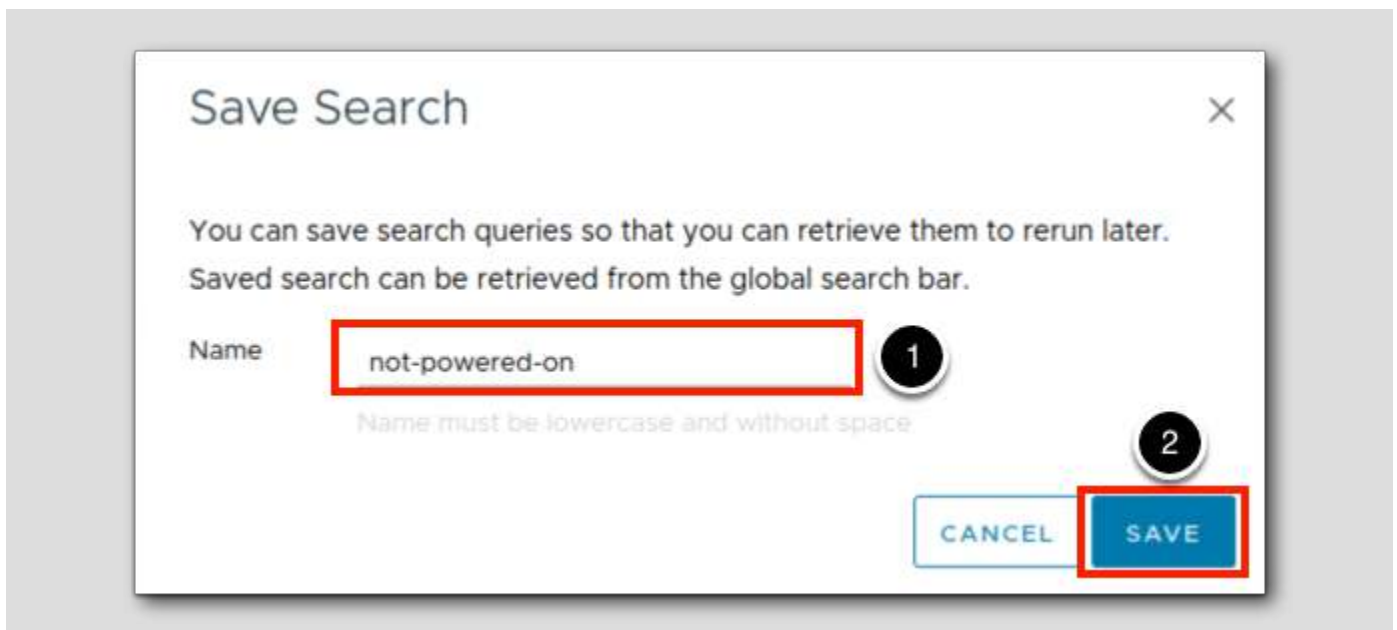


If this is a frequently used search, it can be saved for use in the future.

1. Click the **Save Search** button.

Name Search

[107]

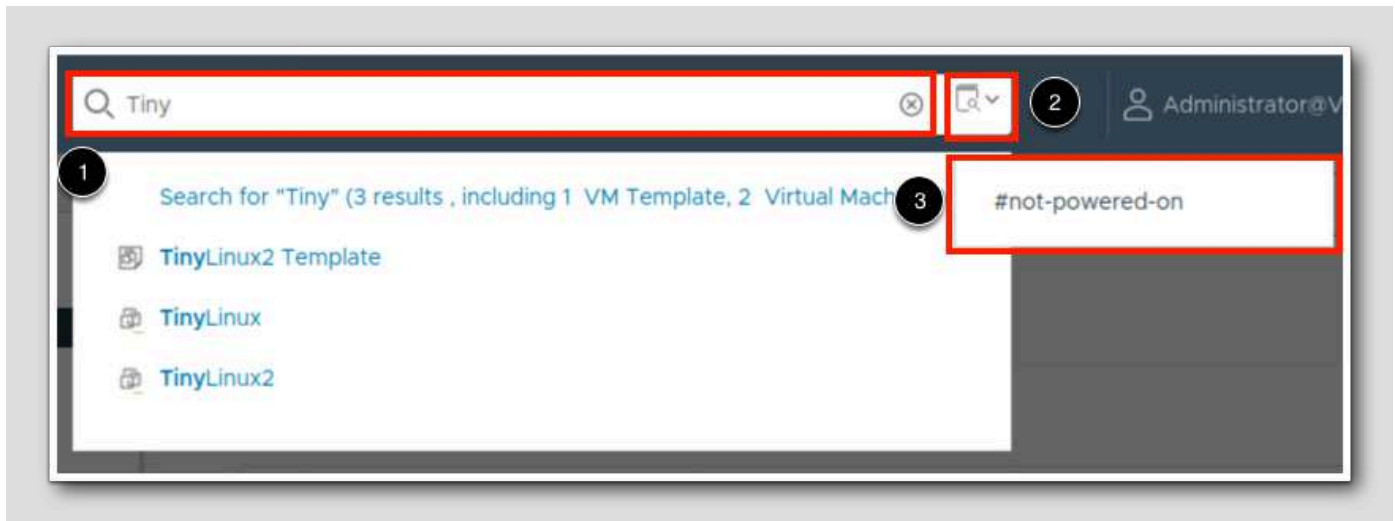


1. Name the search **not-powered-on**
2. Click the **Save** button.

Note that the name must be in lowercase with no spaces between words.

View Saved Search

[108]



1. To view a saved search, click in the Search field.
2. Click on the **drop-down arrow** to see the previously saved search results.
3. Click on **#not-powered-on**.

Not-Powered-On-VMs

[109]

Saved Search: "#not-powered-on"

Search results for: "Tiny"

1 ACTIONS ▾

- Save as
- Rename
- Delete

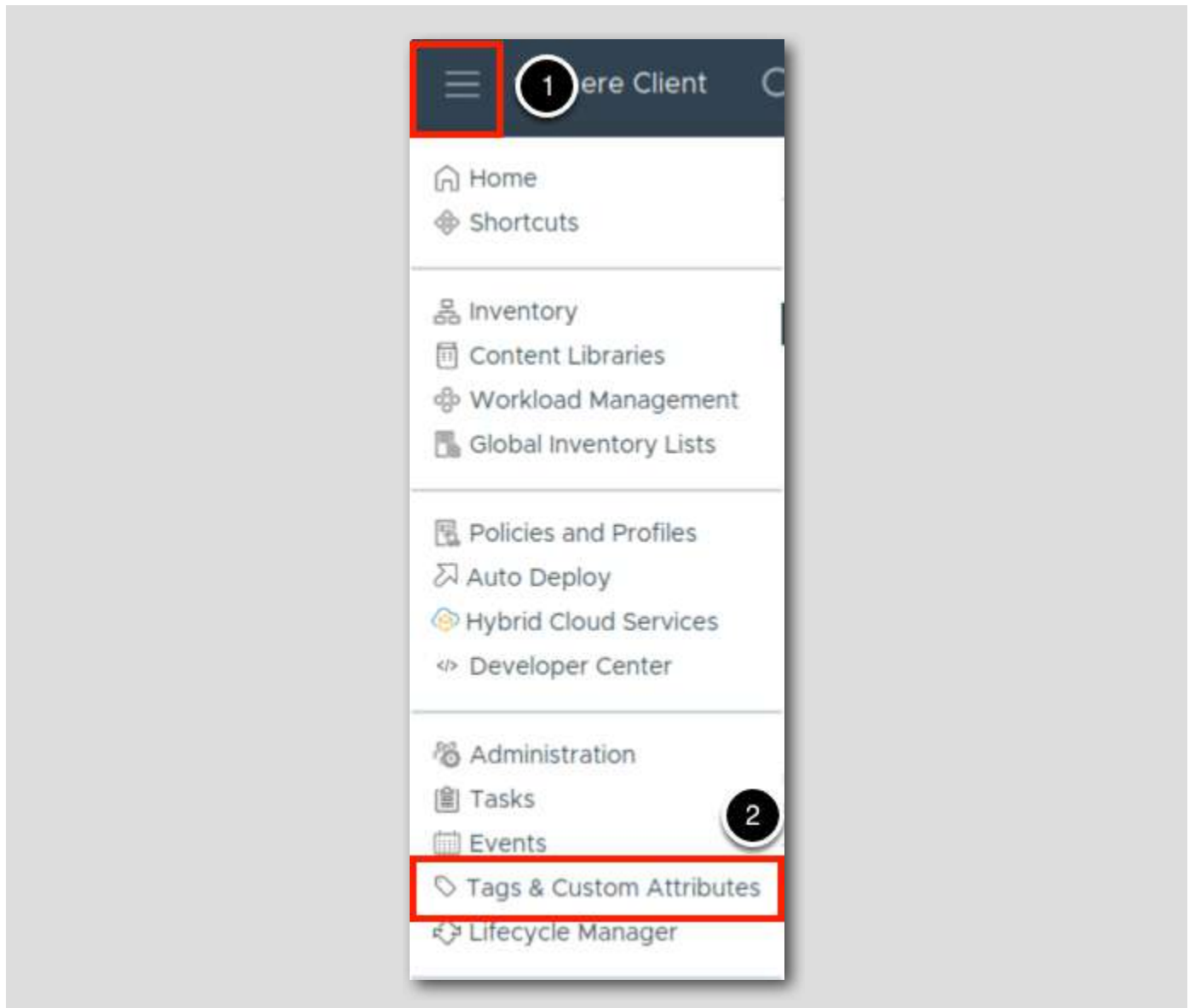
☐	Name	↑	State	Status	Provisioned Space	Used Space	Host CPU
☐	TinyLinux		Powered ...	✓ Normal	733.25 MB	35.43 MB	0 Hz
☐	TinyLinux2		Powered ...	✓ Normal	733.63 MB	35.81 MB	0 Hz
☐	TinyLinux2 Template		Powered ...	✓ Normal	732.82 MB	2 KB	0 Hz

EXPORT 3 items

1. Note that in the Actions menu, this search can be saved as another name and modified. It can also be renamed or deleted.

Tags and Custom Attributes

[110]

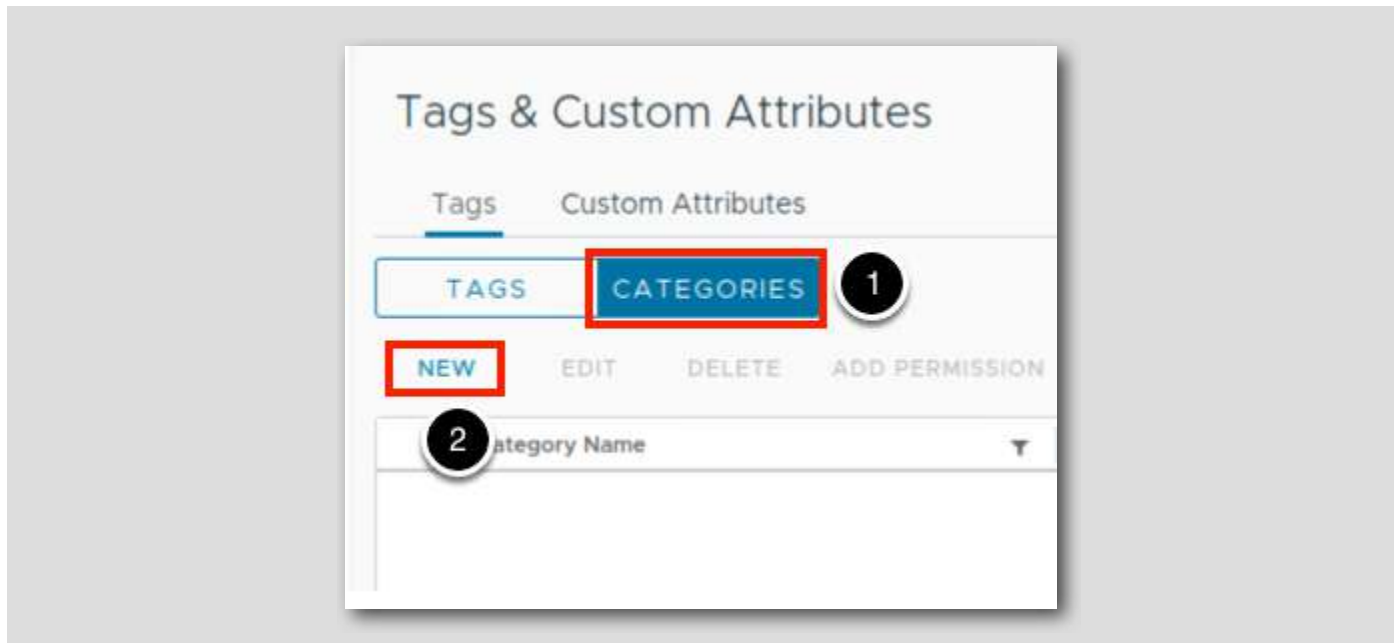


You use tags to add metadata to inventory objects. You can record information about your inventory objects in tags and use the tags in searches.

1. Click **Menu** on the top left corner
2. Select "Tags and Custom Attributes"

Creating Tag Categories

[11]



You use categories to group tags together and define how tags can be applied to objects.

Every tag must belong to one and only one category. You must create at least one category before creating any tags.

1. Click the **Categories** tab.
2. Click **New**.

New Category

Create Category

Category Name: 1

Description: 2

Tags Per Object: ☒ One tag 3 ☐ Many tags

Associable Object Types:

☒ All objects	☒ Cluster
☒ Folder	☒ Datastore
☒ Datacenter	☒ Distributed Port Group
☒ Datastore Cluster	☒ Host
☒ Distributed Switch	☒ Library Item
☒ Content Library	☒ Resource Pool
☒ Network	☒ Virtual Machine
☒ vApp	

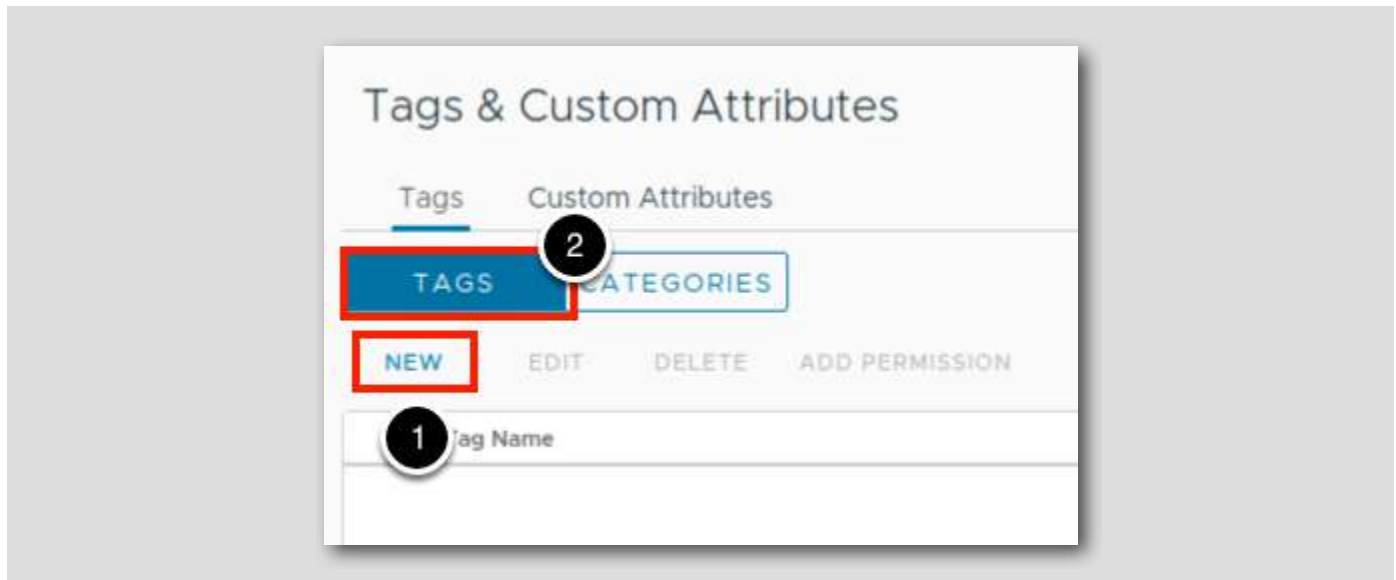
4

Associable Object Types: We will use the default which states that the new tag in this category can be assigned to all objects. The other option is you can specify a specific object, such as virtual machines or datastores.

1. Enter "web tier" for the Category Name.
2. For a description, type "All objects in the web tier."
3. Keep the default One tag tags per object
4. Click Create

Create a New Tag

[113]



The new category has been created.

1. Click the **Tags** tab to create a new a Tag.
2. Click **New**

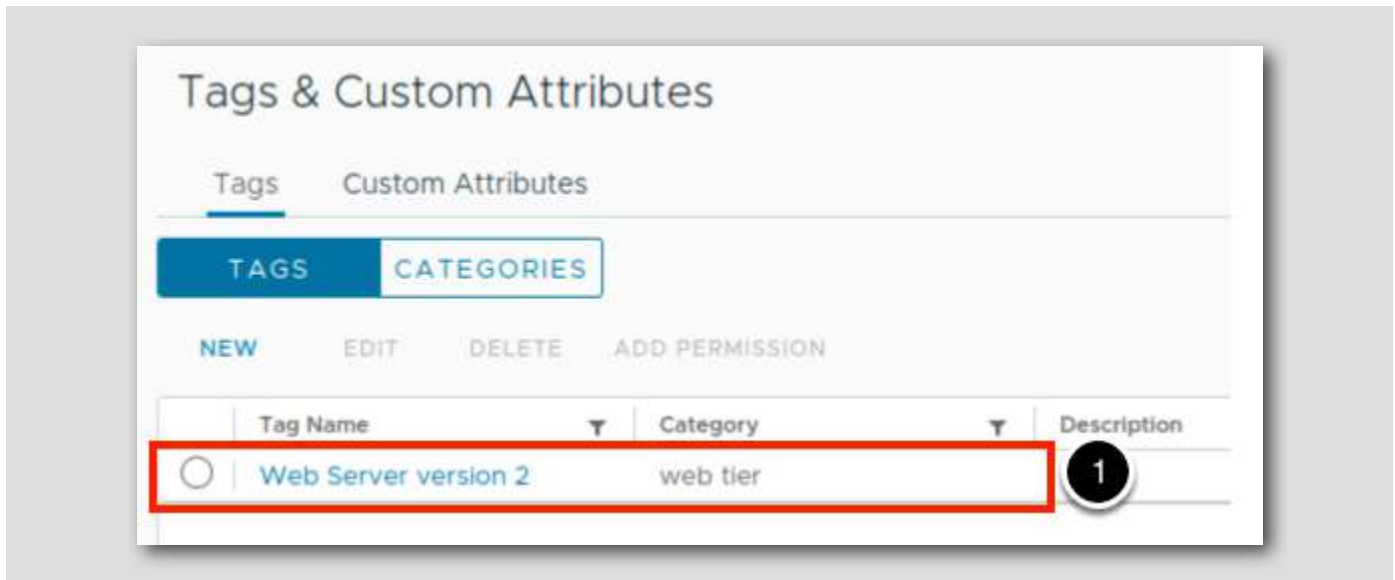
Add Tag

[114]

1. Name the tag **Web Server version 2**
2. Leave the **web tier** category as is.
3. Select **Create**

New Tag

[115]

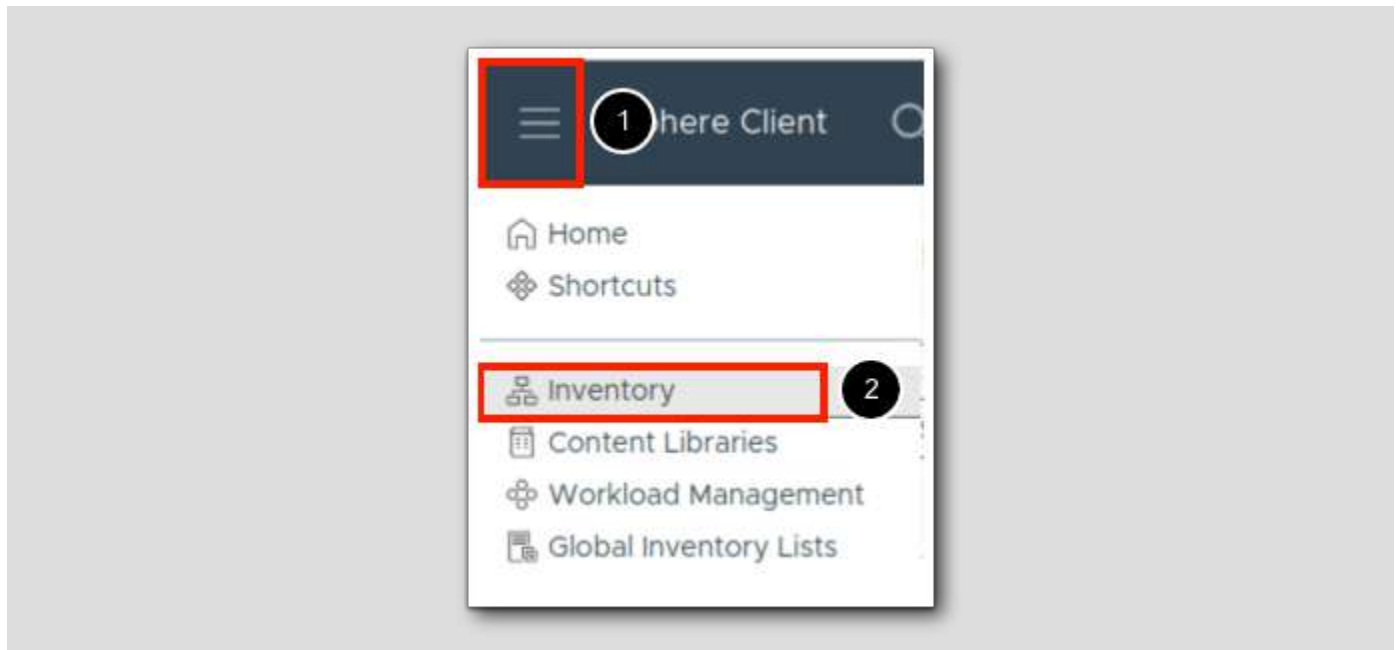


1. The newly created tag has now been added.

In order for these tags to be useful, they need to be assigned to objects. In the next steps, the tag will be assigned to virtual machines.

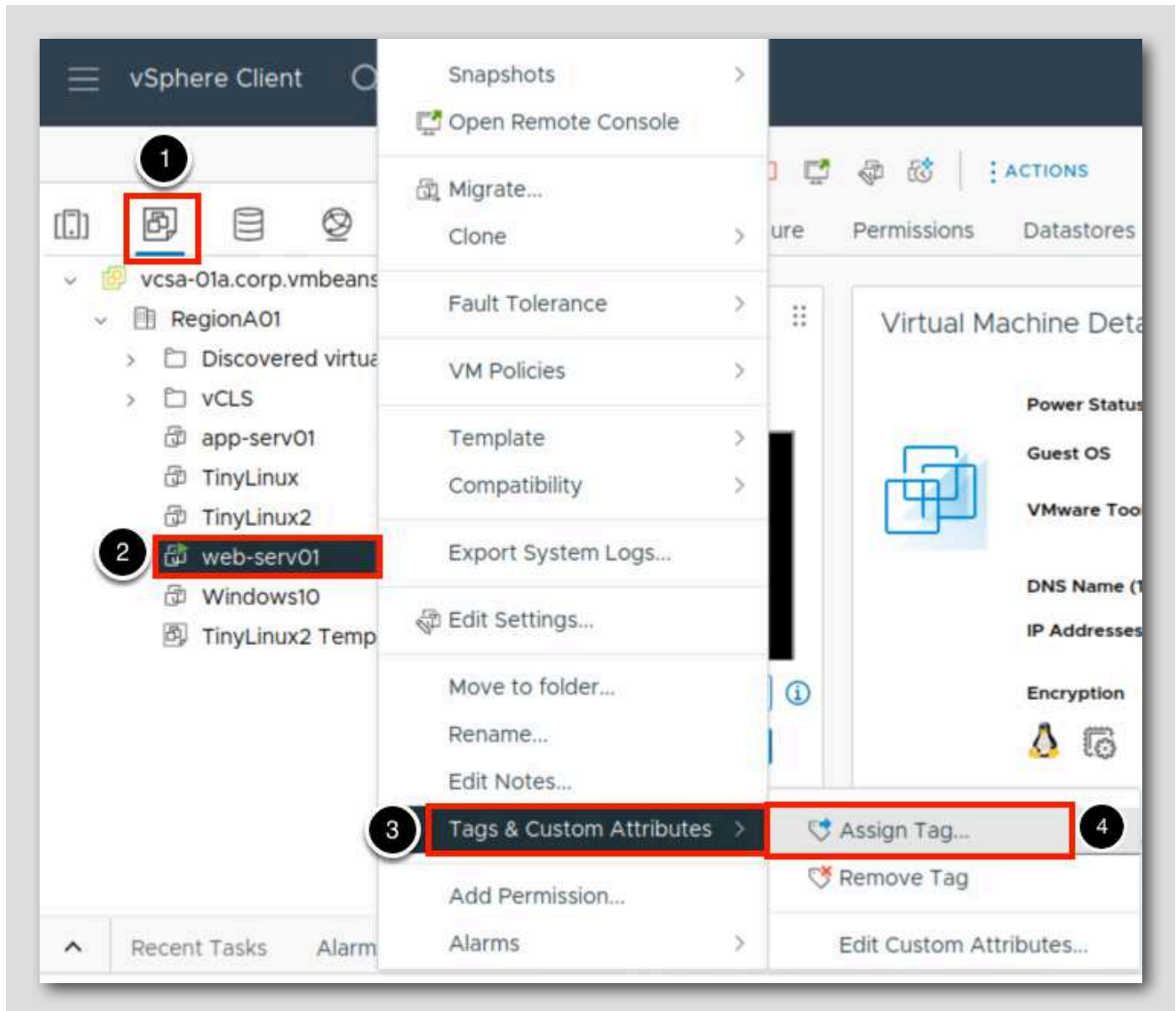
Go to Inventory

[116]



1. Click into the **Menu** on the top left corner
2. Select **Inventory**

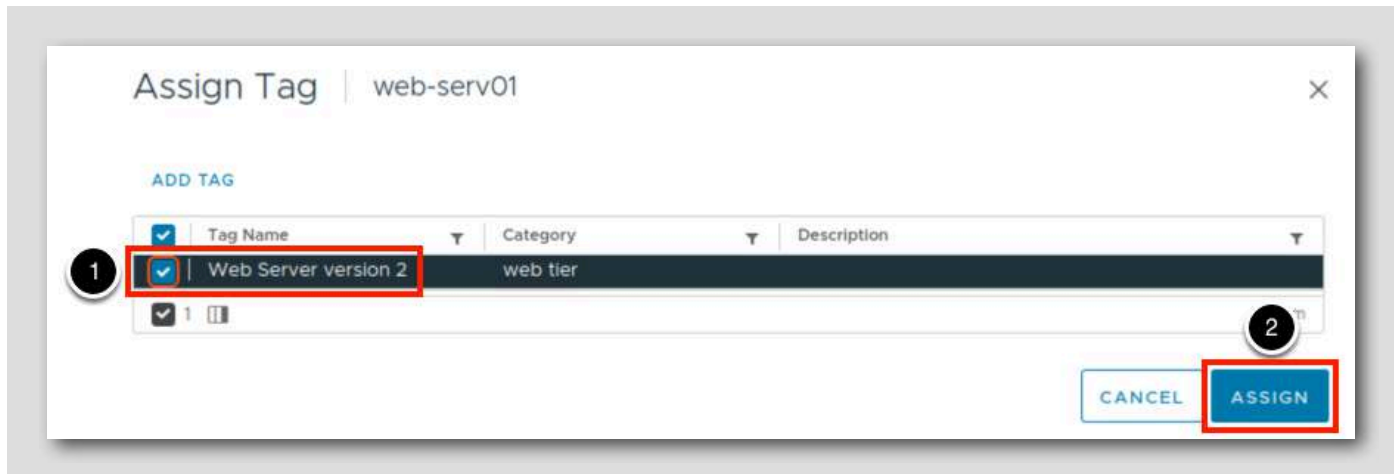
Select a Virtual Machine



1. Select the Virtual Machines and Templates tab
2. Right-click the virtual machine `web-serv01`.
3. Find **Tags & Custom Attributes**
4. Click **Assign Tag...**

Assign Tag

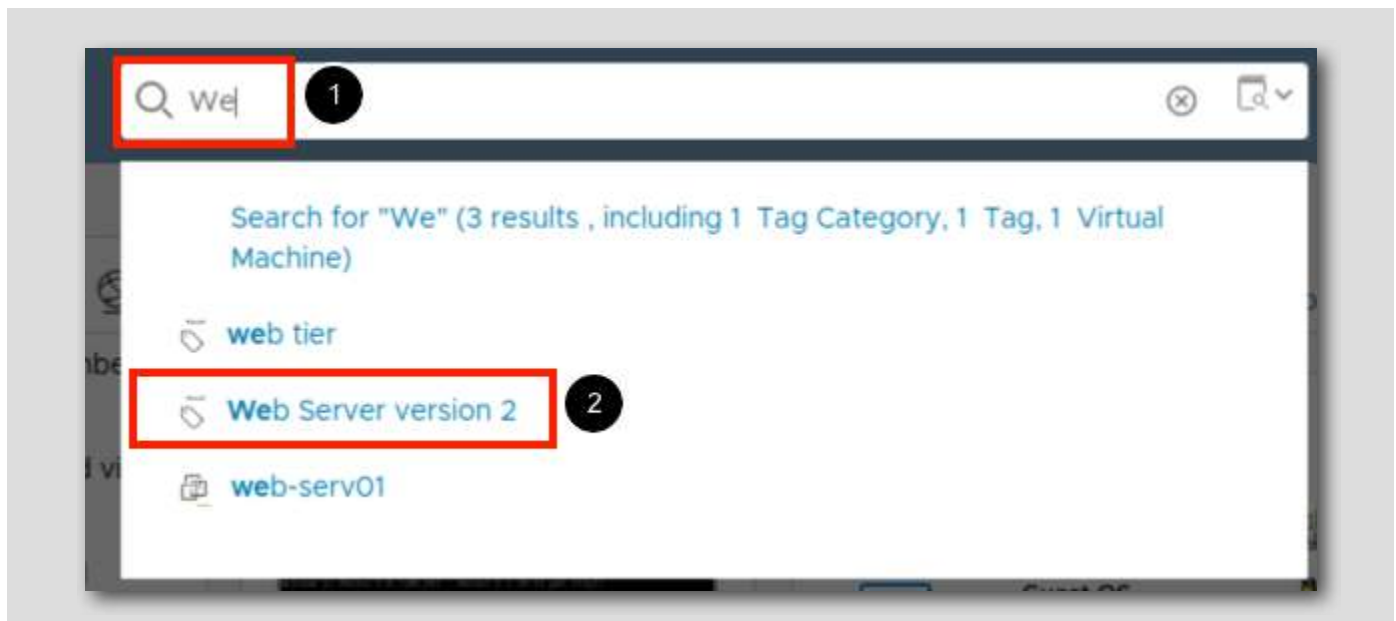
[118]



1. Click the Web Server version 2 tag.
2. Click Assign.

Search Using Tags

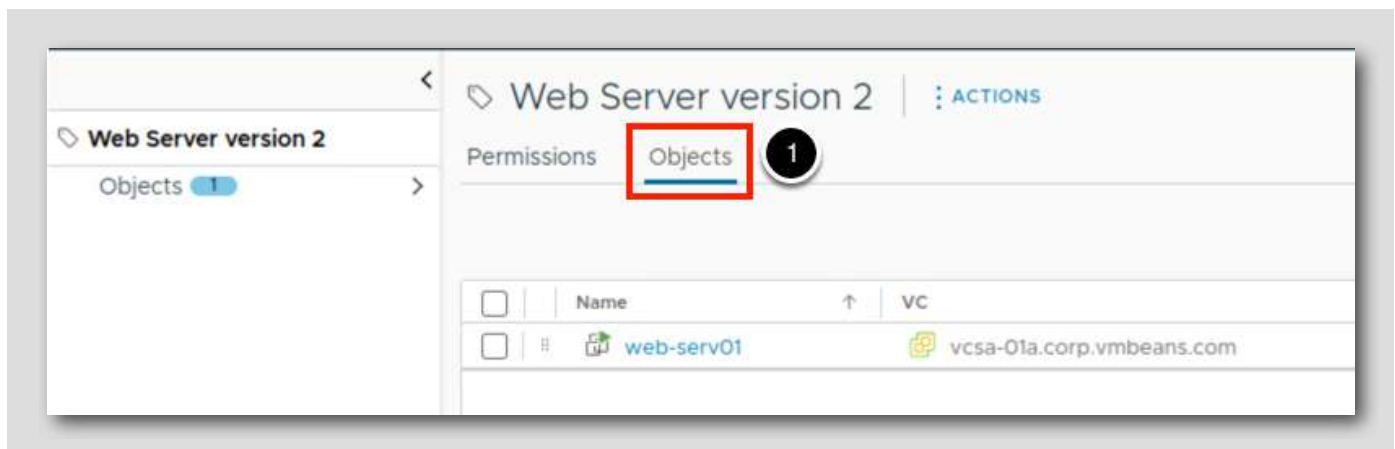
[119]



1. In the Search field enter "we".
2. Select the Tag **Web Server version 2**.

Search Results

[120]



1. Click on the **Objects** tab to find the list of objects which have been assigned the **Web-serv01** tag.

Understanding vSphere High Availability and Distributed Resource Scheduler (DRS)

[121]

This lab shows how to use the VMware vSphere Client to enable and configure vSphere Availability and Dynamic Resource Scheduling (DRS). HA protects from down time by automating recovery in the event of a host failure. DRS ensures performance by balancing virtual machine workloads across hosts a cluster.

What is vSphere High Availability?

[122]

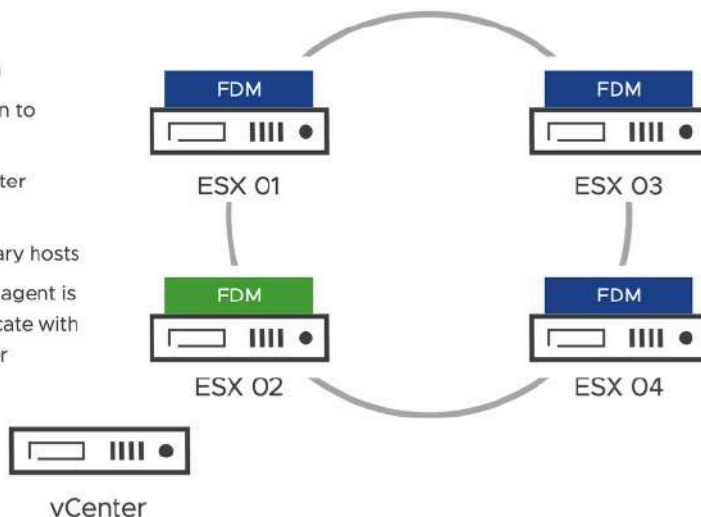
vSphere High Availability provides high availability for virtual machines by pooling the virtual machines and their hosts into a cluster. Hosts in the cluster are monitored and in the event of a failure, the virtual machines on a failed host are restarted on alternate hosts.

When you create a vSphere Availability cluster, a single host is automatically elected as the primary host. The primary host communicates with vCenter Server and monitors the state of all protected virtual machines and of the secondary hosts. Different types of host failures are possible, and the primary host must detect and appropriately deal with the failure. The primary host must distinguish between a failed host and one that is in a network partition or that has become network isolated. The master host uses network and datastore heartbeating to determine the type of failure. Also note that vSphere High Availability is a host function, which means there is no dependency on vCenter to effectively fail over VMs to other hosts in the cluster.

vSphere High Availability Primary Components

[123]

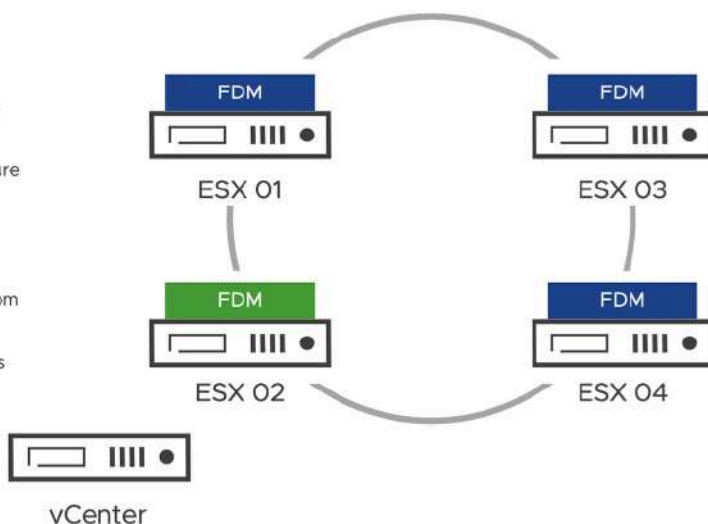
- Every host runs an agent
 - Referred to as Fault Domain Manager (FDM)
 - One of the agents within the cluster is chosen to assume the role of the Primary host
 - There is only one Primary host per cluster during normal operations
 - All other agents assume the roles of secondary hosts
- When you add a host to a vSphere HA cluster, an agent is uploaded to the host and configured to communicate with other agents in the cluster. Each host in the cluster functions as a primary host or a secondary host.



The Primary Host Role

[124]

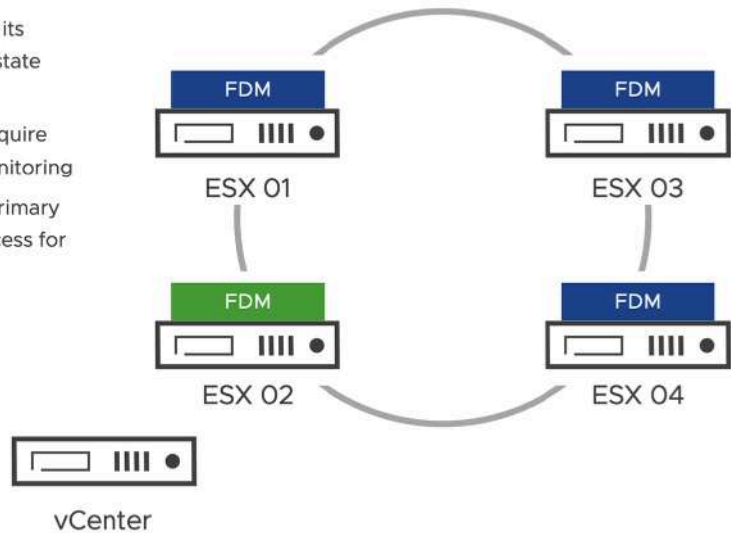
- **The Primary FDM monitors:**
 - ESX Hosts and virtual machine availability
 - All secondary hosts. Upon a secondary host failure, protected VMs on that host will be restarted
 - The power state of all the protected VMs. Upon failure of a protected VM, the primary host will restart it.
- **The primary FDM manages:**
 - The list of hosts that are members of the cluster, updating the lists as hosts are added or removed from the cluster.
 - The list of protected VMs. The primary FDM updates the list after each user-initiated power on/off.



The Secondary Host Role

[125]

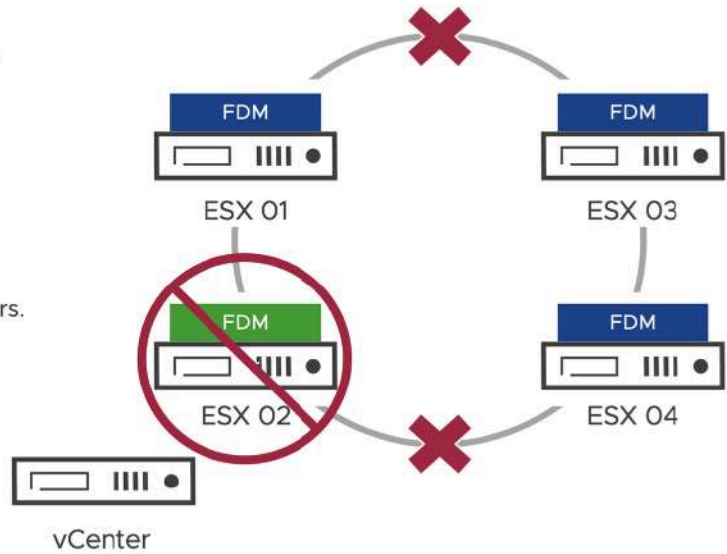
- The secondary host monitors the runtime state of its locally running VMs and forwards any significant state changes to the primary host.
- It implements vSphere HA features that do not require central coordination, most notably VM Health monitoring
- It monitors the health of the primary host. If the primary host should fail, it participates in the election process for a new primary host.
- Maintains list of powered-on VMs.



The Primary Host Election Process

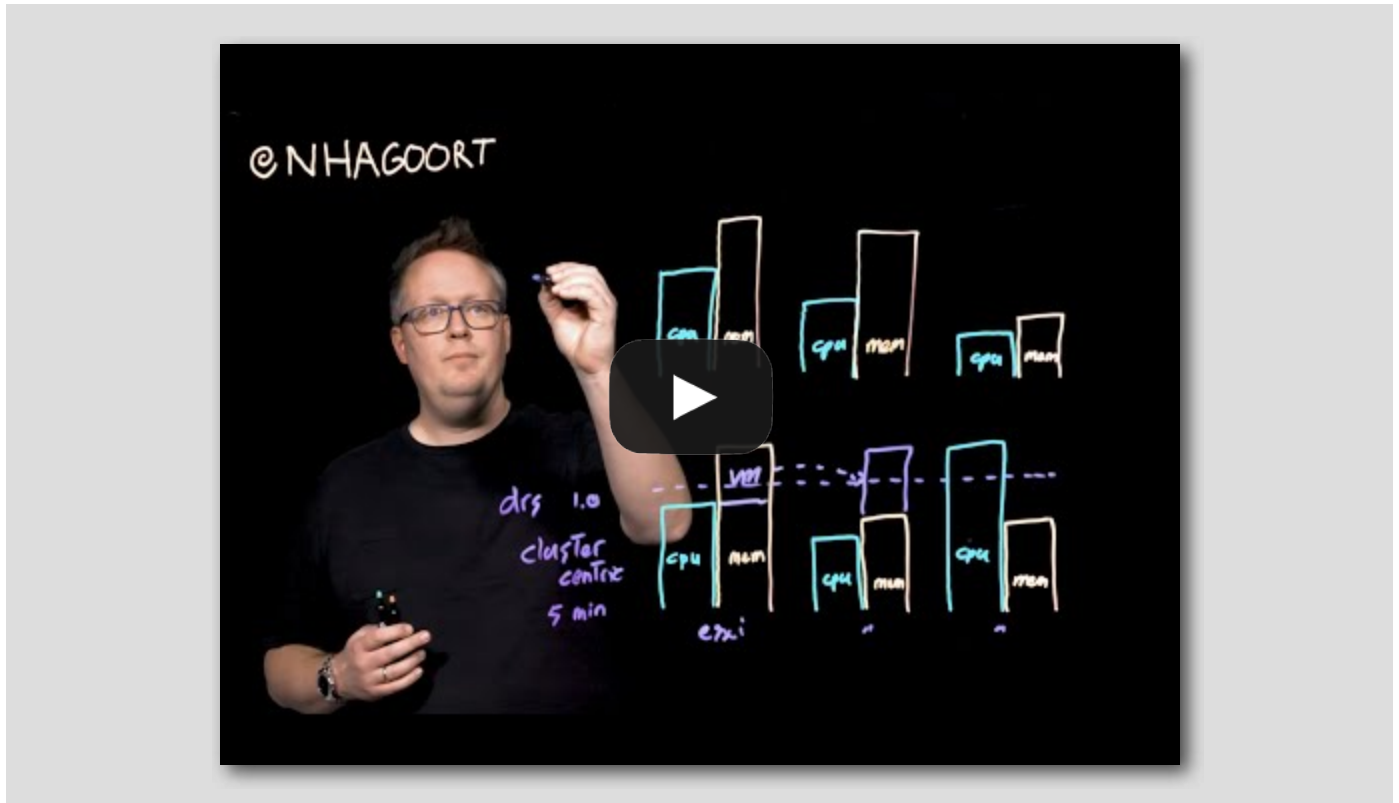
[126]

- The primary host is determined through an election process.
- An election occurs when:
 - vSphere HA is enabled.
 - A primary host fails, is shutdown, or is placed in maintenance mode.
 - A management network partition occurs.



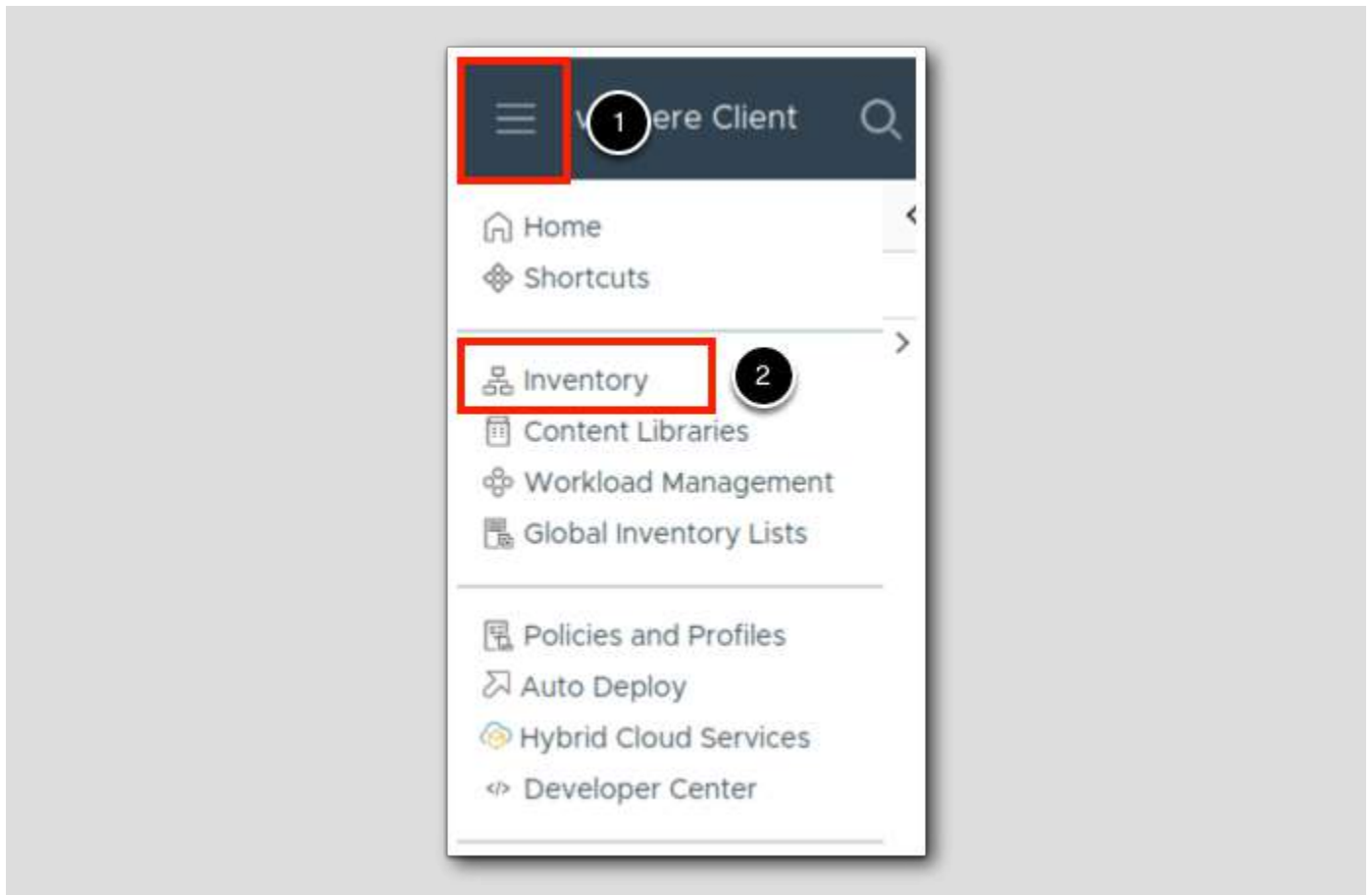
What is New with DRS in vSphere 7 (5:47)

[127]

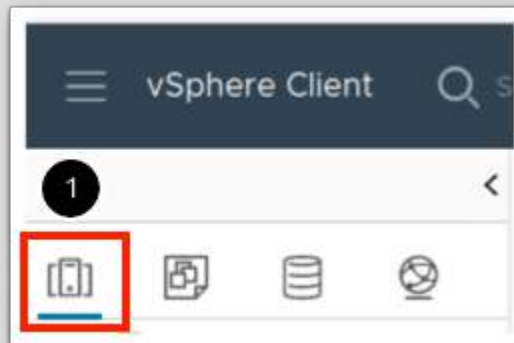
<https://www.youtube.com/watch?v=vnuUzW7Yffo>

Enable and Configure vSphere High Availability

[128]



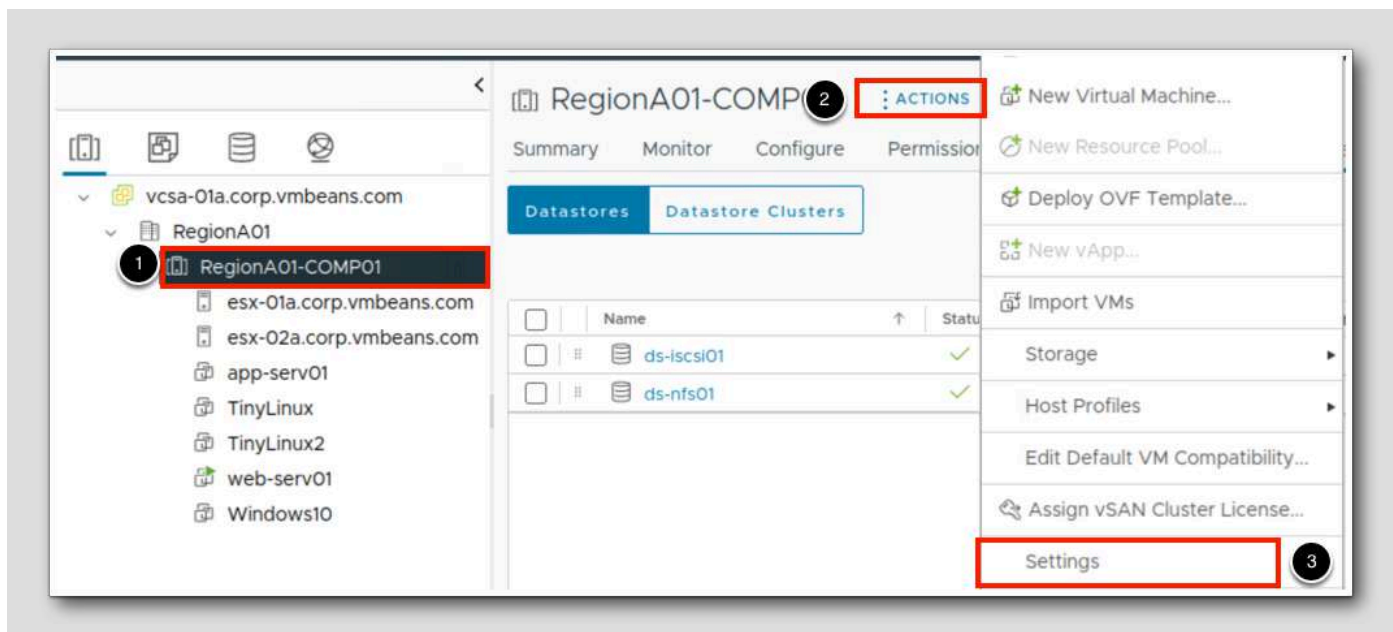
1. First, click on the **Menu** hamburger icon on your top left
2. Select **Inventory**



1. Select Host and Clusters

Settings for vSphere High Availability

[129]



1. Click RegionA01-COMP01 Cluster.
2. Click Actions to bring up the drop down-menu.
3. Click Settings.

Cluster Settings

[130]

RegionA01-COMP01 | ACTIONS

Summary Monitor **Configure** Permissions Hosts VMs Datastores Networks Updates

Services ☒ vSphere DRS **vSphere Availability** ☒ Configuration > Licensing ☒ vSAN Cluster Trust Authority Alarm Definitions

vSphere HA is Turned OFF **2** **EDIT...**

Runtime information for vSphere HA is reported under [vSphere HA Monitoring](#)

Proactive HA is not available **1** **EDIT...**

To enable Proactive HA you must also enable [DRS](#) on the cluster.

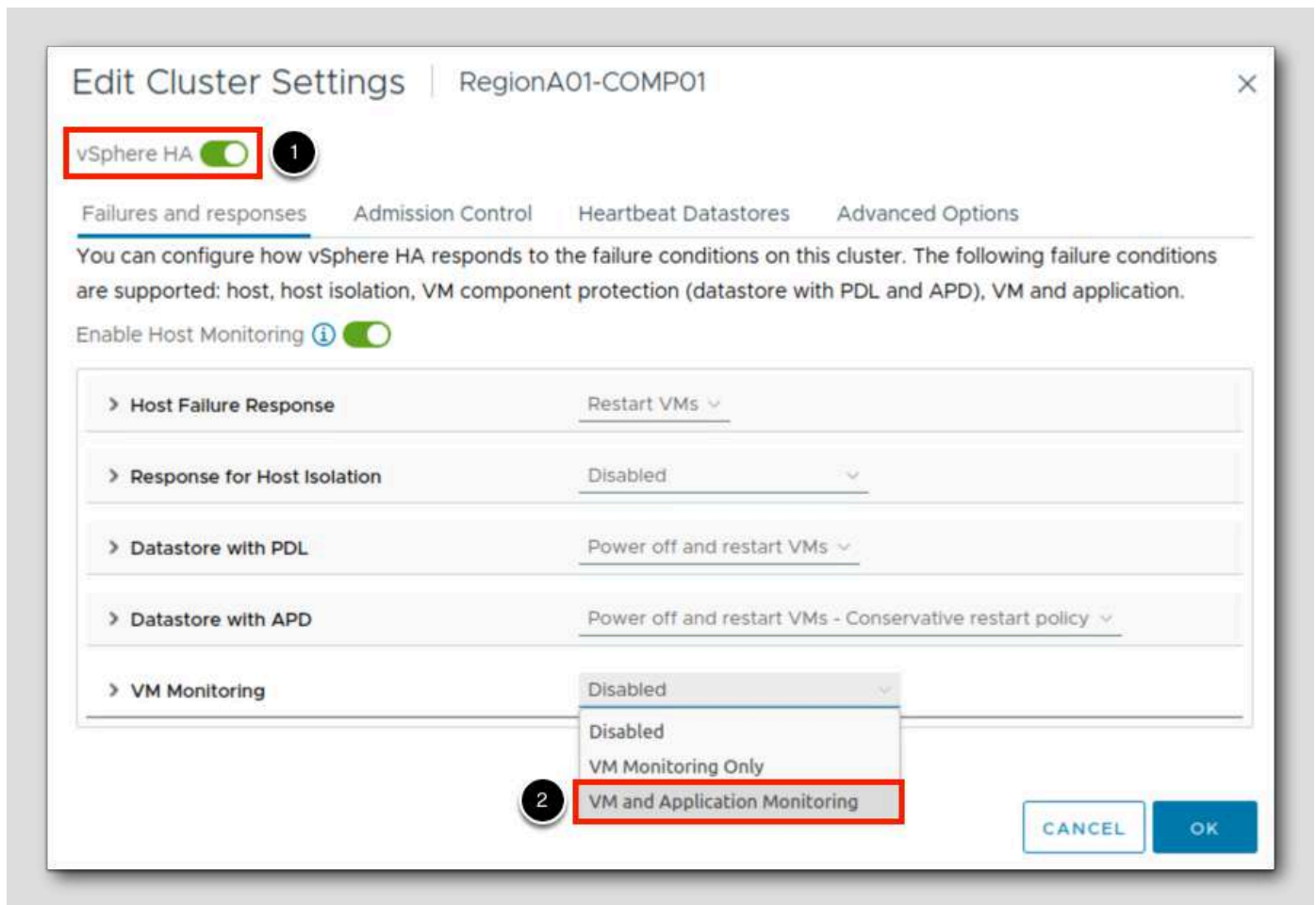
Failure conditions and responses

Failure	Response	Details
Host failure	Disabl...	vSphere HA disabled. VMs are not restarted in the eve
Proactive HA	Disabl...	Proactive HA is not enabled.

1. Click **vSphere High Availability** under **Services** to bring up the settings for high availability. Note that you may need to scroll to the top of the list.
2. Click the **Edit** button next to **vSphere HA is Turned OFF**.

Enable vSphere HA

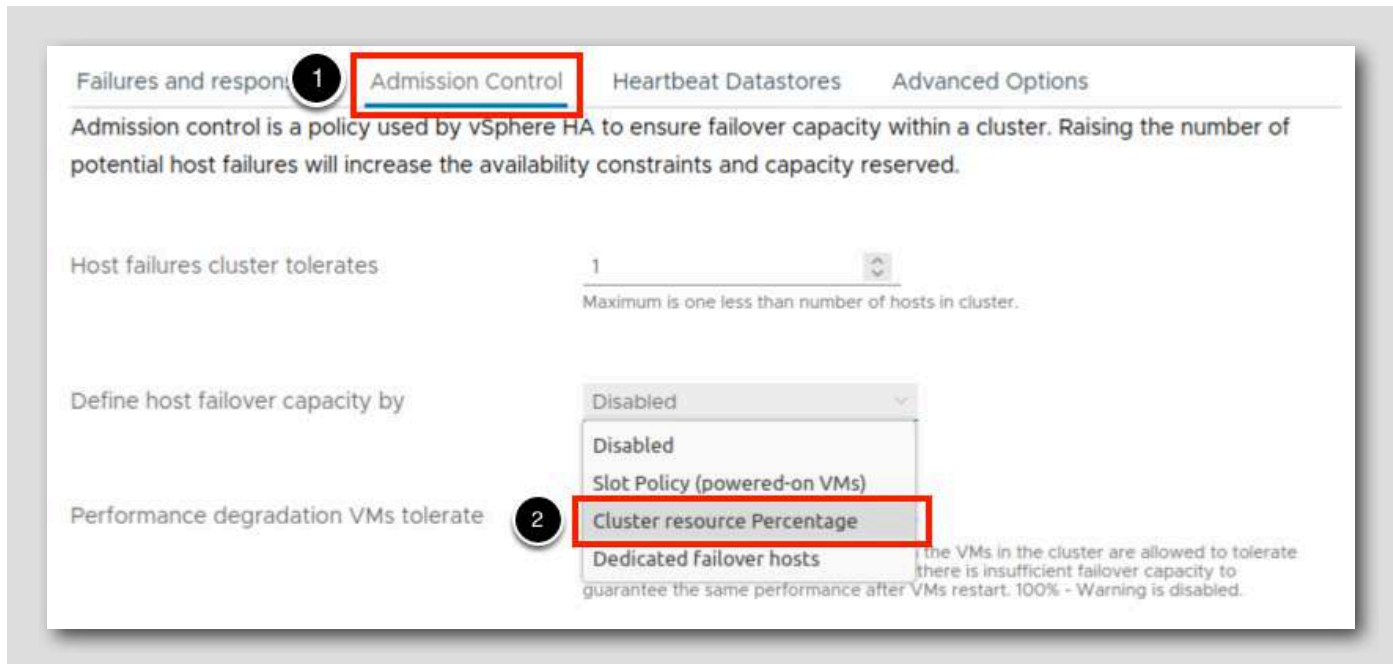
[131]



1. Click the toggle next to vSphere HA to enable it.
2. From the VM Monitoring drop-down list, select VM and Application Monitoring.

By selecting VM and Application Monitoring, a VM will be restarted if heartbeats are not received within a set time, the default is 30 seconds.

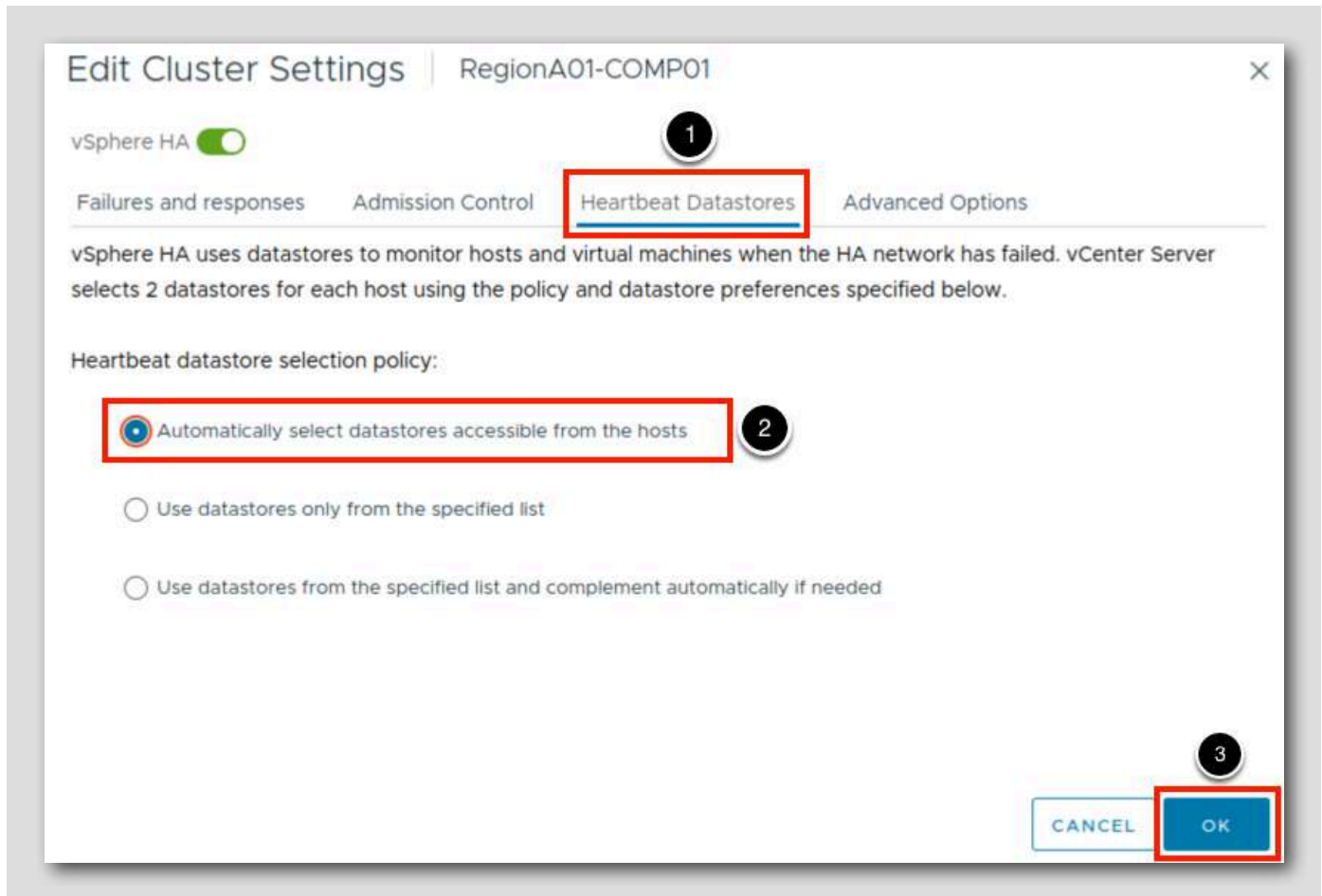
Admission Control



1. Click the **Admission Control** tab.
2. In the **Define host failover capacity by** drop-down menu, select **Cluster resource Percentage**.

We are setting aside a certain percentage of CPU and Memory resources to be used for failover, in the above case 25% for each.

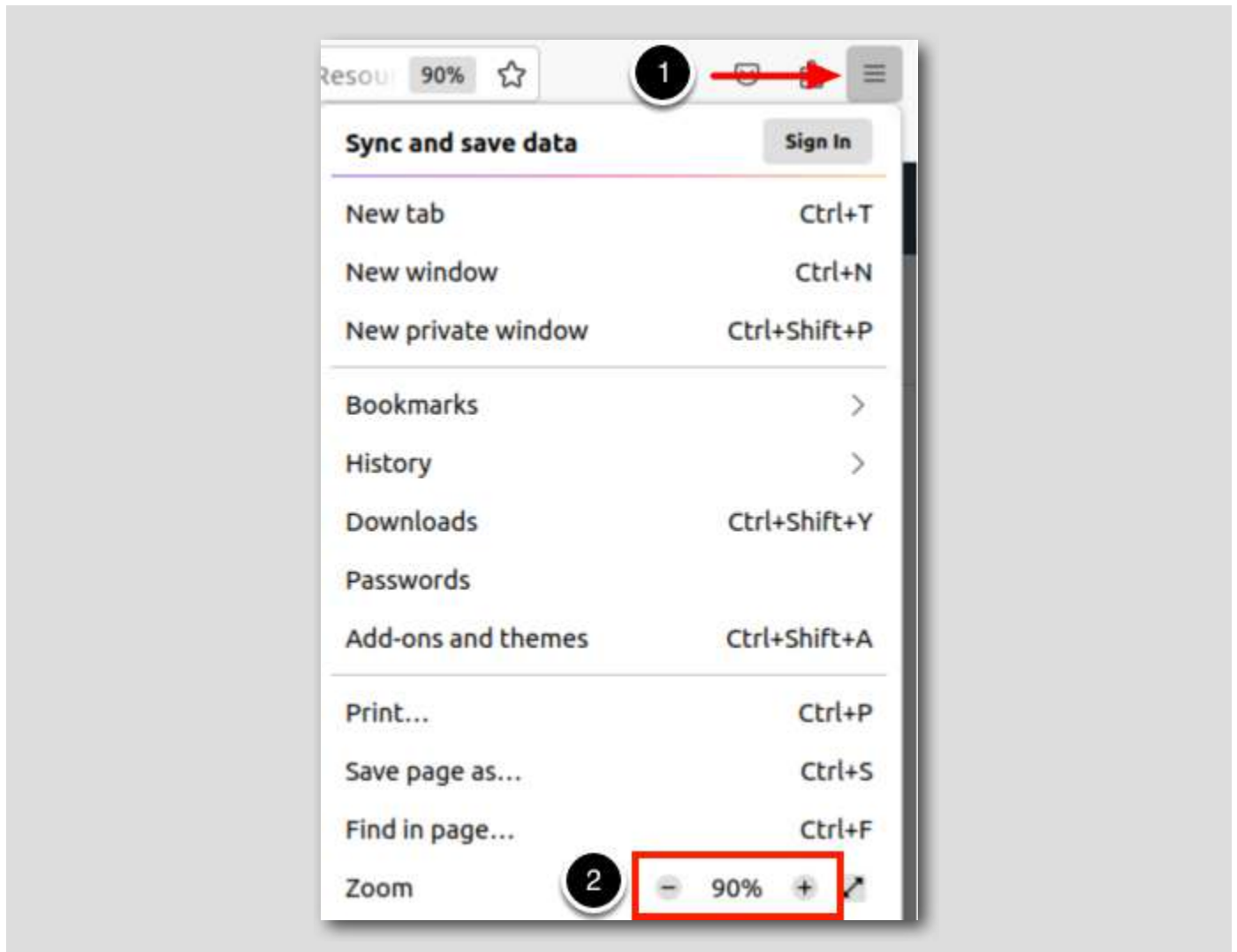
Heartbeat Datastores



1. Click **Heartbeat Datastores**.
2. Select **Automatically select datastores accessible from the hosts**.

This is another layer of protection. Heartbeat Datastores allows vSphere HA to monitor hosts when a management network partition occurs and to continue to respond to failures that occur.

3. Click **OK** to enable vSphere HA.



Note: If you do not see the OK button, you may need to zoom out on the web browser to see it.

Monitor the Task

[134]

Task Name	Target	Status	Details	Initiator	Queued For
Configuring vSphere HA	esx-01a.corp.vmbea...	Completed	Waiting for cluster election t...	System	12 ms
Configuring vSphere HA	esx-02a.corp.vmbea...	Completed	Waiting for cluster election t...	System	4 ms

It will take a minute or two to configure vSphere HA. You can monitor the progress in the Recent Tasks window.

Once the three tasks have been completed, you can move on to the next step.

Use the Summary Tab to Verify that HA Is Enabled

[135]

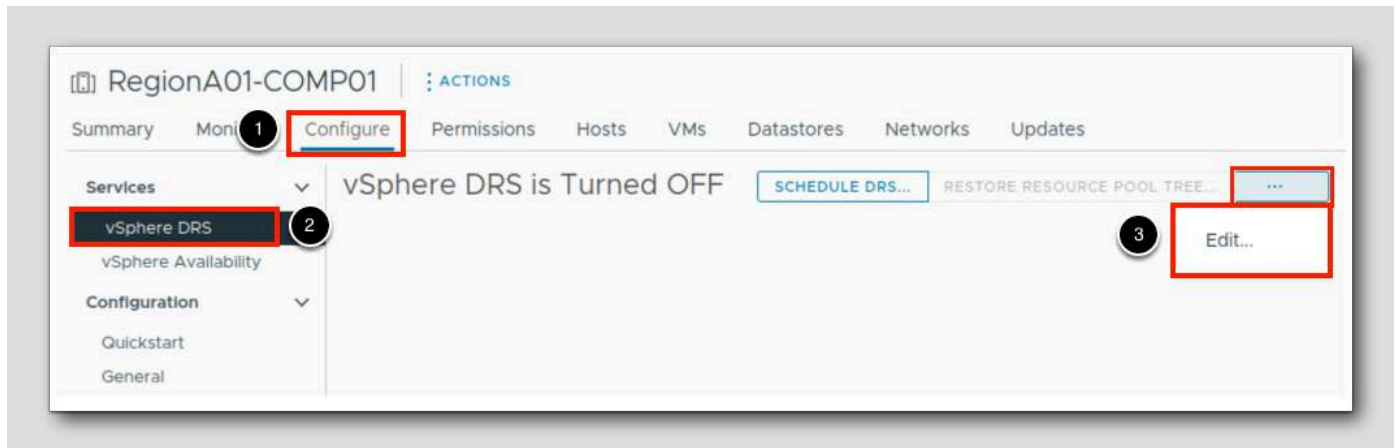
1. Click the **Summary** tab
2. Scroll down
3. Locate **vSphere HA** panel in the data area.

If vSphere HA does not show **Protected** and the tasks completed successfully, you may need to click the refresh button.

Notice the bars that display resource usage in blue, protected capacity in light gray, and reserve capacity using stripes.

Enable Distributed Resource Scheduler (DRS)

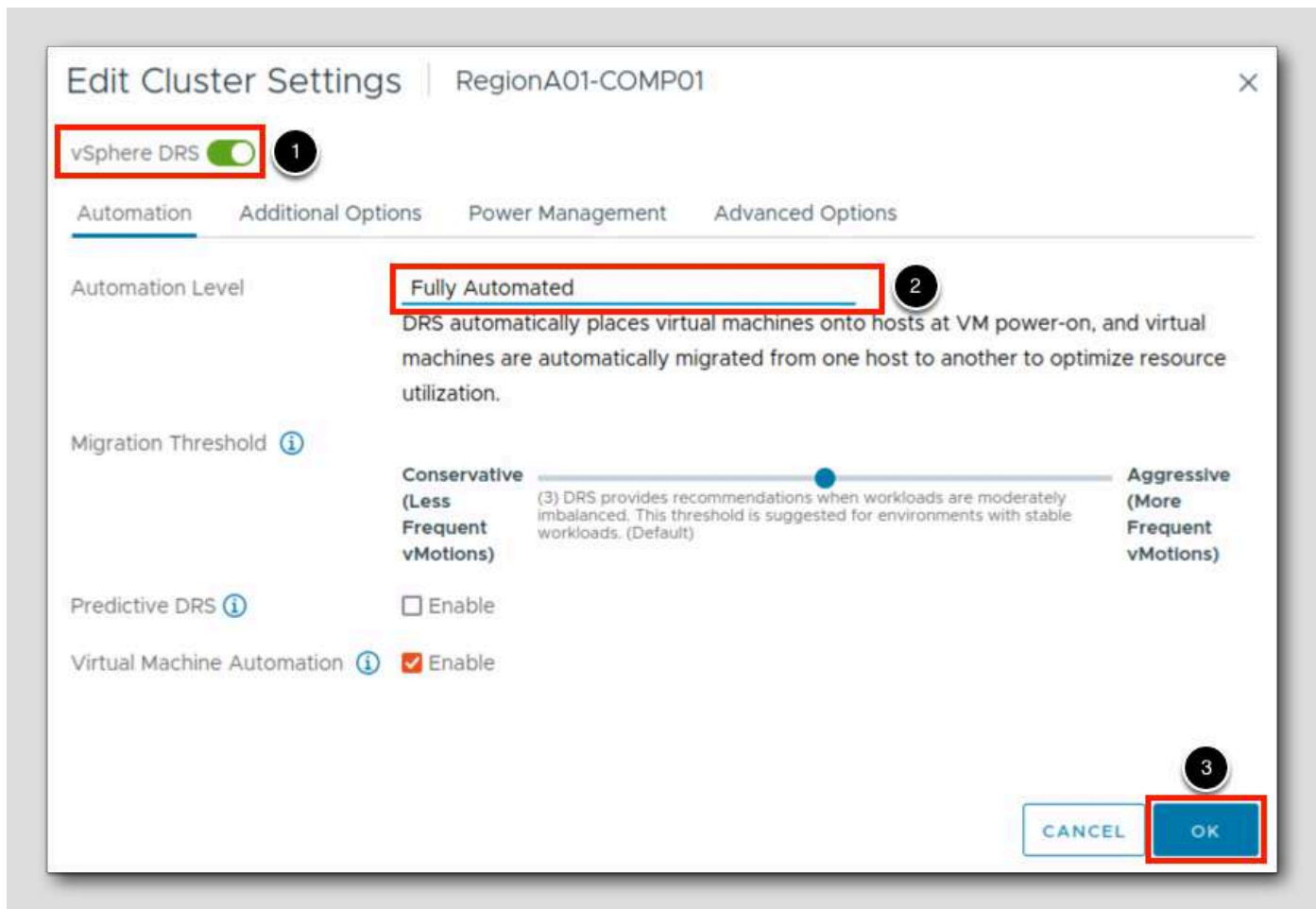
[136]



1. Click on the **Configure** tab to start the process of enabling Distributed Resource Scheduler.
2. Click **vSphere DRS**.
3. Under the "... " button, click on **Edit** to modify the DRS settings.

Enable Distributed Resource Scheduler (DRS)

[137]



1. Verify that vSphere DRS is enabled. If not, click the vSphere DRS to enable.
2. Click the drop-down menu next to Automation Level and select Fully Automated.
3. Click OK.

Automation Levels

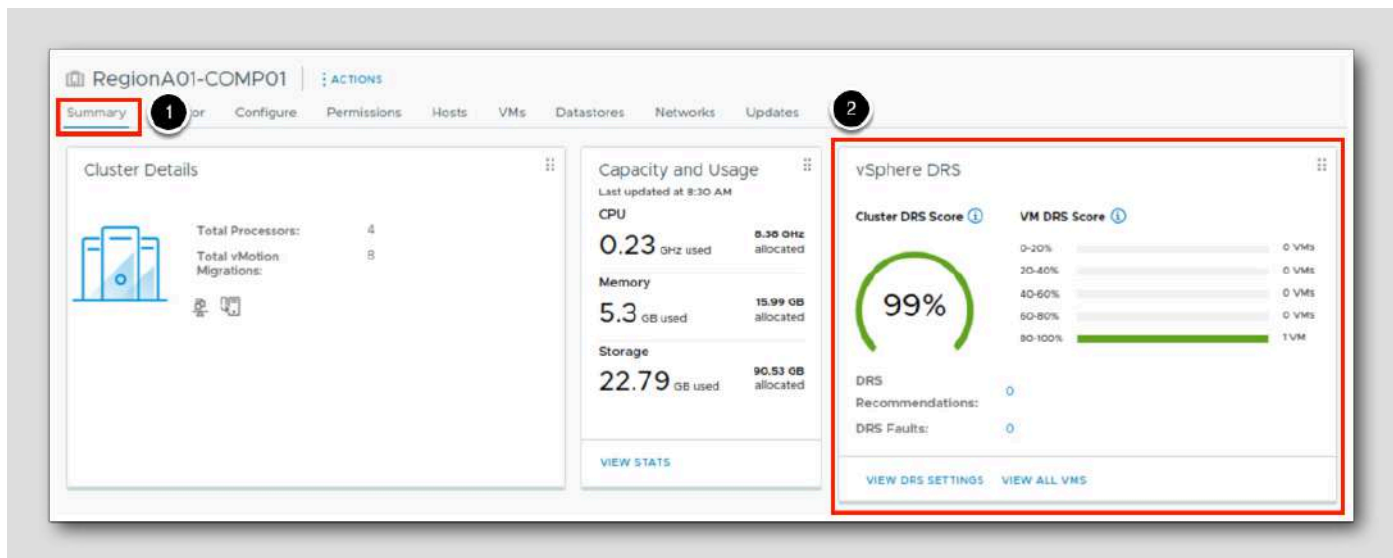
[138]

Automation Level	Action
Manual	- Initial placement: Recommended host(s) is displayed. - Migration: Recommendation is displayed.
Partially Automated	- Initial placement: Automatic. - Migration: Recommendation is displayed.
Fully Automated	- Initial placement: Automatic. - Migration: Recommendation is executed automatically.

The chart shown above is showing how DRS affects placement and migration according to the setting Manual, Partially Automated or Fully Automated.

Check Cluster Balance

[139]



1. Click the **Summary** tab to display the current status of the cluster.
2. The Summary tab of the Cluster RegionA01-COMP01 shows the current balance of the cluster. Also shown in the DRS section is how many recommendations or faults that have occurred with the cluster. (You may have to scroll down to see the vSphere DRS widget).

vSphere 8 Fault Tolerance Provides Continuous Availability

[140]

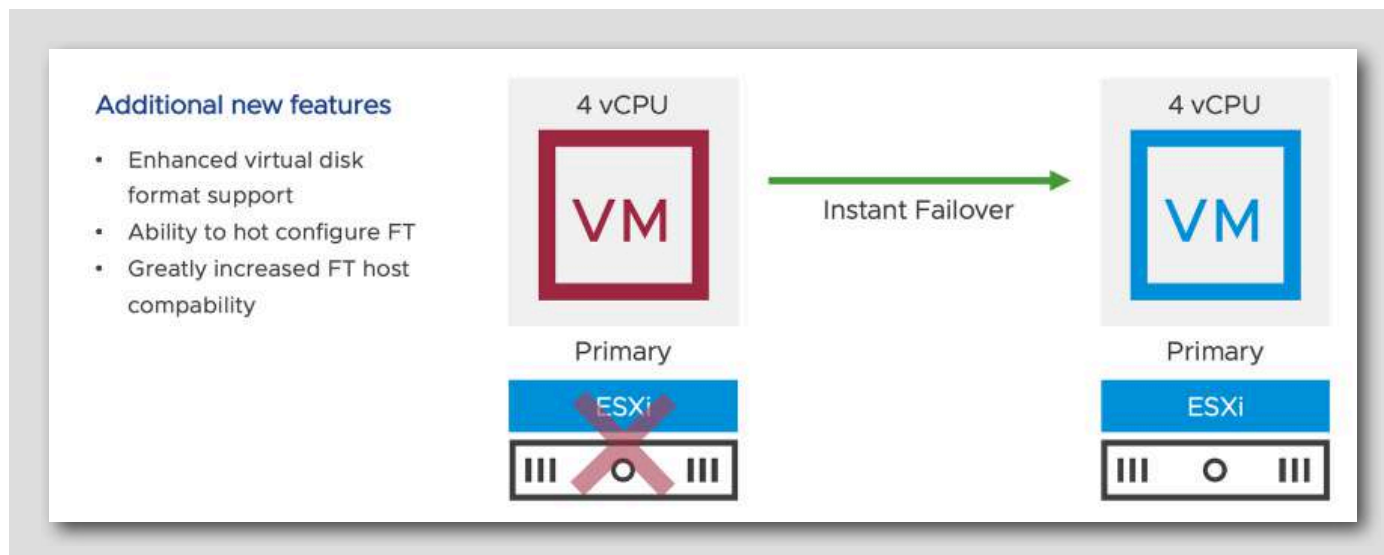
You can use vSphere Fault Tolerance for your virtual machines to ensure continuity with higher levels of availability and data protection. Fault Tolerance is built on the ESXi host platform, and it provides availability by having identical Virtual Machines (VM) run on separate hosts.

vSphere Fault Tolerance (FT) provides continuous availability by creating and maintaining the states of a Primary and Secondary VMs identical. In the event of a failover situation, the Secondary VM will be executed and it will replace the Primary VM (the protected virtual machine). The duplicate virtual machine, the Secondary VM, is created and runs on another host. The primary VM is continuously replicated to the secondary VM so that the secondary VM can take over at any point, thereby providing Fault Tolerant protection. The Primary and Secondary VMs continuously monitor the status of one another to ensure that Fault Tolerance is maintained.

Fault Tolerance avoids "split-brain" situations, which can lead to two active copies of a virtual machine after recovery from a failure. Atomic file locking on shared storage is used to coordinate failover so that only one side continues running as the Primary VM and a new Secondary VM is respawned automatically.

VMware vSphere Fault Tolerance

[141]



Benefits of Fault Tolerance:

- Protect mission critical, high performance applications regardless of OS; No application-specific management and learning
- Continuous availability - Zero downtime, zero data loss for infrastructure failures; no loss of TCP connections
- Fully automated response

Several typical situations can benefit from the use of vSphere Fault Tolerance. Fault Tolerance provides a higher level of business continuity than vSphere HA. When a Secondary VM is called upon to replace its Primary VM counterpart, the Secondary VM immediately takes over the Primary VMs role with the entire state of the virtual machine preserved. Applications are already running, and data stored in memory does not need to be reentered or reloaded. Failover provided by vSphere HA restarts the virtual machines affected by a failure.

This higher level of continuity and the added protection of state information and data informs the scenarios when you might want to deploy Fault Tolerance.

- Applications which must always be available, especially applications that have long-lasting client connections that users want to maintain during hardware failure.
- Custom applications that have no other way of doing clustering.
- Cases where high availability might be provided through custom clustering solutions, which are too complicated to configure and maintain.

Another key use case for protecting a virtual machine with Fault Tolerance can be described as On-Demand Fault Tolerance. In this case, a virtual machine is adequately protected with vSphere HA during normal operation. During certain critical periods, you might want to enhance the protection of the virtual machine. For example, you might be running a quarter-end report which, if interrupted, might delay the availability of critical information. With vSphere Fault Tolerance, you can protect this virtual machine before running this report and then turn off or suspend Fault Tolerance after the report has been produced. You can use On-Demand Fault Tolerance to protect the virtual machine during a critical time period and return the resources to normal during non-critical operation.

See the [Performance Best Practices for VMware vSphere](#) and [vSphere 8.0 Availability](#) for more information.

Monitoring Events and Creating Alarms

[142]

vSphere includes a user-configurable events and alarms subsystem. This subsystem tracks events happening throughout vSphere and stores the data in log files and the vCenter Server database. This subsystem also enables you to specify the conditions under which alarms are triggered. Alarms can change state from mild warnings to more serious alerts as system conditions change and can trigger automated alarm actions. This functionality is useful when you want to be informed, or take immediate action, when certain events or conditions occur for a specific inventory object, or group of objects.

Events are records of user actions or system actions that occur on objects in vCenter Server or on a host. Actions that might be reordered as events include, but are not limited to, the following examples:

- A license key expires
- A virtual machine is powered on
- A user logs in to a virtual machine
- A host connection is lost

Event data includes details about the event such as who generated it, when it occurred, and what type of event.

Alarms are notifications that are activated in response to an event, a set of conditions, or the state of an inventory object. An alarm definition consists of the following elements:

- Name and description - Provides an identifying label and description.
- Alarm type - Defines the type of object that will be monitored.
- Triggers - Defines the event, condition, or state that will trigger the alarm and defines the notification severity.
- Tolerance thresholds (Reporting) - Provides additional restrictions on condition and state triggers thresholds that must be exceeded before the alarm is triggered.
- Actions - Defines operations that occur in response to triggered alarms. VMware provides sets of predefined actions that are specific to inventory object types.

Alarms have the following severity levels:

- Normal – green
- Warning – yellow
- Alert – red

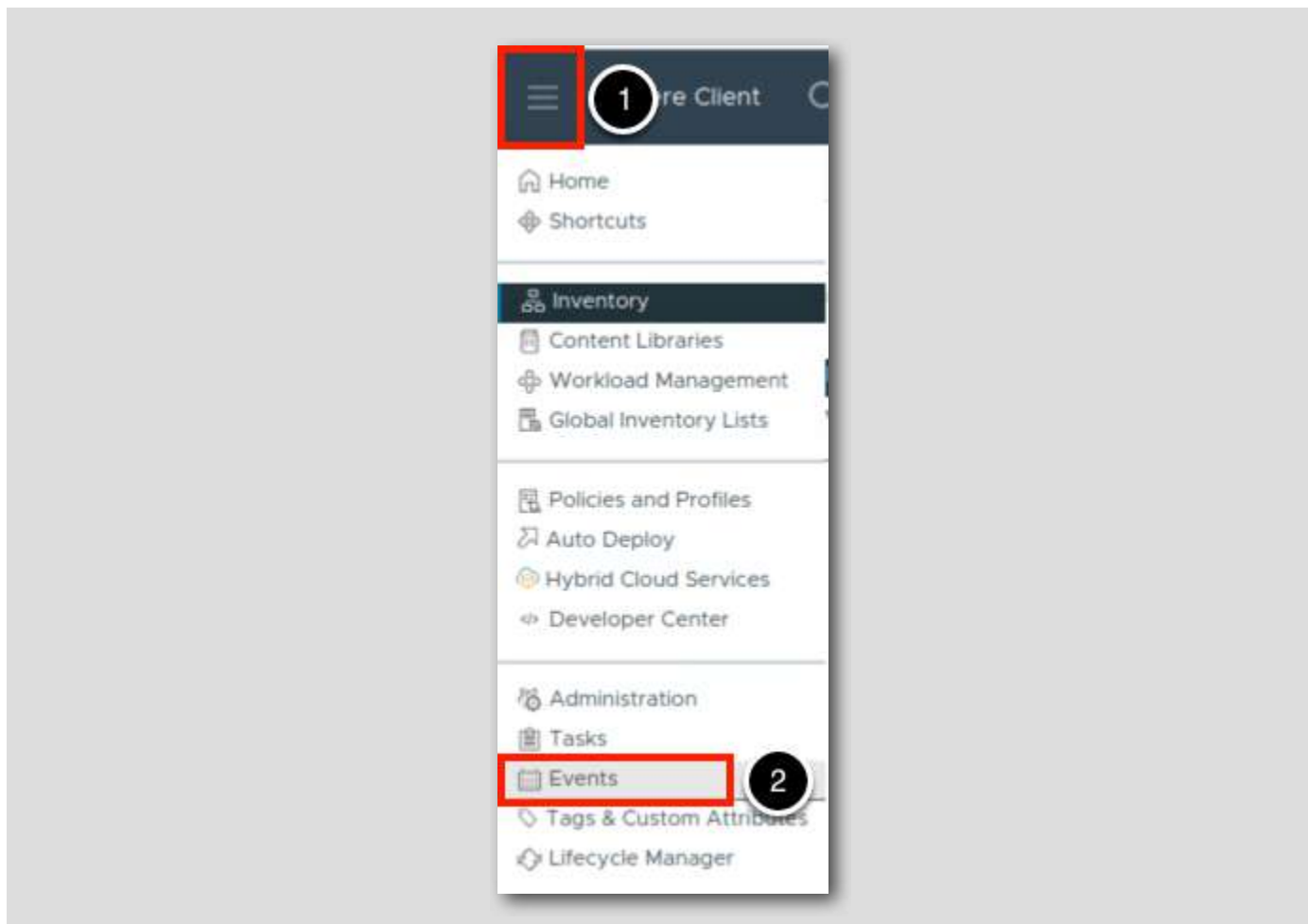
Alarm definitions are associated with the object selected in the inventory. An alarm monitors the type of inventory objects specified in its definition.

For example, you might want to monitor the CPU usage of all virtual machines in a specific host cluster. You can select the cluster in the inventory and add a virtual machine alarm to it. When enabled, that alarm will monitor all virtual machines running in the cluster and will trigger when any one of them meets the criteria defined in the alarm. If you want to monitor a specific virtual machine in the cluster, but not others, you would select that virtual machine in the inventory and add an alarm to it. One easy way to apply the same alarms to a group of objects is to place those objects in a folder and define the alarm on the folder.

In this lab, you will learn how to create an alarm and review the events that have occurred.

Review Default Alerts

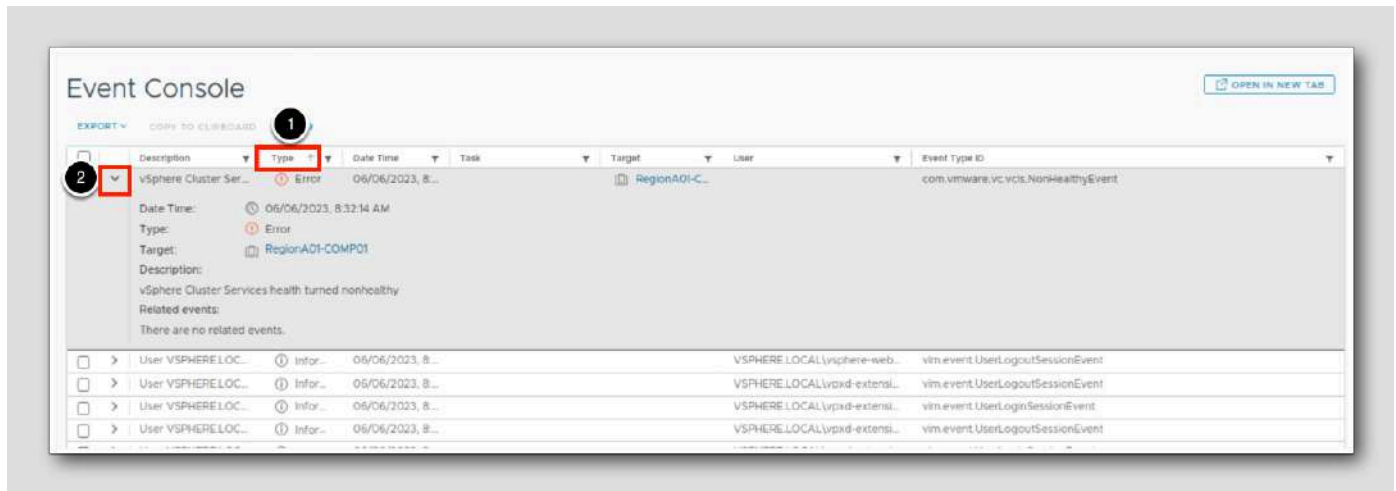
[143]



1. Click on the hamburger Menu on your top left
2. Click on Events

Event Console

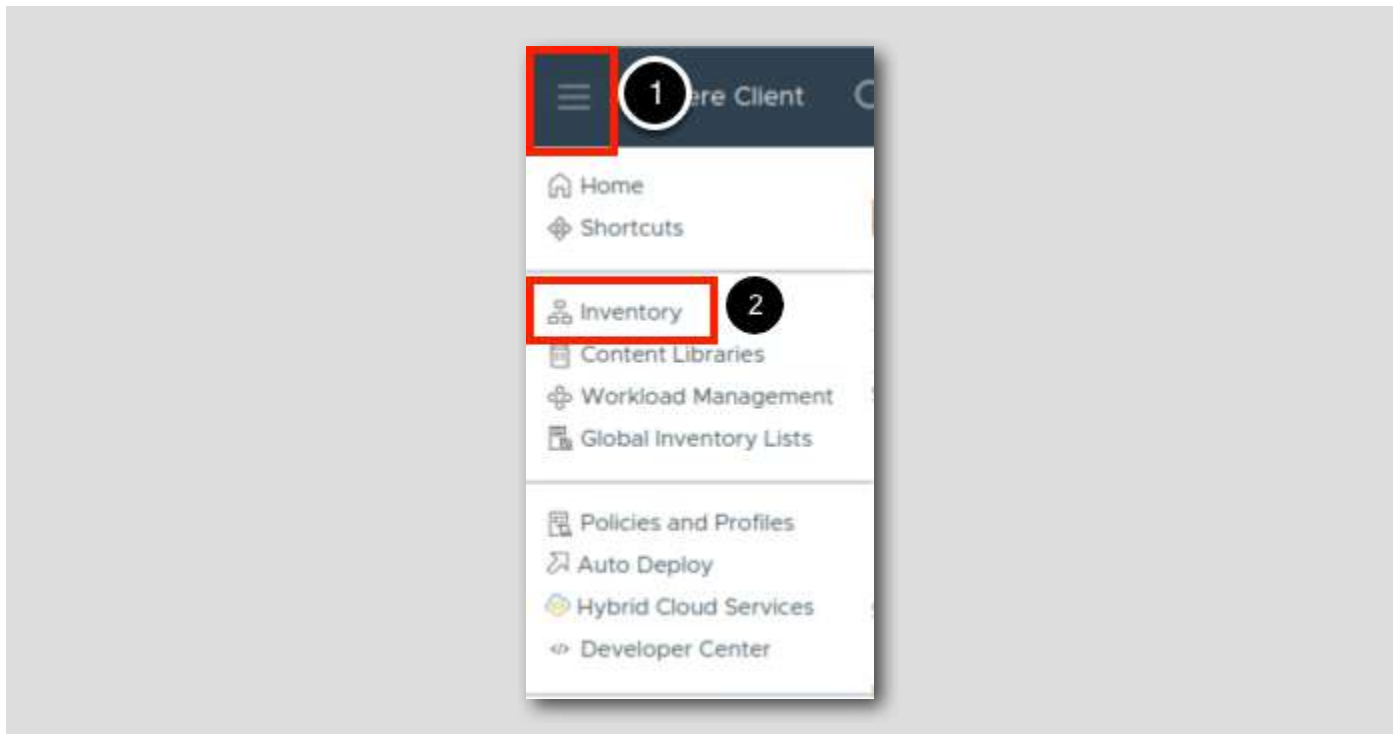
[144]



1. Click on the **Type** column to sort by level of severity.
2. Select an event to review the details of the event. Click on the ">" button to expand its details.

Setup Notifications

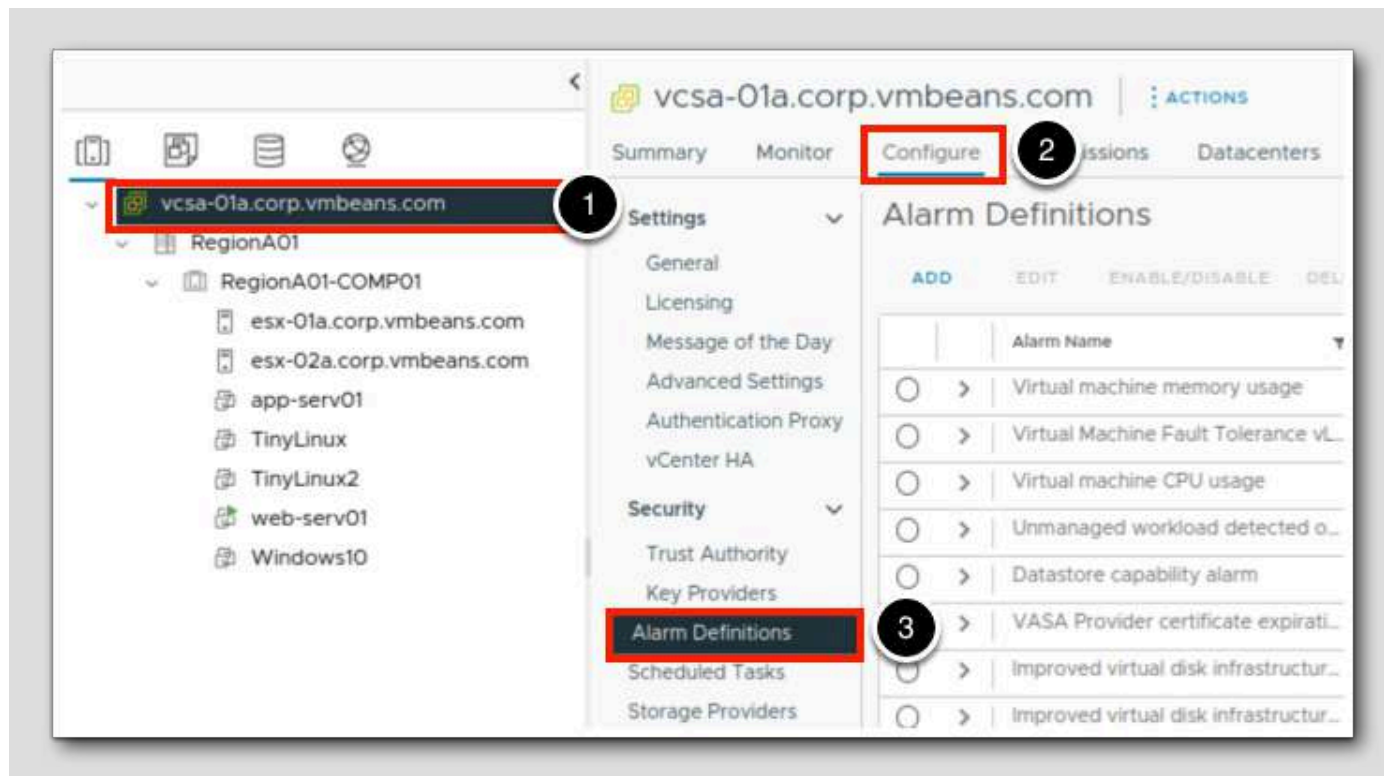
[145]



1. Click on the hamburger **Menu** on your top left
2. Click **Inventory**

Setup Notifications

[146]



1. Select the vCenter - vcsa-01a.corp.vmbeans.com
2. Click the **Configure** tab
3. Click on **Alarm Definitions**. The default alarm definitions are shown.

Alarms can be defined at different levels. In the case of the highlighted alarm, you can see it is defined at the top level. Alarms that are defined at the top level are then inherited by the objects below.

Alarm Definitions

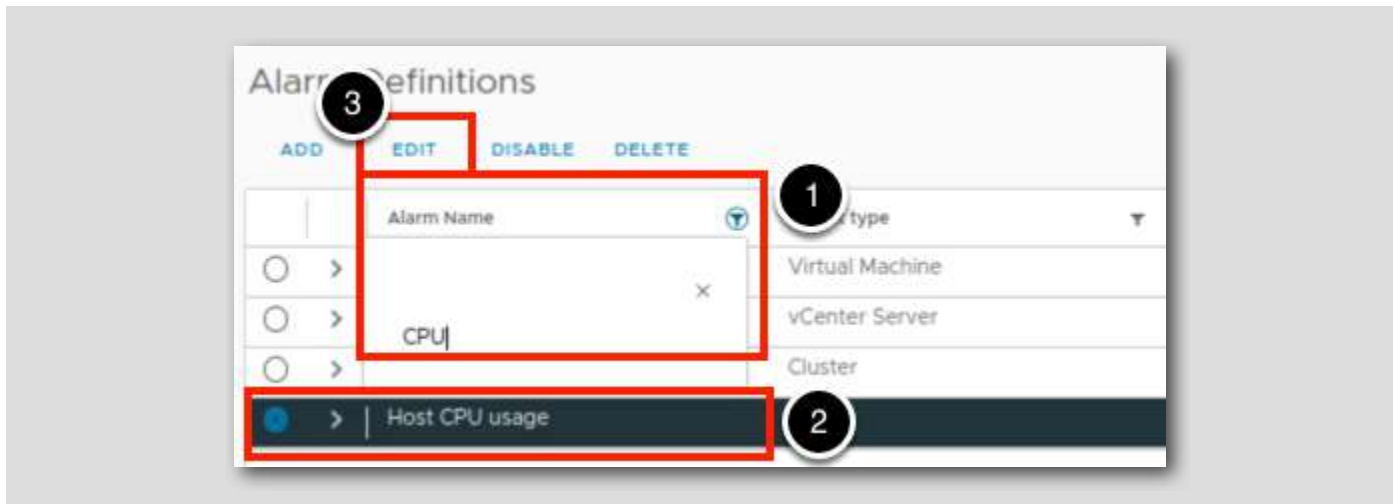
[147]

		Alarm Name	Object type	Defined in	Enabled
☐	>	Virtual machine memory usage	Virtual Machine	 This Object	Enabled
☐	>	Virtual Machine Fault Tolerance vL...	Virtual Machine	 This Object	Enabled
☐	>	Virtual machine CPU usage	Virtual Machine	 This Object	Enabled
☐	>	Unmanaged workload detected o...	Datastore	 This Object	Disabled
☐	>	Datastore capability alarm	Datastore	 This Object	Enabled
☐	>	VASA Provider certificate expirati...	Datastore	 This Object	Enabled
☐	>	Improved virtual disk infrastrucur...	Datastore	 This Object	Enabled
☐	>	Improved virtual disk infrastrucur...	Datastore	 This Object	Enabled
☐	>	Thin-provisioned volume capacity ...	Datastore	 This Object	Enabled
☐	>	Datastore compliance alarm	Datastore	 This Object	Enabled
☐	>	VASA provider disconnected	Datastore	 This Object	Enabled
☐	>	Object type storage alarm	Datastore	 This Object	Enabled
☐	>	License error	Host	 This Object	Enabled

Alarms can be defined at different levels. In the case of the highlighted alarm, you can see it is defined at the top level (vCenter Server). Alarms that are defined at the top level are then inherited by the objects below.

Defining an Alarm

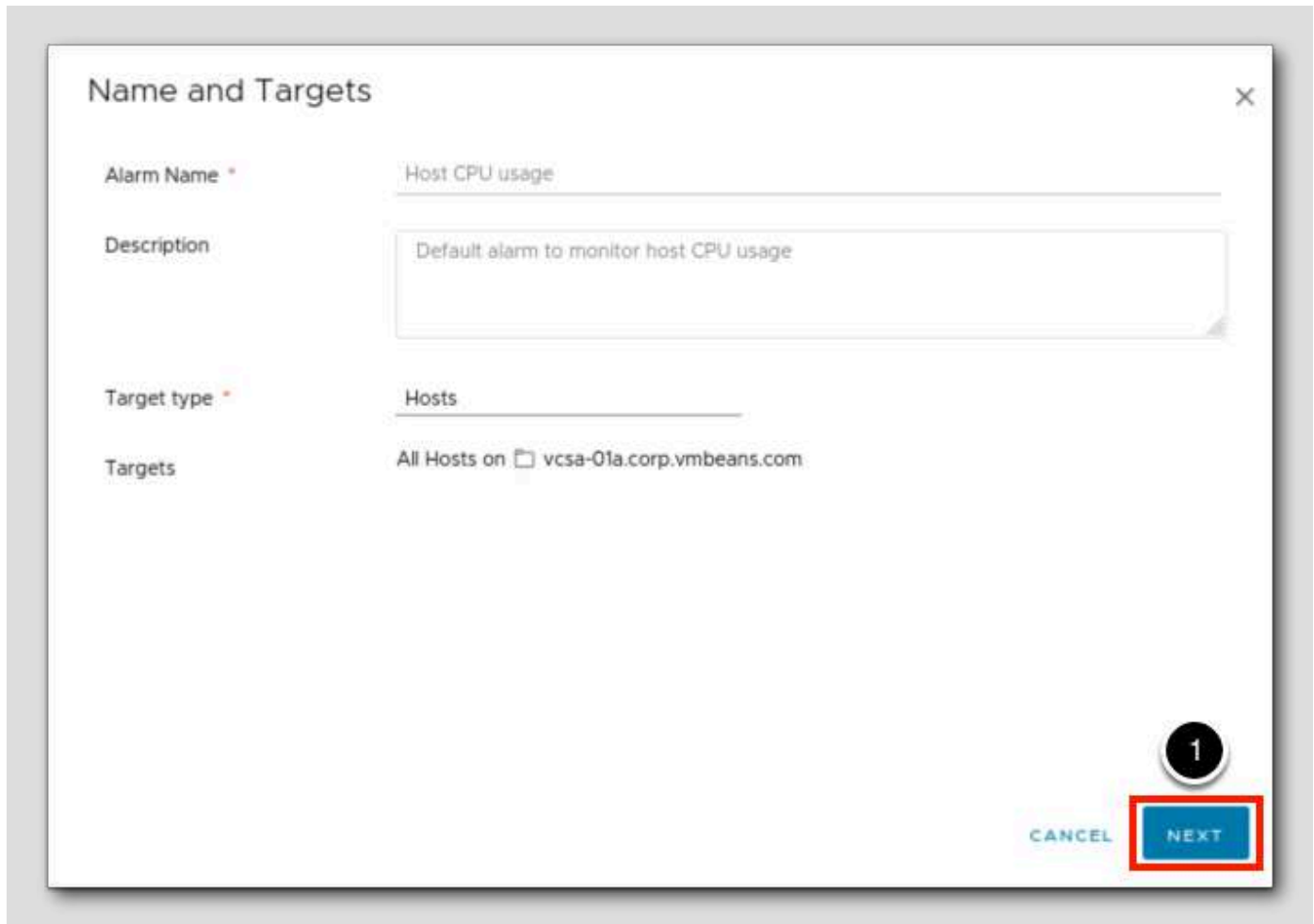
[148]



1. Click on the **Alarm Name** filter field and type "CPU" in the search field.
2. Select the **Host CPU usage** alarm
3. Click the **Edit** button

Name and Targets

[149]



Name and Targets [X]

Alarm Name * Host CPU usage

Description Default alarm to monitor host CPU usage

Target type * Hosts

Targets All Hosts on ☐ vcasa-01a.corp.vmbeans.com

CANCEL NEXT

The Name and Targets screen defines the name of the alarm (Host CPU usage), what object it applies to (Hosts) and where the objects are located.

1. Click **Next**.

Alarm Rule 1

[150]

IF

Host CPU Usage

is above

80 %

5 min

ADD ADDITIONAL TRIGGER

THEN

Trigger the alarm and * Show as Warning

Send email notifications

ADD ANOTHER RULE DUPLICATE RULE REMOVE RULE

1. Change the percentage of 75% to 80%.
2. Use the scroll bar to scroll to the bottom.

Notice this will trigger a **Warning** alarm.

Add Advanced Action

Alarm Rule 1

THEN

Trigger the alarm and ^{*} Show as Warning

Send email notifications ☐

Send SNMP traps ☐

Run script ☐

Enter maintenance mode 2

1 ADD ADVANCED ACTIONS

ADD ANOTHER RULE DUPLICATE RULE REMOVE RULE

CANCEL BACK 3 NEXT

1. Click on **Add Advanced Actions**.
2. From the drop-down menu (Select an advanced action), select **Enter maintenance mode**.
3. Click **Next**

When a Host's CPU runs at or above 80% for more than 5 minutes, a Warning alarm will be triggered, and the Host will be put in Maintenance mode. Maintenance mode is covered in Module 3, but when a host is in this state, it is taken offline and any virtual machines that are running on it will be moved to other hosts in the cluster. This lets maintenance be performed on hosts without suffering downtime.

Alarm Rule 2

[152]

Alarm Rule 2

IF

Host CPU Usage
is above
90 % for
5 min

[ADD ADDITIONAL TRIGGER](#)

THEN

Trigger the alarm and Show as Critical

Send email notifications ☐

[ADD ANOTHER RULE](#) [DUPLICATE RULE](#) [REMOVE RULE](#)

[CANCEL](#) [BACK](#) [NEXT](#)

On this screen we can set additional actions based on when a Host's CPU is about 90% for 5 minutes. In this case, it would trigger a Critical alarm. Additional actions could be taken when a Host is in this state.

1. Click **Next**.

Reset Rule 1

[153]

Reset Rule 1

IF

The warning or critical conditions/states are no longer met

THEN

Reset the alarm to ☒ Normal

Send email notifications ☐

Send SNMP traps ☐

Run script ☐

1

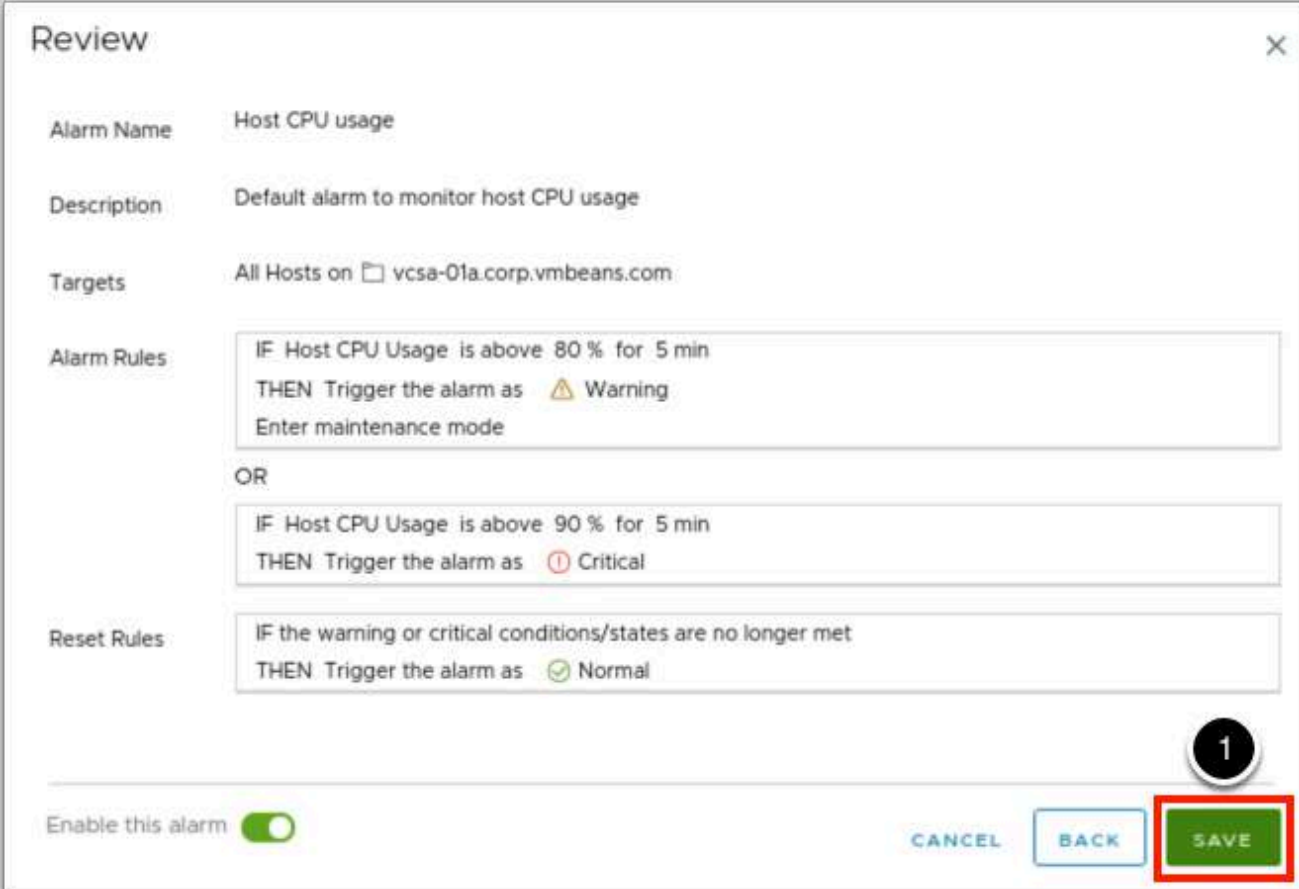
CANCEL BACK **NEXT**

If the conditions that originally triggered the alarm are no longer present, additional actions can take place. As an example, once a Host's CPU is no longer at 80% for more than 5 minutes, an email notification could be sent.

1. Click **Next**.

Review

[154]



The image shows a 'Review' dialog box for an alarm named 'Host CPU usage'. The dialog contains the following fields:

- Alarm Name:** Host CPU usage
- Description:** Default alarm to monitor host CPU usage
- Targets:** All Hosts on `vcsa-01a.corp.vmbeans.com`
- Alarm Rules:**
 - IF Host CPU Usage is above 80 % for 5 min
 - THEN Trigger the alarm as ⚠ Warning
 - Enter maintenance mode
- OR**
- Alarm Rules:**
 - IF Host CPU Usage is above 90 % for 5 min
 - THEN Trigger the alarm as 🔴 Critical
- Reset Rules:**
 - IF the warning or critical conditions/states are no longer met
 - THEN Trigger the alarm as ✅ Normal

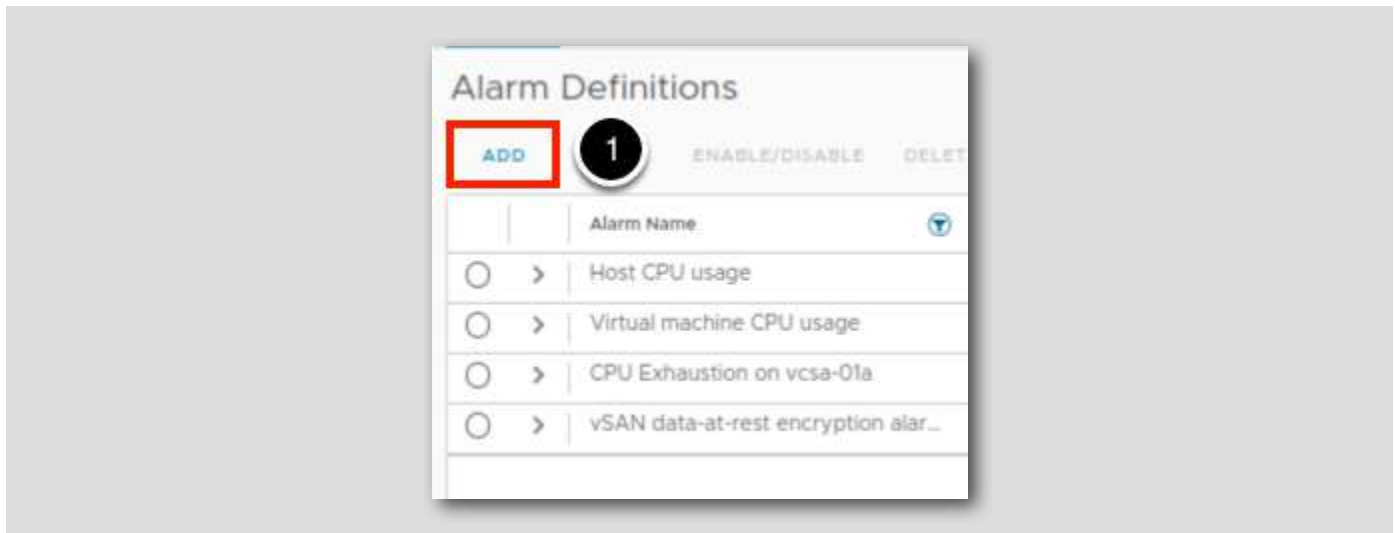
At the bottom, there is a toggle for 'Enable this alarm' which is currently turned on. To the right of the toggle are three buttons: 'CANCEL', 'BACK', and 'SAVE'. The 'SAVE' button is highlighted with a red rectangle and a circled '1' above it, indicating the next step in the process.

The Review screen shows what was configured.

1. Click **Save** to keep the changes made to the Alarm.

Create New Alarm

[155]



1. To add a new alarm, click **Add**.

New Alarm Definition

Name and Targets

Alarm Name * 1 Virtual Machine CPU Ready

Description

Target type * 2 Virtual Machines

Targets All Virtual Machines on ☐ vcsa-01a.corp.vmbeans.com (6)

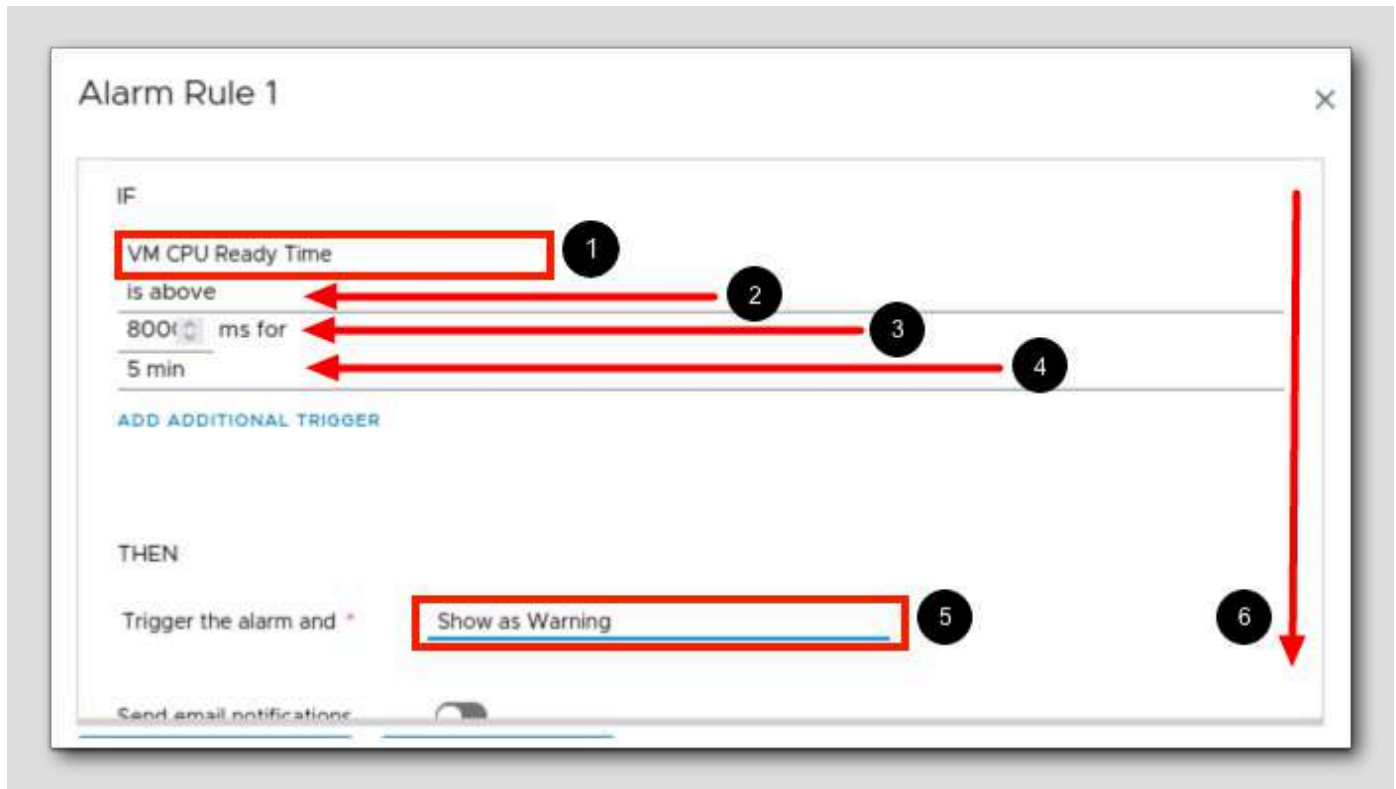
CANCEL 3 NEXT

We will be creating an alarm that will migrate a VM if CPU Ready exceeds an average of 8000ms over the course of 5 minutes.

1. Enter **Virtual Machine CPU Ready** for the Alarm name.
2. Change **Monitor** from vCenter Server to **Virtual Machines**
3. Click **Next** to move to the Alarm Rule 1 screen.

Define CPU Ready Time

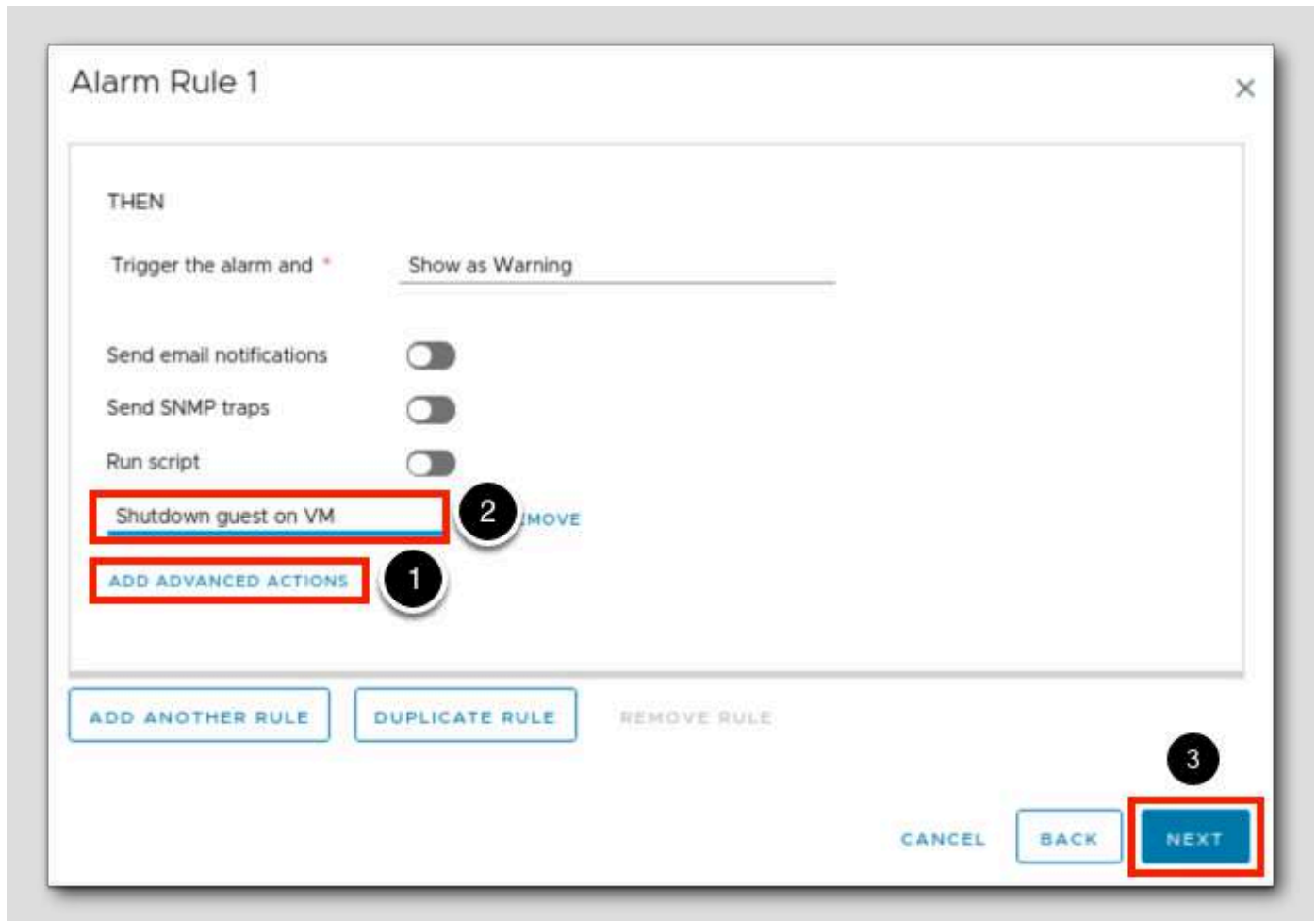
[157]



1. Click in the field under IF and select VM CPU Ready Time.
2. Change the select an operator field to is above.
3. Type 8000 in the ms field
4. Use the drop-down menu to select 5 min.
5. Select Show as Warning in the Trigger the alarm menu.
6. Use the scroll bar to scroll to the Add advanced actions section.

New Alarm Advanced Actions

[158]



1. Click on Add Advanced Actions
2. From the drop-down menu, select Shutdown guest on VM.

This will gracefully shutdown the virtual machine rather than just powering it off.

3. Click Next.

Reset Rule 1

[159]

Reset Rule 1

IF

The warning or critical conditions/states are no longer met

THEN

Reset the alarm to ☒ Normal

Send email notifications ☐

Send SNMP traps ☐

Run script ☐

1

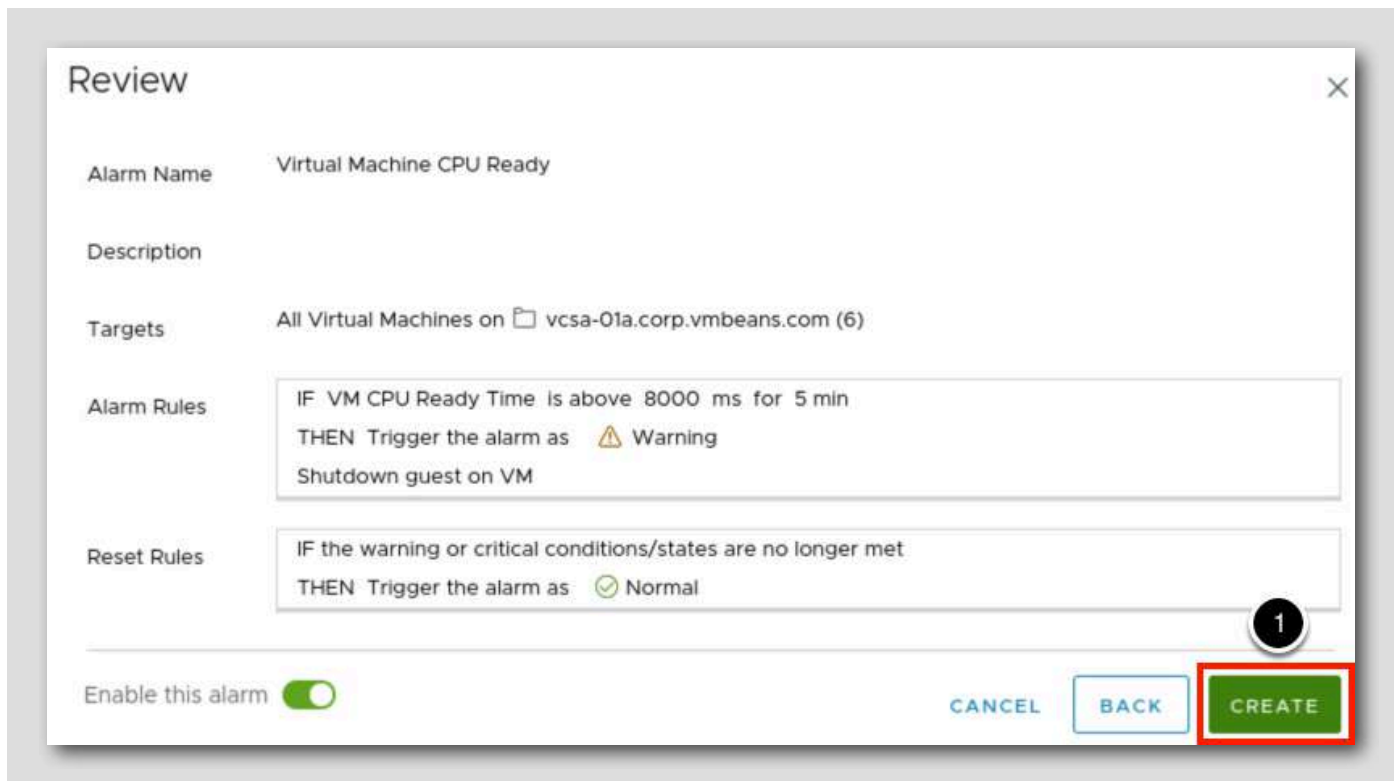
CANCEL BACK **NEXT**

Additional options could be specified once the conditions are clear.

1. Click **Next**

Review

[160]



The image shows a 'Review' dialog box for configuring a new alarm. The dialog has a title bar with 'Review' and a close button. It contains several sections: 'Alarm Name' with the value 'Virtual Machine CPU Ready'; 'Description' which is empty; 'Targets' with the value 'All Virtual Machines on vcsa-01a.corp.vmbeans.com (6)'; 'Alarm Rules' with a rule 'IF VM CPU Ready Time is above 8000 ms for 5 min THEN Trigger the alarm as Warning Shutdown guest on VM'; and 'Reset Rules' with a rule 'IF the warning or critical conditions/states are no longer met THEN Trigger the alarm as Normal'. At the bottom left, there is a toggle switch for 'Enable this alarm' which is turned on. At the bottom right, there are three buttons: 'CANCEL', 'BACK', and 'CREATE'. The 'CREATE' button is highlighted with a red rectangle and a circled number '1' above it, indicating the next step in the process.

Review

Alarm Name: Virtual Machine CPU Ready

Description:

Targets: All Virtual Machines on vcsa-01a.corp.vmbeans.com (6)

Alarm Rules: IF VM CPU Ready Time is above 8000 ms for 5 min
THEN Trigger the alarm as Warning
Shutdown guest on VM

Reset Rules: IF the warning or critical conditions/states are no longer met
THEN Trigger the alarm as Normal

Enable this alarm ☒

CANCEL BACK CREATE

The Review screen shows the details of what was configured for the new alarm.

1. Click **Create**.

New Alarm Created

[161]

Alarm Definitions

[ADD](#) [EDIT](#) [ENABLE/DISABLE](#) [DELETE](#)

		Alarm Name		Object type		Defined In		Enabled
☐	>	Host CPU usage		Host		 This Object		Enabled
☐	>	Virtual machine CPU usage		Virtual Machine		 This Object		Enabled
☐	>	CPU Exhaustion on vcasa-01a		vCenter Server		 This Object		Enabled
☐	>	vSAN data-at-rest encryption alar...		Cluster		 This Object		Enabled
☐	>	Virtual Machine CPU Ready		Virtual Machine		 This Object		Enabled

If the Alarm Name field is still filtering by "cpu", the newly created alarm is displayed. If not, simply click on the Alarm Name field and type cpu ready to see it.

Configure Shares and Resources

[162]

Shares specify the relative importance of a virtual machine (or resource pool). If a virtual machine has twice as many shares of a resource as another virtual machine, it is entitled to consume twice as much of that resource when these two virtual machines are competing for resources. This lab starts with a video walking you through the process of working with shares and resources. The remainder of this module walks you through making the changes to a VM's resources.

Shares are typically specified as High, Normal, or Low

Video: DRS with Scalable Shares in vSphere 7 (4:17)

[163]

This video explains how scalable shares are and how are they used in order to effectively distribute compute and memory resources among virtual machines.

<https://www.youtube.com/watch?v=jkp25l4R0R8>

The diagram shows a VMware resource pool hierarchy. At the top is the **Root Resource Pool** with a capacity of 30 GHz. It is divided into two **Resource Pools**:

- Resource Pool 1** (CPU shares = normal) with an allocation of 6 GHz. It contains **Virtual Machine 1**.
- Resource Pool 2** (CPU shares = high) with an allocation of 24 GHz. It contains **Virtual Machine 2** and **Virtual Machine 3**.

The allocation for Virtual Machine 2 is 12 GHz, and for Virtual Machine 3 is 12 GHz. The total allocation for all VMs is 30 GHz, matching the Root Resource Pool capacity.

Resource Management

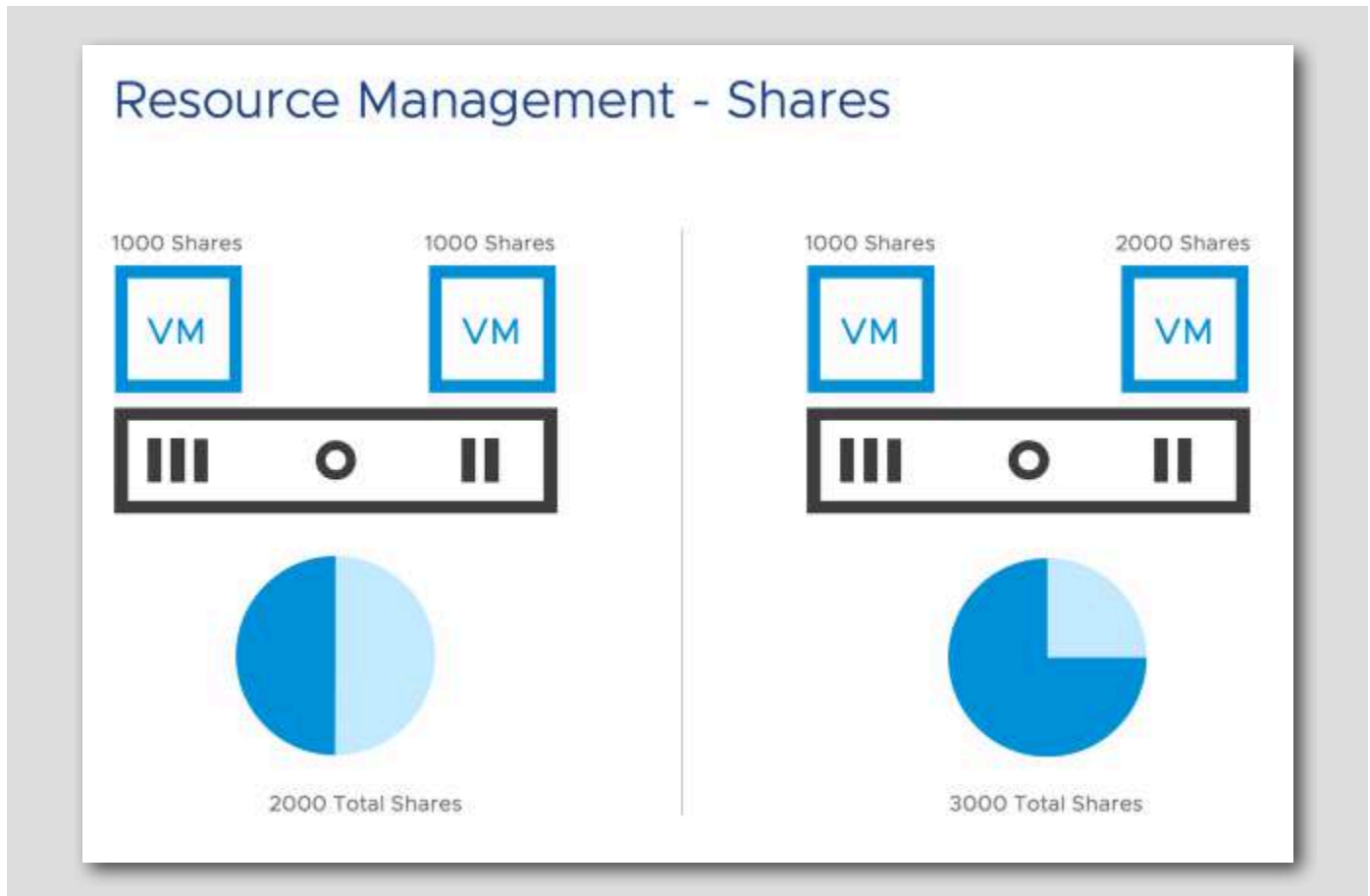
Shares: relative importance of a virtual machine (VM)

Reservation: guaranteed minimum allocation for a VM

Limit: upper bound of resource that can be allocated to a VM

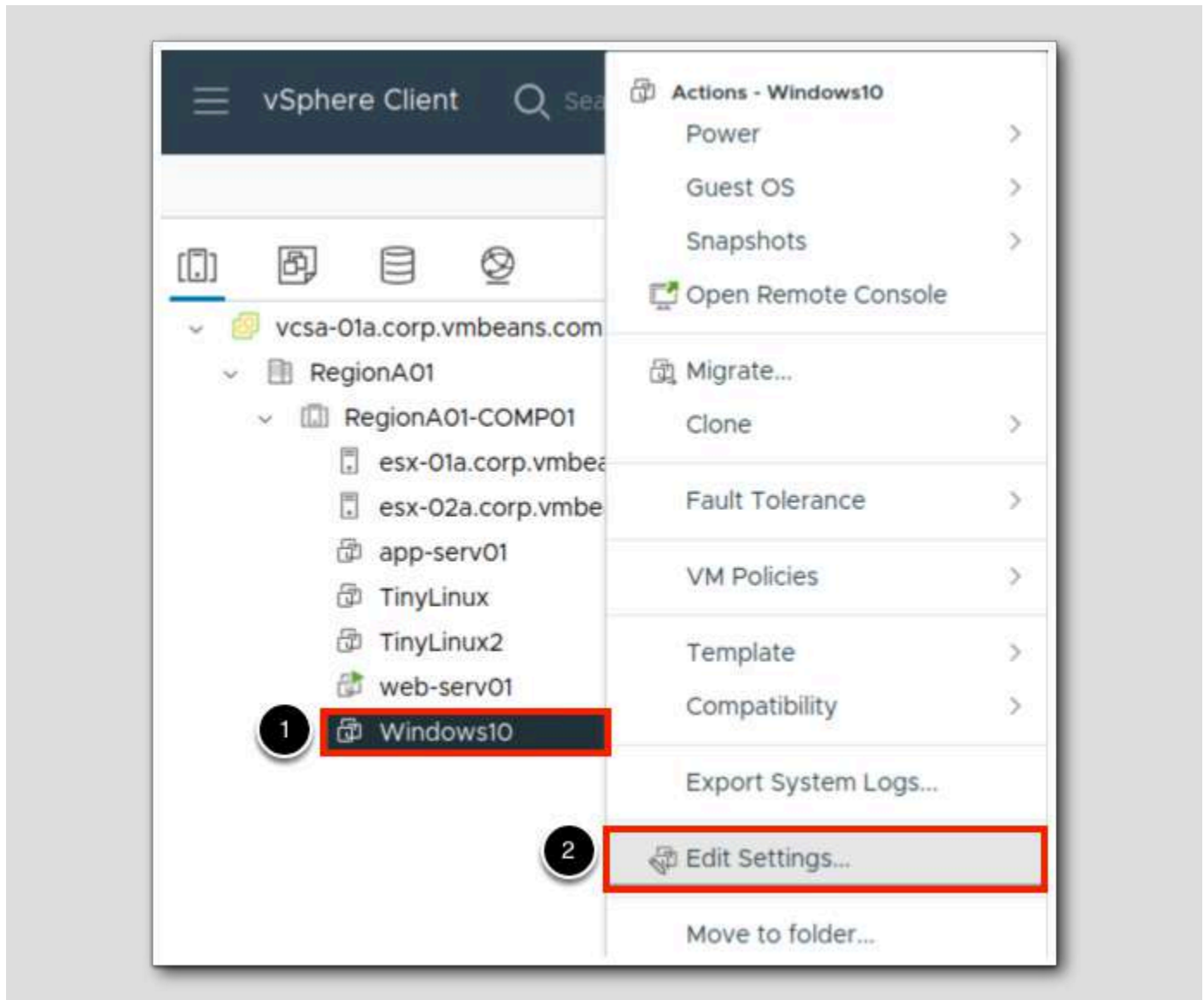
Understanding Shares

[165]



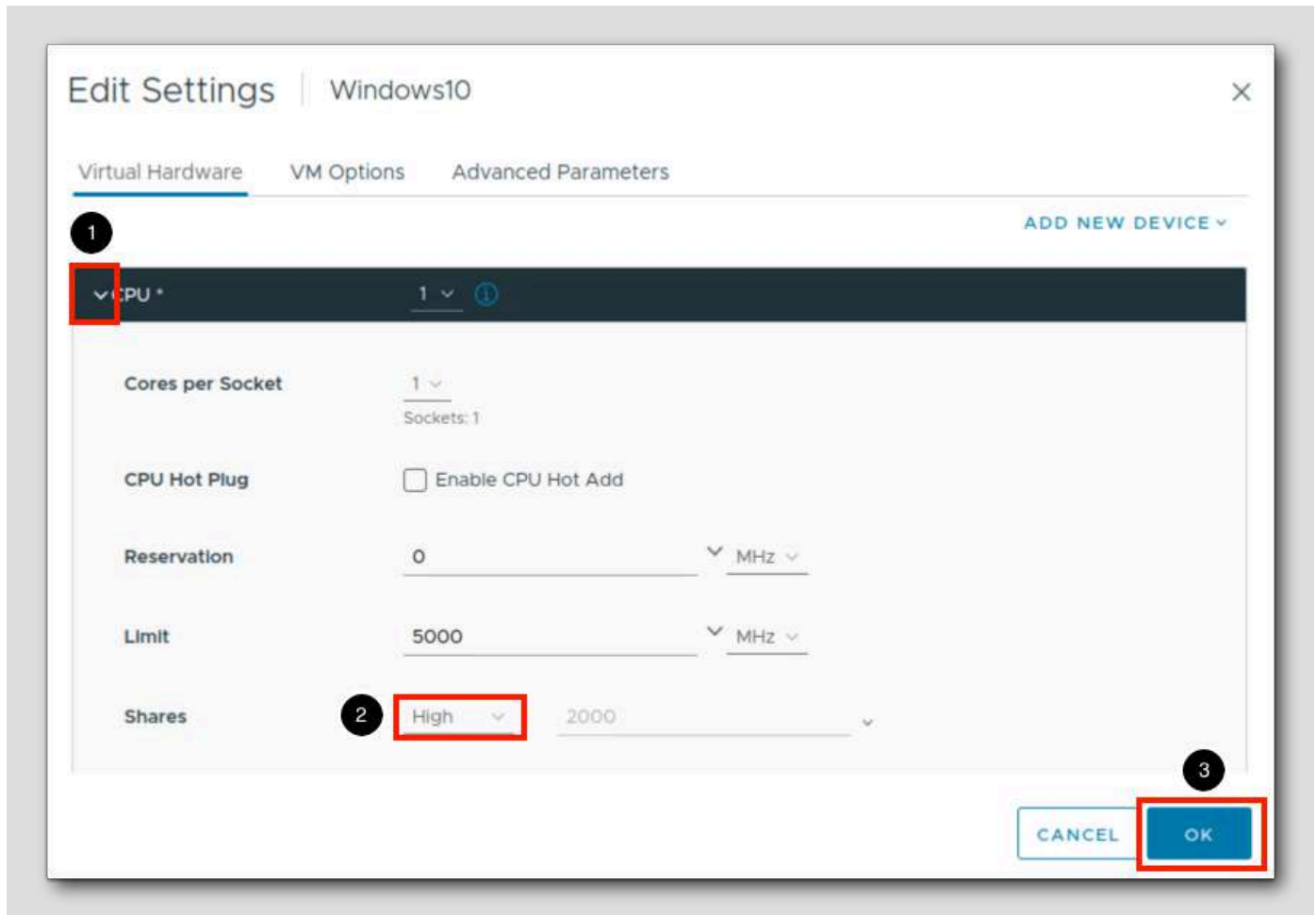
The above example shows 2 VM's, one a development VM and the other a Production VM. On the left-hand side of the diagram, you can see the CPU shares are equal. We want to make sure the Production VM gets the majority of the CPU resources when there is contention for those resources in the environment. Changing the shares for the production VM from 1000 shares to 2000 shares accomplishes this goal. The new settings are shown on the right side of the diagram.

Review CPU settings



1. Right-click the Windows10 virtual machine.
2. Select Edit Settings...

Changing Resource Allocation of CPU shares.

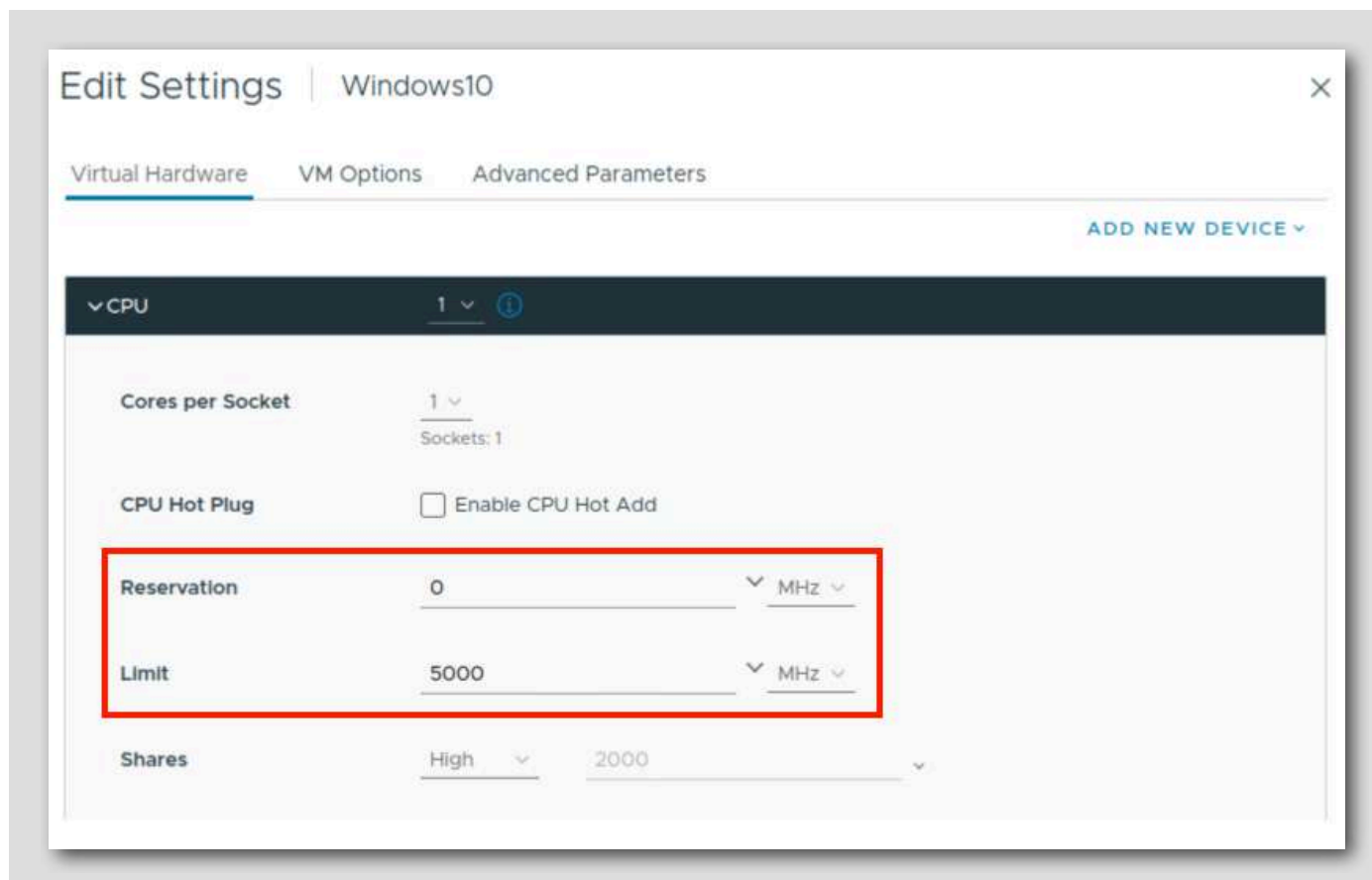


Note the current setting for **Shares** is set to 1000.

1. Expand the **CPU** section of the settings.
2. From the **Shares** drop down box, click **High** to change the setting of the CPU shares.
3. Click **OK**

Settings for Limits and Reservations.

[168]



Limits and Reservations are set with the same procedure. When you click on the "edit" settings for a VM, you will find the ability to set the Limit and Reservations. Limit restricts a VM from using more than the limit setting. Reservations guarantee a minimum amount of a resource be available for the virtual machine. Try out some settings for Limits and Reservations. One note is that if you try to reserve more of a resource such as memory or CPU than is available, the VM may not power on.

Migrating Virtual Machines with VMware vMotion

[169]

Planned downtime typically accounts for over 80% of datacenter downtime. Hardware maintenance, server migration, and firmware updates all require downtime for physical servers. To minimize the impact of this downtime, organizations are forced to delay maintenance until inconvenient and difficult-to-schedule downtime windows.

The vMotion functionality in vSphere makes it possible for organizations to reduce planned downtime because workloads in a VMware environment can be dynamically moved to different physical servers without service interruption. Administrators can perform faster and completely transparent maintenance operations, without being forced to schedule inconvenient maintenance windows. With vSphere vMotion, organizations can:

- Eliminate downtime for common maintenance operations.

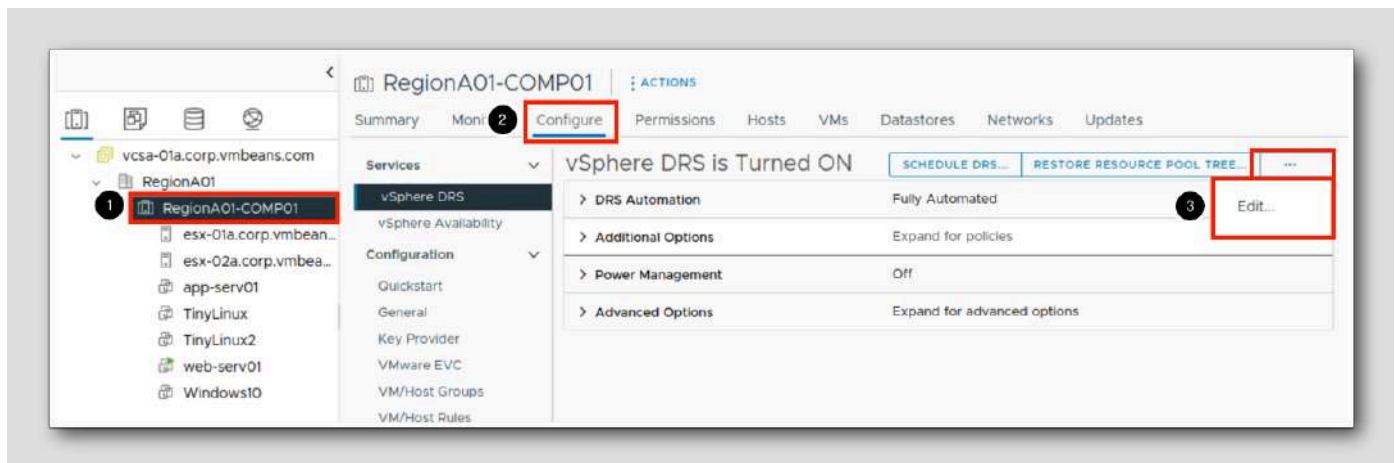
- Eliminate planned maintenance windows.
- Perform maintenance at any time without disrupting users and services.

Another feature of vSphere, Storage vMotion allows a virtual machine to be migrated to different storage devices with zero downtime. This technology is covered in more detail in Module 3.

In this lesson, you will learn how to work with vMotion and move virtual machines to different hosts within the cluster.

Edit Cluster Settings

[170]

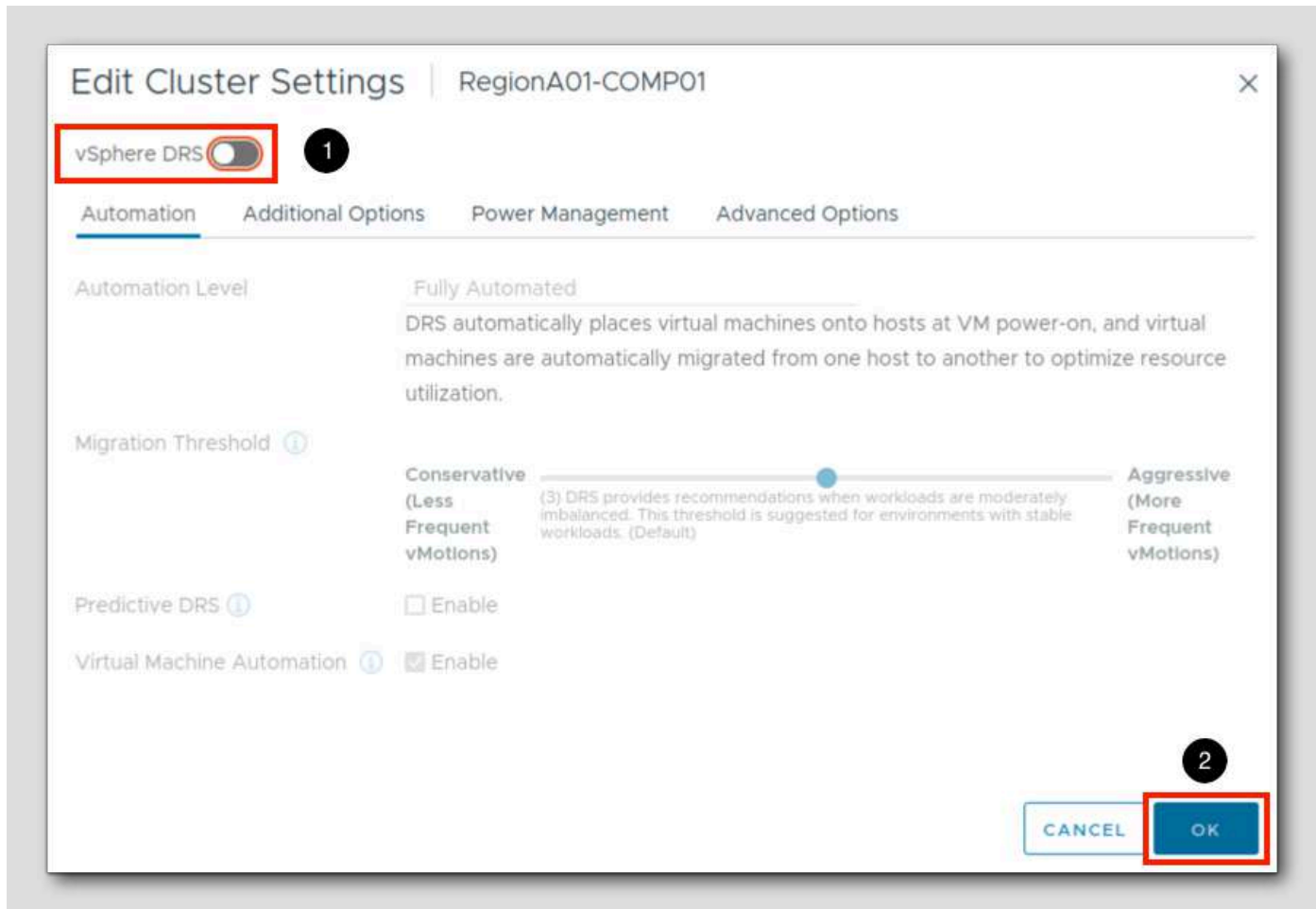


We will disable DRS and then migrate all of the virtual machines `esx-01a.corp.vmbeans.com` hosts over to `esx-02a.corp.vmbeans.com`. This will also help prepare us for the next lesson on Performance.

1. Select `RegionA01-COMP01`
2. Click the `Configure` tab
3. Under the "... " option, click on the `Edit` button

Disable DRS

[171]



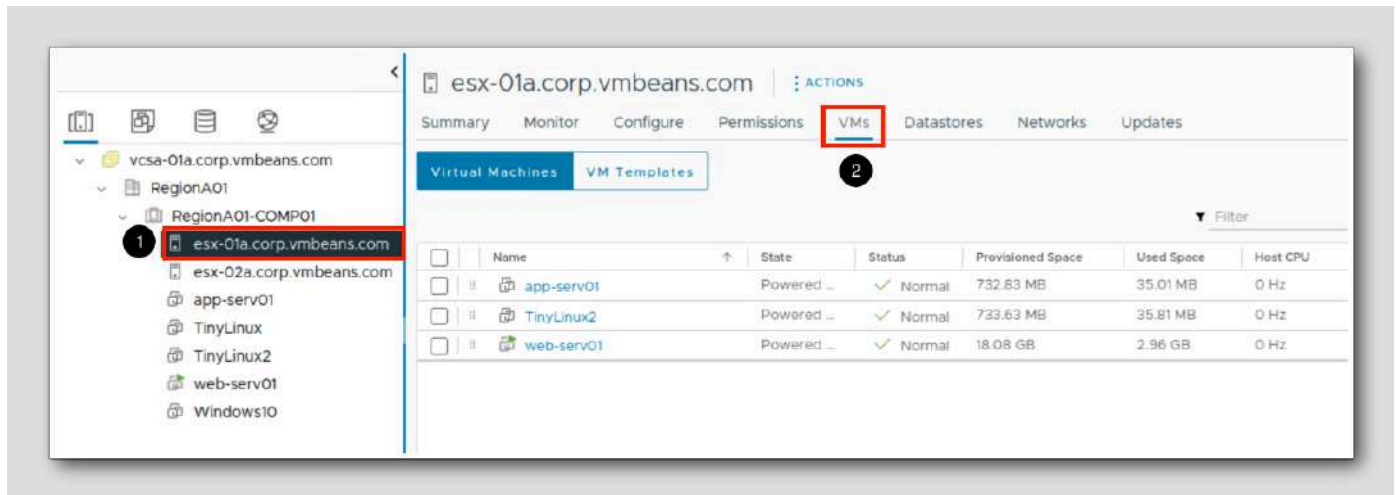
1. Flip the switch to disable vSphere DRS.

2. Click OK

By disabling DRS, this will prevent the virtual machines from being migrated back to esx-01a.corp.vmbeans.com

Migrating to esx-02a.corp.local

[172]

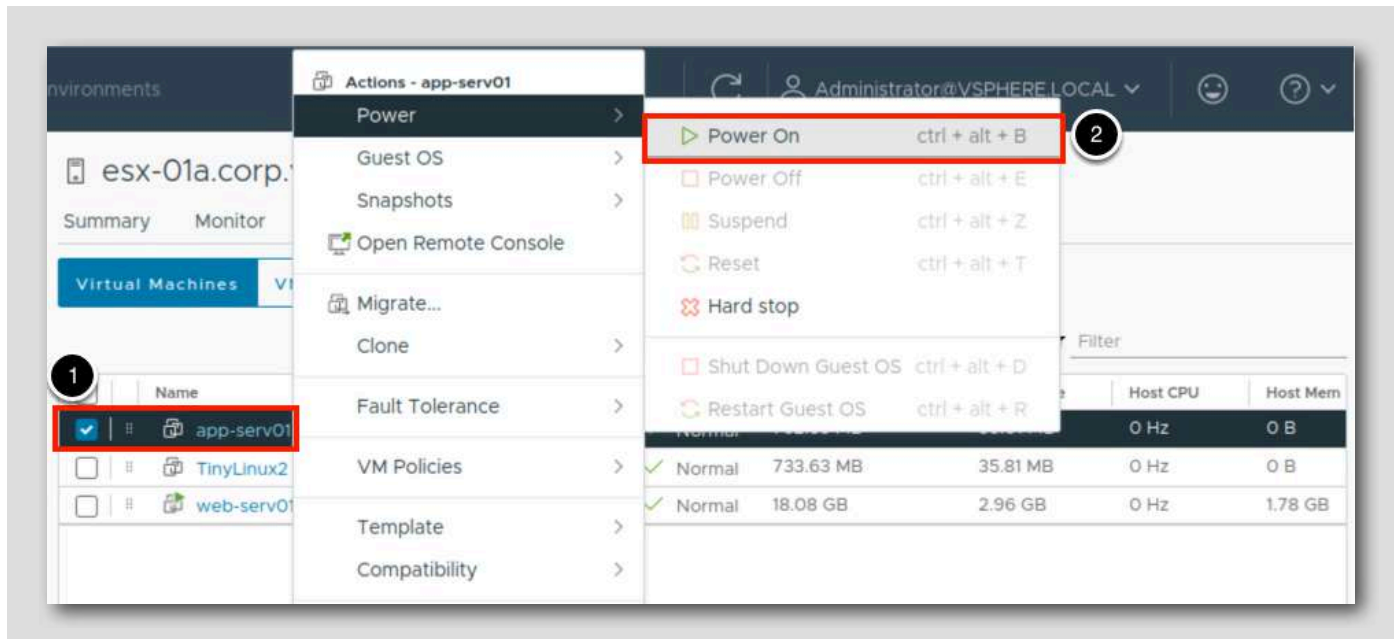


1. Select esx-01a.corp.vmbeans.com
2. Click the VMs tab

Depending on what other modules you have taken, you may see more VMs.

Power on VMs

[173]

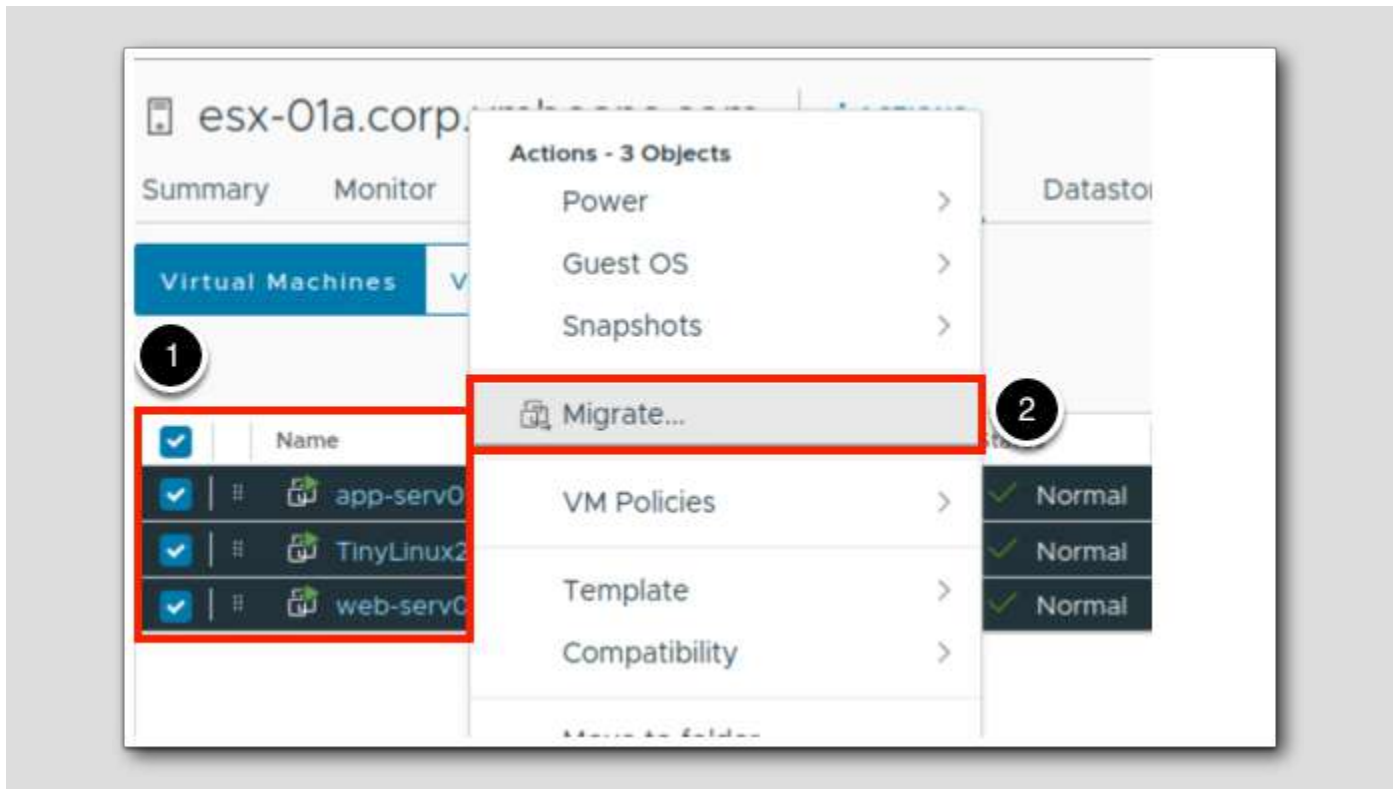


1. Look for any virtual machines that are **Powered Off** and select them. Multiple virtual machines can be selected by holding the Ctrl key and clicking on them.
2. Right-click and select **Power/Power On**

Do this for every powered off virtual machine, otherwise the next step will fail.

Migrate VMs

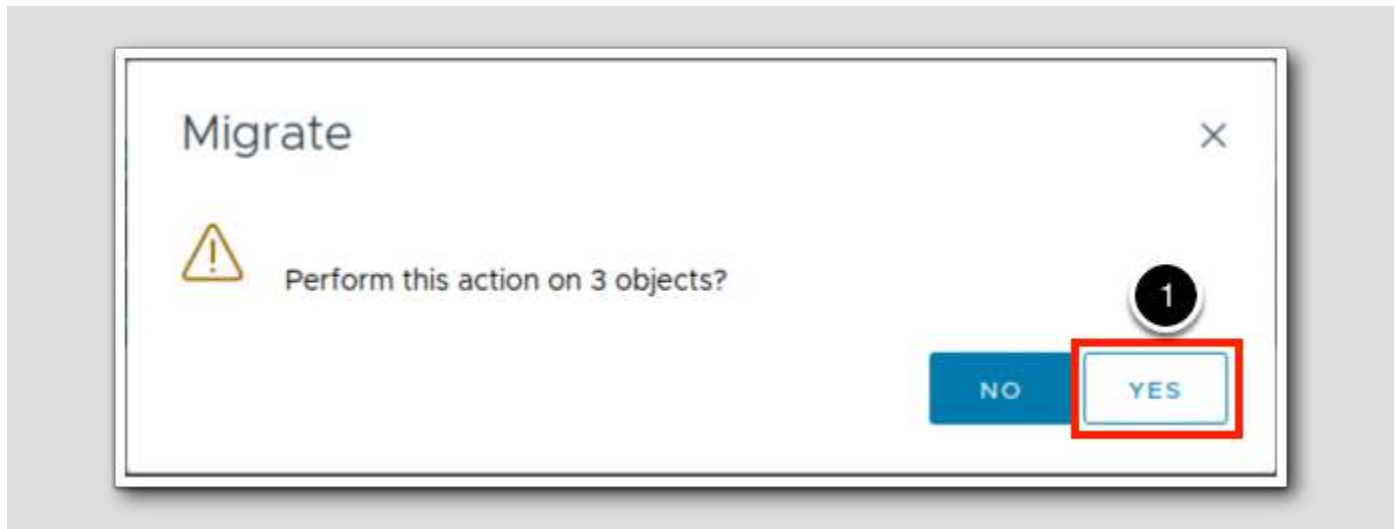
[174]



1. Select all the virtual machines
2. Right-click and select **Migrate...**

Confirm Migration

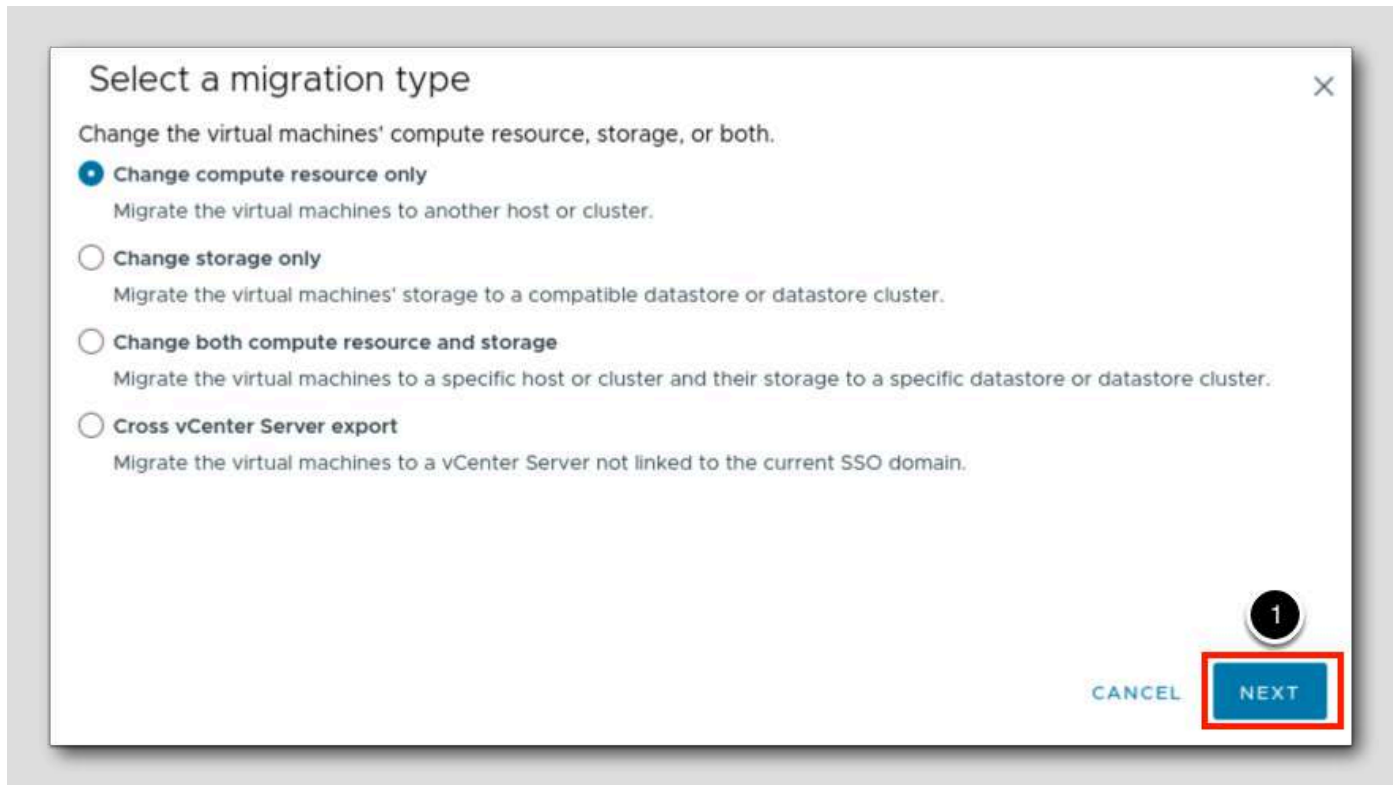
[175]



1. Click **Yes** to start the migration process.

Migration Type

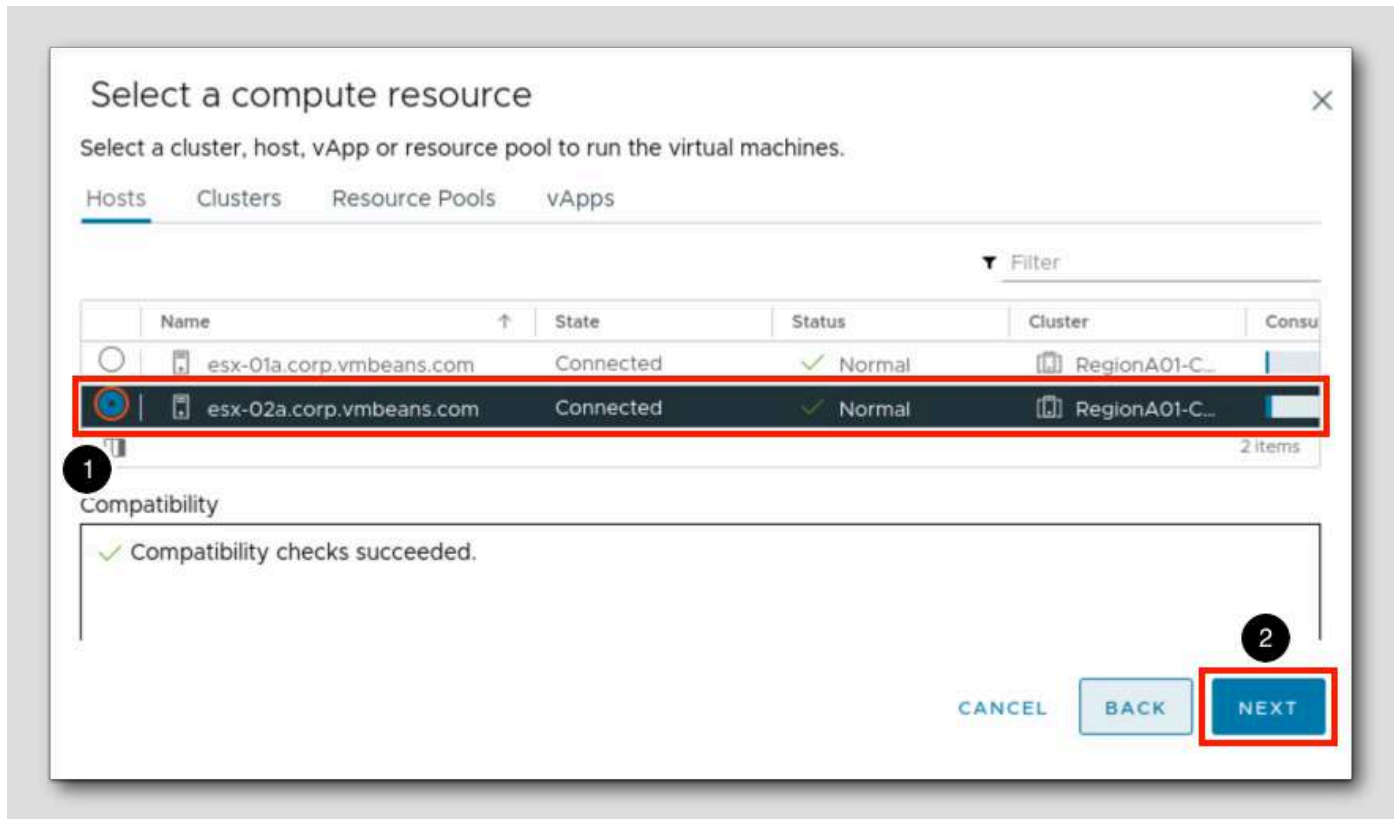
[176]



1. Leave the default setting and click **Next**

In addition to changing what ESXi host the virtual machine will run on (using compute resources), the virtual machine can be moved to different datastores (storage) if needed. A virtual machine can also be moved to a different host and storage at the same time. More on migrating to different storage is covered in Module 3, in the Storage vMotion lesson.

Compute Resource



1. Select esx-02a.corp.vmbeans.com
2. Click Next

Since we want to move all the virtual machines to esx-02a.corp.vmbeans.com, we are selecting a specific host. We could also place it in a Cluster and let DRS decide the best host to move it to.

Select Networks

[178]

Select networks ✕

Select destination networks for the virtual machine migration.
Migrate VM networking by selecting a new destination network for all VM network adapters attached to the same source network.

Source Network	Used By	Destination Network
>> VM Network	3 VMs / 3 Network adapters	VM Network

1 item

ADVANCED >>

Compatibility

✓ Compatibility checks succeeded.

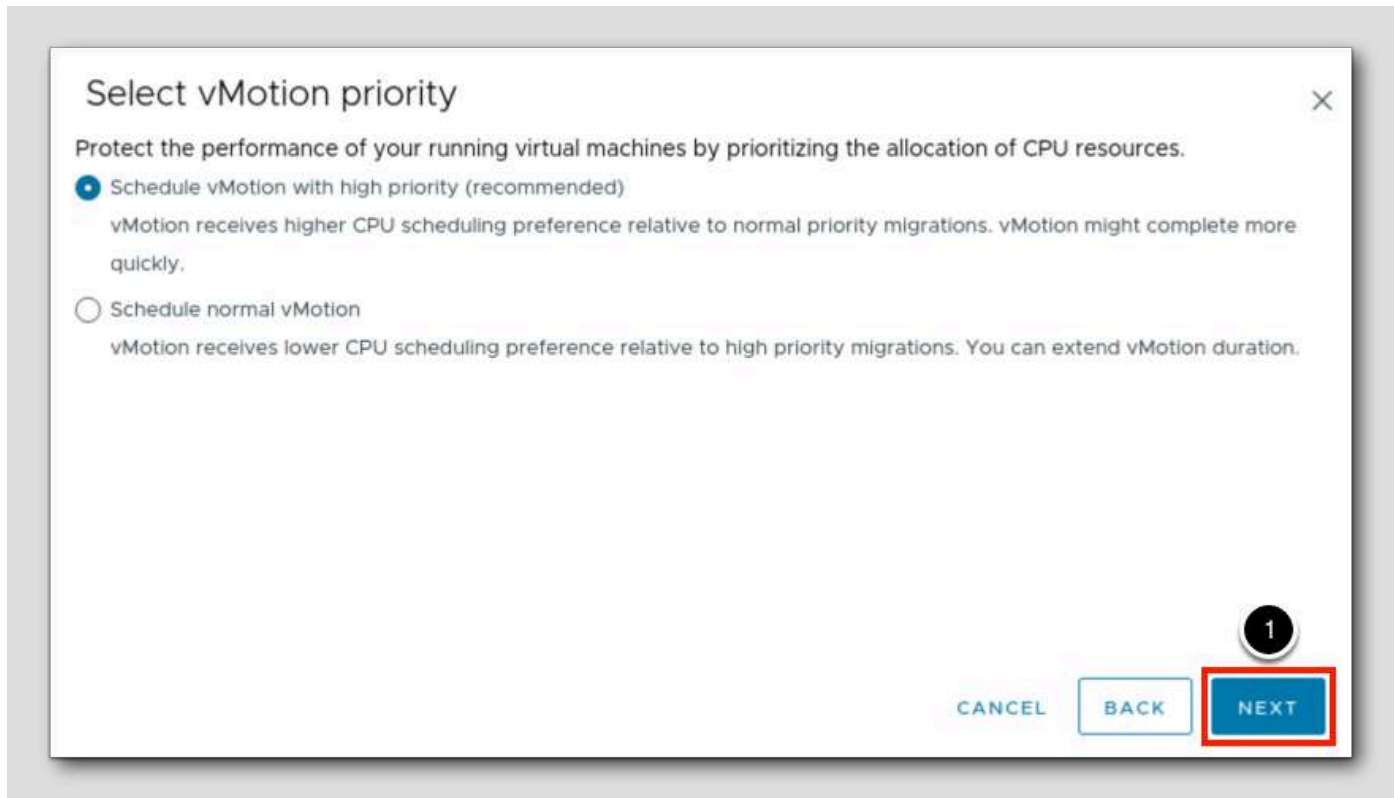
CANCEL **BACK** **NEXT**

In most cases, the network adapter will not need to be changed.

1. Click **Next**

vMotion Priority

[179]

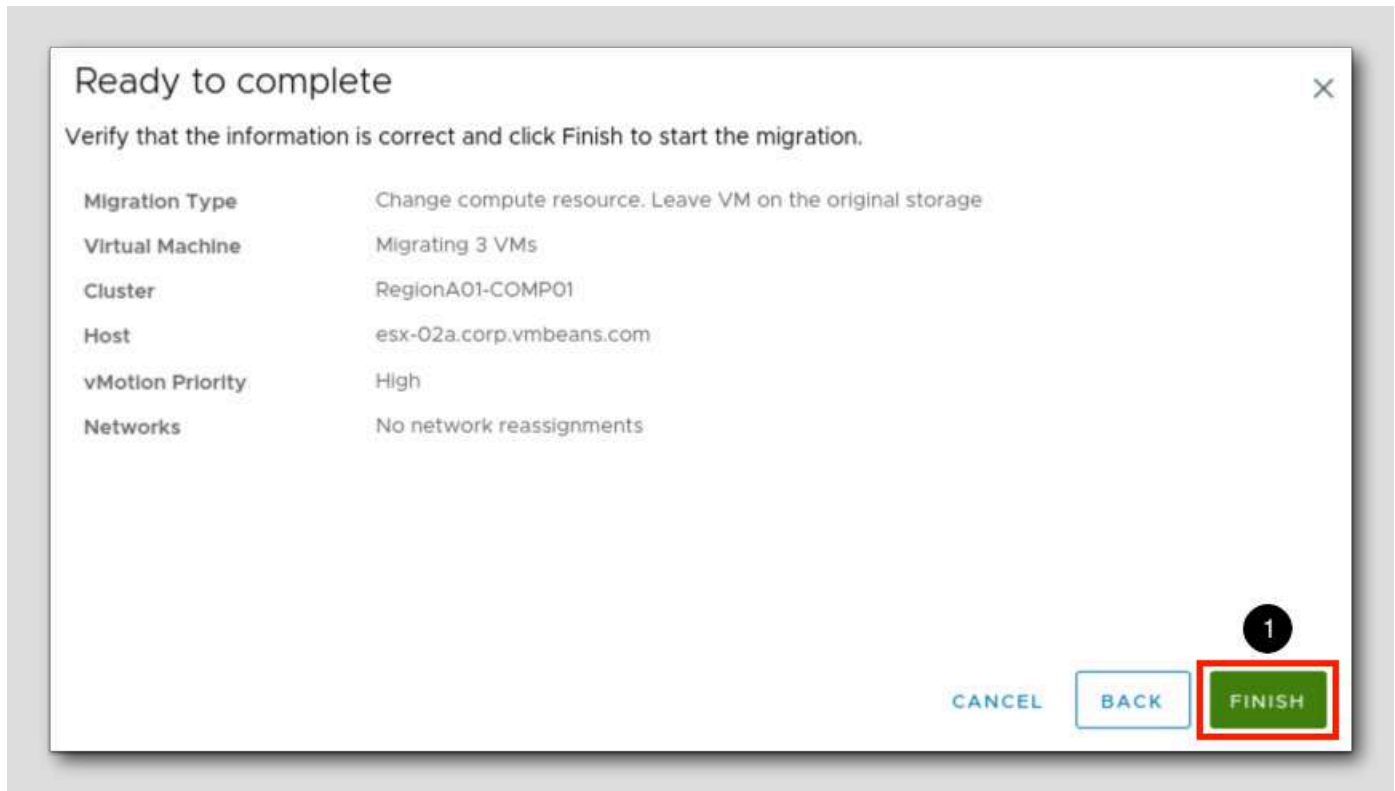


A priority can be set for the vMotion task. In most cases, the default option is OK.

1. Leave the default setting and click **Next**

Ready to Complete

[180]



1. Review the settings and click **Finish** to migrate the virtual machines to `esx-02a.corp.vmbeans.com`

Monitor Progress

[181]

Task Name	Target	Status
Relocate virtual machine	app-serv01	Completed
Relocate virtual machine	TinyLinux2	Completed
Relocate virtual machine	web-serv01	28%

You can monitor progress using Recent Tasks.

Migration Complete

[182]

Name	State	Status	Provisioned Space
app-serv01	Powered ...	✓ Normal	436.15 MB
TinyLinux	Powered ...	✓ Normal	733.25 MB
TinyLinux2	Powered ...	✓ Normal	436.95 MB
web-serv01	Powered ...	✓ Normal	18.08 GB
Windows10	Powered ...	✓ Normal	32.23 GB

When the task has been completed successfully, you should see all of the virtual machines moved over to esx-02a.corp.vmbeans.com

1. Click on esx-02a.corp.vmbeans.com
2. Click on VMs

vSphere Monitoring and Performance

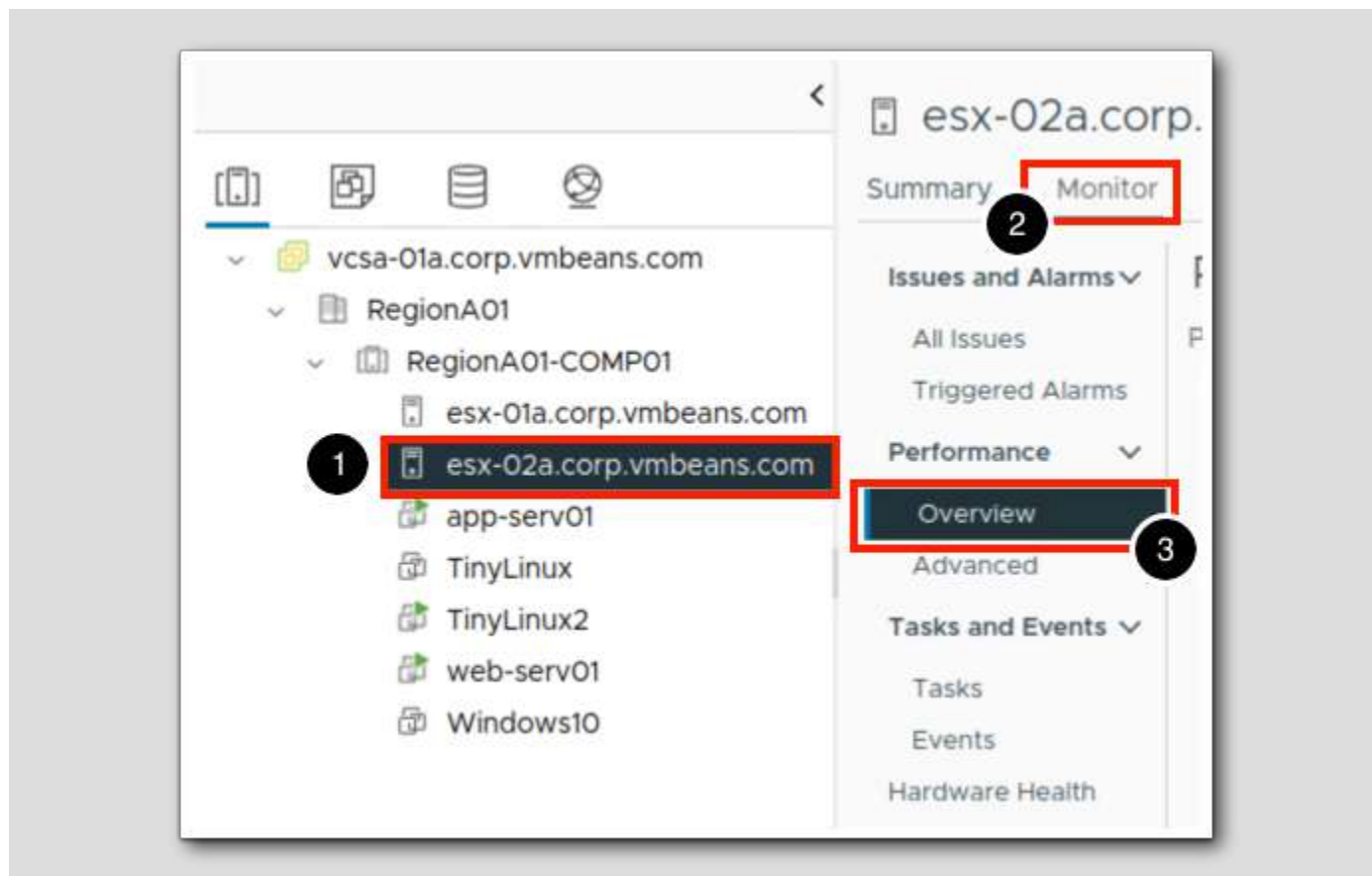
[183]

VMware provides several tools to help you monitor your virtual environment and to locate the source of potential issues and current problems. This lesson will walk through using the performance charts and graphs in the vSphere Client.

For a more advanced look at monitoring and performance, consider taking one of the Aria Operations Hands-on Labs. Aria Operations provides a more dynamic, proactive approach to monitoring your virtual infrastructure.

Select `esx-02a.corp.vmbeans.com`

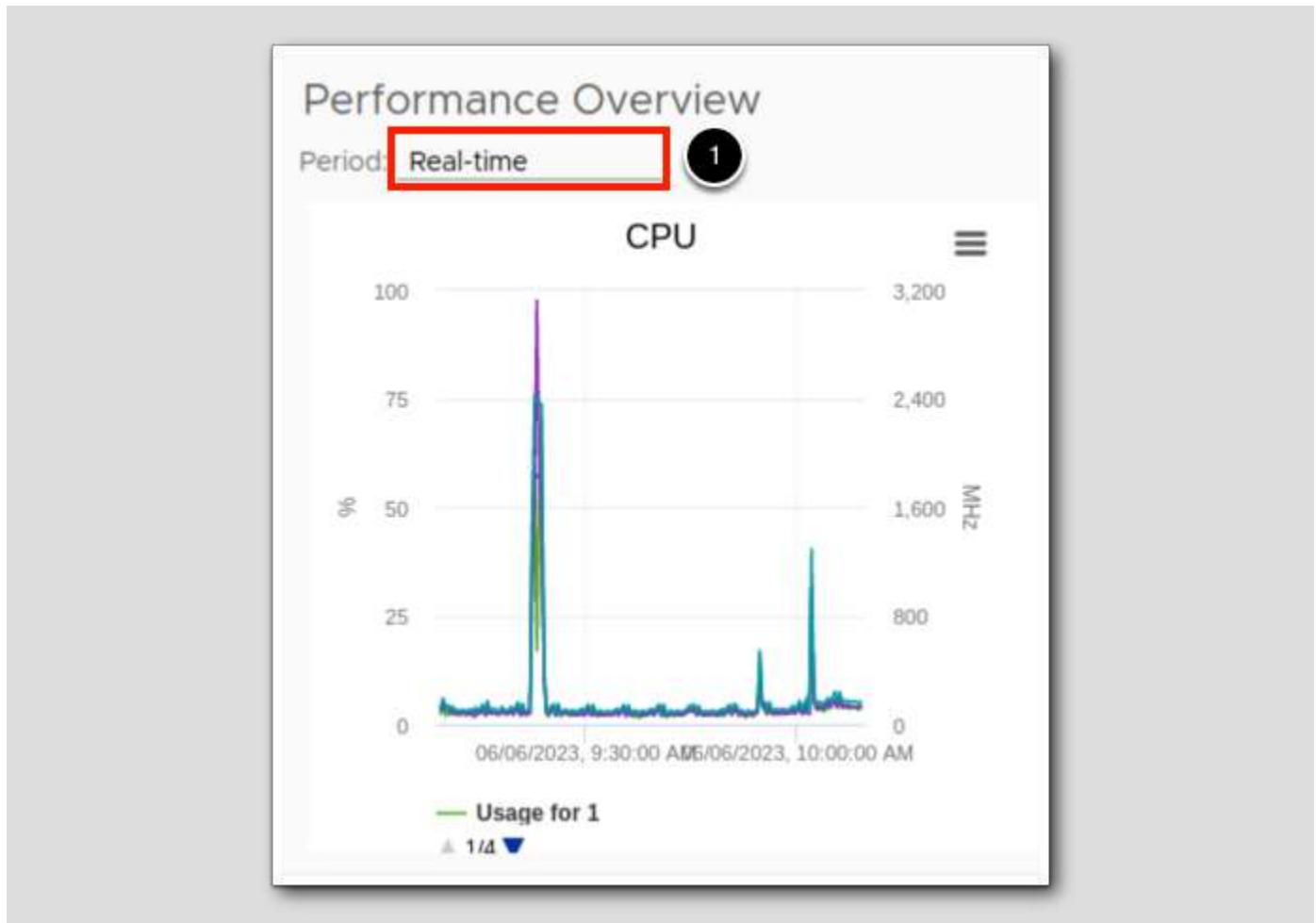
[184]



1. Select `esx-02a.corp.vmbeans.com`
2. Click the **Monitor** tab
3. Click **Overview** under the **Performance** section.

Host CPU Usage

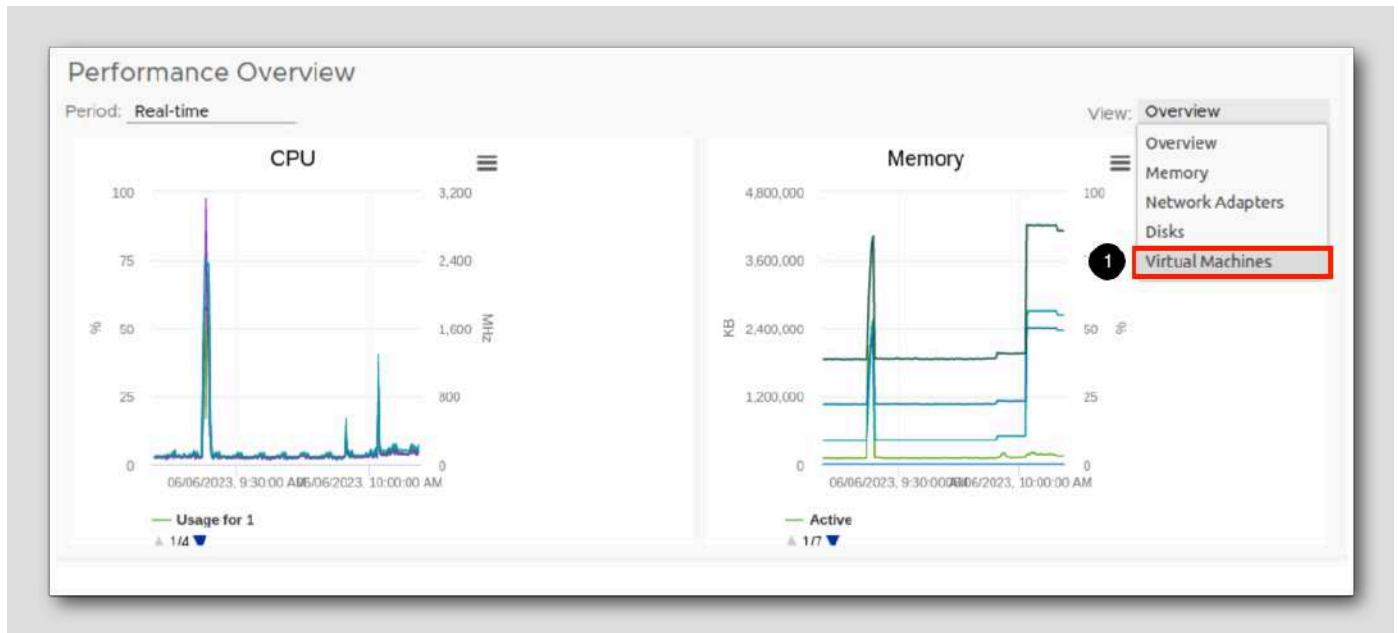
[185]



1. Ensure **Real-time** has been selected from the Time Range drop-down menu.

Here we can see in real time the CPU usage in percent for esx-02a.corp.vmbeans.com. By default, the chart will refresh every 20 seconds. The amount of data you see will depend on how long you have been taking the lab.

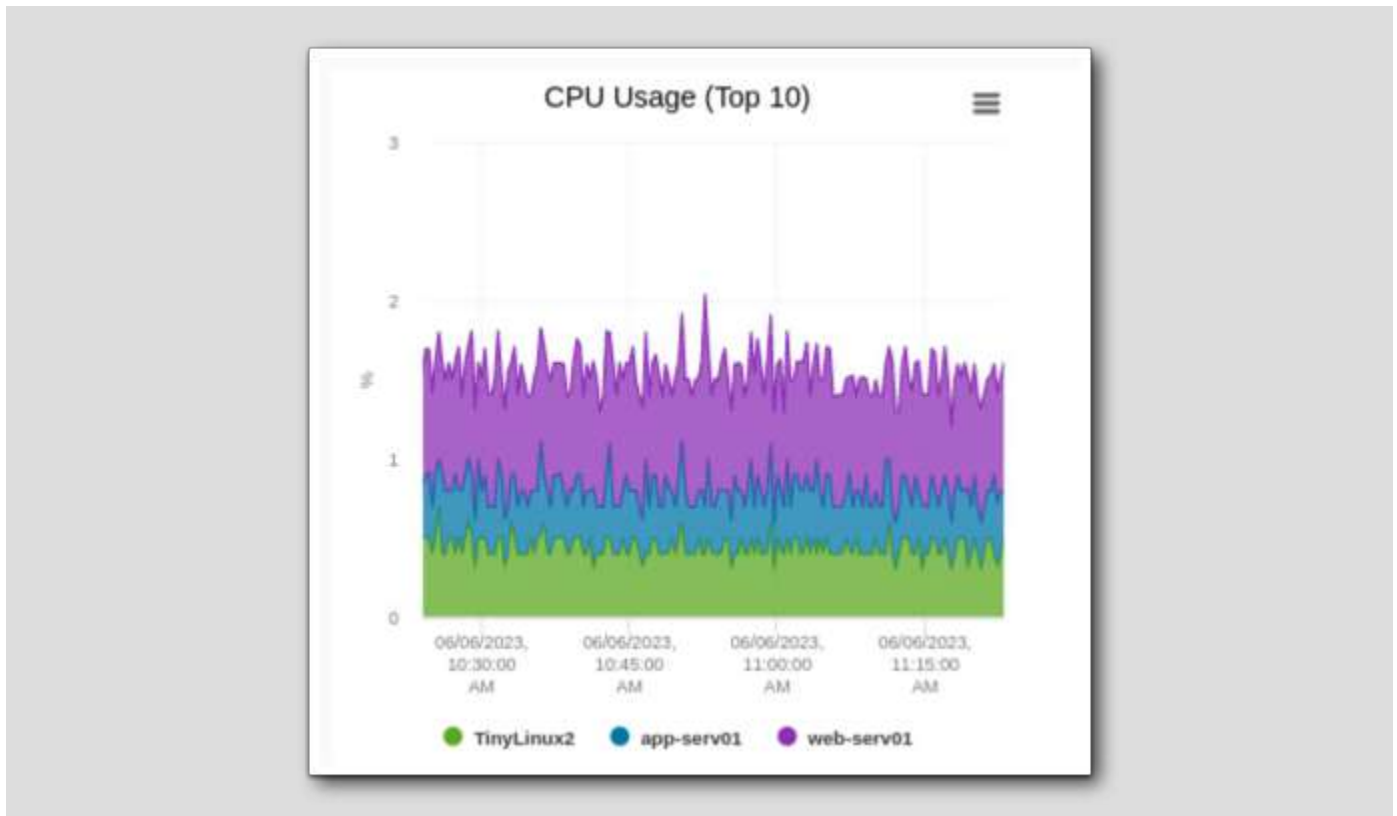
Virtual Machine CPU Usage



1. Now click the View drop-down box and select Virtual Machines.

Combined CPU Usage

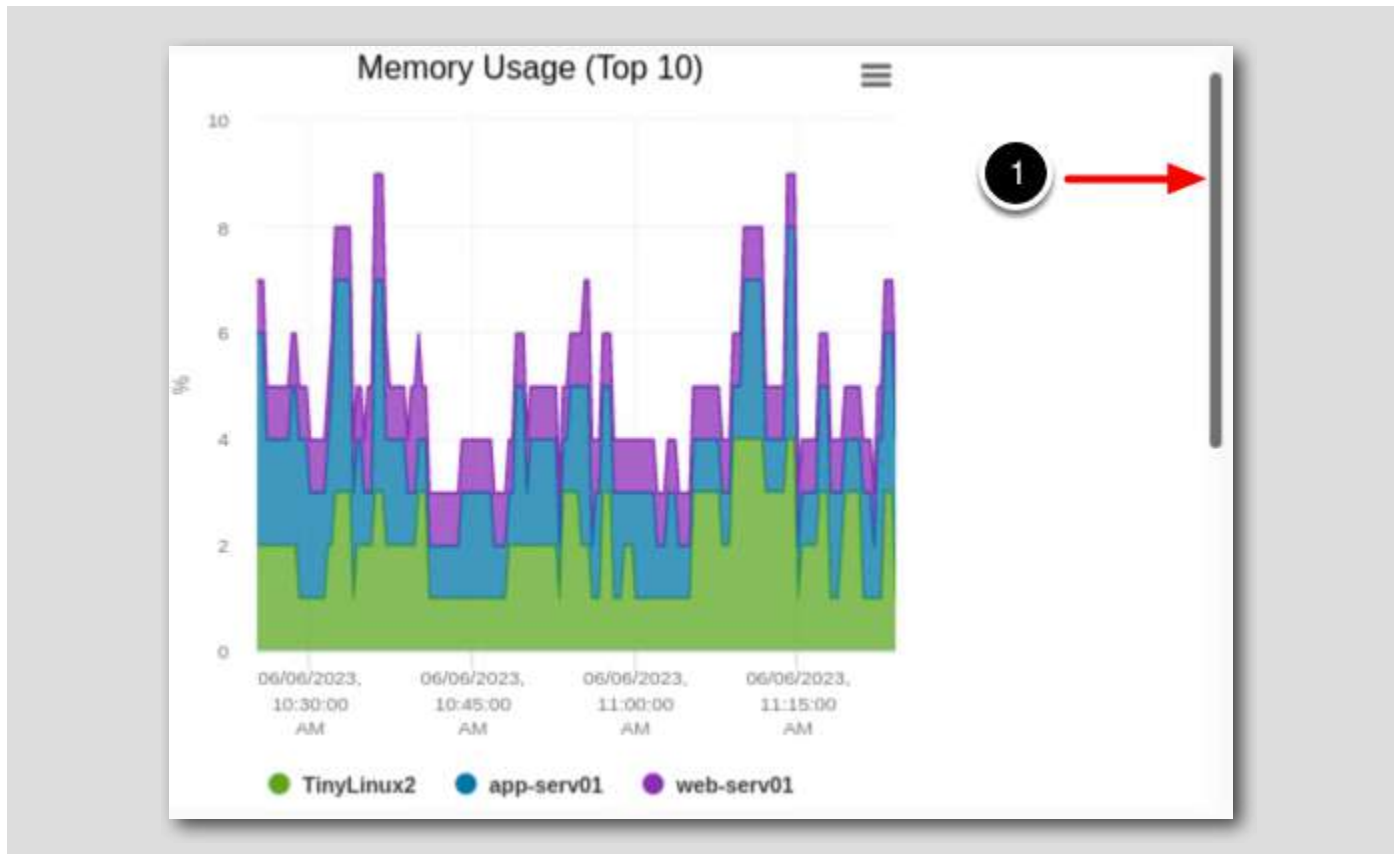
[187]



This chart shows the real-time CPU usage of each virtual machine. Each VM is represented by a different color in the graph and you can see at the bottom, which VM is represented by what color. Combined, they give you an idea of overall CPU usage on the host.

Other Available Graphs

[188]



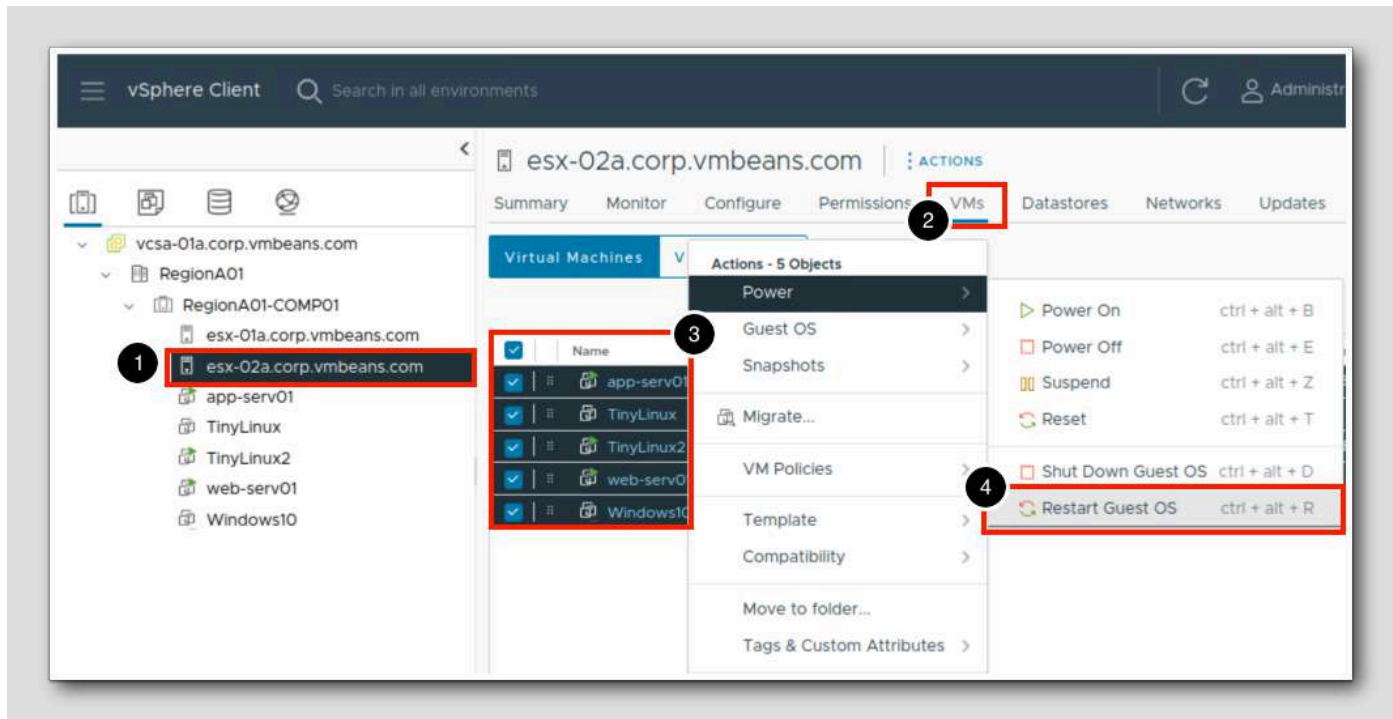
There are other graphs available to show host and virtual machine memory usage, network (Mbps) and disk (KBps).

1. Use the **scroll bars** to access the additional charts.

The graphs we have looked at so far will give you an overview of the four main components, CPU, memory, disk and storage. The advanced graphs will give you more detailed information on each of these.

Before we look at these charts, let's generate some CPU activity on `esx-02a.corp.vmbeans.com` by restarting all of the virtual machines it hosts.

Select the VMs to be Restarted

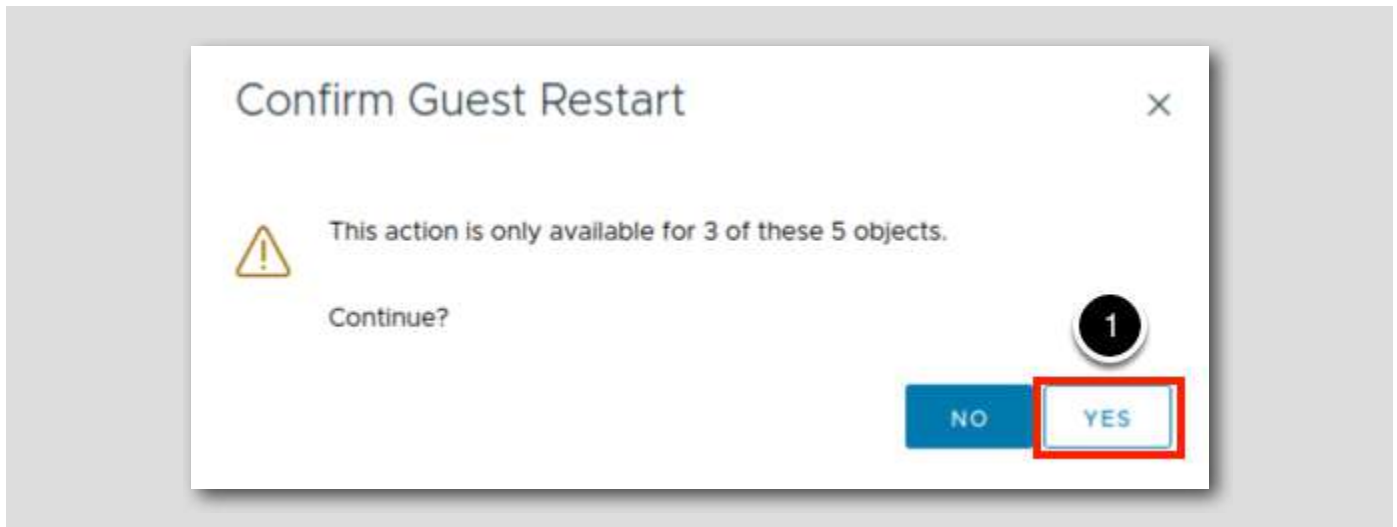


To generate some activity on esx-02a.corp.vmbeans.com, the virtual machines will be rebooted.

1. Select **esx-02a.corp.vmbeans.com**
2. Click on the **VMs** tab
3. Select **all the VMs**. Alternatively, you can also click on the **first VM** that is listed, hold down the **Shift** key and select the **last VM** on the list
4. Right-click on one of the VMs, select **Power** and click the **Restart Guest OS** button

Confirm Restart

[190]

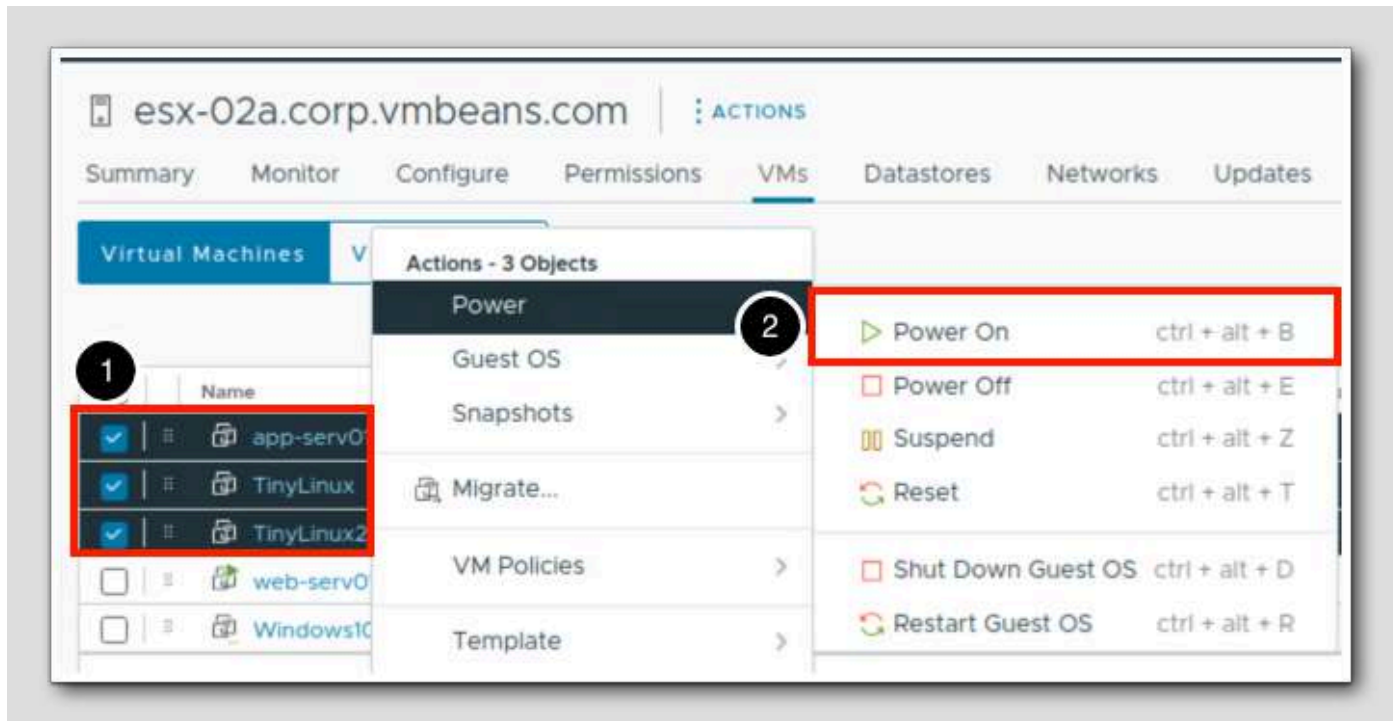


1. Click **Yes** to continue.

Note: You may also receive a warning that only X of X virtual machines will be restarted. This depends on what other modules and/or lessons have been completed in the lab previously.

Manually Start VMs

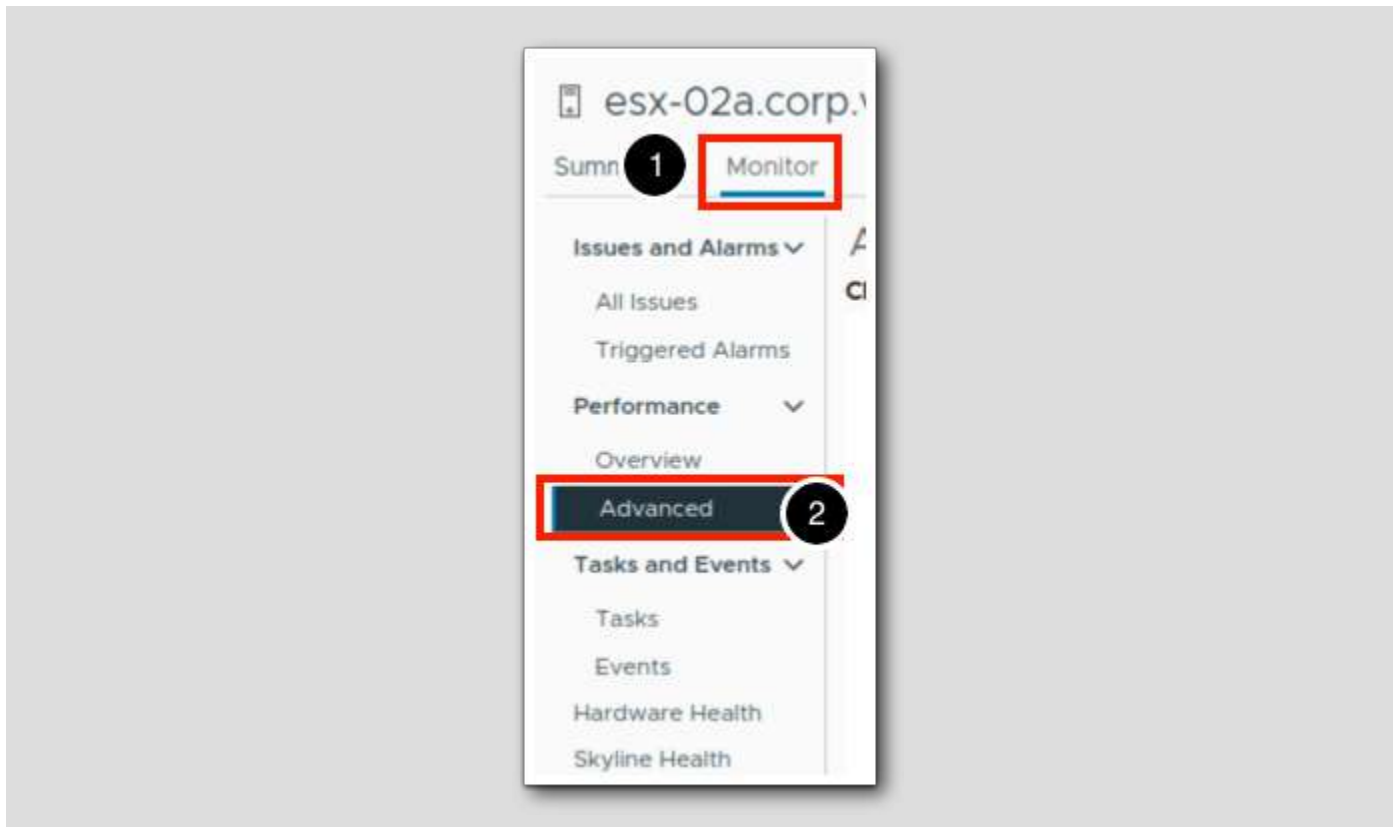
[191]



1. If `TinyLinux`, `TinyLinux2`, or `app-serv01` did not restart, but instead shut down,
2. Select them and power them on manually.

Monitor Performance

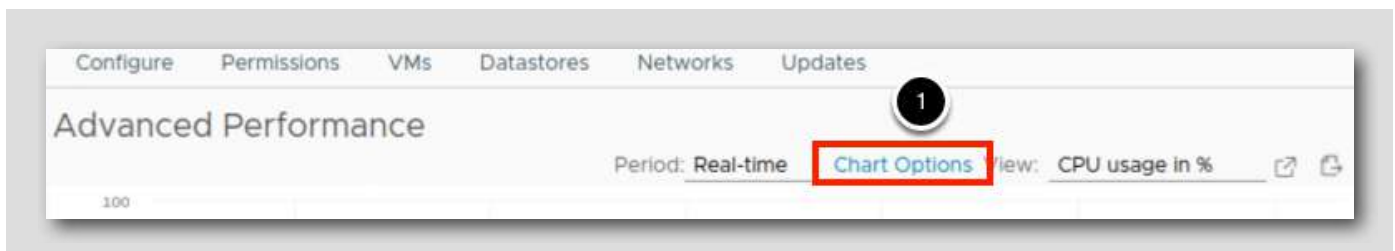
[192]



1. Click on the **Monitor** tab.
2. Click **Advanced** in the Performance section.

Chart Options

[193]



1. Click the Chart Options link.

This will bring up options to customize the chart.

Stacked Graph per VM

[194]

Select counters for this chart:

☐	Counters	Rollups	Units	Internal Name	Stat Type	Description
☐	Co-stop	Summation	ms	costop	Delta	Time the virtual machine is ...
☐	Core Utilization	Average	%	coreUtilization	Rate	CPU utilization of the corre...
☐	Demand	Average	MHz	demand	Absolute	The amount of CPU resour...
☐	Idle	Summation	ms	idle	Delta	Total time that the CPU spe...
☐	Latency	Average	%	latency	Rate	Percent of time the virtual ...

Timespan: Real-time

Last: 1 Hour(s)

From: 06/05/2023 11:33:24

To: 06/06/2023 11:33:24

(time is in ISO 8601 format)

Chart Type: Line Graph

- Line Graph
- Stacked Graph
- Stacked Graph per VM** 1

Select object for this chart:

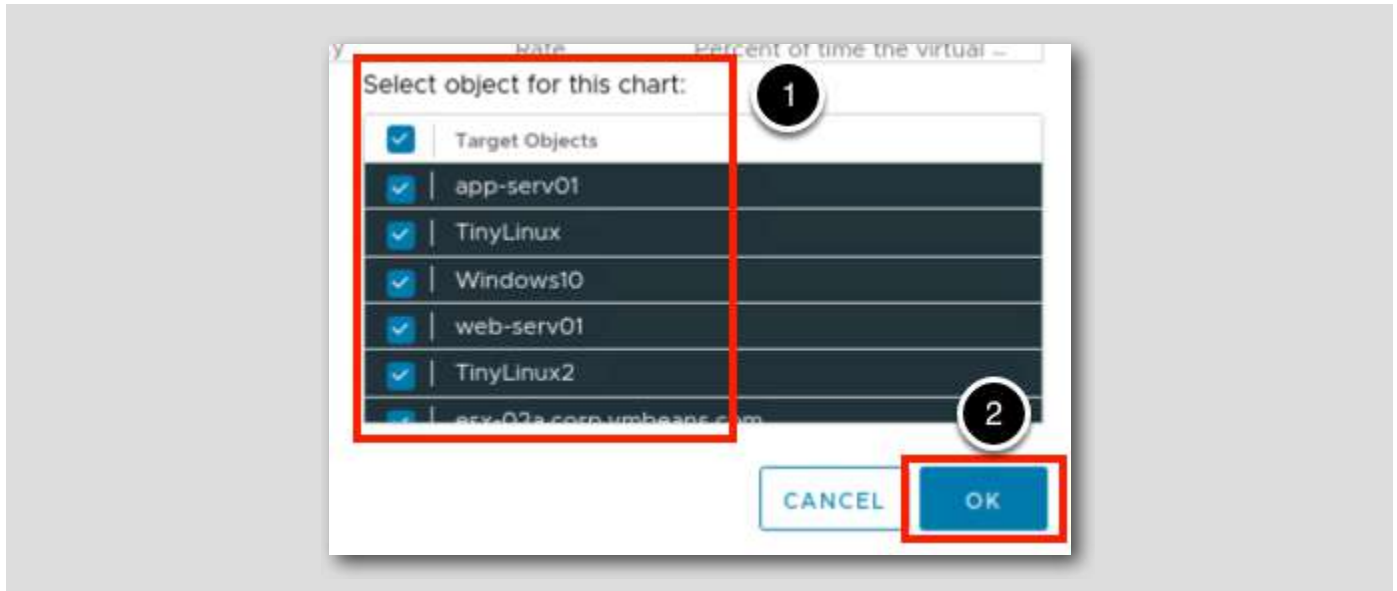
- ☒ Target Objects
- ☒ esx-02a.corp.vmbeans.com
- ☒ 0
- ☒ 1

CANCEL OK

1. From the Chart Type drop-down menu, select Stacked Graph per VM.

Select Objects

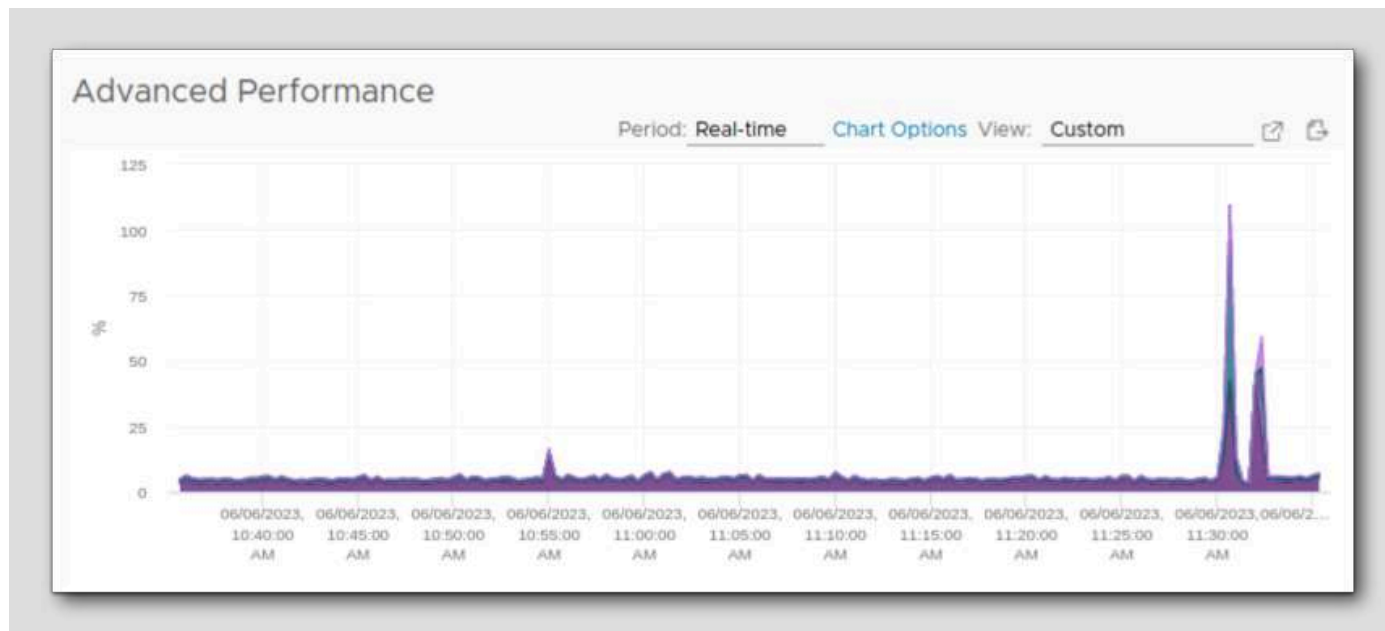
[195]



1. Under the Select objects for this chart box, verify **all** the virtual machines are selected.
2. Click the **OK** button to see the newly customized chart.

CPU Usage in Real-time

[196]



Here we can see the CPU usage of each virtual machine and esx-02a.corp.local.

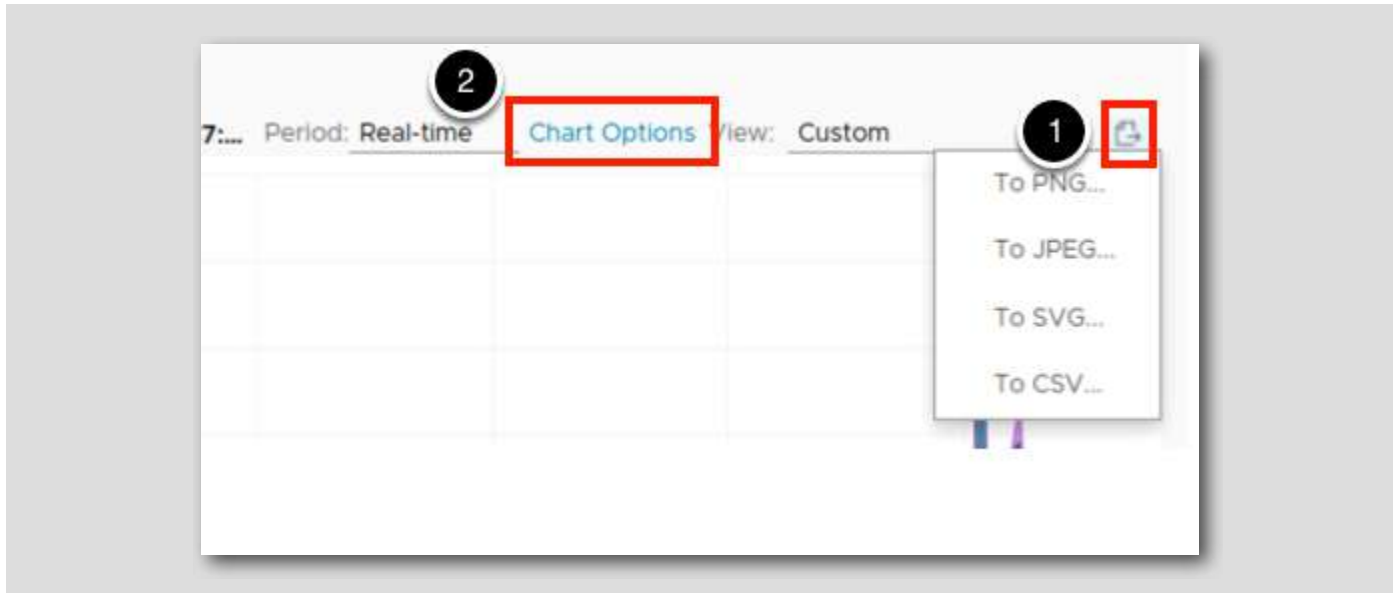
Performance Chart Legend



1. Scroll down and you will see the Performance Chart Legend. You can click on any of the virtual machines or esx-01a.corp.vmbeans.com to highlight it on the chart.

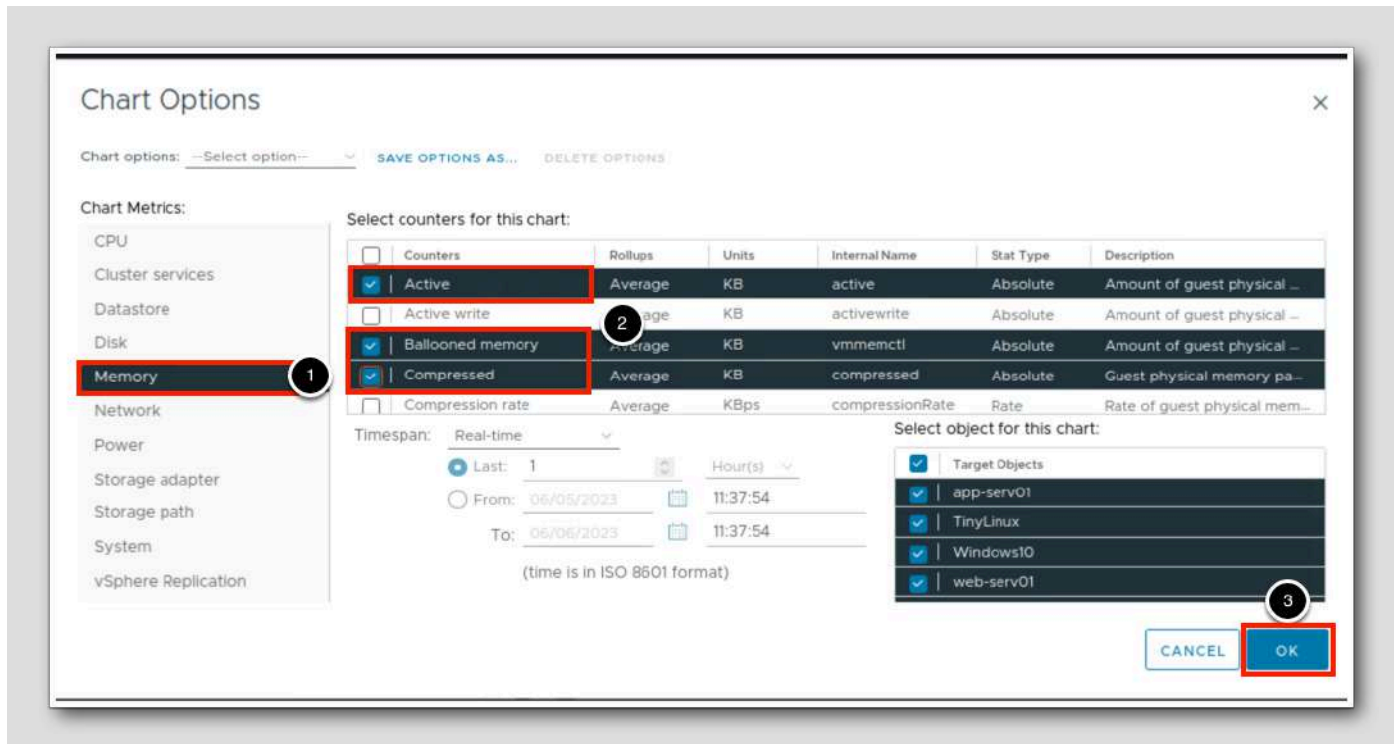
Exporting a Chart Image

[198]



1. You can export the chart in multiple formats, either as a graphic or CSV file by clicking the Export button.
2. Click the Chart Options link

Chart Metrics

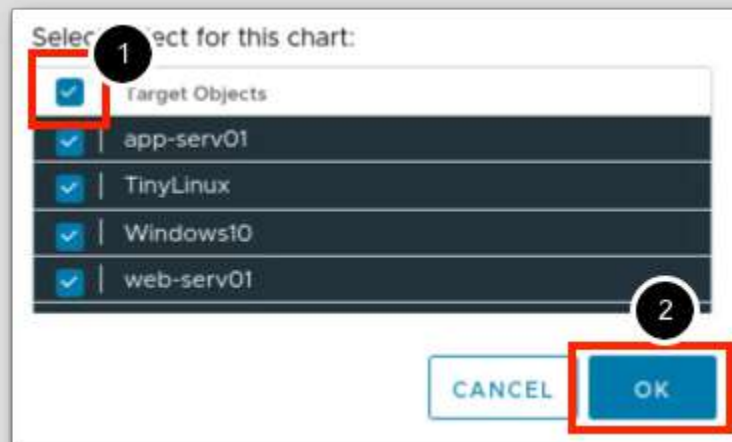


On the left-hand side, you will see a list of all the available chart metrics that can be viewed. The counters will update based on what metric you select.

1. Select **Memory** under Chart metrics.
2. Select **Active**, **Ballooned memory**, and **Compressed** for Counters to add.

Notice the counters section updates and now we have additional counters to view for this chart.

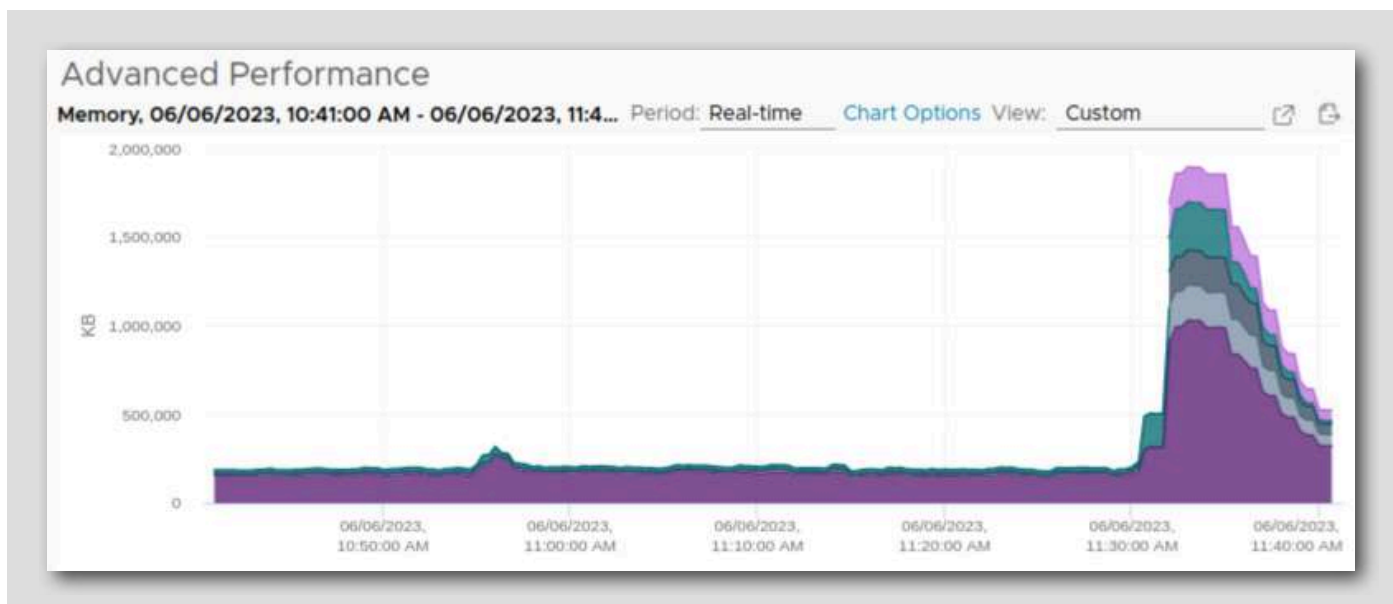
3. Click OK.



Note: If you receive an error that No Counter were selected, uncheck and check Target Objects, then click OK.

Memory Real-time

[200]

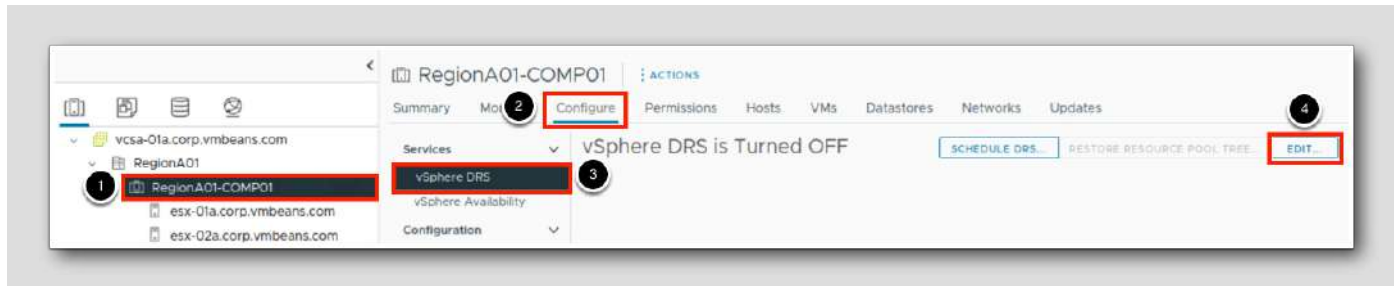


This chart shows the memory counters relative to memory for esx-02a.corp.vmbeans.com. Scroll down the Performance Chart Legend to see the counter each line represents.

Feel free to explore the various chart options and/or continue to the next step.

Enable DRS

[201]

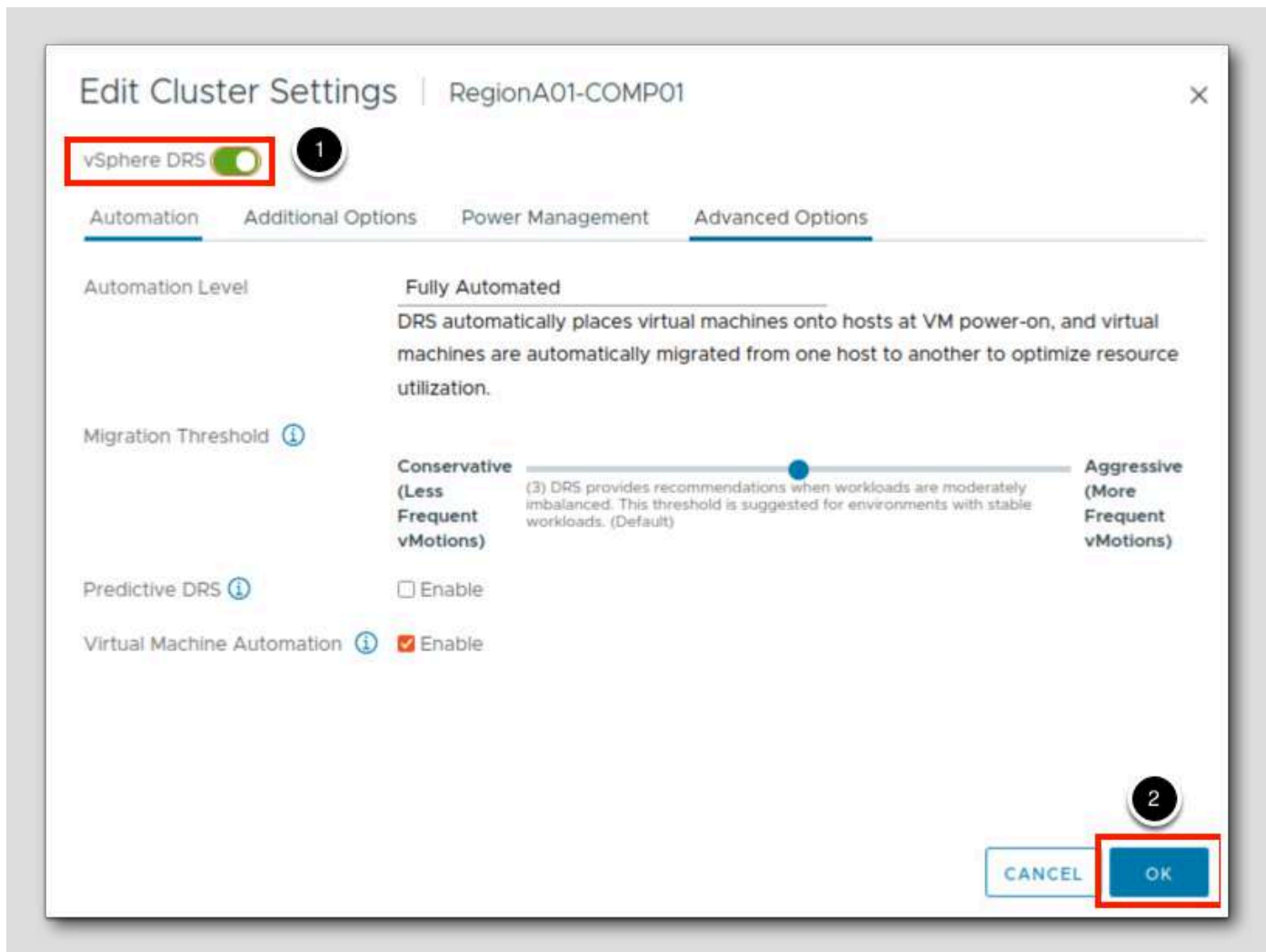


Once you have finished viewing the charts, DRS needs to be enabled again.

1. Select **RegionA01-COMP01**.
2. Click the **Configure** tab.
3. Click on **vSphere DRS**.
4. Click the **Edit** button.

Turn ON vSphere DRS

[202]



1. Check the Turn ON vSphere DRS box to enable DRS.
2. Click OK.

Further Information

[203]

For more information on performance charts, you can view the [vSphere Monitoring and Performance](#) guide.

Introduction to vSphere with Tanzu

[204]

Tanzu Kubernetes Grid on vSphere 8 consolidates the Tanzu Kubernetes offerings into a **single unified Kubernetes runtime** from VMware. VMware vSphere 8 includes the next generation of Tanzu Kubernetes (Tanzu Kubernetes Grid 2.0). It makes possible for customers to transform their existing virtual compute infrastructure into an enterprise-ready Kubernetes operating model that extends across data centers and clouds.

Common Platform for Running both Kubernetes/Containerized Workloads and VMs

[205]

Kubernetes is now built into vSphere which allows developers to continue using the same industry-standard tools and interfaces they've been using to create modern applications. vSphere Admins also benefit because they can help manage the Kubernetes infrastructure using the same tools and skills they have developed around vSphere. To help bridge these two worlds we've introduced a new vSphere construct called Namespaces, allowing vSphere Admins to create a logical set of resources, permissions, and policies that enable an application-centric approach.

Agile Operations for Kubernetes Applications

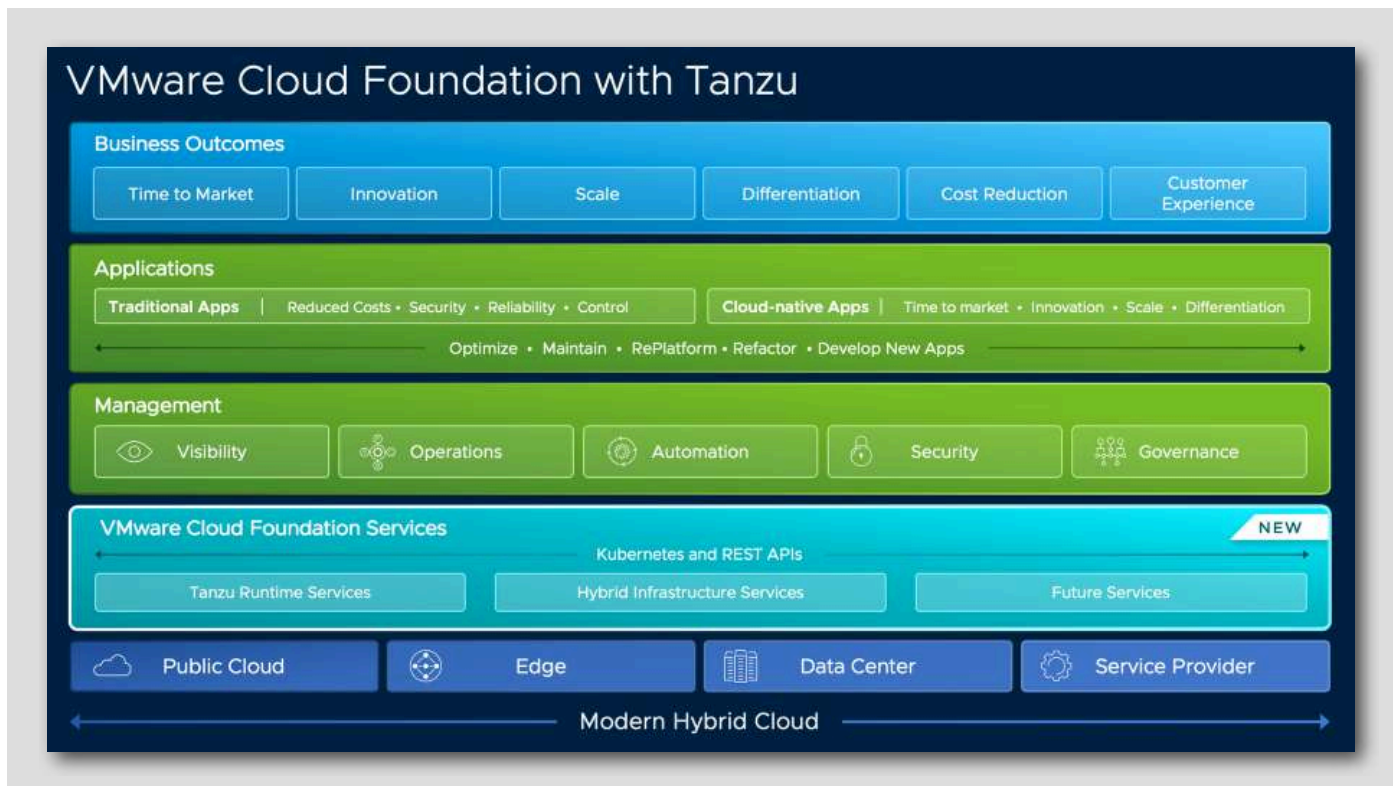
[206]

We are introducing a lot of value in vSphere with Tanzu for the VI admin. We deliver a new way to manage infrastructure, called **application-focused management**. This enables VI admins to organize multiple objects into a logical group and then apply policies to the entire group. For example, an administrator can apply security policies and storage limits to a group of virtual machines and Kubernetes clusters that represent an application, rather than to all of the VMs and clusters individually.

VMware Cloud Foundation Services

[207]

vSphere with Tanzu is available through VMware Cloud Foundation 4 with Tanzu. One key innovation available only in VMware Cloud Foundation is a set of developer-facing services and a Kubernetes API surface that IT can provision, called VMware Cloud Foundation Services.



It consists of two families of services: Tanzu Runtime Services and Hybrid Infrastructure Services.

- **Tanzu Runtime Services**– deliver core Kubernetes development services, including an up-to-date distribution of:
 - **Tanzu Kubernetes Grid Service**– which allows developers to manage consistent, compliant, and conformant Kubernetes clusters to build their modern applications.
- **Hybrid Infrastructure Services**– include full Kubernetes and REST API access that spans creating and manipulating virtual machines, containers, storage, networking, and other core capabilities. It includes the following services today:
 - **vSphere Pod Service** – extends Kubernetes with the ability to run pods directly on the hypervisor. When developers deploy containers using the vSphere Pod Service, they get the same level of security isolation, performance guarantees, and management capabilities that VMs enjoy.
 - **Storage service** – allows developers to manage persistent disks for use with containers, Kubernetes, and virtual machines.
 - **Network service** – allows developers to manage Virtual Routers, Load Balancers, and Firewall Rules.
 - **Registry service** – allows developers to store, manage, and better secure Docker and OCI images using Harbor.

Conclusion

[208]

VMware vSphere 8 is the efficient and secure platform for the hybrid cloud. It provides a powerful, flexible, and secure foundation for business agility that accelerates the digital transformation to the hybrid cloud as well as success in the digital economy.

Here are the other vSphere labs to take to get familiar with the latest vSphere 8 release:

- [HOL-2311-01-SDC - VMware vSphere - What's New](#)
- [HOL-2313-01-SDC - vSphere 8 with Tanzu](#)

Certification Path

[209]

Learn and practice with Hands-On Labs to help prepare for several VMware Certifications.



This lab can help you study for the industry-recognized VCAP-DCV Deploy 2023 certification which validates that you know how to deploy and optimize VMware vSphere infrastructures.

Learn more here: <https://www.vmware.com/learning/certification/vcap-dcv-deploy.html>



Hands-on Labs Interactive Simulation and Videos: ESXi Installation and Configuration

[210]

Due to the environment the Hands on Labs are running in and the high I/O it would cause, we are not able to install software. Please use the following videos or simulation to walk through the process.

Hands-on Labs Interactive Simulation: ESXi Installation and Configuration

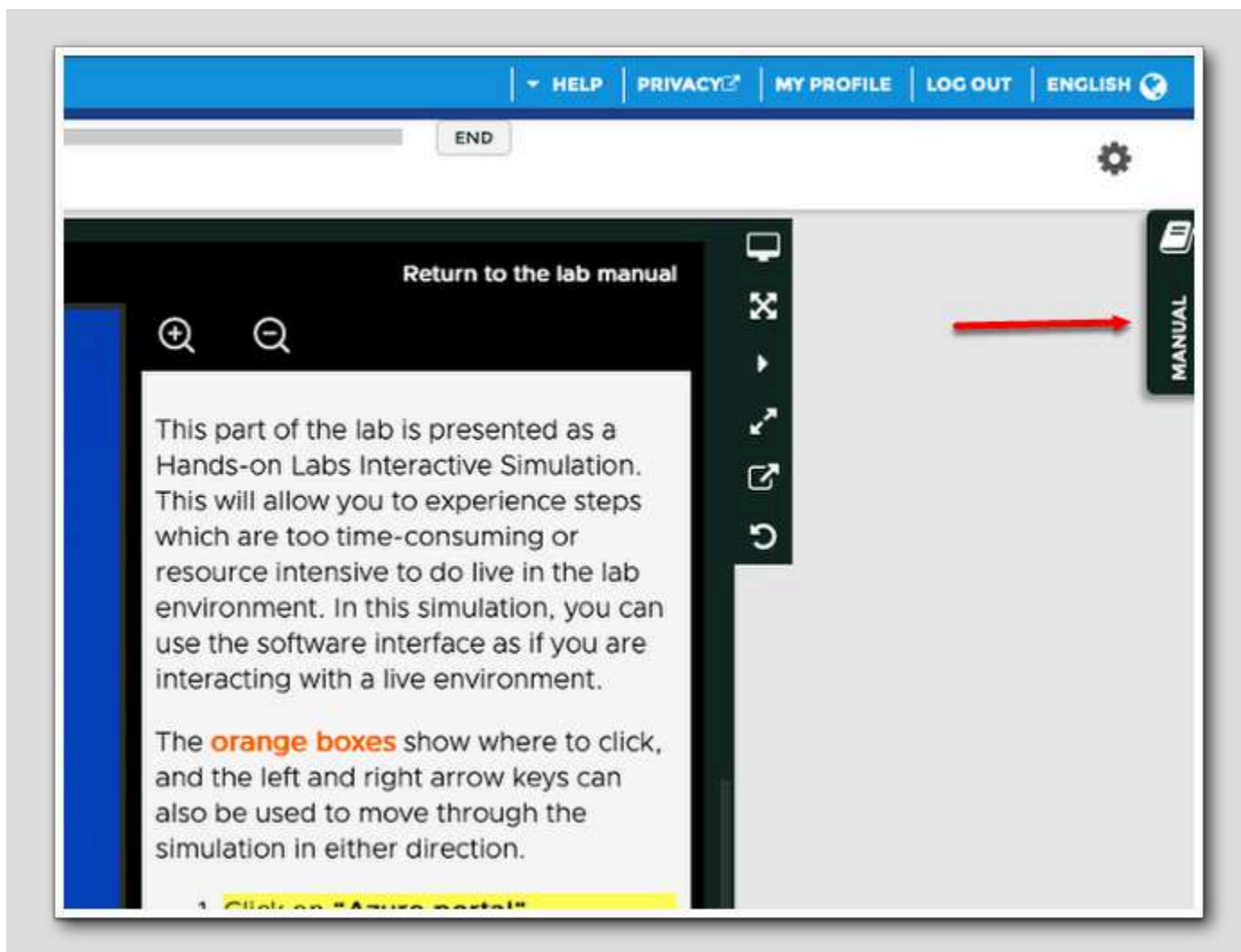
[211]

This part of the lab is presented as a **Hands-on Labs Interactive Simulation**. This will allow you to experience steps which are too time-consuming or resource intensive to do live in the lab environment. In this simulation, you can use the software interface as if you are interacting with a live environment.

Click the button below to start the simulation!

[vlp:switch-console|HOL-2410-01-ISM_Mod1|Start the Interactive Simulation]

You can hide the manual to use more of the screen for the simulation.



NOTE: When you have completed the simulation, click on the Manual tab to open it and continue with the lab.

[vlp:close-panell|manual|Close Instructions Panel]

Open the Lab Main Console and Manual

[212]

Click the button below to switch back to the Main Console

[vlp:switch-console|Main Console|Switch Back to the Main Console]

Click on the button below to open back the main manual

[vlp:close-panel|manual|Close Instructions Panel]

Video: Installing and Configuring vSphere (4:36)

[213]

The following video will walk through the process of installing and configuring vSphere.

<https://www.youtube.com/watch?v=naK5opxyKWA>



Video: Overview of the DCUI (4:58)

[214]

This video will walk you through the Direct Console User Interface (DCUI).

<https://www.youtube.com/watch?v=CPsX3Sx7XpI>



Module 2 - Introduction to vSphere Networking and Security (60 minutes)

Introduction

[216]

The ability to connect virtual machines through a logical switch that is part of the vSphere hypervisor is a necessity for operating systems and applications to communicate on the physical network. Traditionally this was done through a Standard vSwitch, configured individually at each ESXi host in the datacenter.

Since its introduction, the vSphere Distributed Switch quickly became the recommended type of virtual switch to use for most if not all types of network traffic in and out of the ESXi host. This is due mostly in part to its ability to be created and managed centrally through vCenter, as well as the advanced networking features it provides.

Let's spend some time reviewing the similarities and differences between the two types of switches.

Types of Virtual Switches

[217]

There are two types of virtual switches in ESXi 6.x, ESXi 7.x and ESXi 8:

- vSphere Standard Switches (VSS)
- vSphere Distributed Switch (VDS)

vSphere Standard Switch (vSwitch, VSS)

[218]

It works much like a physical Ethernet switch. It detects which virtual machines are logically connected to each of its virtual ports and uses that information to forward traffic to the correct virtual machines. A vSphere standard switch can be connected to physical switches by using physical Ethernet adapters, also referred to as uplink adapters, to join virtual networks with physical networks. This type of connection is similar to connecting physical switches together to create a larger network. Even though a vSphere standard switch works much like a physical switch, it does not have some of the advanced functionality of a physical switch.

As in VMware Infrastructure 3, the configuration of each vSwitch resides on the specific ESXi/ESX host. The VI administrators have to manually maintain consistency of the vSwitch configuration across all ESXi/ESX hosts to ensure that they can perform operations such as vMotion.

vSphere Distributed Switch (dvSwitch, VDS)

[219]

A vSphere distributed switch acts as a single switch across all associated hosts in a data center to provide centralized provisioning, administration, and monitoring of virtual networks. You configure a vSphere distributed switch on the vCenter Server system and the configuration is propagated to all hosts that are associated with the switch. This lets virtual machines maintain consistent network configuration as they migrate across multiple hosts.

The configuration of vDS is centralized to vCenter Server. The ESXi 6.x, ESXi 7.x and ESXi 8 hosts that belong to a dvSwitch do not need further configuration to be compliant.

Starting with vSphere 8.0, vSphere Distributed Services Engine (vDSE) introduces virtual infrastructure as a distributed architecture with the addition of data processing units (DPUs) also known as SmartNic that enable offloading infrastructure functions from the host or server CPUs to data processing units (DPUs).

Comparing vSphere Standard Switch with vSphere Distributed Switch

These features are available with both types of virtual switches:

- Can forward L2 frames
- Can segment traffic into VLANs
- Can use and understand 802.1q VLAN encapsulation
- Can have more than one uplink (NIC Teaming)
- Can have traffic shaping for the outbound (TX) traffic

These features are available only with a Distributed Switch:

- Can shape inbound (RX) traffic
- Has a central unified management interface through vCenter Server
- Supports Private VLANs (PVLANS)
- Provides potential customization of Data and Control Planes

vSphere 6.x provides these improvements to Distributed Switch functionality:

- Network IO Control - New support for per virtual machine Distributed vSwitch bandwidth reservations to guarantee isolation and enforce limits on bandwidth.
- Multicast Snooping - Supports IGMP snooping for IPv4 packet and MLD snooping for IPv6 packets in VDS. Improves performance and scale with multicast traffic.
- Multiple TCP/IP Stack for vMotion - Allows vMotion traffic a dedicated networking stack. Simplifies IP address management with a dedicated default gateway for vMotion traffic.

vSphere 8.x provides these improvements to Distributed Switch functionality backed by DPUs:

- Creation and deletion of the vSphere distributed switch.
- Configuration management.
- vSphere distributed switch health check.
- Link aggregation control protocol (LACP).
- Port mirroring.
- Private LANs.
- Link layer discovery protocol.

For more information on the new VMware vSphere Distributed Services Engine back by DPUs, refer to the [vSphere Distributed Services Engine and Networking Acceleration](#) guide.

vSS vs vDS architecture

[221]

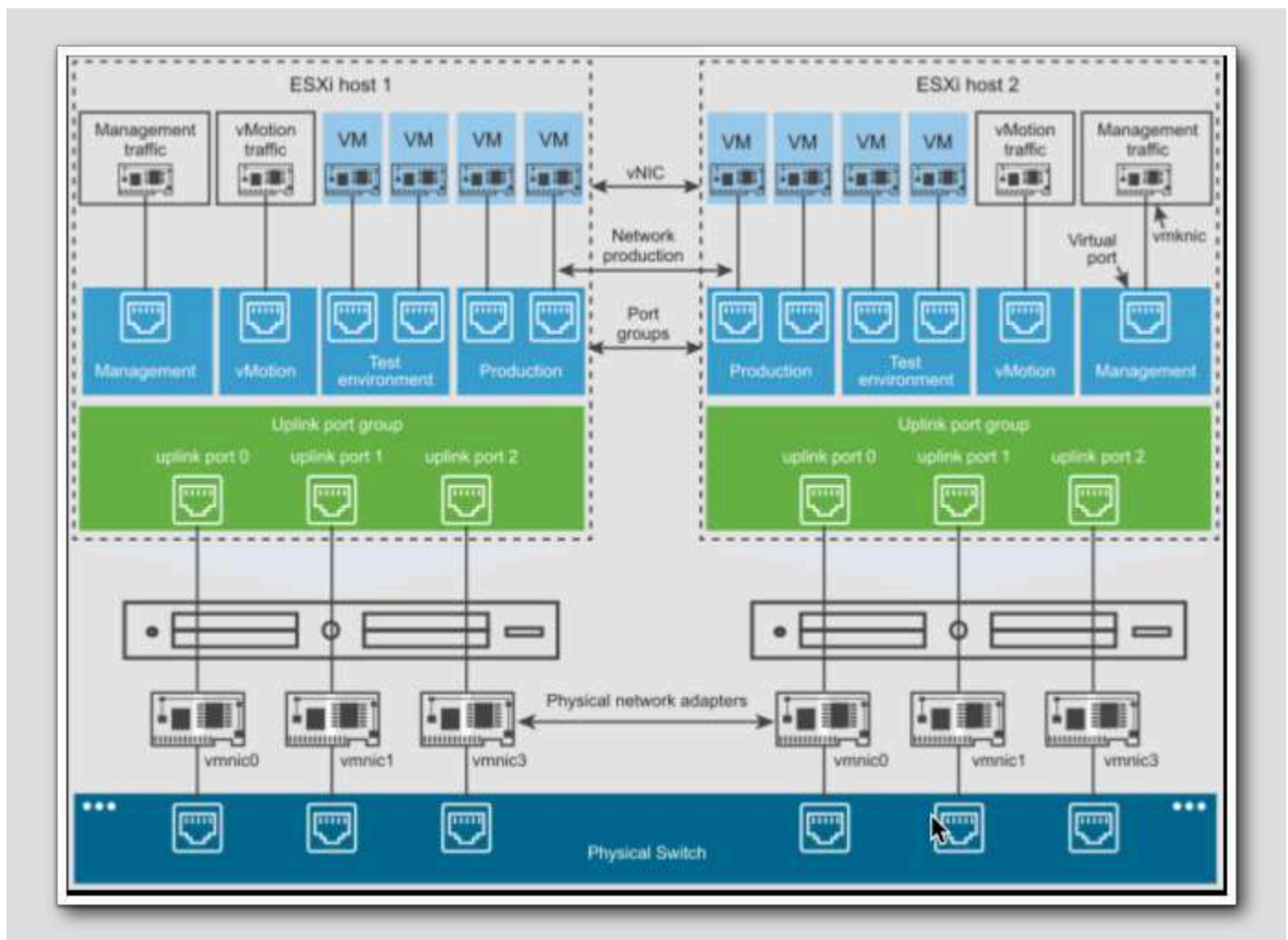
Spend a few minutes reviewing the differences between the [Standard vSwitch](#) and [Distributed vSwitch](#) architectures.

Pay special attention to how the port groups and uplinks are designed.

vSphere Standard Switch Architecture

[222]

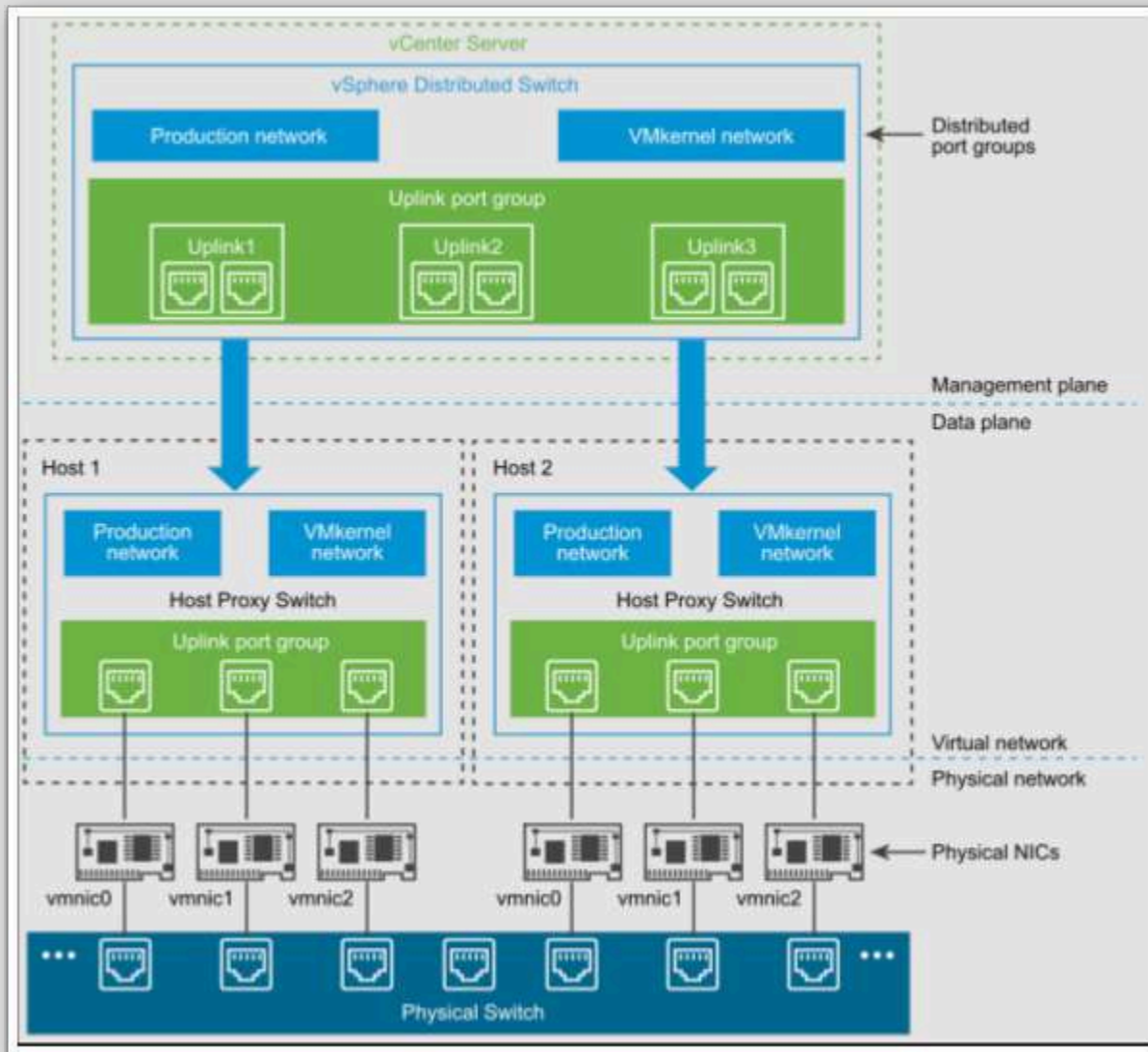
To provide network connectivity to hosts and virtual machines, you connect the physical NICs of the hosts to uplink ports on the standard switch. Virtual machines have network adapters (vNICs) that you connect to port groups on the standard switch. Every port group can use one or more physical NICs to handle their network traffic. If a port group does not have a physical NIC connected to it, virtual machines on the same port group can only communicate with each other but not with the external network.



A vSphere Standard Switch is very similar to a physical Ethernet switch. Virtual machine network adapters and physical NICs on the host use the logical ports on the switch as each adapter uses one port. Each logical port on the standard switch is a member of a single port group.

vSphere Distributed Switch Architecture

[223]



A network switch in vSphere consists of two logical sections that are the data plane and the management plane. The data plane implements the packet switching, filtering, tagging, and so on. The management plane is the control structure that you use to configure the data plane functionality. A vSphere Standard Switch contains both data and management planes, and you configure and maintain each standard switch individually.

A vSphere Distributed Switch separates the data plane and the management plane. The management functionality of the distributed switch resides on the vCenter Server system that lets you administer the networking configuration of your environment on a data center level. The data plane remains locally on every host that is associated with the distributed switch. The data plane section of the distributed switch is called a host proxy switch. The networking configuration that you create on vCenter Server (the management plane) is automatically pushed down to all host proxy switches (the data plane).

The vSphere Distributed Switch introduces two abstractions that you use to create consistent networking configuration for physical NICs, virtual machines, and VMkernel services.

Let us get started!

[224]

Now that we have a better understanding of what a Distributed vSwitch is and why we would want to use it, let's spend a little time exploring an example of one.

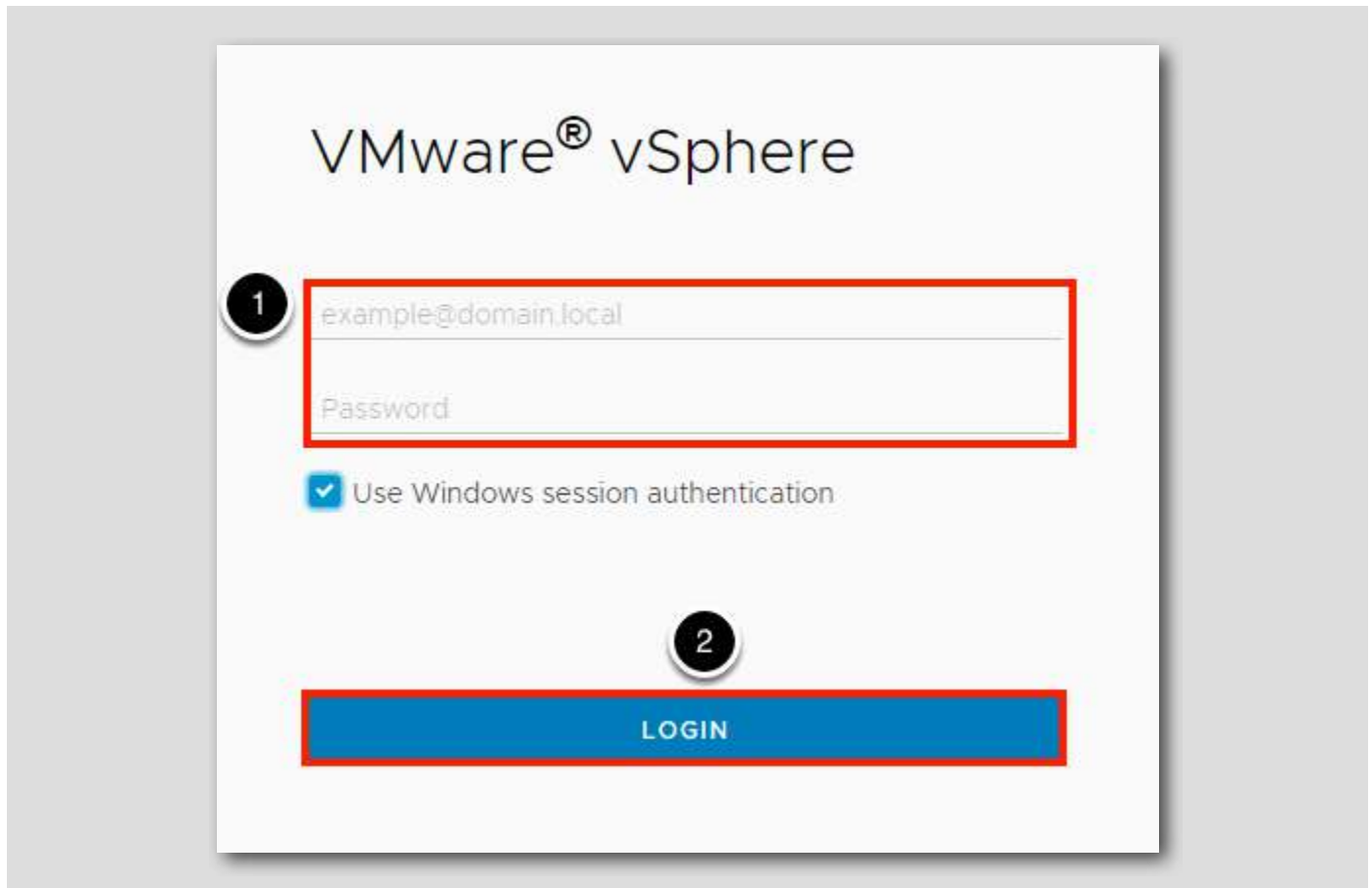
Adding and Configuring vSphere Standard Switch

[225]

The following lesson will walk you through the process of creating and configuring the vSphere Standard Switch.

Adding a Virtual Machine Port Group with the vSphere Client

[226]



If you are not already logged in, launch Firefox from the desktop and log in to the vSphere Client.

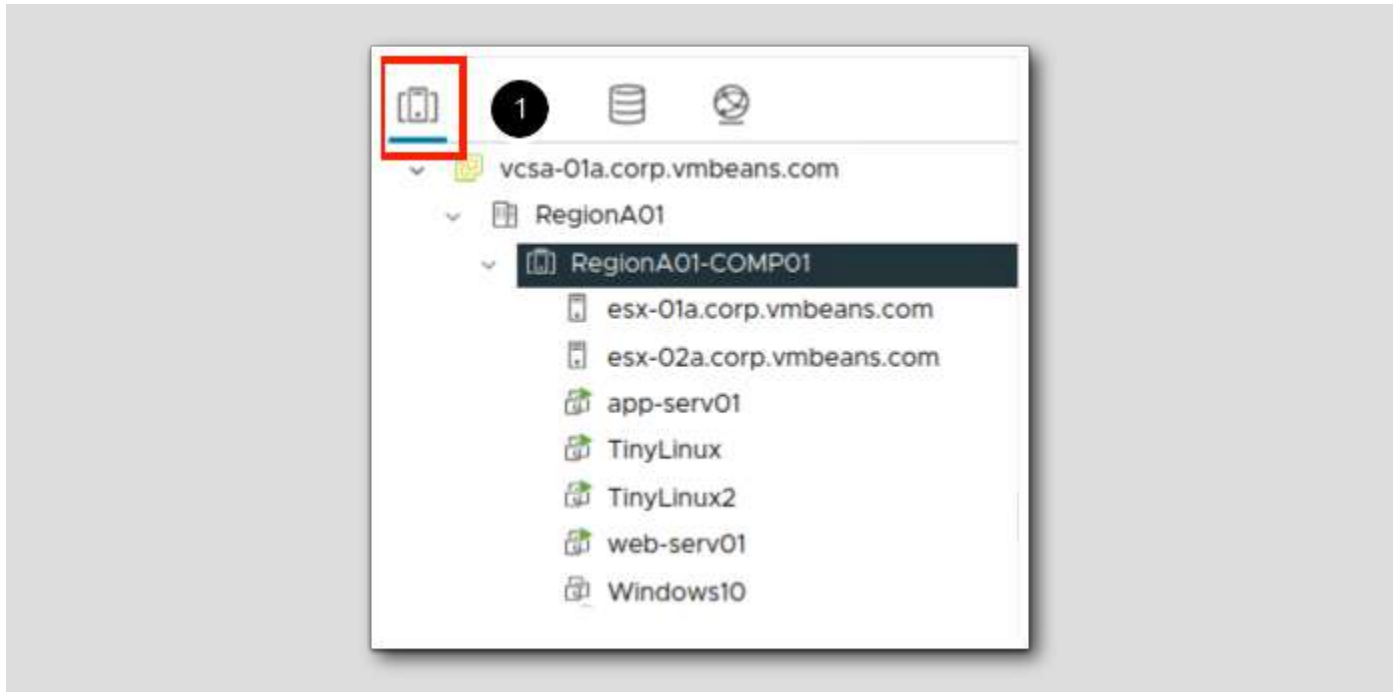
1. Log in with the following credentials:

- administrator@vsphere.local
- VMware1!

2. Click "Login"

Select Hosts and Clusters

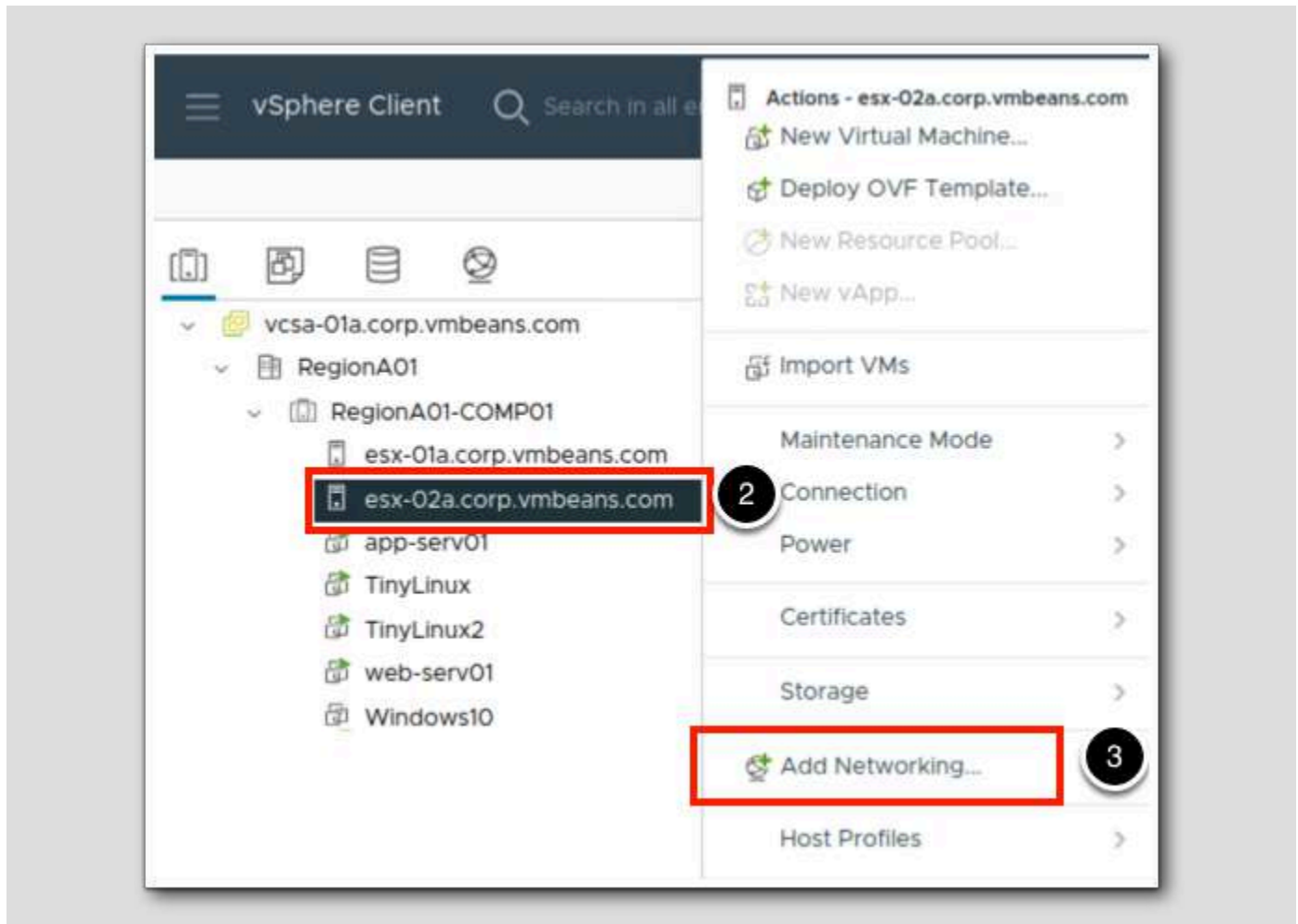
[227]



1. If you are not directed to "Hosts and Clusters", click the icon for it.

Add Networking

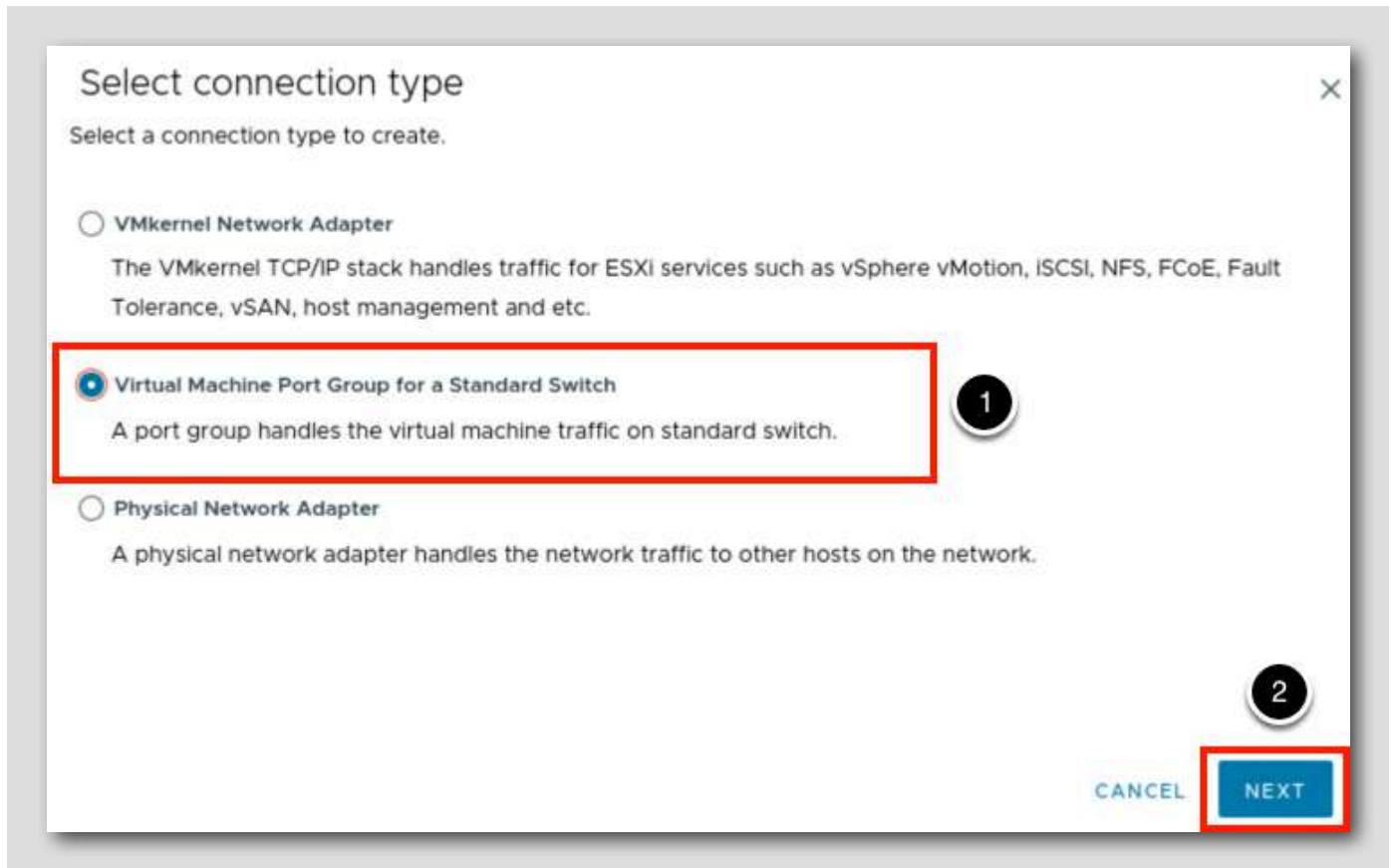
[228]



1. Under vcsa-01a.corp.vmbeans.com, expand RegionA01 and then RegionA01-COMP01.
2. Next, right-click on esx-02a.corp.vmbeans.com in the Navigator.
3. Select Add Networking....

Connection Type

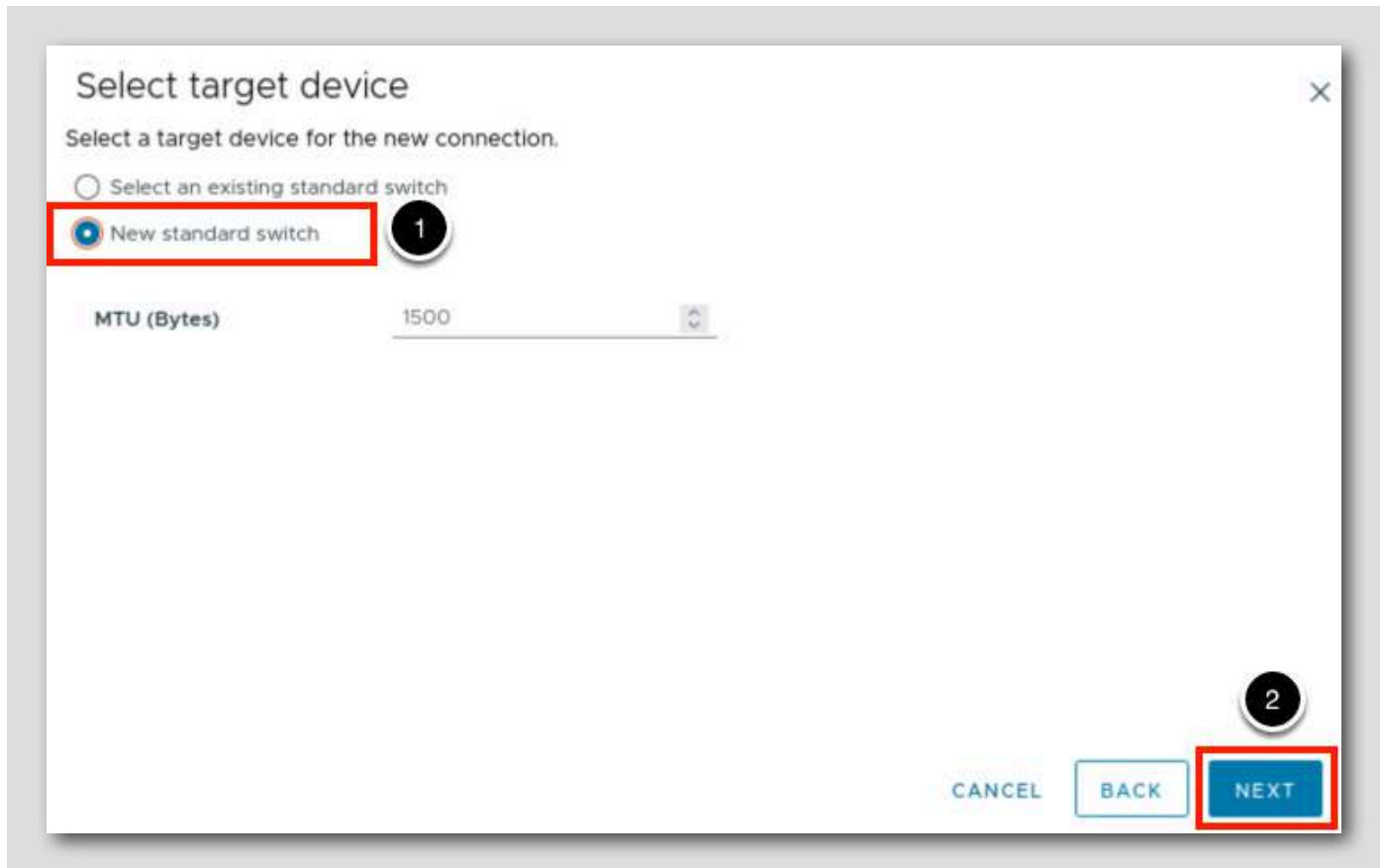
[229]



1. When asked to select connection type, choose **Virtual Machine Port Group for a Standard Switch**.
2. Click **Next**.

Target Device

[230]



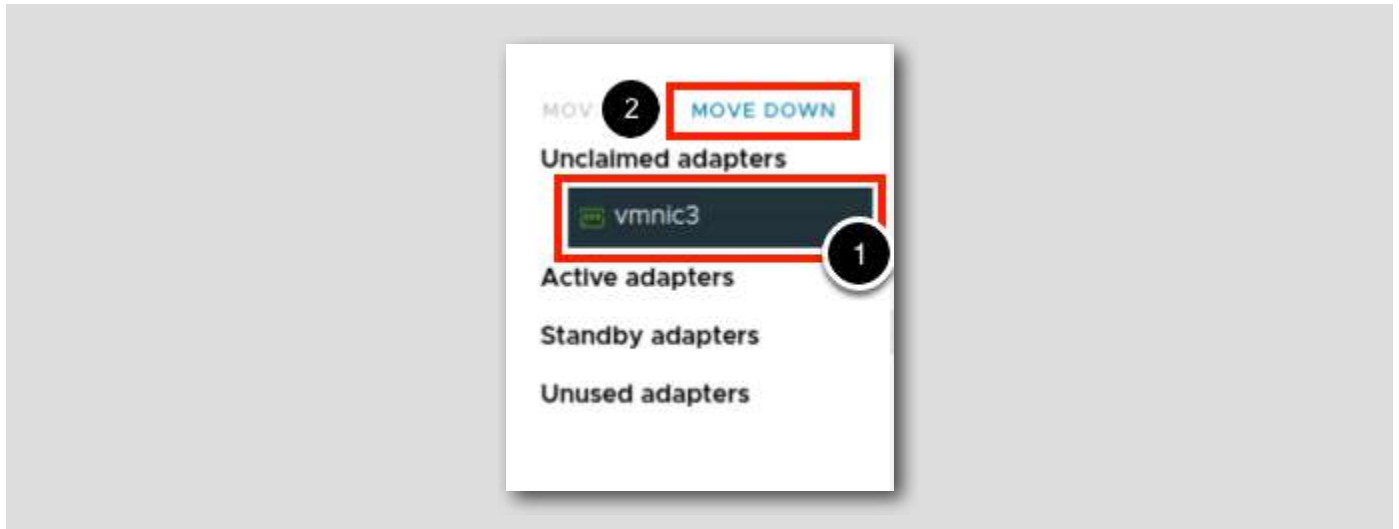
1. When asked to select a target device, choose **New Standard Switch**.

Note that a larger MTU size can be specified if needed.

2. Click **Next**.

Create a Standard Switch

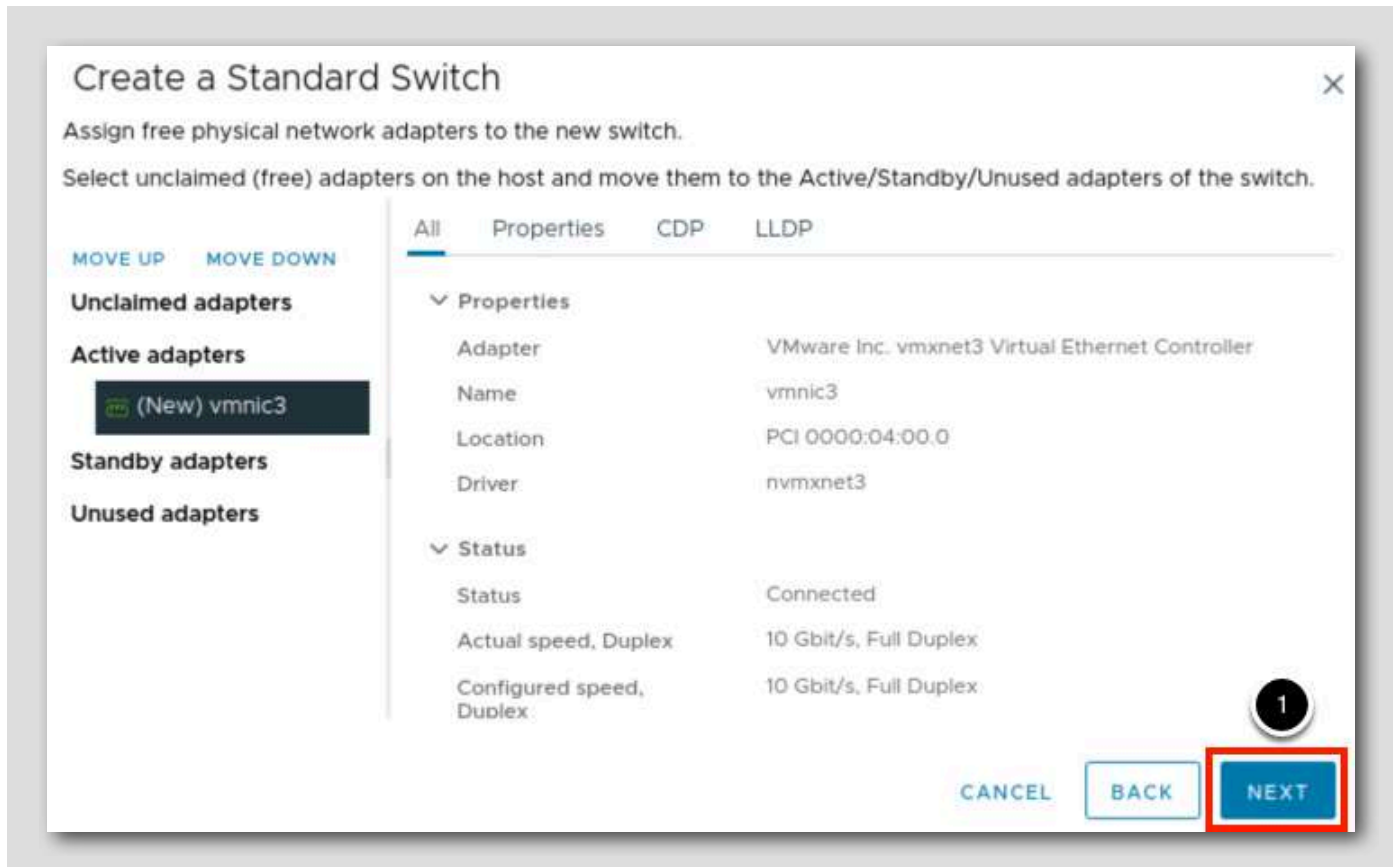
[231]



1. Select the `vmnic3`
2. Press on **Move Down** until `vmnic3` is under **Active adapters**

Add Physical Adapter

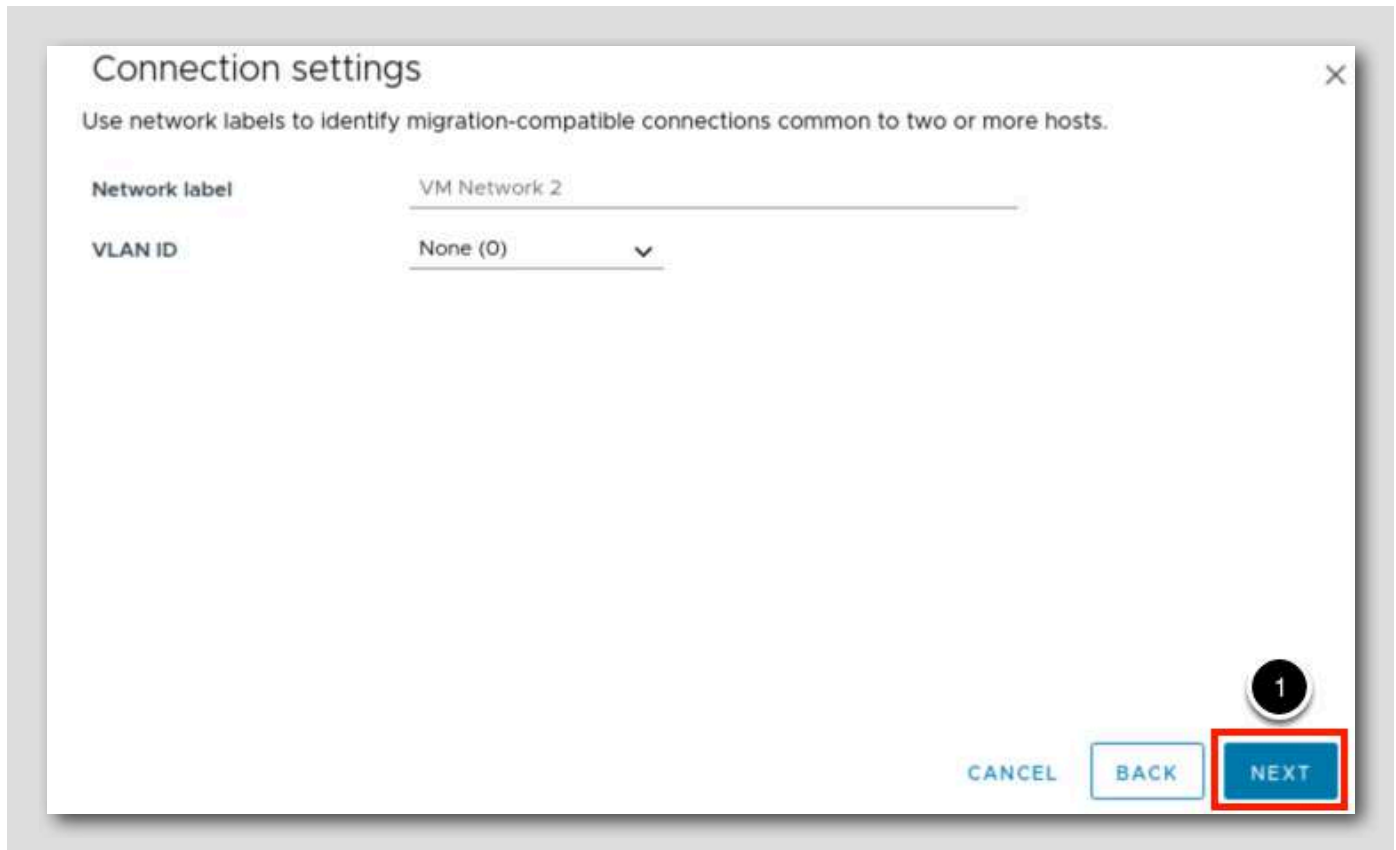
[232]



1. Click **Next** to continue.

Connection Settings

[233]



Connection settings

Use network labels to identify migration-compatible connections common to two or more hosts.

Network label VM Network 2

VLAN ID None (0) ▼

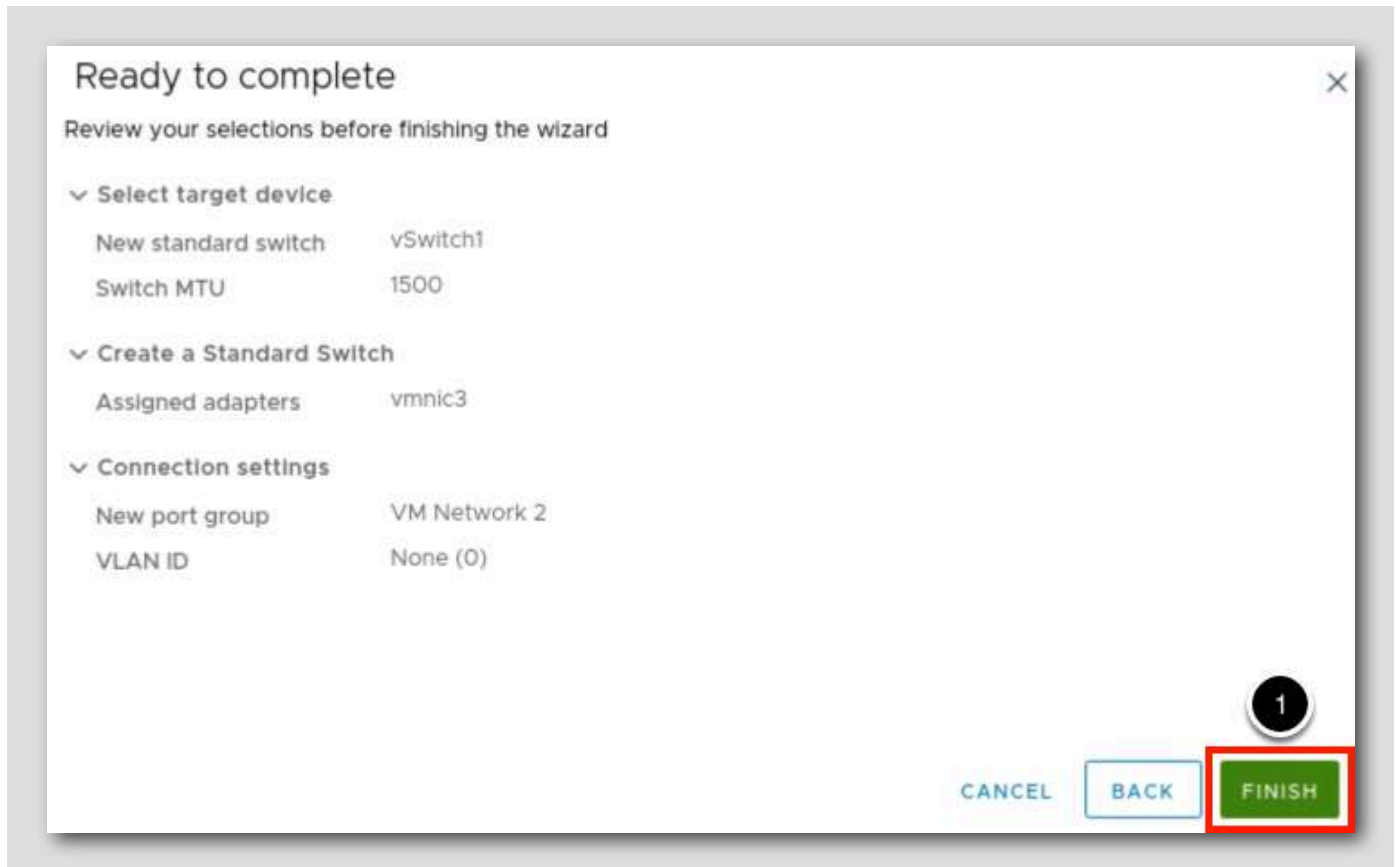
CANCEL BACK NEXT

At the Connection settings step of the wizard, for Network label, leave the default name of **VM Network 2**. Do not change the VLAN ID; leave this set to **None (0)**.

1. Click **Next**

Complete the Wizard

[234]



1. Review the port group settings in Ready to complete and click Finish.

Virtual Switches

[235]

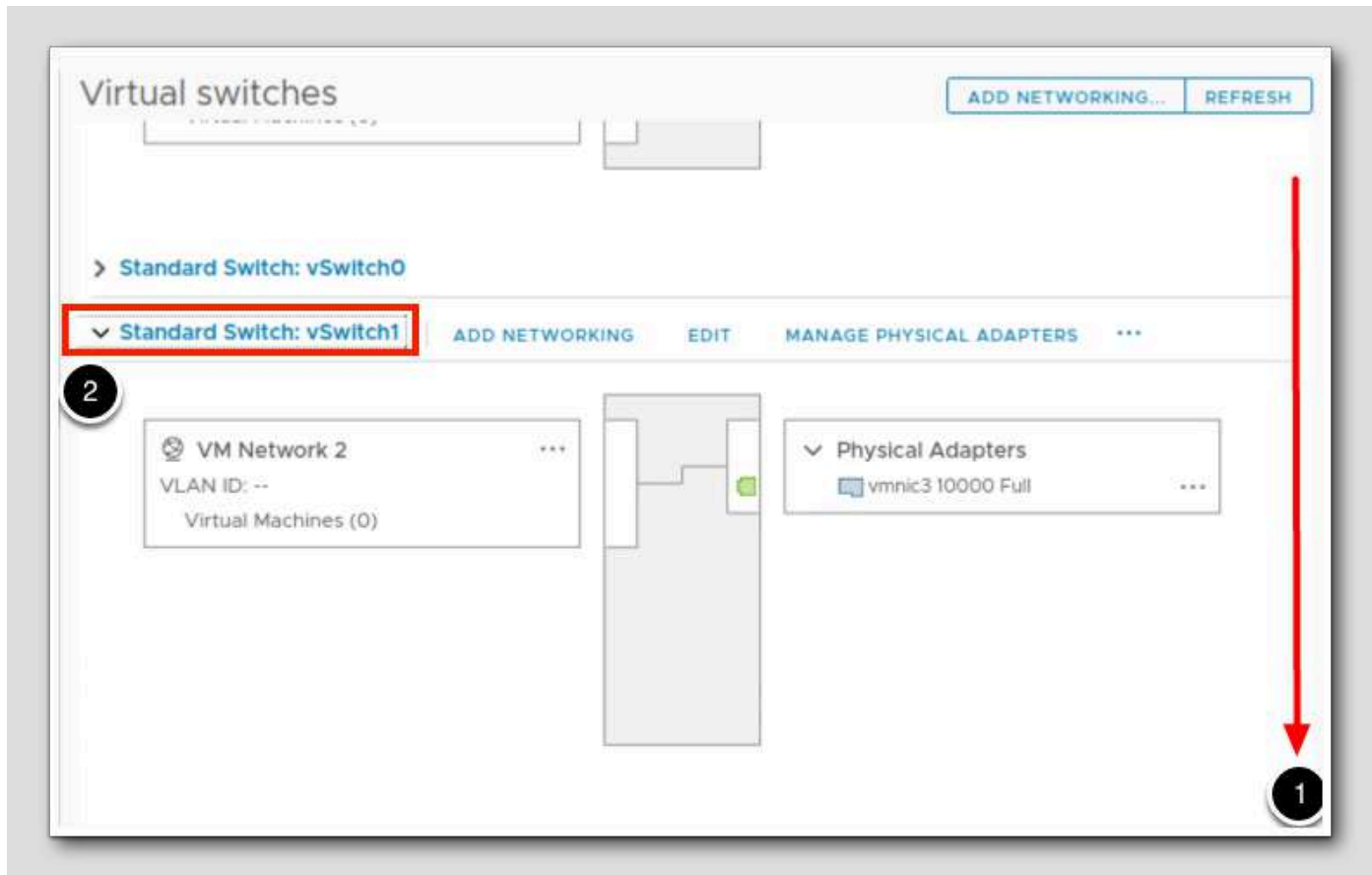


Next, we will verify the switch has been created.

1. Within the same esx-02a.corp.vmbeans.com host, click Configure.
2. Click on **Virtual Switches**.

Standard Switch: vSwitch1

[236]



1. Scroll down until you see **Standard Switch: vSwitch1**.
2. If needed, **expand** the section.

You should see the above diagram showing a virtual port group (VM Network 2) that is on vSwitch1 and it is using vmnic3 as an uplink.

Editing a Standard Switch in vSphere Client

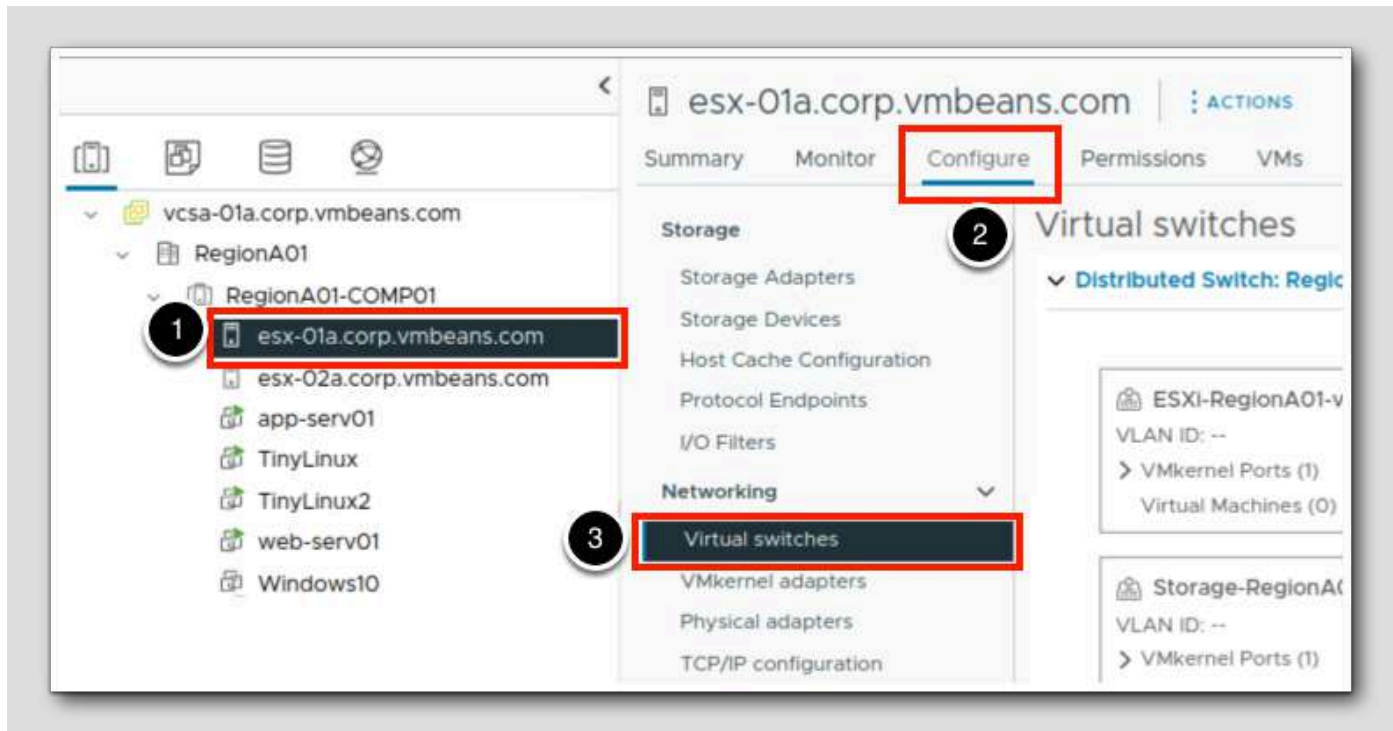
[237]

In this lesson, we will review the various properties of a Standard Switch.

vSphere Standard Switch settings control switch-wide defaults and switch properties such as the uplink configuration.

Select esx-01a.corp.vmbeans.com

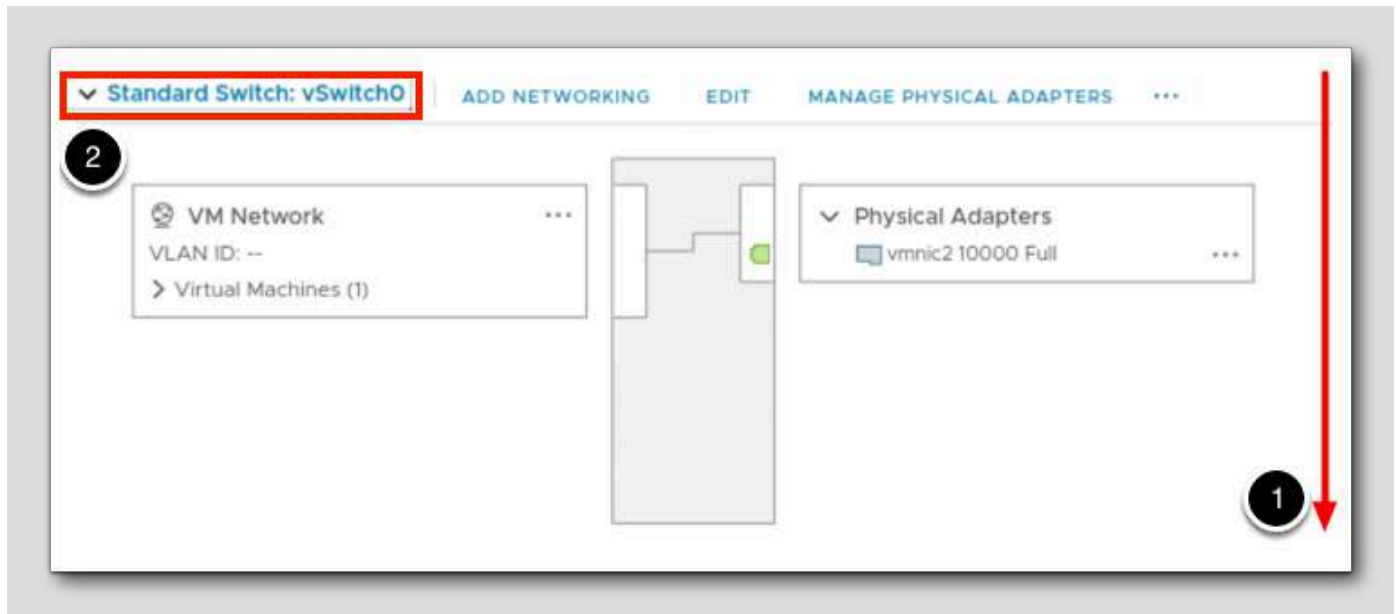
[238]



1. Select esxi-01a.corp.vmbeans.com
2. Ensure the Configure tab is selected.
3. Click Virtual switches.

Select vSwitch0

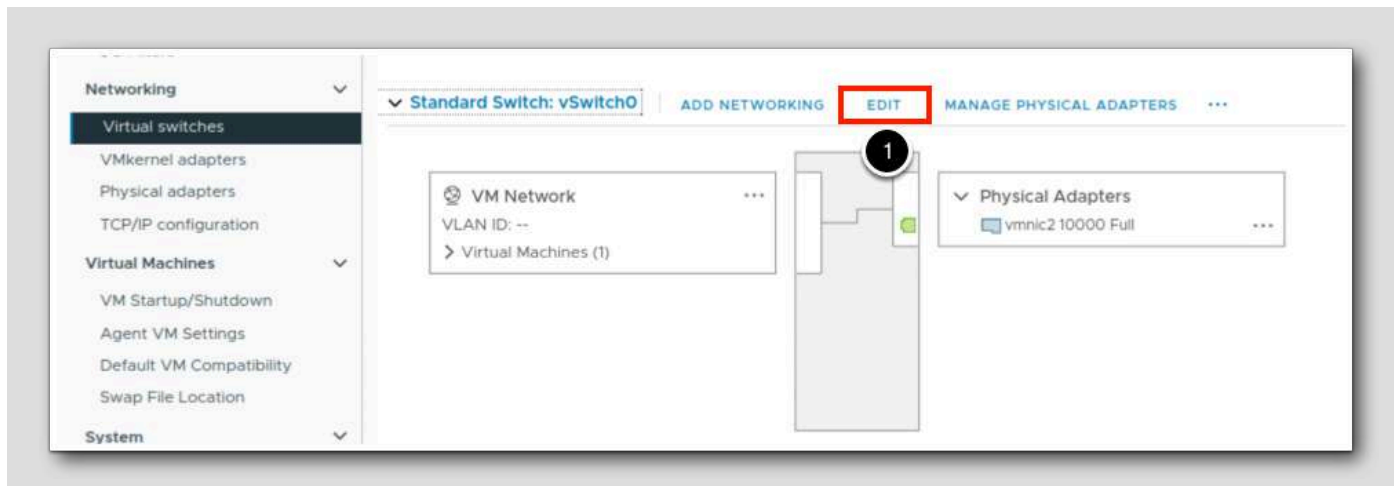
[239]



1. You will need to scroll down until you reach the **Standard Switch: vSwitch0** section.
2. Expand the section to view the layout of the switch.

Edit vSwitch1

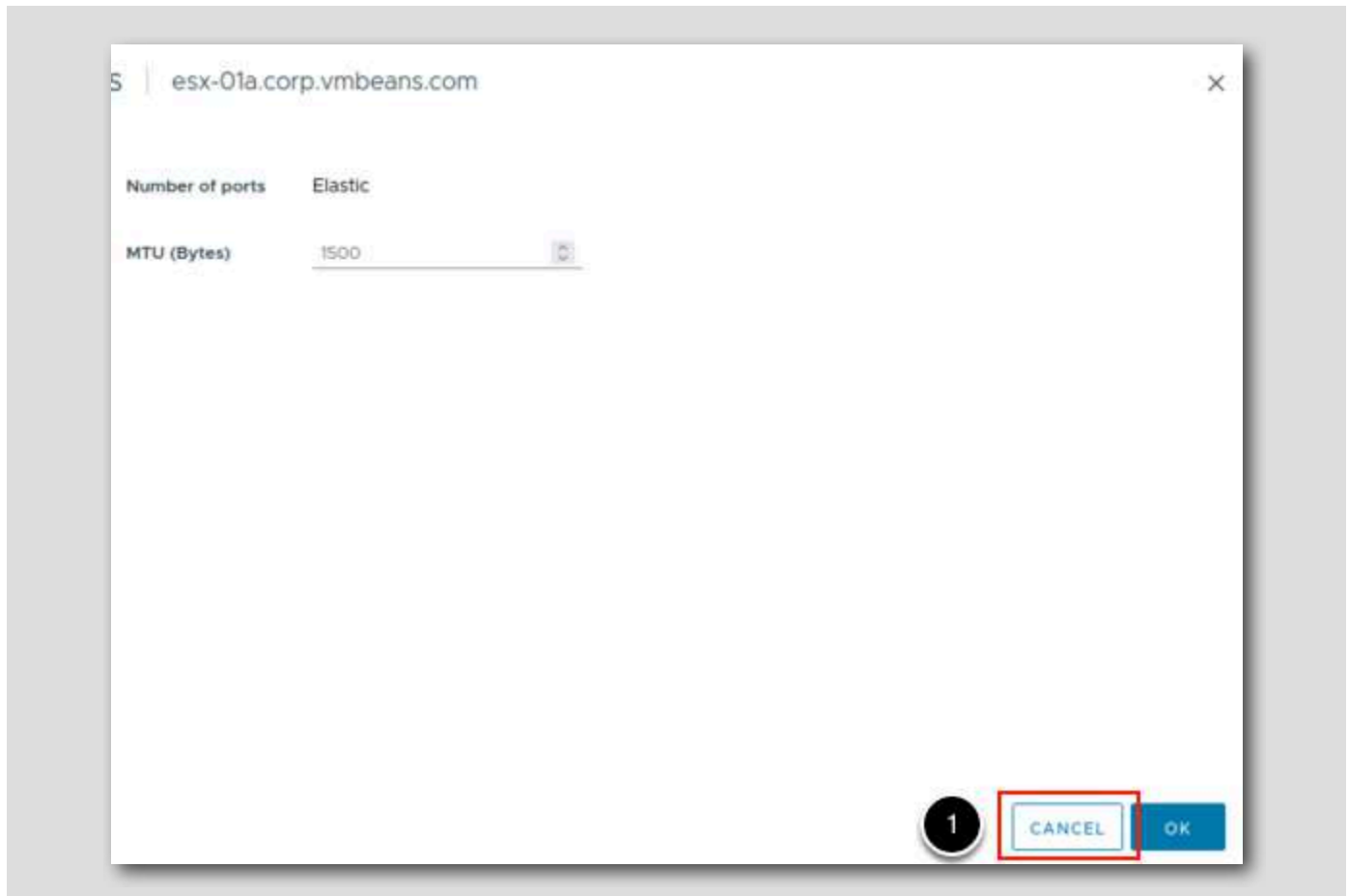
[240]



1. Click **Edit**.

Properties (MTU Setting)

[241]



If you are using jumbo frames in your environment and want to leverage this on a vSphere Standard Switch, you can change the MTU setting here.

You can change the size of the maximum transmission unit (MTU) on a vSphere Standard Switch to increase the amount of payload data transmitted with a single packet, that is, enabling jumbo frames. **Be sure to check with your Networking team prior to making any modifications here.** To realize the benefit of this setting and prevent performance issues, compatible MTU settings are required across all virtual and physical switches and end devices such as hosts and storage arrays.

You will also notice the Security, Traffic shaping, and Team and Failover options. This is where the default settings for the virtual switch would be set. As you will see later, these defaults may be overridden at the port group level as required.

1. Click the **Cancel** button.

Next, an additional uplink will be added to the switch and the other options will be reviewed.

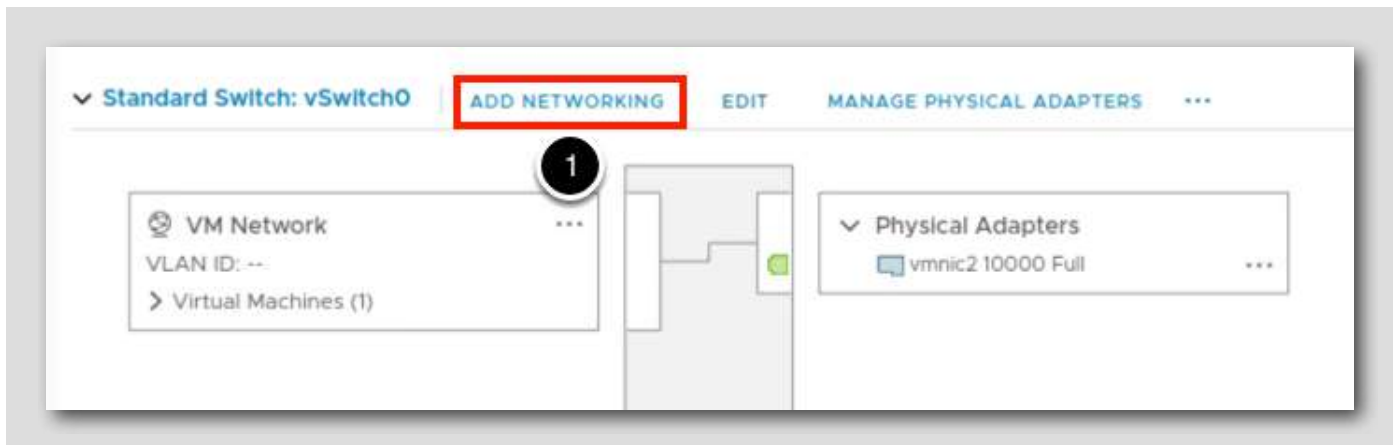
Add Uplink Adapters in vSphere Client

[242]

You can associate multiple adapters to a single vSphere standard switch to increase throughput and provide redundancy should a link fail. This is known as "NIC Teaming."

Select Virtual switches

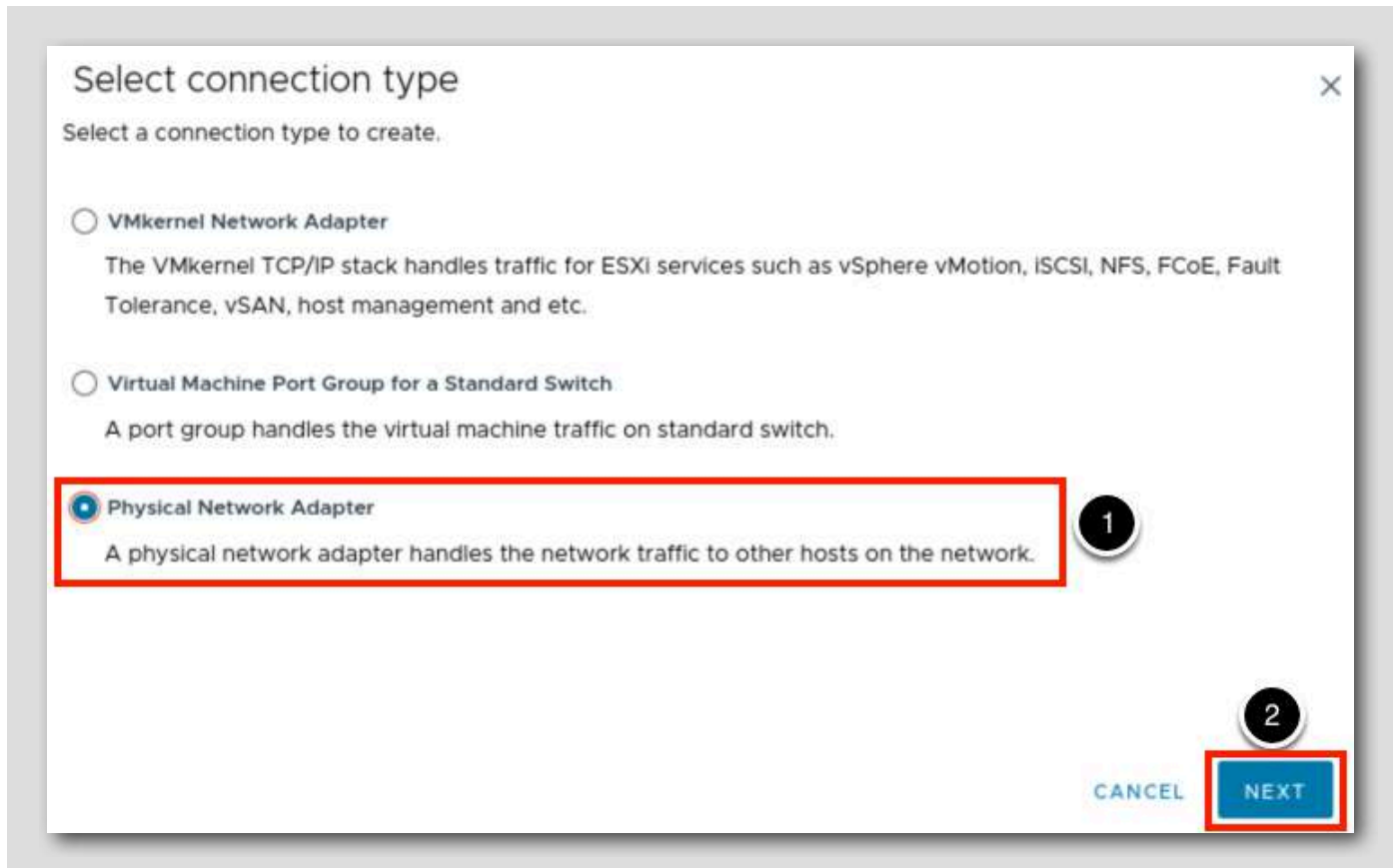
[243]



1. Click Add Networking

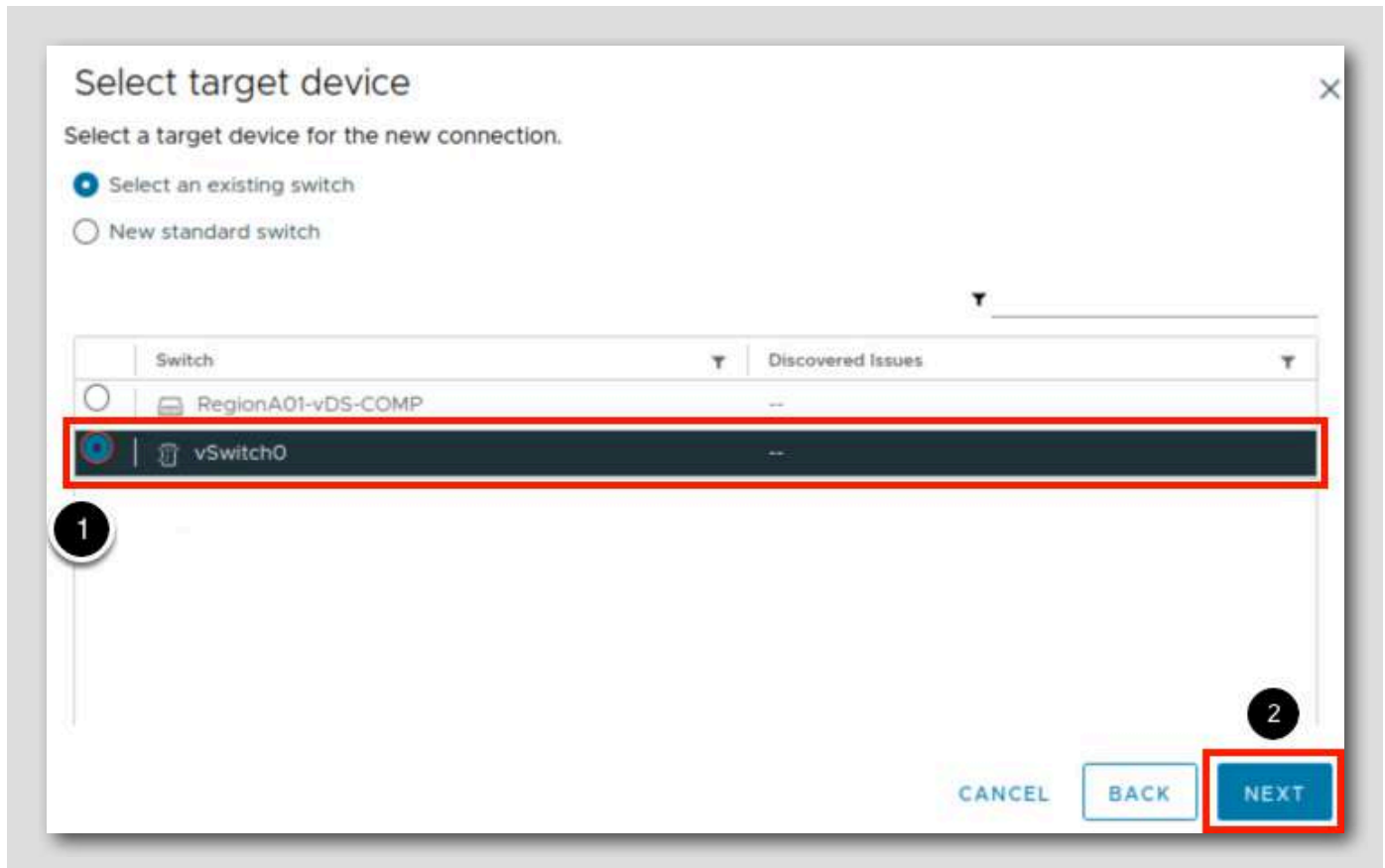
Select Connection Type

[244]



1. Select Physical Network Adapter.
2. Click Next.

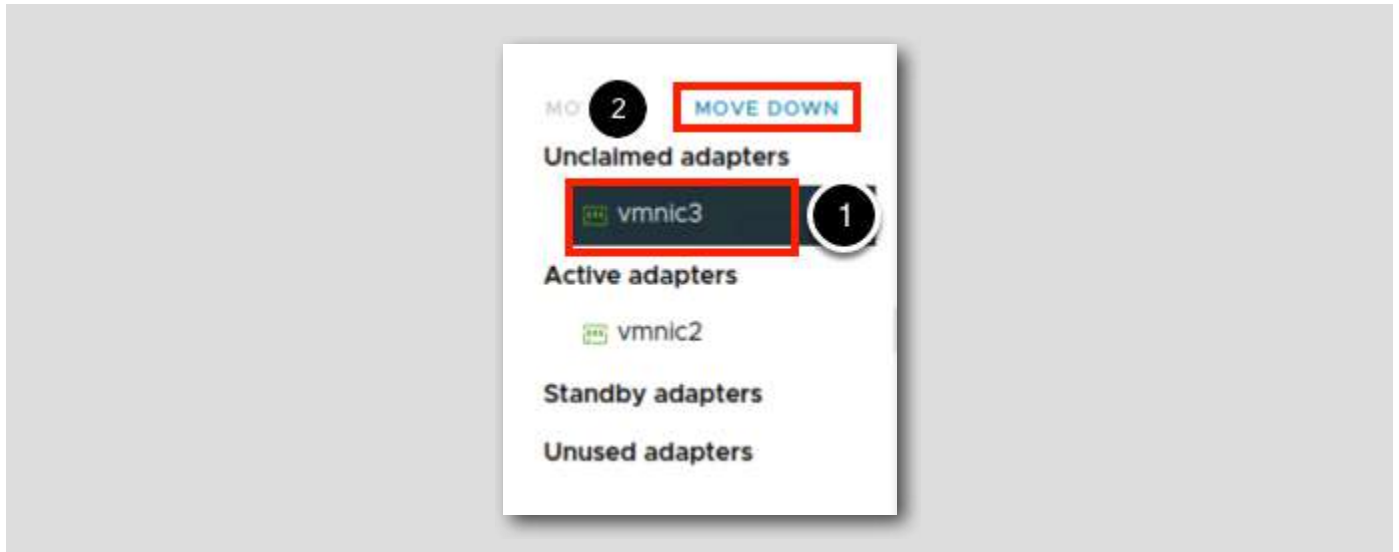
Select Target Device



1. Select vSwitch0 as the existing switch for the new connection
2. Click **Next**.

Add Networking

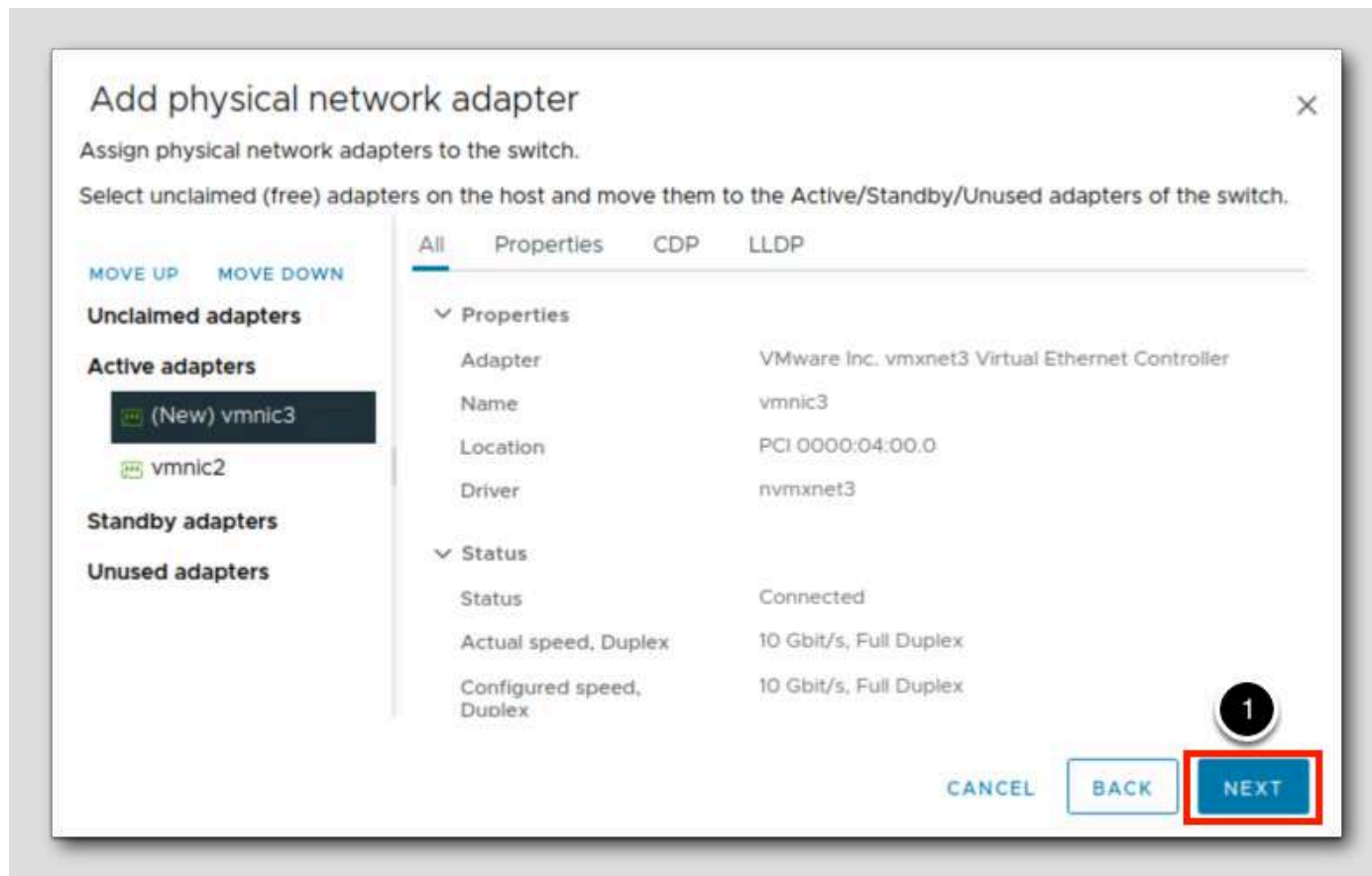
[246]



1. Select vmnic3
2. Click on **Move Down** until vmnic3 is under **Active adapters**

Assigned Adapters

[247]

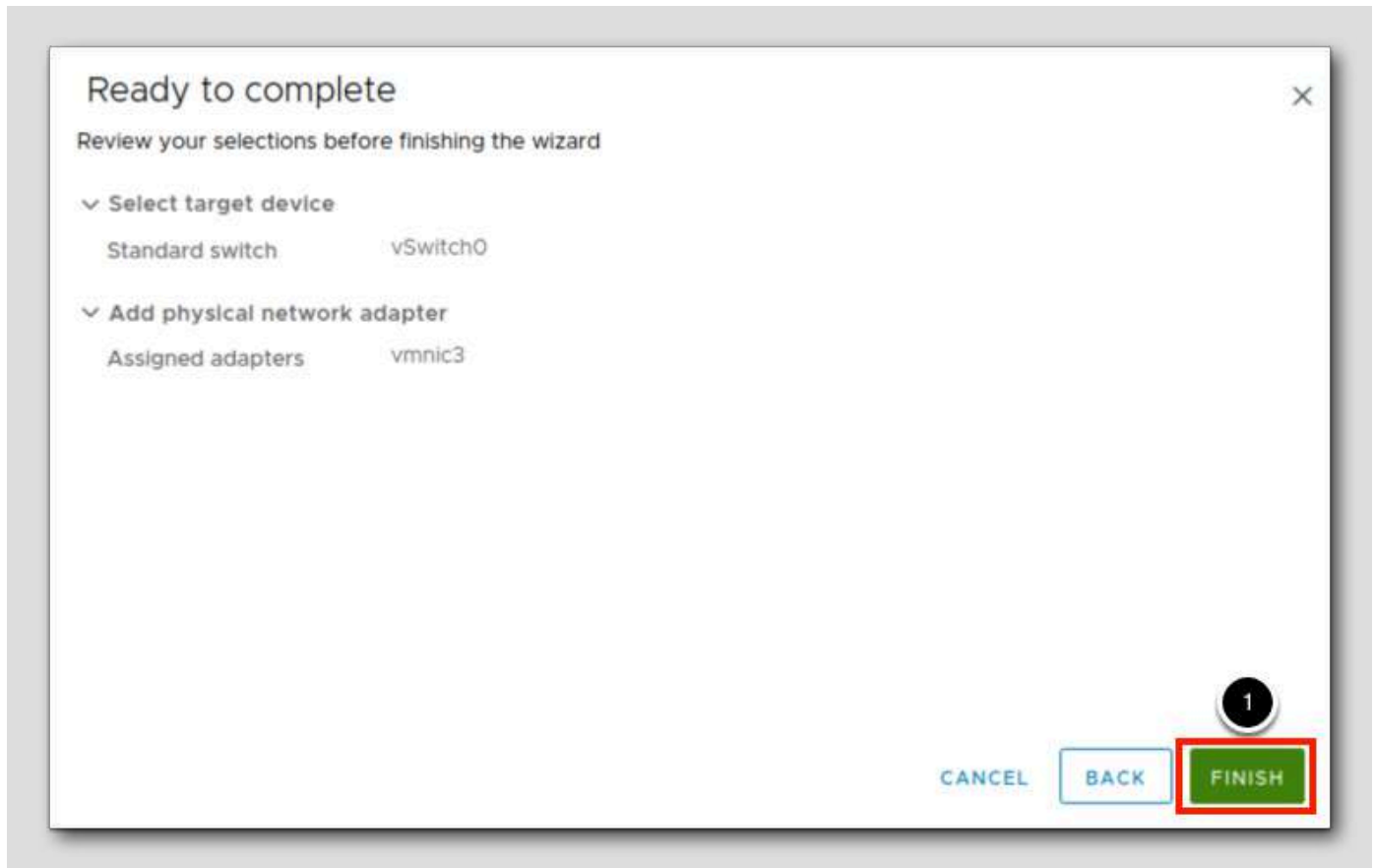


The new adapter has been added in the Active Adapters section. An adapter could also be moved to the Standby Adapters section to be used for failover. The Unused Adapters section can be used when there are multiple portgroups on a switch and you would like the ability to control what traffic flows through which physical adapter. It can be used to segment traffic or be used for individual VLAN traffic.

1. Click **Next**.

Ready to Complete

[248]



1. Click **Finish** to add vmnic3 to vSwitch0.

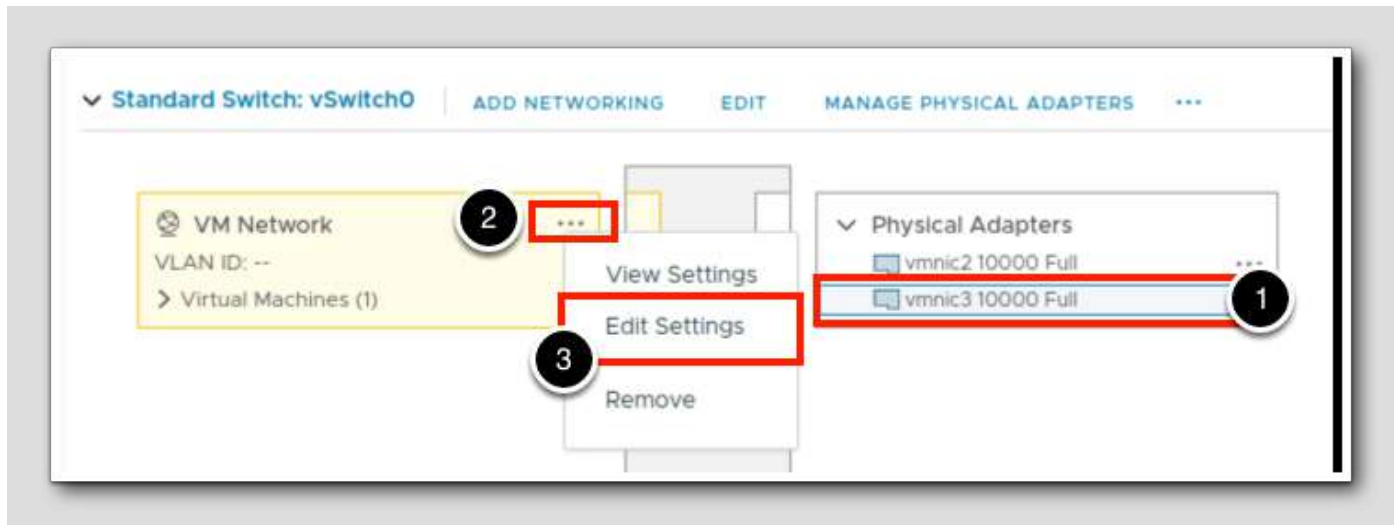
Editing a Standard Switch Port Group

[249]

Once the vSwitch has been configured and its defaults have been set, the port group can be configured. The port group is the construct that is connected to virtual machine NICs and usually represents a VLAN or physical network partition such as Production, Development, Desktop or DMZ.

New vmnic Added

[250]



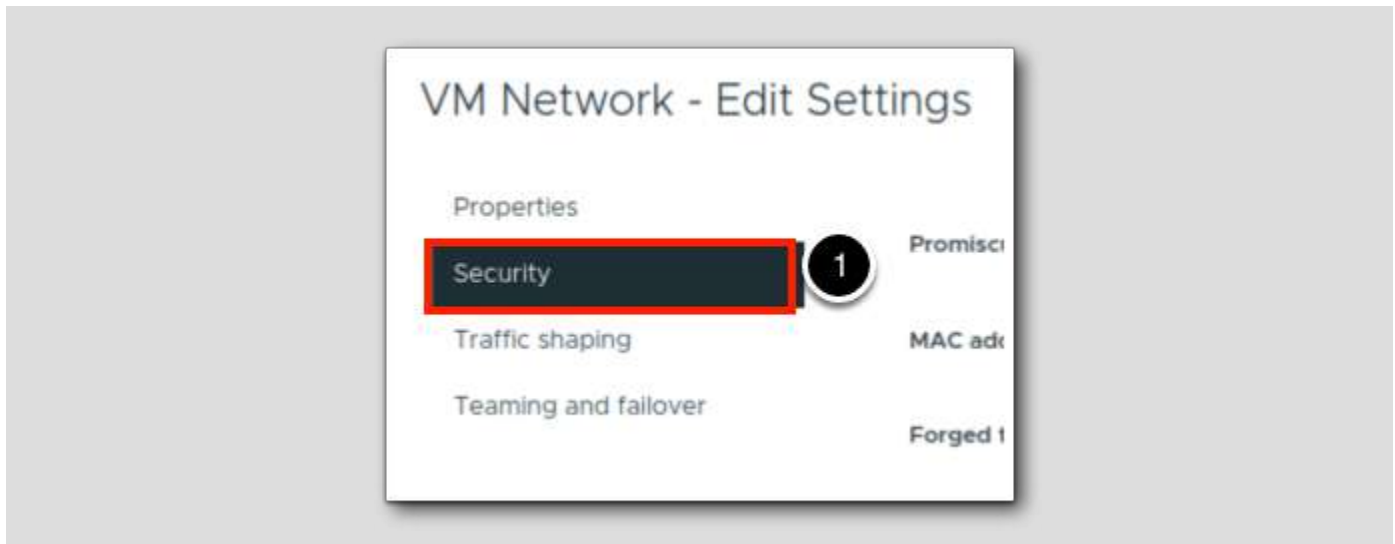
1. In the Physical Adapters section, vmnic3 has been added to the switch.

Now we will look at some of the options that can be selected at the port group level of a Standard Switch.

2. Click on the drop-down menu for the **VM Network** port group.
3. Select **Edit Settings**.

Port Group Properties

[251]



The Properties setting section is where the name or VLAN ID of the port group can be modified.

There is no need to modify these settings for this part of the lab.

1. Click Security.

Security Settings

By ticking the Override box, you can override the default setting of the Standard Switch for just this port group.

In this section, you can configure the following:

Promiscuous Mode

- Reject — Placing a guest adapter in promiscuous mode has no effect on which frames are received by the adapter.
- Accept — Placing a guest adapter in promiscuous mode causes it to detect all frames passed on the vSphere standard switch that are allowed under the VLAN policy for the port group that the adapter is connected to.

MAC Address Changes

- Reject — If you set the MAC Address Changes to Reject and the guest operating system changes the MAC address of the adapter to anything other than what is in the .vmx configuration file, all inbound frames are dropped. If the Guest OS changes the MAC address back to match the MAC address in the .vmx configuration file, inbound frames are passed again.
- Accept — Changing the MAC address from the Guest OS has the intended effect: frames sent to the altered MAC address are received by the virtual machine.

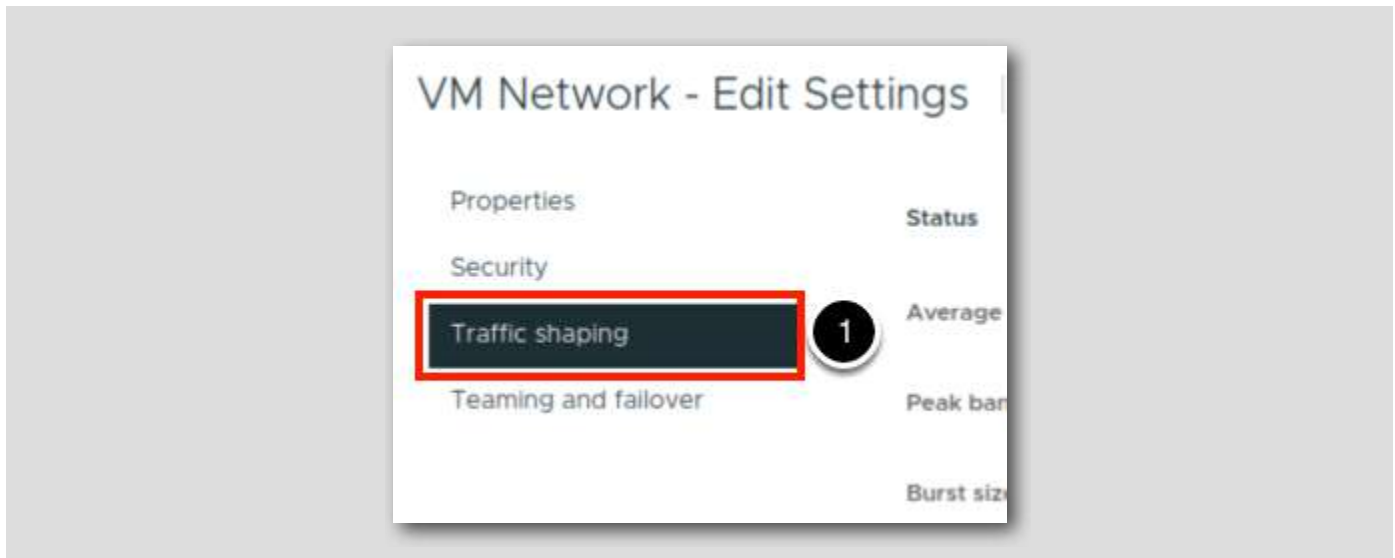
Forged Transmits

- Reject — Any outbound frame with a source MAC address that is different from the one currently set on the adapter are dropped.
- Accept — No filtering is performed and all outbound frames are passed.

No changes are needed here.

Port Group Traffic Shaping

[253]



1. Click Traffic shaping.

Traffic Shaping

[254]



Just like in the Security settings, you can override the default policy set at the switch level to apply to just this port group.

A traffic shaping policy is defined by average bandwidth, peak bandwidth, and burst size. You can establish a traffic shaping policy for each port group.

ESXi shapes outbound network traffic on standard switches. Traffic shaping restricts the network bandwidth available on a port, but can also be configured to allow bursts of traffic to flow through at higher speeds.

Average Bandwidth

- Establishes the number of bits per second to allow across a port, averaged over time. This number is the allowed average load.

Peak Bandwidth

- Maximum number of bits per second to allow across a port when it is sending or receiving a burst of traffic. This number limits the bandwidth that a port uses when it is using its burst bonus.

Burst Size

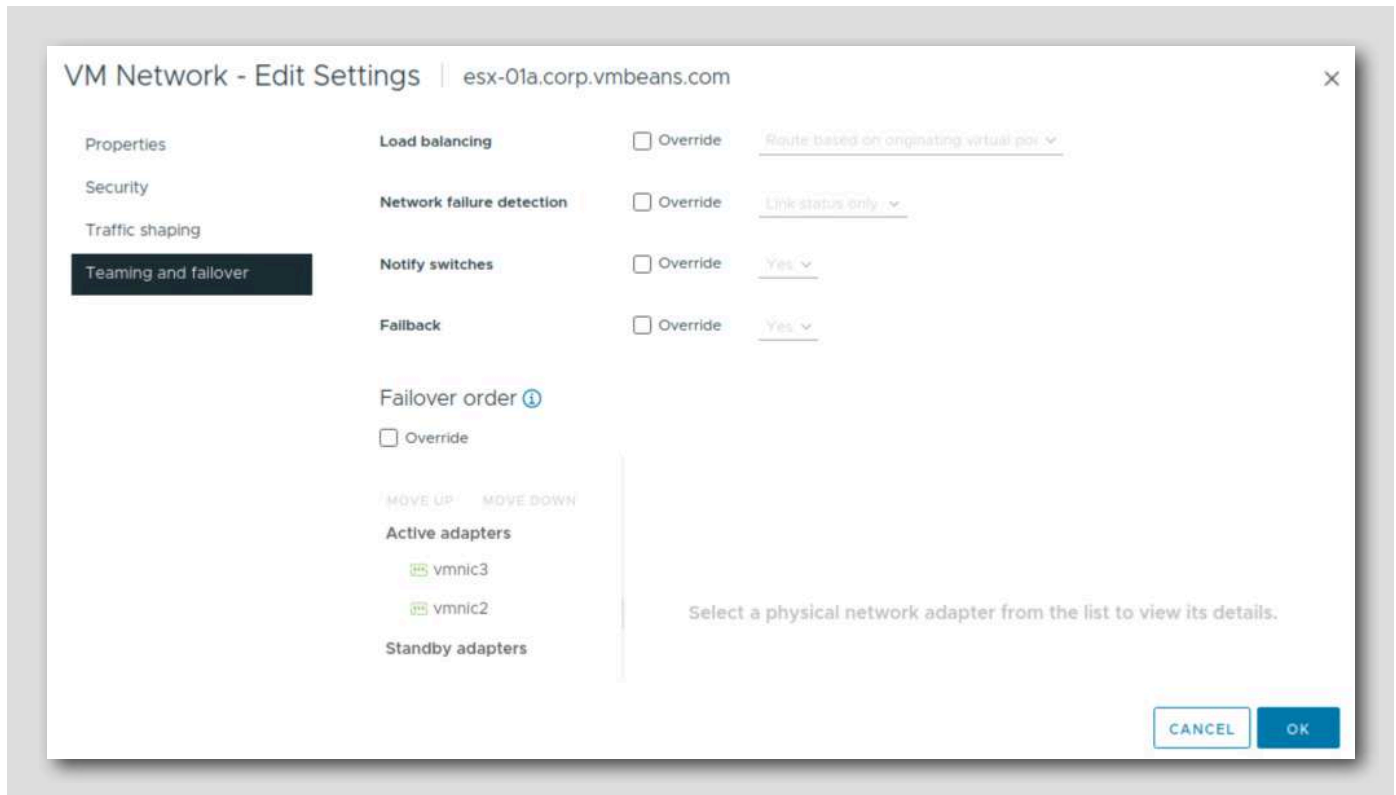
- Maximum number of bytes to allow in a burst. If this parameter is set, a port might gain a burst bonus if it does not use all its allocated bandwidth. When the port needs more bandwidth than specified by the average bandwidth, it might be allowed to temporarily transmit data at a higher speed if a burst bonus is available. This parameter limits the number of bytes that have accumulated in the burst bonus and transfers traffic at a higher speed.

No changes are needed here.

1. Clicking **Teaming and failover**.

Teaming and Failover

[255]



Again, we have the option to override the default virtual switch settings.

Load Balancing Policy - The Load Balancing policy determines how network traffic is distributed between the network adapters in a NIC team. vSphere virtual switches load balance only the outgoing traffic. Incoming traffic is controlled by the load balancing policy on the physical switch.

- Route based on the originating virtual port - Select an uplink based on the virtual port IDs on the switch. After the virtual switch selects an uplink for a virtual machine or a VMkernel adapter, it always forwards traffic through the same uplink for this virtual machine or VMkernel adapter.
- Route based on IP hash - Select an uplink based on a hash of the source and destination IP addresses of each packet. For non-IP packets, the switch uses the data at those fields to compute the hash. IP-based teaming requires that the physical switch is configured with EtherChannel.
- Route based on source MAC hash - Select an uplink based on a hash of the source Ethernet.
- Route based on physical NIC load - Available for distributed port groups or distributed ports. Select an uplink based on the current load of the physical network adapters connected to the port group or port. If an uplink remains busy at 75 percent or higher for 30 seconds, the host proxy switch moves a part of the virtual machine traffic to a physical adapter that has free capacity.
- Use explicit failover order - From the list of active adapters, always use the highest order uplink that passes failover detection criteria. No actual load balancing is performed with this option.

Network Failure Detection - The method the virtual switch will use for failover detection.

- Link Status only - Relies only on the link status that the network adapter provides. This option detects failures such as removed cables and physical switch power failures.
- Beacon Probing - Sends out and listens for beacon probes on all NICs in the team, and uses this information, in addition to link status, to determine link failure. ESXi sends beacon packets every second. The NICs must be in an active/active or active/standby configuration because the NICs in an unused state do not participate in beacon probing.

Notify Switches - specifies whether the virtual switch notifies the physical switch in case of a failover.

Failover - specifies whether a physical adapter is returned to active status after recovering from a failure.

- If failback is set to Yes, the default selection, the adapter is returned to active duty immediately upon recovery, displacing the standby adapter that took over its slot, if any.
- If failback is set to No for a standard port, a failed adapter is left inactive after recovery until another currently active adapter fails and must be replaced.

You can also override the default virtual switch setting for the Failover order of the physical adapters.

No changes are needed here and you may proceed to the next step.

Cancel the Changes

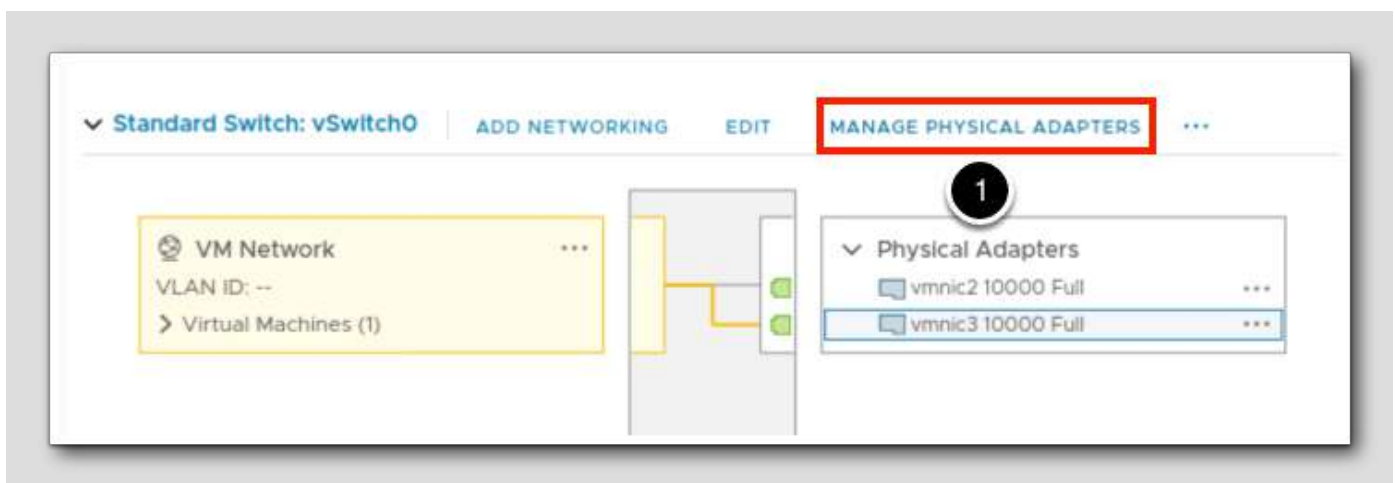
[256]



1. Since we don't want to make any changes to the port group, click the Cancel button.

Removing a Physical Adapter

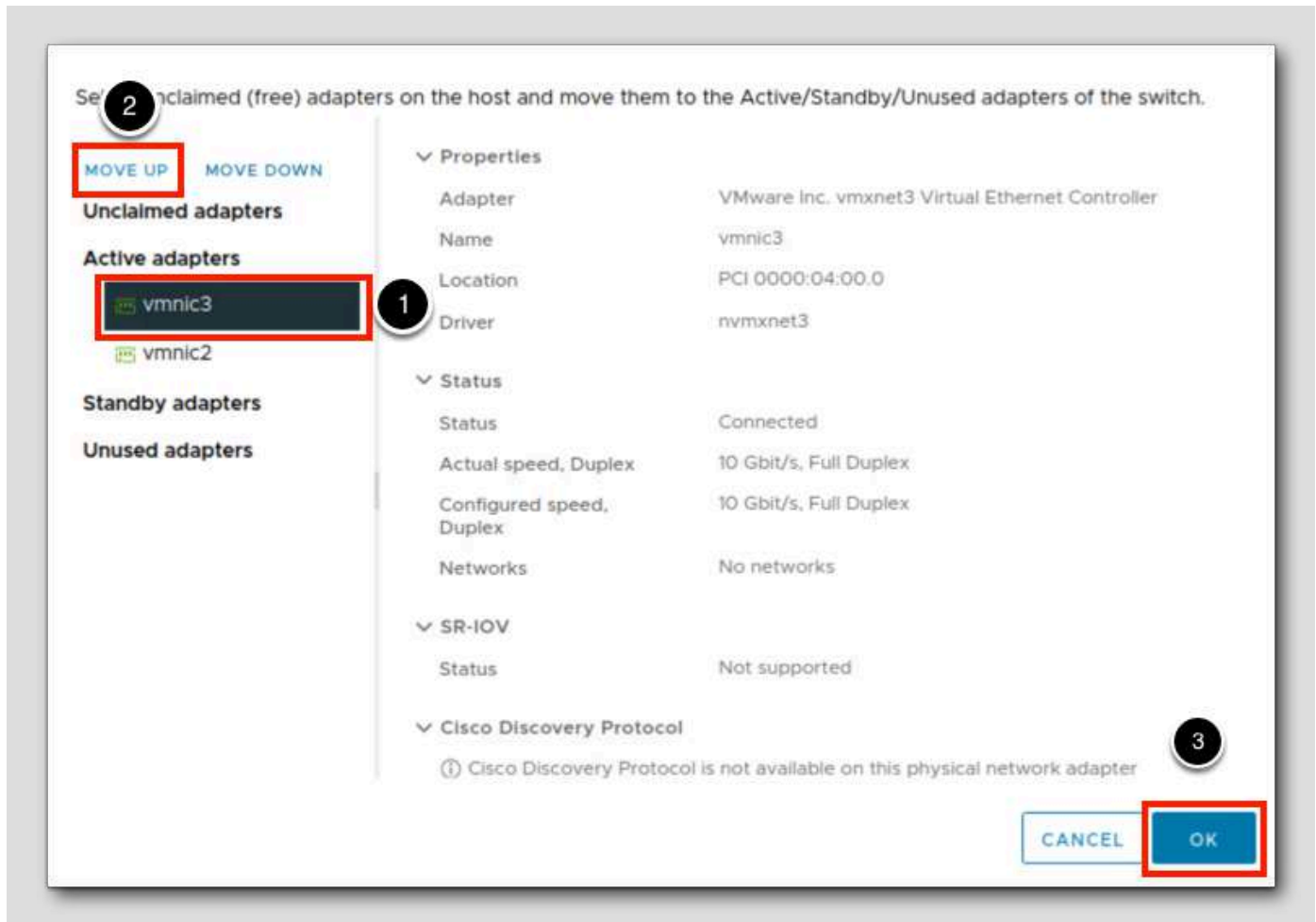
[257]



1. Click **Manage Physical Adapters** for vSwitch0.

Remove vmnic3

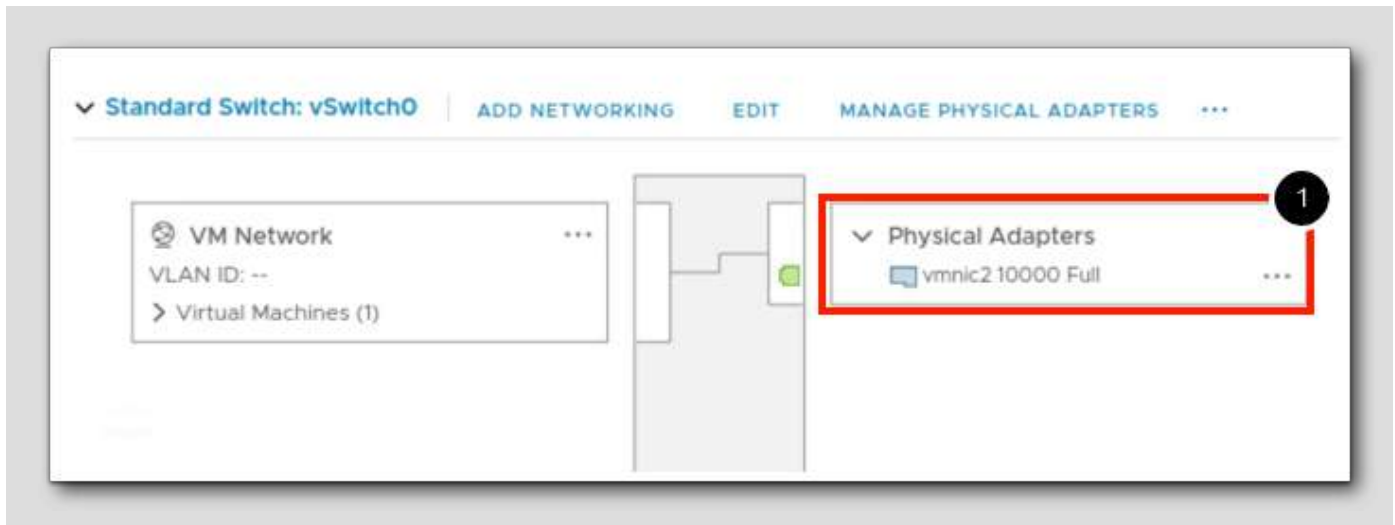
[258]



1. Click on vmnic3.
2. Click on Move Up until the adapter is under Unclaimed adapters
3. Click OK.

Adapter Removed

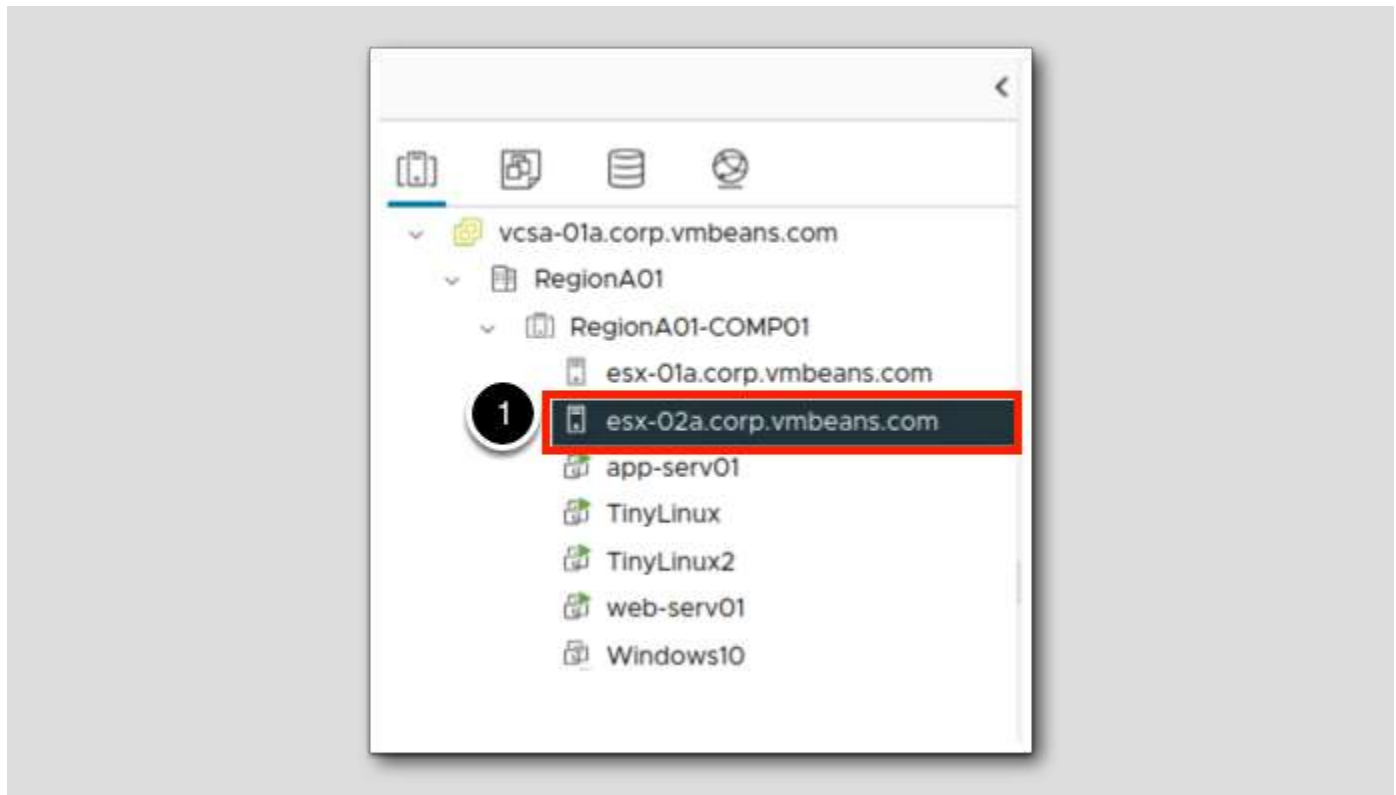
[259]



1. The adapter, `vmnic3` has been removed from the list of physical adapters.

Deleting a Standard Switch

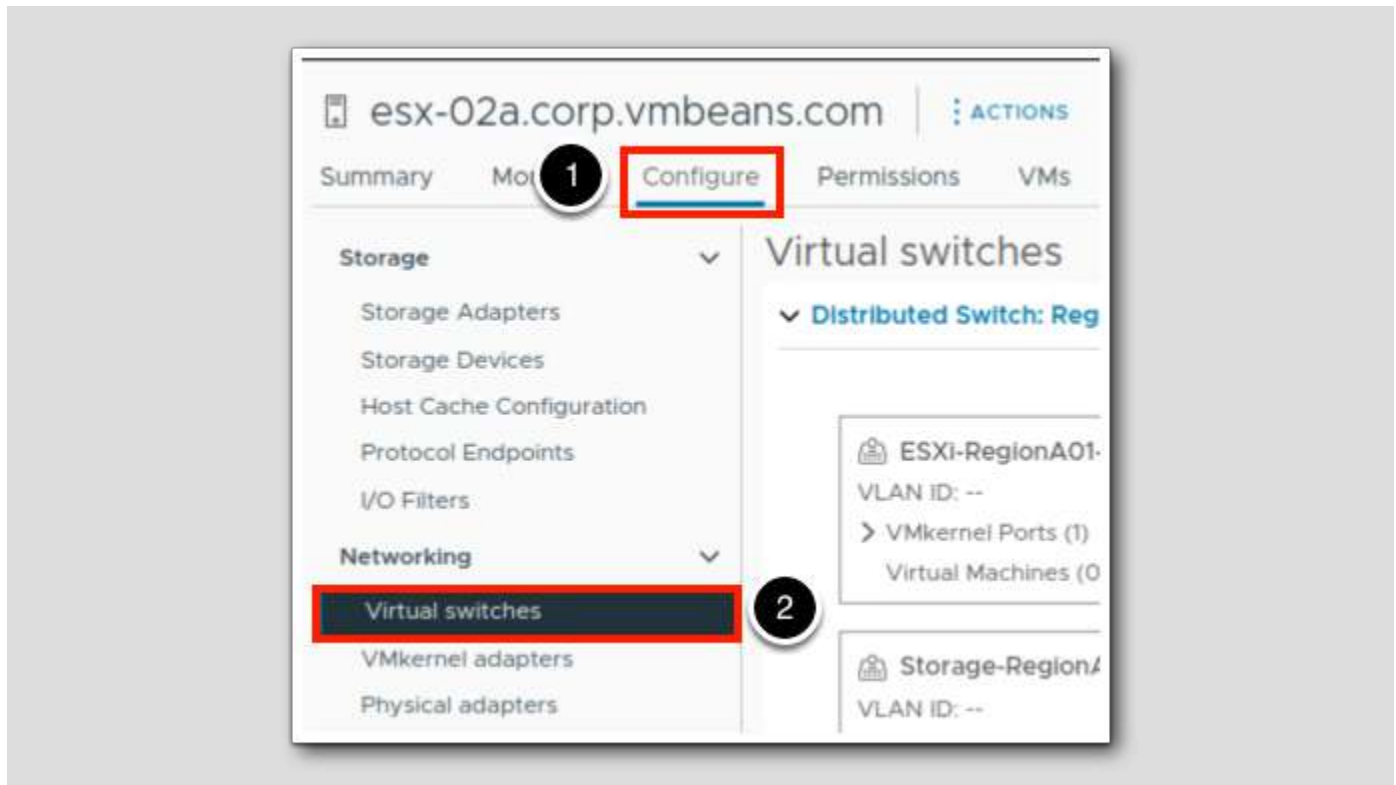
[260]



1. Click on esx-02a.corp.vmbeans.com

Virtual Switches

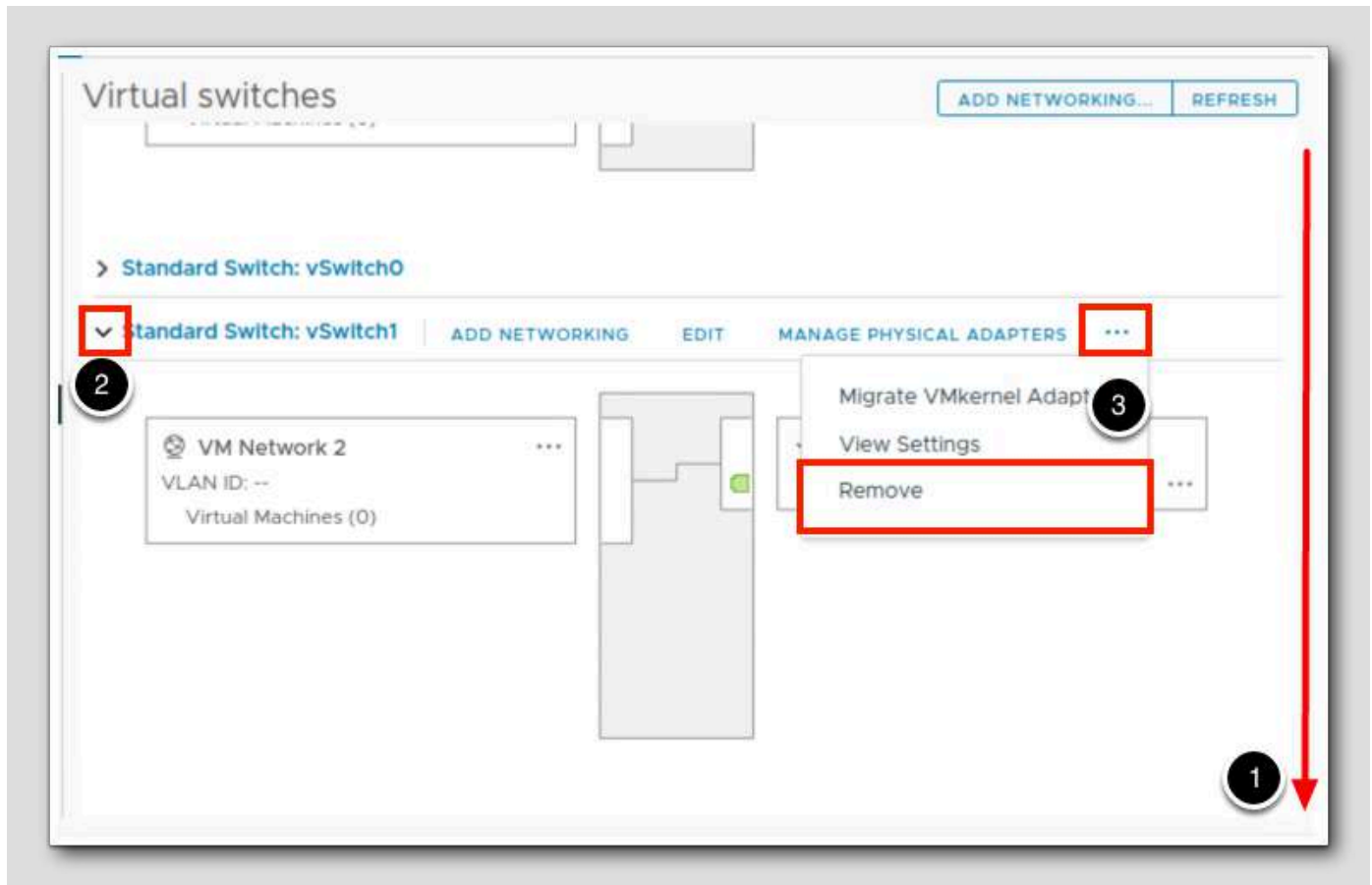
[261]



In preparation for the next lesson, we will delete the Standard Switch we created on esx-02a.corp.vmbeans.com

1. Click the **Configure** tab.
2. Select **Virtual switches** in the Networking section.

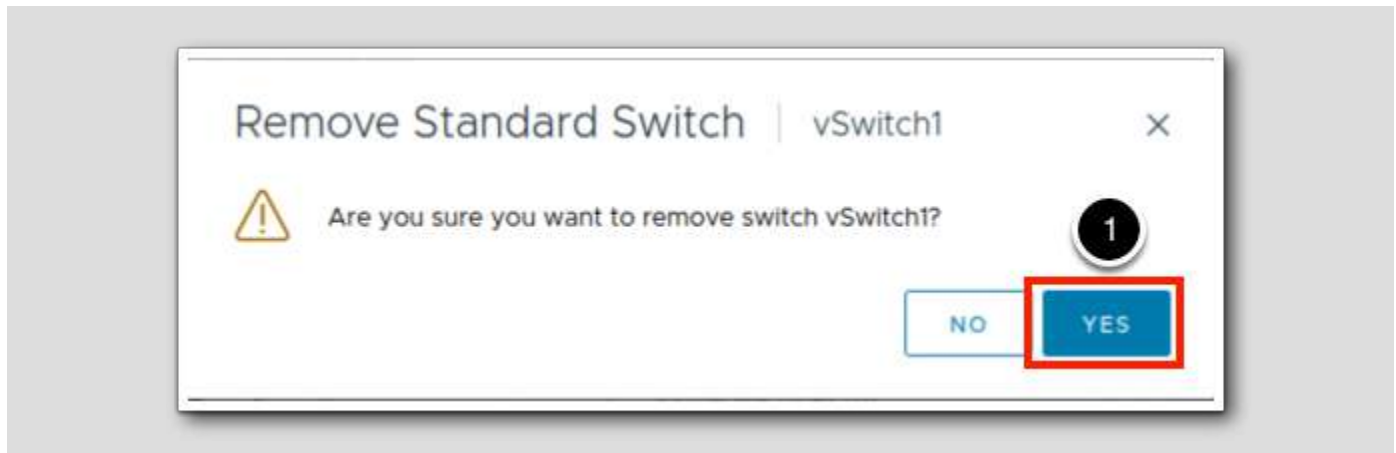
Standard Switch: vSwitch1



1. Scroll **down** until you see the Standard Switch: vSwitch1 section
2. Expand the section, if needed.
3. Click the '...' menu and select **Remove**

Remove Standard Switch

[263]



1. Click **Yes** to remove vSwitch1.

Conclusion

[264]

The vSphere Standard Switch is a simple virtual switch configured and managed at the host level. This switch provides access, traffic aggregation and fault tolerance by allowing multiple physical adapters to be bound to each virtual switch.

The VMware vSphere Distributed Switch builds on the capabilities of the vSS and simplifies management in large deployments by appearing as a single switch spanning multiple associated hosts. This allows changes to be made once and propagated to every host that is a member of the switch.

Working with the vSphere Distributed Switch

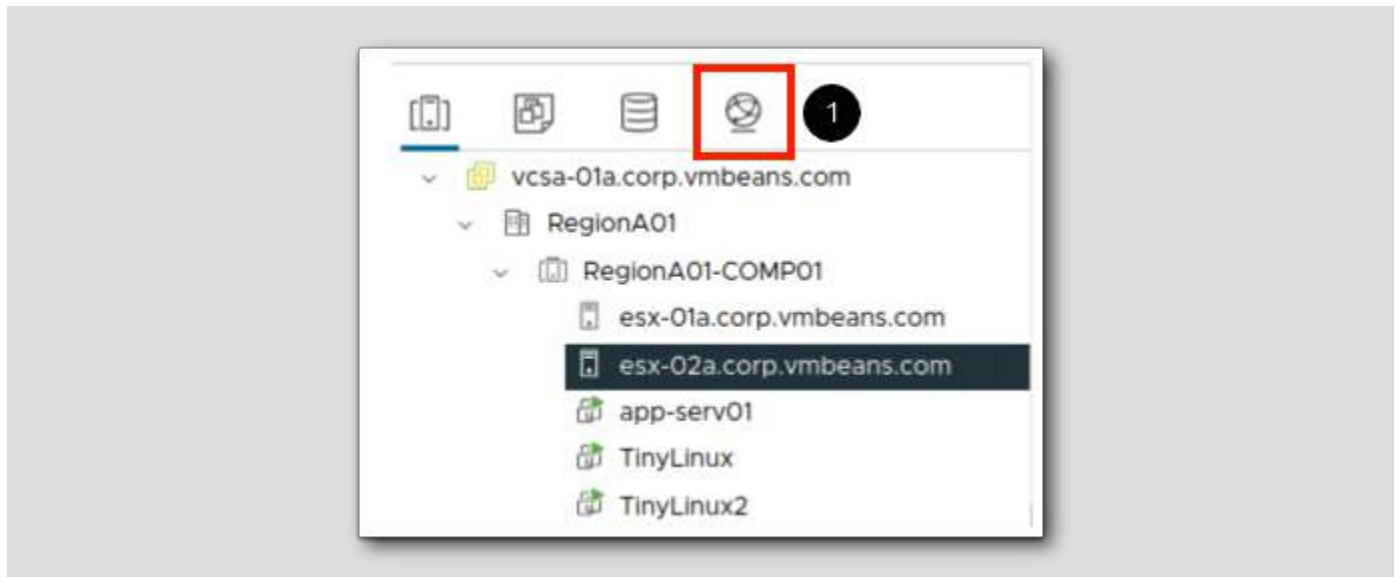
[265]

Before we walk through the process of building our own Distributed vSwitch, let's take a minute to explore an existing vDS.

In this lab we will see how a Distributed vSwitch compares to a Standard vSwitch, how it is configured, and how it is connected to a running virtual machine.

Navigate to Networking

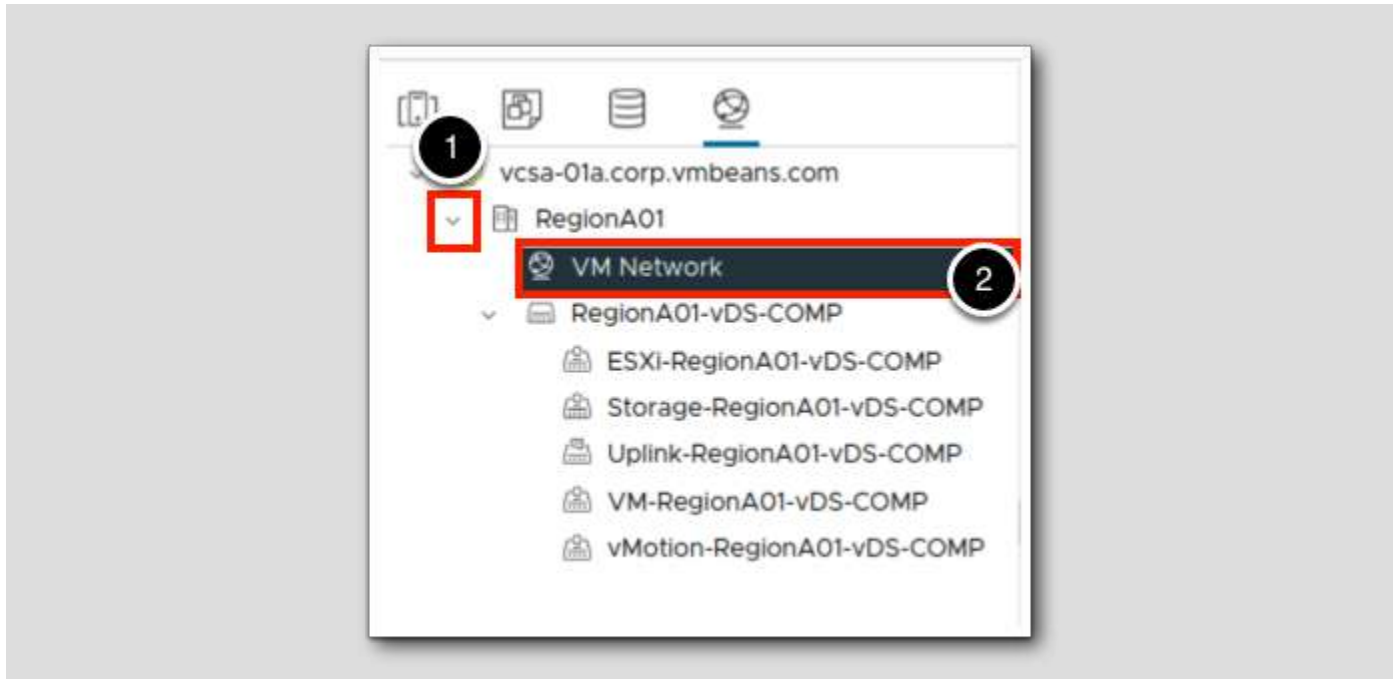
[266]



1. Click on the **Networking** icon

View Standard vSwitch

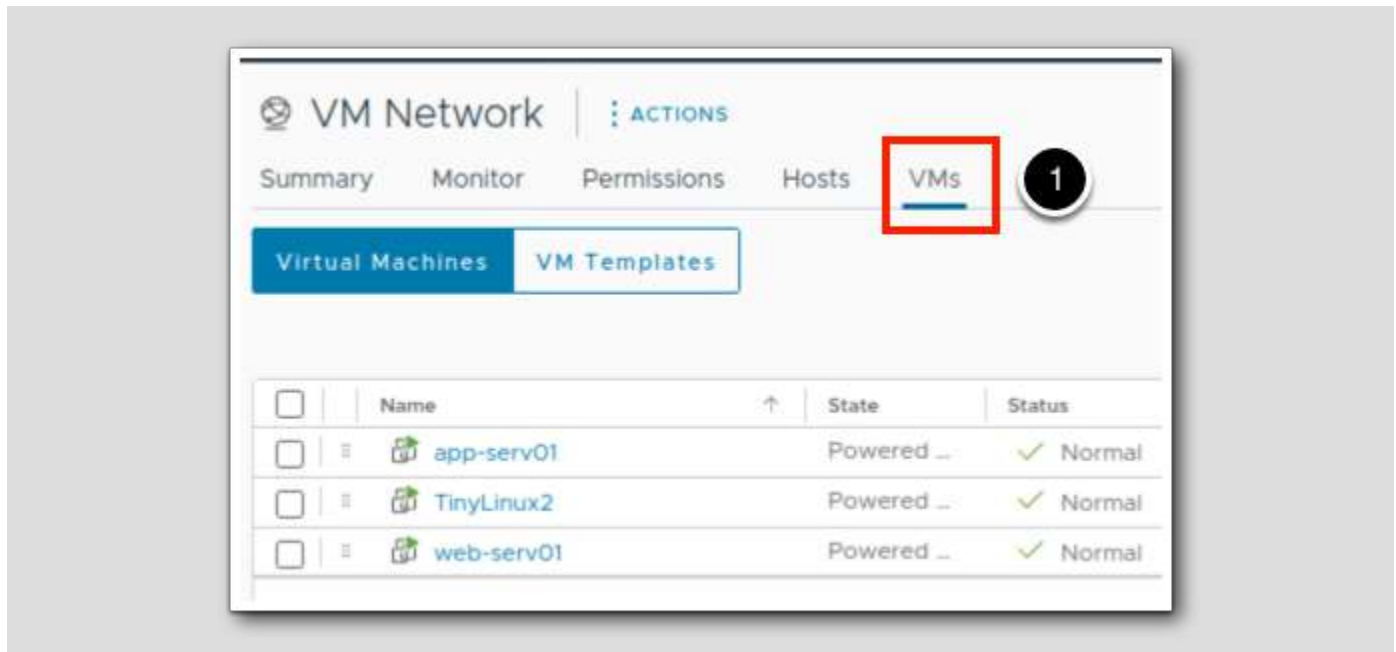
[267]



1. Expand RegionA01
2. Select VM Network

VM Network

[268]



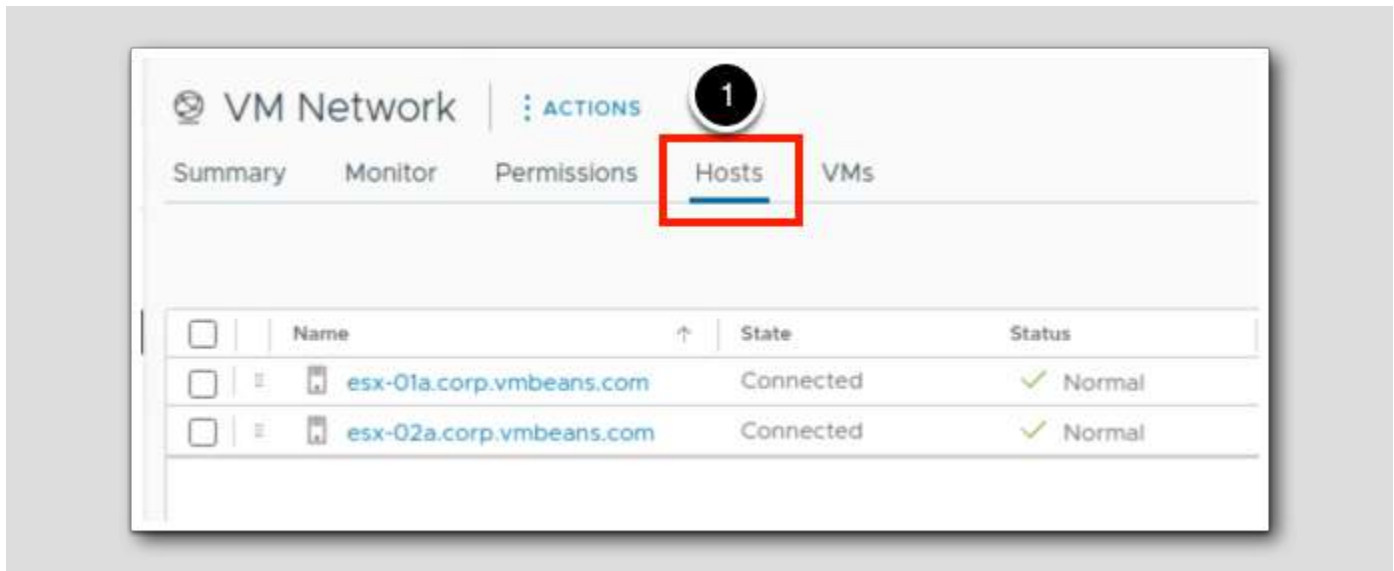
1. Click on VMs tab

Take note of the virtual machines that are connected to this vSwitch. You should see a VM called TinyLinux2.

Note: You may see different results based on what lessons or modules you have already completed.

Hosts

[269]

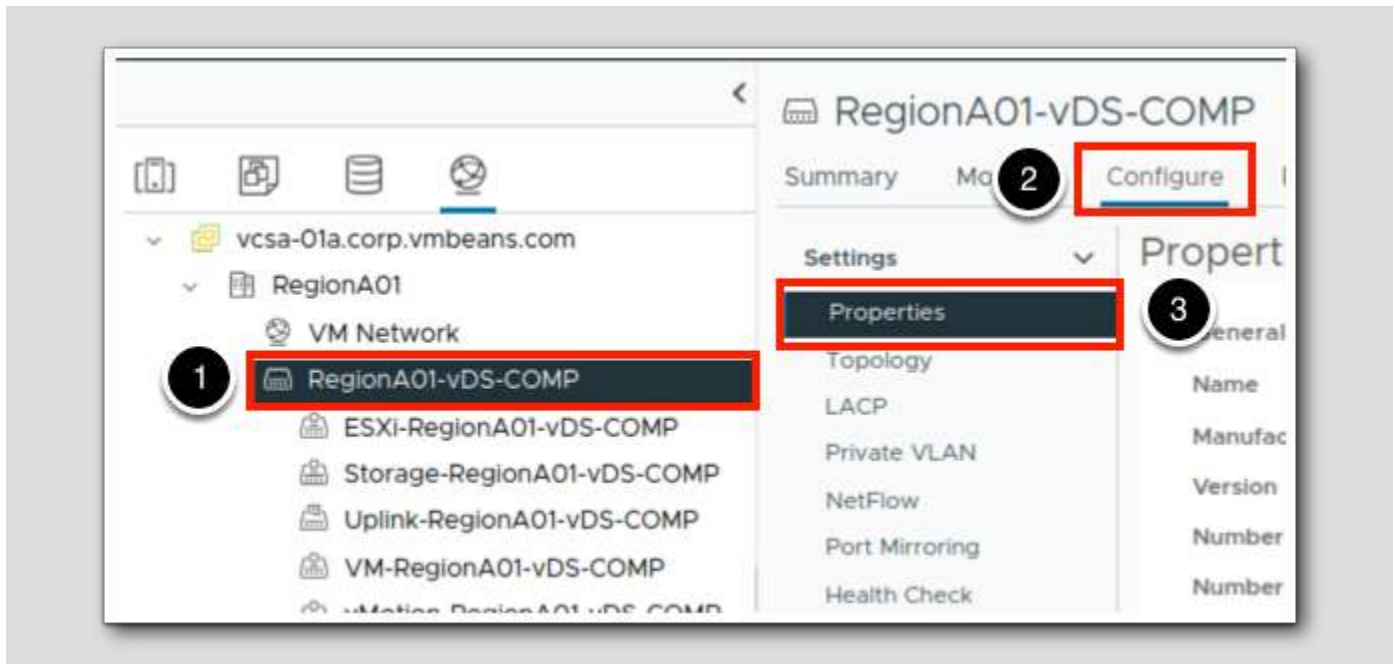


1. Click on Hosts tab

Take note of the hosts connected to the VM Network vSwitch. You should see `esx-01a.corp.vmbeans.com` and `esx-02a.corp.vmbeans.com`

View Distributed Switch

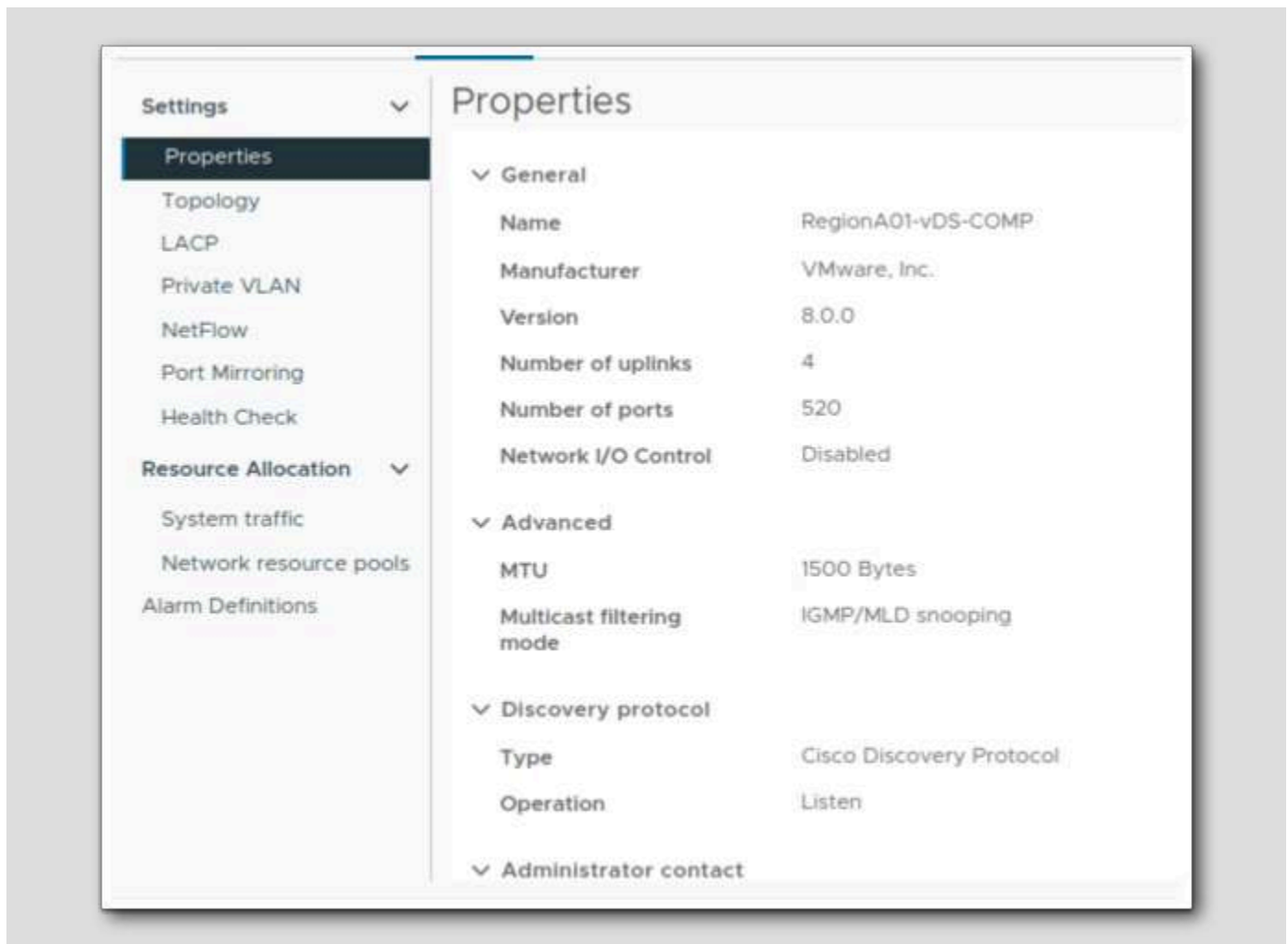
[270]



1. Click on RegionA01-vDS-COMP
2. Select the Configure tab
3. Select Properties

Review vDS configuration

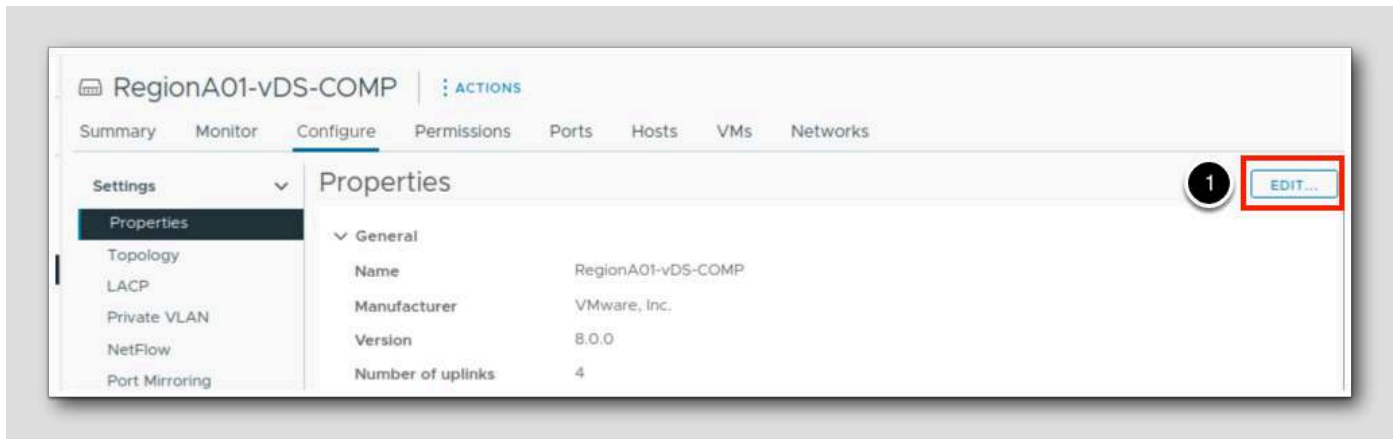
[271]



Basic settings of Distributed Switch are displayed. Such as MTU settings, the version of the switch and discovery protocol being used.

Edit the Switch Properties

[272]



Next, we will explore the various properties of the switch.

1. Click **Edit**

General Settings

[273]

Distributed Switch - Edit Settings | RegionA01-vDS- X
COMP

General | Advanced | Uplinks

1

Name | RegionA01-vDS-COMP

Network offloads compatibility | None ⓘ

Network I/O Control | Disabled

Description |

CANCEL **OK**

1. You should land in the **General** tab. Here, you can review the vSphere distributed switch settings. Here you can modify the following:

Name: You can modify the name of your distributed switch.

Network I/O Control: You can use the drop-down menu to enable or disable Network I/O Control on the switch.

Description: You can use this field to give a meaningful description of the switch.

Advanced Settings

[274]

Distributed Switch - Edit Settings | RegionA01-vDS- x COMP

1

General **Advanced** Uplinks

MTU (Bytes) 1500

Multicast filtering mode IGMP/MLD snooping

Discovery protocol

Type Cisco Discovery Protocol

Operation Listen

Administrator contact

Name

Other details

CANCEL OK

1. Click **Advanced** to view the vSphere distributed switch settings. Here you will find the following advanced settings for the switch:

MTU (Bytes): Maximum MTU size for the vSphere Distributed Switch. To enable jumbo frames, set a value greater than 1500 bytes. Make sure you check with your Networking team prior to modifying this setting in your environment.

Multicast filtering mode

- Basic - The distributed switch forwards traffic that is related to a multicast group based on a MAC address generated from the last 23 bits of the IPv4 address of the group.
- IGMP/MLD snooping - The distributed switch forwards multicast traffic to virtual machines according to the IPv4 and IPv6 addresses of subscribed multicast groups by using membership messages defined by the Internet Group Management Protocol (IGMP) and Multicast Listener Discovery protocol.

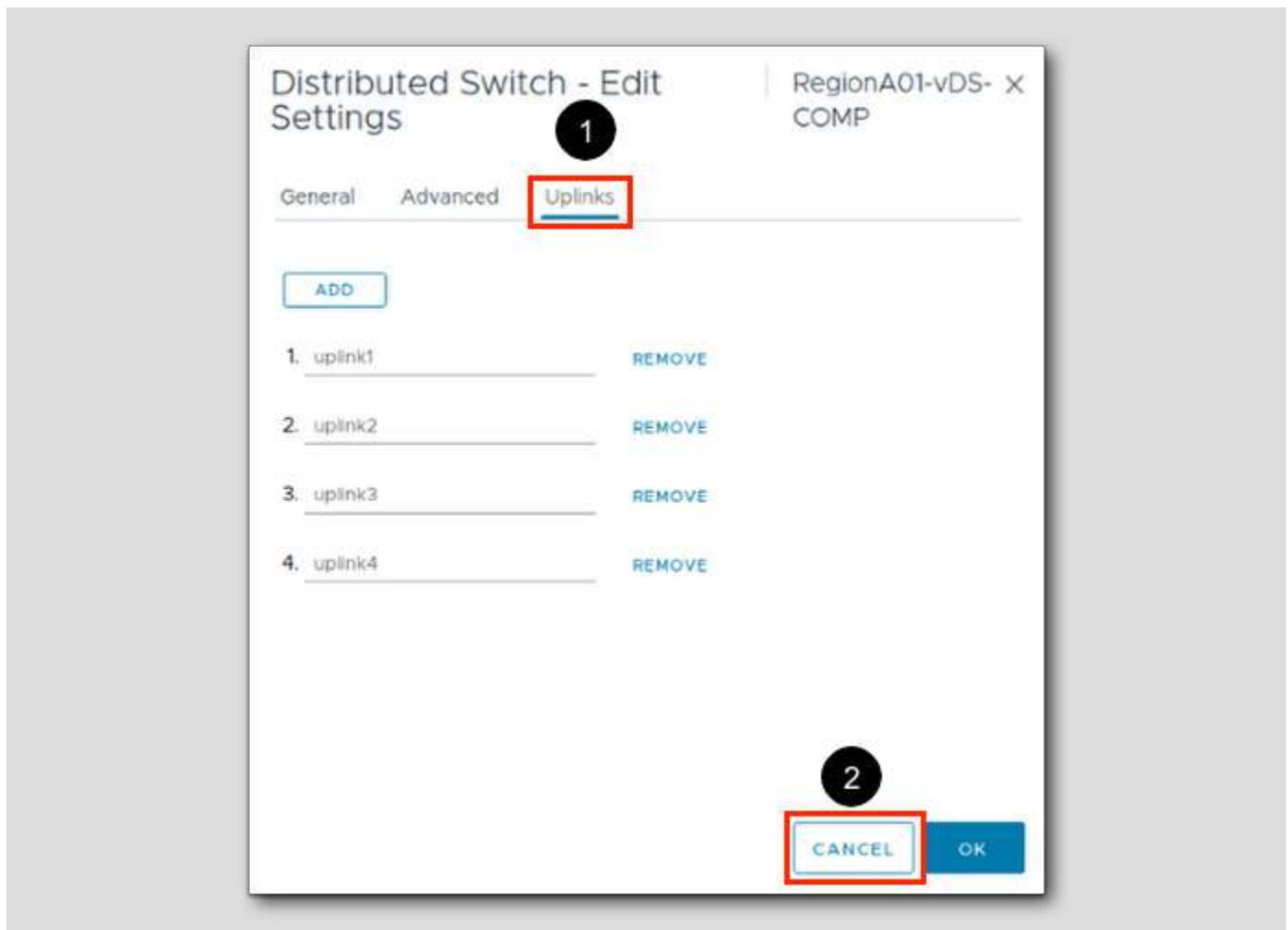
Discovery Protocol

- Type - Cisco Discovery Protocol, Link Layer Discovery Protocol, or disabled.
- Operation - to Listen, Advertise, or Both.

Administrator Contact: Type the name and other details of the administrator for the distributed switch.

Uplinks Settings

[275]



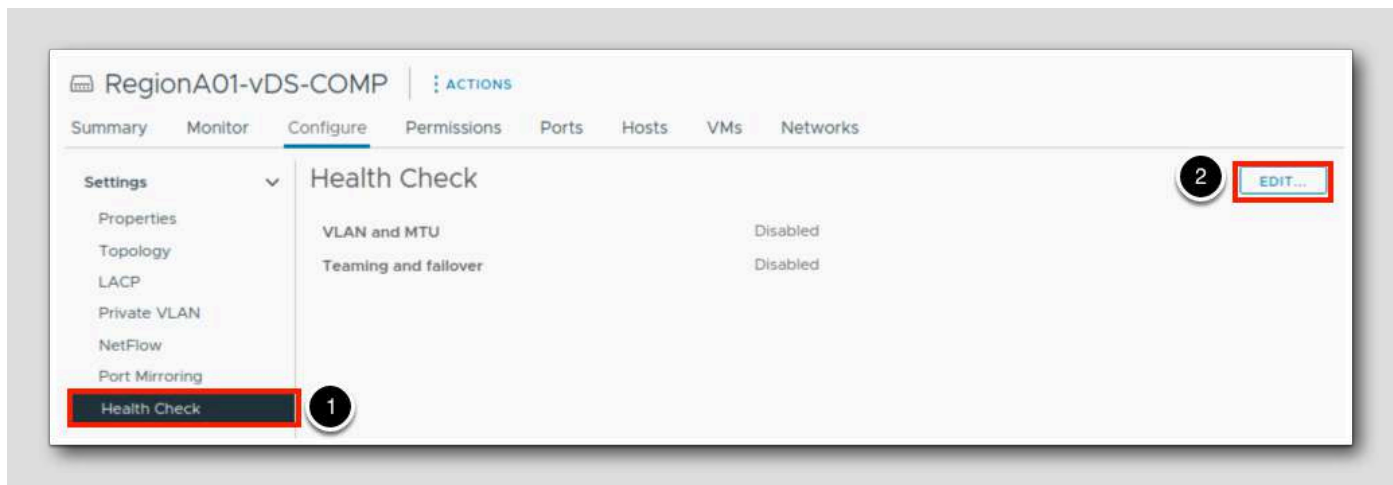
1. Click Uplinks

Number of Uplinks: Add or remove the number uplink ports attached to the distributed switch. Note that you can also click the Edit uplink names button to give the uplinks meaningful names.

2. We don't want to make any changes here, just click Cancel.

Enable or Disable vSphere Distributed Switch Health Check in the vSphere Client

[276]



The Distributed Switch Health Check monitors for changes in vSphere Distributed Switch configurations. You must enable vSphere Distributed Switch Health Check to perform checks on Distributed Switch configurations.

Health Check is available on ESXi 5.1 Distributed Switches and higher.

1. Click on the **Health check** tab for Distributed Switch

We can see that Health check is disabled for VLAN and MTU as well as Teaming and failover.

2. Click the **Edit** button

Edit Health Check Settings

[277]

Edit Health Check Settings | RegionA01-vDS-CO x MP

VLAN and MTU

State: **Enabled** ▾

Interval: 1 minutes

Teaming and failover

State: **Enabled** ▾

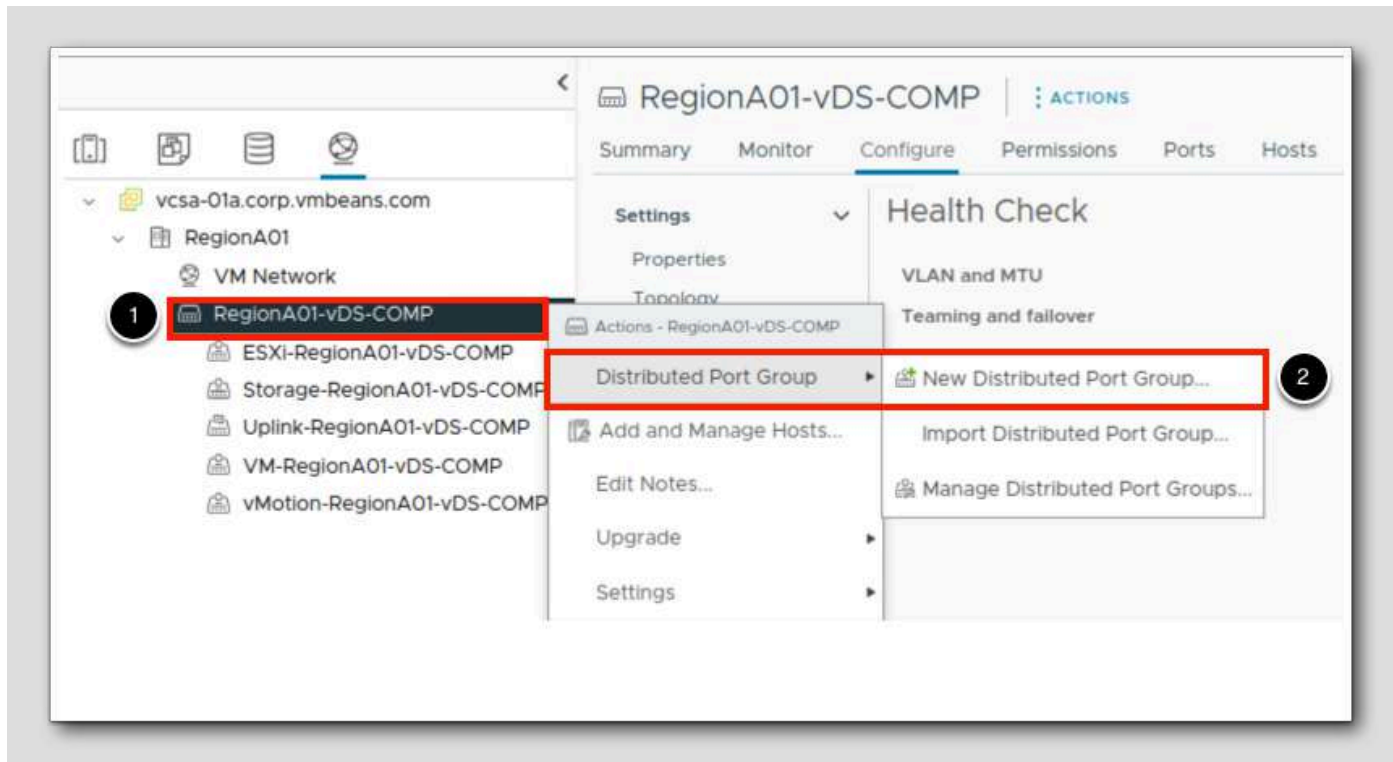
Interval: 1 minutes

CANCEL **OK**

1. Select **Enabled** for both
2. Click **OK** button

Distributed Port Groups

[278]



A distributed port group specifies port configuration options for each member port on a vSphere distributed switch. Distributed port groups define how a connection is made to a network.

1. Right-click **RegionA01-vDS-COMP** in the navigator
2. Select **Distributed Port Group** and then **New Distributed Port Group...**

Select name and location section

[279]

Name and location X

Specify distributed port group name and location.

Name WebVMTraffic 1

Location RegionA01-vDS-COMP

CANCEL NEXT 2

1. Name the new port group **WebVMTraffic**
2. Click **Next**

Configure settings

[280]

Configure settings [X]

Set general properties of the new port group.

Port binding Static binding [v]

Port allocation Elastic [v] ⓘ

Number of ports 8 [up/down]

Network resource pool (default) [v]

VLAN

VLAN type None [v]

Advanced

CANCEL BACK **NEXT** 1

When creating a Distributed Port Group, you have the following options available:

Port binding - Choose when ports are assigned to virtual machines connected to this distributed port group.

- Static binding - Assign a port to a virtual machine when the virtual machine connects to the distributed port group.
- Ephemeral - No port binding. You can assign a virtual machine to a distributed port group with ephemeral port binding also when connected to the host.

Port allocation

- Elastic - The default number of ports is eight. When all ports are assigned, a new set of eight ports is created. This is the default.
- Fixed - The default number of ports is set to eight. No additional ports are created when all ports are assigned.

Number of ports: Enter the number of ports on the distributed port group.

Network resource pool: If you have created network pool to help control network traffic, you can select it here.

VLAN: Use the Type drop-down menu to select VLAN options:

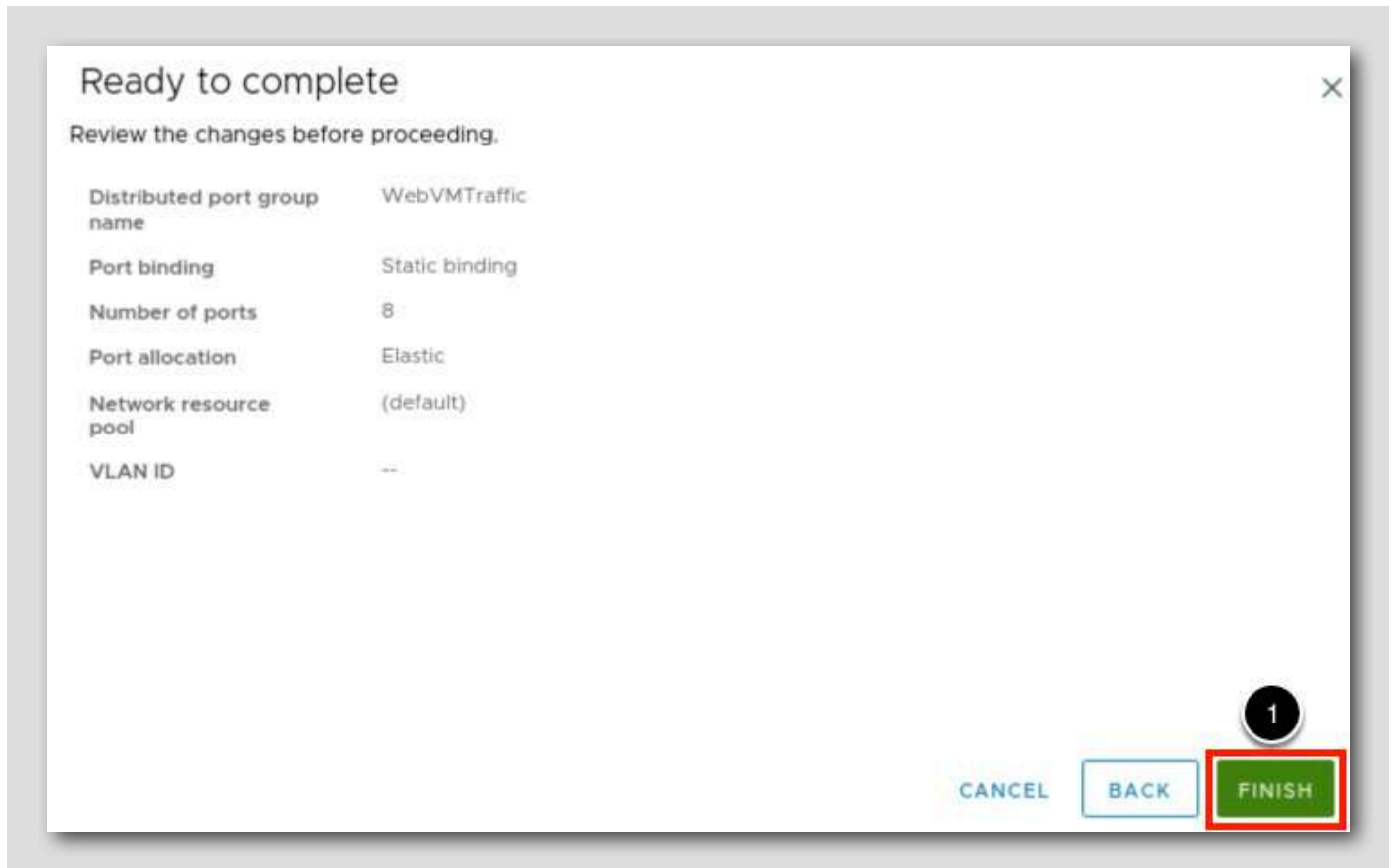
- None - Do not use VLAN.
- VLAN - In the VLAN ID field, enter a number between 1 and 4094.
- VLAN Trunking - Enter a VLAN trunk range.
- Private VLAN - Select a private VLAN entry. If you did not create any private VLANs, this menu is empty.

Advanced: Select this check box to customize the policy configurations for the new distributed port group.

1. Accept the defaults and click **Next** to continue.

Ready to complete

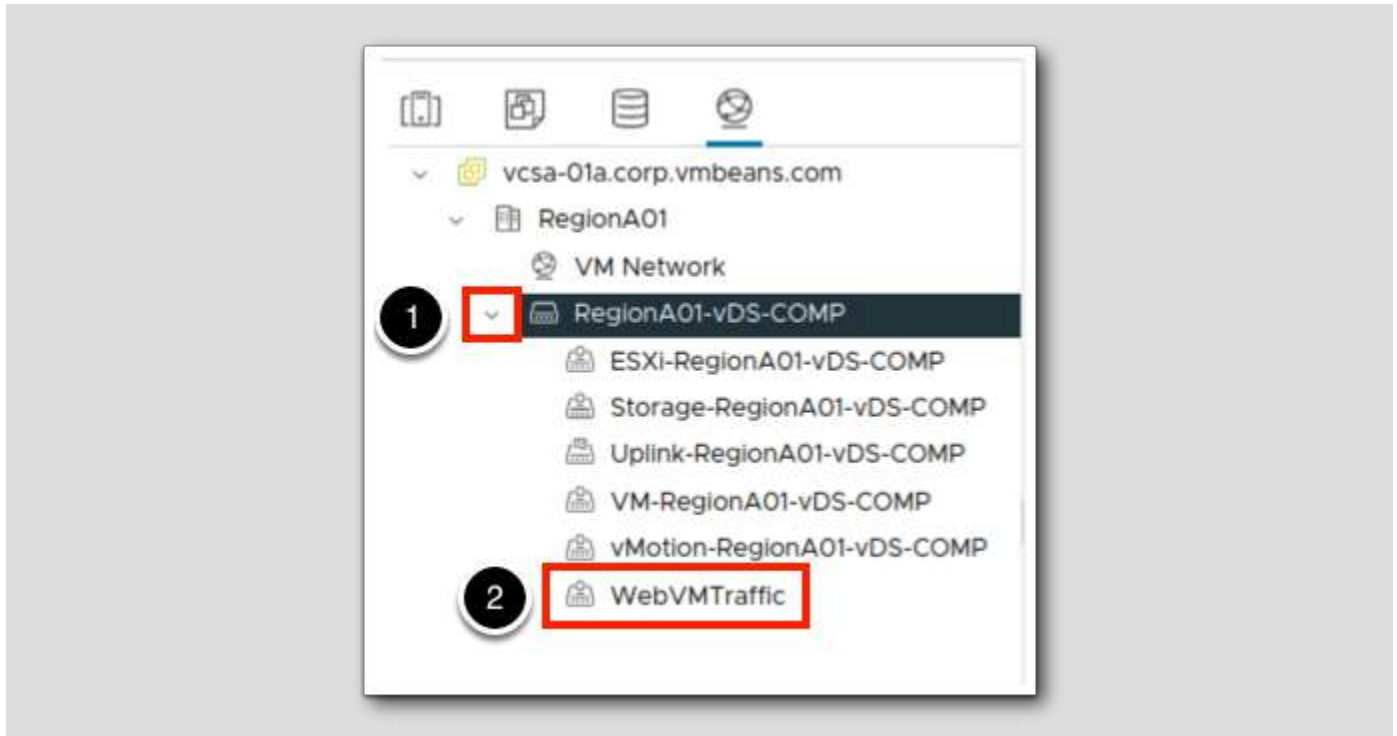
[281]



1. Review your settings and click Finish

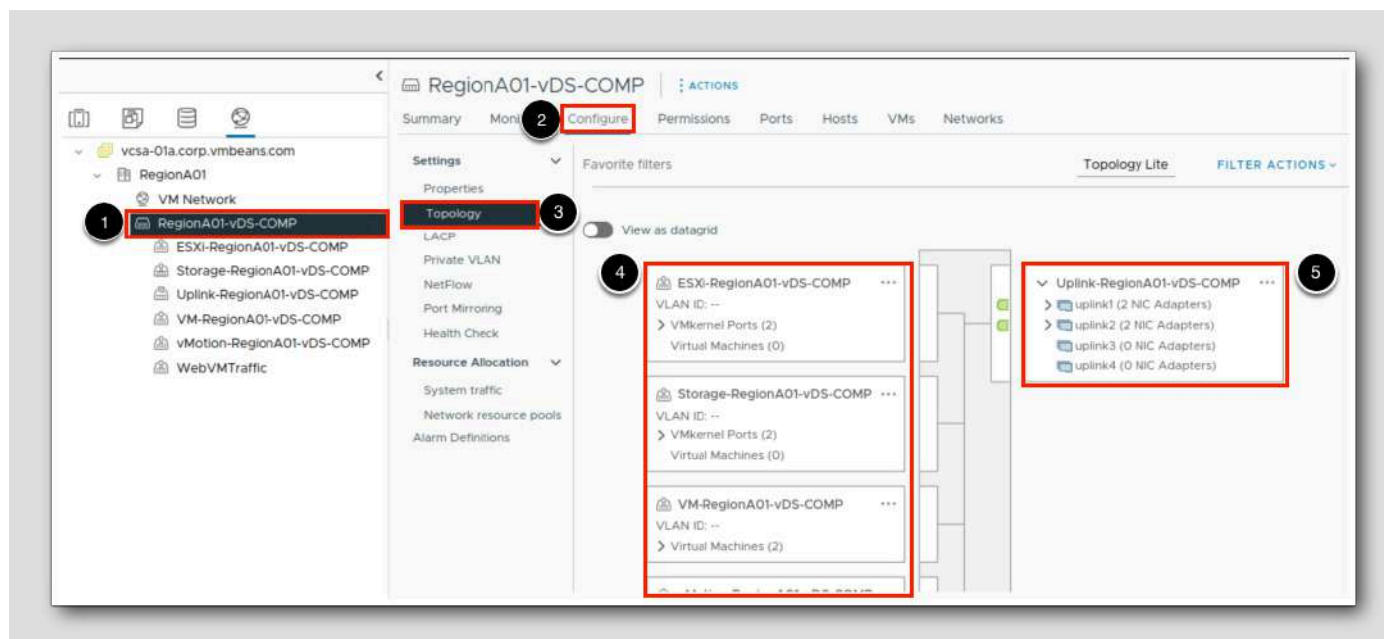
View the New Distributed Port Group

[282]



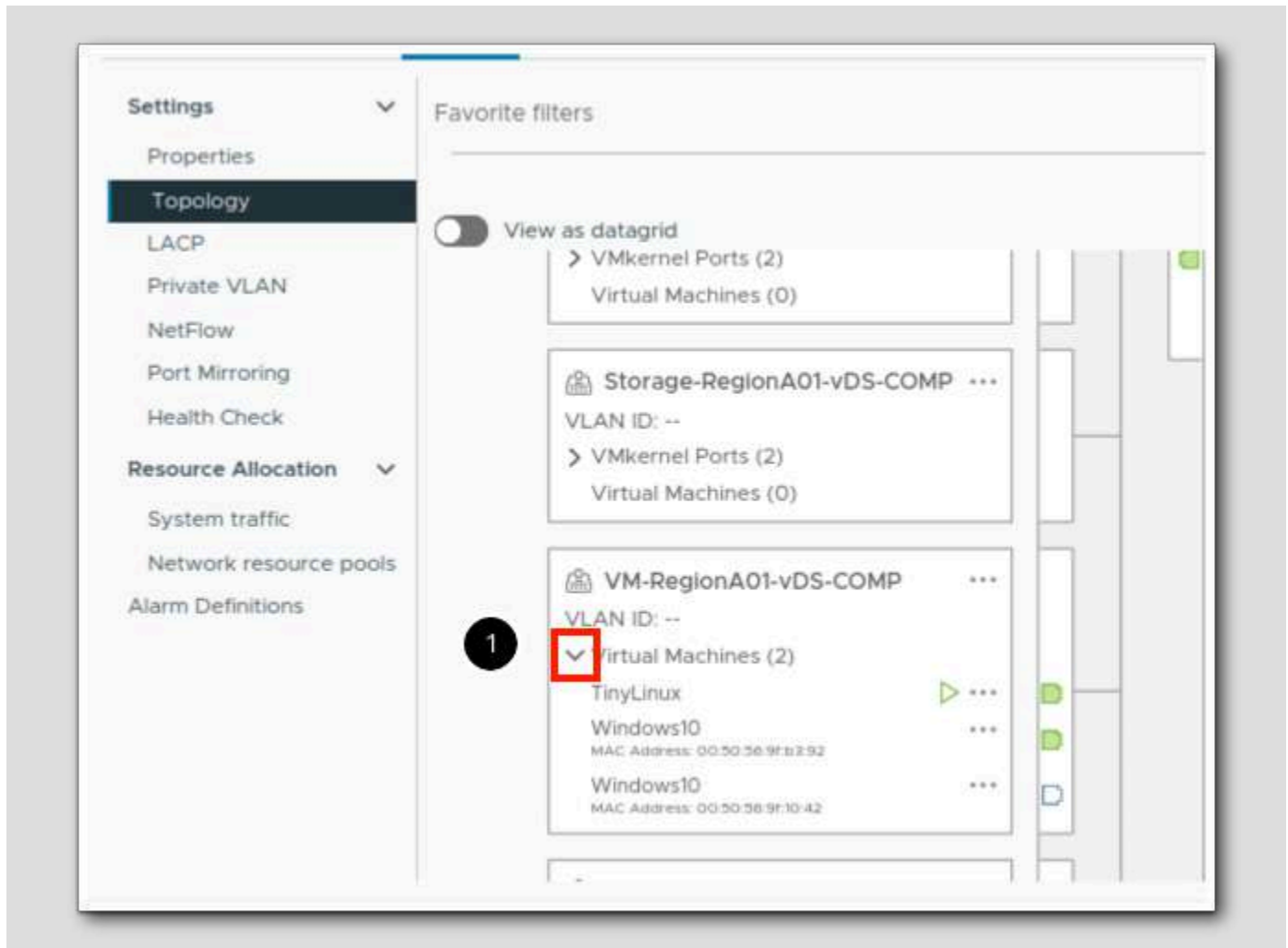
1. In the Navigator, expand out RegionA01-vDS-COMP
2. The newly created **WebVMTraffic** Distributed Port Group has been created

Topology



1. Click on **RegionA01-vDS-COMP**
2. Select **Configure**
3. Click on **Topology**
4. On the left side of the diagram you will see the ports groups associated with the distributed switch **RegionA01-vDS-COMP**. These port groups are how the virtual machines and kernel ports are connected to the vDS. Note how there are VMkernel ports for Management, Storage and vMotion. This is very similar to the configuration you would see on a Standard vSwitch, except that these are defined and configured in one central location instead of individually at each host.
5. On the right you will see the uplinks associated with this vDS. These are used to connect the vDS directly to the physical NICs on the hosts that are tied to this Distributed vSwitch.

VM Port Group

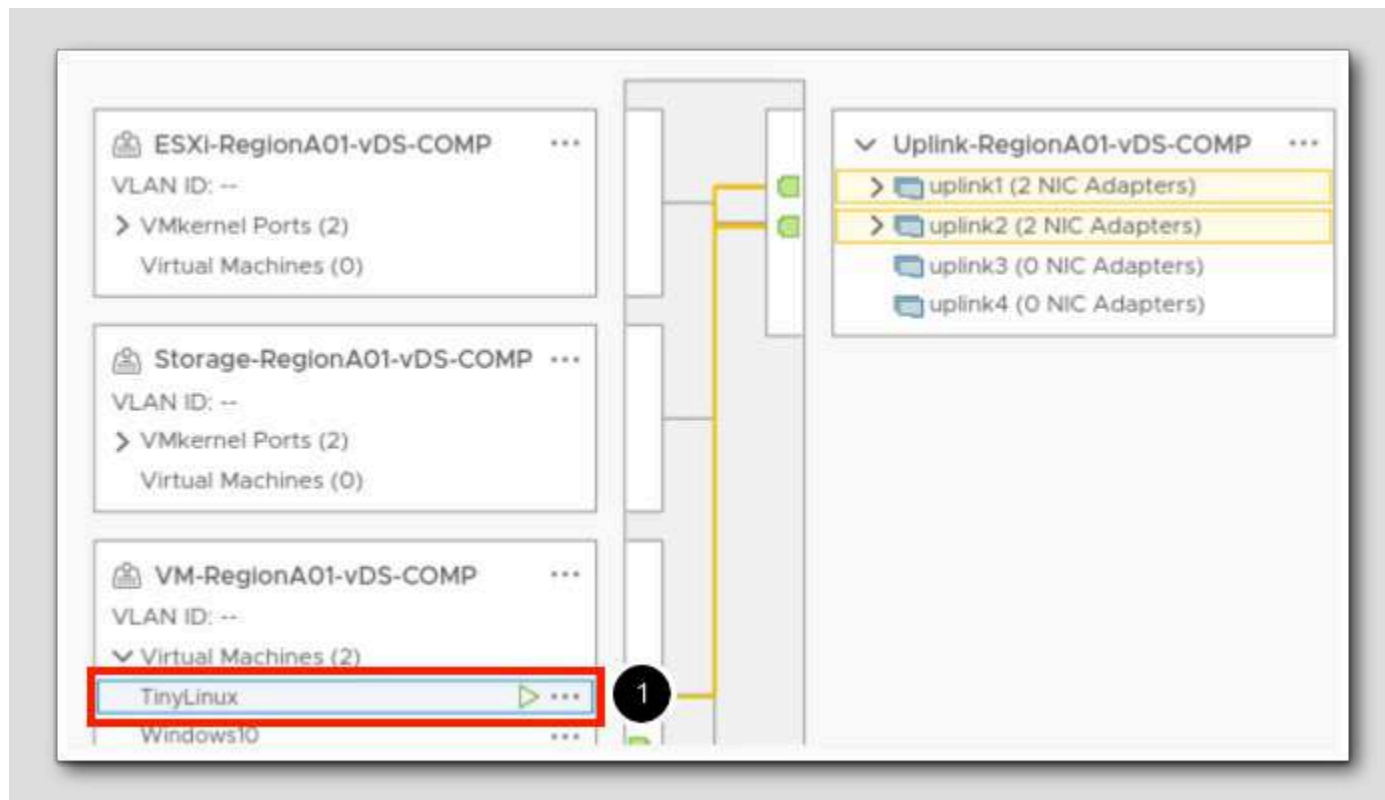


1. Expand **Virtual Machines** on the VM-RegionA01-vDS-COMP port group

Again, note how there are virtual machines tied to this distributed port group just like you would see in a port group on a standard vSwitch.

Path to Uplinks

[285]



1. Click on TinyLinux

Note that a path to an uplink is drawn out and highlighted in orange to show the uplinks, hosts and vmnics it is associated with.

Creating a New Distributed Switch

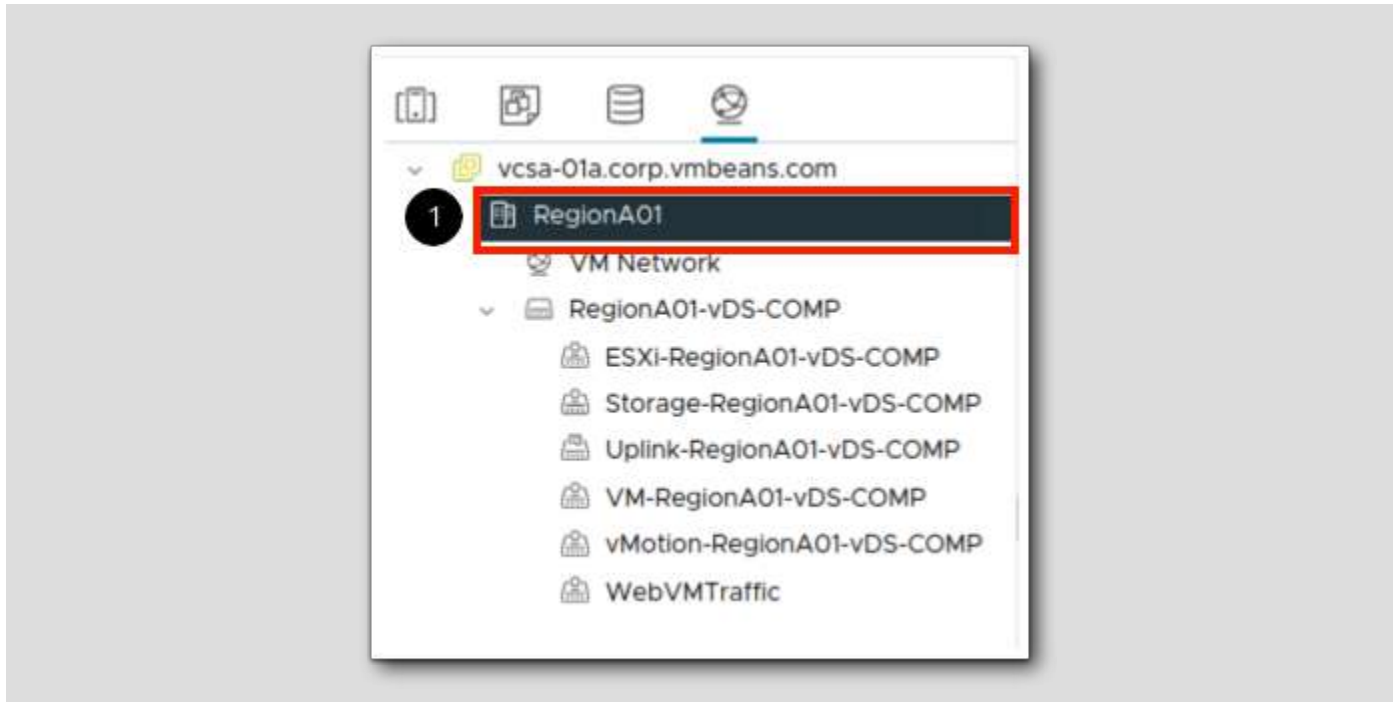
[286]

Now that we have had a chance to explore an existing vDS, let's build one of our own.

In this lab we will create a new Distributed vSwitch, add ESXi hosts to it, build port groups and connect them to uplinks so that we can use it to forward virtual machine traffic on to the physical network.

Navigate to RegionA01 Datacenter

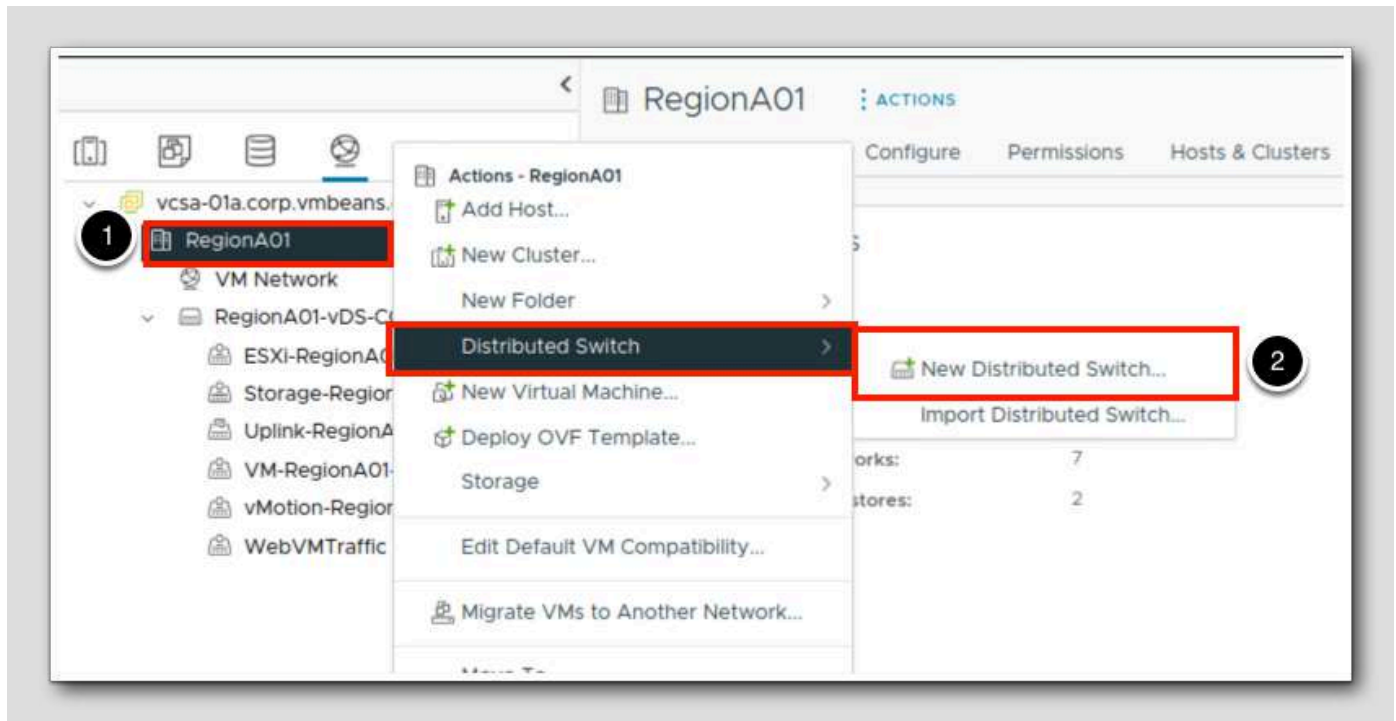
[287]



1. In the vSphere Web Client, click on RegionA01

Create a New Distributed Switch

[288]



1. In the navigator, right-click the **RegionA01**
2. Select **Distributed Switch** and then **New Distributed Switch**

This will open the New Distributed Switch wizard.

Name the Distributed Switch

[289]

The screenshot shows the 'New Distributed Switch' wizard with the 'Name and location' step selected. The 'Name' field contains 'New-vDS' and is highlighted with a red box and a circled '1'. The 'Location' is set to 'RegionA01'. The 'NEXT' button is highlighted with a red box and a circled '2'. The 'CANCEL' button is also visible.

New Distributed Switch

1 Name and location

2 Select version

3 Ready to complete

Name and location

Specify distributed switch name and location.

Name New-vDS

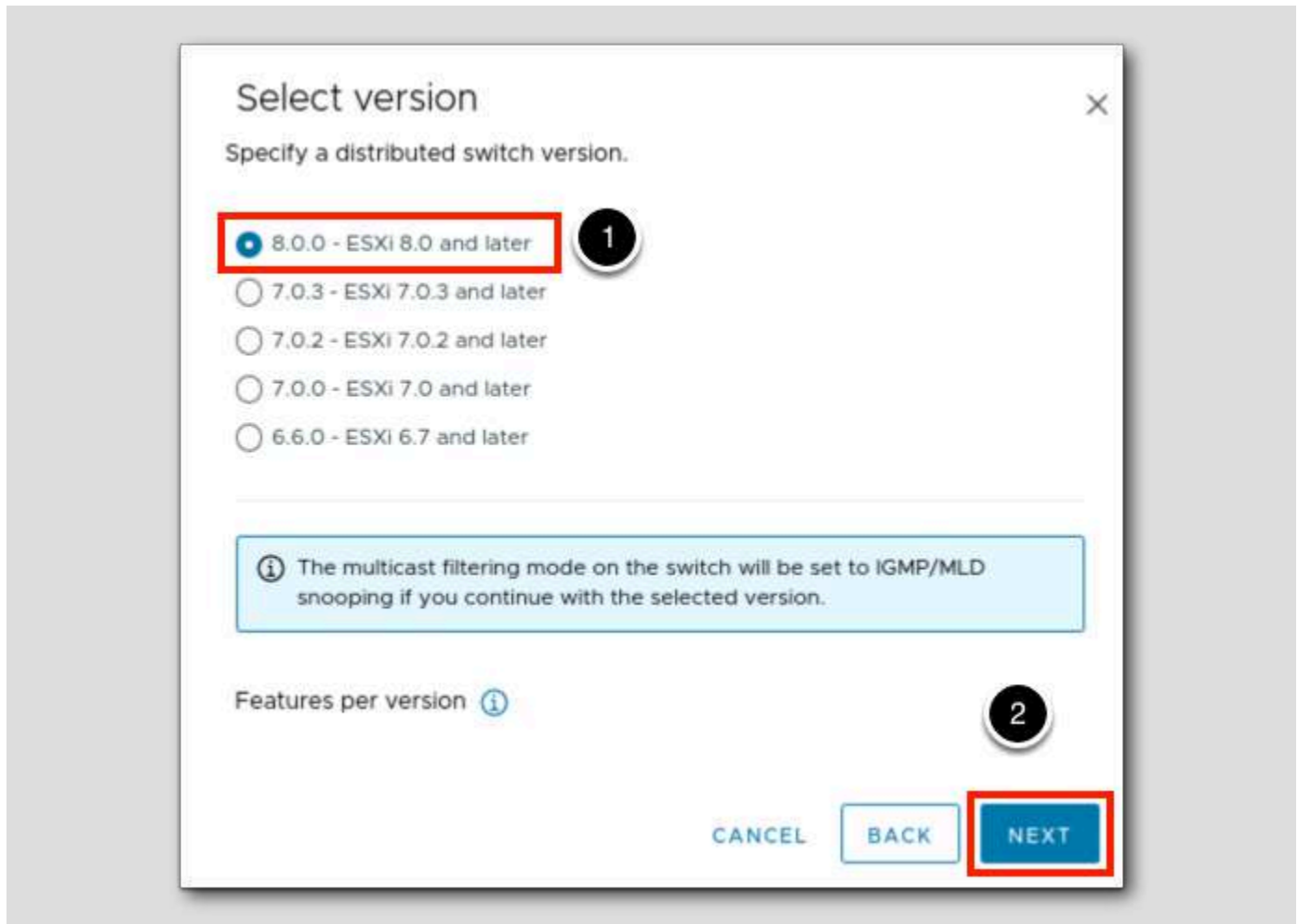
Location RegionA01

CANCEL NEXT

1. Type **New-vDS** in the Name field
2. Click **Next**

Select the Version

[290]



1. Select 8.0.0 - ESXi 8.0 and later
2. Click Next

Configure Settings

[291]

Configure settings [X]

Specify network offloads compatibility, number of uplink ports, resource allocation and default port group.

Network Offloads compatibility None ⓘ

Number of uplinks 4 ↕

Network I/O Control Enabled ▾

Default port group ☒ Create a default port group

Port group name DPortGroup

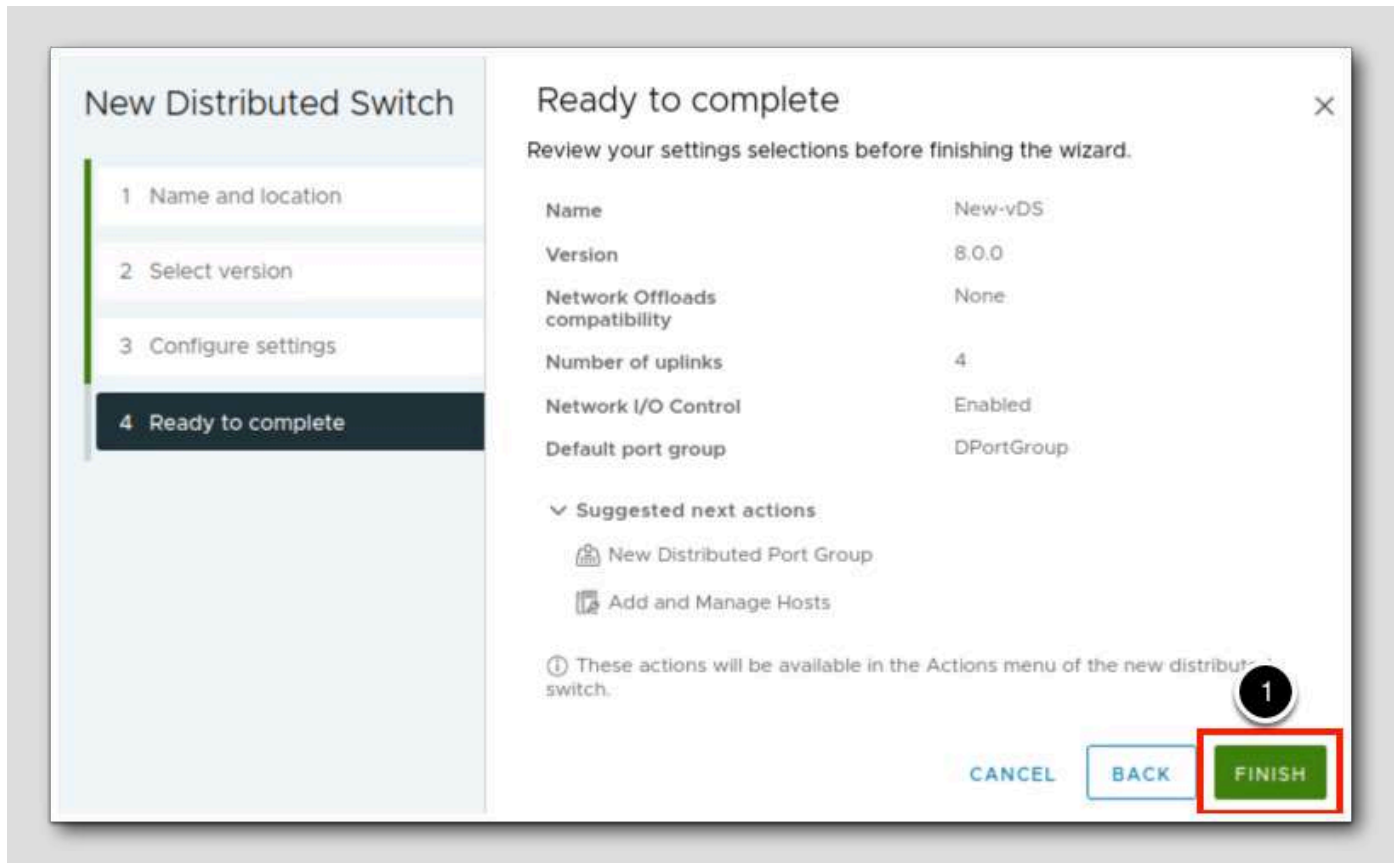
CANCEL BACK **NEXT**

1

1. Leave the default options and click Next

Complete the Build

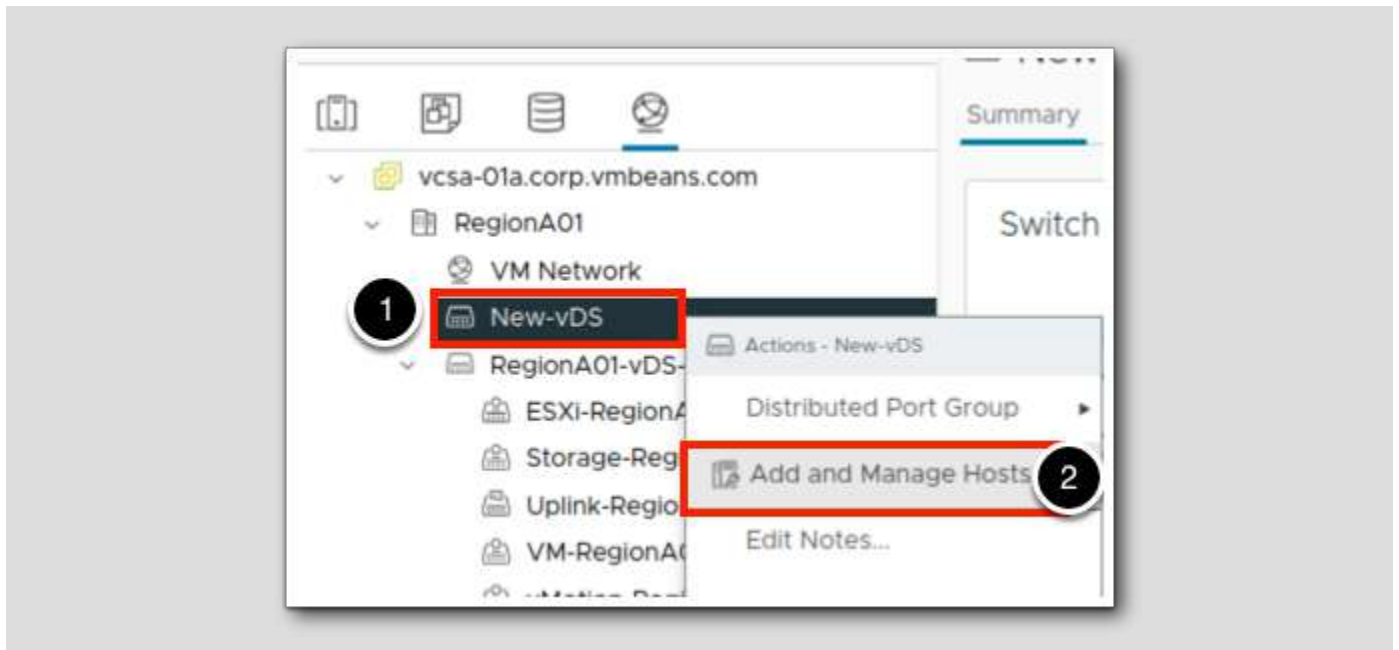
[292]



1. Review your settings and click Finish

Add Hosts to New Distributed Switch

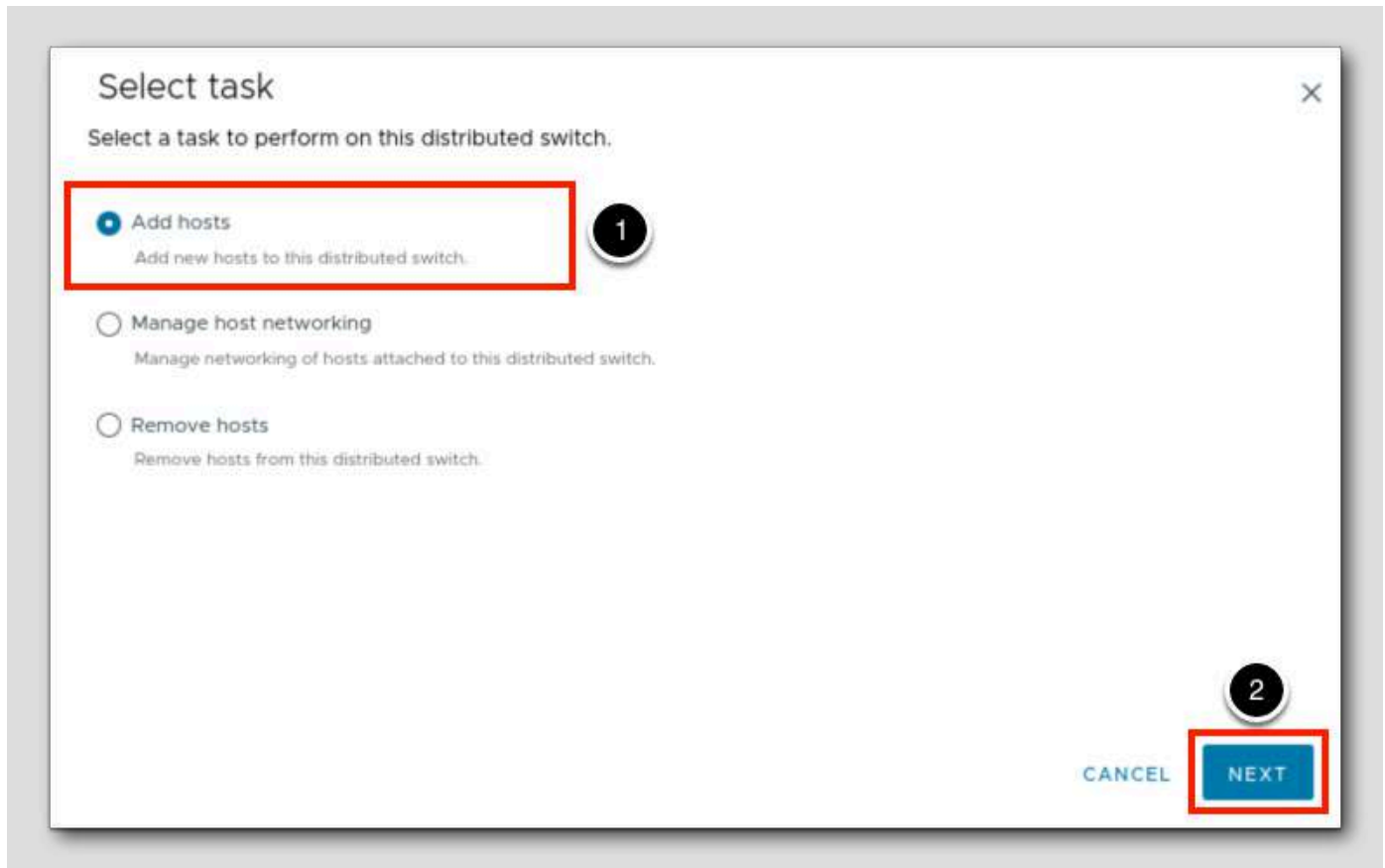
[293]



1. Right-click on the newly created switch, New vDS
2. Select Add and Manage Hosts

Select Task

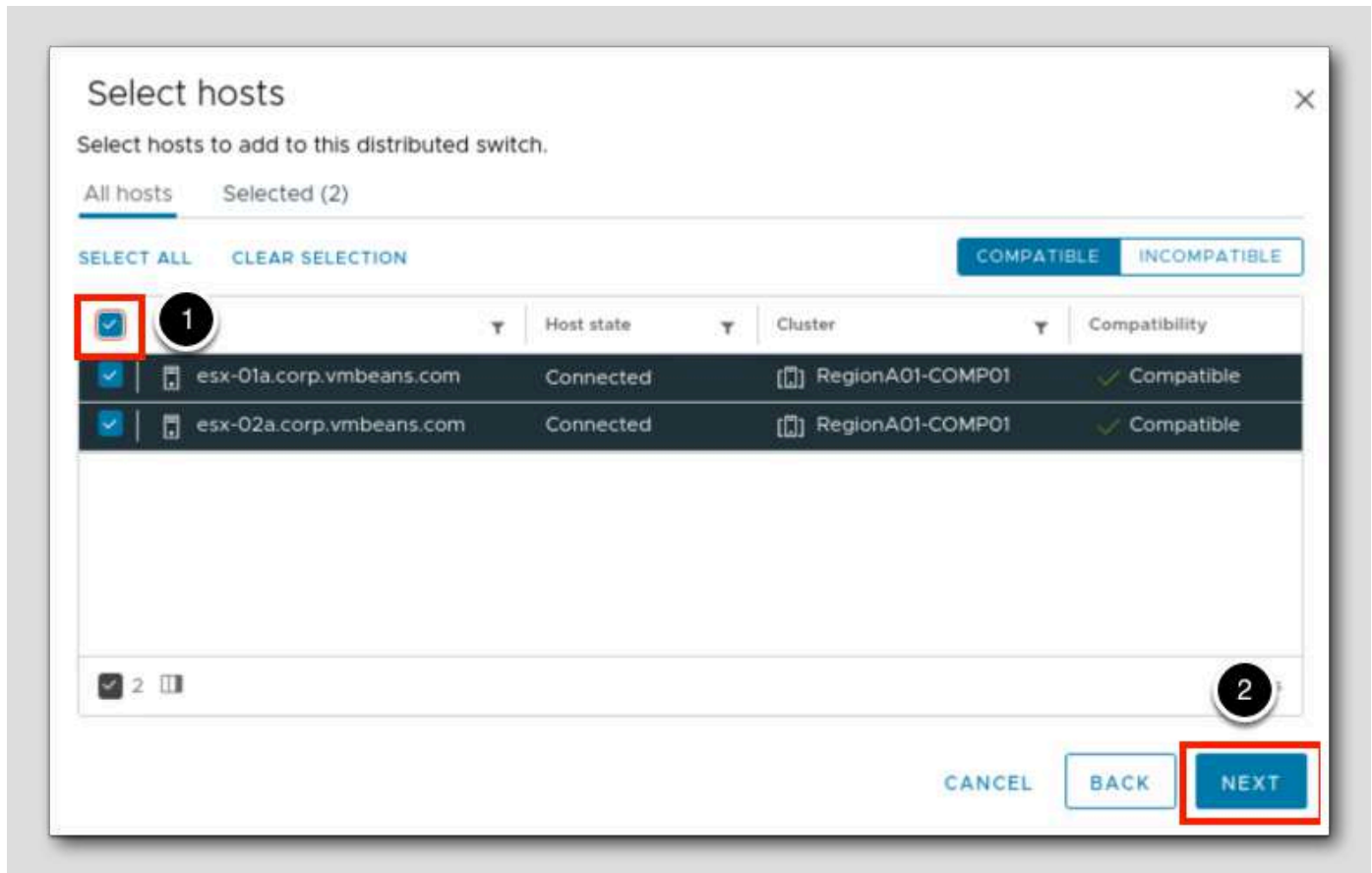
[294]



1. On the Select task page, select **Add hosts**

2. Click **Next**

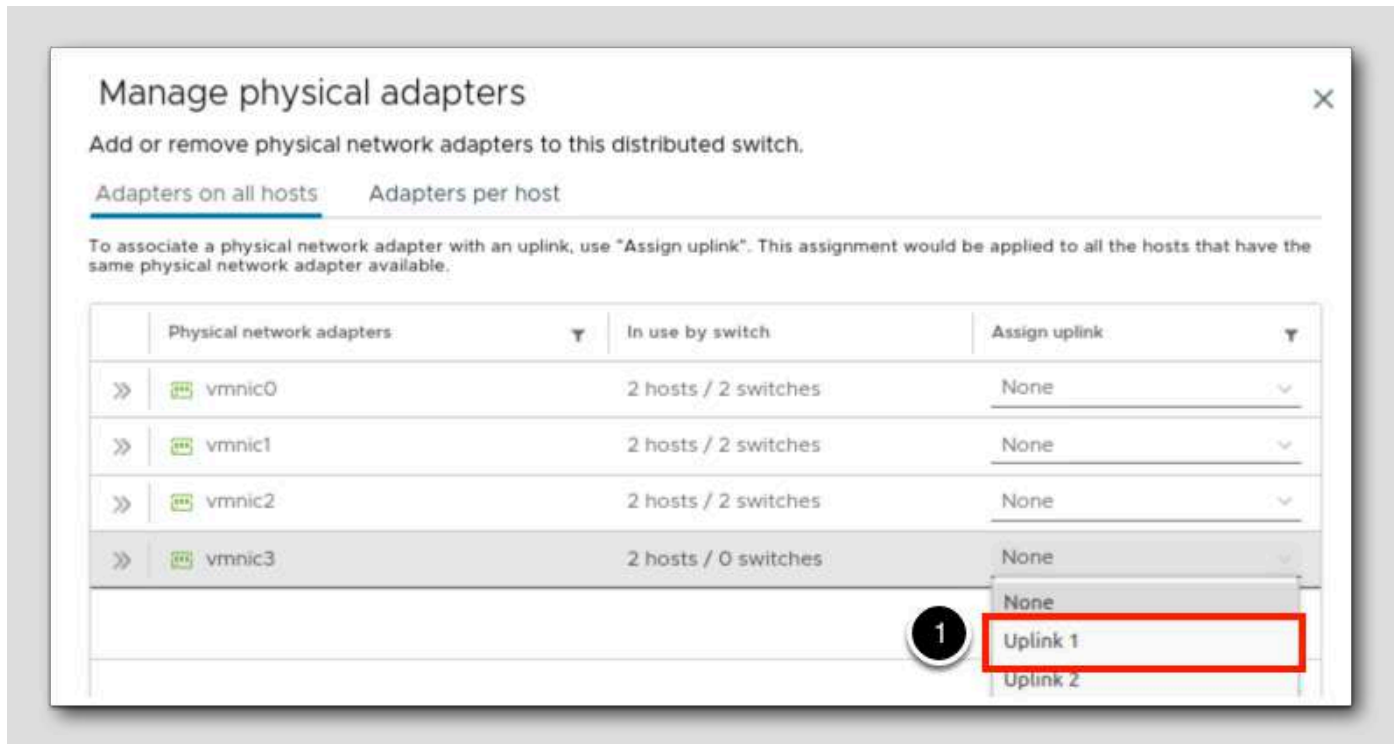
Select New Hosts



1. Click the check box on the left to select **both** hosts in the datacenter
2. Click OK

Assign Physical Adapters

[296]



On the Manage physical network adapters page, we want to configure which physical NICs will be used on the distributed switch.

1. Click Assign uplink drop-down menu for vmnic3 and select Uplink1

Review Settings

[297]

Manage physical adapters [X]

Add or remove physical network adapters to this distributed switch.

Adapters on all hosts Adapters per host

To associate a physical network adapter with an uplink, use "Assign uplink". This assignment would be applied to all the hosts that have the same physical network adapter available.

Physical network adapters	In use by switch	Assign uplink
>> vmnic0	2 hosts / 2 switches	None
>> vmnic1	2 hosts / 2 switches	None
>> vmnic2	2 hosts / 2 switches	None
>> vmnic3	This switch	Uplink 1

4 physical network adapters

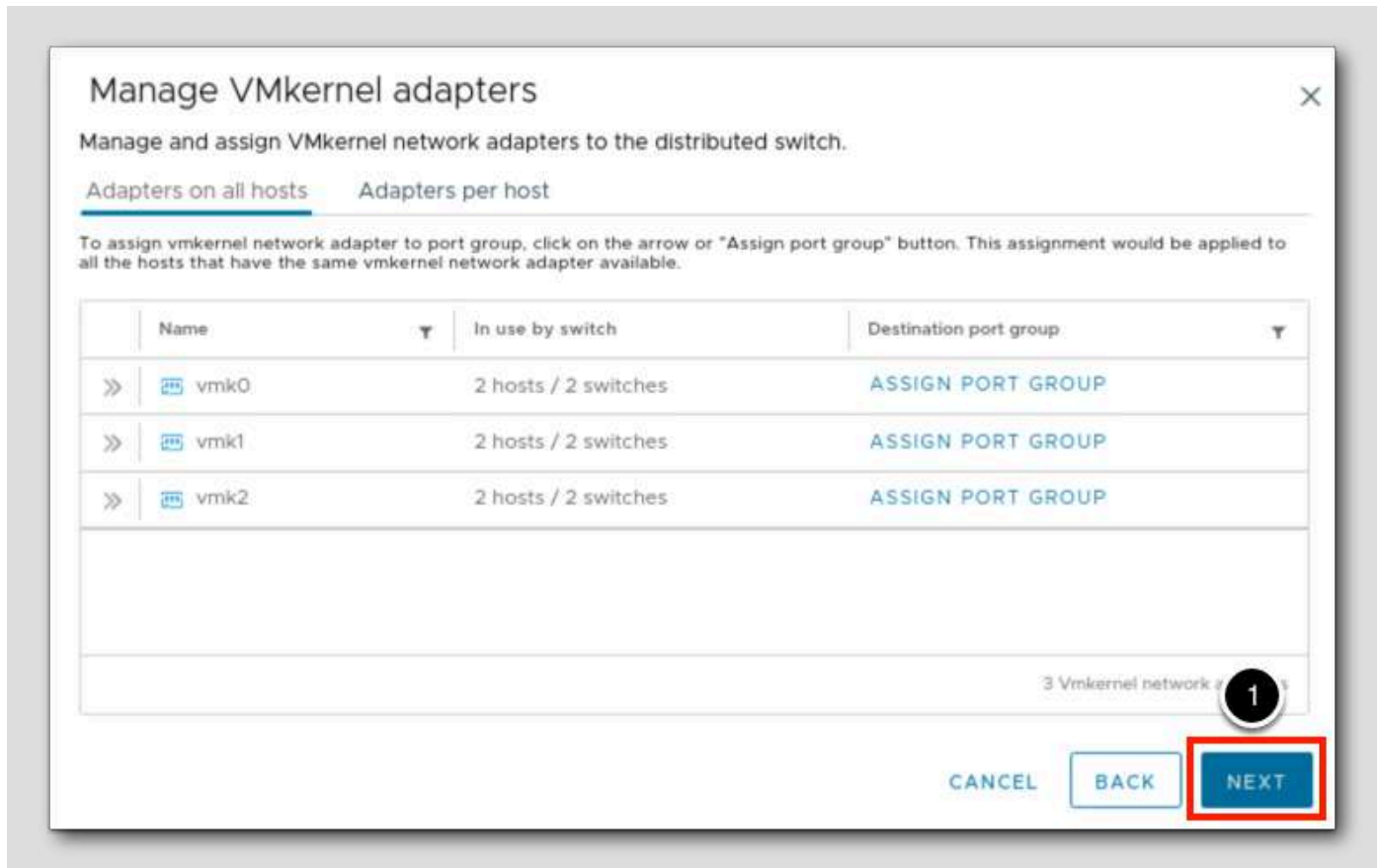
CANCEL BACK **NEXT**

1

1. Review vmnic and uplink settings for the hosts you are adding and click **Next**

Manage VMkernel Adapters

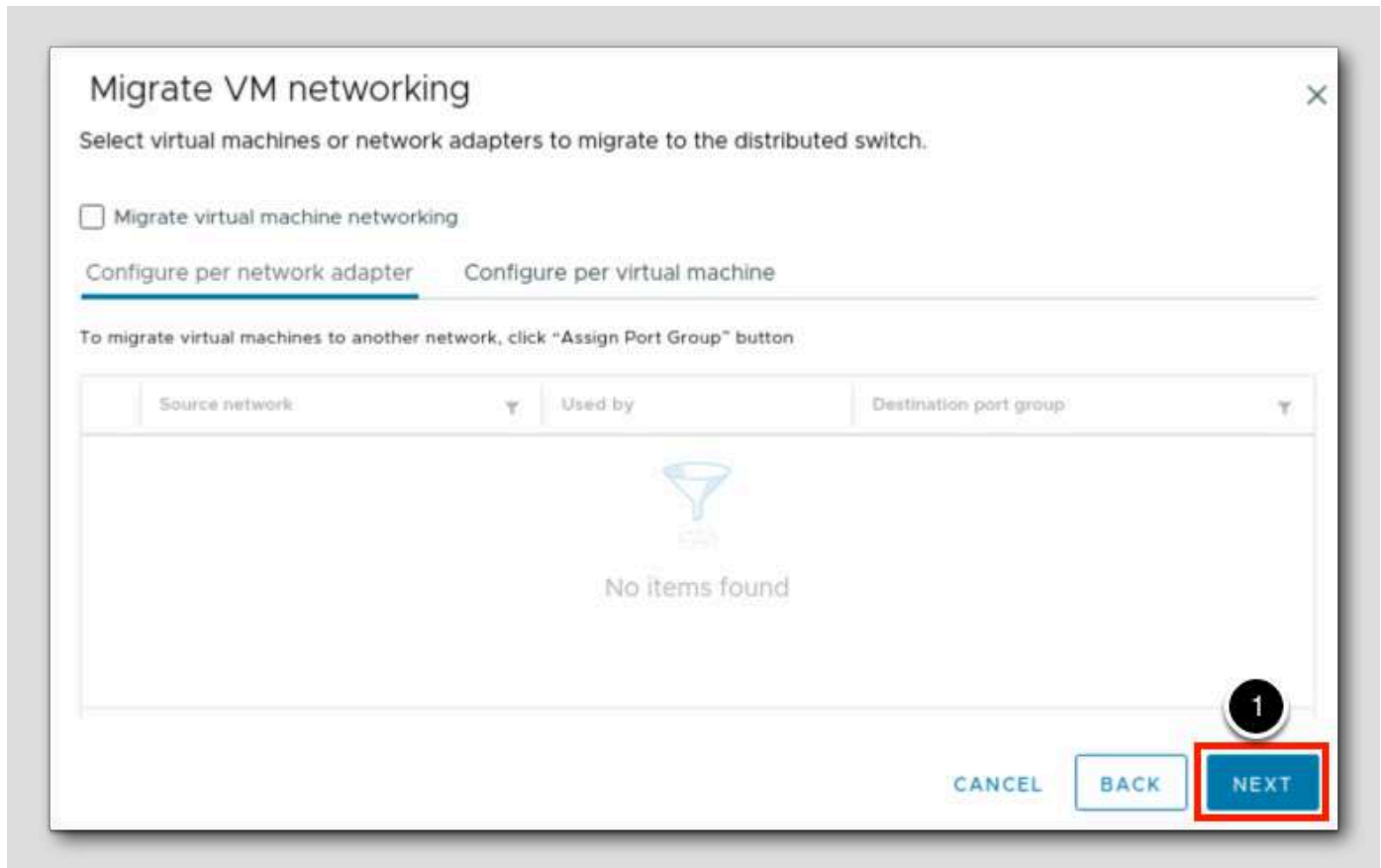
[298]



1. Since we will not be using this distributed switch for any VMkernel functions, click Next

Migrate VM Networking

[299]



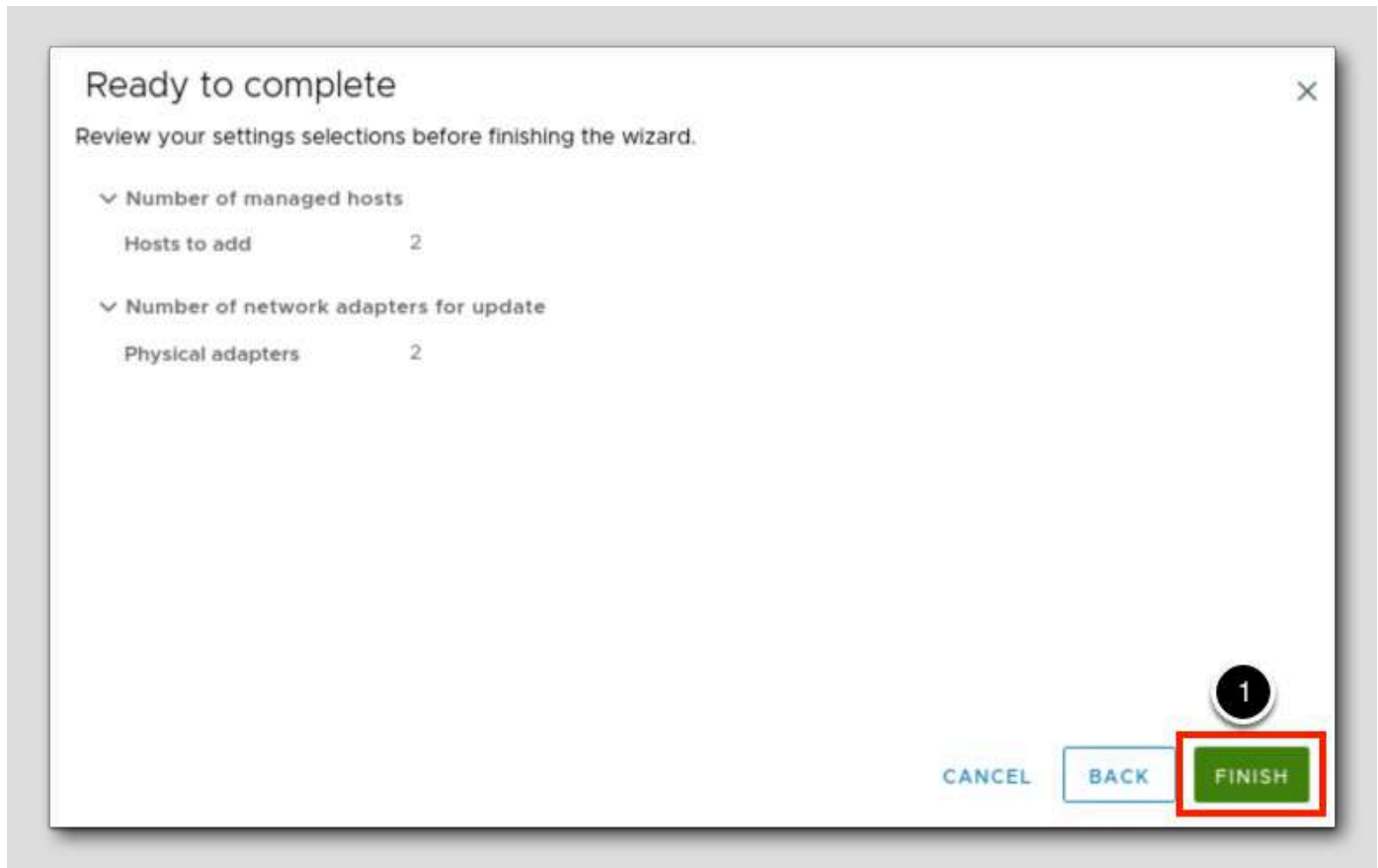
The add hosts wizard also gives us the ability to migrate VMs from one distributed switch to another on this page. While this action can be done here, we will be doing this in the next lesson.

1. Click **Next**

Also note that this wizard is not the typical place where you would migrate VMs from one virtual switch to another. The process we will be using later is the recommended method.

Complete the Host Add Wizard

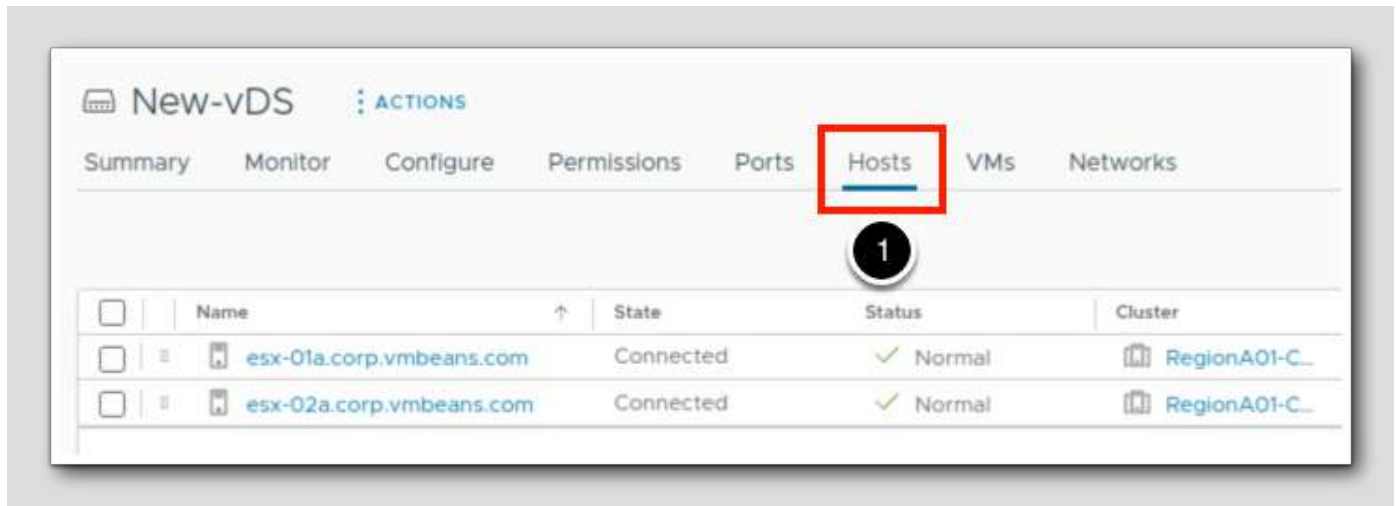
[300]



1. On the Ready to Complete page, click Finish

Explore your New vDS

[301]

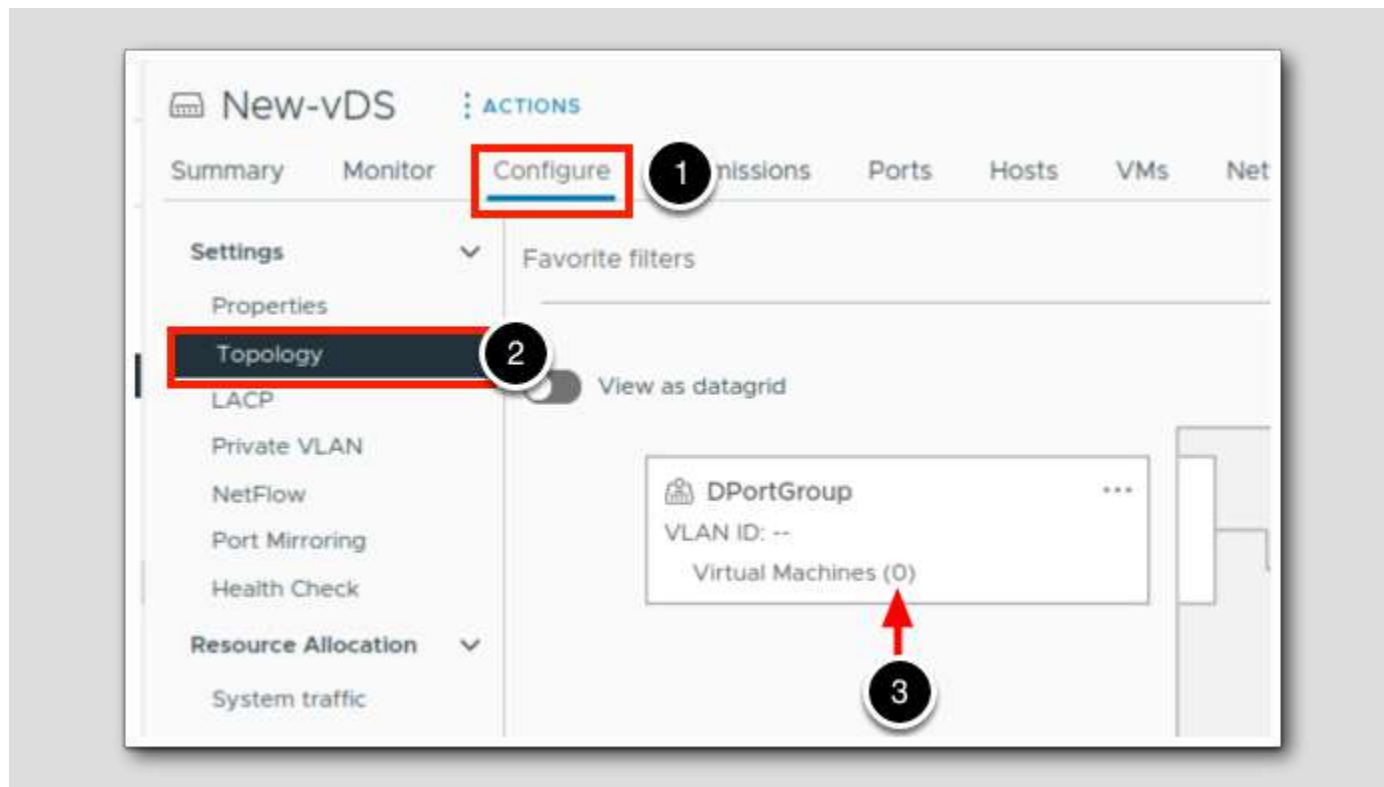


With your new Distributed Switch highlighted, feel free to explore the associated tabs to get a feel for the setup and configuration.

1. Click on the **Hosts** tab to see the newly connected hosts

Topology

[302]



1. Click Configure
2. Click Topology

Note that your distributed port group DPortGroup does not have any VMs connected to it. The next lesson will walk through the process of migrating VMs to the new vDS.

Migrating VMs from vSS to vDS

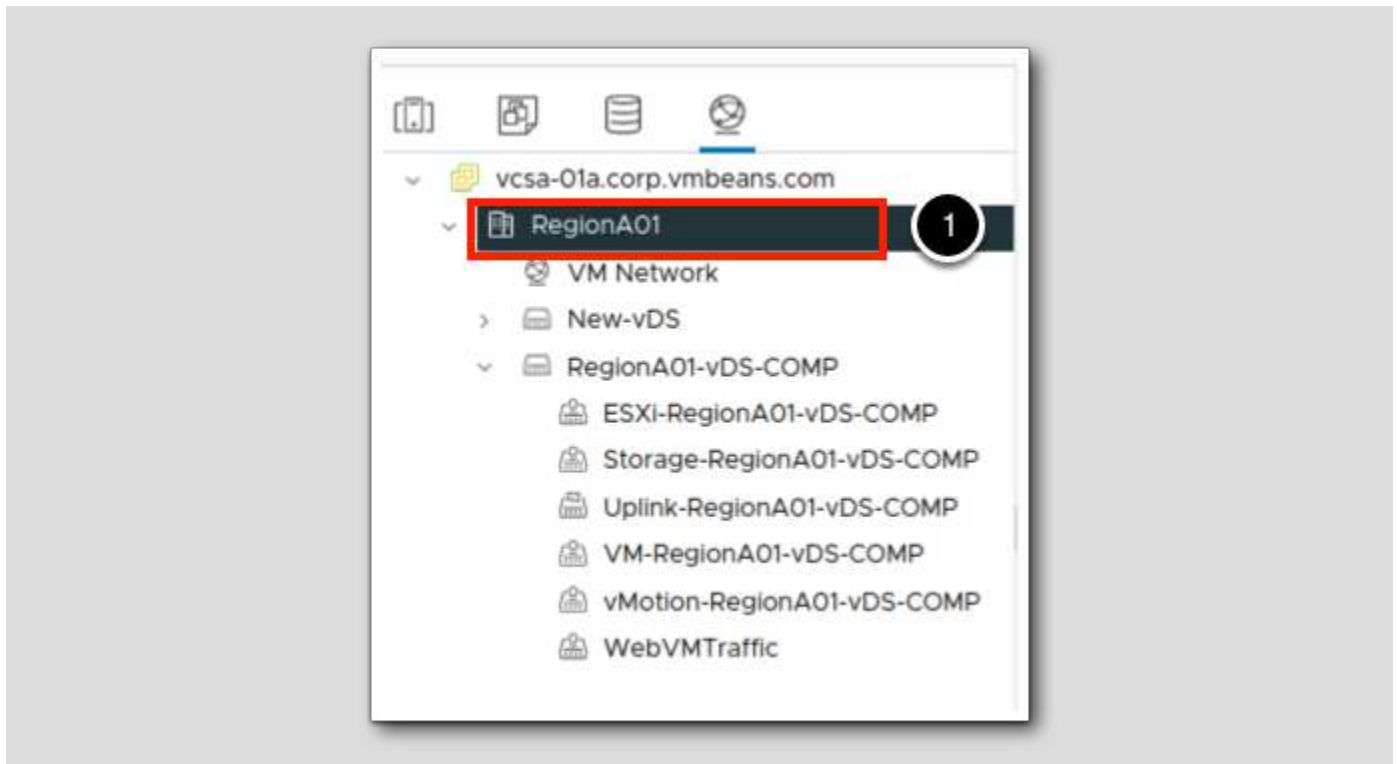
[303]

Now that we have created a new vDS, we want to take advantage of its capabilities. In this lab we will migrate a running virtual machine from a virtual standard switch to the newly created distributed virtual switch.

In the vSphere Client, there are numerous ways to accomplish the task of VM network migration. However, we will be walking through the procedures specifically outlined in the vSphere product documentation.

Navigate to your Datacenter

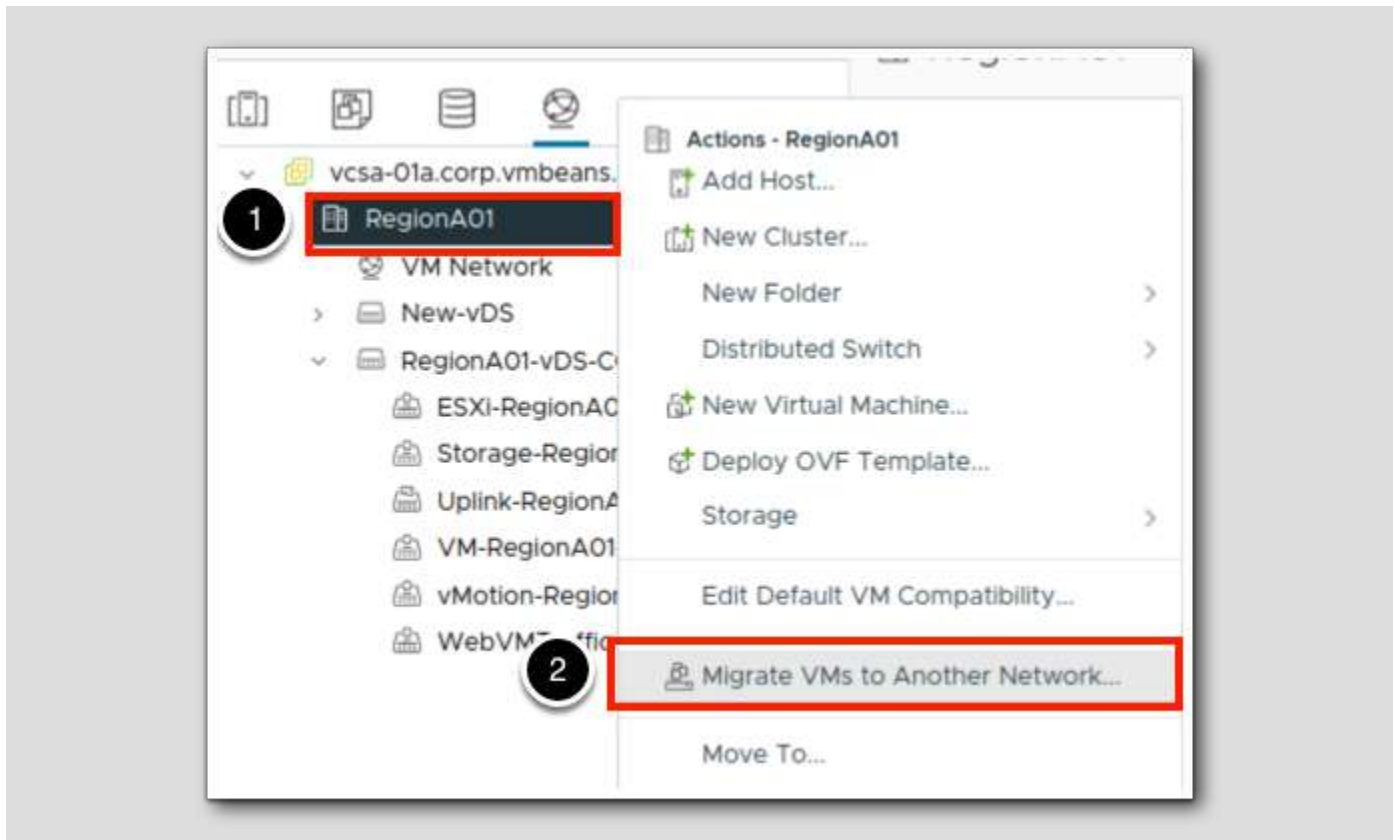
[304]



1. To get started, click on RegionA01

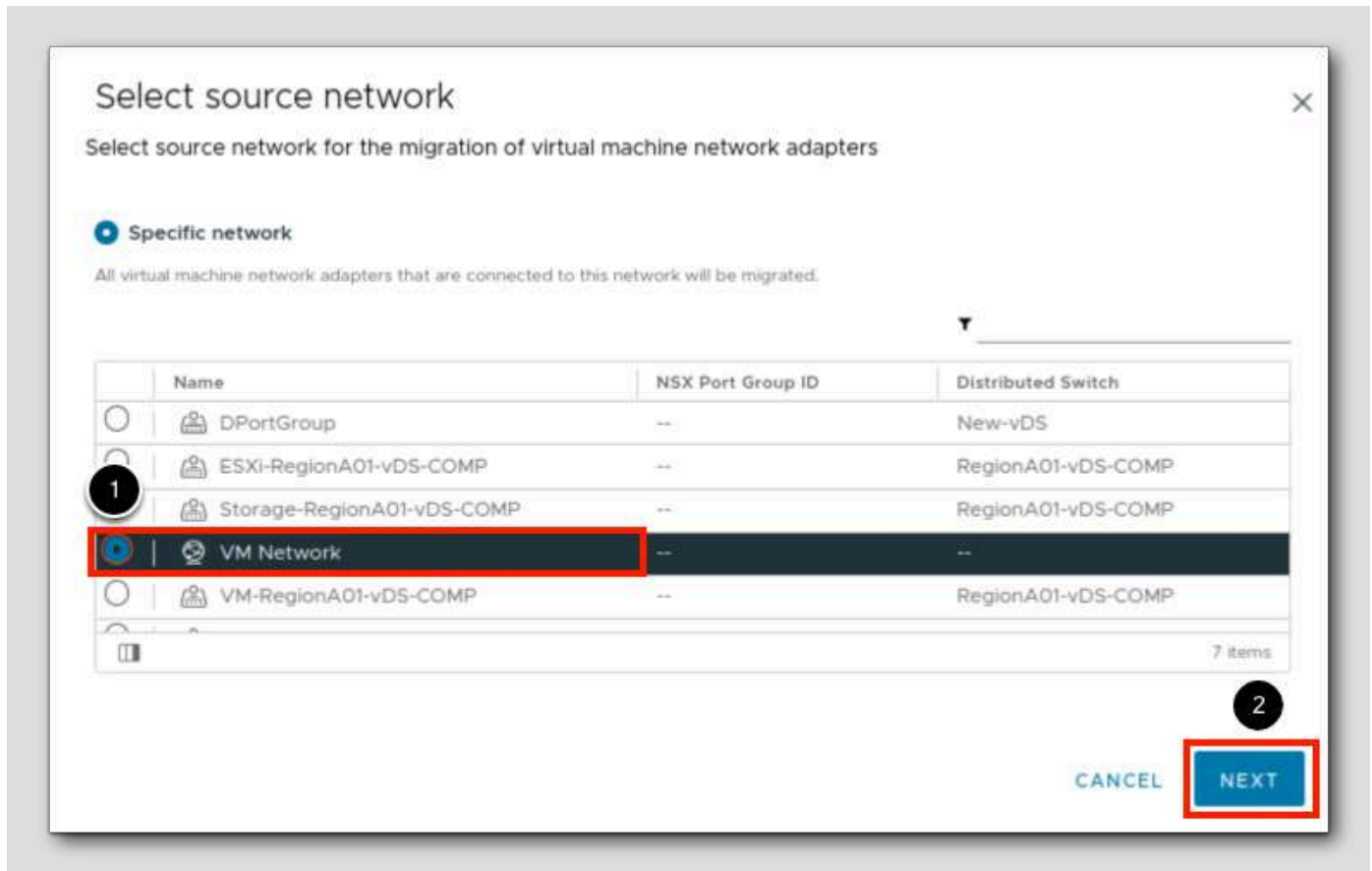
Migrate VMs

[305]



1. Right-click on RegionA01
2. Select Migrate VMs to Another Network

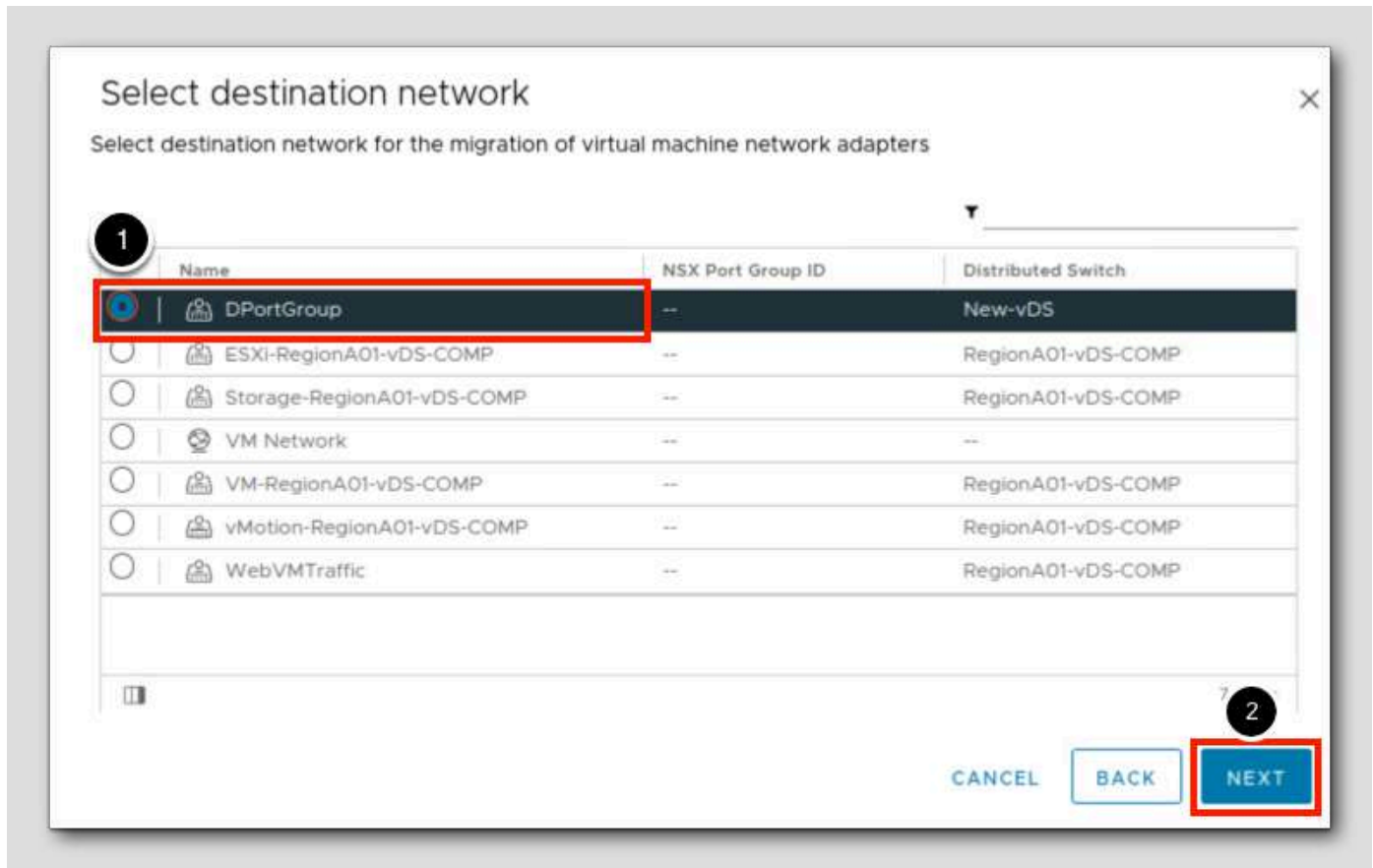
Select Source Network



1. Select **VM Network** as your source network
2. Click **Next**

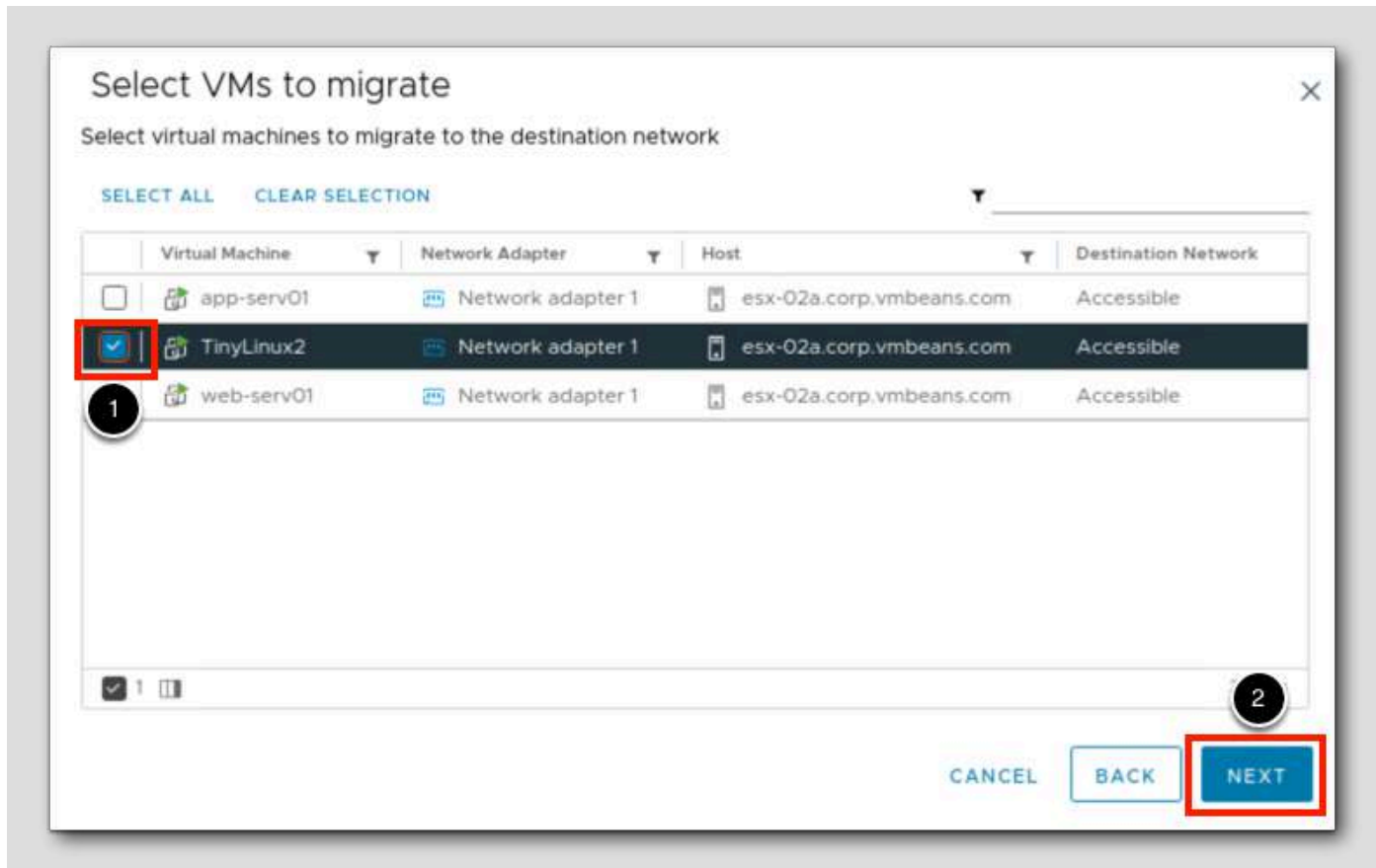
Select Destination Network

[307]



1. Select DPort Group as your destination network
2. Click Next

Select VM to Migrate



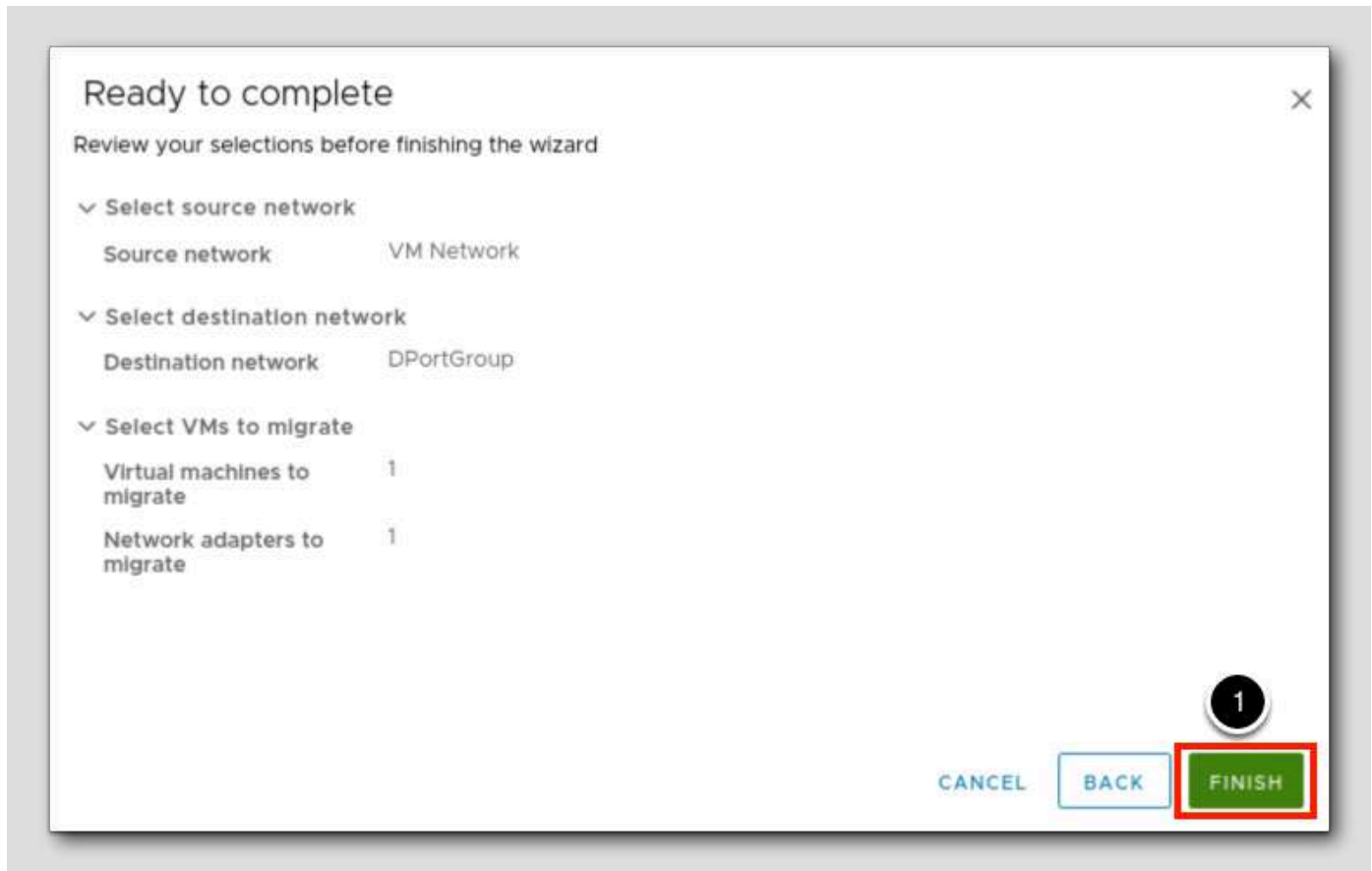
1. Click on TinyLinux2

Note that there is only one adapter associated with this VM. If there was more than one, you would have the option of choosing which one you would want to connect to the new vDS.

2. Click Next.

Ready to Complete

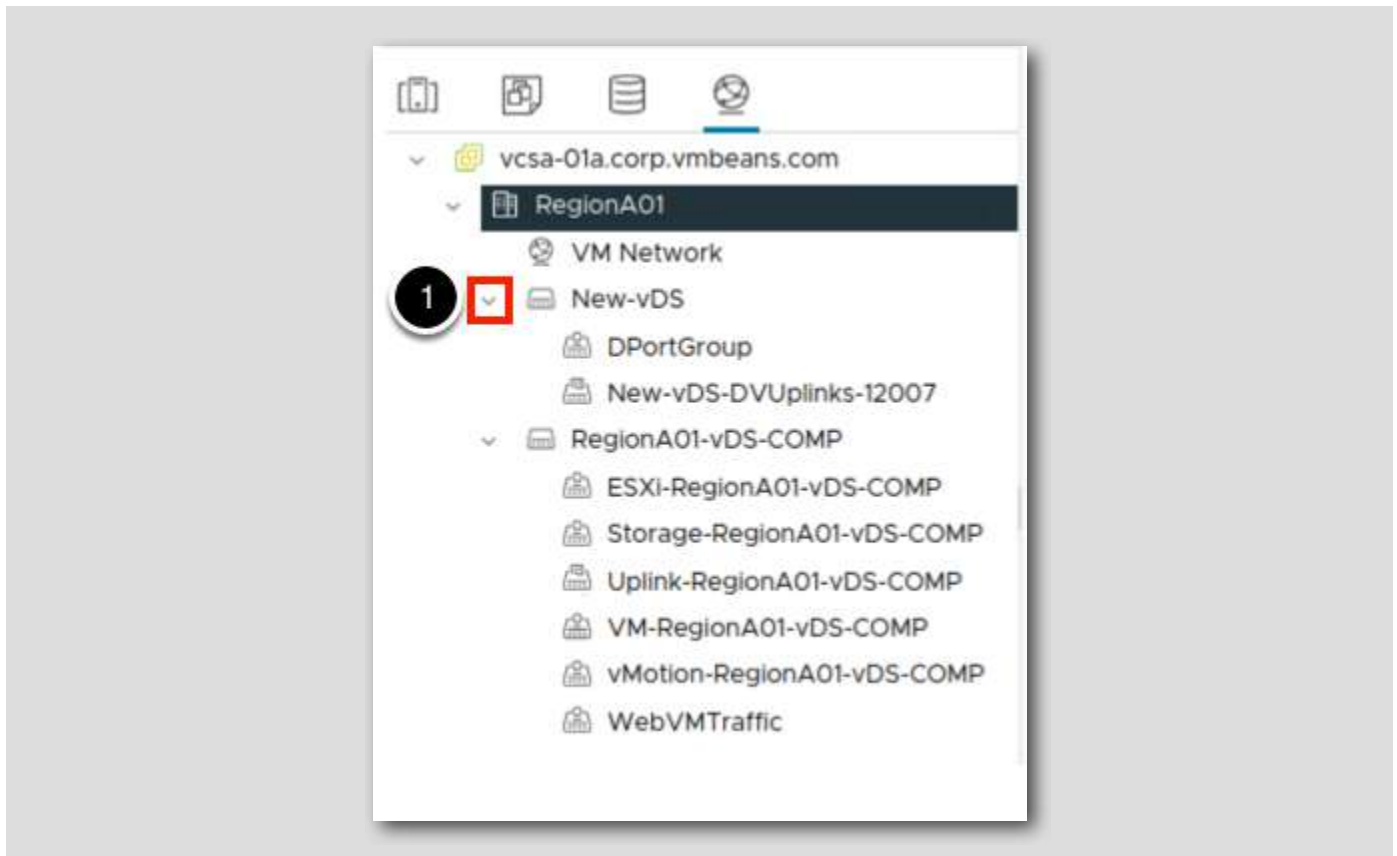
[309]



1. Click **Finish** to migrate the VM from a Standard Switch to the new Distributed Switch

Explore your Changes

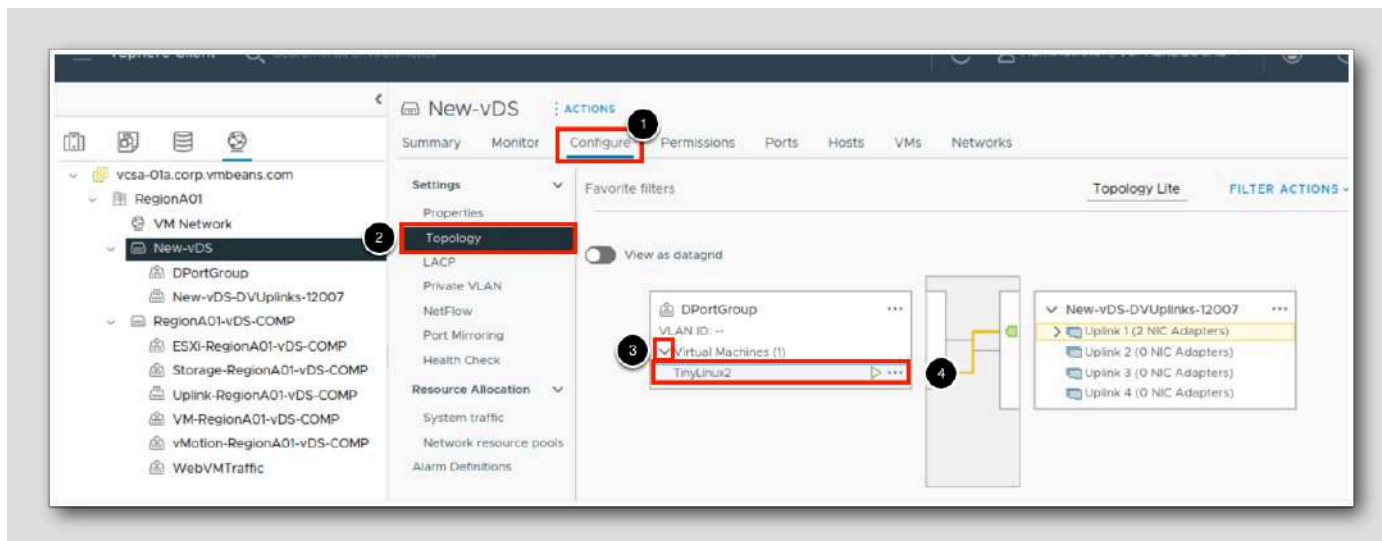
[310]



1. Click on the **New-vDS** and **expand** it to see all associated port groups and uplink

New-vDS Topology Map

[311]



1. Click **Configure**
2. Click **Topology**
3. Under **DPortGroup**, click on the drop-down **arrow** to **expand** the view
4. Select the **TinyLinux2** VM and note the highlighted path through the new vDS and Uplink.

Adding and Configuring a vSphere Distributed Switch

[312]

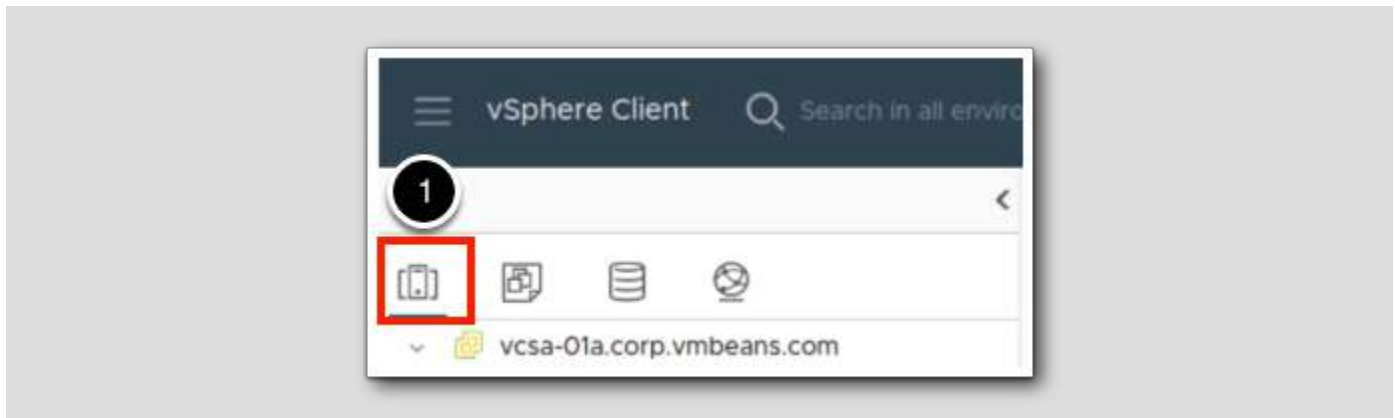
This lesson will walk you through adding and configuring a Distributed Switch.

Create a vSphere Distributed Switch on a vSphere datacenter to handle networking traffic for all associated hosts in the datacenter. If your system has many hosts and complex port group requirements, creating distributed port groups rather than a standard port groups can go a long way towards easing the administrative burden.

1. Keep the default values and click **Next**

Select Host and Clusters

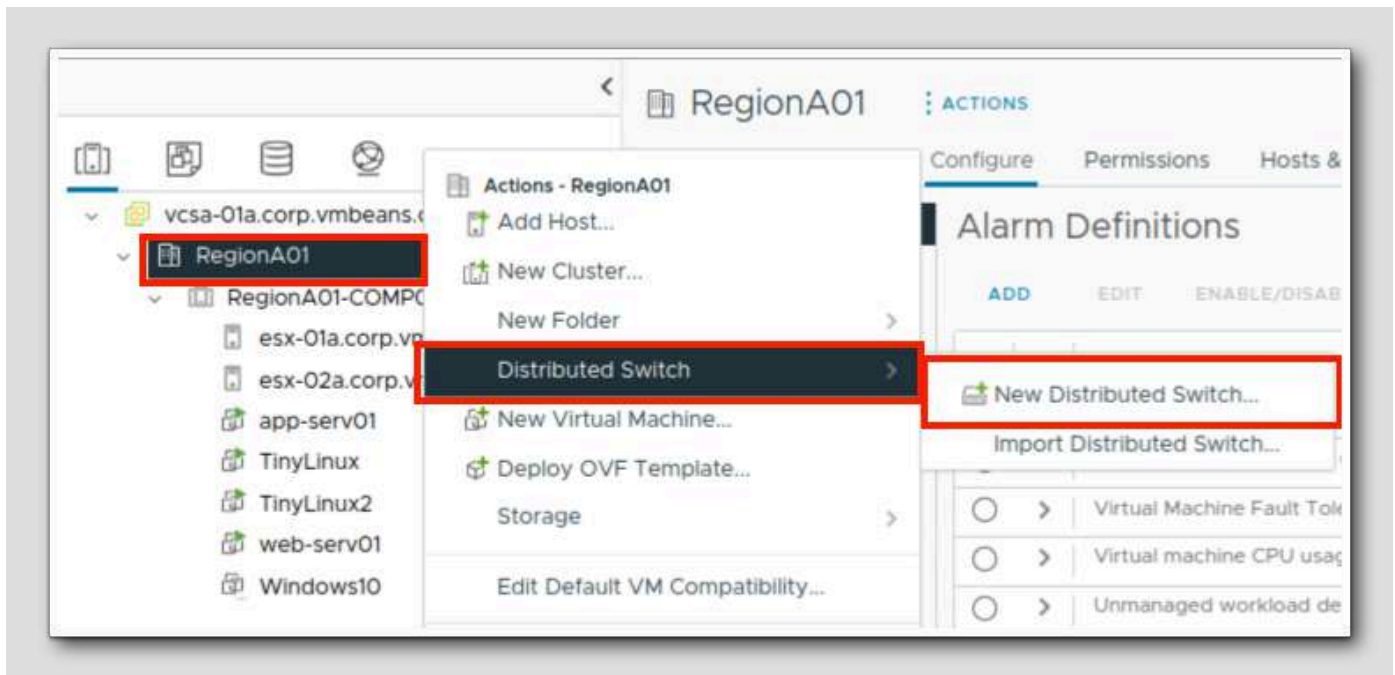
[313]



1. Select Host and Clusters

Add a vSphere Distributed Switch Using the vSphere Web Client

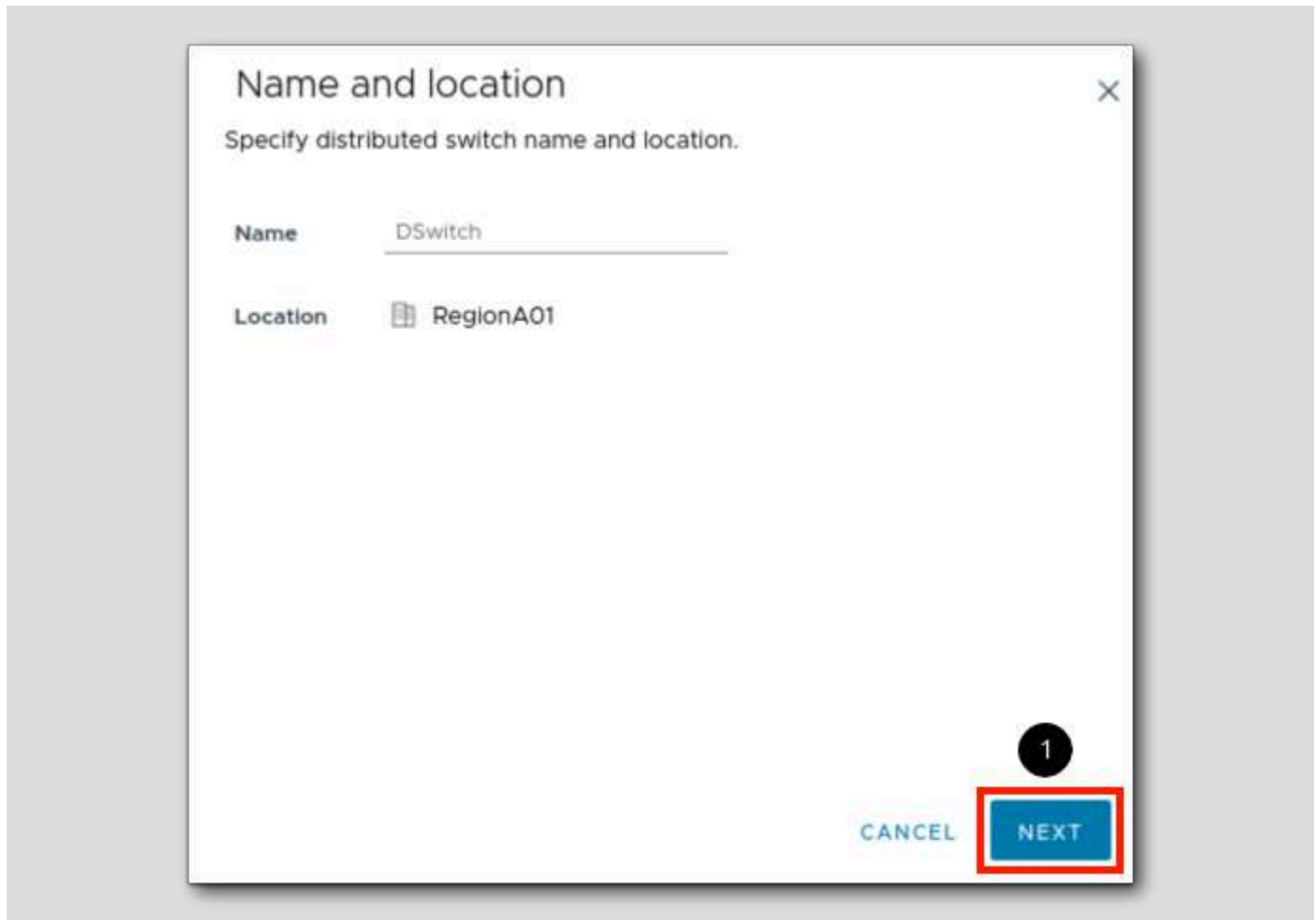
[314]



1. Under vcsa-01a.corp.vmbeans.com, right-click RegionA01
2. Select Distributed Switch and then click New Distributed Switch

Name and Location

[315]

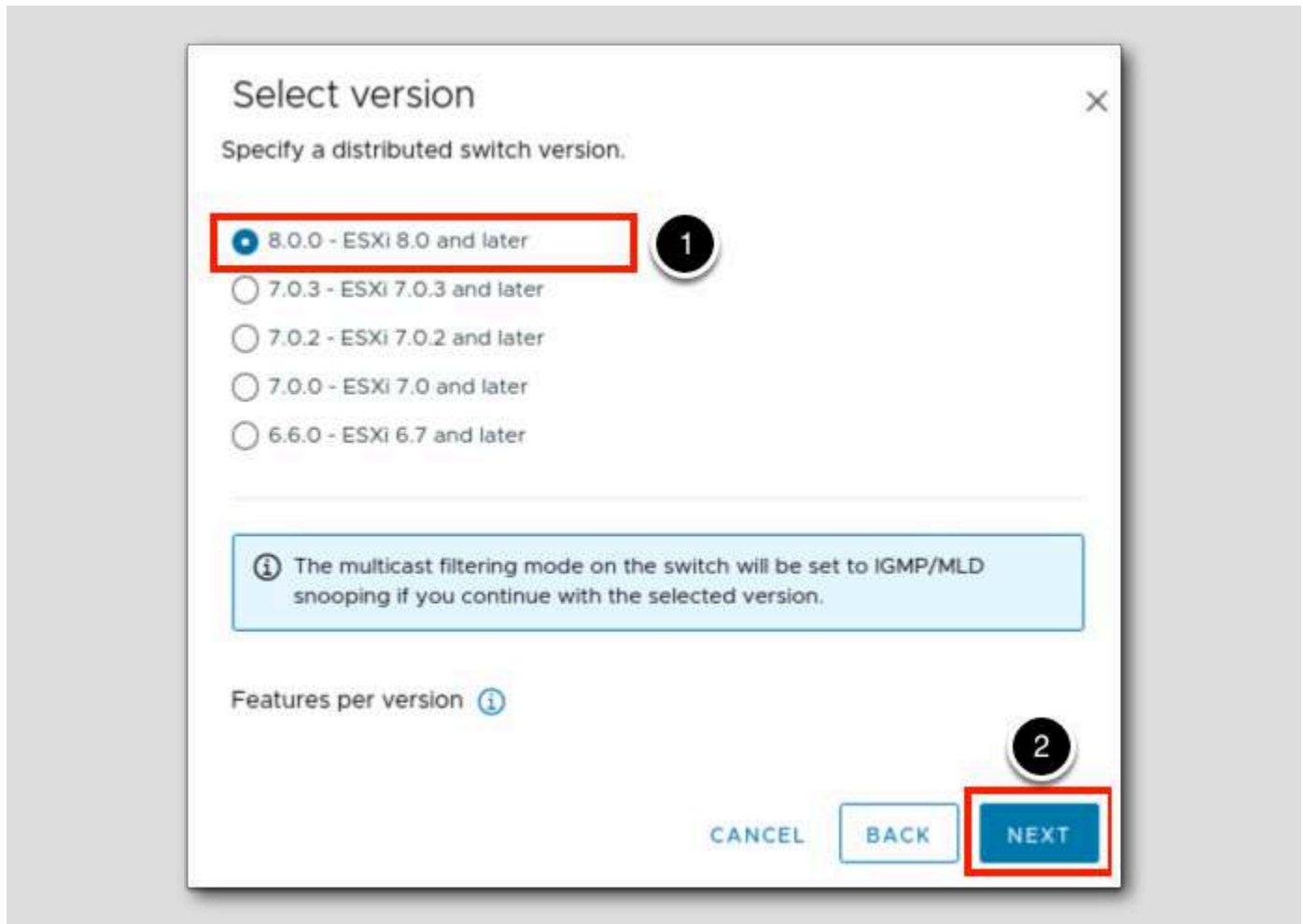


Keep the default name for the new distributed switch.

1. Click **Next**

Select Version

[316]



1. Select 8.0.0 - ESXi 8.0 and later

2. Click **Next**

Note that the version of the Distributed Switch determines which ESXi host versions are able to join the switch. Once all hosts that are a member of a Distributed Switch have been upgraded, the switch may be upgraded to the matching version.

Edit Settings

[317]

Configure settings [X]

Specify network offloads compatibility, number of uplink ports, resource allocation and default port group.

Network Offloads compatibility None ⓘ

Number of uplinks 4 ⌵⌶

Network I/O Control Enabled ⌵

Default port group ☒ Create a default port group

Port group name DPortGroup 1

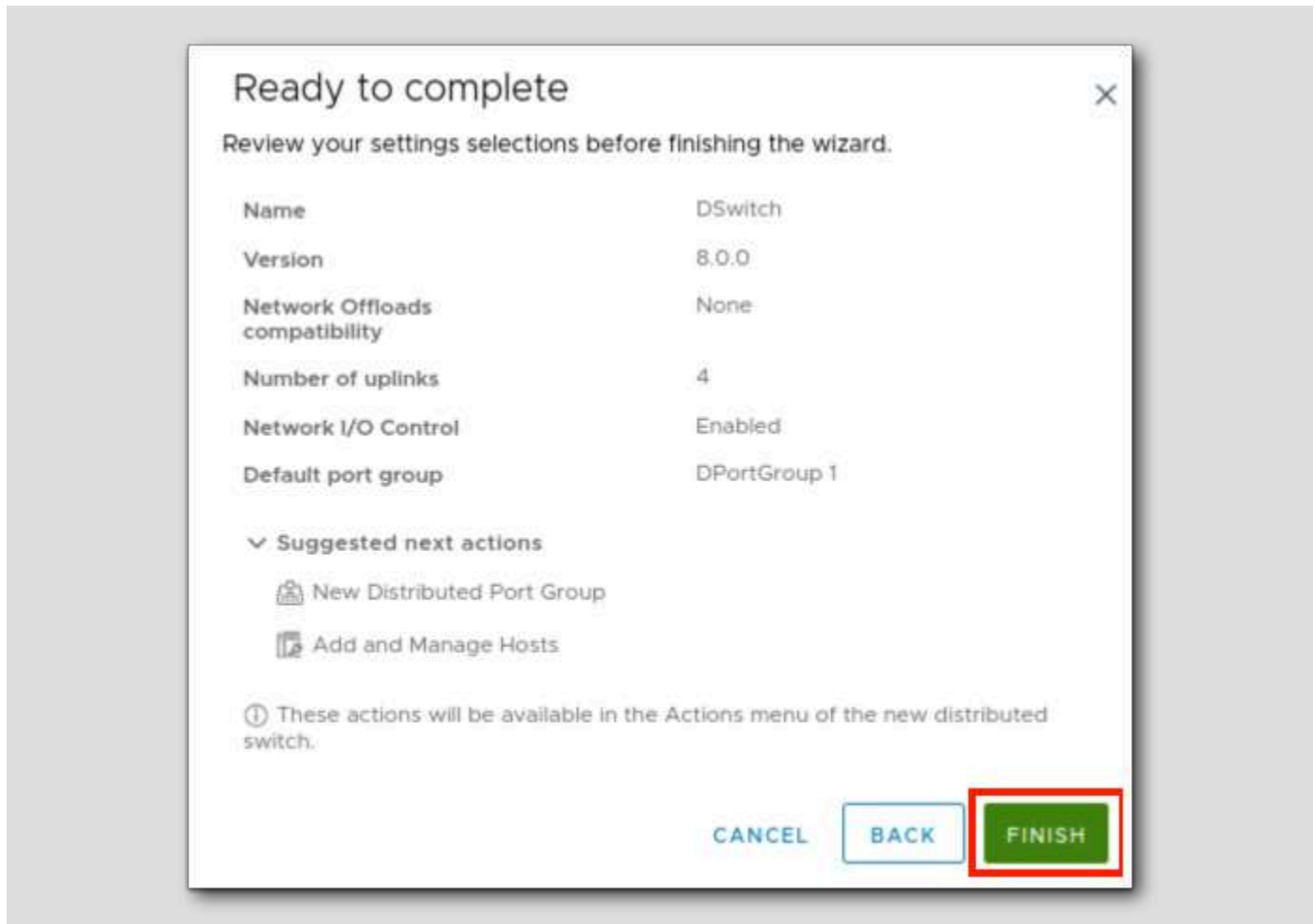
CANCEL BACK **NEXT**

1

1. Leave the default options and click Next

Ready to Complete

[318]

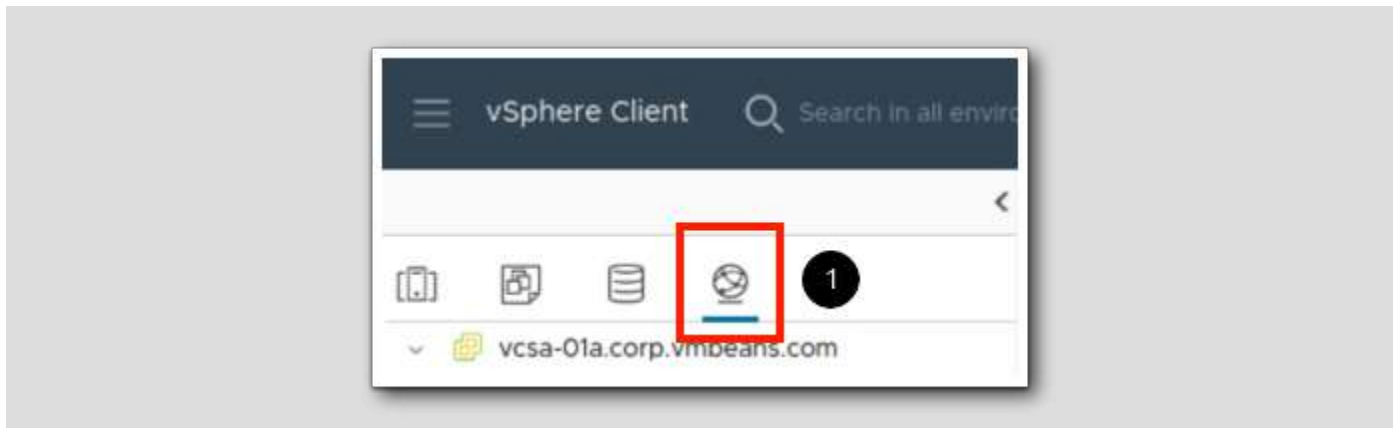


1. Review the settings and click **Finish**

Notice the next suggested steps are to create Distributed Port Groups and adding Hosts.

Add Hosts to a vSphere Distributed Switch in the vSphere Web Client

[319]

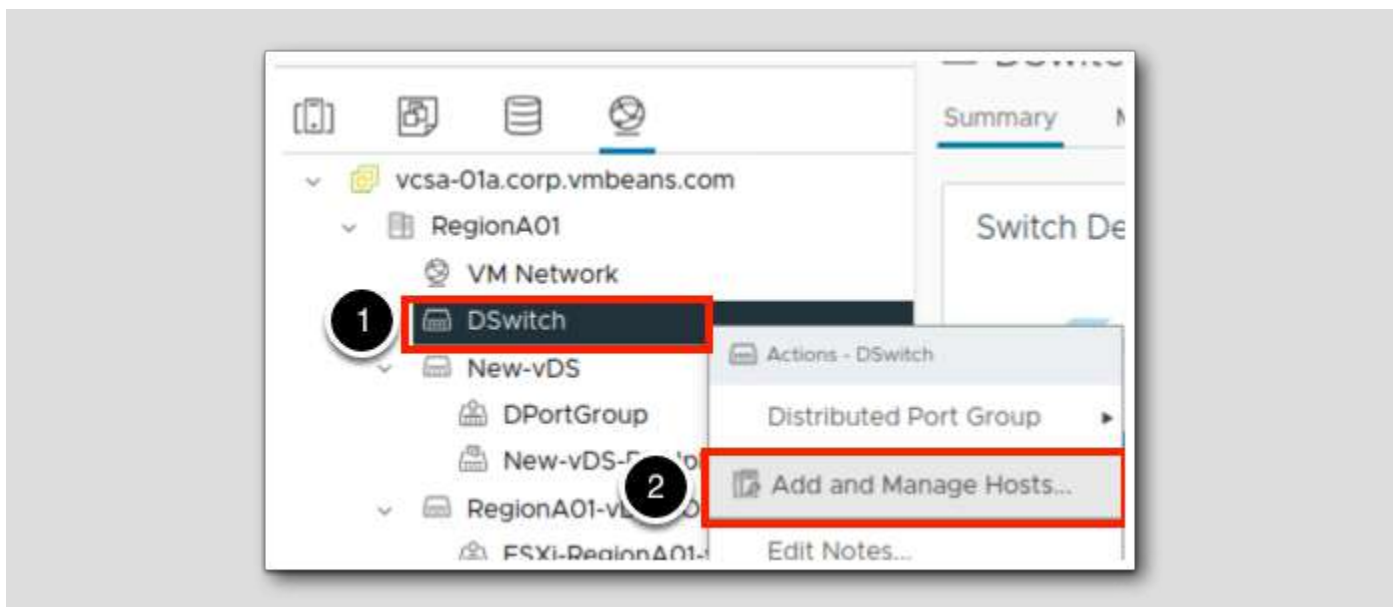


Now that we have created a vSphere distributed switch, let's add hosts and physical adapters to create a virtual network.

1. Click on the **Networking** icon

Add Hosts

[320]

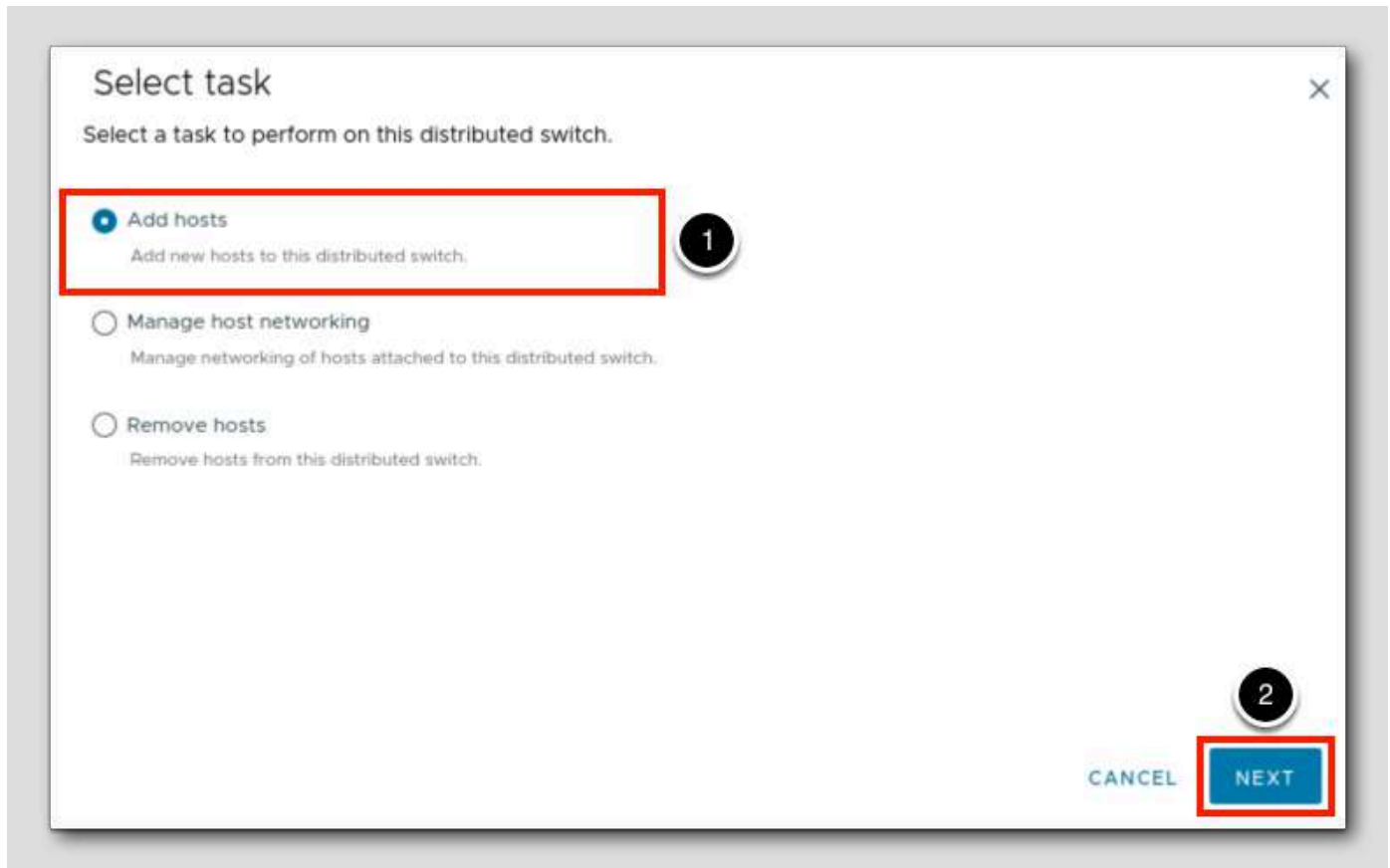


Expand **RegionA01** until you see the Distributed Switch we just created, **DSwitch**.

1. Right-click on **DSwitch**
2. Select **Add and Manage Hosts...**

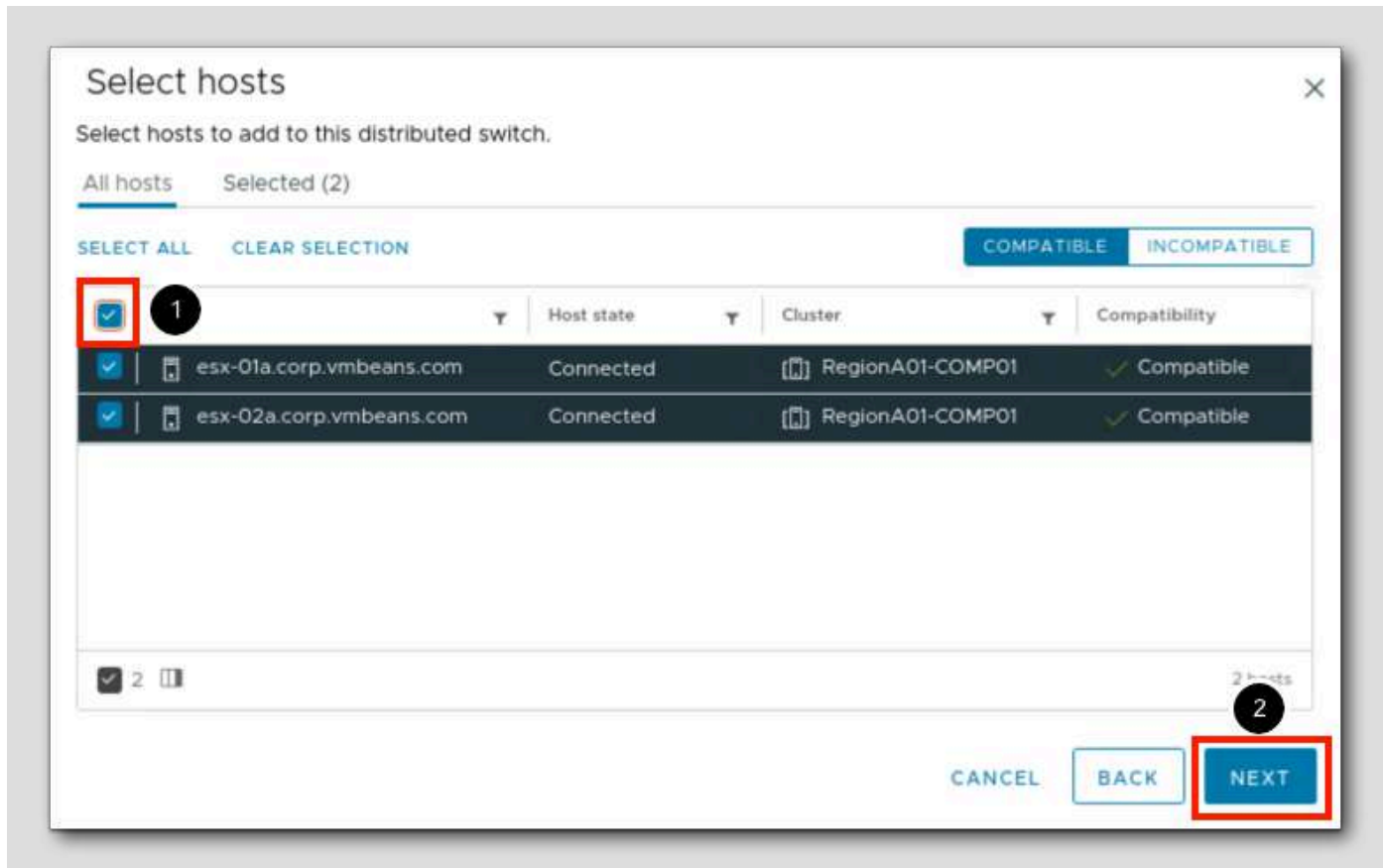
Select Task

[321]



1. Select **Add hosts**
2. Click **Next**

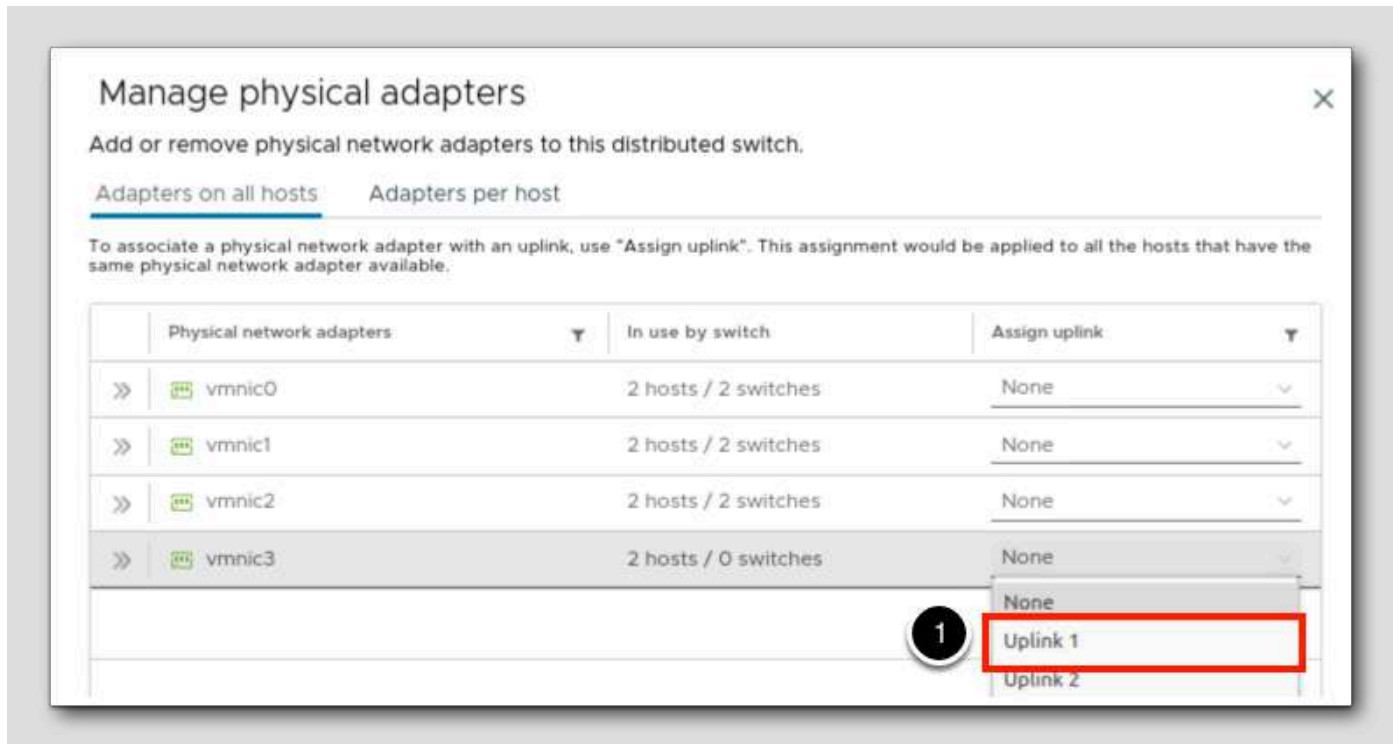
Select Hosts



1. Select all hosts, in this case, it would be esx-01a.corp.vmbeans.com and esx-02a.corp.vmbeans.com
2. Click Next

Assign Physical Adapters

[323]



On the Manage physical network adapters page, we want to configure which physical NICs will be used on the distributed switch.

1. Click Assign uplink drop-down menu for vmnic3 and select Uplink1

Review Settings

Manage physical adapters

Add or remove physical network adapters to this distributed switch.

Adapters on all hosts Adapters per host

To associate a physical network adapter with an uplink, use "Assign uplink". This assignment would be applied to all the hosts that have the same physical network adapter available.

Physical network adapters	In use by switch	Assign uplink
>> vmnic0	2 hosts / 2 switches	None
>> vmnic1	2 hosts / 2 switches	None
>> vmnic2	2 hosts / 2 switches	None
>> vmnic3	This switch	Uplink 1

4 physical network adapters

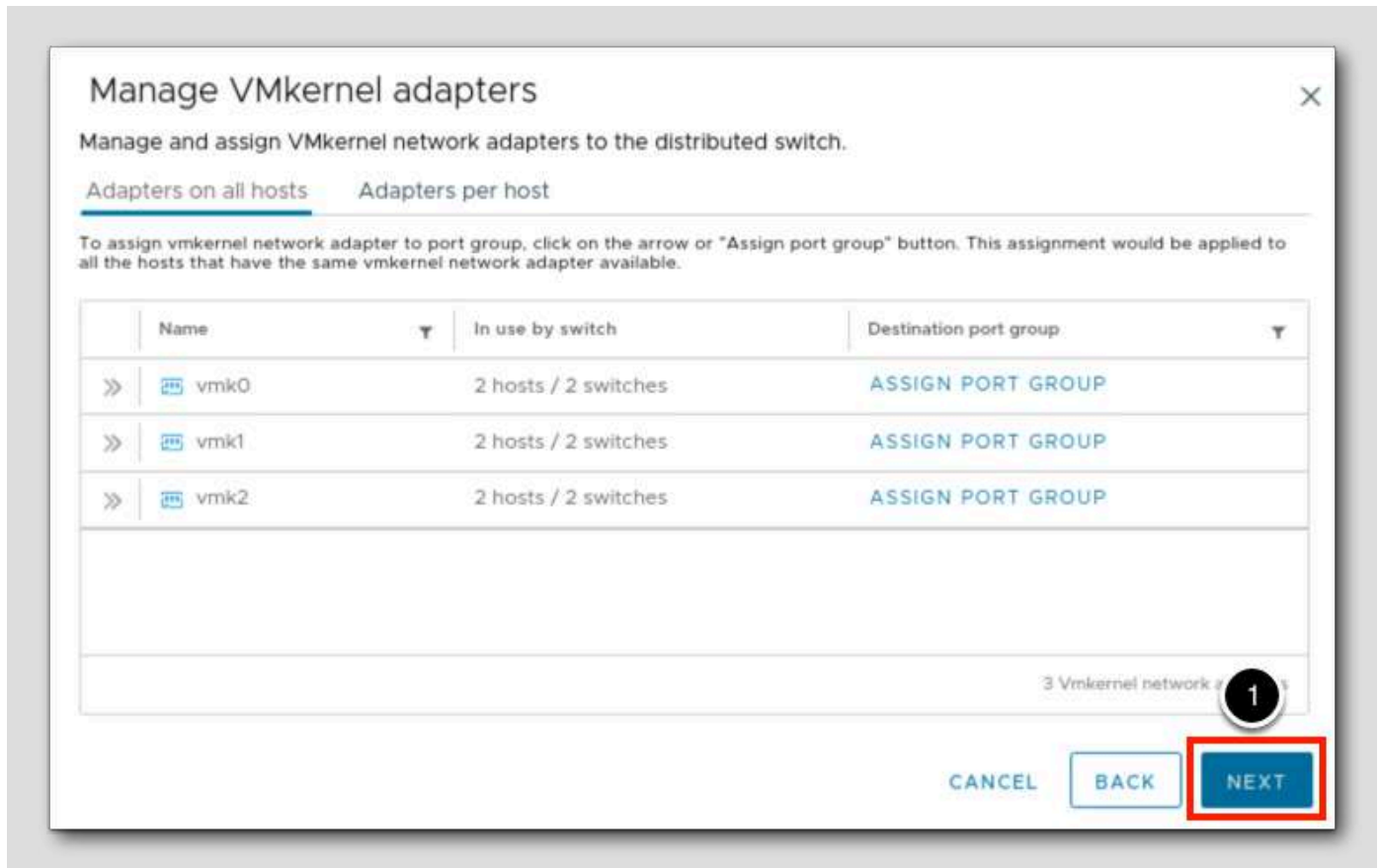
1

CANCEL BACK **NEXT**

1. Review vmnic and uplink settings for the hosts you are adding and click **Next**

Manage VMkernel Adapters

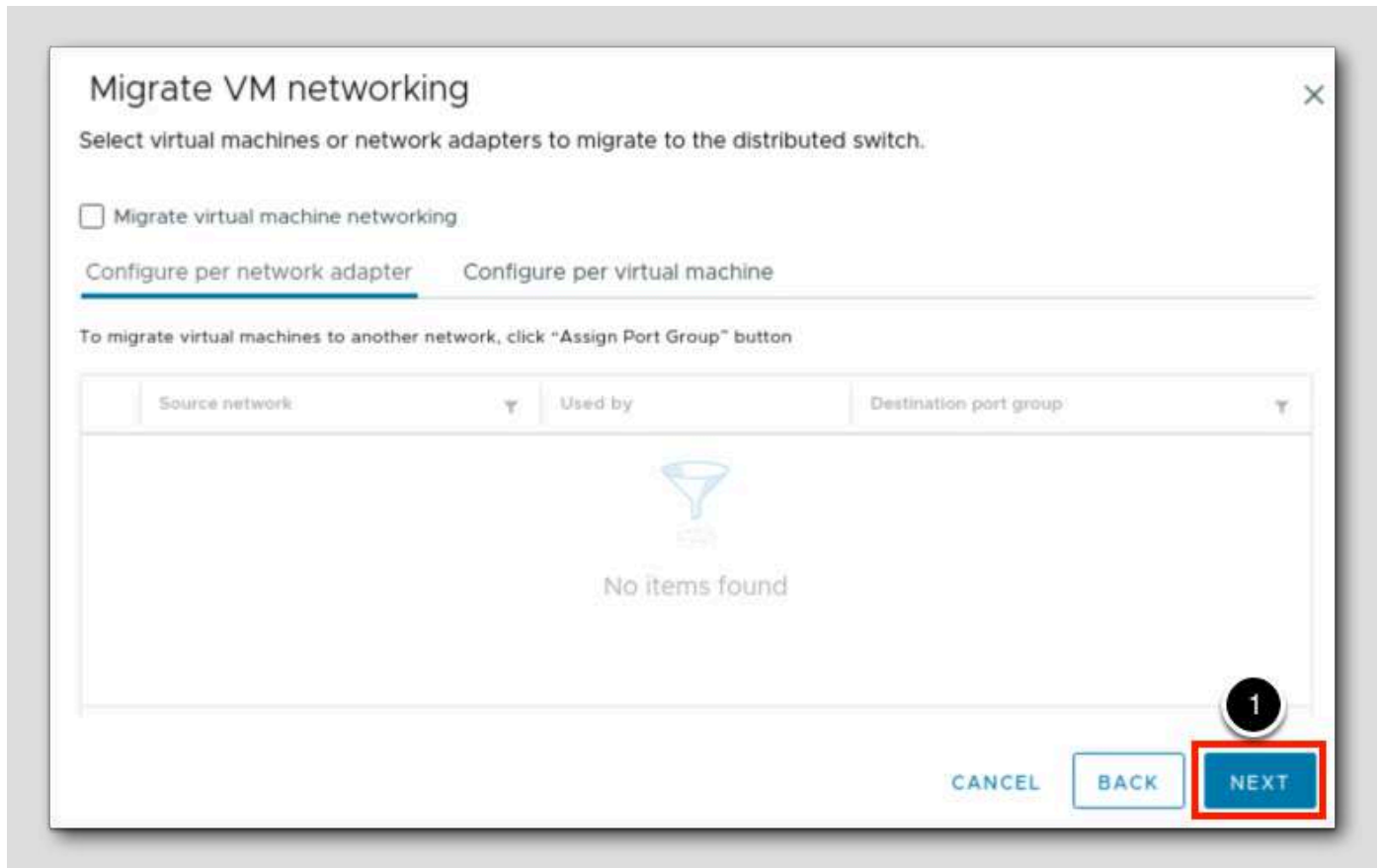
[325]



1. Since we will not be using this distributed switch for any VMkernel functions, click Next

Migrate VM Networking

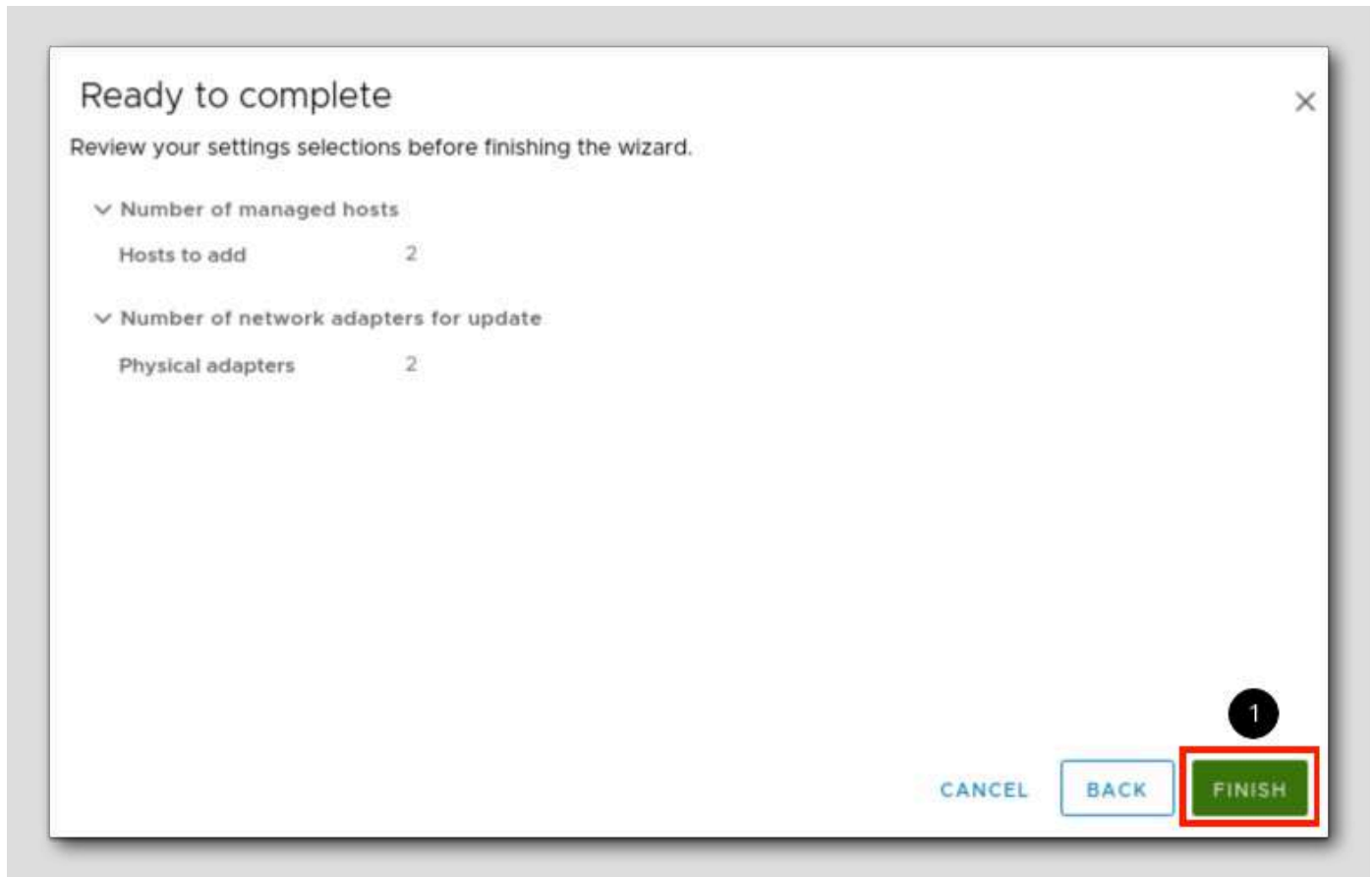
[326]



1. Click Next

Ready to Complete

[327]

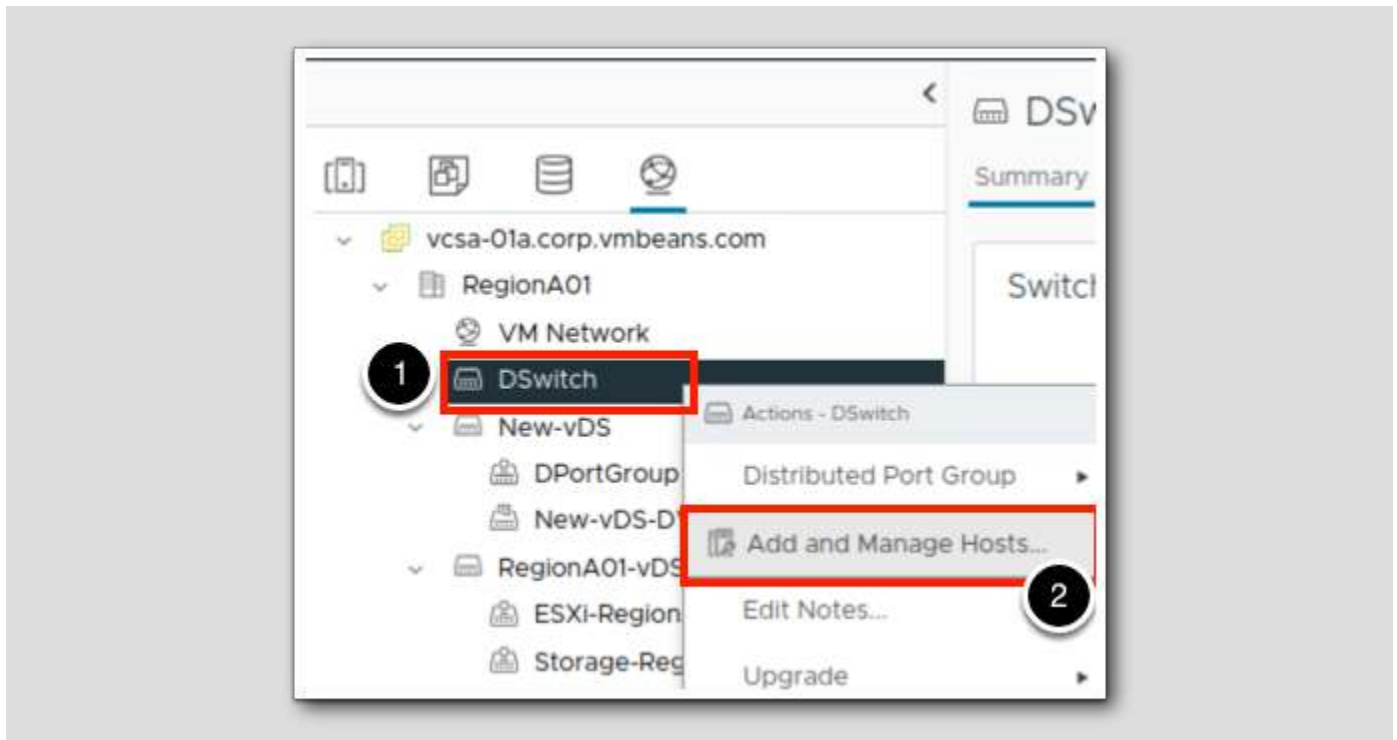


You are now asked to verify the changes you are about to make.

1. Click **Finish** to commit the changes

Manage Hosts on a vSphere Distributed Switch in the vSphere Web Client

[328]

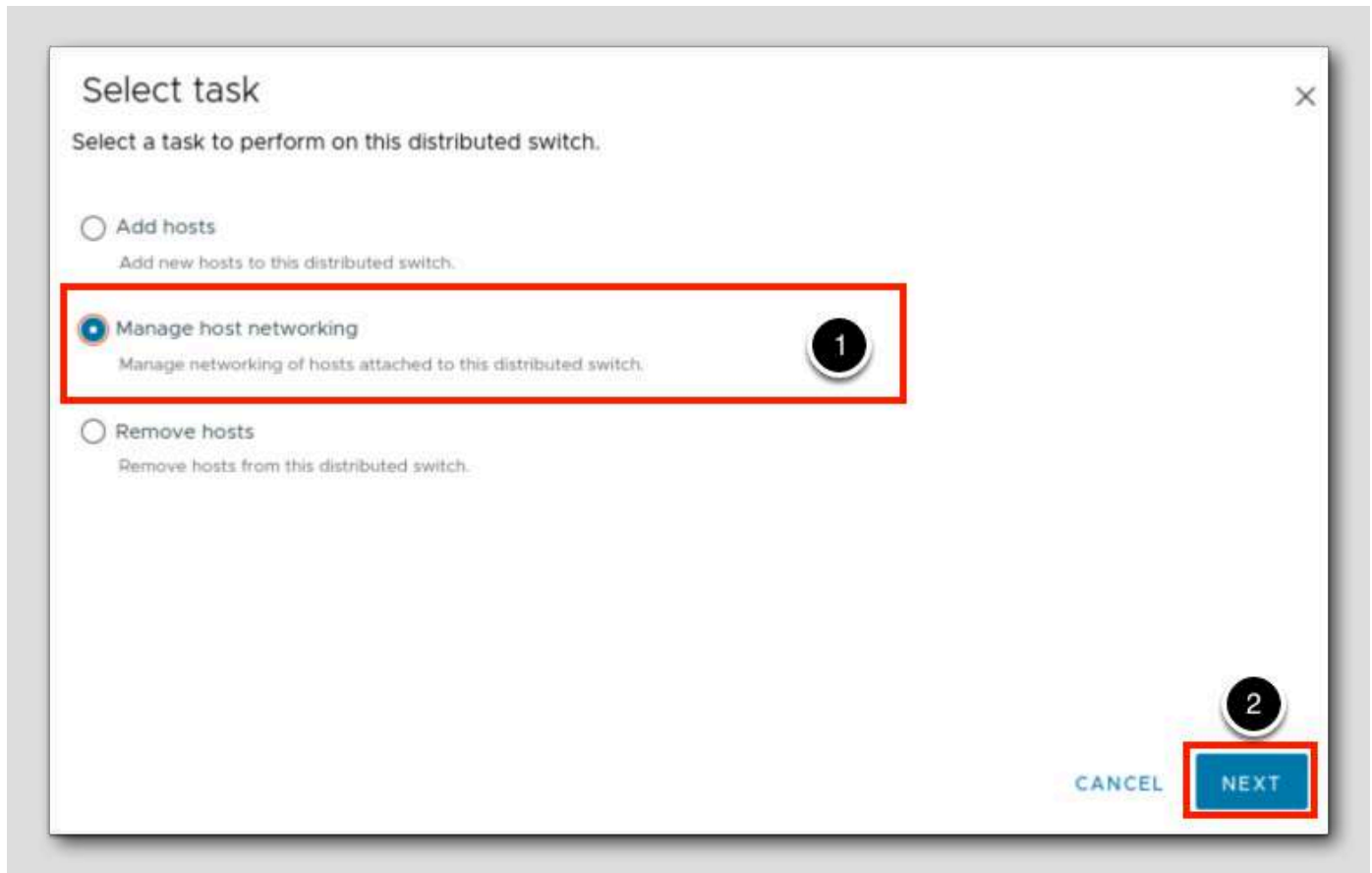


You can change the configuration for hosts and physical adapters on a vSphere Distributed Switch after they are added to the distributed switch.

1. Right-click DSwitch in the navigator
2. Select Add and Manage Hosts.

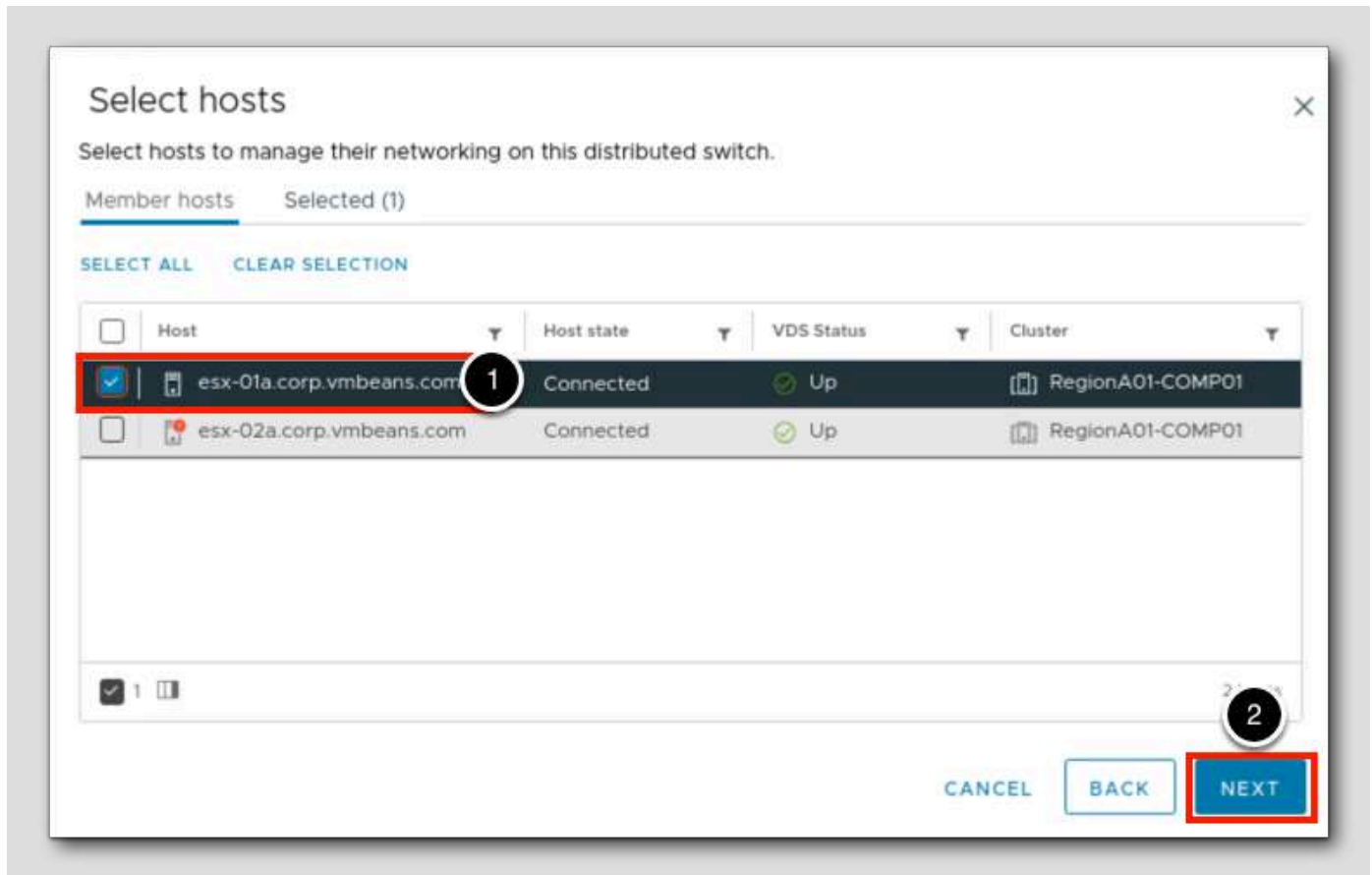
Select Task

[329]



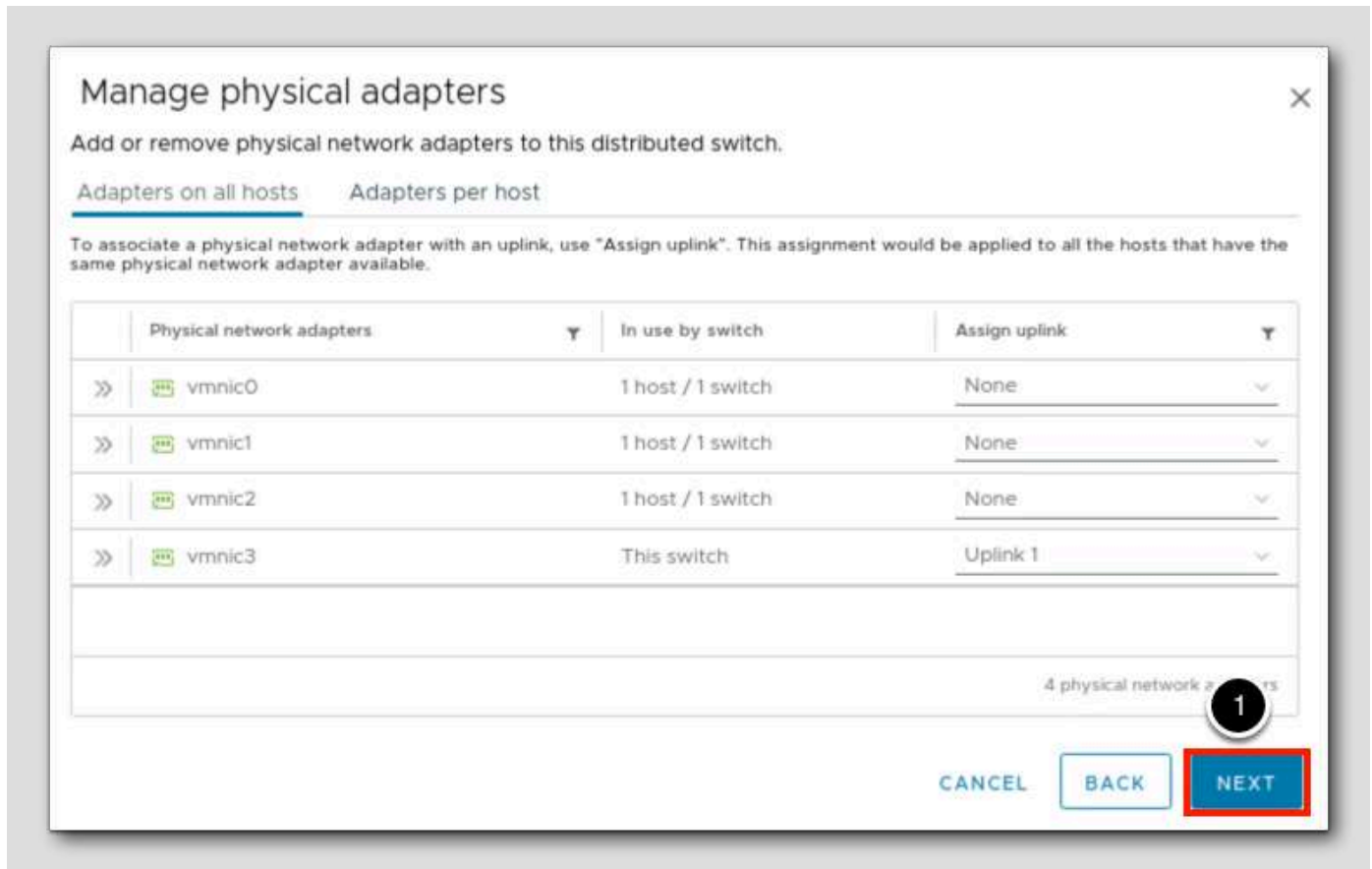
1. On the 'Select tasks' page, select **Manage host networking**
2. Click **Next**

Select Member Hosts



1. On the "Select member hosts" page, select esx-01a.corp.vmbeans.com
2. Click OK

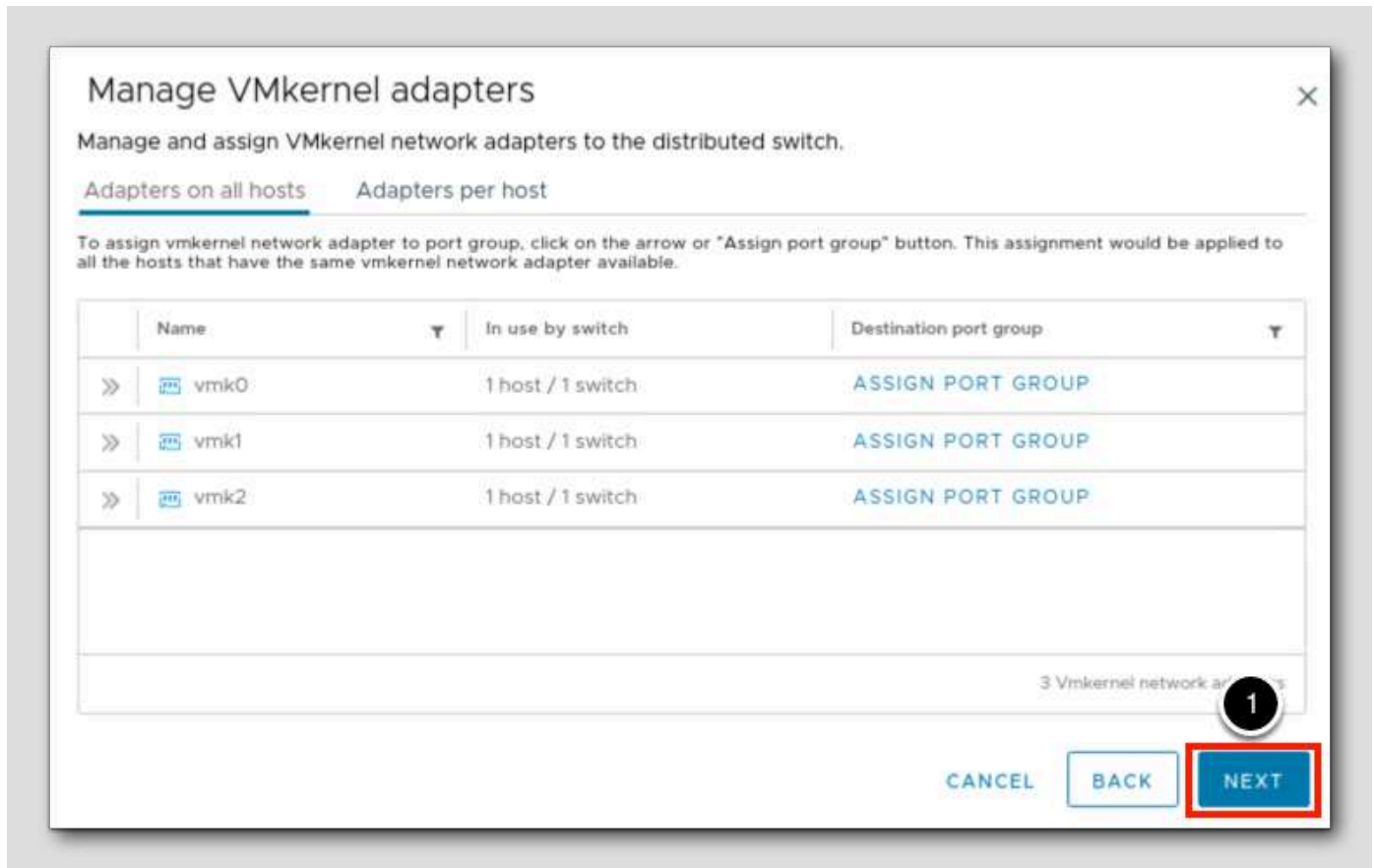
Manage Physical Network Adapters



1. Click **Next** to continue

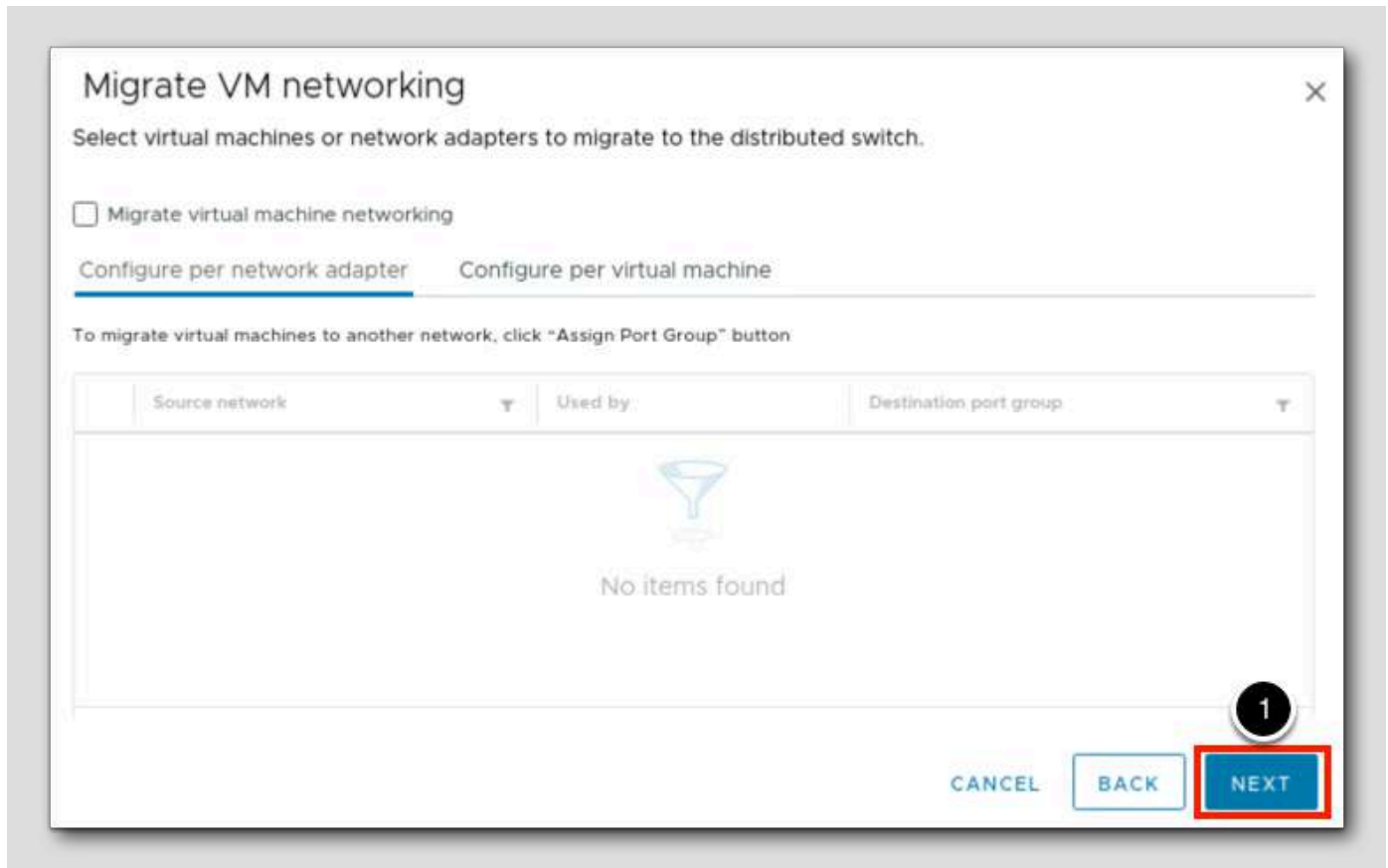
Manage VMkernel Adapters

[332]



Migrate VM Networking

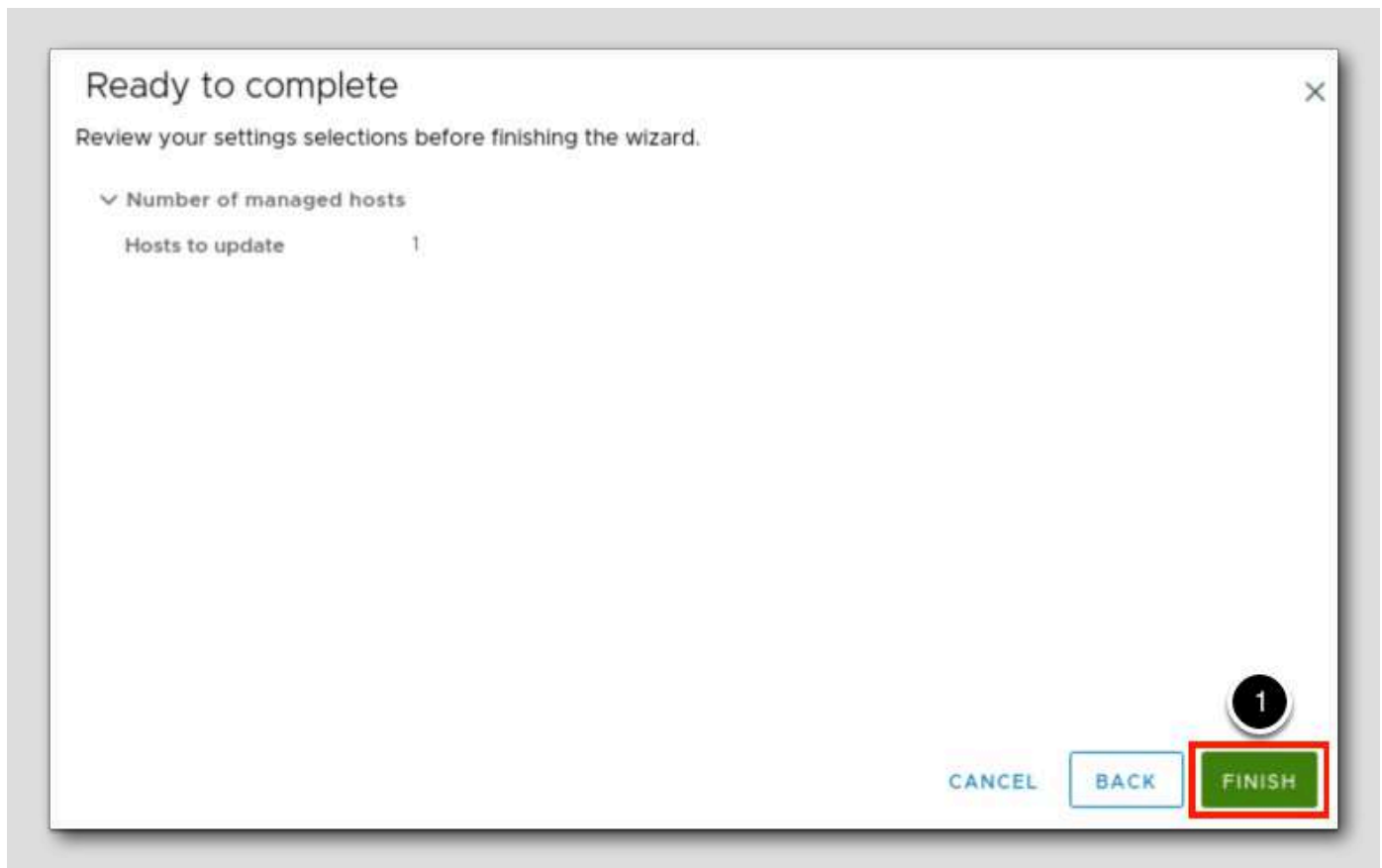
[333]



1. Click **Next** to continue

Ready to Complete

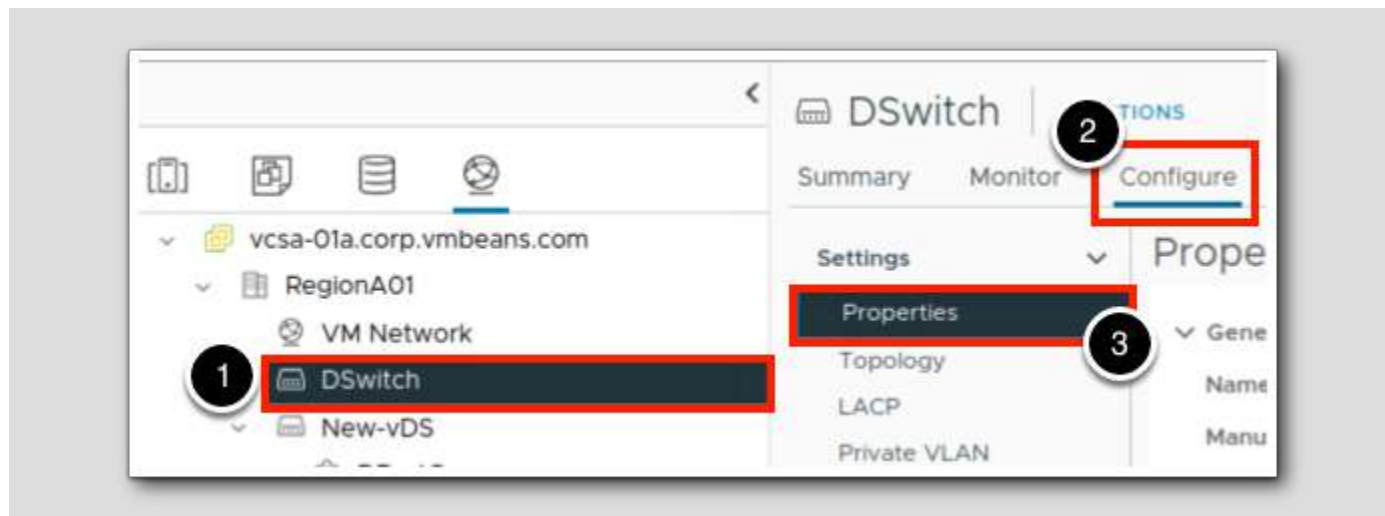
[334]



1. Click Finish

Edit General and Advanced vSphere Distributed Switch Settings in the vSphere Web Client

[335]

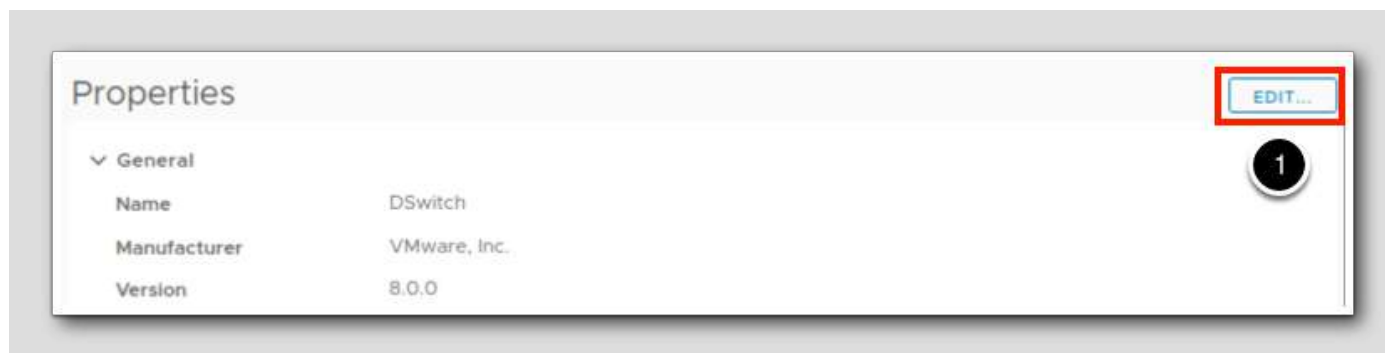


General settings for a vSphere Distributed Switch include the distributed switch name and the number of uplink ports on the distributed switch. Advanced settings for a vSphere Distributed Switch include the Discovery Protocol configuration and the maximum MTU for the switch. Both general and advanced settings can be configured using the vSphere Web Client.

1. Make sure the DSwitch is selected under the Navigator pane
2. Click the Configure tab
3. Click Properties, under Settings

Edit the Switch Properties

[336]



1. Click Edit

General Settings

[337]

Distributed Switch - Edit Settings | DSwitch X

General Advanced Uplinks

Name DSwitch

Network offloads compatibility None ⓘ

Network I/O Control Enabled ▾

Description

CANCEL OK

Click General to view the vSphere distributed switch settings. Here you can modify the following:

Name: You can modify the name of your distributed switch.

Number of Uplinks: Increase or decrease the number uplink ports attached to the distributed switch. Note that you can also click the Edit uplink names button to give the uplinks meaningful names.

Number of Ports: This setting cannot be modified. The port count will dynamically be scaled up or down by default.

Network I/O Control: You can use the drop-down menu to enable or disable Network I/O Control on the switch.

Description: You can use this field to give a meaningful description of the switch.

Advanced Settings

[338]

Distributed Switch - Edit Settings | DSwitch X

General **Advanced** Uplinks

MTU (Bytes) 1500

Multicast filtering mode IGMP/MLD snooping

Discovery protocol

Type Cisco Discovery Protocol

Operation Listen

Administrator contact

Name

Other details

CANCEL **OK**

1. Click Advanced to view the vSphere distributed switch settings. Here you will find the following advanced settings for the switch:

MTU (Bytes): Maximum MTU size for the vSphere Distributed Switch. To enable jumbo frames, set a value greater than 1500 bytes. Make sure you check with your Networking team prior to modifying this setting in your environment.

Multicast filtering mode

- Basic - The distributed switch forwards traffic that is related to a multicast group based on a MAC address generated from the last 23 bits of the IPv4 address of the group.
- IGMP/MLD snooping - The distributed switch forwards multicast traffic to virtual machines according to the IPv4 and IPv6 addresses of subscribed multicast groups by using membership messages defined by the Internet Group Management Protocol (IGMP) and Multicast Listener Discovery protocol.

Discovery Protocol

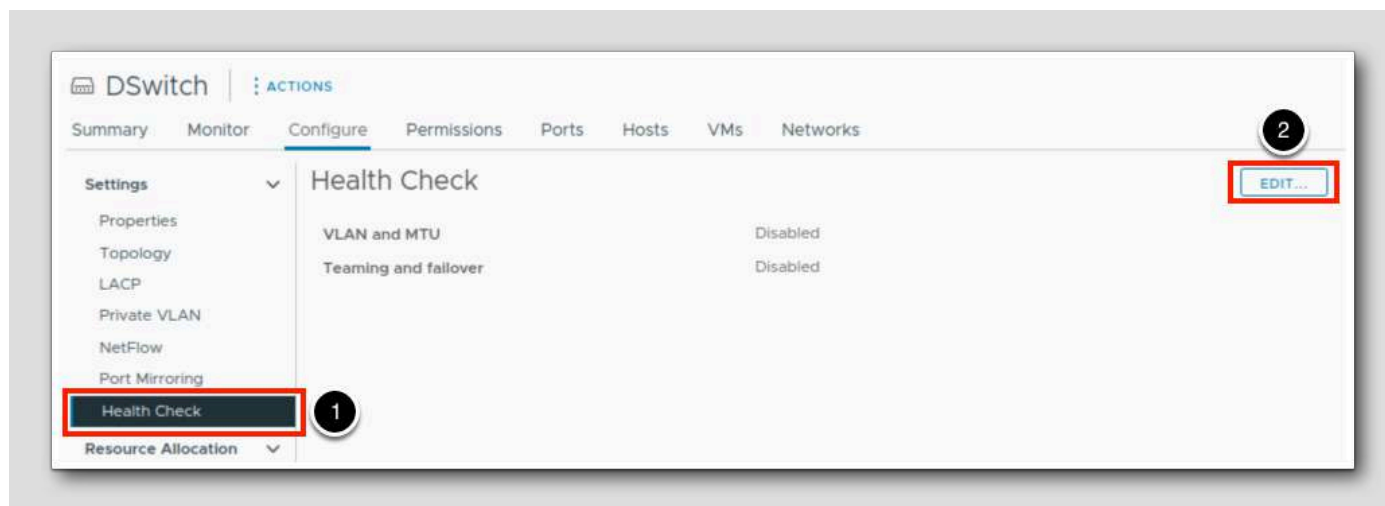
- Type - Cisco Discovery Protocol, Link Layer Discovery Protocol, or disabled..
- Operation - to Listen, Advertise, or Both.

Administrator Contact: Type the name and other details of the administrator for the distributed switch.

1. We don't want to make any changes here, just click **Cancel**.

Enable or Disable vSphere Distributed Switch Health Check in the vSphere Web Client

[339]



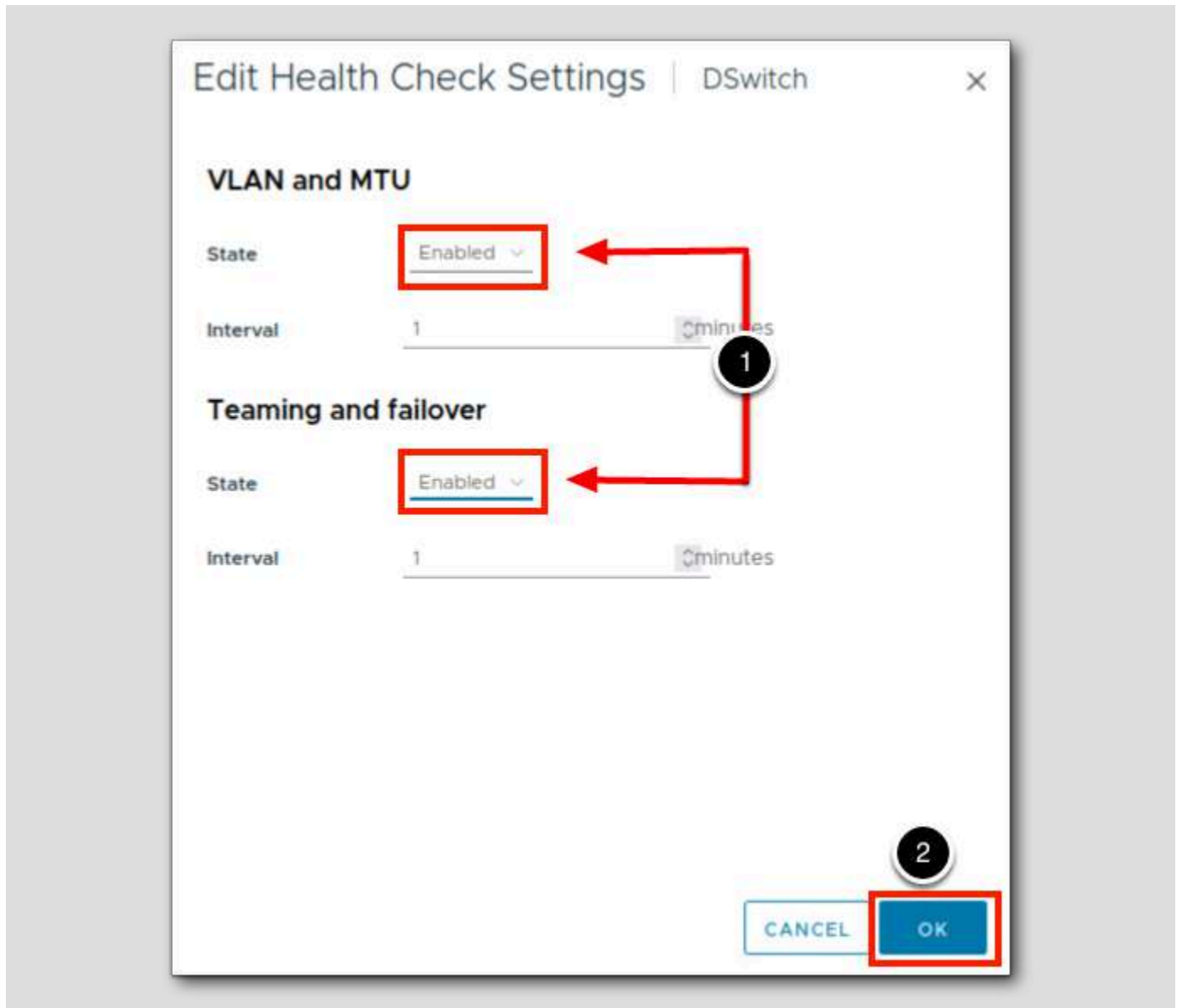
The Distributed Switch Health Check monitors for changes in vSphere Distributed Switch configurations. You must enable vSphere Distributed Switch Health Check to perform checks on Distributed Switch configurations.

Health Check is available on ESXi 5.1 Distributed Switches and higher. Also, you can only view Health Check information through the vSphere Web Client 5.1 or later.

1. Click on the **Health Check** tab for DSwitch. We can see that Health check is disabled for VLAN and MTU as well as Teaming and failover.
2. Click the **Edit** button

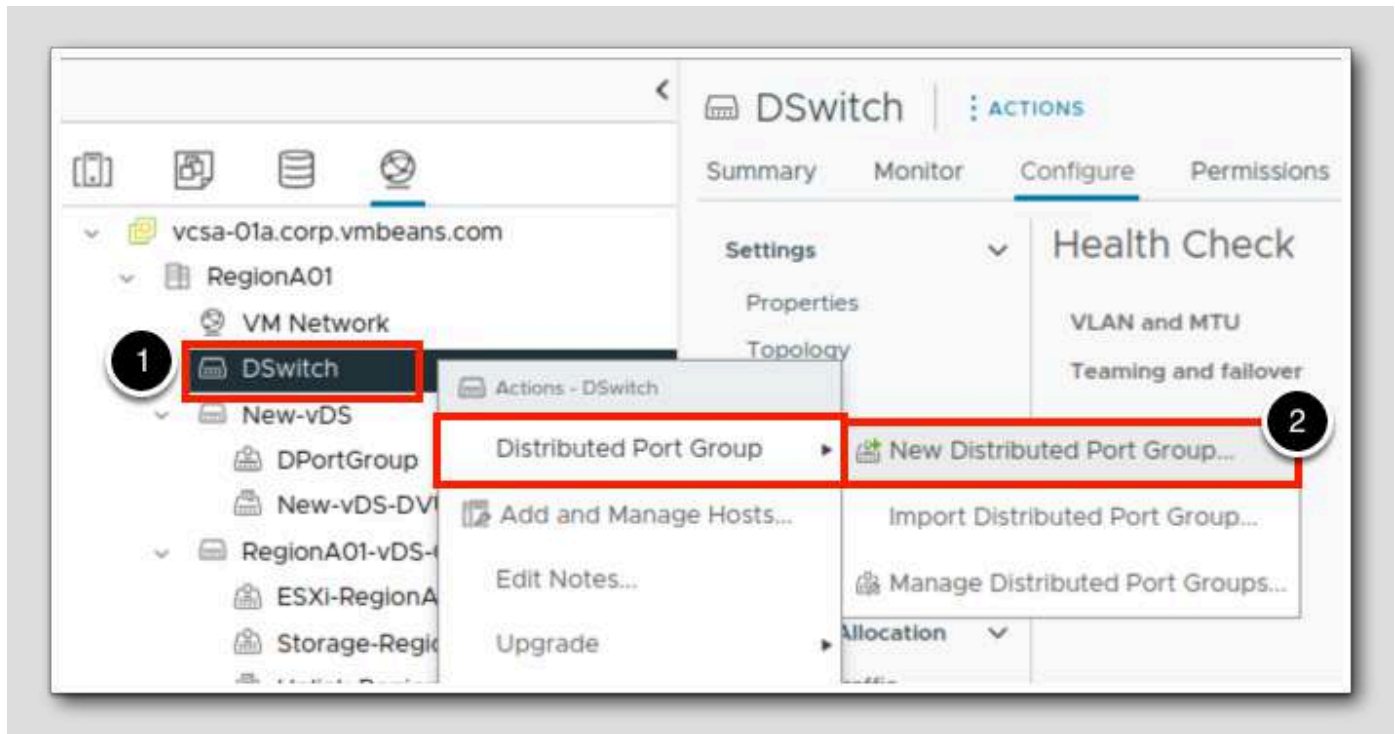
Edit Health Check Settings

[340]



1. Select **Enabled** for both
2. Click **OK**

Distributed Port Groups



A distributed port group specifies port configuration options for each member port on a vSphere distributed switch. Distributed port groups define how a connection is made to a network.

1. Right-click the DSwitch in the navigator
2. Select **Distributed Port Group** and then click **New Distributed Port Group**

Select Name and Location Section

[342]

Name and location [X]

Specify distributed port group name and location.

Name 1

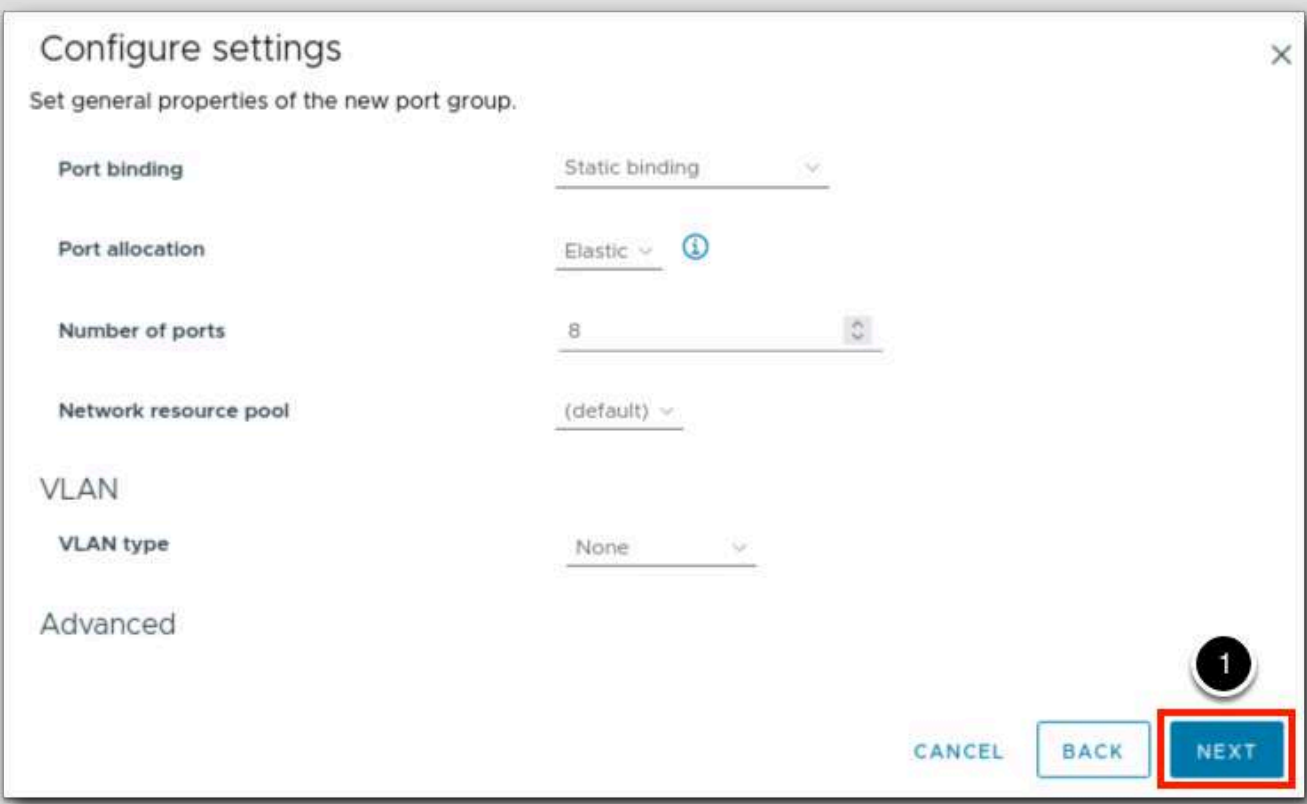
Location DSwitch

CANCEL NEXT 2

1. Name the new port group **WebVMTraffic2**
2. Click **Next**

Configure Settings

[343]



Configure settings ✕

Set general properties of the new port group.

Port binding Static binding ▼

Port allocation Elastic ▼ ⓘ

Number of ports 8 ↕

Network resource pool (default) ▼

VLAN

VLAN type None ▼

Advanced

CANCEL BACK **NEXT** 1

1. Keep default settings and click **Next**

When creating a Distributed Port Group, you have the following options available:

Port binding - Choose when ports are assigned to virtual machines connected to this distributed port group.

- Static binding - Assign a port to a virtual machine when the virtual machine connects to the distributed port group.
- Dynamic binding - Assign a port to a virtual machine the first time the virtual machine powers on after it is connected to the distributed port group. Dynamic binding has been deprecated since ESXi 5.0.
- Ephemeral - No port binding. You can assign a virtual machine to a distributed port group with ephemeral port binding also when connected to the host.

Port allocation

- Elastic - The default number of ports is eight. When all ports are assigned, a new set of eight ports is created. This is the default.
- Fixed - The default number of ports is set to eight. No additional ports are created when all ports are assigned.

Number of ports: Enter the number of ports on the distributed port group.

Network resource pool: If you have created network pool to help control network traffic, you can select it here.

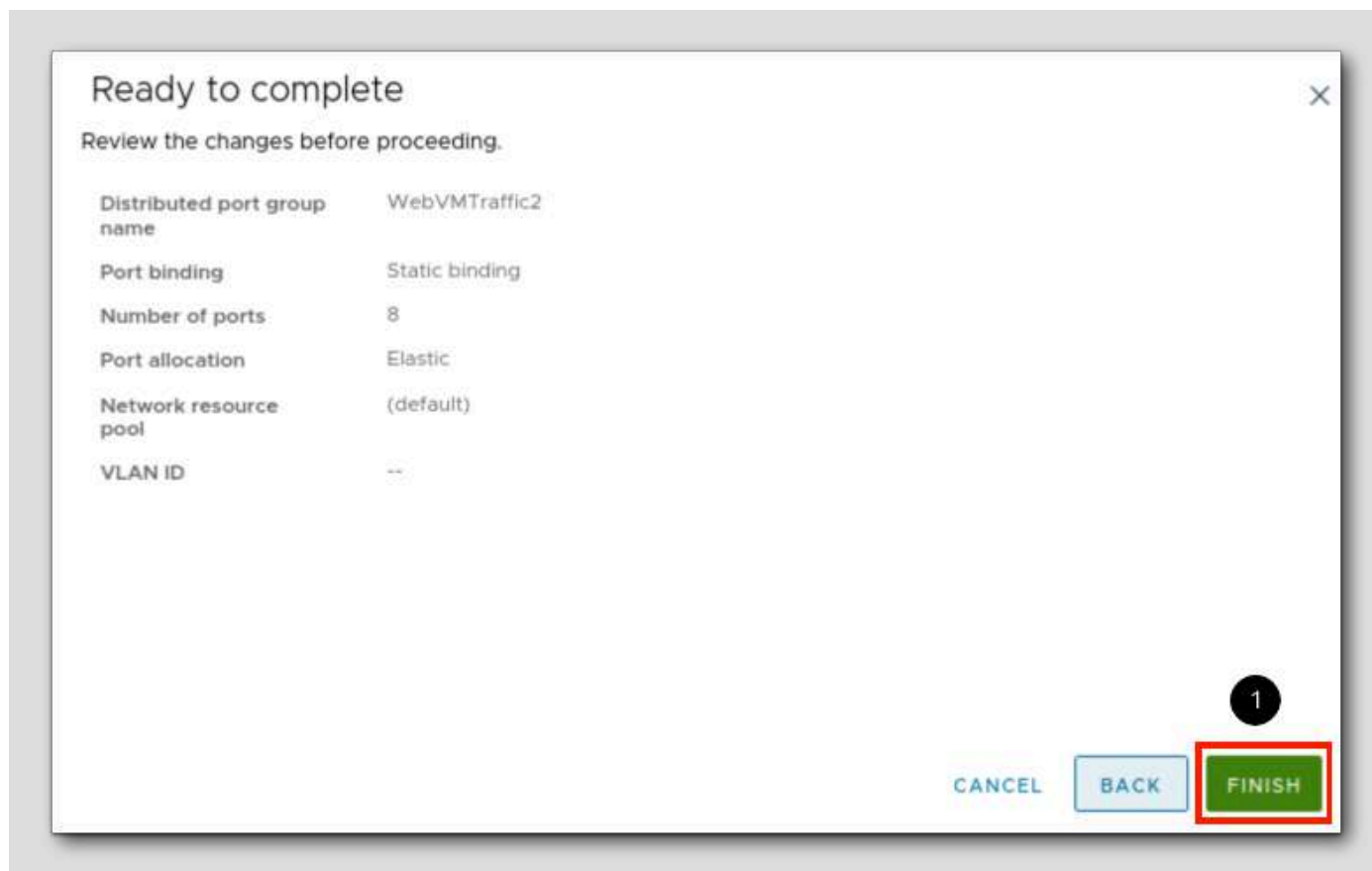
VLAN: Use the Type drop-down menu to select VLAN options:

- None - Do not use VLAN.
- VLAN - In the VLAN ID field, enter a number between 1 and 4094.
- VLAN Trunking - Enter a VLAN trunk range.
- Private VLAN - Select a private VLAN entry. If you did not create any private VLANs, this menu is empty.

Advanced: Select this check box to customize the policy configurations for the new distributed port group.

Ready to Complete

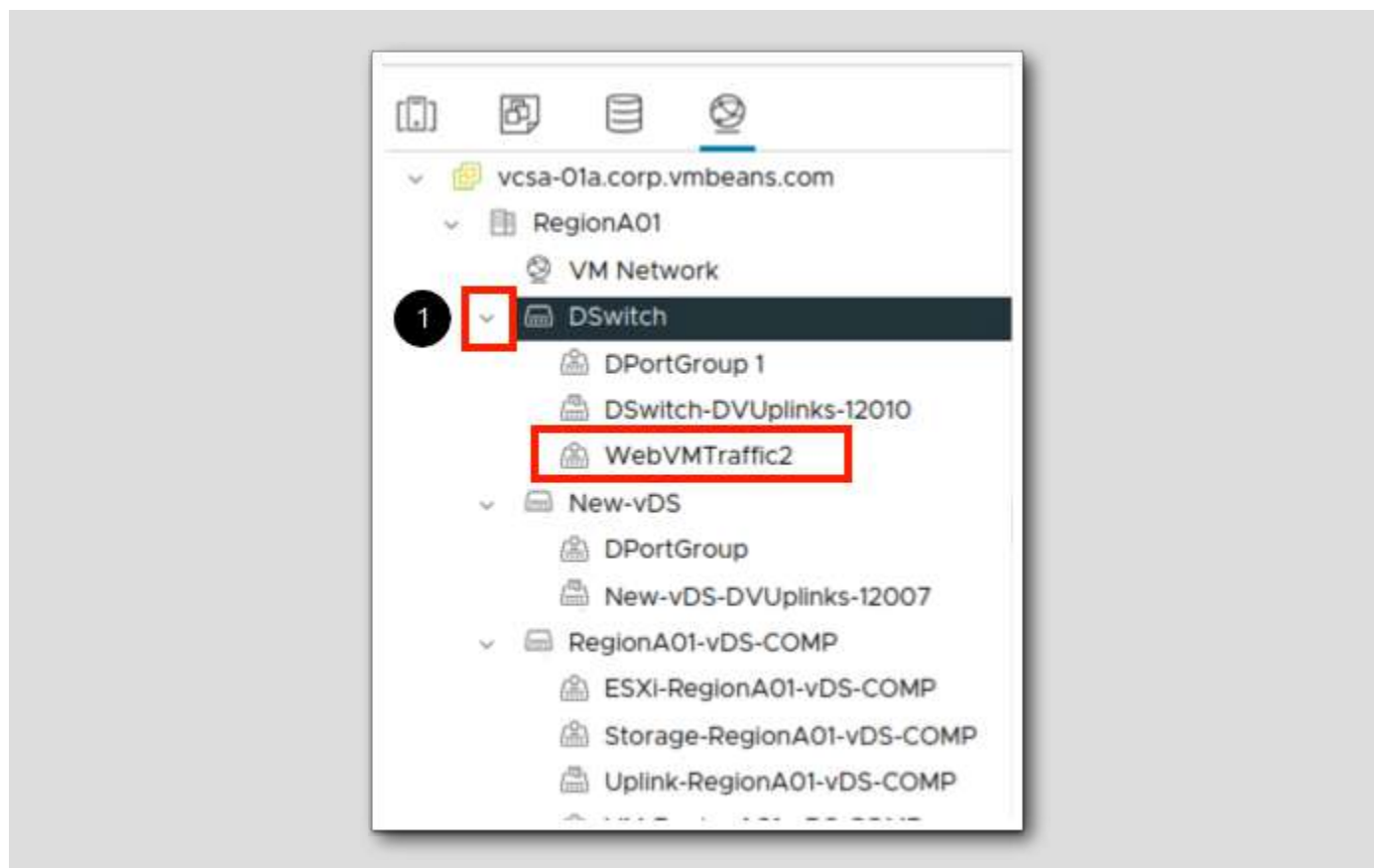
[344]



1. Review the settings and click Finish

View the New Distributed Port Group

[345]



In the Navigator, expand out **DSwitch** and you will see the newly created **WebVMTraffic** Distributed Port Group.

Using Host Lockdown Mode

[346]

To increase the security of your ESXi hosts, you can put them in lockdown mode.

When you enable lockdown mode, no users other than vpxuser have authentication permissions, nor can they perform operations against the host directly. Lockdown mode forces all operations to be performed through vCenter Server.

When a host is in lockdown mode, you cannot run vSphere CLI commands from an administration server, from a script or from vSphere Management Assistant (vMA) against the host. External software or management tools might not be able to retrieve or modify information from the ESXi host.

Lockdown mode is only available on ESXi hosts that have been added to vCenter Server. You can enable lockdown mode using the Add Host wizard to add a host to vCenter Server, using the vSphere Web Client to manage a host or using the Direct Console User Interface (DCUI).

NOTES:

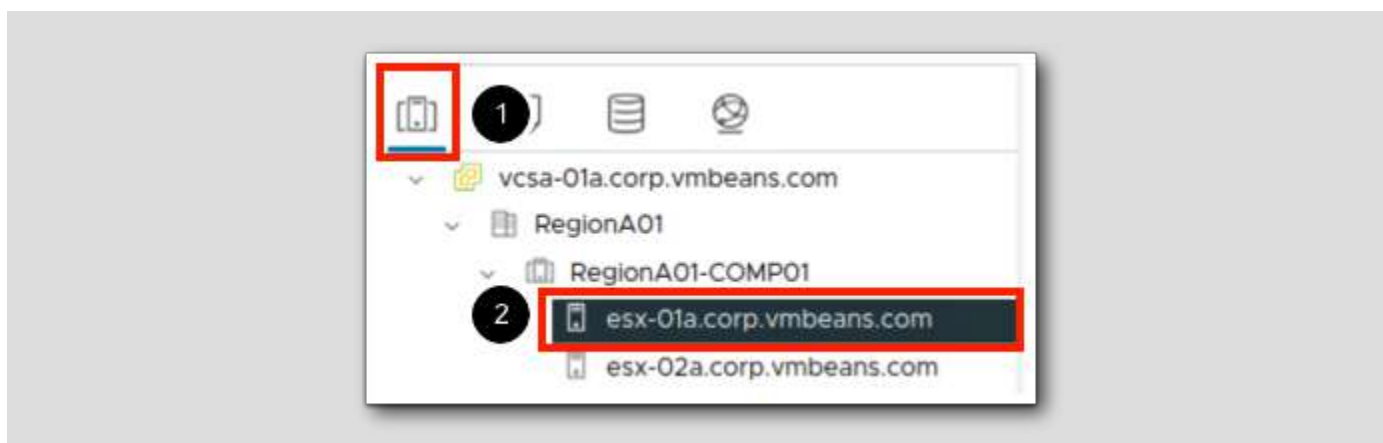
Users with the DCUI Access privilege are authorized to log in to the Direct Console User Interface (DCUI) when lockdown mode is enabled. When you disable lockdown mode using the DCUI, all users with the DCUI Access privilege are granted the Administrator role on the host. The DCUI Access privilege is granted in Advanced Settings on the host.

If you enable or disable lockdown mode using the Direct Console User Interface (DCUI), permissions assigned to users and groups on the host are discarded. To preserve these permissions, you must enable and disable lockdown mode using the vSphere Client connected to vCenter Server.

Enabling or disabling lockdown mode affects which types of users are authorized to access host services, but it does not affect the availability of those services. In other words, if the ESXi Shell, SSH, or Direct Console User Interface (DCUI) services are enabled they will continue to run whether or not the host is in lockdown mode.

Select Hosts and Clusters

[347]

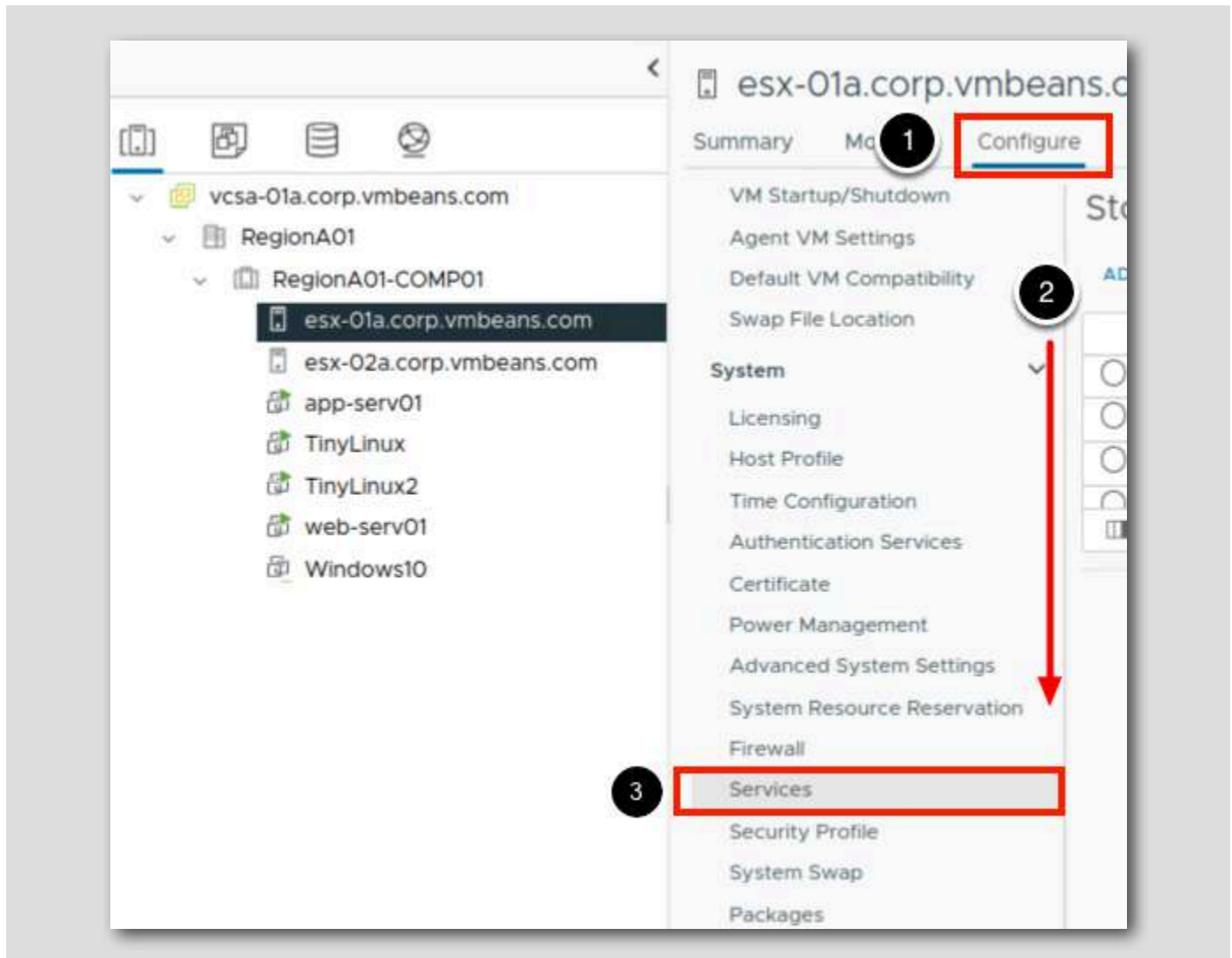


First, you will enable Host Lockdown Mode with the Normal setting on **esx-01a.corp.vmbeans.com**. This will mean the host will be accessible from vCenter and through the DCUI, but not remotely over SSH.

1. From the Navigator, select the **Hosts and Clusters** tab
2. Next, select **esx-01a.corp.vmbeans.com**

Security Profile

[348]

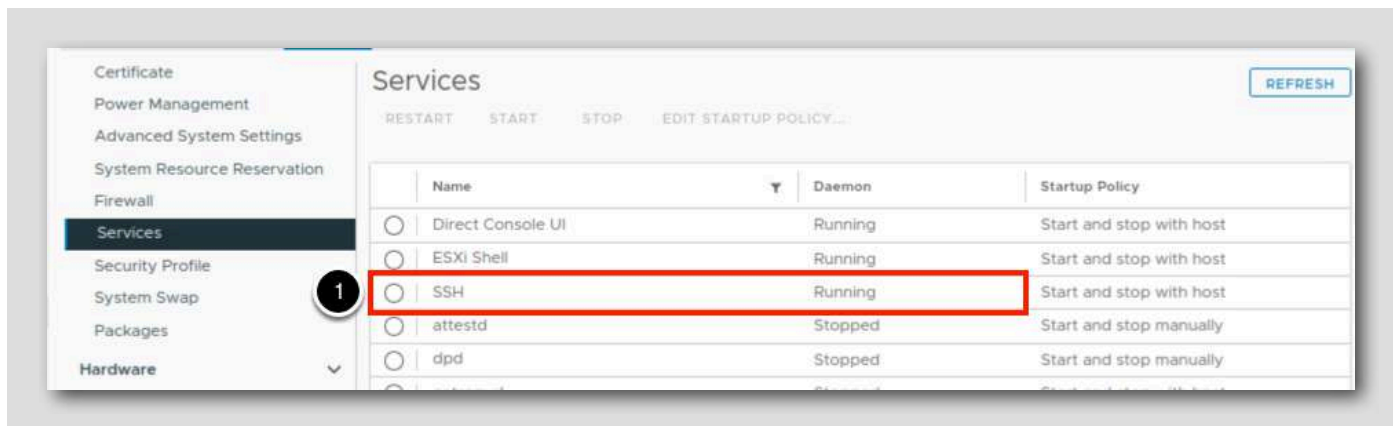


Before we configure Host Lockdown Mode, let's verify the SSH service is running on esx-01a.corp.vmbeans.com

1. Clicking **Configure** tab
2. Scroll down until you find the **System** section
3. Click **Services**

Verify SSH is Enabled

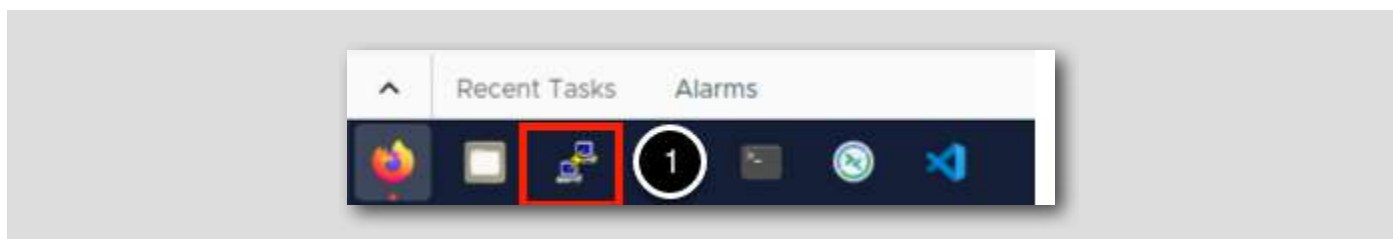
[349]



1. We can see that the SSH service is enabled and Running on esx-01a.corp.vmbeans.com

Open an SSH session to esx-01a

[350]

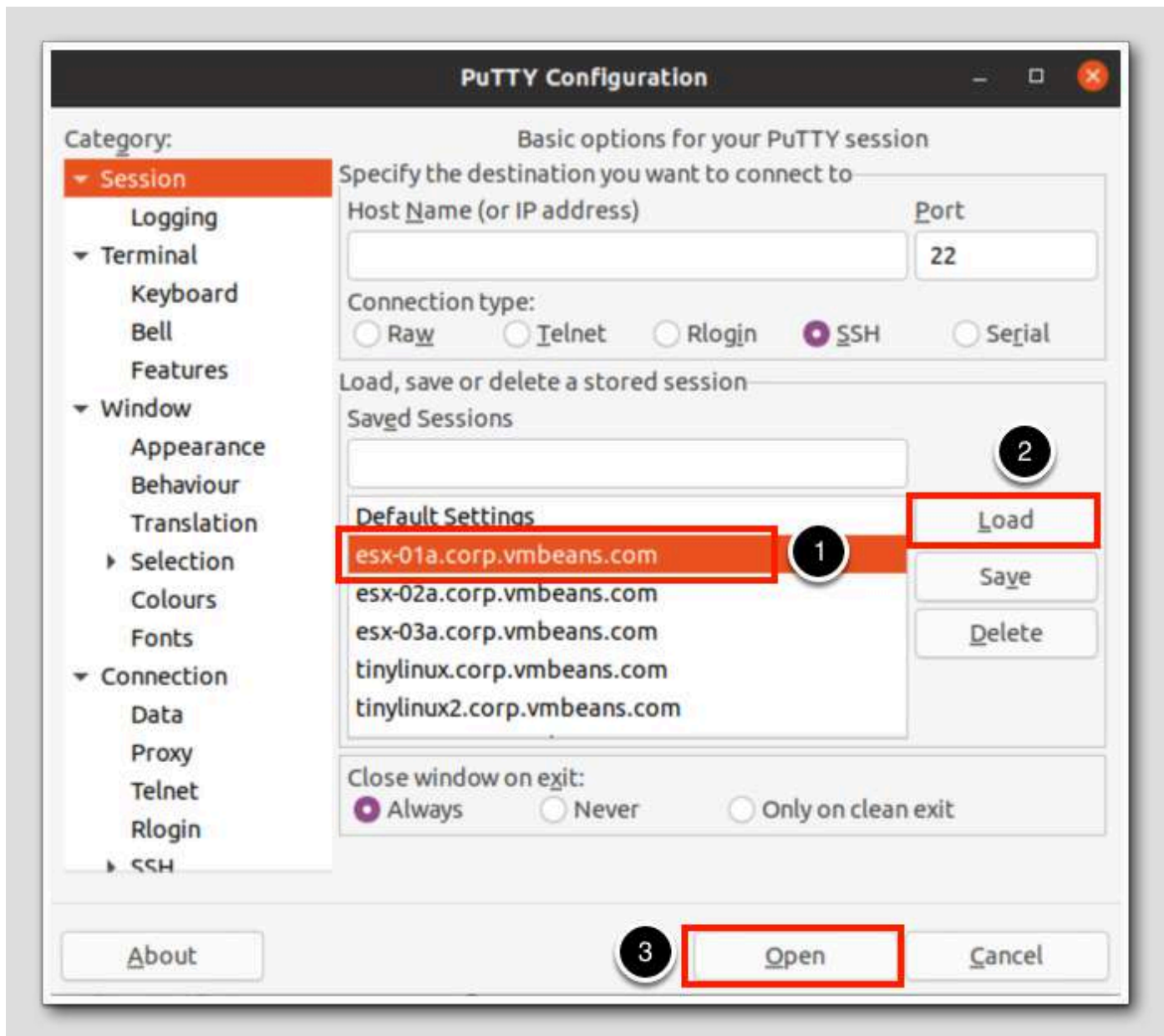


First, verify you can login to esx-01a using an SSH connection.

1. From the Linux Taskbar, click on the PuTTY icon

Connect to esx-01a

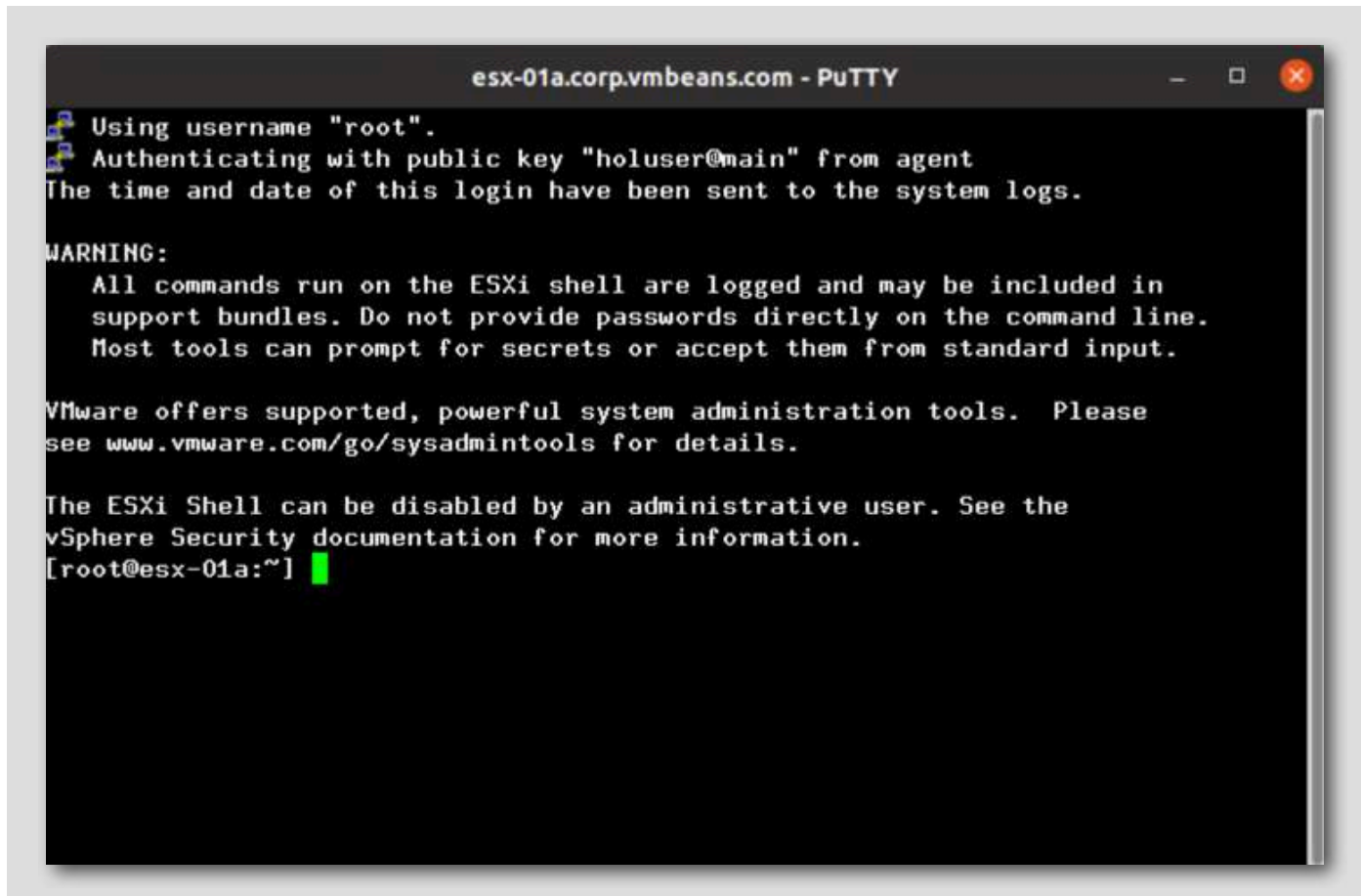
[351]



1. Under Saved Sessions, click on `esx-01a.corp.vmbeans.com`
2. Click Load
3. Click the Open button

Logged into esx-01a

[352]



```
esx-01a.corp.vmbeans.com - PuTTY

Using username "root".
Authenticating with public key "holuser@main" from agent
The time and date of this login have been sent to the system logs.

WARNING:
  All commands run on the ESXi shell are logged and may be included in
  support bundles. Do not provide passwords directly on the command line.
  Most tools can prompt for secrets or accept them from standard input.

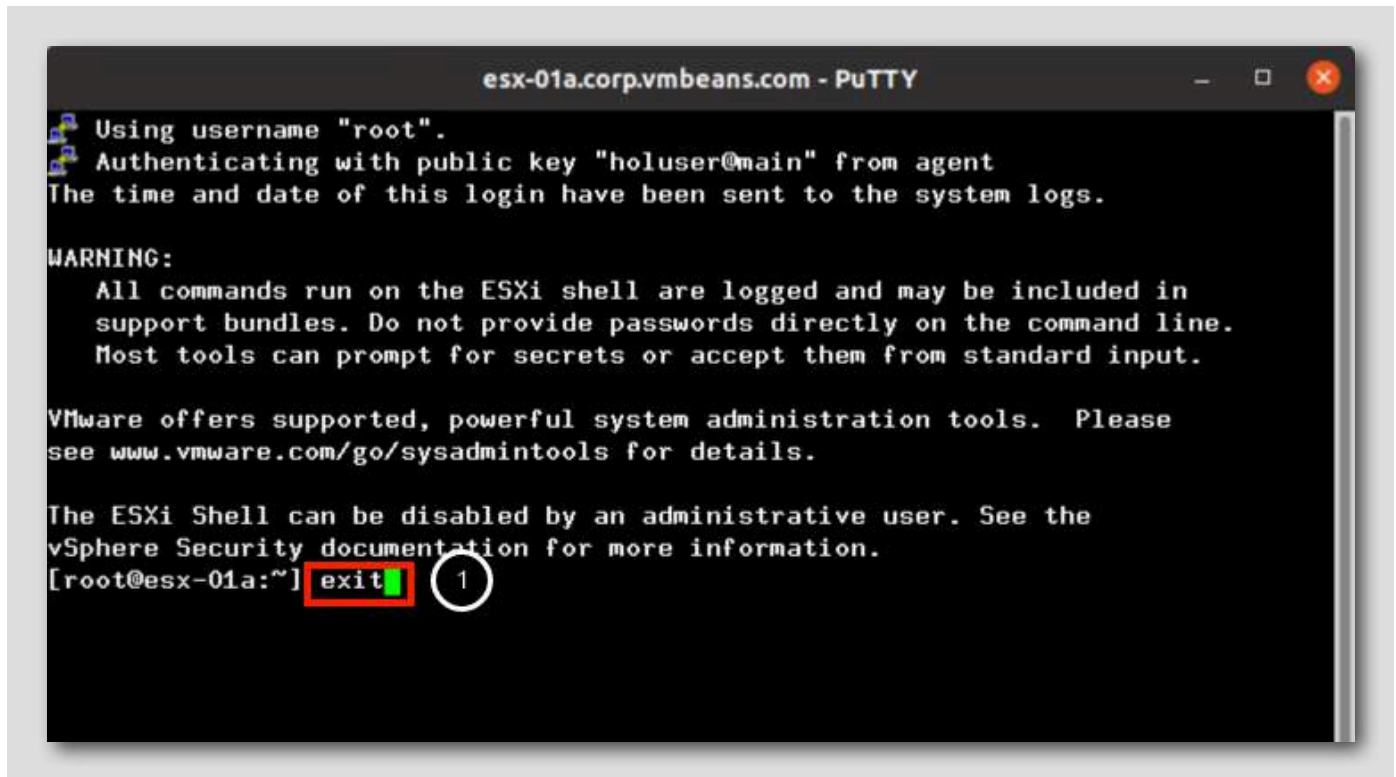
VMware offers supported, powerful system administration tools. Please
see www.vmware.com/go/sysadmintools for details.

The ESXi Shell can be disabled by an administrative user. See the
vSphere Security documentation for more information.
[root@esx-01a:~] █
```

You will be automatically logged in to esx-01a.corp.vmbeans.com because we have configured public-key authentication from the Main Console machine to the ESXi host.

Close the PuTTY Session

[353]

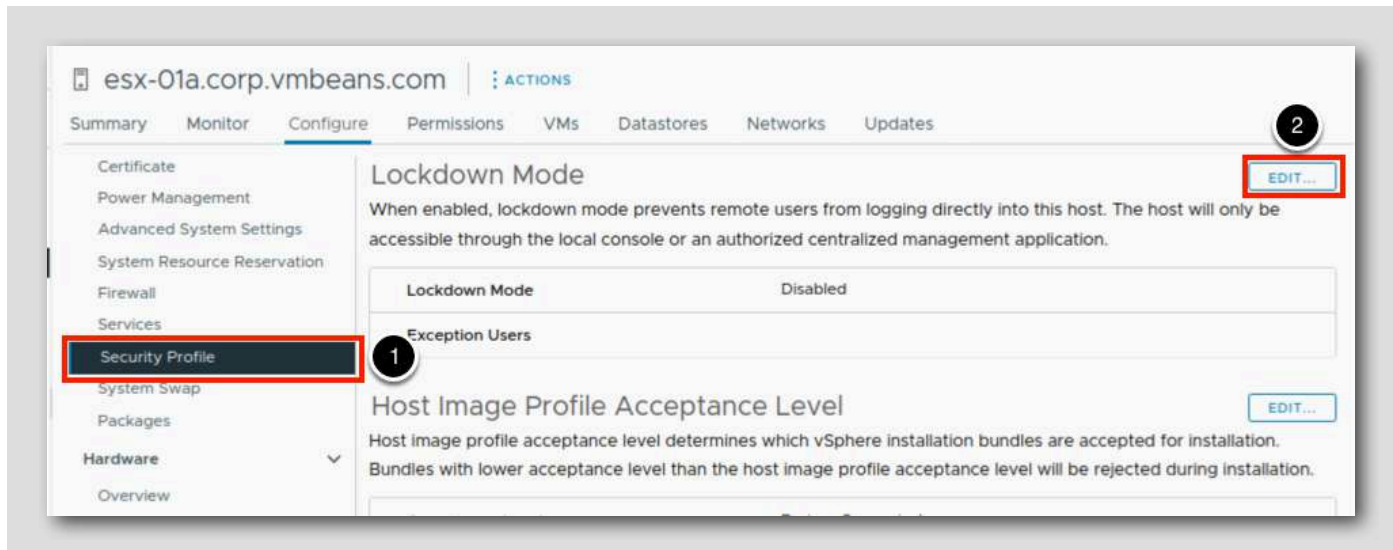


1. Close the PuTTY session by typing 'exit' and pressing Enter

Once you hit Enter, the PuTTY window will disappear.

Enabling Lockdown Mode

[354]

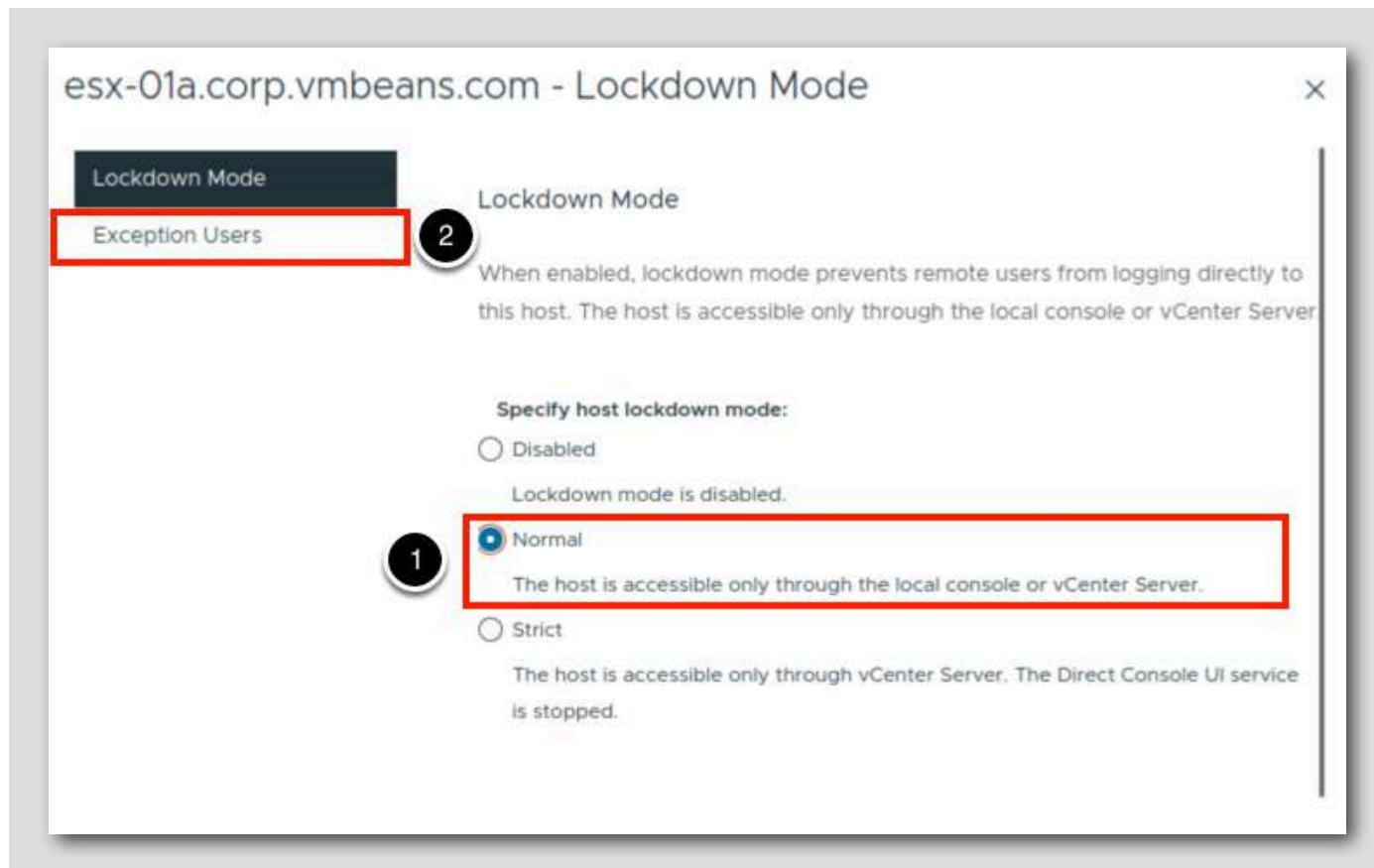


Go back to the vSphere Client

1. Click **Security Profile**
2. Click on the **Edit** button next to Lockdown Mode

Lockdown Mode

[355]

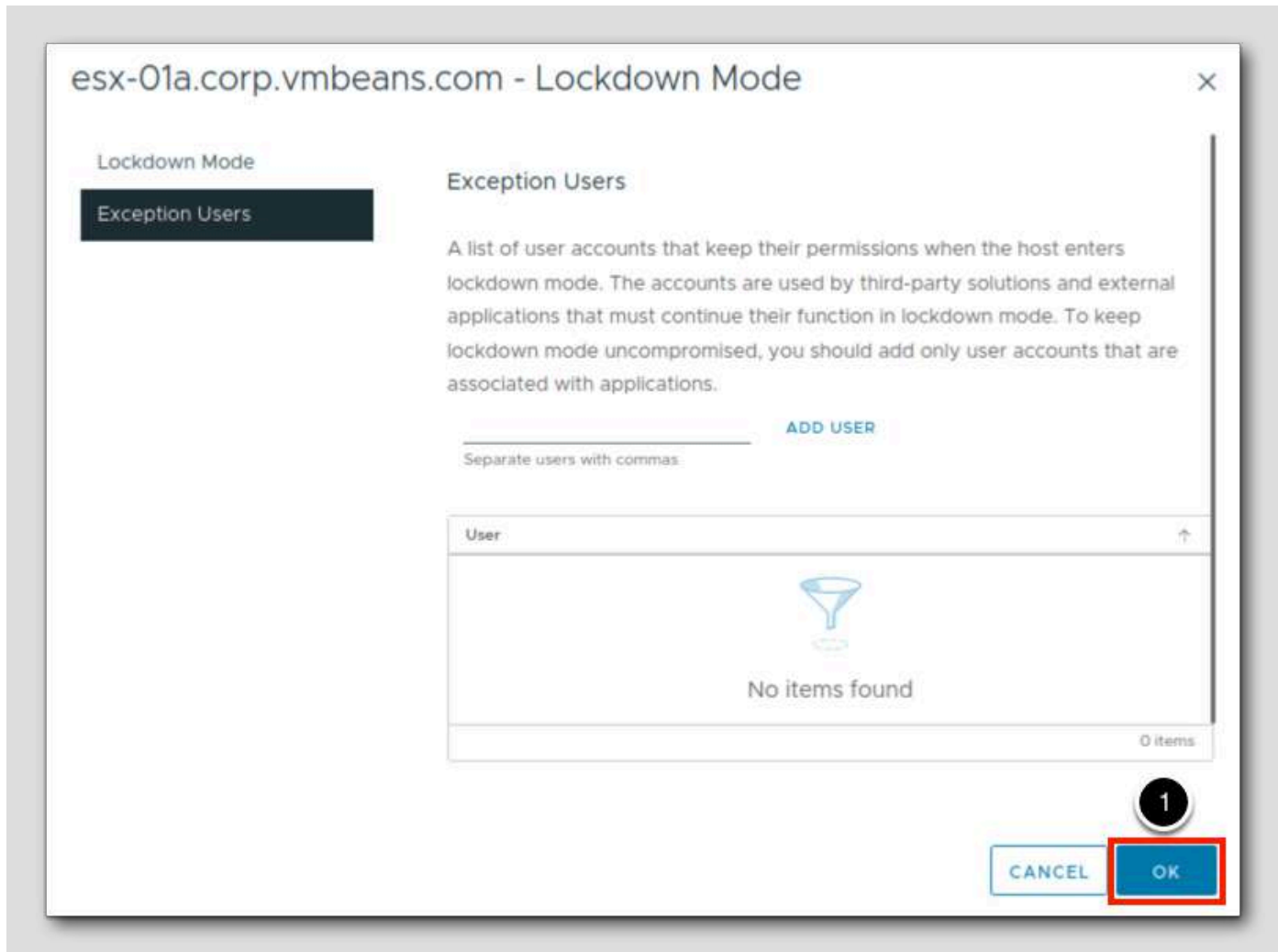


Lockdown Mode is currently disabled. If we set it to Normal, we will not be able to access the host over SSH and only through vCenter or the local console (physically in front of the host). Lockdown Mode can also be set to Strict, meaning only vCenter can access the host and SSH and the local console are disabled.

1. Click the **Normal** radio button
2. Click on **Exception Users**

Exception Users

[356]

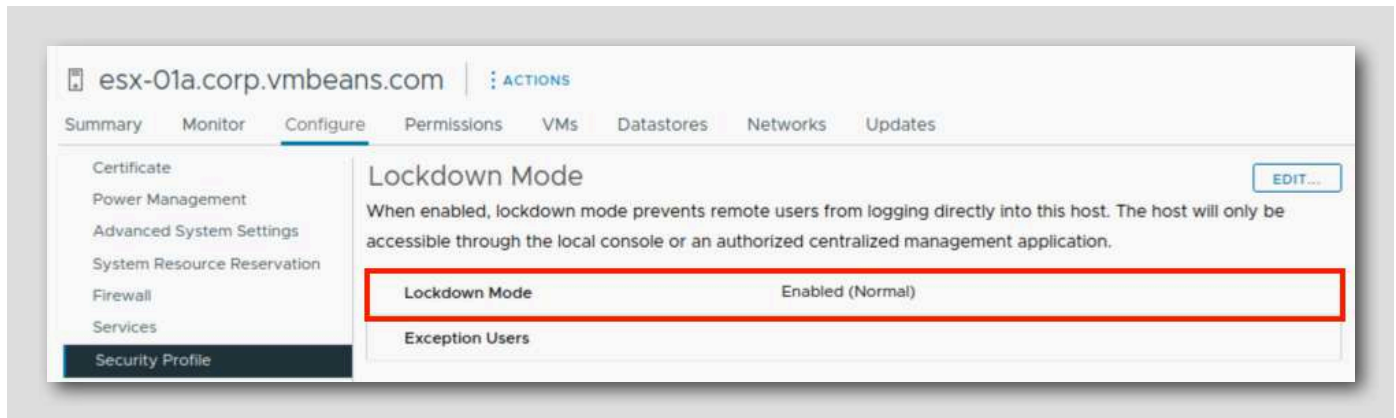


As previously noted, when Lockdown Mode is enabled, remote access to the host is disabled. Some third-party applications rely on this access and it can be granted by adding the accounts they use to the Exception List. This should not be a way for specific users to bypass security and should only be used for applications that require access.

1. Click OK to enable Lockdown Mode

Lockdown Mode Enabled

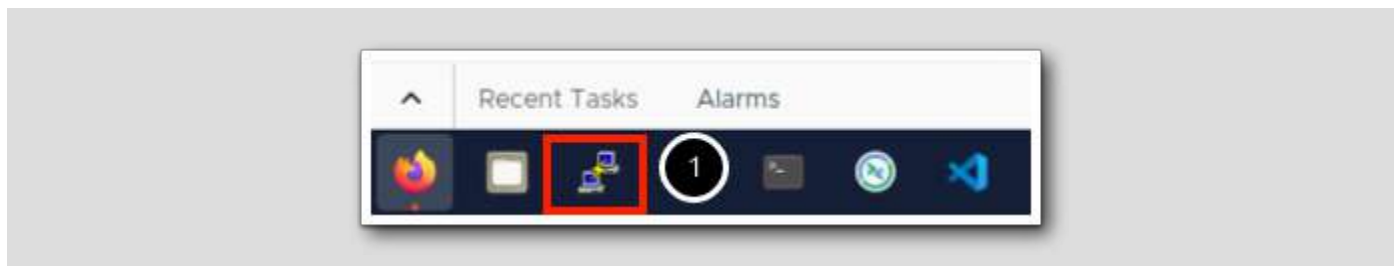
[357]



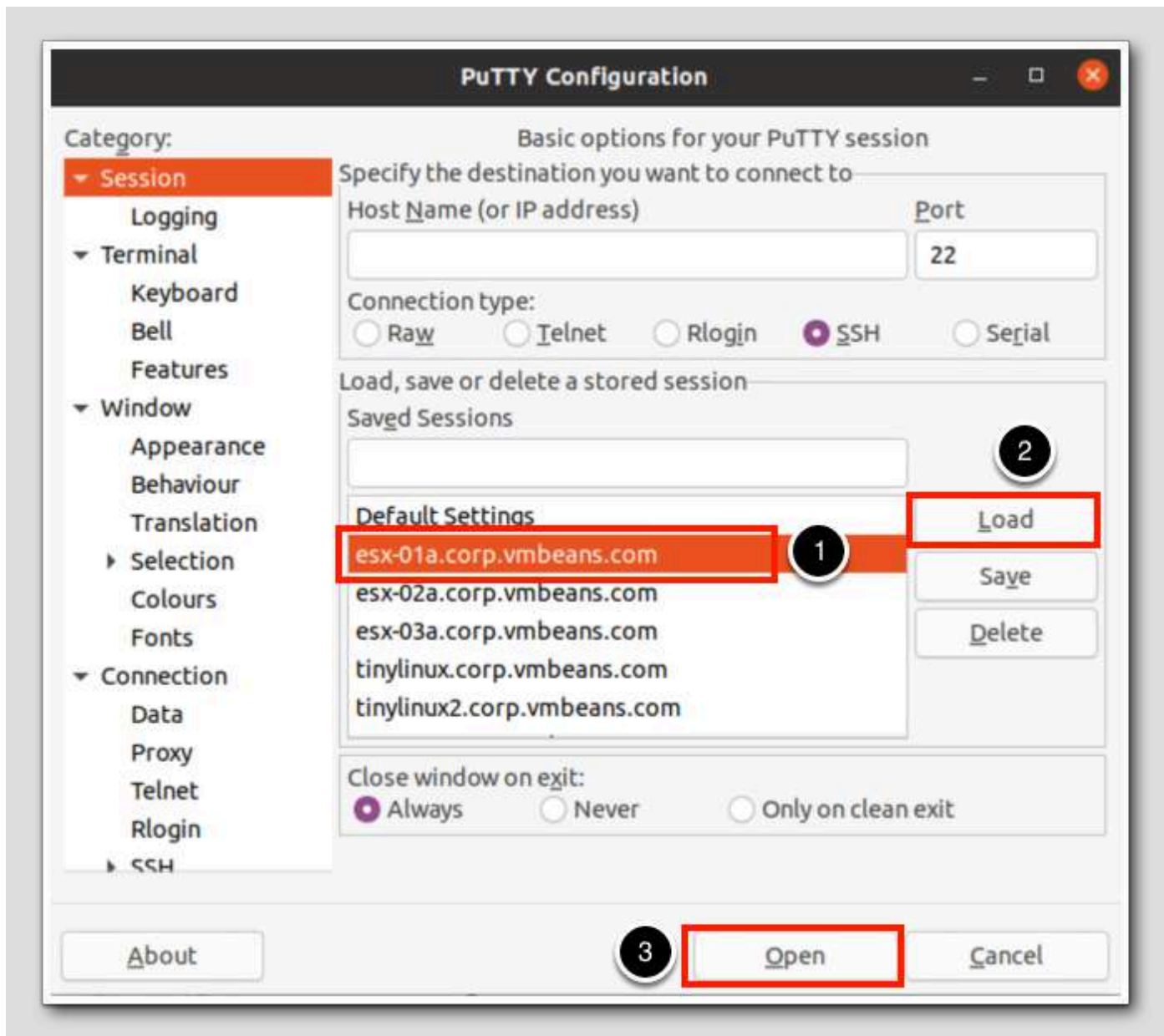
Wait for the vSphere Client to refresh to see that Lockdown Mode has been enabled.

PuTTY Session to esx-01a

[358]



1. Open a PuTTY again

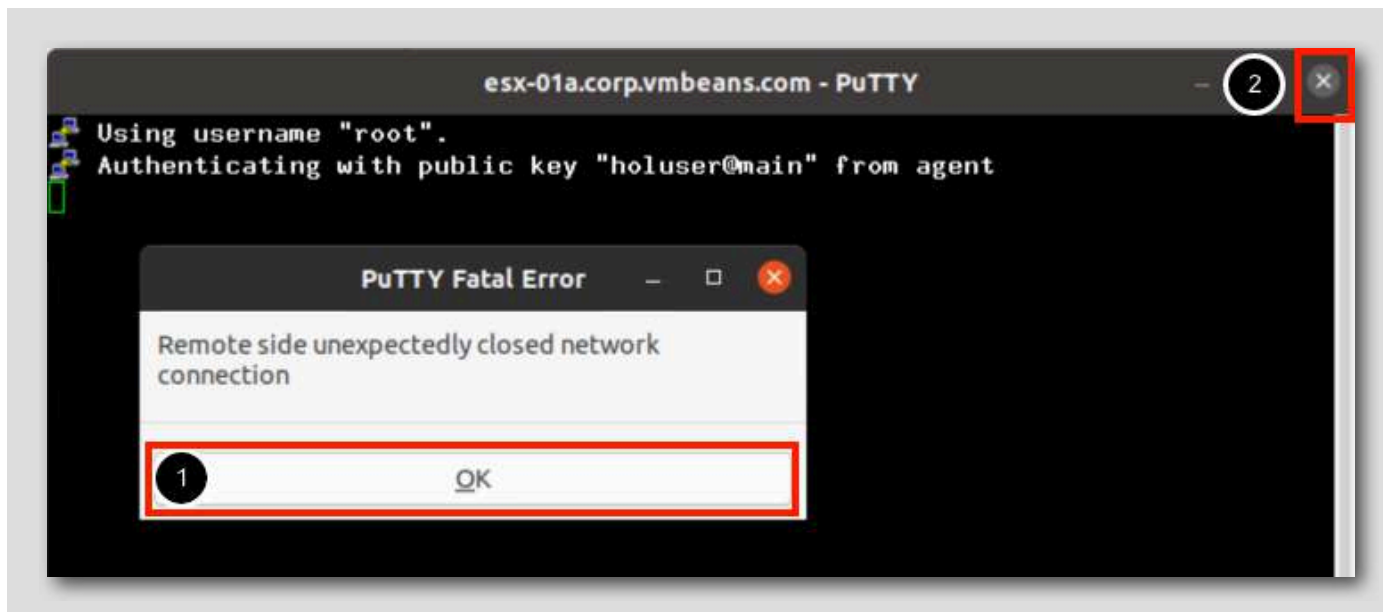


Using the same steps we used above, open the PuTTY application from the Windows Taskbar.

1. Click on esx-01a.corp.vmbeans.com under Saved Sessions
2. Click Load
3. Click Open

Denied!

[359]

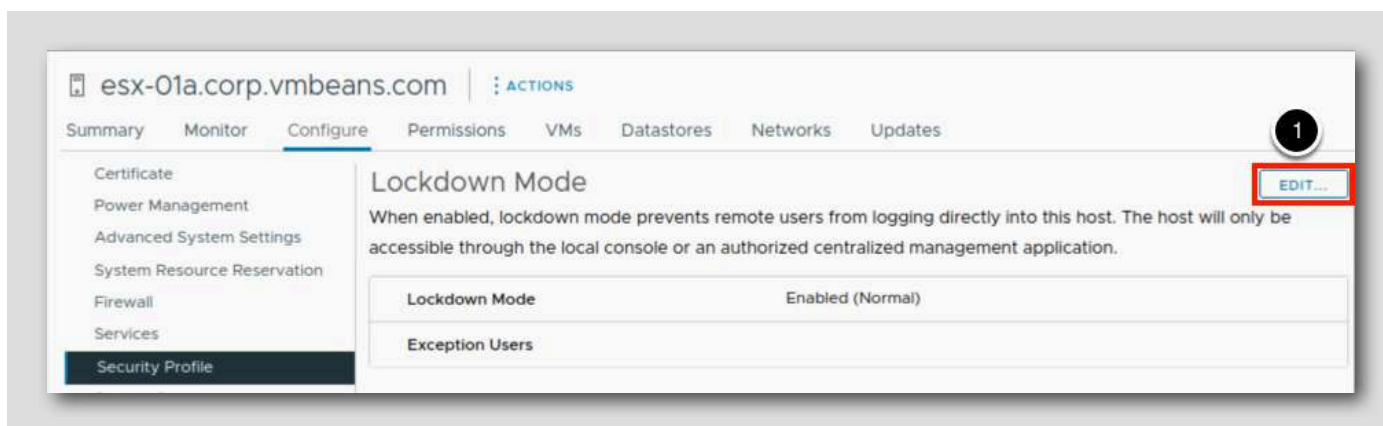


You should receive an error when trying to connect to esx-01a.corp.vmbeans.com. The host has been configured with Host Lockdown Mode and will refuse any remote connections, unless those users were added to the Exception User list.

1. Click OK
2. Close PuTTY by clicking the 'X' in the top right-hand corner of the window

Disable Lockdown Mode

[360]

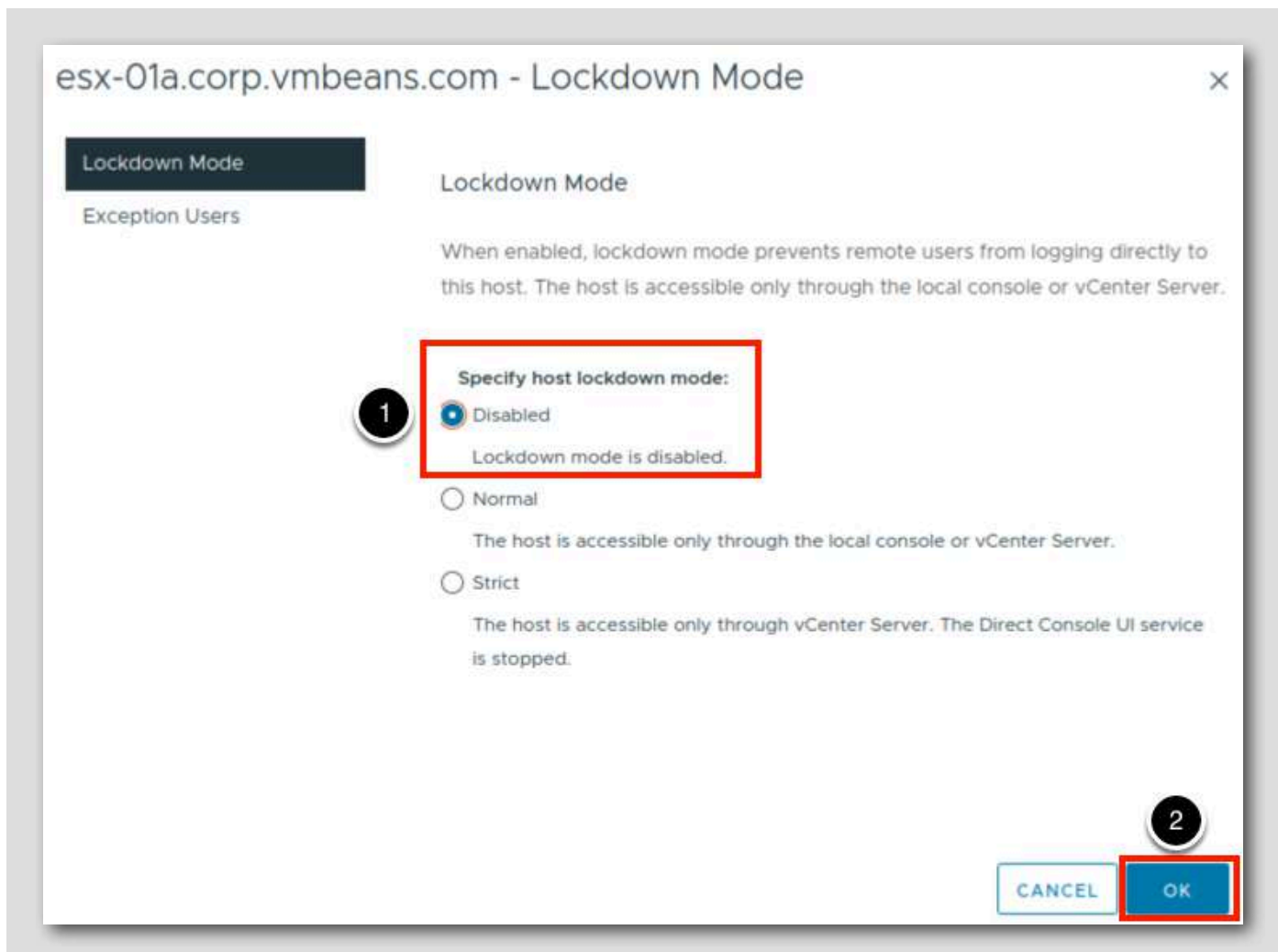


Go back to the vSphere Client.

1. Click on the **Edit** button again under Lockdown Mode

Lockdown Mode

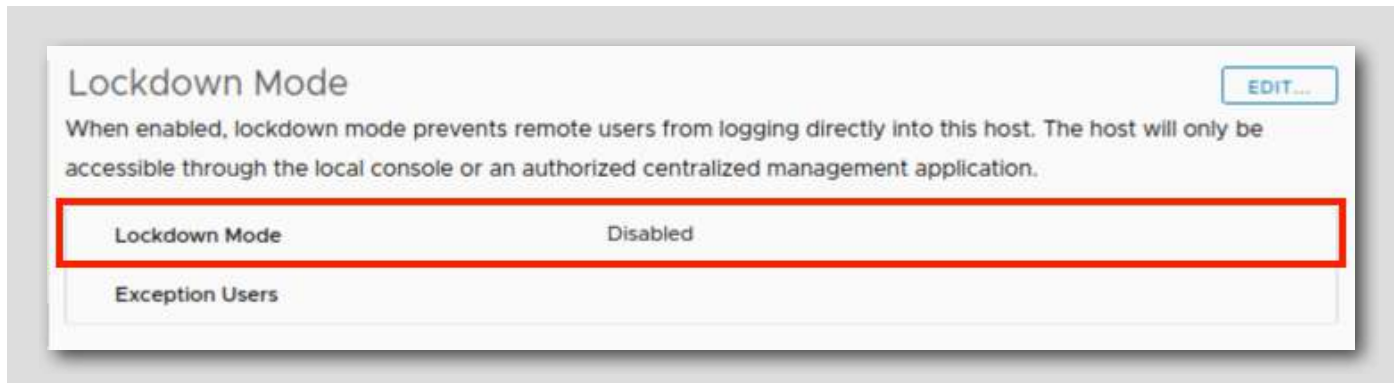
[361]



1. Check the **Disabled** radio button
2. Click **OK** to continue

Host Lockdown Mode Disabled

[362]

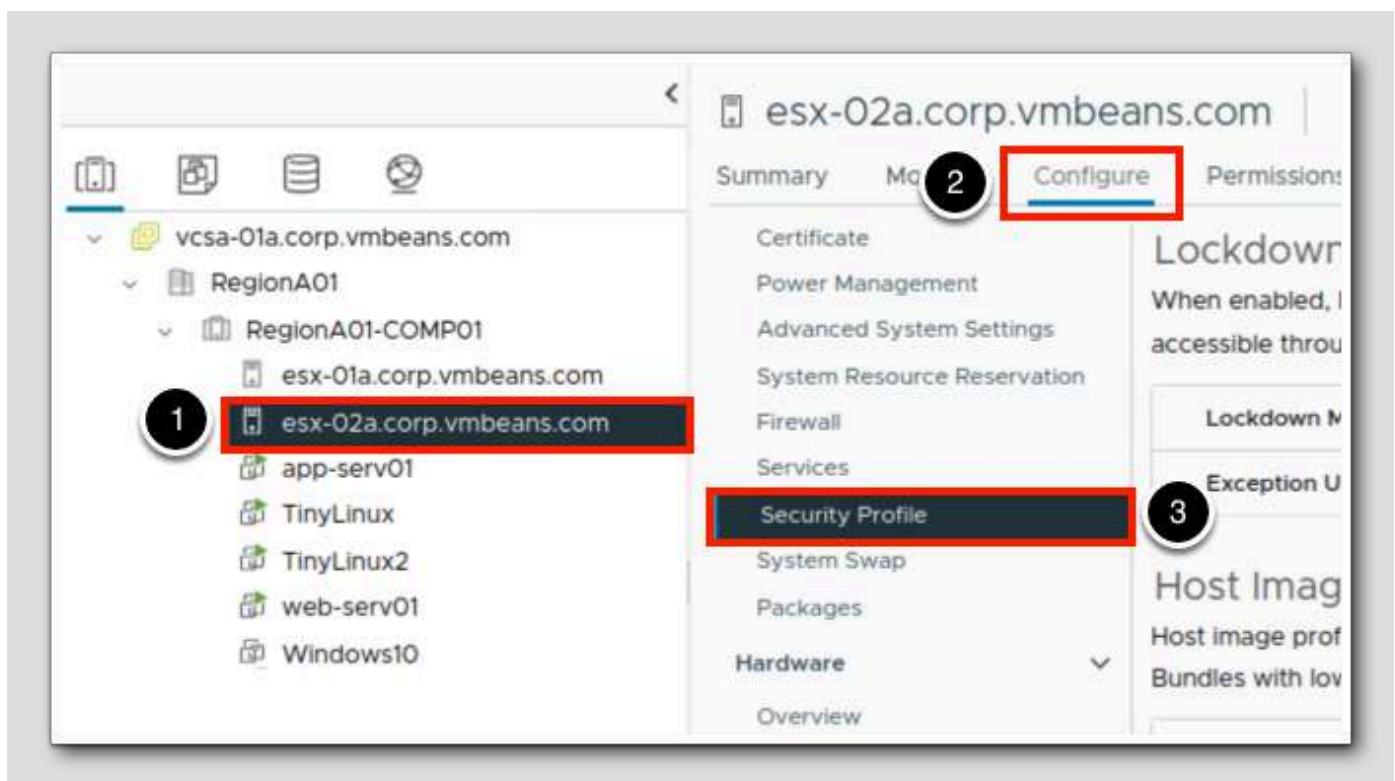


Lockdown Mode for the host should now be **disabled**.

Host Lockdown Mode provides an excellent way to further secure your vSphere hosts.

Strict Mode

[363]

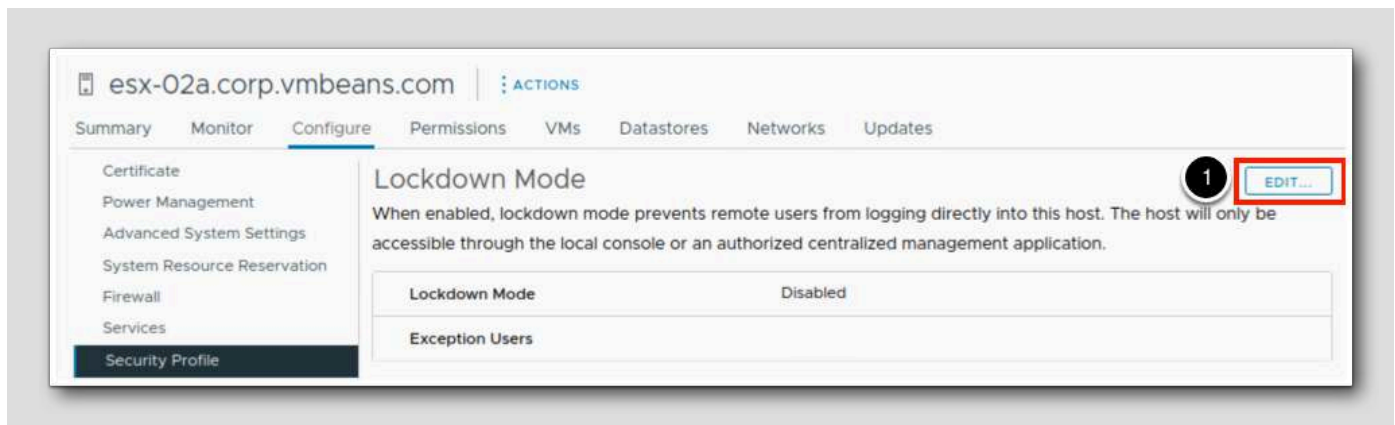


Now you will set esx-02a.corp.vmbeans.com to use the Strict Mode of Host Lockdown. This means the host is only available through vCenter Server and access to the DCUI and SSH are disabled.

1. Click on `esx-02a.corp.vmbeans.com`
2. Click the **Configure** tab, if it is not already selected
3. Click on **Security Profile** under the **System** section

Enable Lockdown Mode

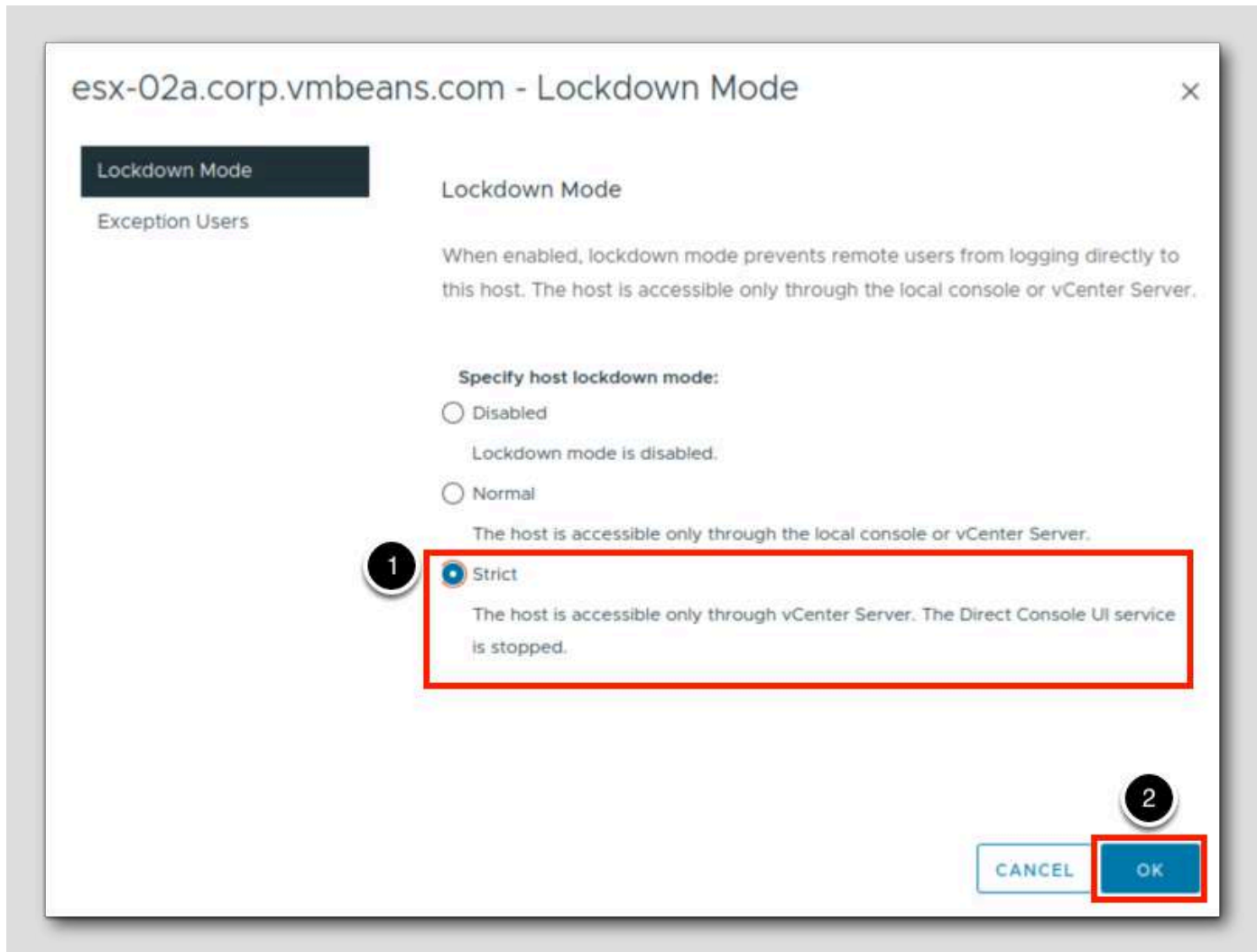
[364]



1. Click on the Edit button

Lockdown Mode - Strict

[365]

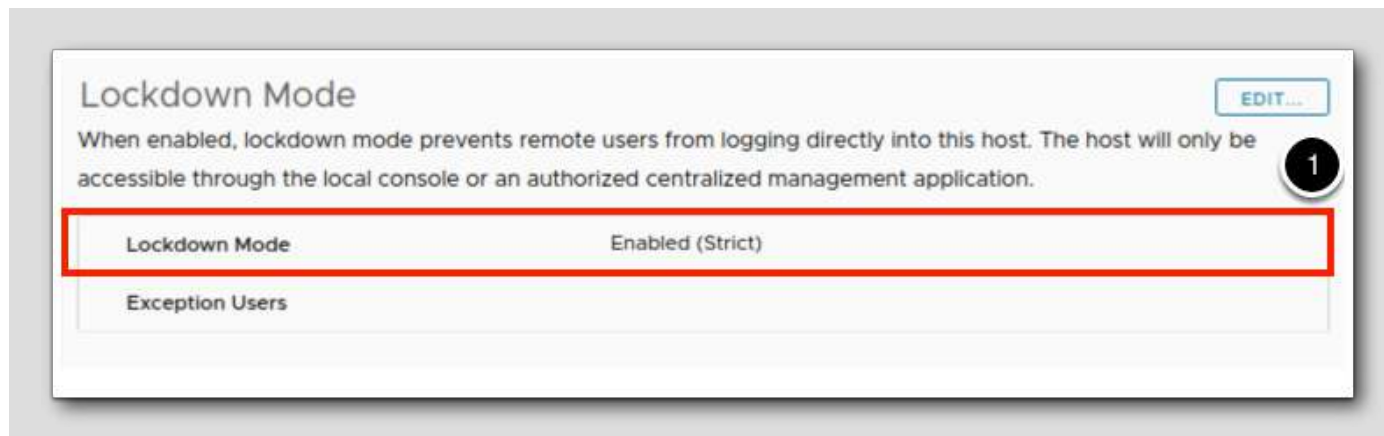


1. Click button next to **Strict**
2. Click **OK**

Again, note that users can be added to the exception list. This will only apply to SSH and not the DCUI.

Strict Mode - Enabled

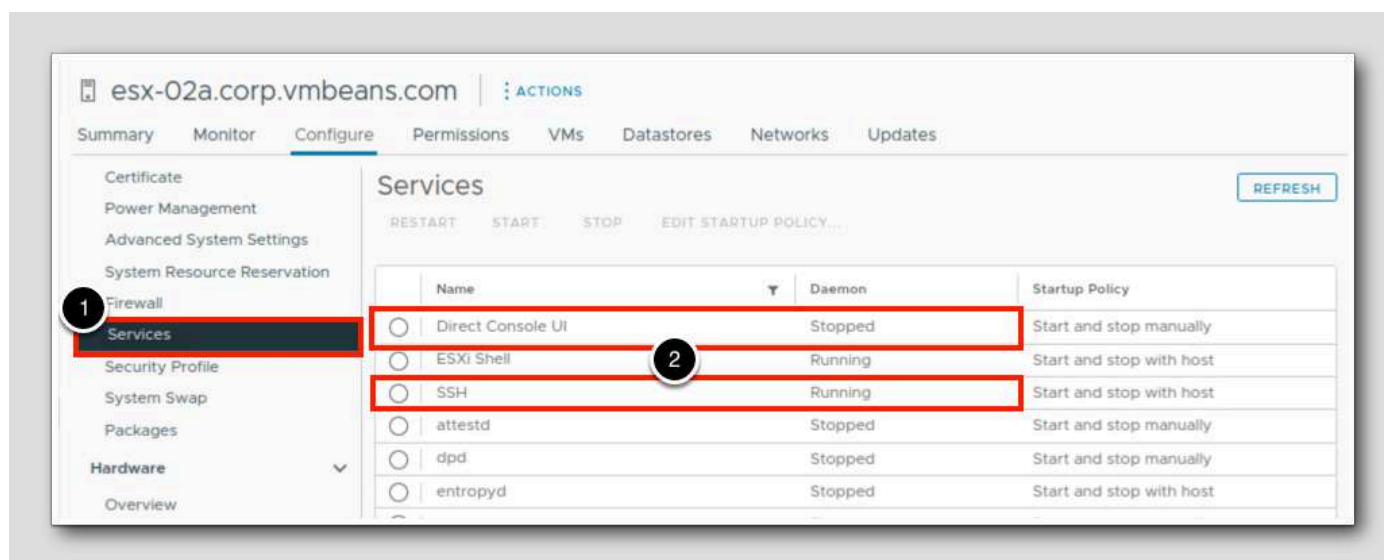
[366]



1. Notice Lockdown Mode is now Enabled

Services

[367]



1. Click on **Services**.
2. You can see the Direct Console UI (DCUI) service has been stopped. Note that the SSH service is still running in case users have been added to the Exception List.

Select Consoles

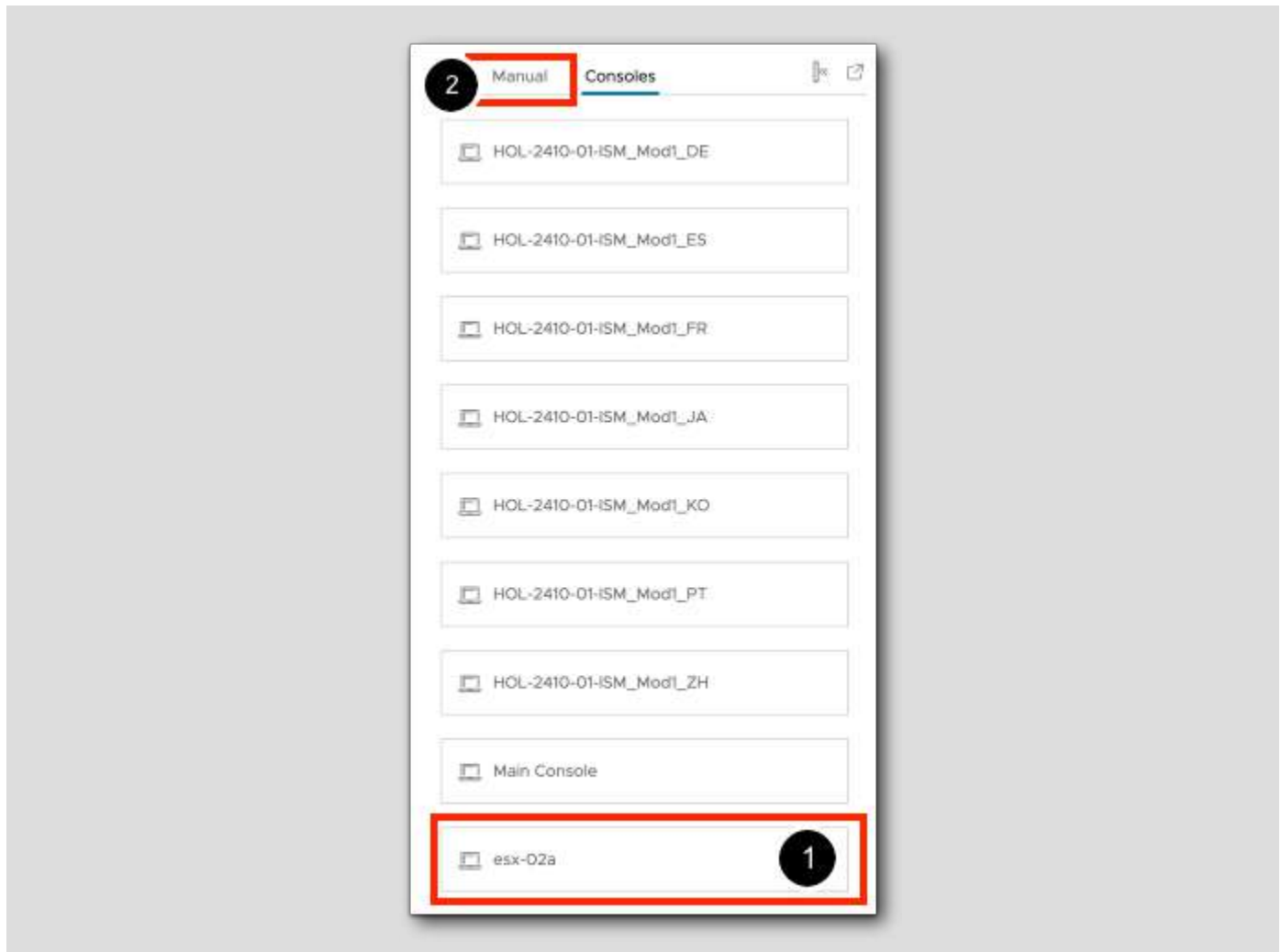
[368]



1. On the far, right-hand side of the web page, look for the Consoles tab and click on it.

This will give us access to the DCUI on `esx-02a-corp.vmbeans.com`

Select esx-02a

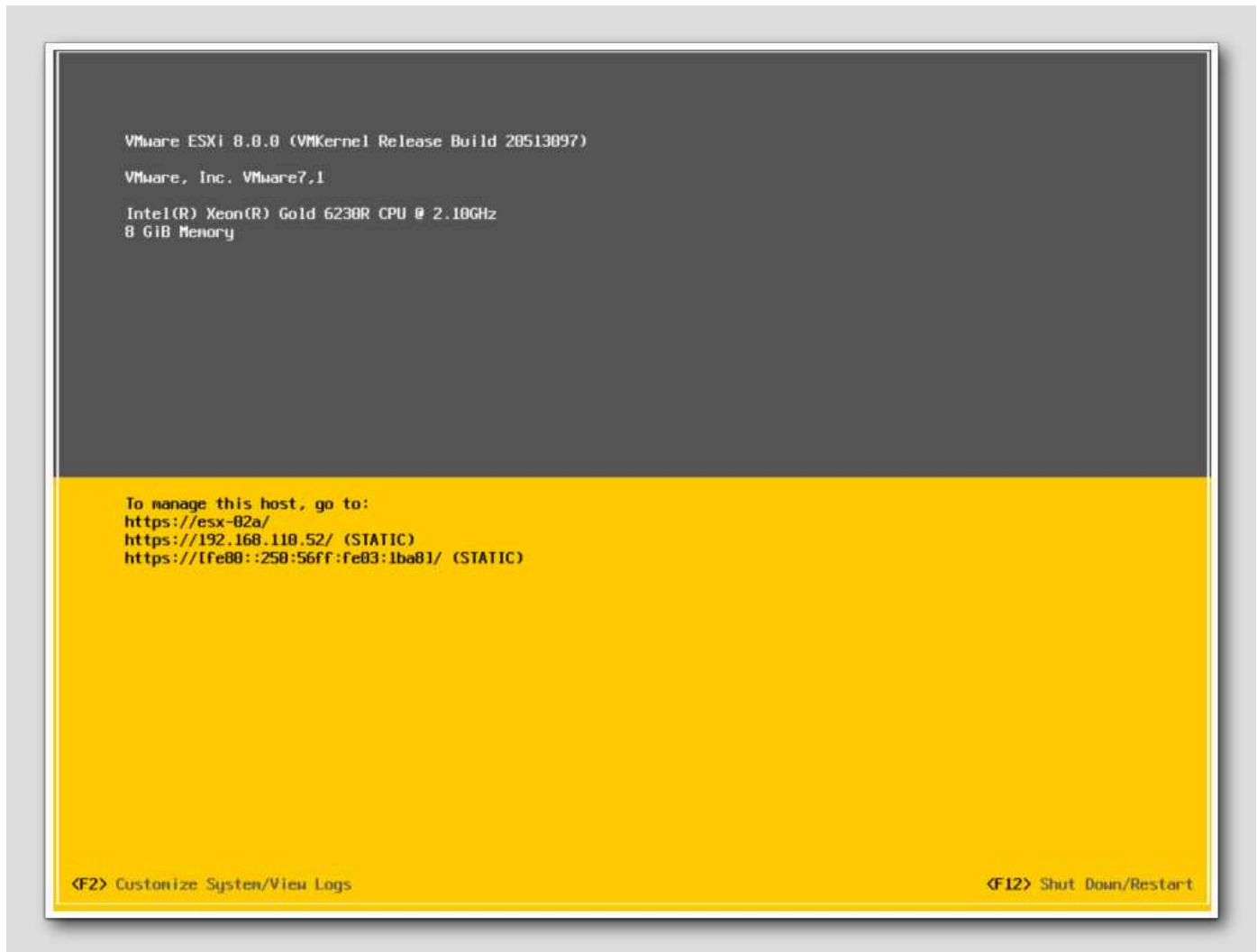


1. Scroll down and select **esx-02a**.
2. Click on **Manual** to return to the steps.

The console window will load the DCUI for `esx-02a.corp.vmbeans.com`

Click in the Console

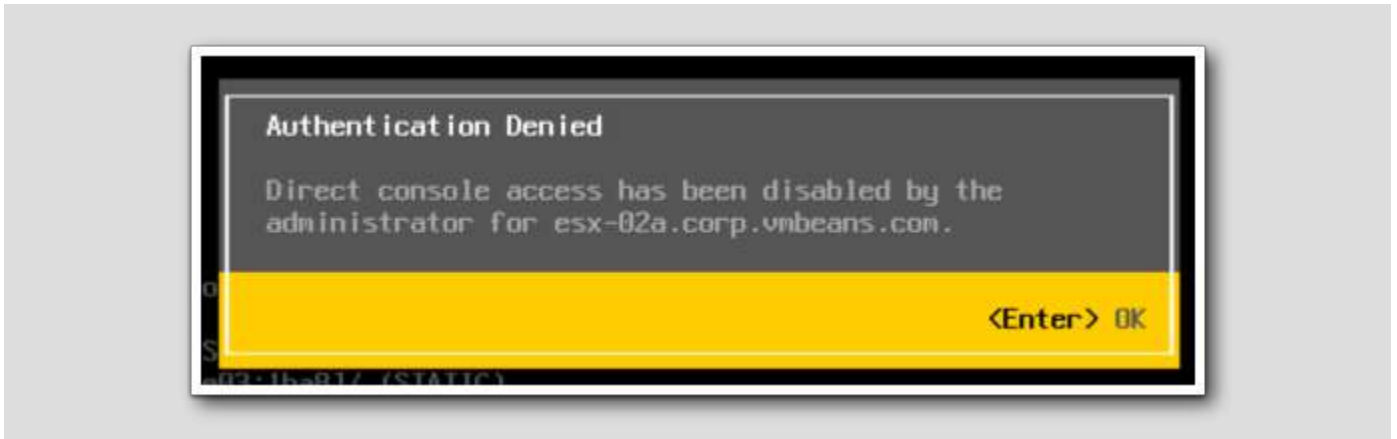
[370]



1. Click in the console and press the **space** bar to wake up the host.

Press F2

[371]



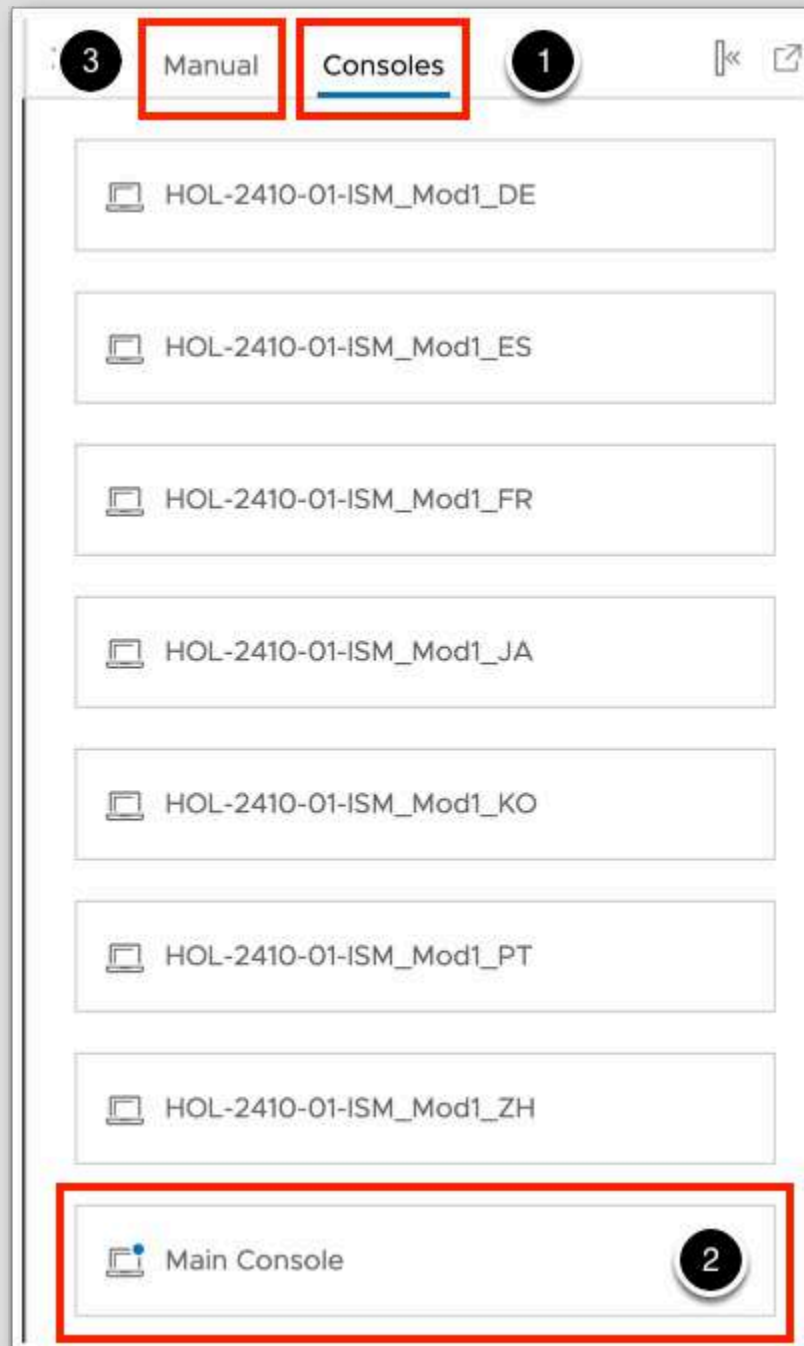
1. Now press the **F2** key to log in to the DCUI.

You should receive an error that access to the DCUI has been disabled.

2. Press the **Enter** key to dismiss the message.

Main Console

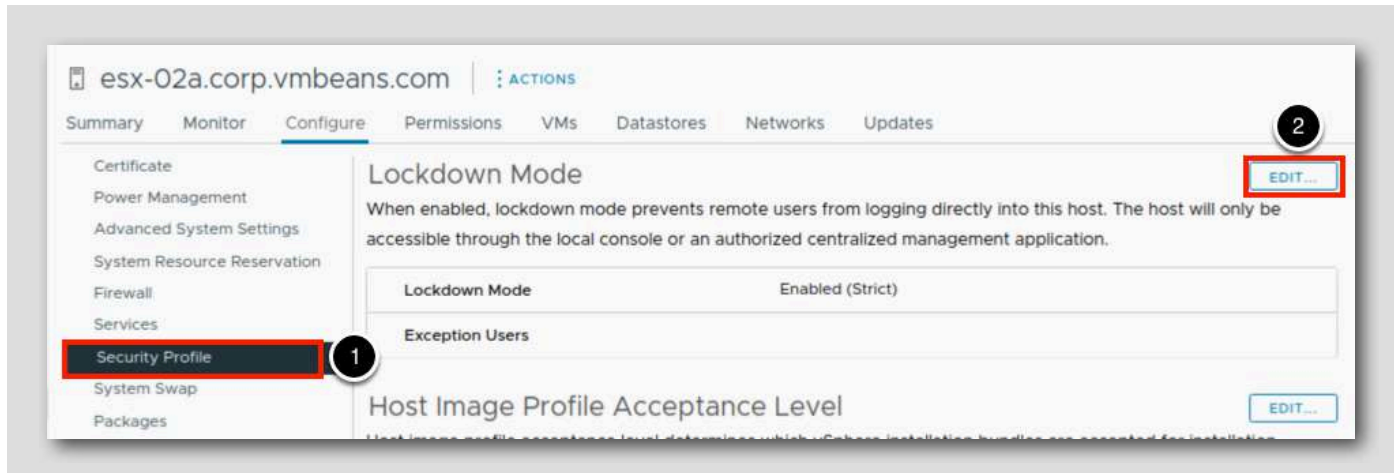
[372]



1. Click on **Consoles** again.
2. Select **Main Console** to return to the main desktop.
3. Click on **Manual** to continue.

Disable Lockdown Mode

[373]

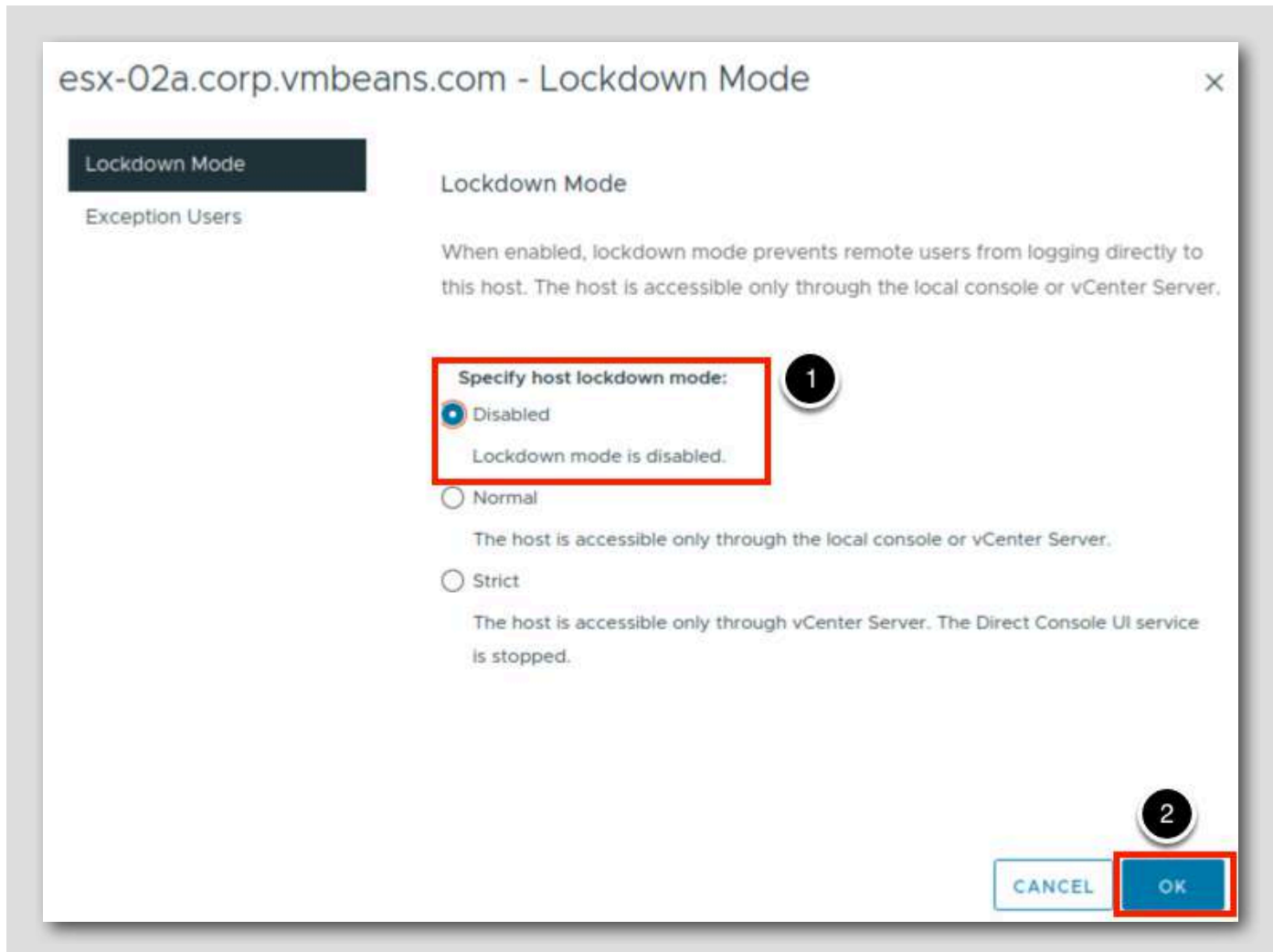


Go back to the vSphere Client.

1. Click on **Security Profile**.
2. Click on the **Edit** button again under Lockdown Mode.

Lockdown Mode

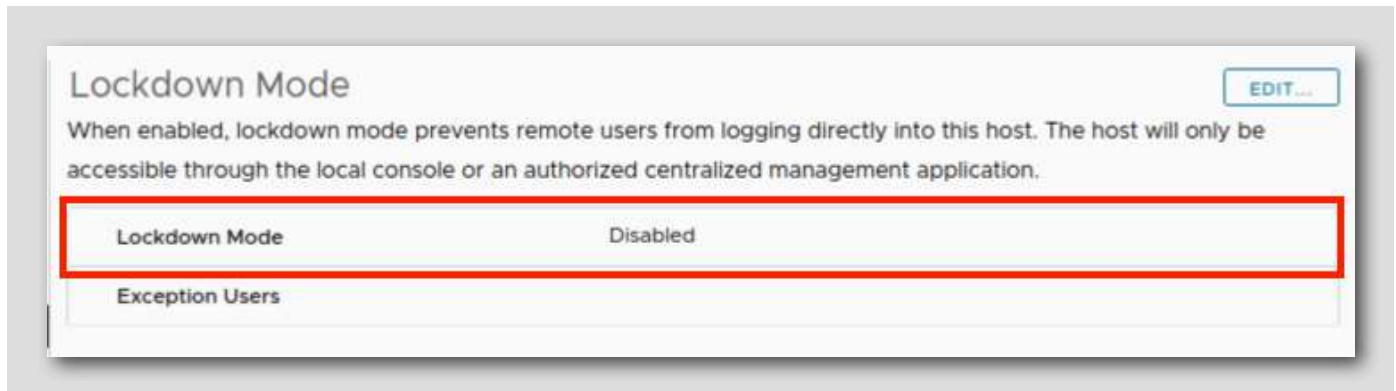
[374]



1. Check the **Disabled** radio button
2. Click **OK** to continue.

Host Lockdown Mode Disabled

[375]



Lockdown Mode for the host should now be disabled.

Host Lockdown Mode provides an excellent way to further secure your vSphere hosts.

Configuring the Host Services and Firewall

[376]

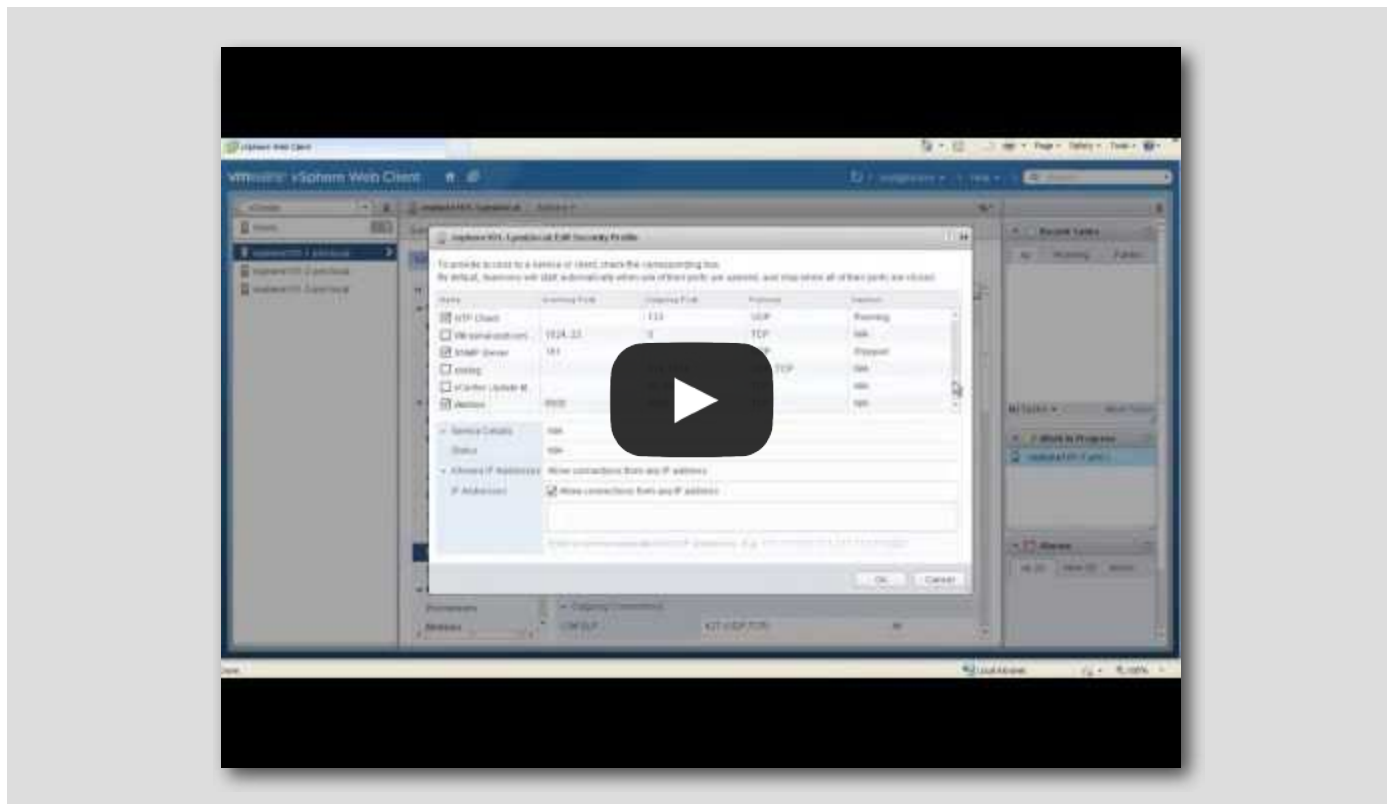
This lesson includes a short video on how to use the VMware ESXi firewall.

Video: Configure vSphere Host Firewall for VMware vSphere (4:34)

[377]

This video shows how to use the VMware ESXi Firewall on the vSphere host to block incoming and outgoing communication and to manage the services running on the host.

<https://www.youtube.com/watch?v=bzjsjQdnTuk>



User Access and Authentication Roles

[378]

VMware recommends that you create roles to suit the access control needs of your environment. If you create or edit a role on a vCenter Server system that is part of a connected group in Linked Mode, the changes that you make are propagated to all other vCenter Server systems in the group.

Linked Mode connects multiple vCenter Server systems together by using one or more Platform Services Controllers. It lets you view and search across all linked vCenter Servers and replicate roles, permissions, licenses, policies and tags.

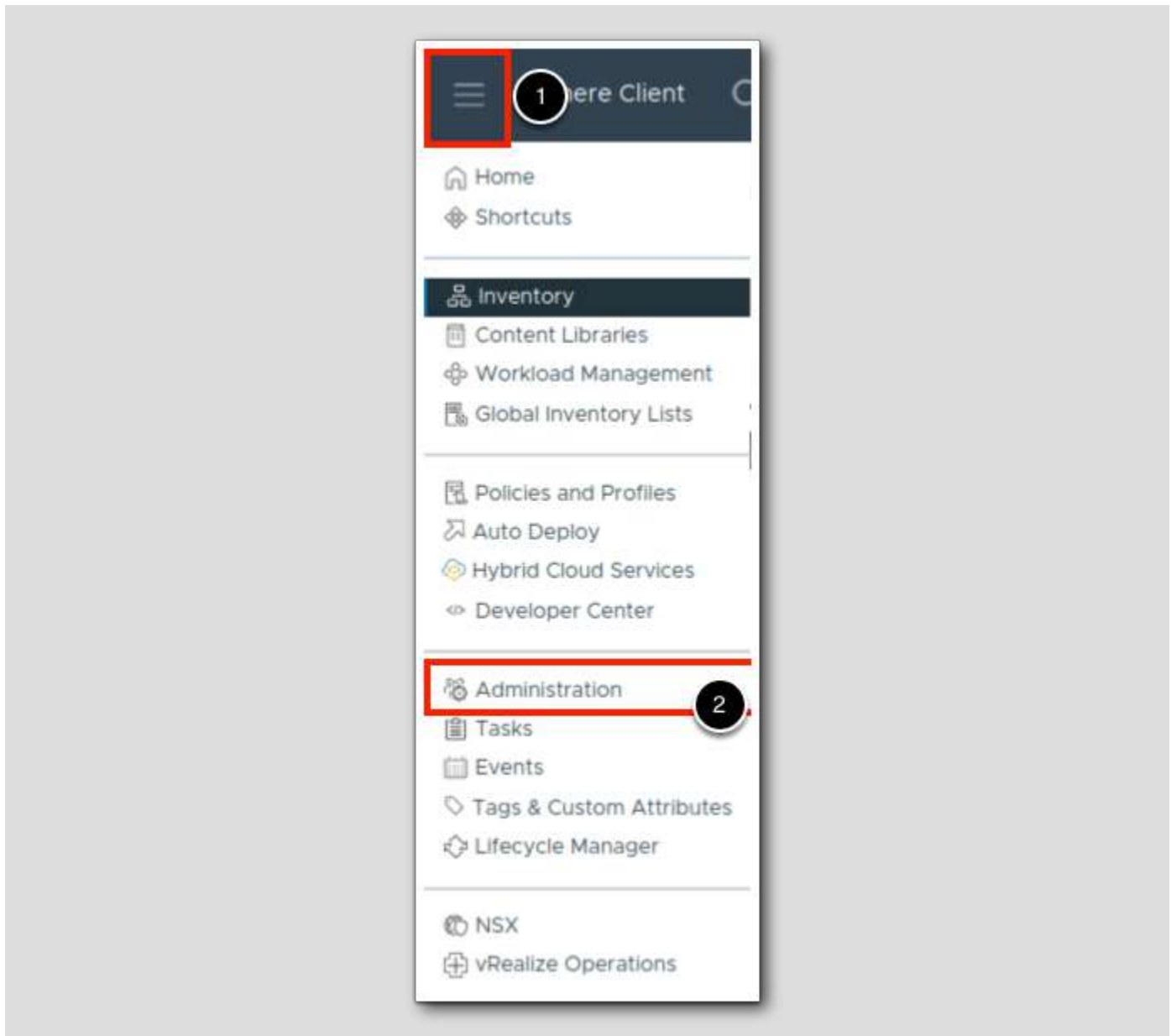
Create a Role in the vSphere Client

[379]

In the following steps, we will create a role in the vSphere Client that we can assign rights for the role.

Administration

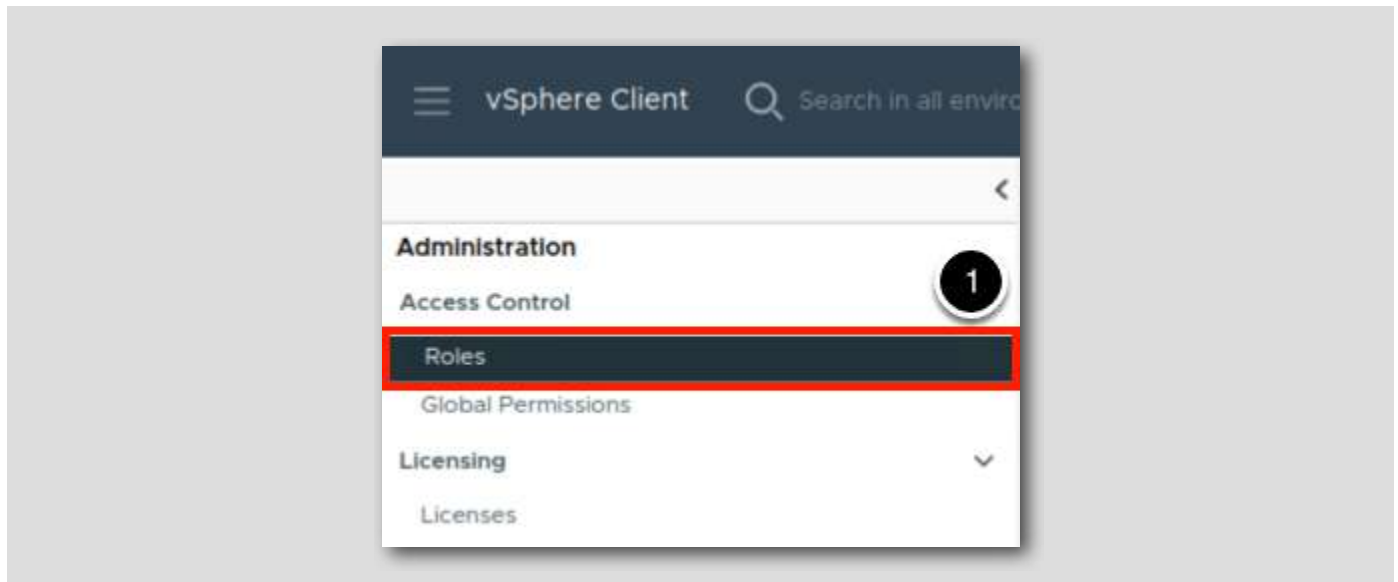
[380]



1. In the vSphere Client, click on **Menu**
2. Select **Administration**

Roles

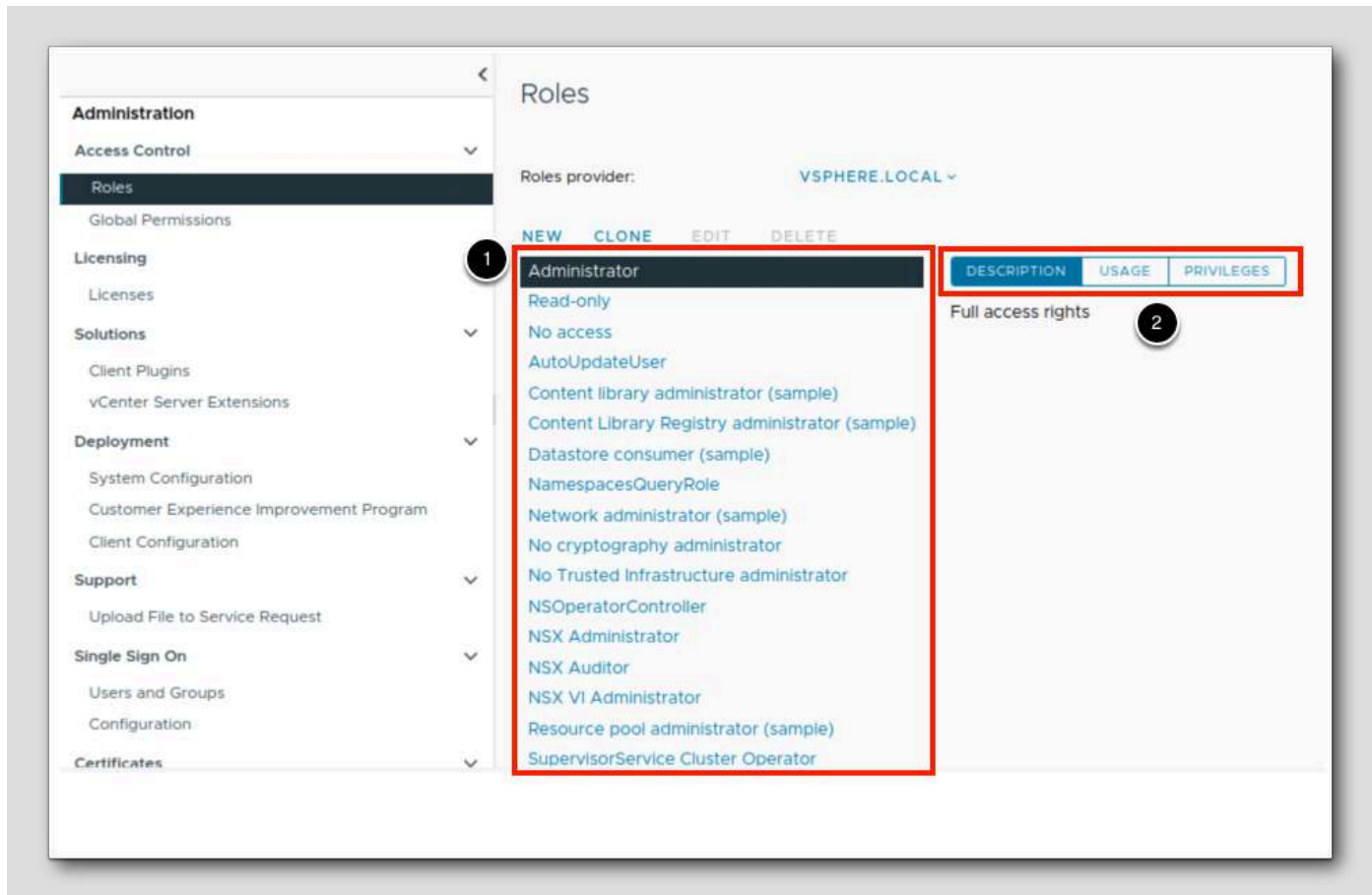
[381]



1. Verify the **Roles** tab is selected

Roles Overview

[382]

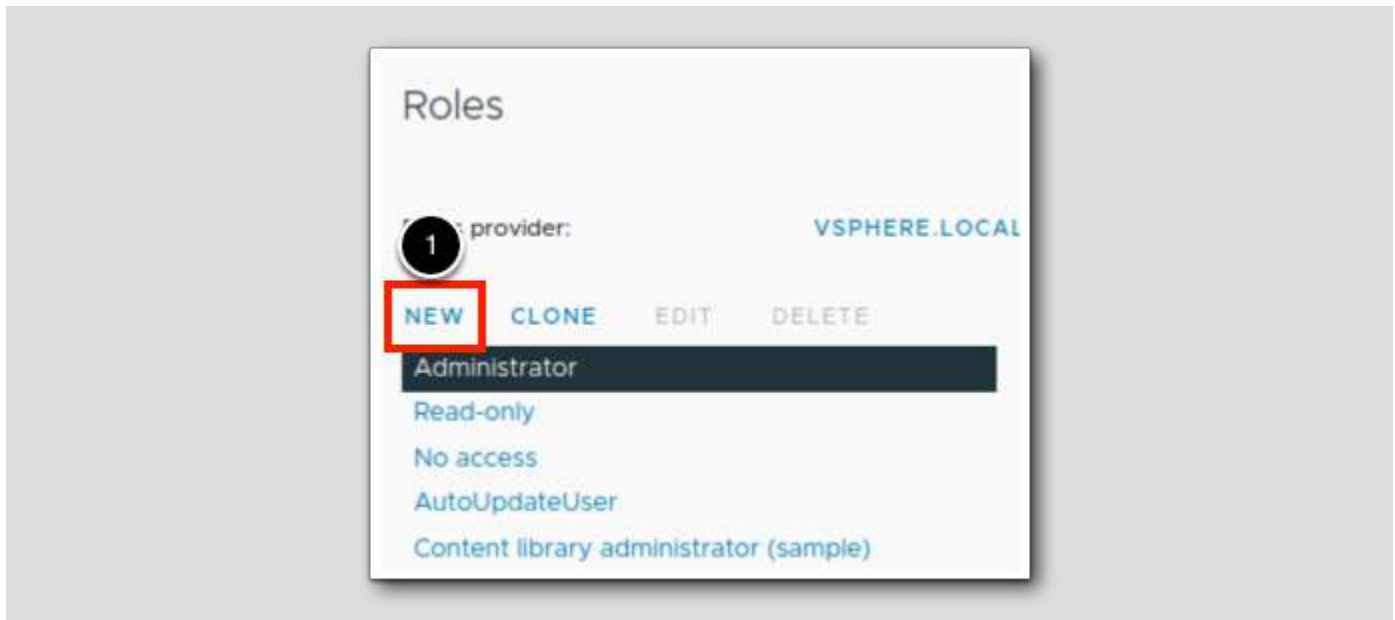


1. The "Roles" panel shows various roles that already exist or are provided as sample to use or create roles from
2. When a role is selected, information such as Description, Usage, and Privileges will be displayed by clicking the corresponding buttons

You can use one of the provided roles as a starting point to create your own or in some cases, it may make sense to create a new rule with zero permissions and only add the one the role will need.

Add a Role

[383]



In this first example, a role will be created for a new contractor that will only be performing networking tasks.

1. Click on **New** to add a new role

New Role

[384]

1. Type "Role1" on the Role name field
2. Use the scrollbar to scroll down until you see **Network**
3. Click **Network**
4. Tick the box for **Select all** to select all network privileges
5. Click **Create**

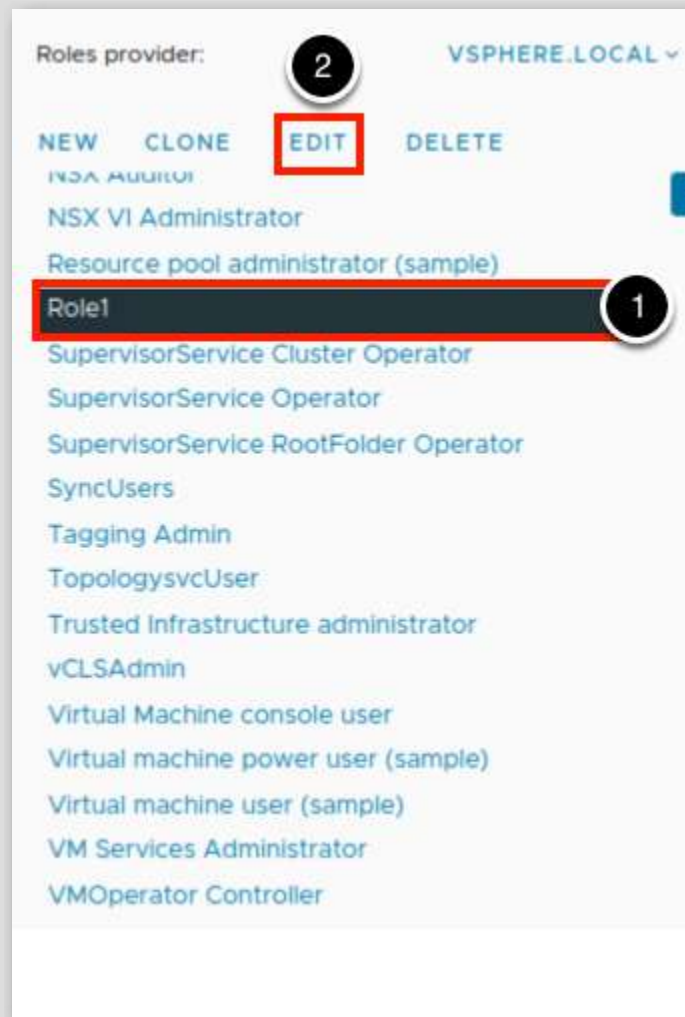
Edit a Role in the vSphere Client

[385]

When you edit a role, you can change the privileges selected for that role. When completed, these privileges are applied to any user or group that is assigned the edited role. In Linked Mode, the changes you make are propagated to all other vCenter Server systems in the group. However, assignments of roles to specific users and objects are not shared across linked vCenter Server systems.

Edit Role

[386]

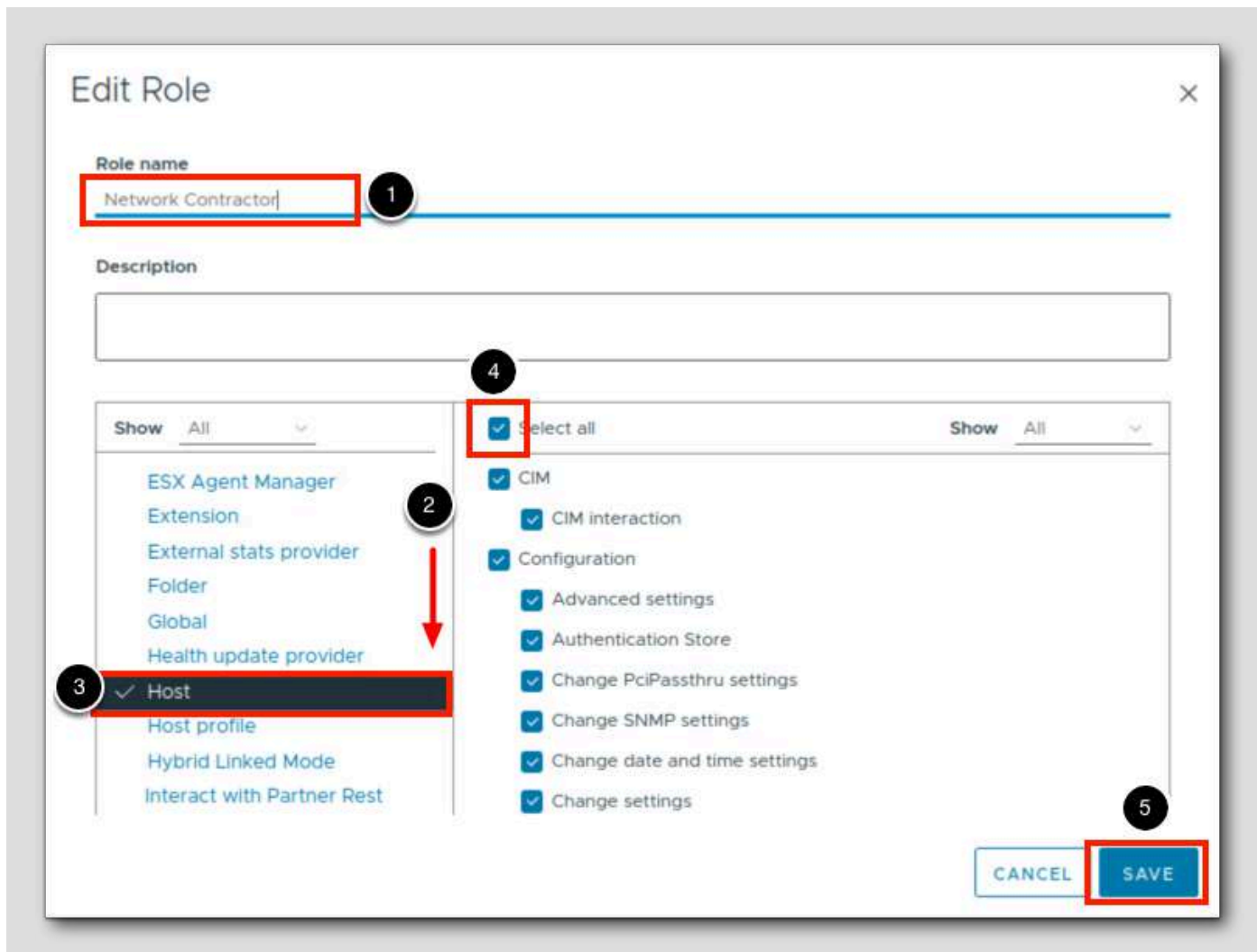


Sometimes a role may need to be updated for access to additional objects or tasks in vCenter. As an example, say the Network Contractor now needs access to the ESXi Hosts.

1. Scroll down if necessary, and click on Role1
2. Click the **Edit** button to modify the role

Add Permissions

[387]



1. Change the name from "Role1" to "Network Contractor"
2. Scroll down until you see Host
3. Click Host
4. Tick the **Select all** box to select all host privileges
5. Click **Save**

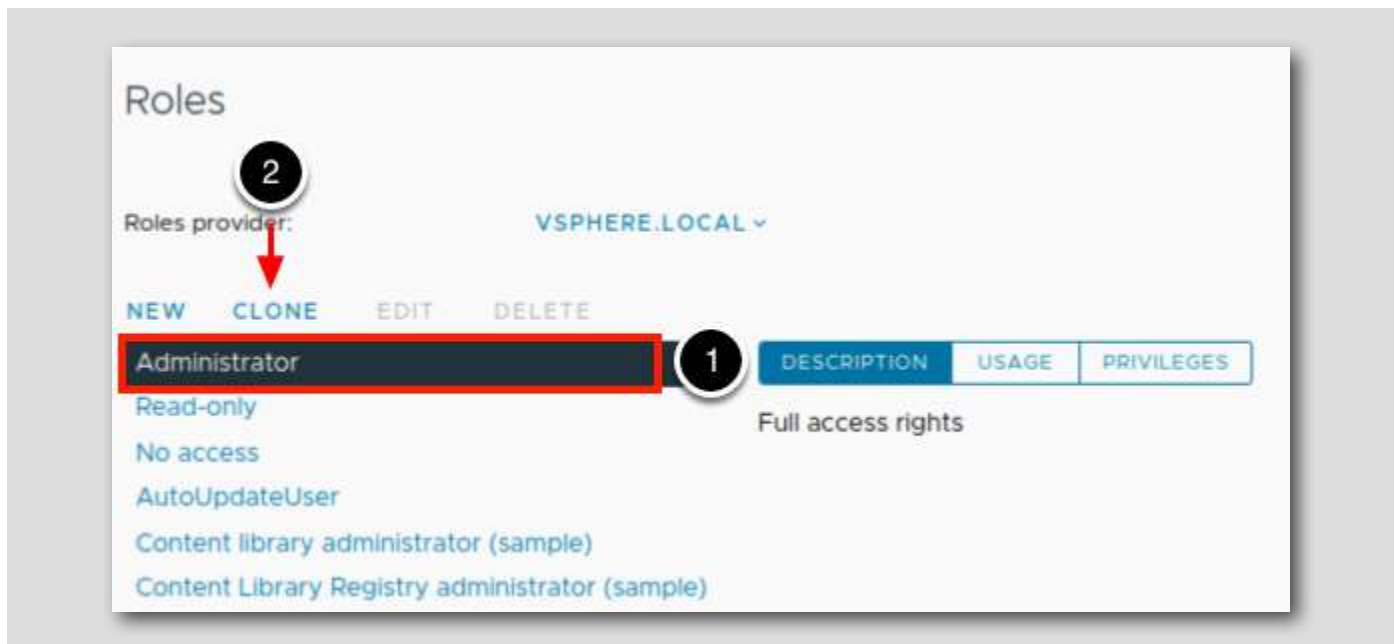
Clone a Role in the vSphere Client

[388]

You can make a copy of an existing role, rename it, and edit it. When you make a copy, the new role is not applied to any users, groups or objects -- it does not inherit anything from the parent except the settings. In Linked Mode, the changes are propagated to all other vCenter Server systems in the group, but assignments of roles to specific users and objects are not shared across linked vCenter Server systems.

Clone a Role

[389]



In this next example, the Administrator role will be cloned and the privileges that are not needed will be removed.

1. Click on the **Administrator** role
2. Click the **Clone** button

Clone Role

[390]

Clone Role

Role name: vSphere Administrator 1

Description: Full rights to all but networking 2

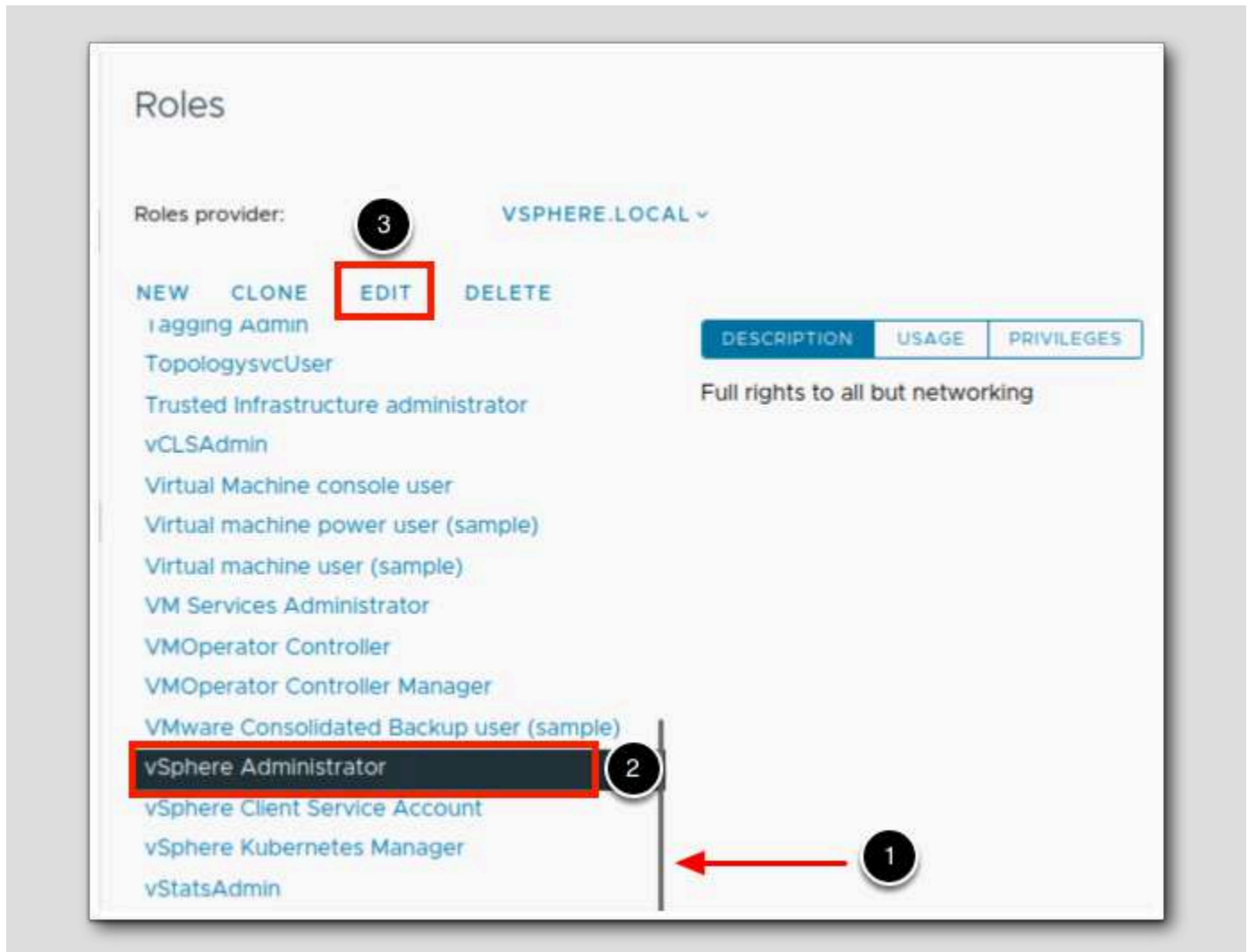
CANCEL OK 3

As an example, a new vSphere Admin is hired and they only need access to the compute and storage infrastructure, with no access to networking components.

1. For the Role name, type vSphere Administrator
2. In the Description field, type Full rights to all but Networking
3. Click OK

New Role Cloned

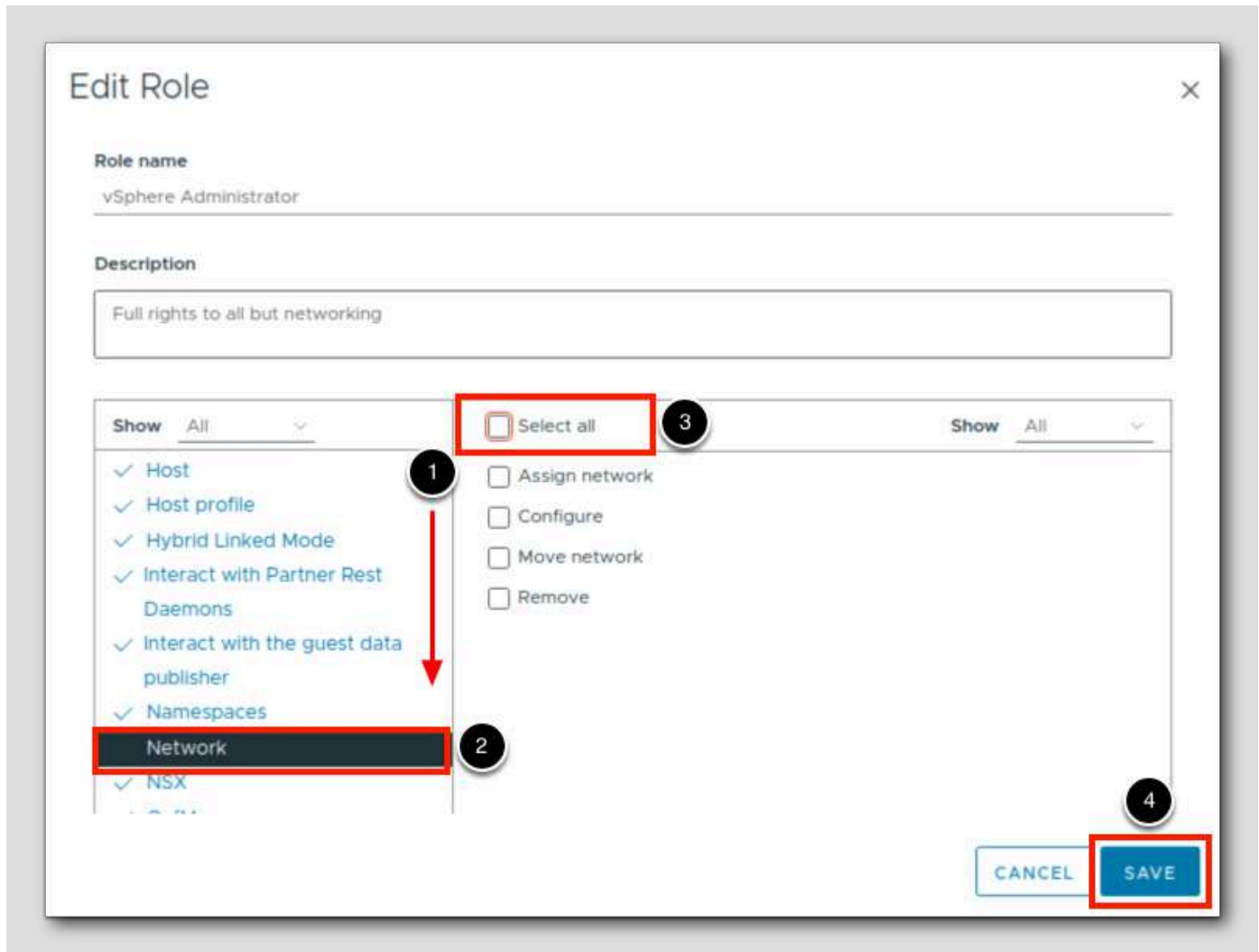
[391]



1. Scroll to the bottom of the list to find the newly created role
2. Click on vSphere Administrator
3. Click the Edit button to modify the role

Edit Role - Network

[392]



Remove a Role in the vSphere Client

[393]

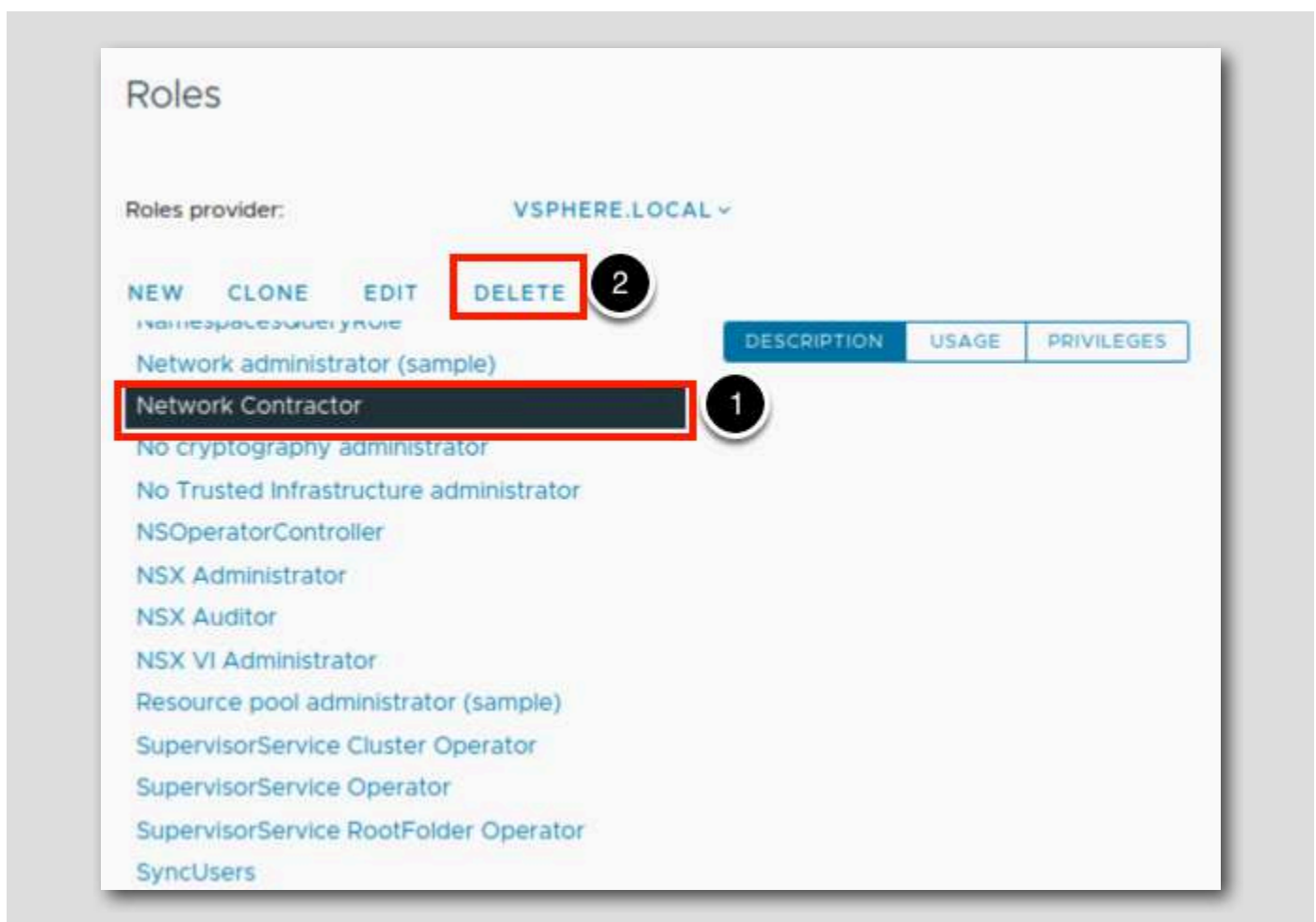
When you remove a role that is not assigned to any users or groups, the definition of the role is removed from the list of roles. When you remove a role that is assigned to a user or group, you can remove assignments or replace them with an assignment to another role.

NOTE:

Before removing a role from a vCenter Server system that is part of a connected group in Linked Mode, check the use of that role on the other vCenter Server systems in the group. Removing a role from one vCenter Server system also removes that role from all other vCenter Server systems in the group, even if you reassign permissions to another role on the current vCenter Server system.

Delete Role

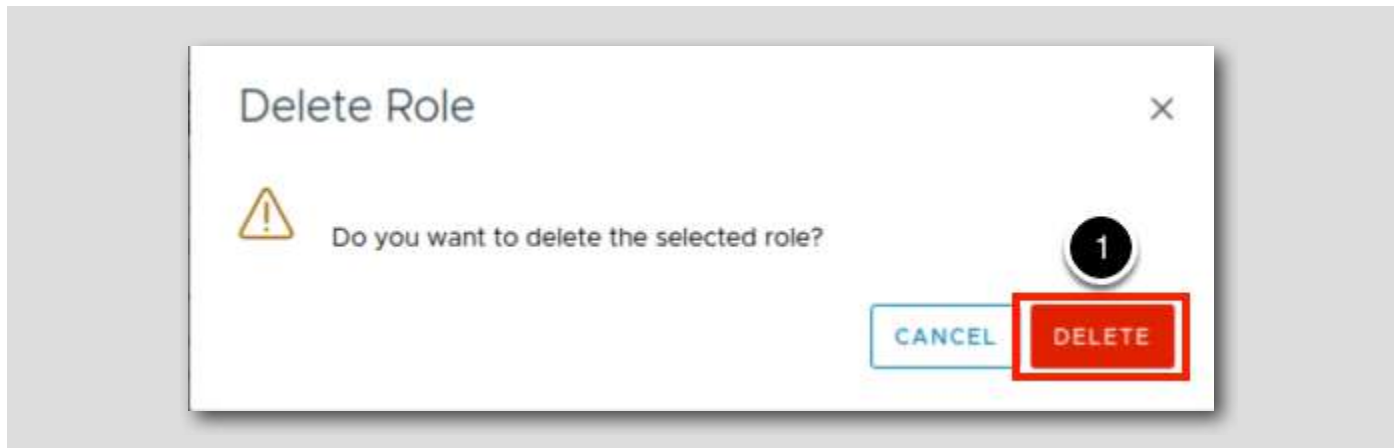
[394]



1. Click on the **Network Contractor** role to select it
2. Click the **Delete** button

Confirm Deletion

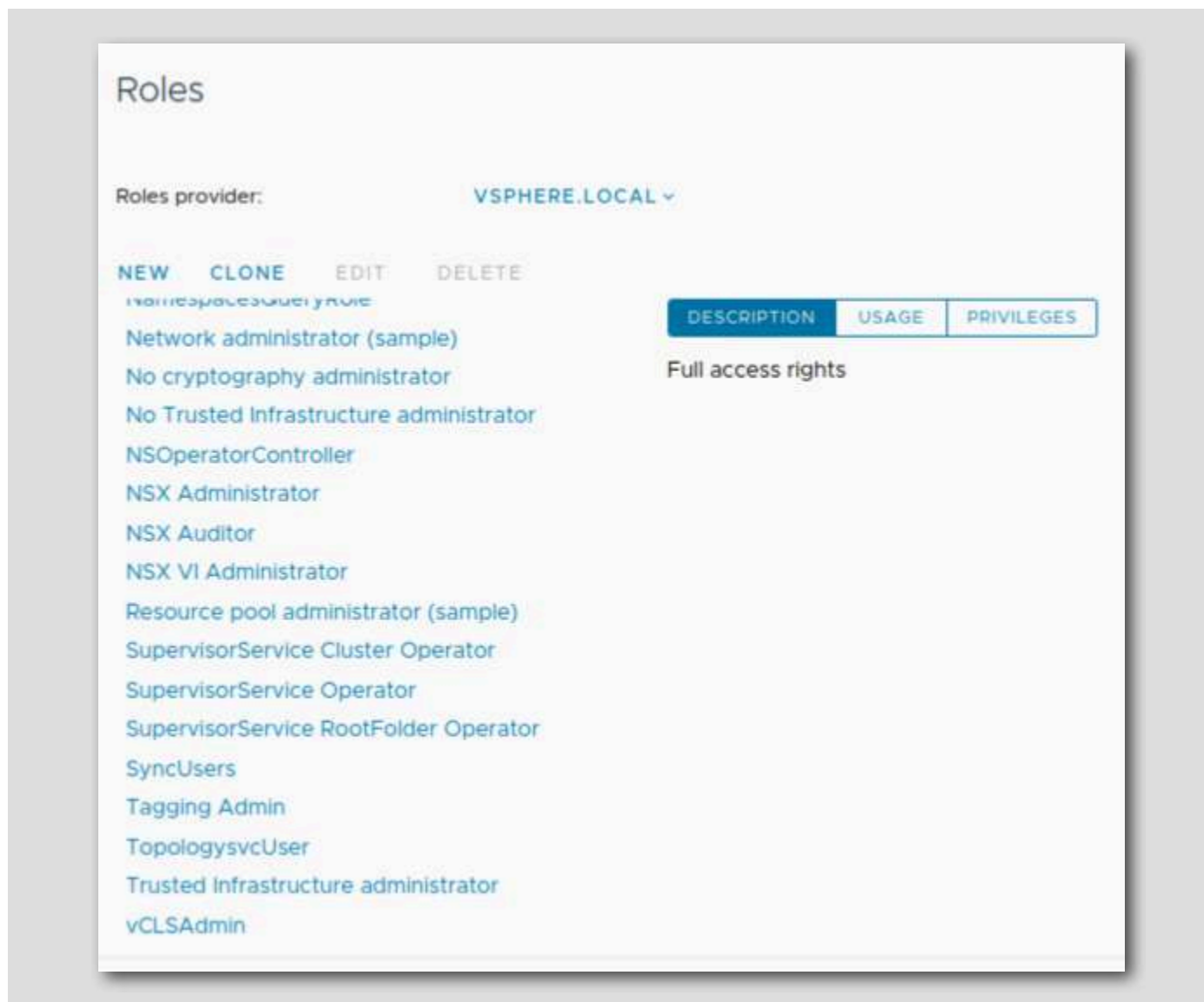
[395]



1. Click **Delete** to confirm you want to delete this role

Role Deleted

[396]



We can see that the role named **Network Contractor** has been deleted.

Creating unique and granular roles for users in your organization enables better security for your vSphere infrastructure.

Understanding Single Sign On

[397]

You use vCenter Single Sign-On to authenticate and manage vCenter Server users.

The Single Sign-On administrative interface is part of the vSphere Web Client. To configure Single Sign-On and manage Single Sign-On

users and groups, you log in to the vSphere Web Client as a user with Single Sign-On administrator privileges. This might not be the same user as the vCenter Server administrator. Enter the credentials on the vSphere Web Client login page and upon authentication, you can access the Single Sign-On administration tool to create users and assign administrative permissions to other users.

In vSphere versions prior to 5.1, users were authenticated when vCenter Server validated their credentials against an Active Directory domain or the list of local operating system users. As of vSphere 5.1, users authenticate through vCenter Single Sign On. The default Single Sign-On administrator for vSphere 5.1 is `admin@System-Domain` and `administrator@vsphere.local` for vSphere 5.5 and higher. The password for this account is the one you specified at installation. These credentials are used to log in to the vSphere Web Client to access the Single Sign-On administration tool. You can then assign Single Sign-On administrator privileges to specific users who are allowed to manage the Single Sign-On server. These users might be different from the users that administer vCenter Server.

NOTE: Logging in to the vSphere Web Client with Windows LDAP credentials is supported only for Active Directory users of the domain to which the Single Sign On system belongs.

Single Sign-On Identity Sources

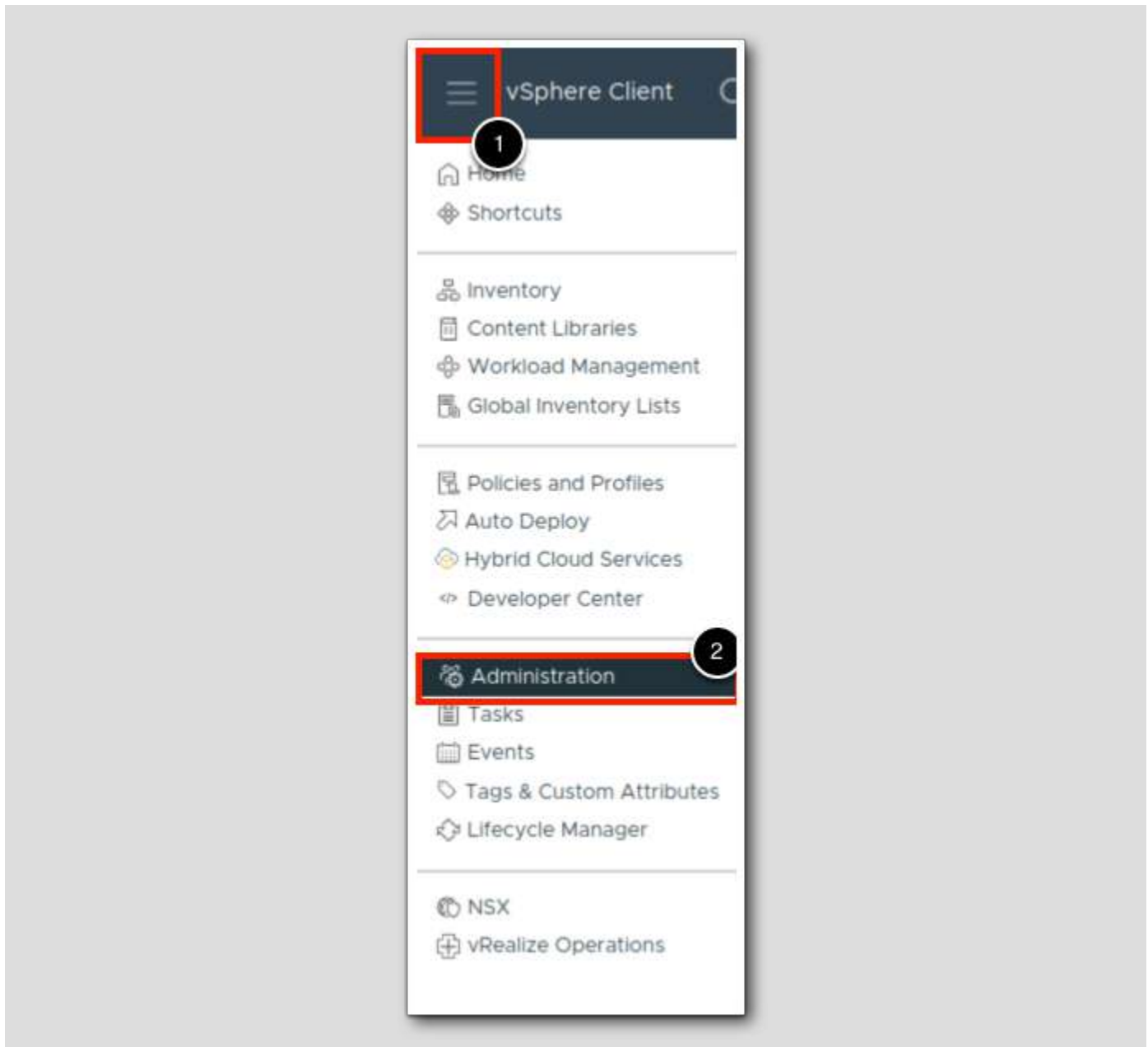
[398]

In most cases, vSphere SSO will be deployed to use an external Identity Source for primary authentication.

In this section, we will look at the configured Identity Sources within Single Sign-on.

Navigate to Administration

[399]

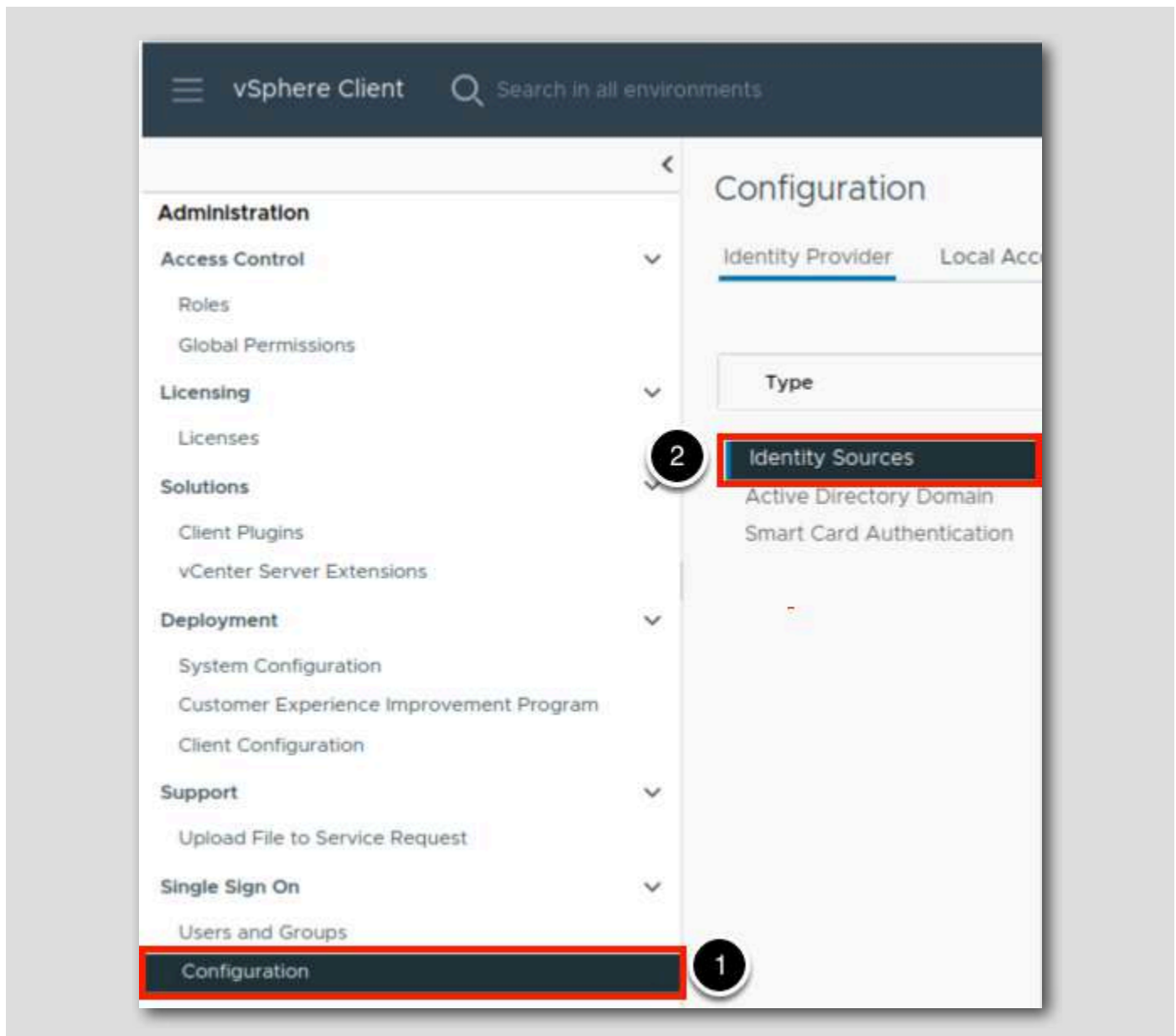


Navigate to Administration if you're not there. If you are already in the screen, skip this step.

1. Click **Menu**
2. Select **Administration**

vSphere Single Sign-on

[400]

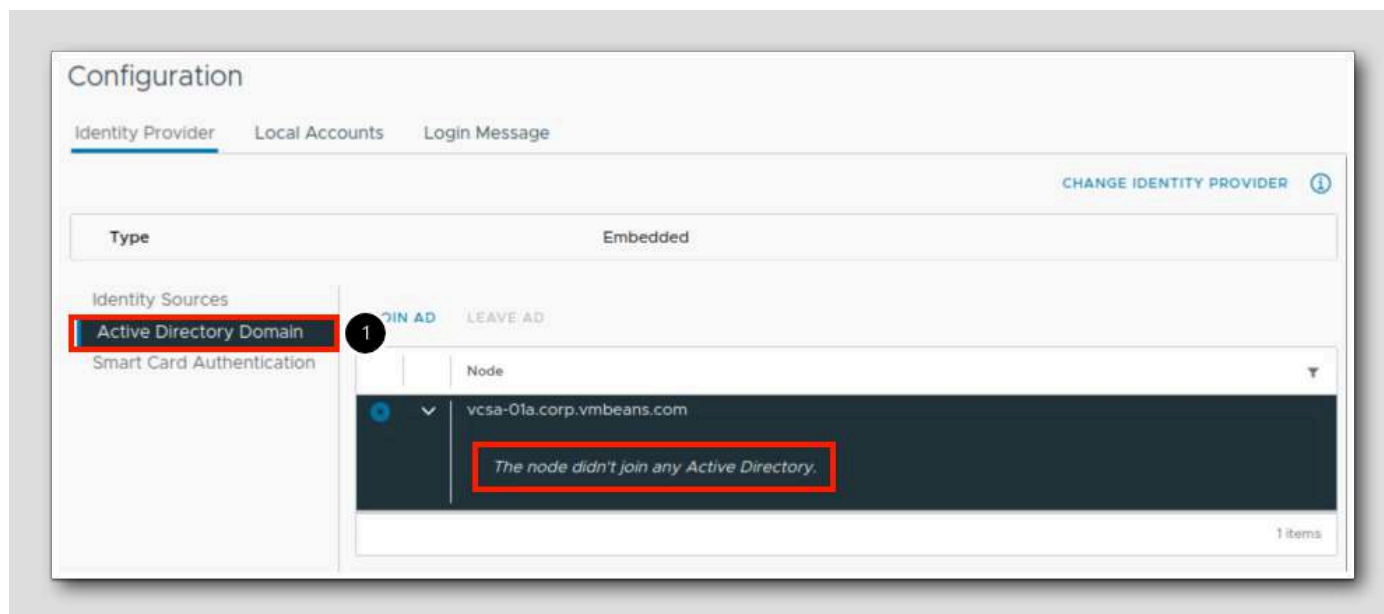


When the machine with the Platform Services Controller (PSC), which runs the Single Sign-On component, is added to an Active Directory domain, the Identity Source for that domain is automatically added to SSO.

1. Click on **Configuration** in the Single Sign-On section of the Navigator
2. Click on the **Identity Sources** tab

Identity Sources

[401]



1. Notice that `vcsa-01a.corp.vmbeans.com` is listed as an Active Directory under Active Directory Domain

However, notice the "The node didn't join any Active Directory" messaging as for this lab purposes, we did not include any Active Directory Sources.

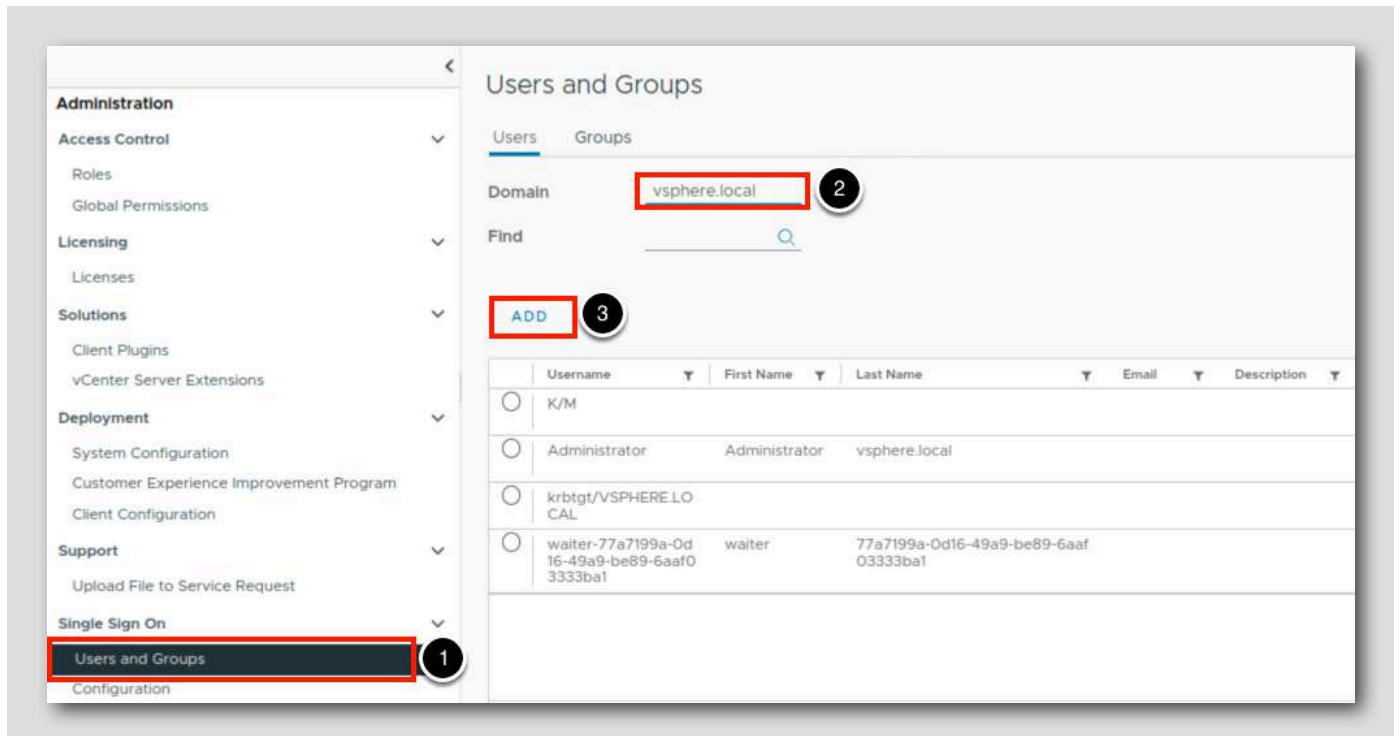
Add a vCenter Single Sign On User with the vSphere Client

[402]

In the vSphere Client, users listed on the Users tab are internal to vCenter Single Sign On. These users are not the same as local operating system users, which are local to the operating system of the machine where Single Sign On is installed (for example, Windows). When you add a Single Sign On user with the Single Sign On administration tool, that user is stored in the Single Sign On database, which runs on the system where Single Sign On is installed. These users are part of the SSO domain, by default, "vsphere.local" -- or "System-Domain" for vSphere 5.1. Exactly one system identity source is associated with an installation of Single Sign On.

List Current Users and Add New User

[403]



1. Click on **Users and Groups** under Single Sign-On
2. From the drop-down list, select **vsphere.local** for the Domain
3. On the Users tab, click the **Add**

Enter Properties for New User

[404]

Add User

Username: holadmin

Password: (toggle icon)

Confirm Password: (toggle icon)

First Name: HOL

Last Name: Admin

Email: holadmion@vsphere.local

Description: [Empty text box]

1

2

CANCEL ADD

1. Fill out the New User form as follows:

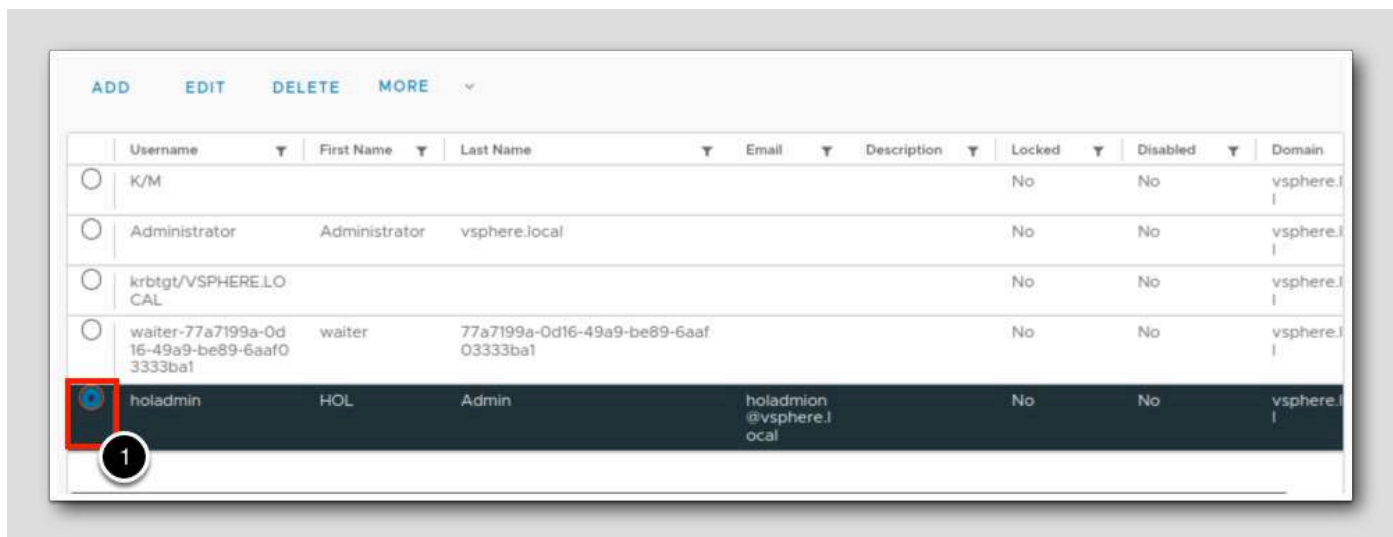
- Username: **holadmin**
- Password: **VMware1!**
- Confirm password: **VMware1!**
- First name: **HOL**
- Last name: **Admin**
- Email address: **holadmin@vsphere.local**

2. Click **ADD** to create the user

NOTE: You cannot change the user's name after you create the user. First and Last name are optional parameters.

New User Added

[405]



	Username	First Name	Last Name	Email	Description	Locked	Disabled	Domain
☐	K/M					No	No	vsphere.local
☐	Administrator	Administrator	vsphere.local			No	No	vsphere.local
☐	krbtgt/VSHERE.LOCAL					No	No	vsphere.local
☐	waiter-77a7199a-0d16-49a9-be89-6aaf03333ba1	waiter	77a7199a-0d16-49a9-be89-6aaf03333ba1			No	No	vsphere.local
☒	holadmin	HOL	Admin	holadmion@vsphere.local		No	No	vsphere.local

Here we can see the new user has been added.

1. By selecting the **radio button** next to the username, it allows for editing, deleting or disabling the user.

Add a vCenter Single Sign On Group with the vSphere Client

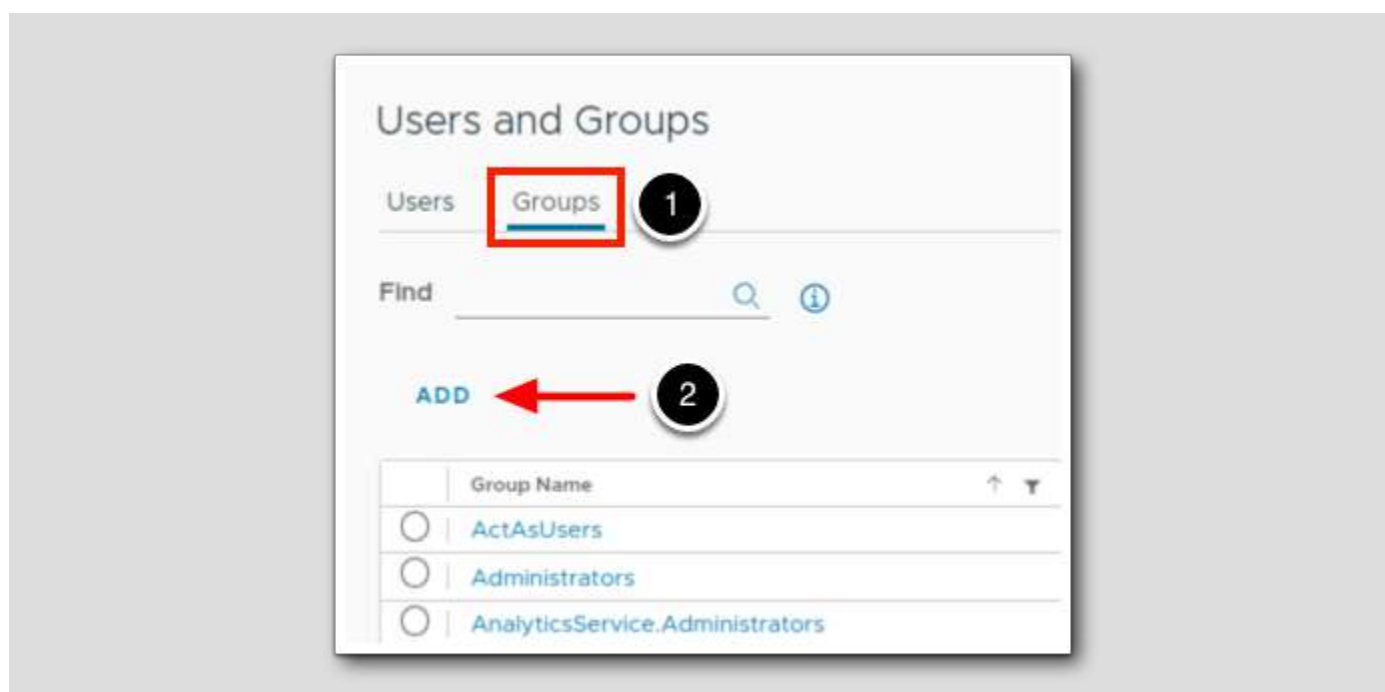
[406]

In the vSphere Client, groups listed on the Groups tab are internal to vCenter Single Sign On. A group lets you create a container for a collection of group members called principals. When you add a Single Sign On group with the Single Sign On administration tool, the group is stored in the Single Sign On database. The database runs on the system where Single Sign On is installed. These groups are part of the identity source domain vsphere.local (the default for vSphere 5.5 and higher), or System-Domain for vSphere 5.1.

Group members can be users or other groups, and a group can contain members from across multiple identity sources. After you create a group and add principals, you apply permissions to the group. Members of the group inherit the group permissions.

Click Groups

[407]



1. Click Groups
2. Click Add Group

Create the New Group

[408]

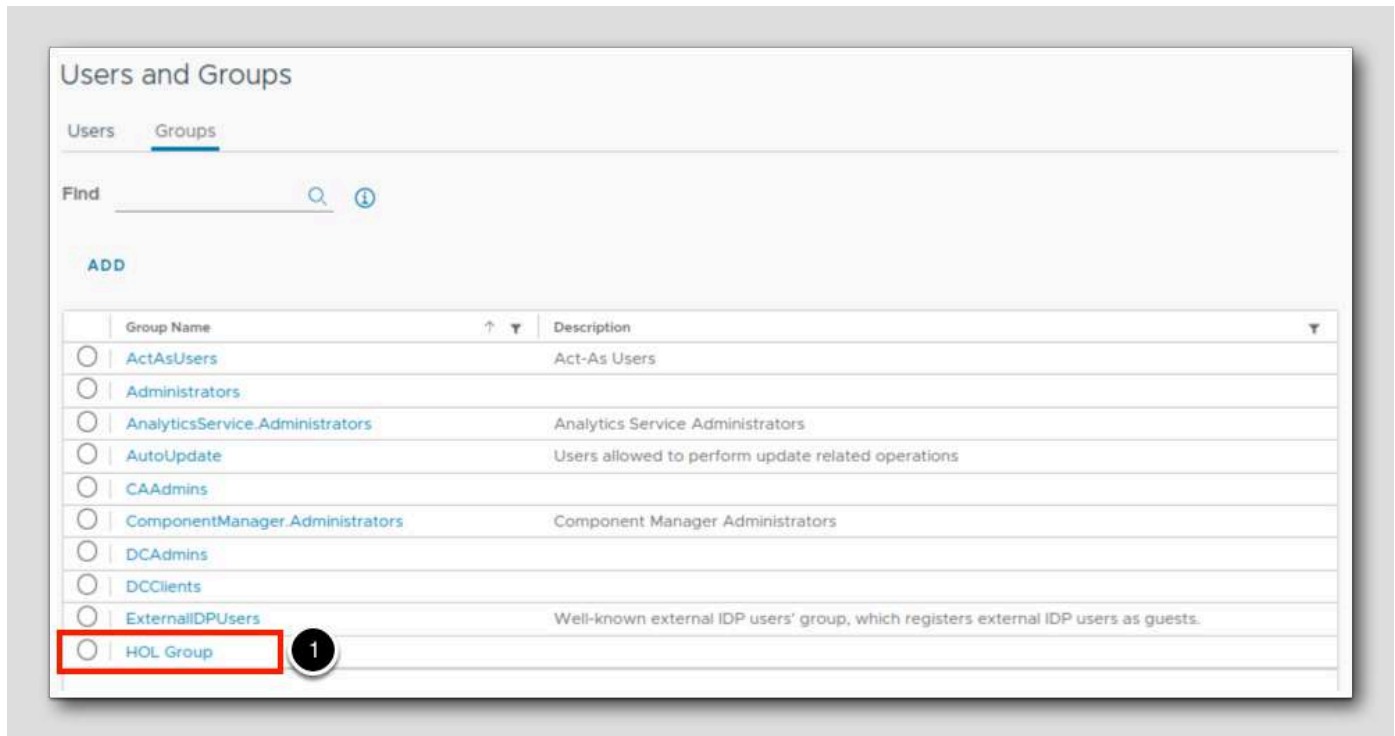
The screenshot shows the 'Add Group' dialog box with the following fields and actions:

- Group Name:** A text field containing 'HOL Group' (marked with a red box and a circled '1').
- Description:** An empty text field.
- Add Members:** A section with a domain field containing 'vsphere.local' and a search dropdown (marked with a circled '2'). The dropdown list shows 'holadmin' (marked with a circled '3' and a red arrow pointing to it).
- Buttons:** 'CANCEL' and 'ADD' buttons at the bottom right. The 'ADD' button is highlighted with a red box and a circled '4'.

1. For the Group Name, type HOL Group
2. Add the user that was previously created by typing holadmin
3. Click **holadmin** from the drop-down list
4. Click the **Add** button

New Group Added

[409]



1. Here is the HOL Group that was just created

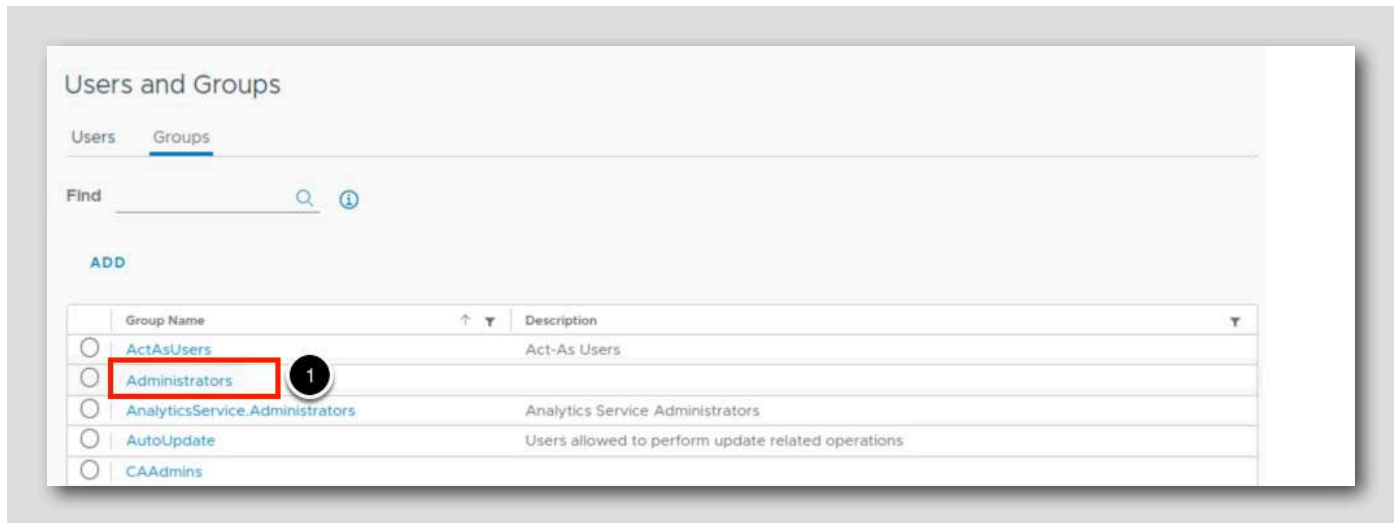
Add Members to a vCenter Single Sign On Group in the vSphere Client

[410]

Members of a vCenter Single Sign On group can be users or other groups from one or more identity sources. Members of a group are called principals. Groups listed on the Groups tab in the vSphere Client are internal to Single Sign On and are part of the identity source System-Domain. You can add group members from other domains to a local group. You can also nest groups.

Add Members to Users and Groups

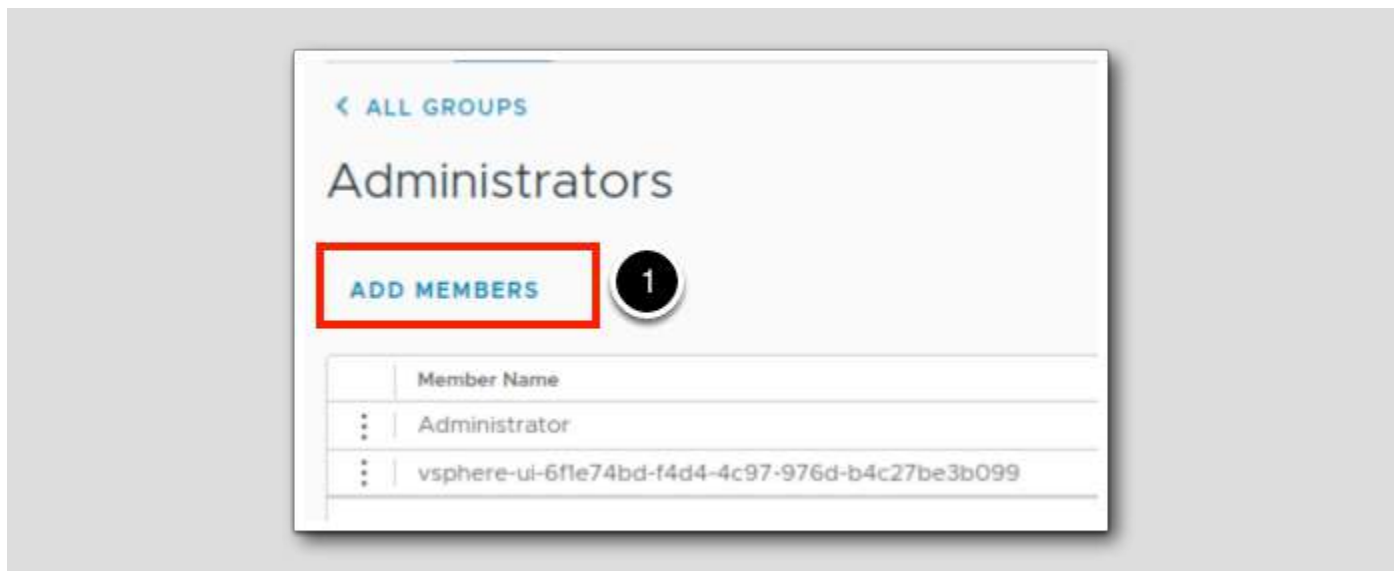
[411]



1. Click on the **Administrators** group under the Group Names table

Add Members

[412]



The Administrator account for the vsphere.local and corp.local domains are members.

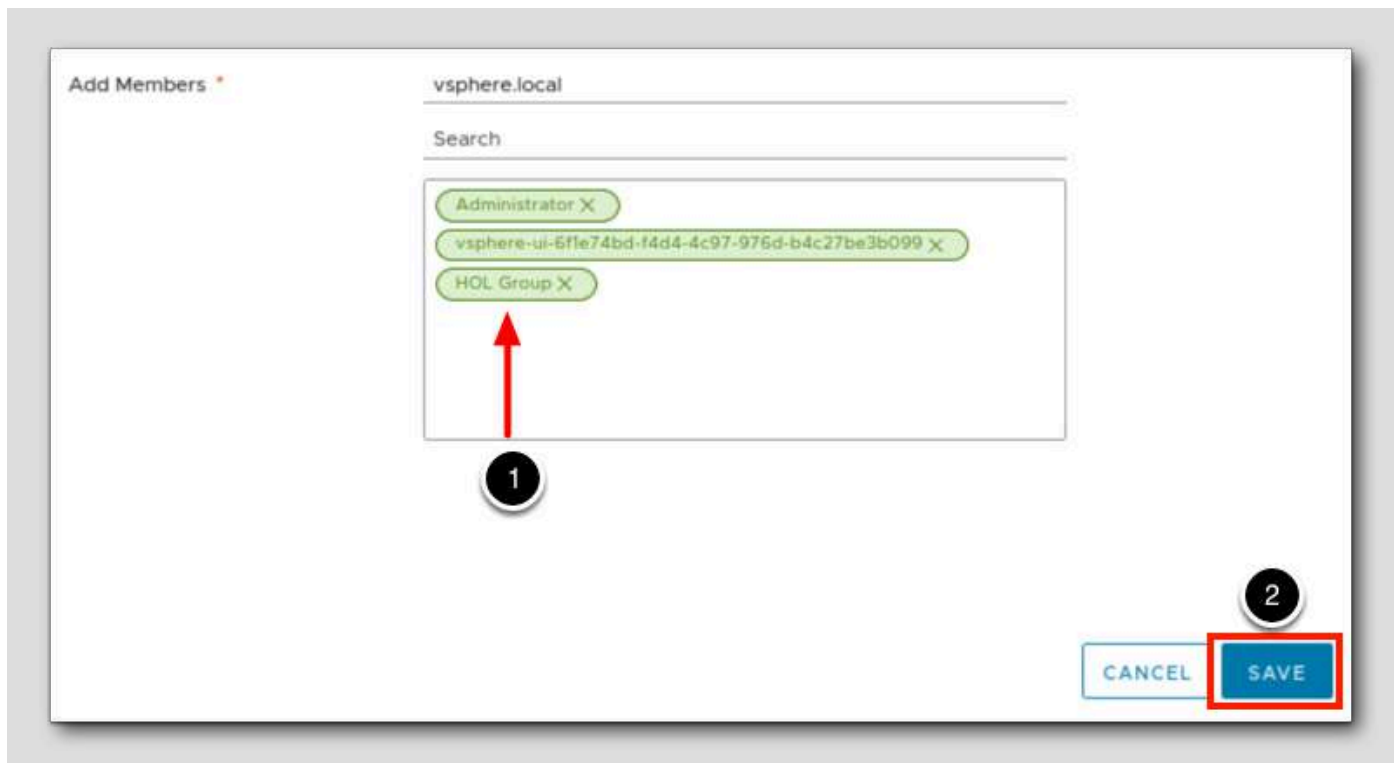
1. Click **Add Members**

Edit Group

[413]

The screenshot shows the 'Edit Group' dialog box. On the left, there are labels for 'Group Name', 'Description', and 'Add Members'. On the right, there are input fields and a list of members. The 'Add Members' section has a search bar containing 'HOL Group'. Below the search bar, a list of results is shown, with 'HOL Group' selected. Three numbered callouts are present: 1 points to the domain 'vsphere.local' in the search bar; 2 points to the search text 'HOL Group'; 3 points to the selected 'HOL Group' entry in the results list. At the bottom right, there are 'CANCEL' and 'SAVE' buttons.

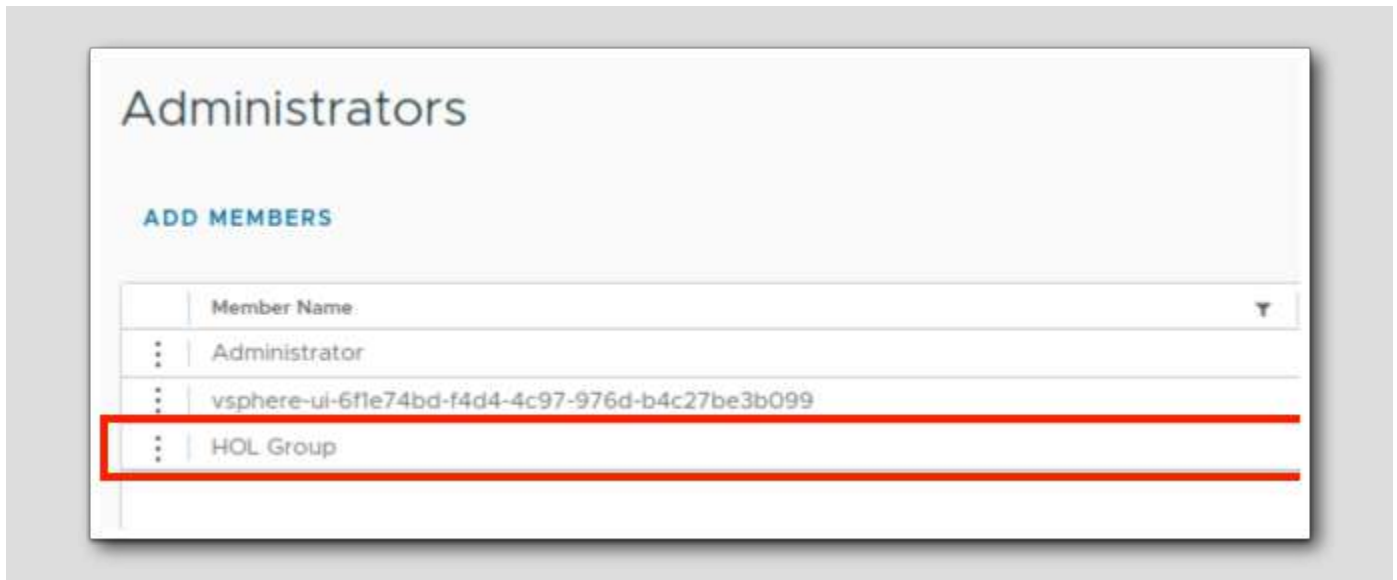
1. Make sure the domain selected is `vsphere.local`
2. Type `HOL Group` in the search box
3. Click on `HOL Group` to add it to the member list



1. You should see HOL Group added to the list.
2. Click Save.

New Member Added

[414]



The HOL Group has now been added to the Administrator group.

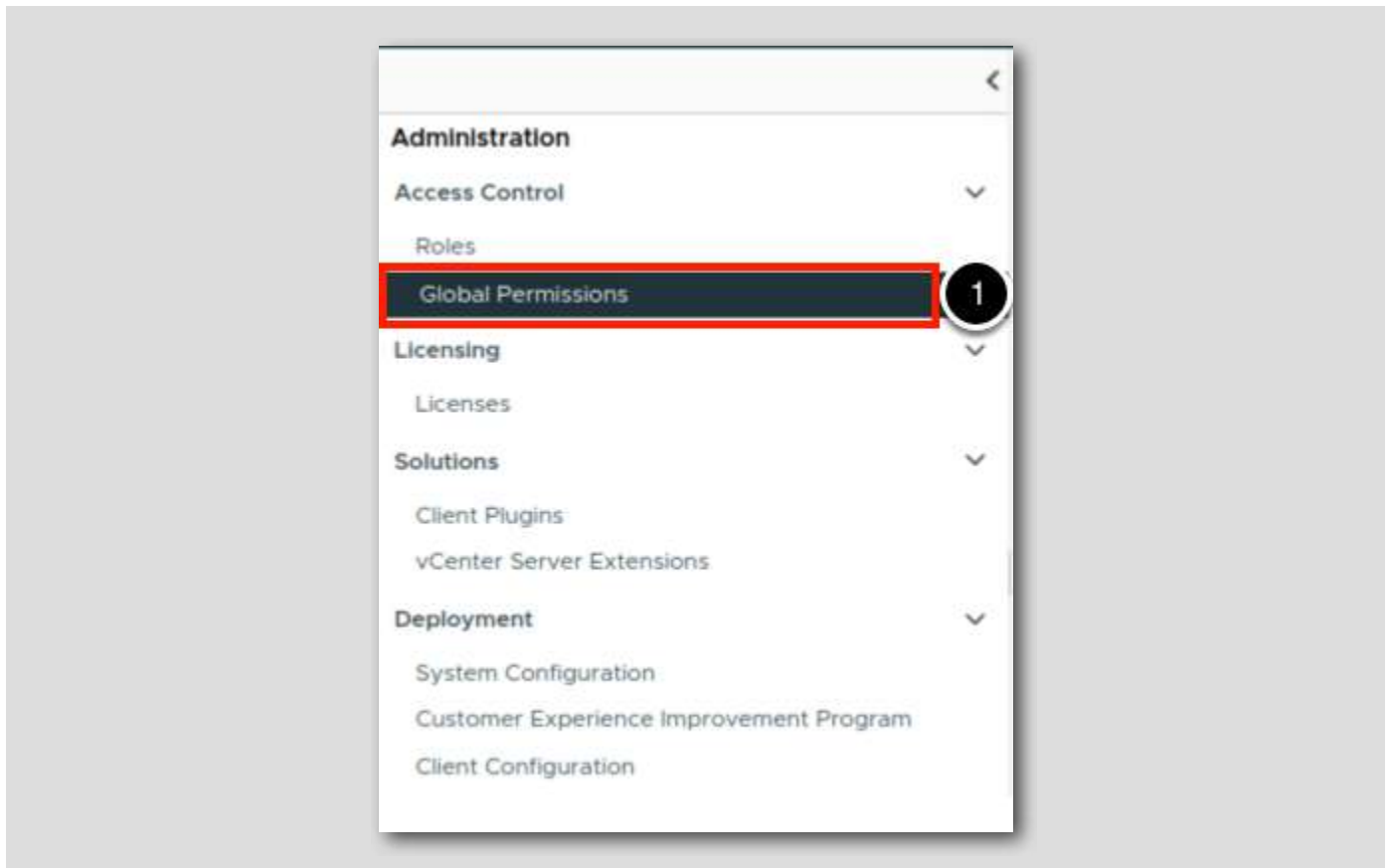
Assign Global Permissions

[415]

Once identity sources, users and groups have been configured, they must be assigned permissions in order to be useful in vSphere.

List Global Permissions

[416]

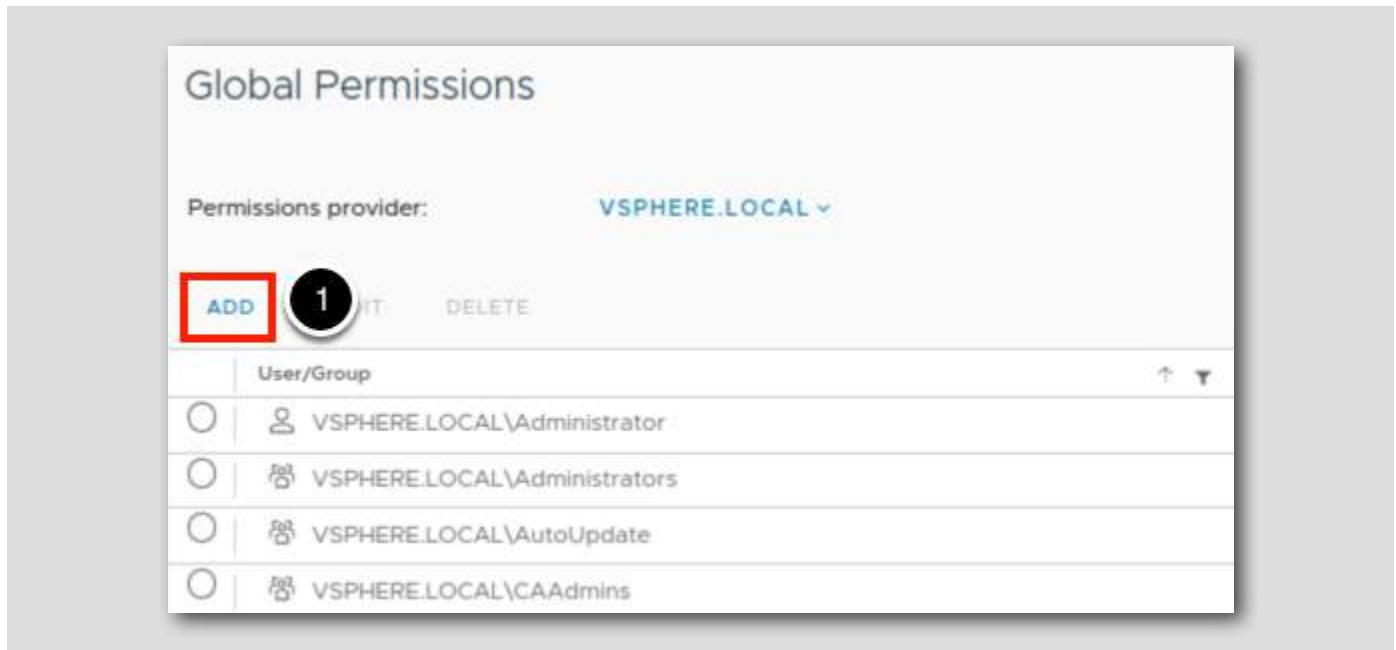


1. Click on the **Global Permissions** item under Access Control

SSO provides the ability to grant Global Permissions to an account by specifying the required access here. In the lab, this list represents the default permissions granted, with the exception of the `VSPHERE.LOCAL\Administrator` user that we have added with Administrator permissions to the entire vSphere infrastructure.

Add New Global Permission

[417]

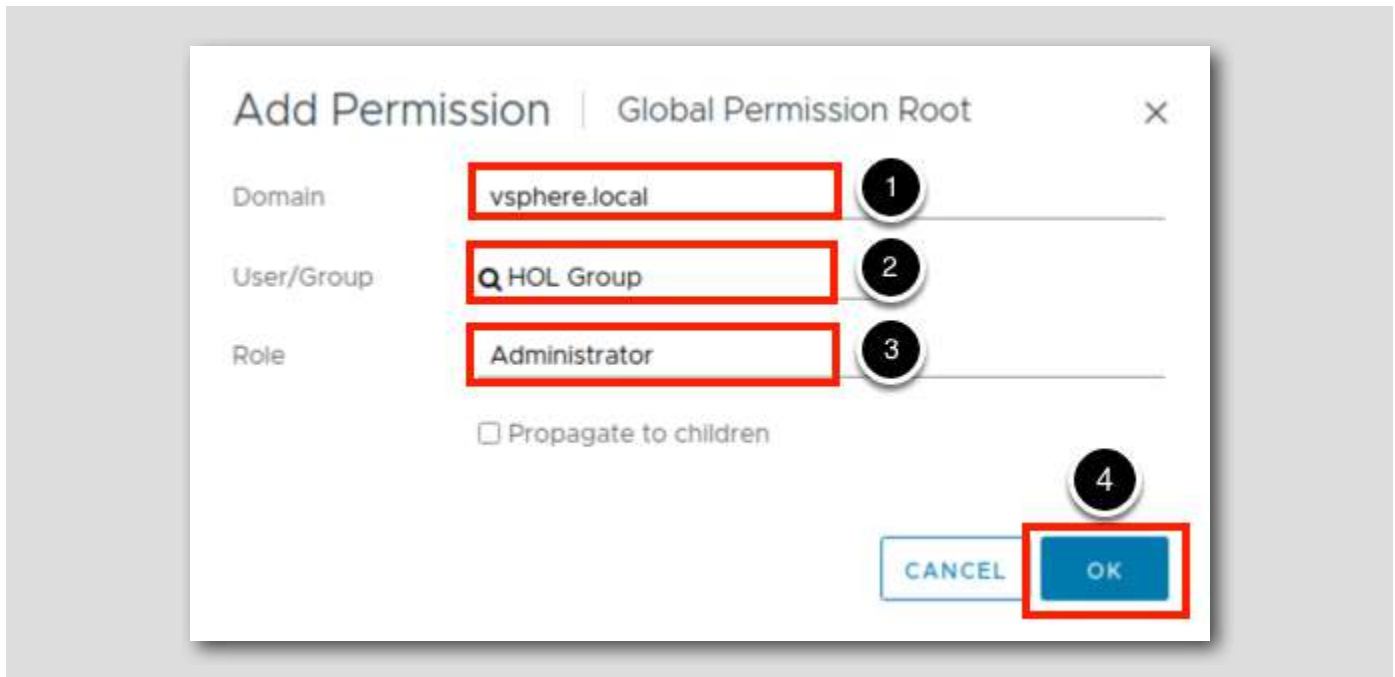


The members of the HOL Group will need to manage all virtual machines in the environment, so we will configure permissions here.

1. Click **Add** button to open the Add New Permission window

Locate the HOL Group

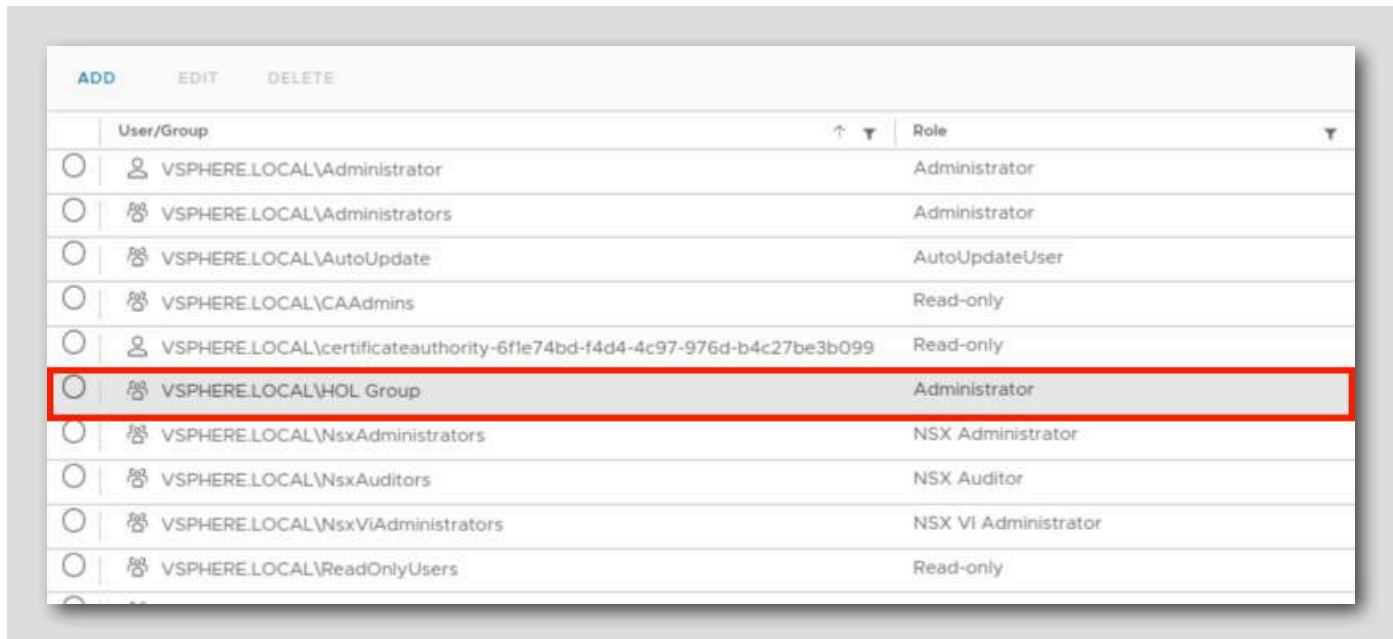
[418]



1. Ensure that the `vsphere.local` domain is selected
2. Type `HOL Group` in the search field
3. For the Role, select the `Administrator` group
4. Click the OK button

New Global Permission

[419]



	ADD	EDIT	DELETE		User/Group	Role
☐					VSPHERE.LOCAL\Administrator	Administrator
☐					VSPHERE.LOCAL\Administrators	Administrator
☐					VSPHERE.LOCAL\AutoUpdate	AutoUpdateUser
☐					VSPHERE.LOCAL\CAAdmins	Read-only
☐					VSPHERE.LOCAL\certificateauthority-6f1e74bd-f4d4-4c97-976d-b4c27be3b099	Read-only
☐					VSPHERE.LOCAL\HOL Group	Administrator
☐					VSPHERE.LOCAL\NsxAdministrators	NSX Administrator
☐					VSPHERE.LOCAL\NsxAuditors	NSX Auditor
☐					VSPHERE.LOCAL\NsxViAdministrators	NSX Vi Administrator
☐					VSPHERE.LOCAL\ReadOnlyUsers	Read-only

The newly created `vsphere.local` Global Permission has been created.

Conclusion

[420]

Typically, user accounts will not be managed naively within the SSO domain, but will be handled by an external directory source like Microsoft Active Directory or OpenLDAP. Understanding how SSO handles accounts and where to look for account-to-permission binding is useful for managing a vSphere implementation.

Certification Path

[421]

Learn and practice with Hands-On Labs to help prepare for several VMware Certifications.



This lab can help you study for the industry-recognized VCAP-DCV Deploy 2023 certification which validates that you know how to deploy and optimize VMware vSphere infrastructures.

Learn more here: <https://www.vmware.com/learning/certification/vcap-dcv-deploy.html>

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Data Center
Virtualization Deploy
2023

Module 3 - Introduction to vSphere Storage (60 minutes)

vSphere Storage Overview

[423]

The following lesson provides an overview of the different types of storage available in vSphere.

The vSphere Hypervisor, ESXi, provides host-level storage virtualization, which logically abstracts the physical storage layer from virtual machines.

A vSphere virtual machine uses a virtual disk to store its operating system, program files, and other data associated with its activities. A virtual disk is a large physical file, or a set of files, that can be copied, moved, archived, and backed up as easily as any other file. You can configure virtual machines with multiple virtual disks.

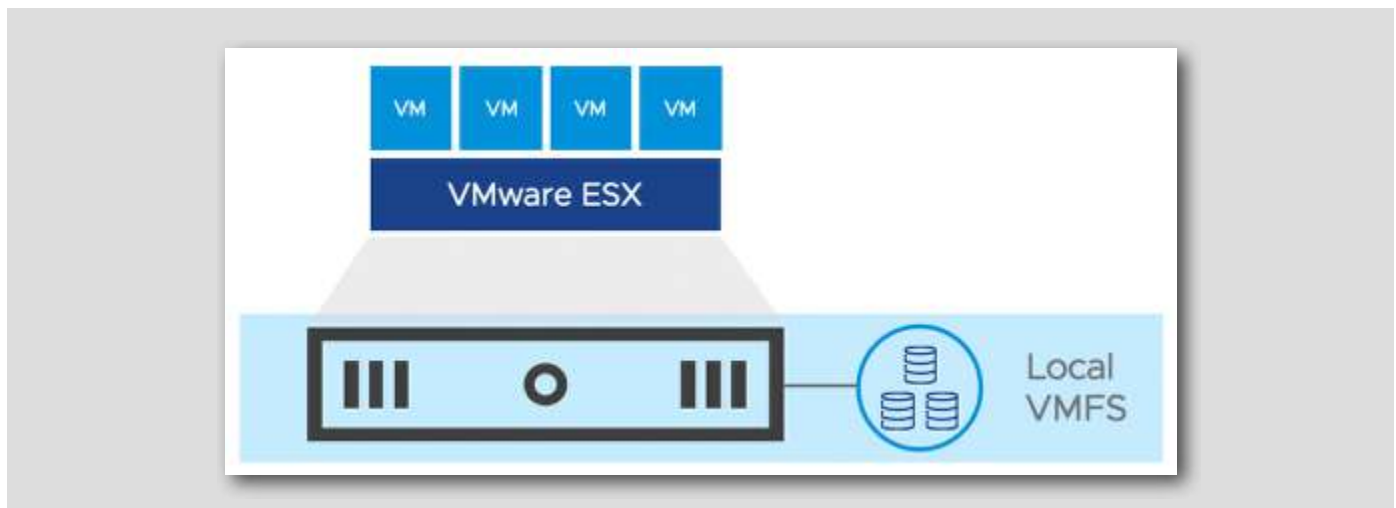
To access virtual disks, a virtual machine uses virtual SCSI controllers. These virtual controllers include BusLogic Parallel, LSI Logic Parallel, LSI Logic SAS, and VMware Paravirtual. These controllers are the only types of SCSI controllers that a virtual machine can see and access.

Each virtual disk resides on a vSphere Virtual Machine File System (VMFS) datastore or an NFS-based datastore that are deployed on physical storage. From the standpoint of the virtual machine, each virtual disk appears as if it were a SCSI drive connected to a SCSI controller. Whether the actual physical storage device is being accessed through parallel SCSI, iSCSI, network, Fibre Channel, or FCoE adapters on the host is transparent to the guest operating system and to applications running on the virtual machine.

The vSphere storage management process starts with storage space that your storage administrator allocates on different storage systems prior to vSphere ESXi assignment. vSphere supports two types of storage - Local and Networked. Each type is detailed in the following lesson steps.

Local Storage

[424]

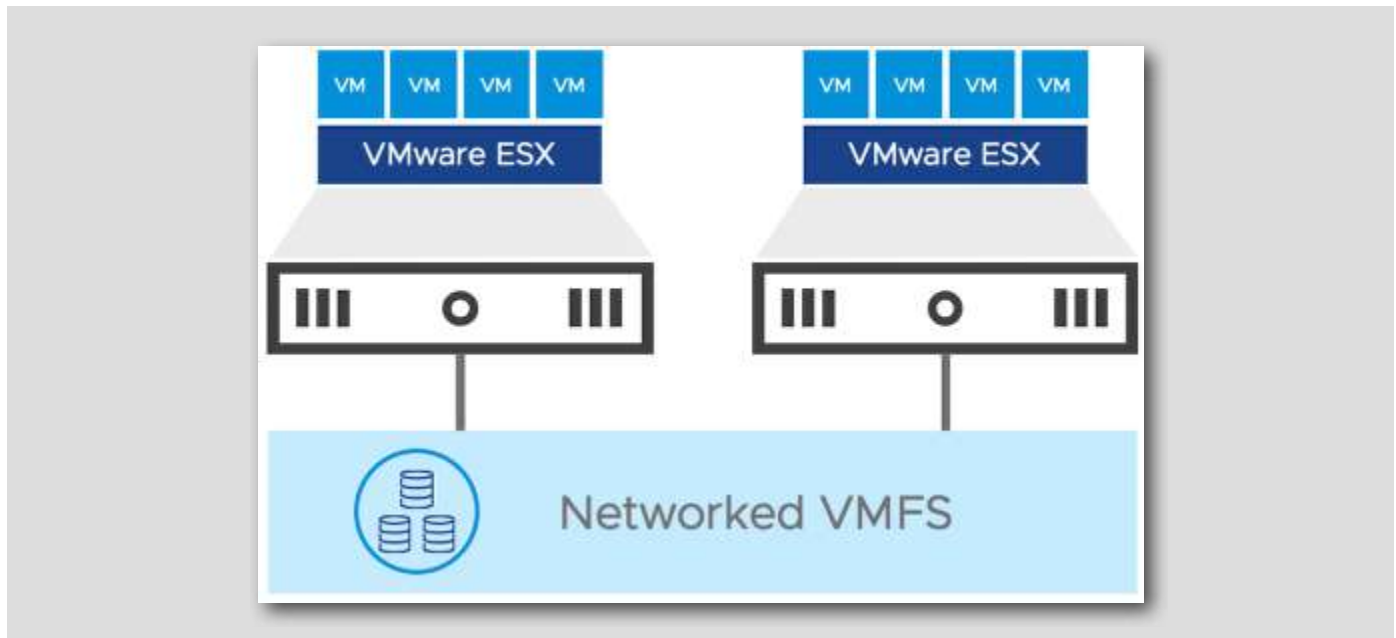


The illustration above depicts virtual machines using Local VMFS storage directly attached to a single ESXi host.

Local storage can be internal hard disks located inside your ESXi host, or it can be external storage systems located outside and connected to the host directly through protocols such as SAS or SATA.

Networked Storage

[425]

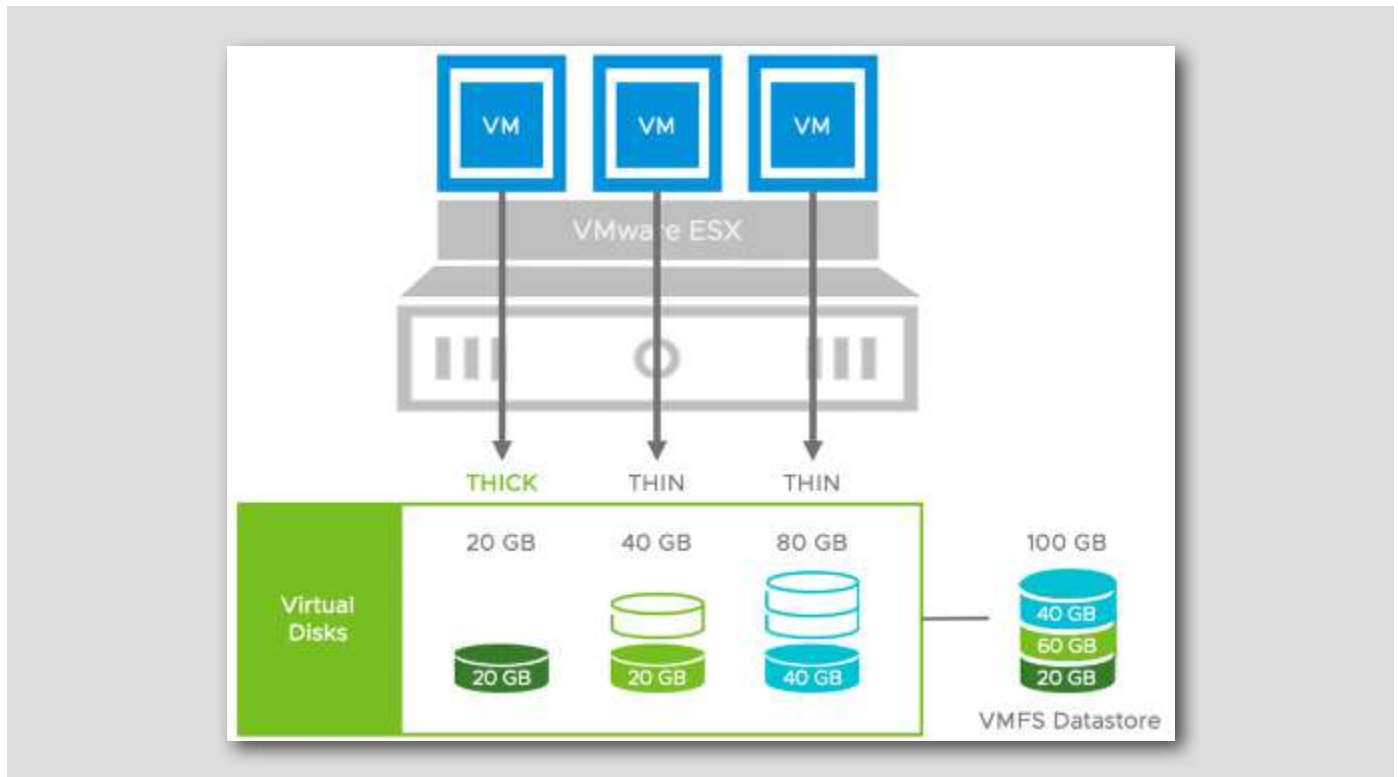


The illustration above depicts virtual machines using networked VMFS storage presented to multiple ESXi hosts.

Networked storage consists of external storage systems that your ESXi host uses to store virtual machine files remotely. Typically, the host accesses these systems over a high-speed storage network. Networked storage devices are typically shared. Datastores on networked storage devices can be accessed by multiple hosts concurrently, and as a result, enable additional vSphere technologies such as High Availability host clustering, Distributed Resource Scheduling, vMotion and Virtual Machines configured with Fault Tolerance. ESXi supports several networked storage technologies - Fiber Channel, iSCSI, NFS, and Shared SAS.

Virtual Machine Disks

[426]



The illustration above depicts virtual machines using different types of virtual disk formats against a shared VMFS Datastore.

When you perform certain virtual machine management operations, such as creating a virtual disk, cloning a virtual machine to a template, or migrating a virtual machine, you can specify a provisioning policy for the virtual disk file format. There are three types of virtual disk formats:

Thin Provision

Use this format to save storage space. For the thin disk, you provision as much datastore space as the disk would require based on the value that you enter for the disk size. However, the thin disk starts small and at first, uses only as much datastore space as the disk needs for its initial operations.

Thick Provision Lazy Zeroed

Creates a virtual disk in a default thick format. Space required for the virtual disk is allocated when the virtual disk is created. Data remaining on the physical device is not erased during creation, but is zeroed out on demand at a later time on first write from the virtual machine.

Using the thick-provision, lazy-zeroed format does not zero out or eliminate the possibility of recovering deleted files or restoring old data that might be present on this allocated space. You cannot convert a thick-provisioned, lazy-zeroed disk to a thin disk.

Thick Provision Eager Zeroed

A type of thick virtual disk that supports clustering features such as Fault Tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the thick-provision, lazy-zeroed format, the data remaining on the physical device is zeroed out when the virtual disk is created. In general, it takes much longer to create disks in this format than to create other types of disks.

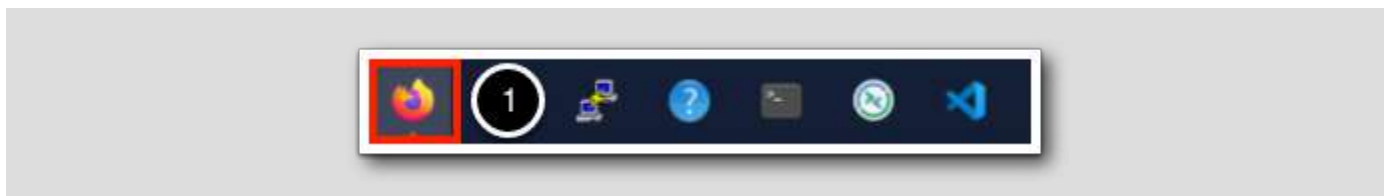
Creating and Configuring vSphere Datastores

[427]

This lesson will walk you through creating and configuring an NFS, and an iSCSI vSphere Datastore. Also adding and configuring an iSCSI software adapter.

Launch Firefox web browser

[428]

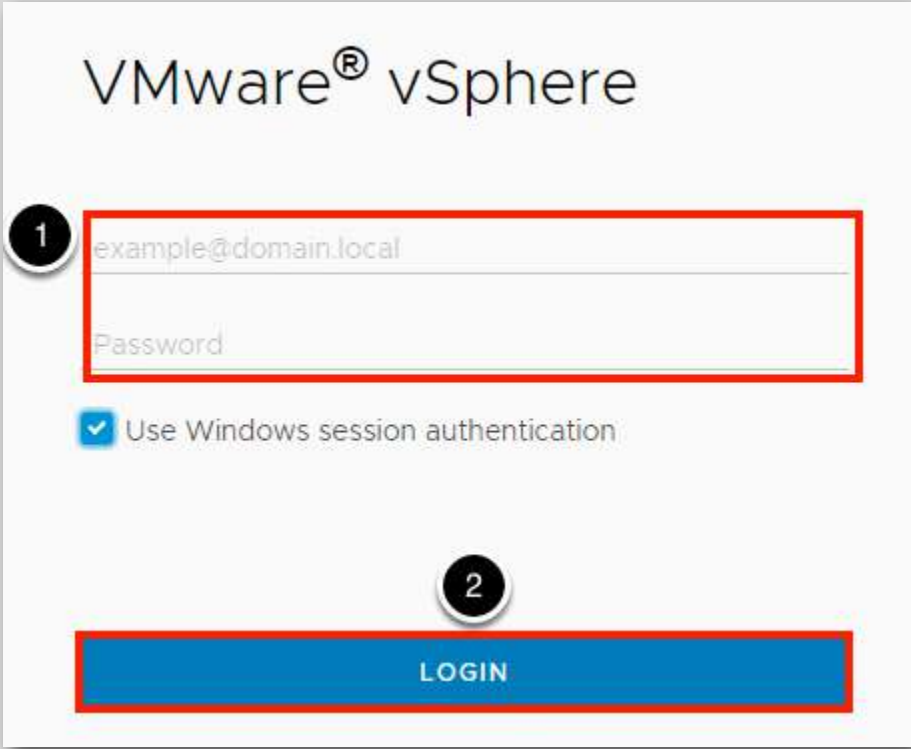


1. Click on the **Firefox** on the **Linux Task Bar**

If you're coming from the previous module, feel free to skip this step and the next one.

Enter credentials and log in

[429]



The image shows the VMware vSphere login interface. At the top, the text "VMware® vSphere" is displayed. Below this, there is a login form. A red rectangular box highlights the username and password fields, with a black circle containing the number "1" next to it. The username field contains the text "example@domain.local" and the password field is labeled "Password". Below these fields, there is a checkbox with a blue checkmark and the text "Use Windows session authentication". At the bottom of the form, there is a blue button with the text "LOGIN" in white, with a black circle containing the number "2" above it.

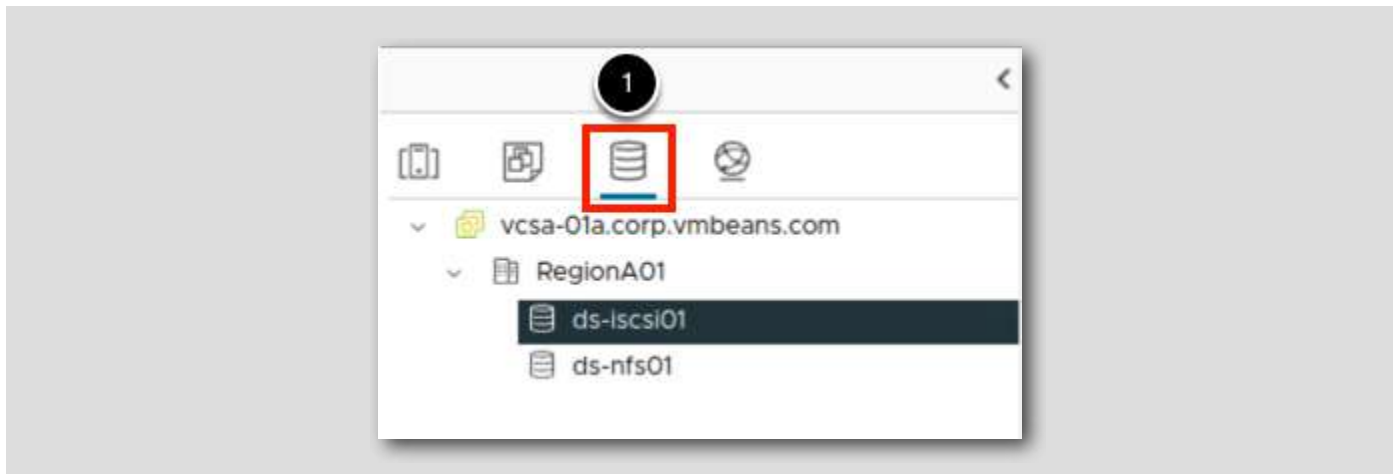
1. Use the below credentials to login:

- username: administrator@vsphere.local
- password: VMware1!

2. Select Login

Navigate to Storage Management

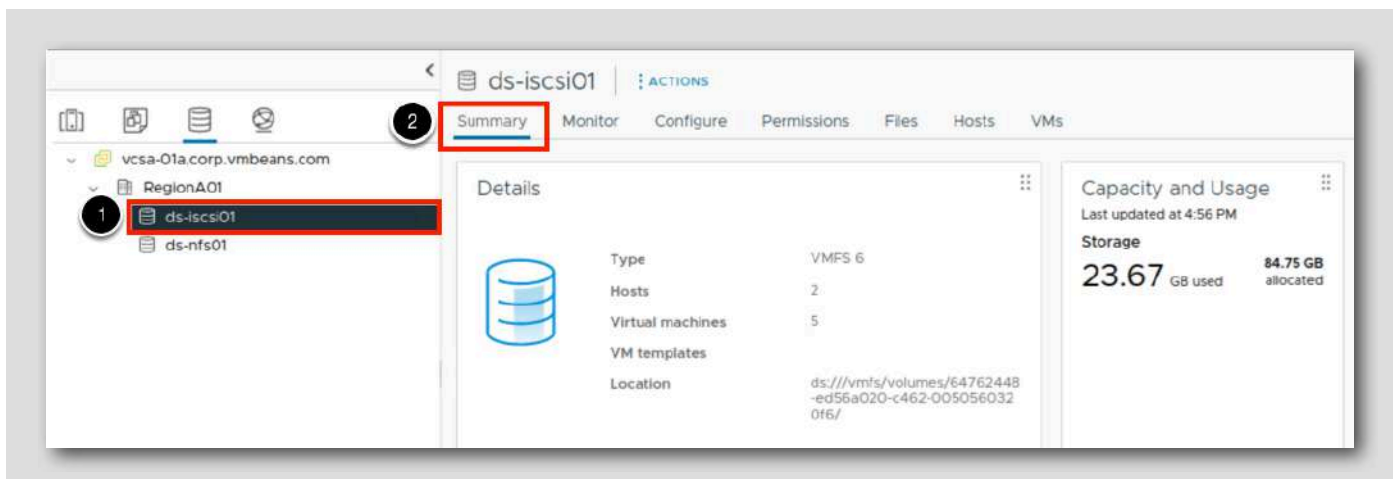
[430]



1. Select the Storage tab.

Expand RegionA01 Datacenter

[431]



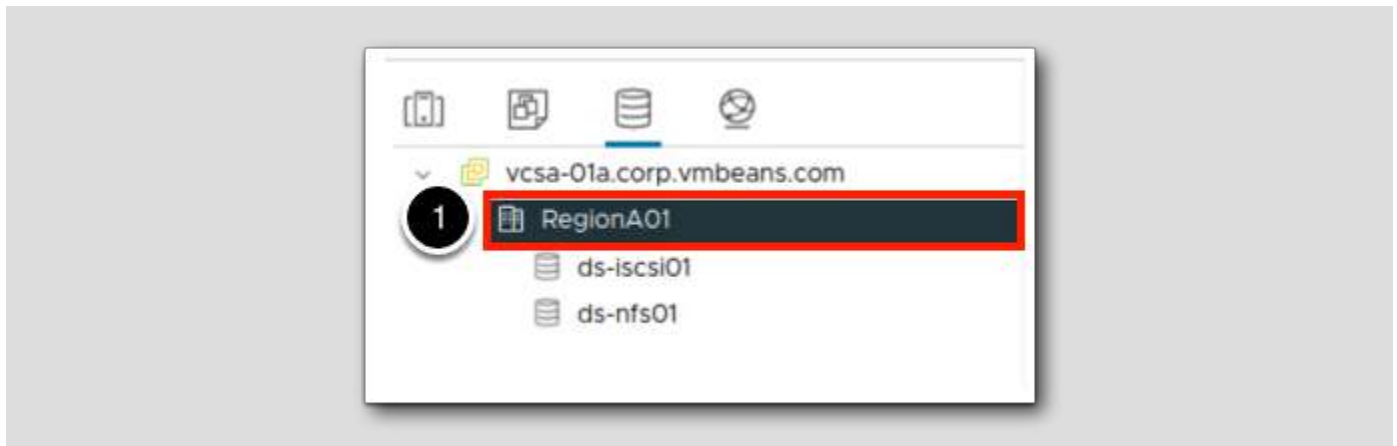
There are 2 storage datastores configured, an iSCSI datastore and an NFS datastore.

1. Select the **ds-iscsi01** datastore.
2. Click on **Summary** for summary details of the datastore.

Repeat the steps for the **ds-nfs01** datastore.

Create a vSphere NFS Datastore

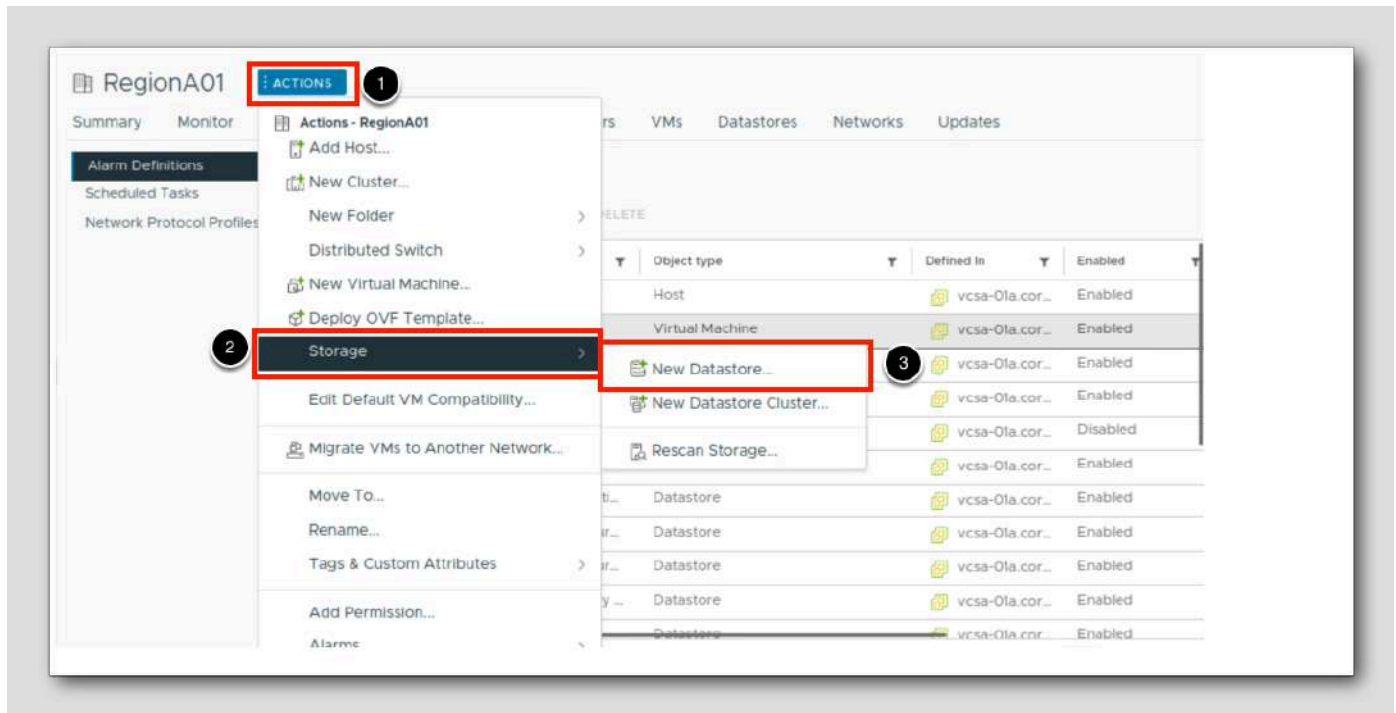
[432]



In this section, you will create a new vSphere NFS Datastore using a pre-provisioned NFS mount.

1. Select **RegionA01** Datacenter.

New Datastore

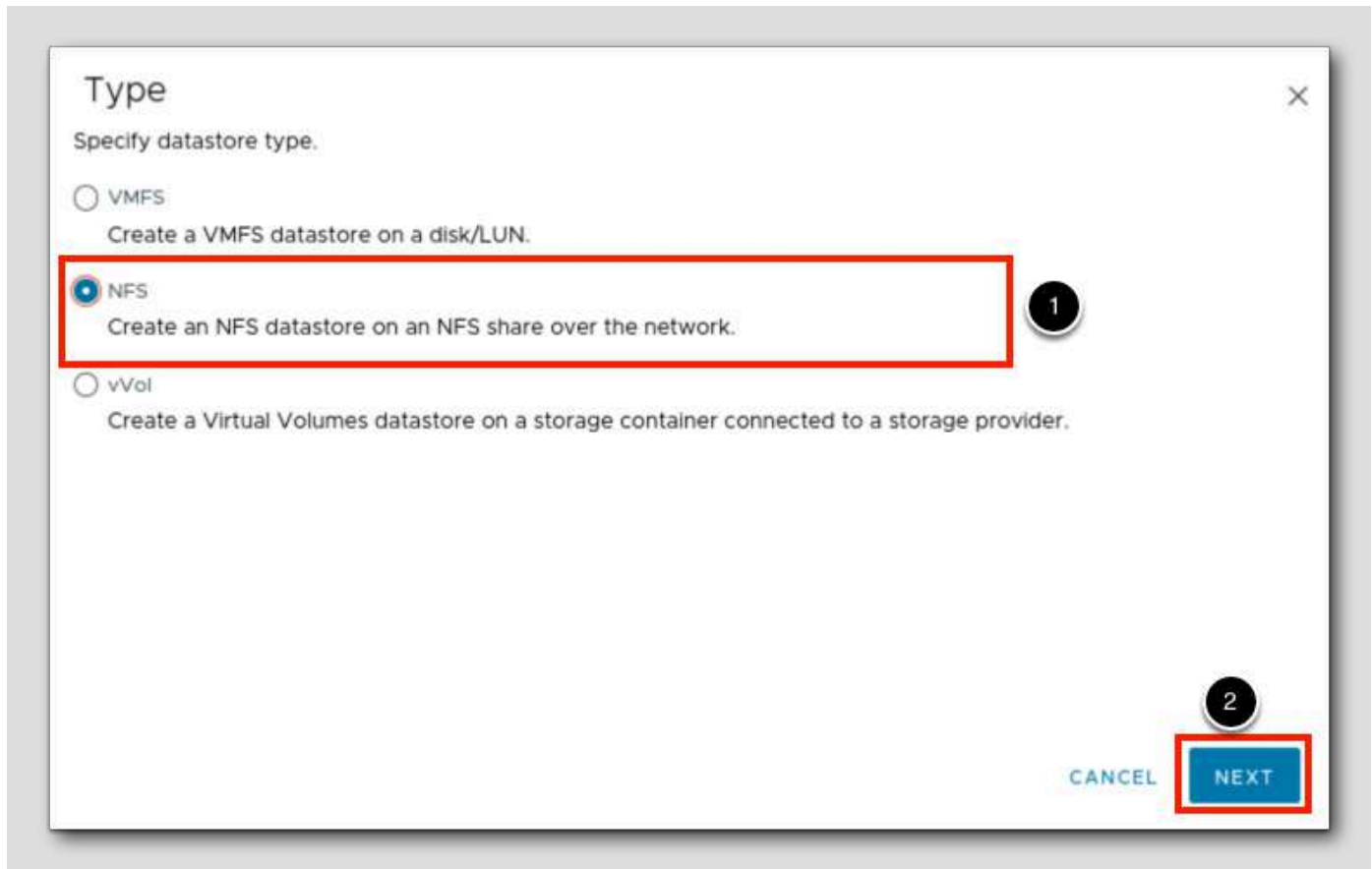


In this section, you will create a new vSphere NFS Datastore using a pre-provisioned NFS mount.

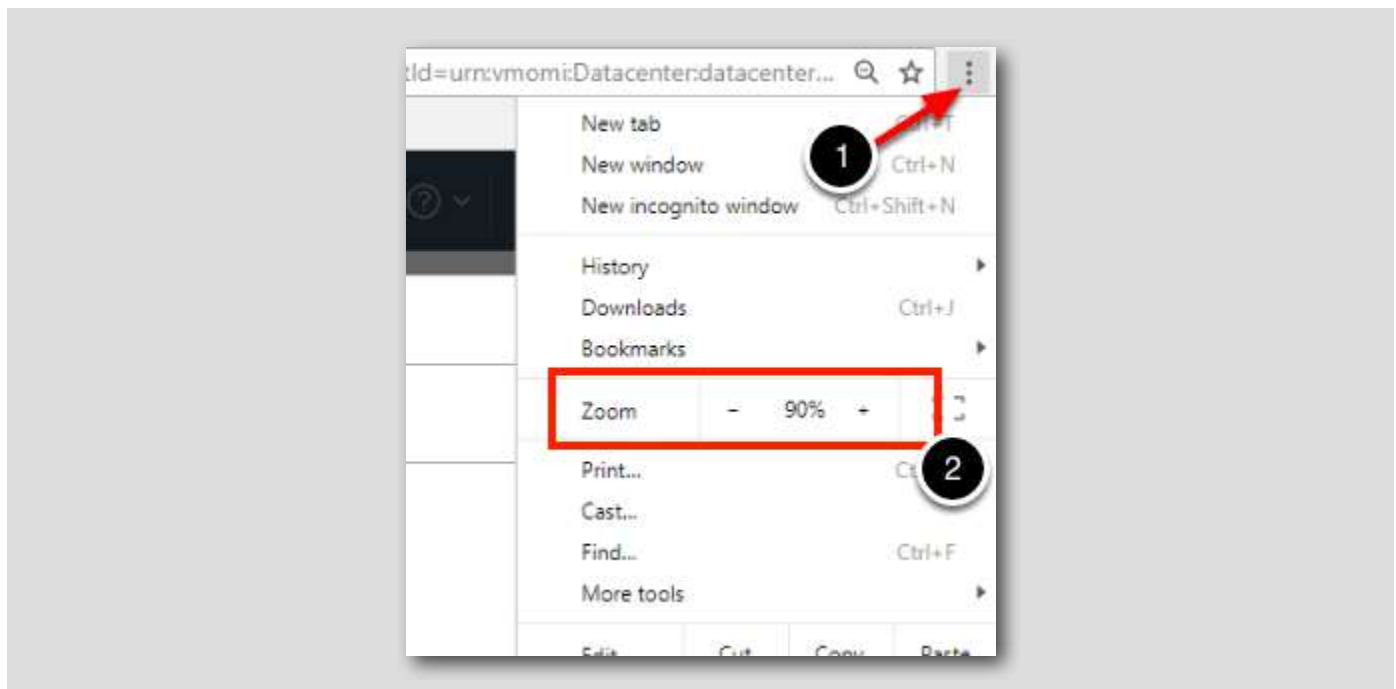
1. Select Actions.
2. Select Storage.
3. Select New Datastore.

New Datastore - Type

[434]



1. Select **NFS** for the new Datastore type
2. Click **Next**



Note: You may need to zoom out in order to see the **Next** button.

New Datastore - NFS Version

[435]

NFS version

Select the NFS version.

☒ NFS 3
NFS 3 allows the datastore to be accessed by ESX/ESXi hosts of version earlier than 6.0

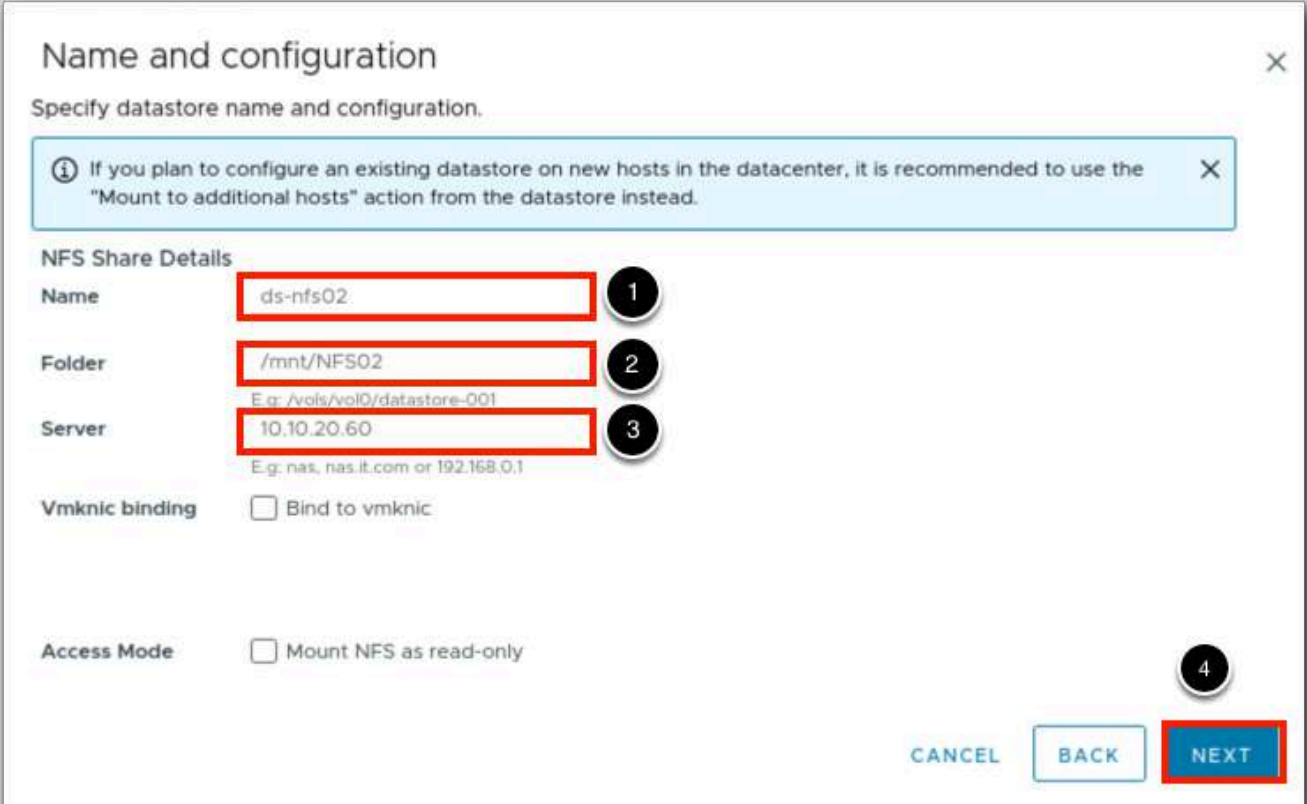
☐ NFS 4.1
NFS 4.1 provides multipathing for servers and supports the Kerberos authentication protocol

CANCEL BACK NEXT

1. Select NFS 3 as your NFS Version
2. Click **Next**

New Datastore - Name and Configuration

[436]



Name and configuration ×

Specify datastore name and configuration.

ⓘ If you plan to configure an existing datastore on new hosts in the datacenter, it is recommended to use the "Mount to additional hosts" action from the datastore instead. ×

NFS Share Details

Name 1

Folder 2
E.g. /vols/vol0/datastore-001

Server 3
E.g. nas, nas.it.com or 192.168.0.1

Vmknfc binding ☐ Bind to vmknfc

Access Mode ☐ Mount NFS as read-only 4

CANCEL BACK NEXT

1. Give the new Datastore a name, **ds-nfs02**.
2. Enter the Folder **/mnt/NFS02** in the NFS Share Details area.
3. Enter the Server **10.10.20.60** in the NFS Share Details area.
4. Click **Next**.

New Datastore - Host Accessibility

[437]

Hosts accessibility

Select the hosts that require access to the datastore.

COMPATIBLE (2 HOSTS) INCOMPATIBLE (0 HOSTS)

Filter

	Host	Cluster
☒	esx-01a.corp.vmbeans.com	RegionA01-COMP01
☒	esx-02a.corp.vmbeans.com	RegionA01-COMP01

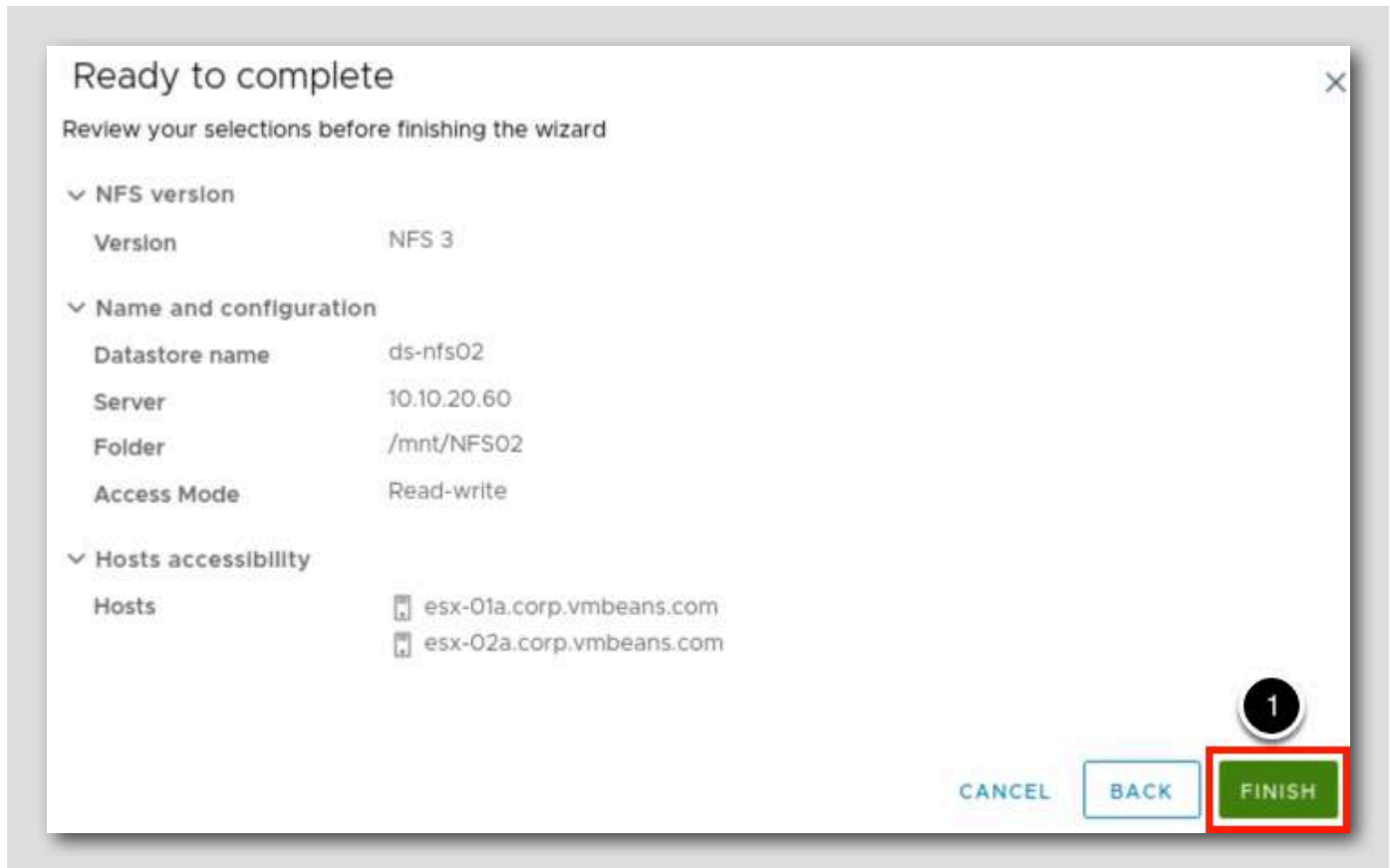
☒ 2

CANCEL BACK **NEXT**

1. Select the **check box** to include all hosts.
2. Click **Next**.

New Datastore - Ready to complete

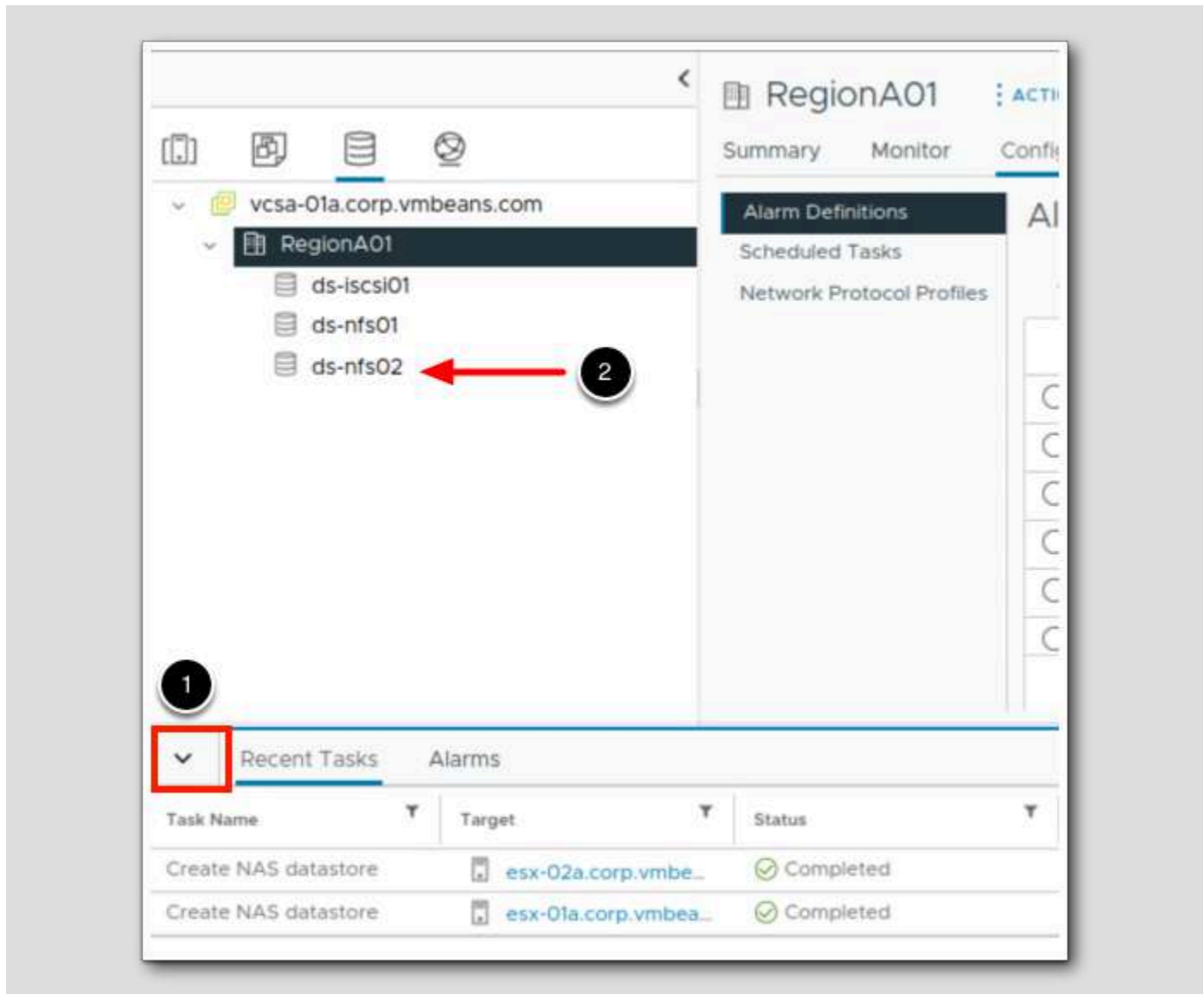
[438]



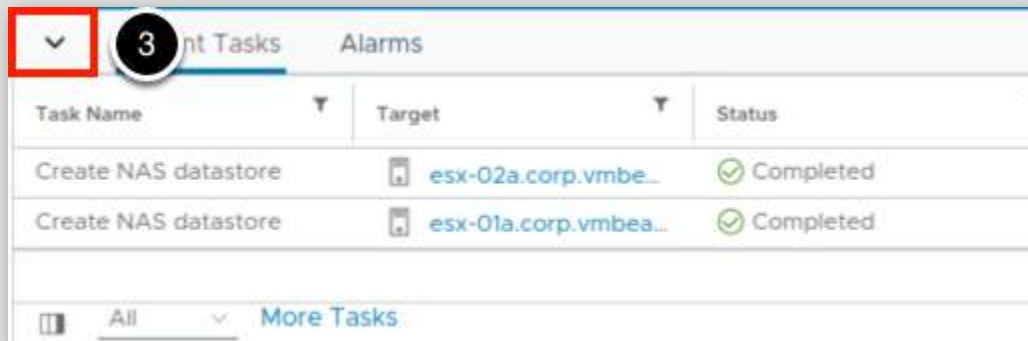
1. Review New Datastore configuration and click Finish.

Monitor task progress

[439]



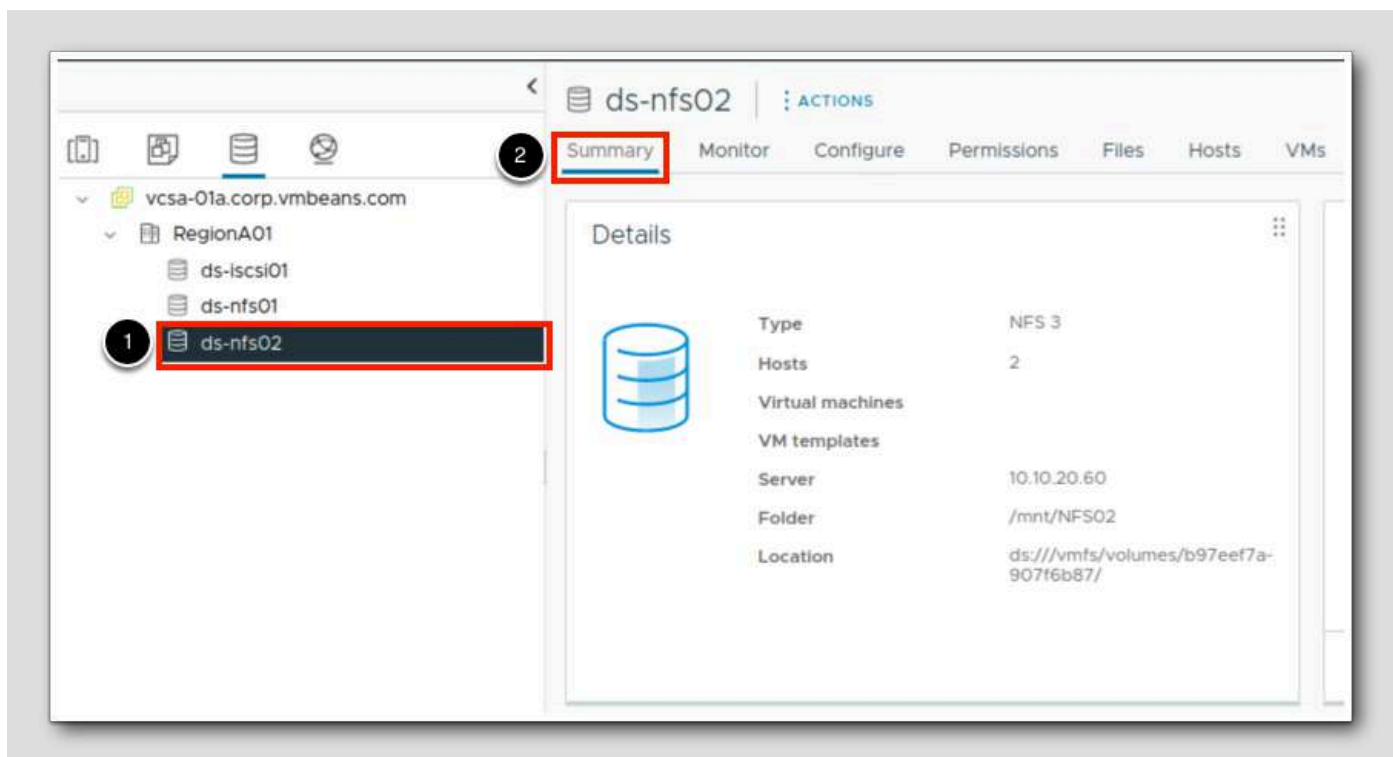
1. You can follow the progress in the Recent Tasks pane (by clicking on Recent Tasks)
2. When complete, you should see the new ds-nfs02 Datastore available for use



3. Minimize the Recent Tasks pane before continuing to the next step

Review new Datastore Settings

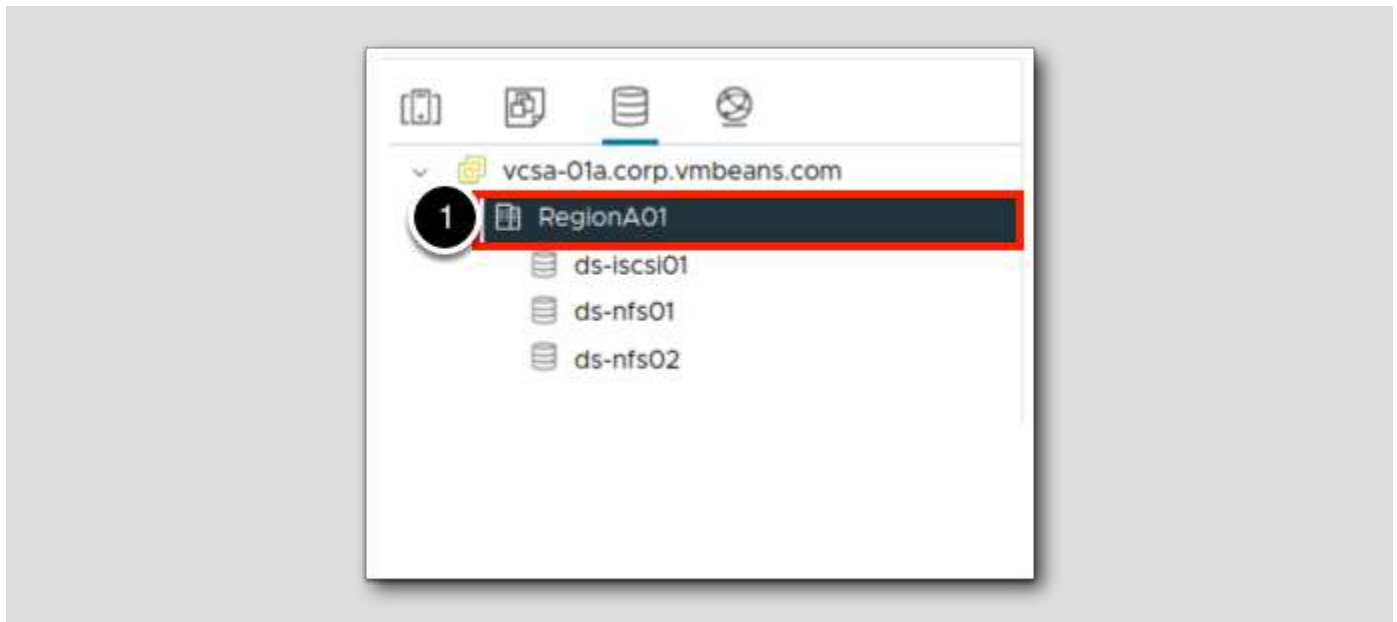
[440]



1. Select the datastore **ds-nfs02** from the inventory list
2. Select **Summary** to review capacity and configuration details

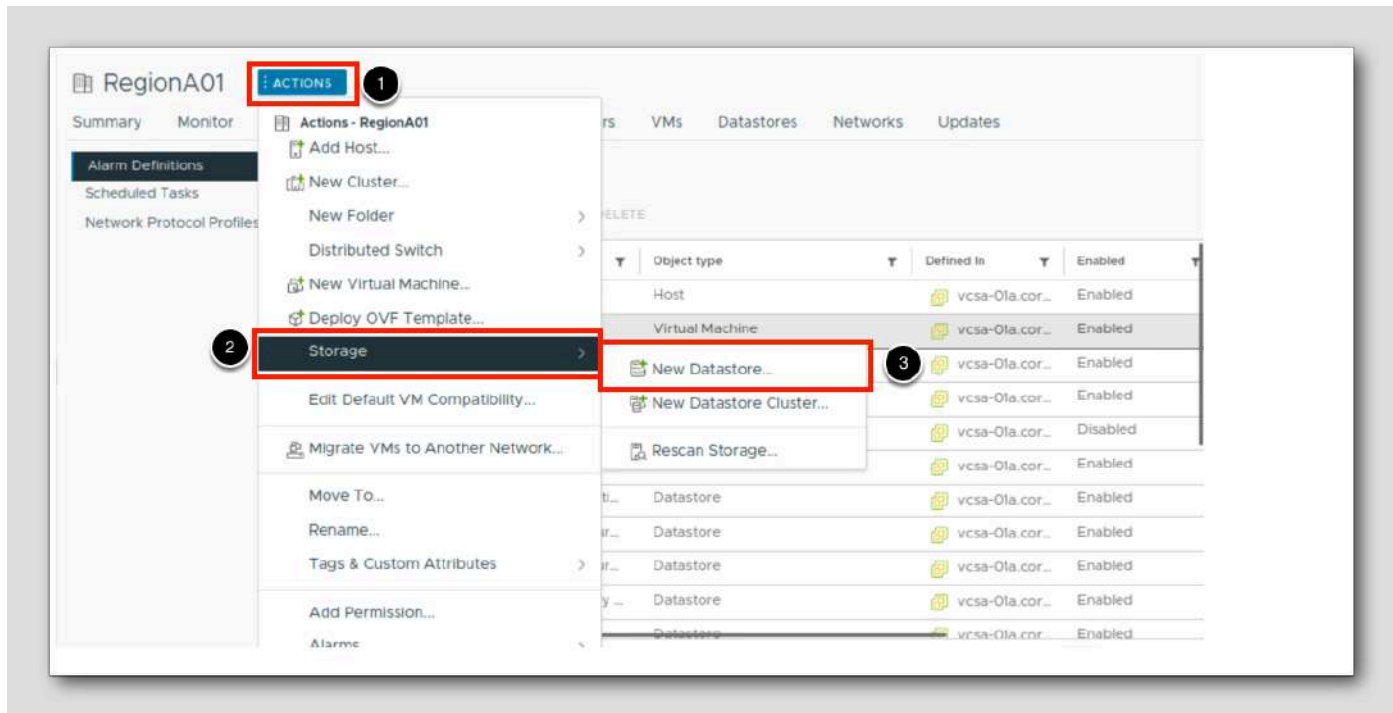
Create a vSphere iSCSI Datastore

[441]



1. Select **RegionA01** Datacenter.

New Datastore

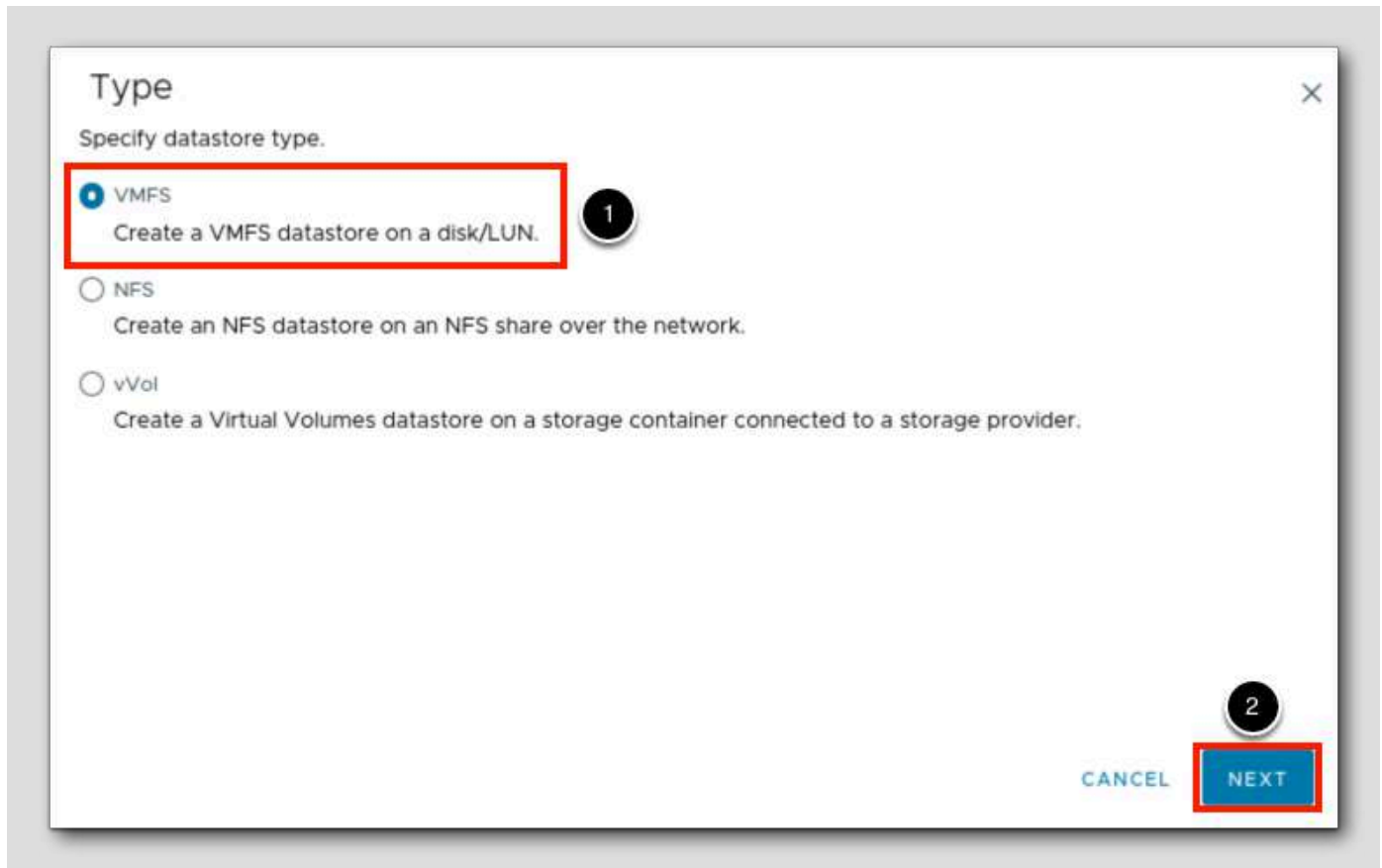


In this section, you will create a new vSphere iSCSI Datastore with a pre-provisioned iSCSI LUN.

1. Select Actions.
2. Select Storage.
3. Select New Datastore.

New Datastore - Type

[443]



1. Select VMFS
2. Click Next.

New Datastore - Name and Device configuration

Name and device selection

Specify datastore name and a disk/LUN for provisioning the datastore.

Name

The datastore will be accessible to all the hosts that are configured with access to the selected disk/LUN. If you do not find the disk/LUN that you are interested in, it might not be accessible to that host. Try changing the host or configure accessibility of that disk/LUN.

Select a host

Select a host to view its accessible disks/LUNs:

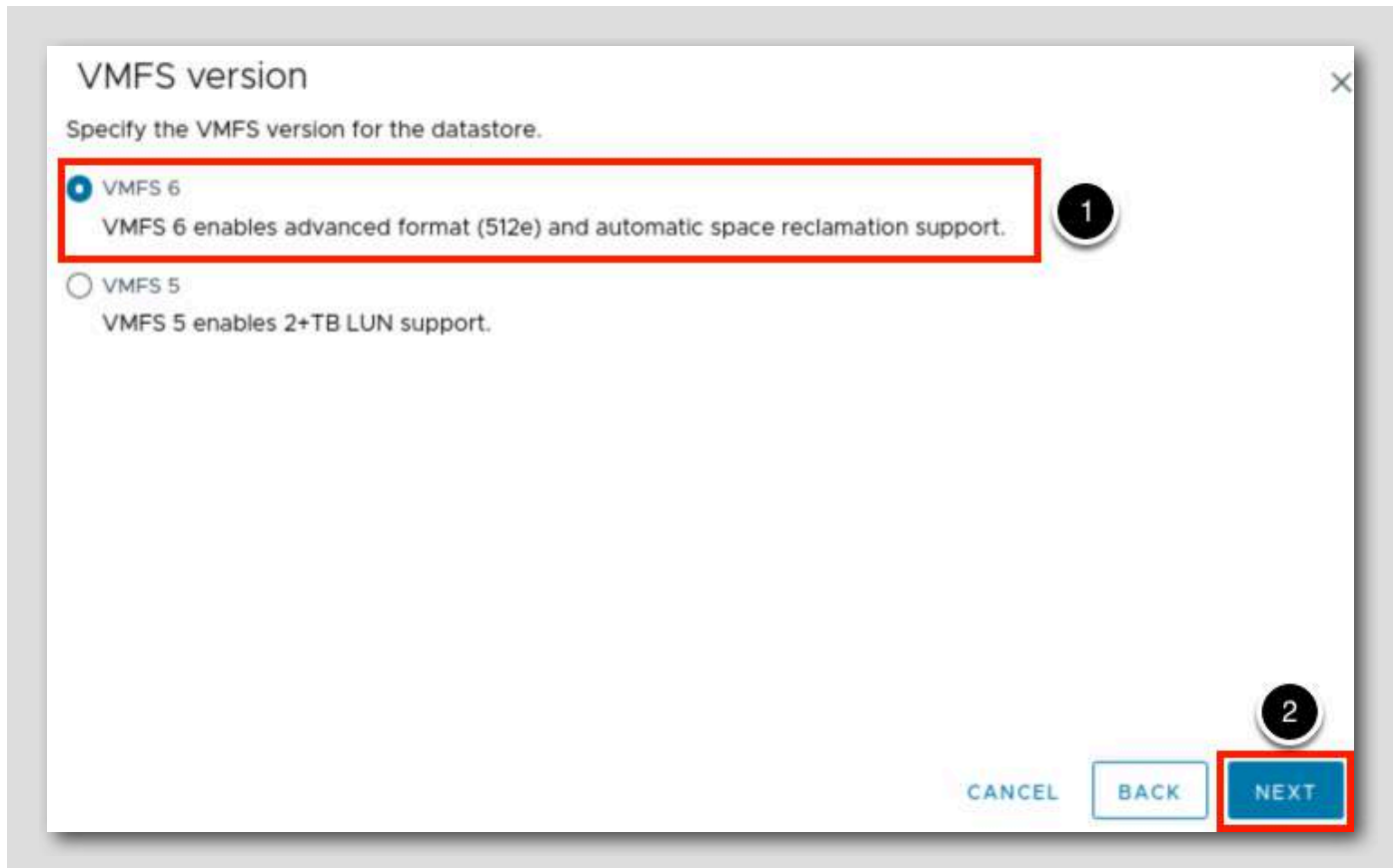
	Name	LUN	Capacity	Hardware Acceleration	Drive Type	Sector Format	Clust VMD Supp
☒	FreeNAS iSCSI Disk (naa.6...)	2	45.00 GB	Supported	Flash	--	No
☐	Local VMware Disk (mpx.v...)	0	10.00 GB	Not supported	HDD	512n	No

CANCEL **BACK** **NEXT**

1. Give the new Datastore the name `ds-iscsi02`.
2. Select a Host to view the accessible disks/LUNs and select `esx-01a.corp.vmbeans.com` in the drop-down box.
3. Select the device with LUN ID 2. In this case, it should be the only device visible with a **FreeNAS** prefix.
4. Click **Next**.

New Datastore - VMFS Version

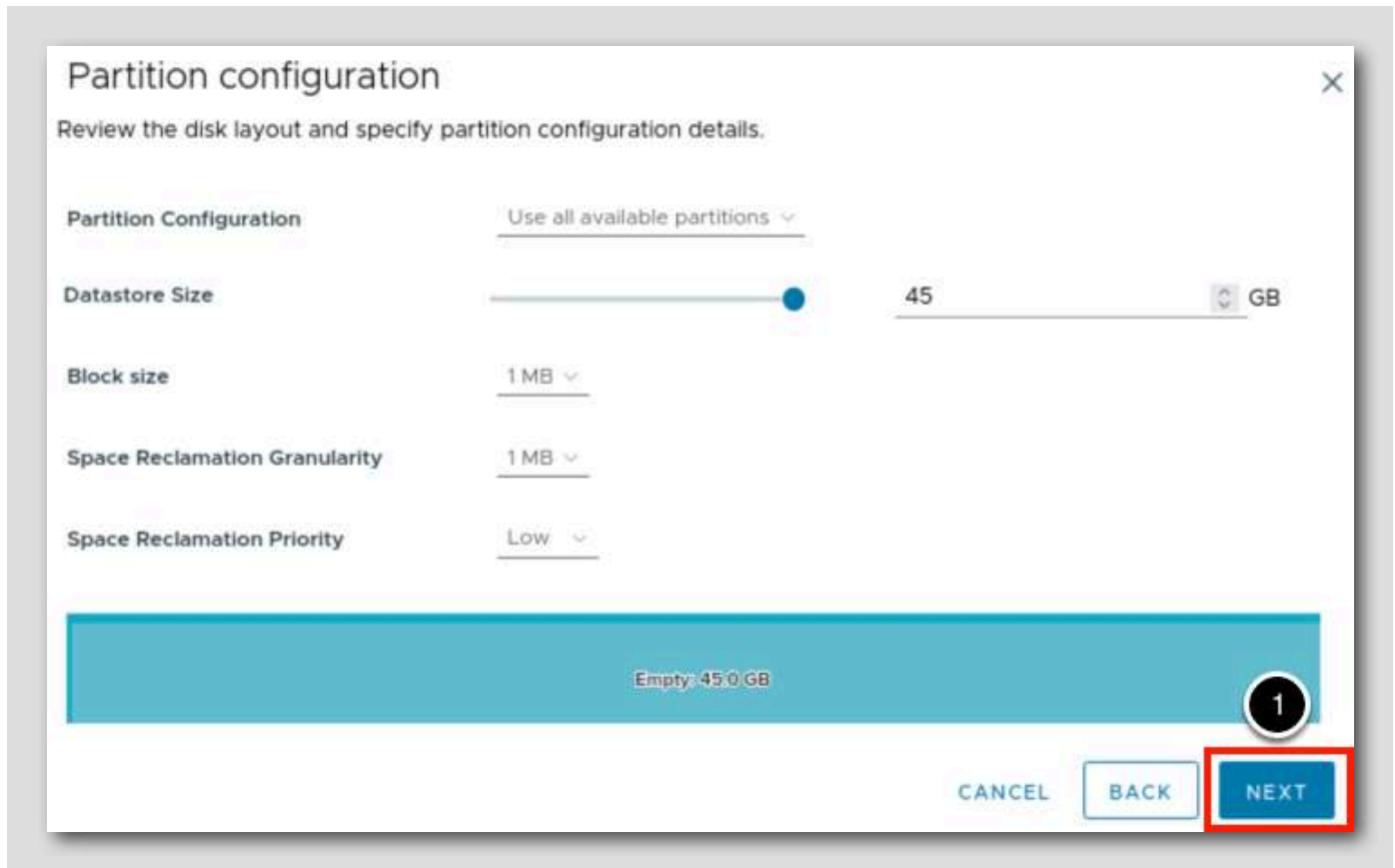
[445]



1. Leave the default of **VMFS 6** selected.
2. Click **Next**.

New Datastore - Partition Configuration

[446]



The image shows a 'Partition configuration' dialog box with a close button (X) in the top right corner. Below the title is the instruction 'Review the disk layout and specify partition configuration details.' The dialog contains several configuration options: 'Partition Configuration' is set to 'Use all available partitions' with a dropdown arrow; 'Datastore Size' is set to 45 GB, with a slider and a small up/down arrow icon; 'Block size' is set to 1 MB with a dropdown arrow; 'Space Reclamation Granularity' is set to 1 MB with a dropdown arrow; and 'Space Reclamation Priority' is set to Low with a dropdown arrow. At the bottom, there is a large blue bar representing the disk layout, with the text 'Empty: 45.0 GB' in the center. To the right of this bar is a circular icon with the number '1'. Below the blue bar are three buttons: 'CANCEL', 'BACK', and 'NEXT'. The 'NEXT' button is highlighted with a red rectangular border.

Partition configuration

Review the disk layout and specify partition configuration details.

Partition Configuration Use all available partitions ▼

Datastore Size 45 GB

Block size 1 MB ▼

Space Reclamation Granularity 1 MB ▼

Space Reclamation Priority Low ▼

Empty: 45.0 GB

1

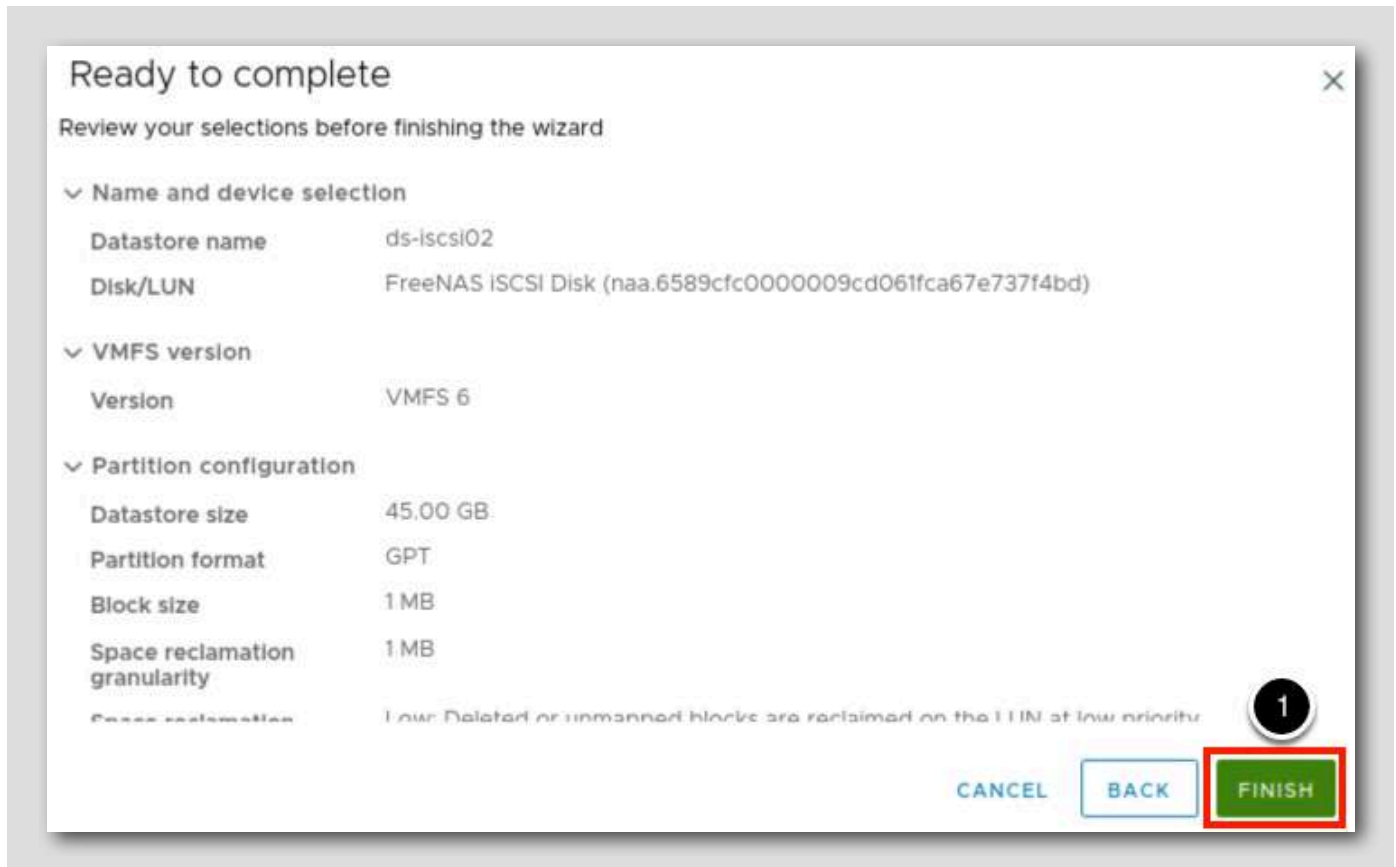
CANCEL BACK NEXT

We can use all available capacity for this datastore or change the size if needed. The defaults are fine for this step.

1. Select Next.

New Datastore - Ready to complete

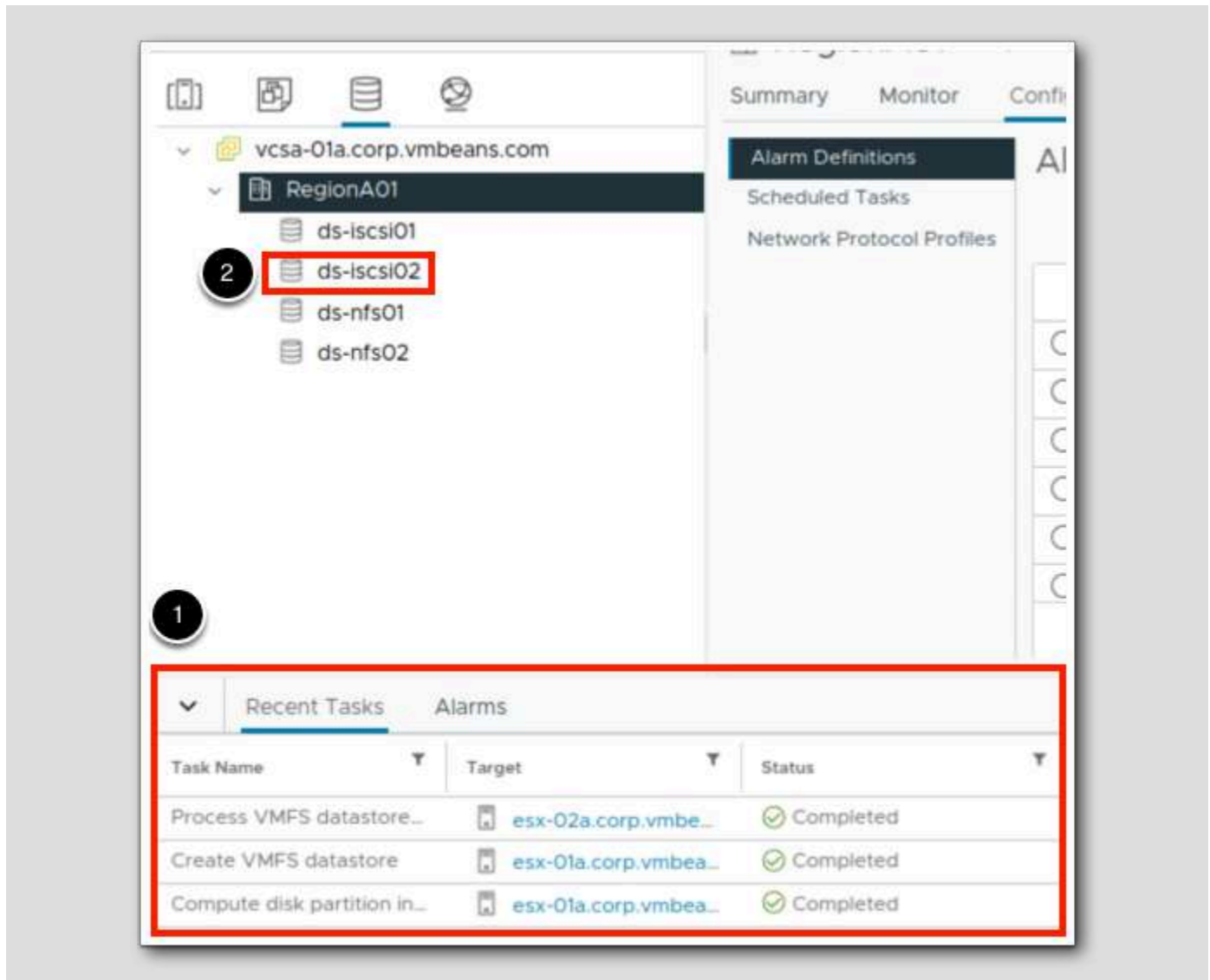
[447]



1. Review New Datastore configuration and click Finish.

New Datastore - Monitor task progress

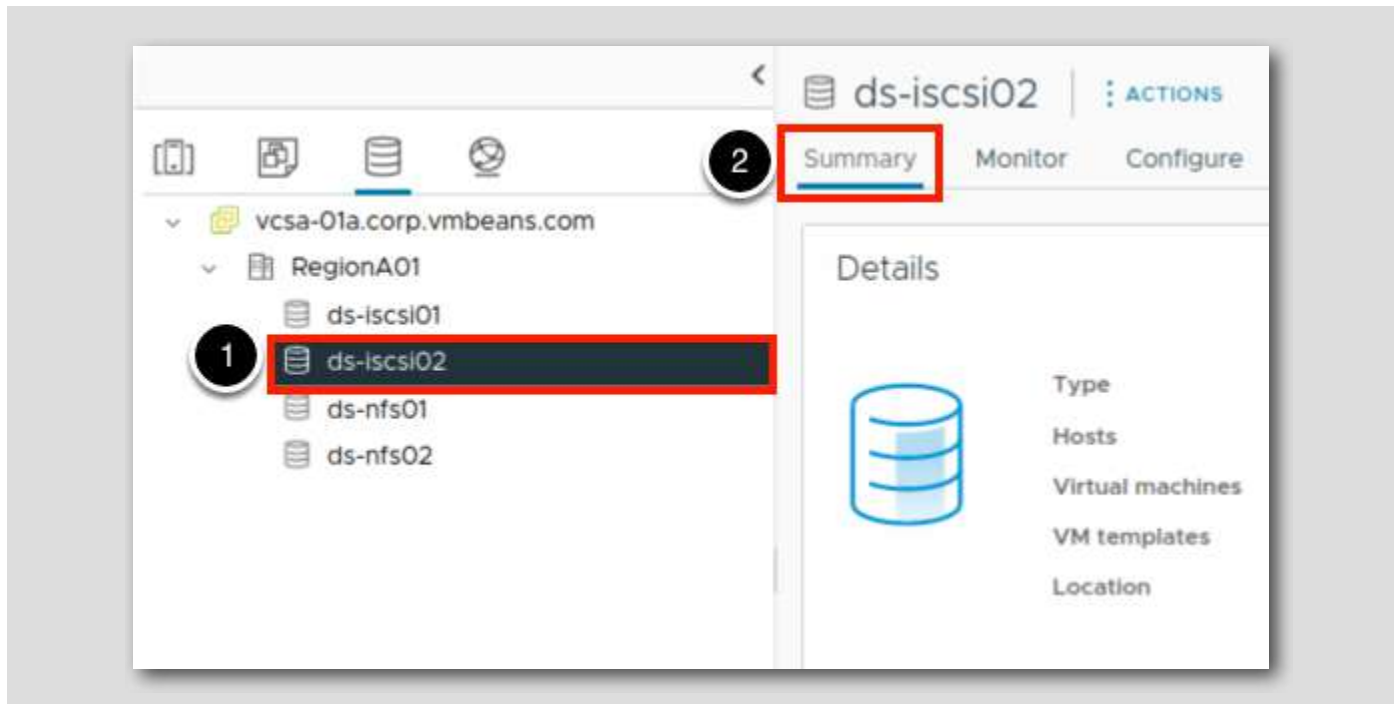
[448]



1. Note the progress in the **Recent Tasks** pane
2. When complete, you should see the **ds-iscsi02** Datastore available for use

New Datastore - Review Settings

[449]



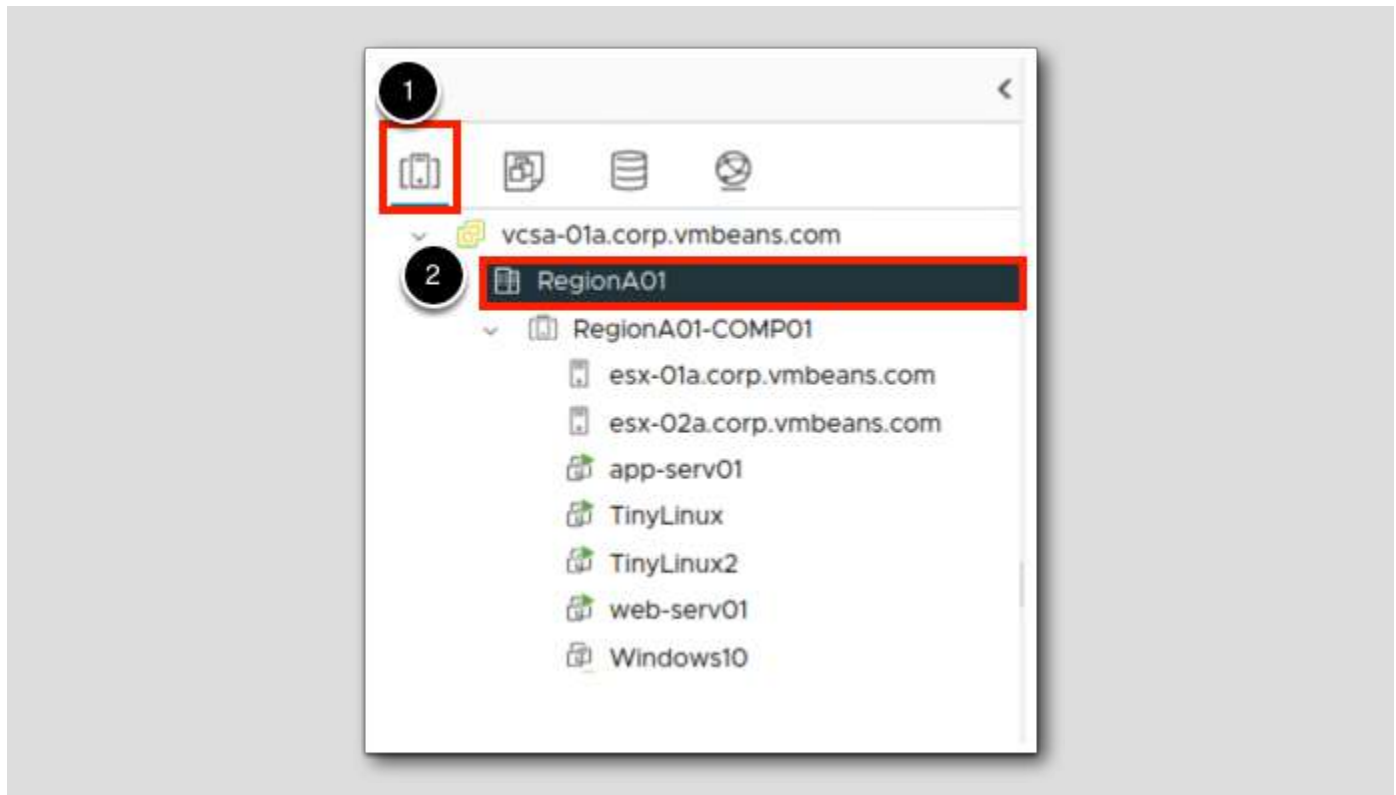
1. Select the datastore `ds-iscsi02` from the inventory list
2. Select **Summary** to review capacity and configuration details

Add a New ESXi Host

[450]

In this section, we will add a new ESXi host, `esx-03a.corp.local`, to the environment in RegionA01 and ensure that it has the appropriate storage configured so that it can become a productive member of the cluster.

Hosts and Clusters View

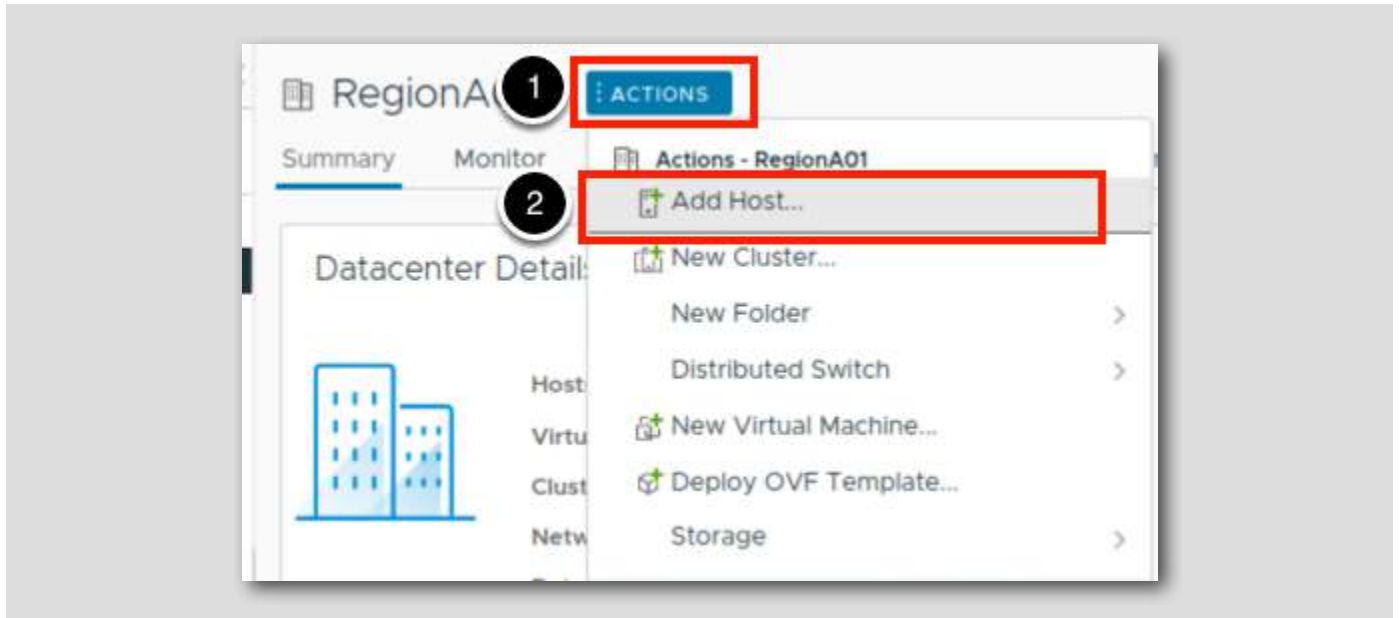


1. Click on the **Hosts and Clusters** icon to return to that Inventory view.
2. Select **RegionA01** Datacenter.

It is a best practice to bring hosts into a datacenter first before adding them to a cluster. If a host is added to a cluster first, by not having access to the cluster's storage volumes, it could impact High Availability (see Module 1 for more details on High Availability).

Begin the Add Host workflow

[452]



1. Go to the Actions menu.
2. Select Add Host...

Enter the Hostname

[453]

Name and location [X]

Enter the name or IP address of the host to add to vCenter Server.

Host name or IP address: 1

Location:

CANCEL NEXT 2

1. In the Host name or IP address, enter: esx-03a.corp.vmbeans.com
2. Click **Next**.

Connection Settings

[454]

Connection settings

Enter the host connection details

User name: root

Password:

CANCEL BACK NEXT

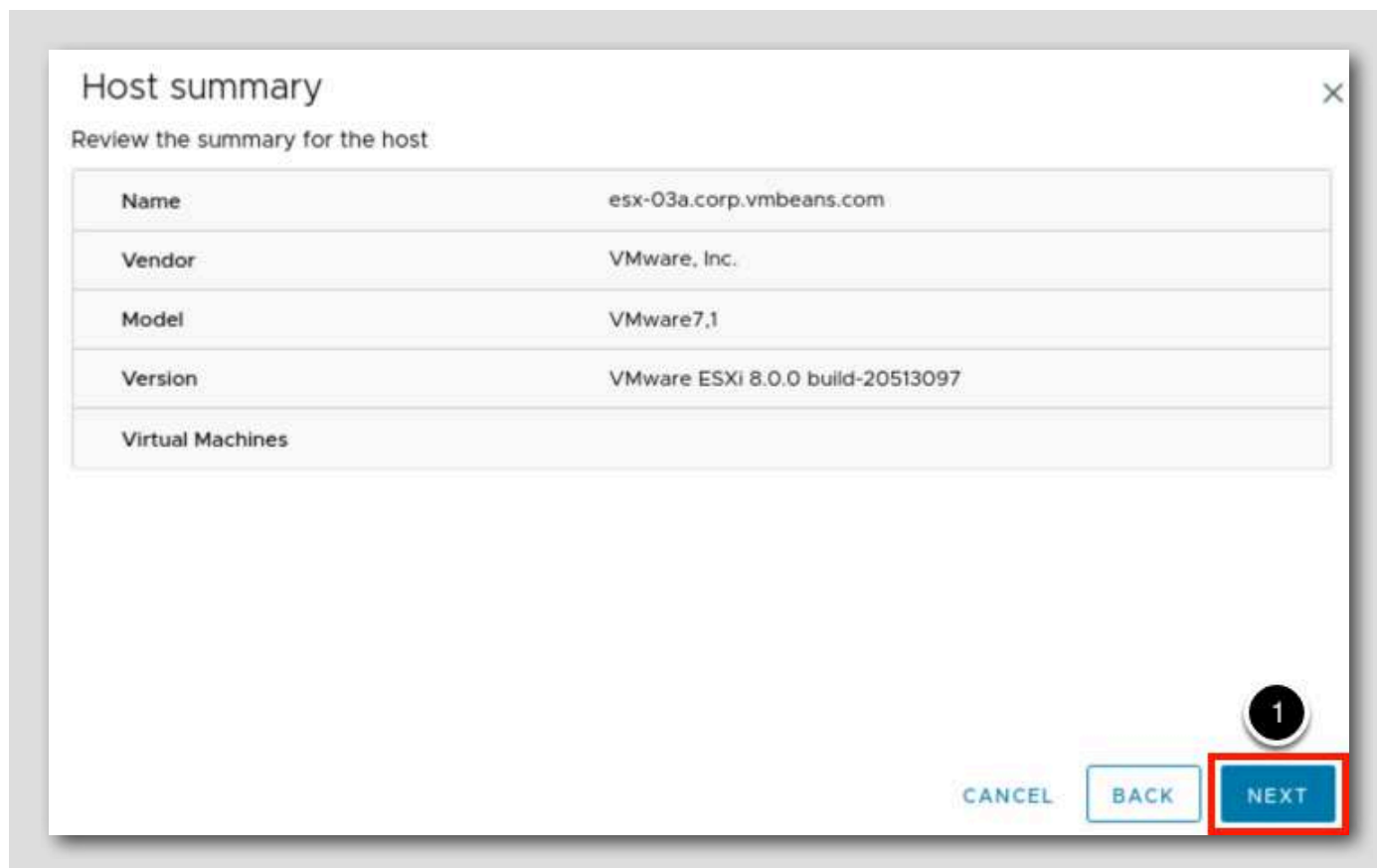
1. Enter the following login details:

- User name: root
- Password: VMware1!

2. Click **Next**.

Host Summary

[455]

A screenshot of a 'Host summary' dialog box. The dialog has a title bar with 'Host summary' and a close button. Below the title bar, it says 'Review the summary for the host'. There is a table with five rows: 'Name' (esx-03a.corp.vmbeans.com), 'Vendor' (VMware, Inc.), 'Model' (VMware7,1), 'Version' (VMware ESXi 8.0.0 build-20513097), and 'Virtual Machines'. At the bottom right, there are three buttons: 'CANCEL', 'BACK', and 'NEXT'. The 'NEXT' button is highlighted with a red border and a black circle with the number '1' above it.

Host summary	
Review the summary for the host	
Name	esx-03a.corp.vmbeans.com
Vendor	VMware, Inc.
Model	VMware7,1
Version	VMware ESXi 8.0.0 build-20513097
Virtual Machines	

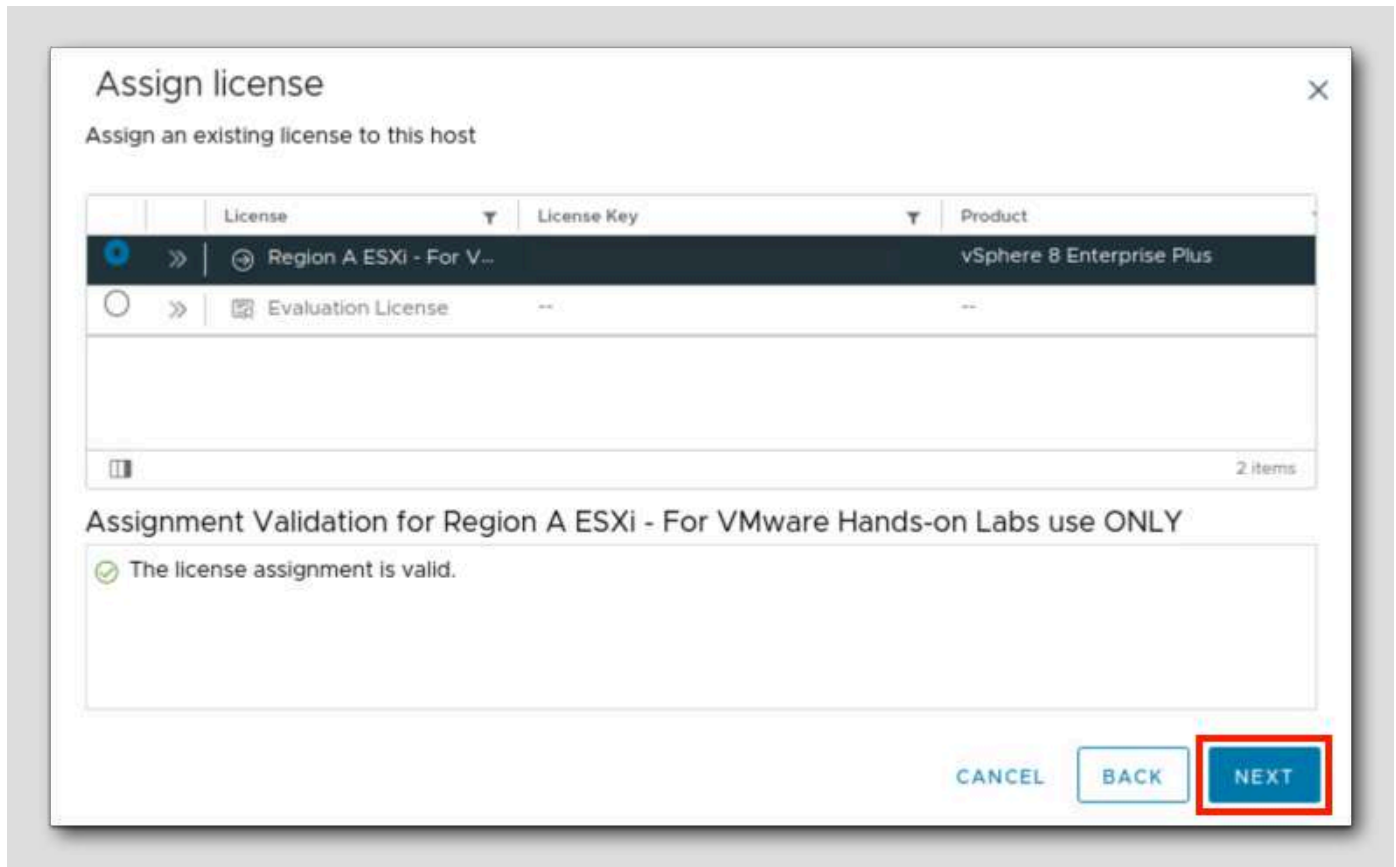
CANCEL BACK NEXT

This screen shows the details of the host.

1. Click **Next**.

Assign License

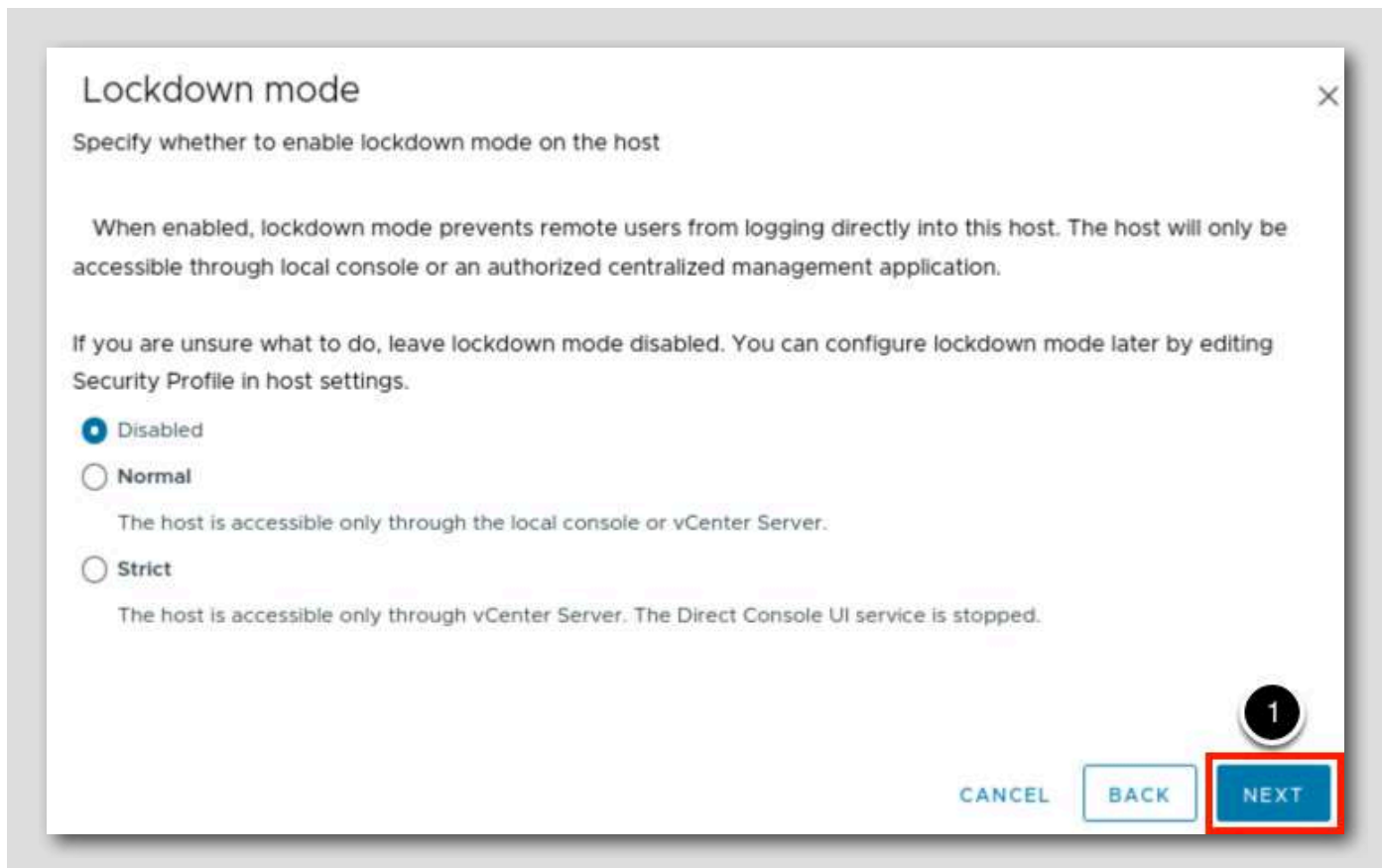
[456]



1. Leave the default license choice and click **Next**.

Lockdown Mode

[457]

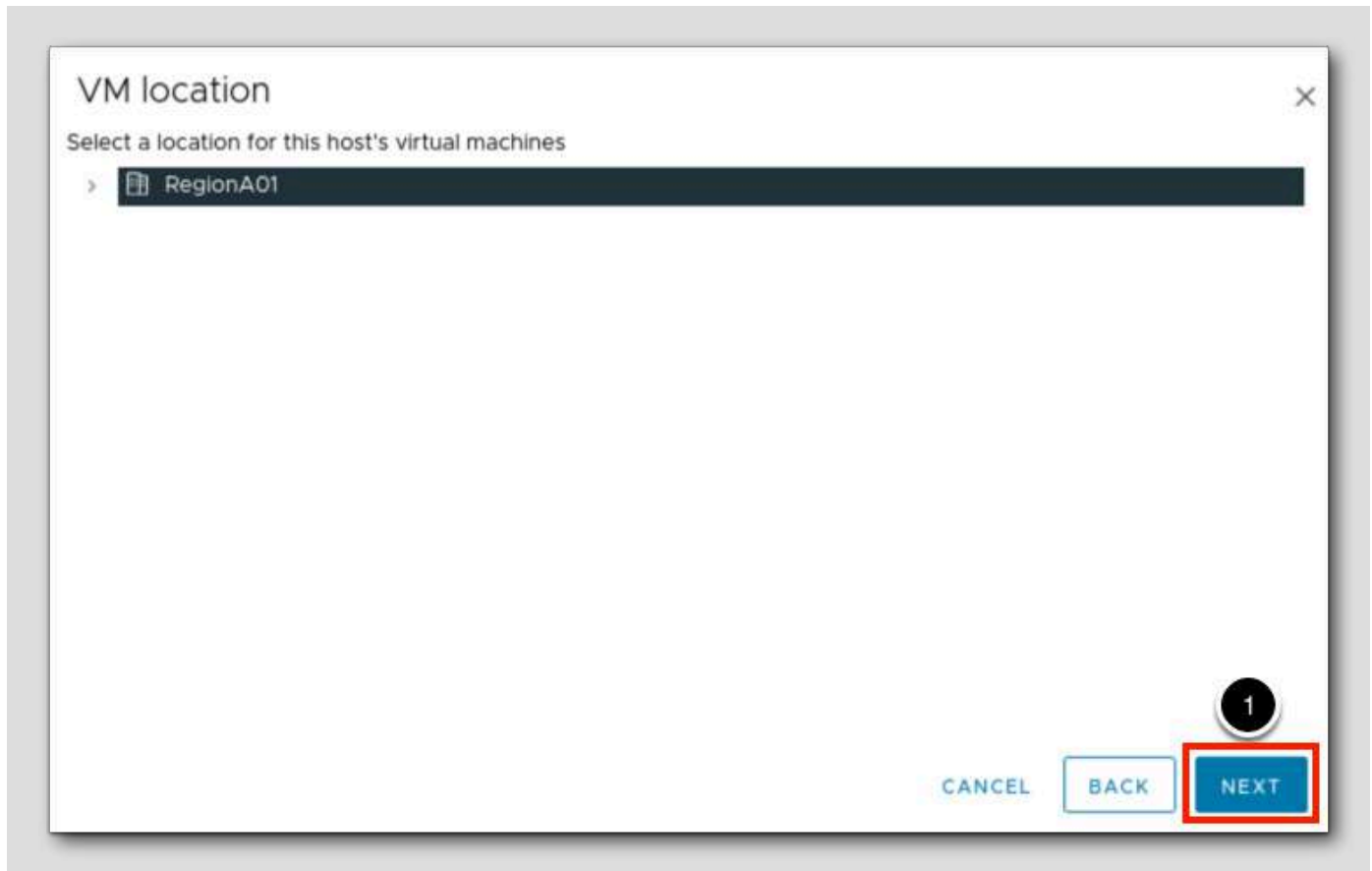


When a host is being added to a Datacenter, it can be placed in what is called Lockdown mode. This can prevent unauthorized users from gaining access to the ESXi host either through local console access or remotely by way of SSH. If you are interested in Lockdown Mode, the details are covered in Module 1.

1. Leave the default setting and click **Next**.

VM Location

[458]

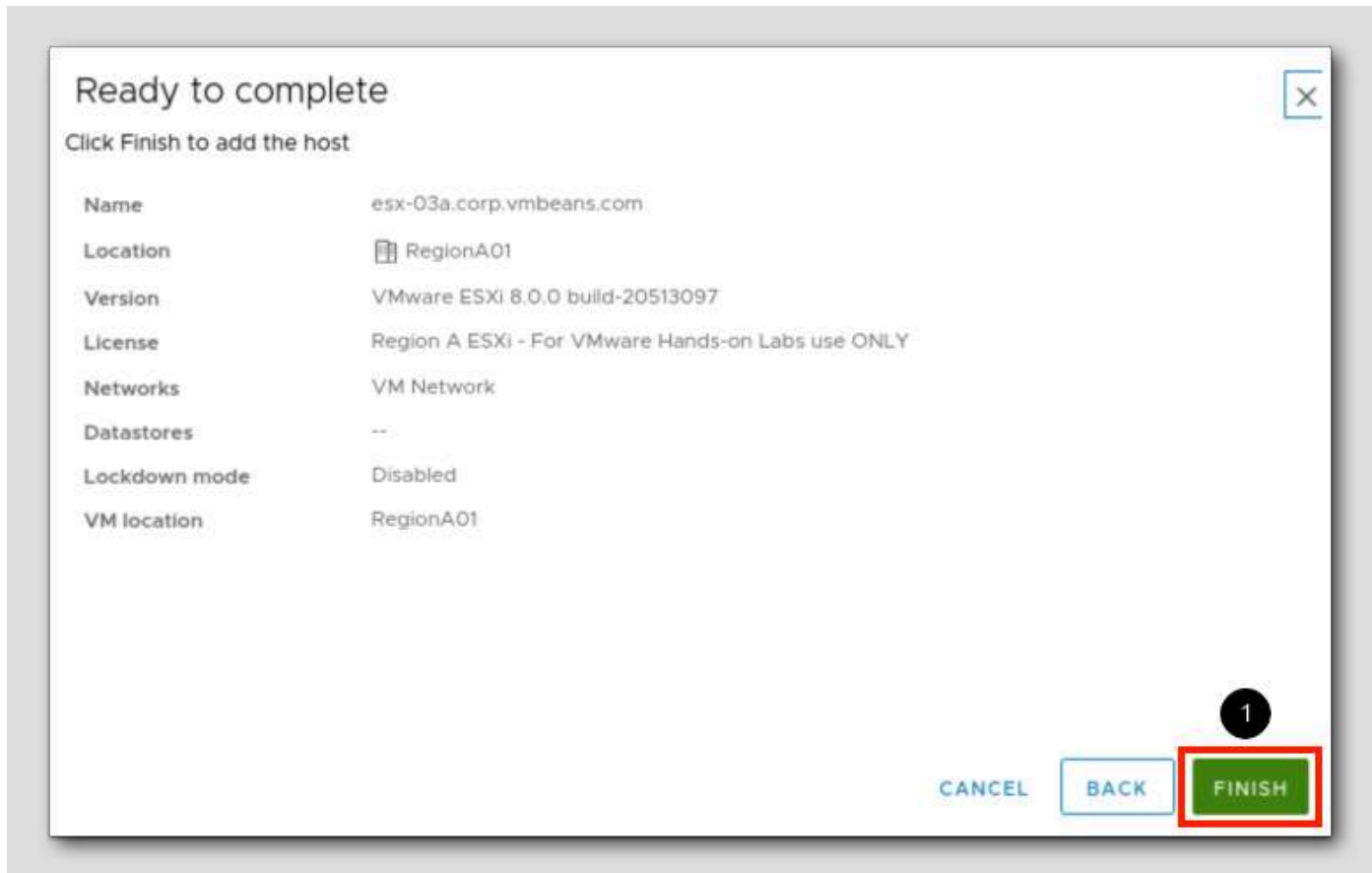


The virtual machines currently on the ESXi host being imported can be placed in either the Datacenter itself or in the default Discovered virtual machines folder.

1. Since there are no virtual machines on esx-03a.corp.vmbeans.com, leave the default setting and click **Next**.

Ready to Complete

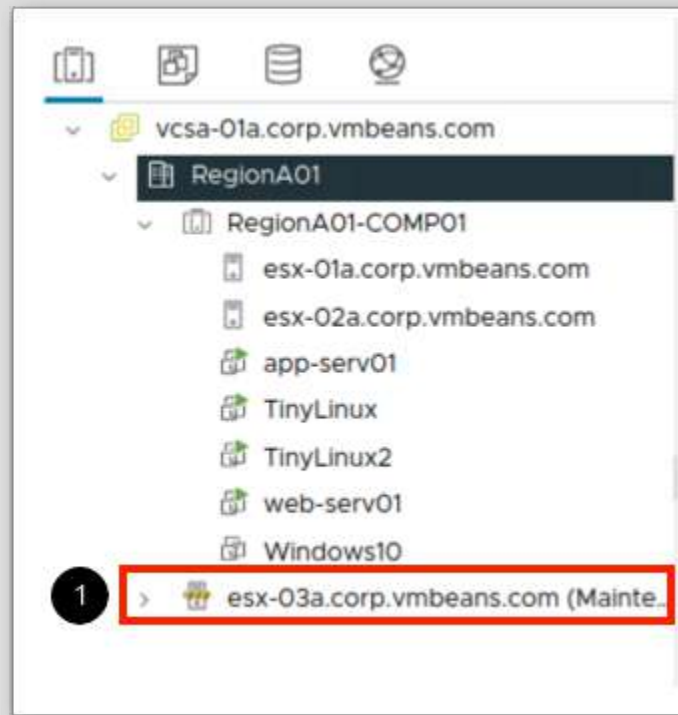
[459]



1. Review the settings and click **Finish** to add the esx-03a.corp.vmbeans.com to the datacenter.

Host Added to Datacenter

[460]

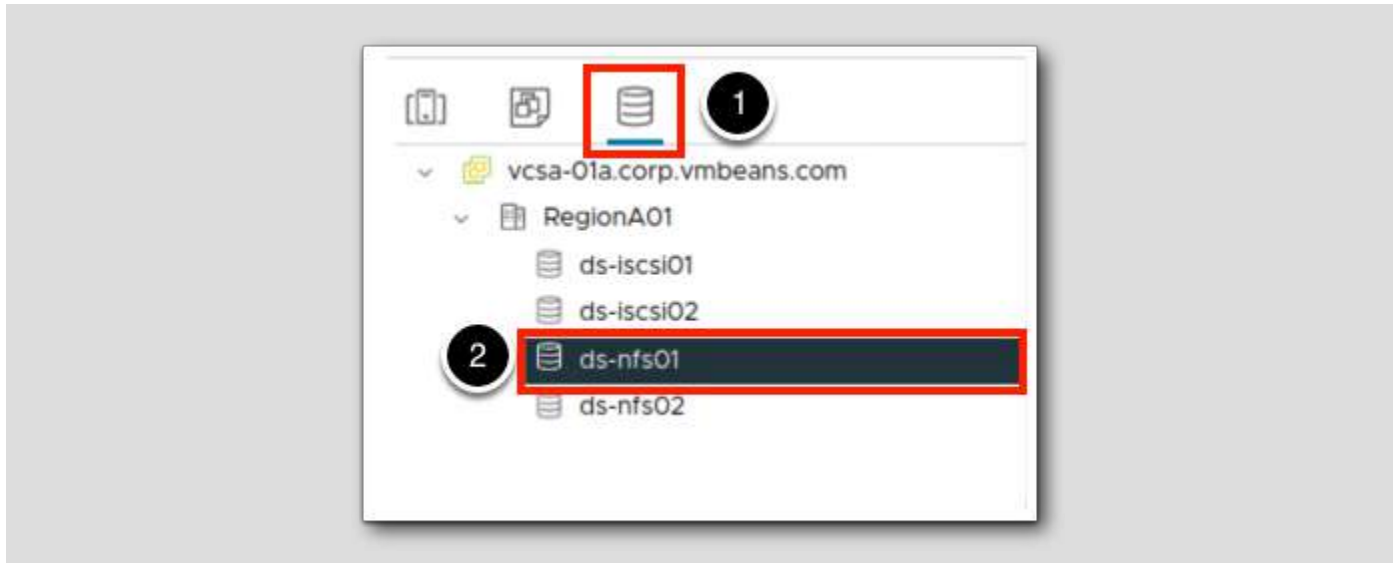


Here you can see esx-03a.corp.vmbeans.com has been added to the datacenter and is in Maintenance Mode.

Maintenance Mode is used for hosts that service. A host could enter Maintenance Mode so that it can be brought offline in order for additional memory to be added to the physical host. In our case, it is in Maintenance Mode once it has been added to the datacenter so that we can verify its settings prior to bringing it online and potentially conflicting with other hosts in the environment.

Datastore View

[461]

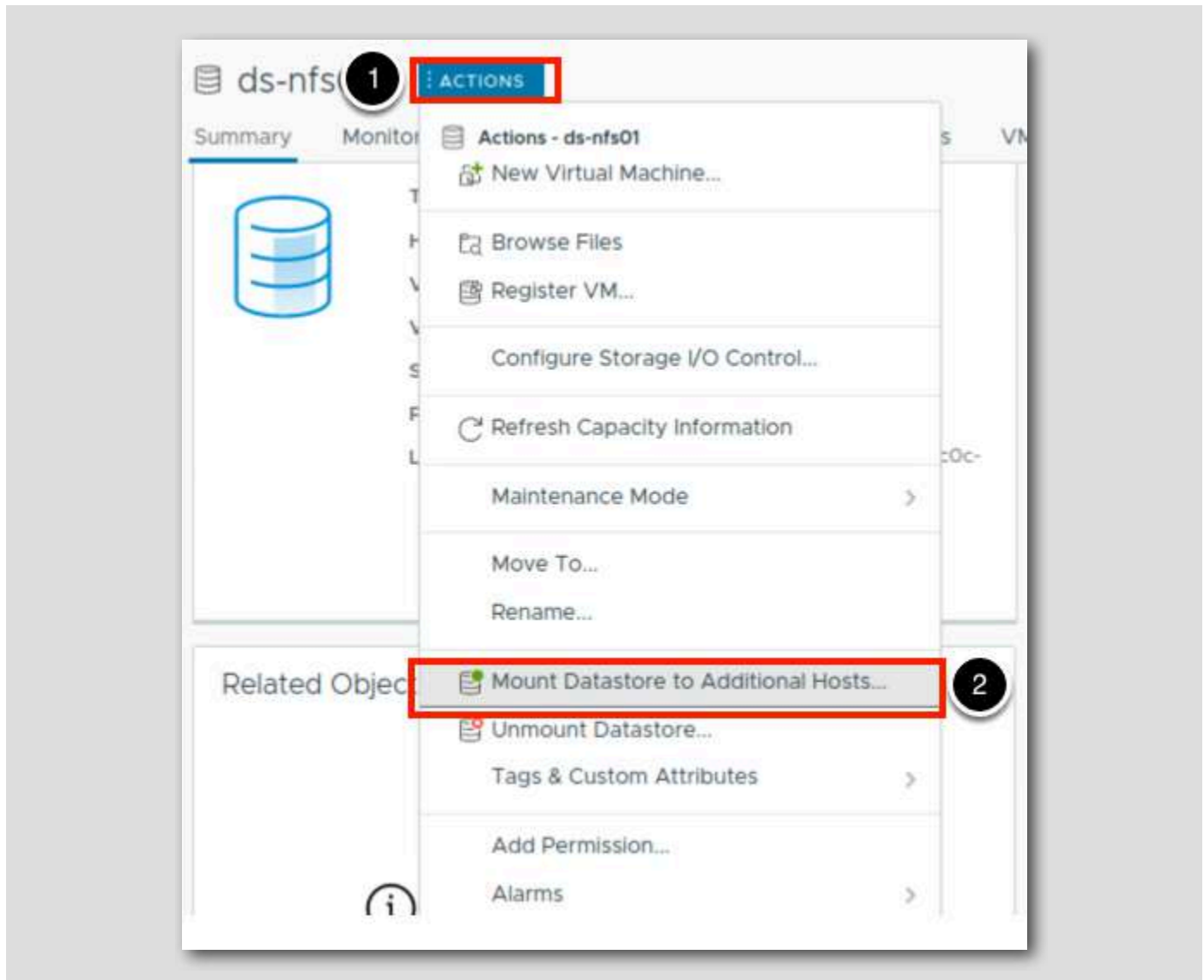


Prior to adding the new host to the cluster, an NFS datastore will be added to the host.

1. Click on the **Datastore** icon to switch to the Datastores view.
2. Select the **ds-nfs01** datastore in the Inventory.

Mount NFS Datastore to New Host Wizard

[462]

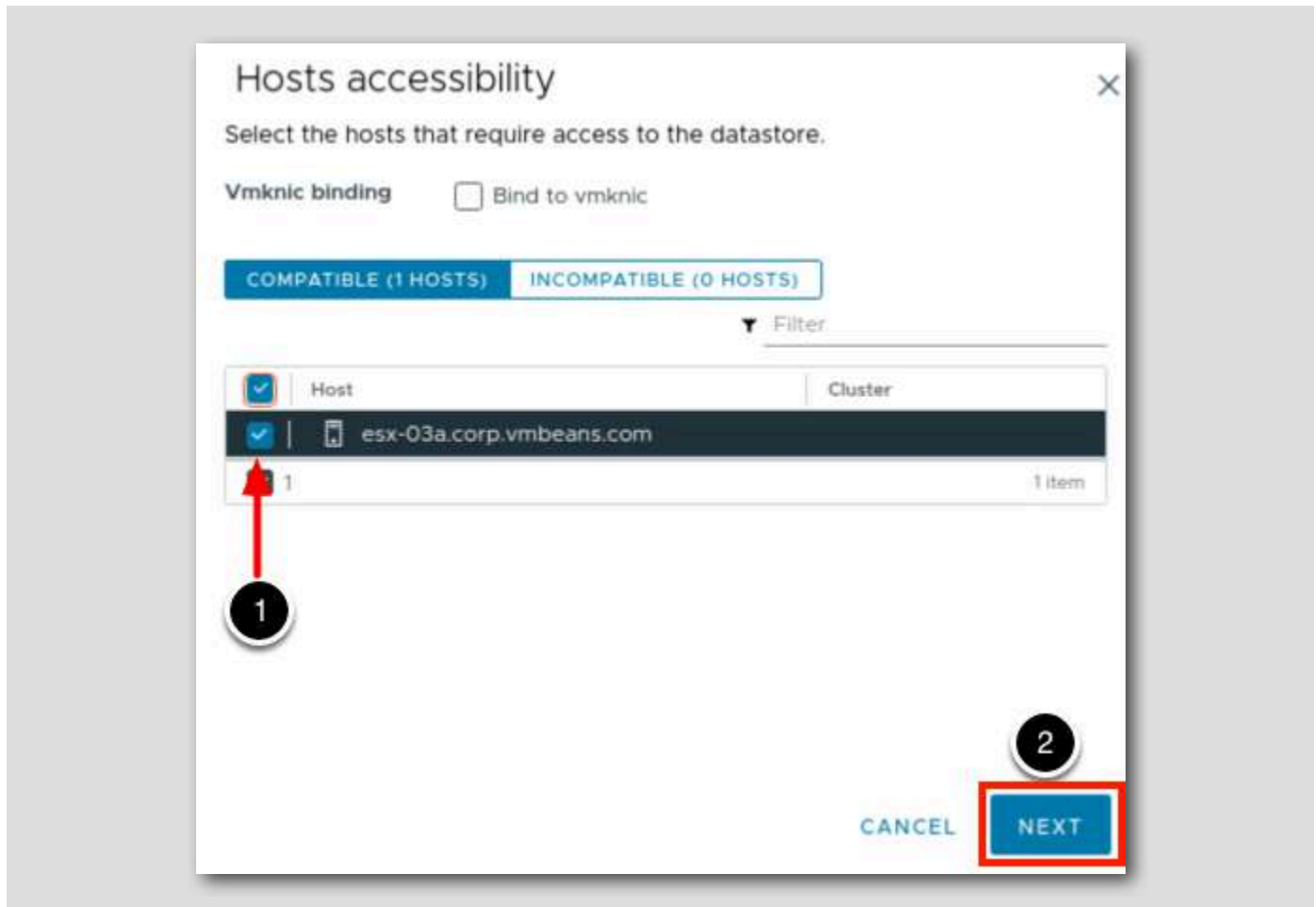


In this case, there are two NFS datastores used by RegionA01 cluster. Adding an existing NFS datastore to a new host is a simple process.

1. Click on the **Actions** menu.
2. Select **Mount Datastore to Additional Hosts...**

Mount NFS Datastore - Select Host

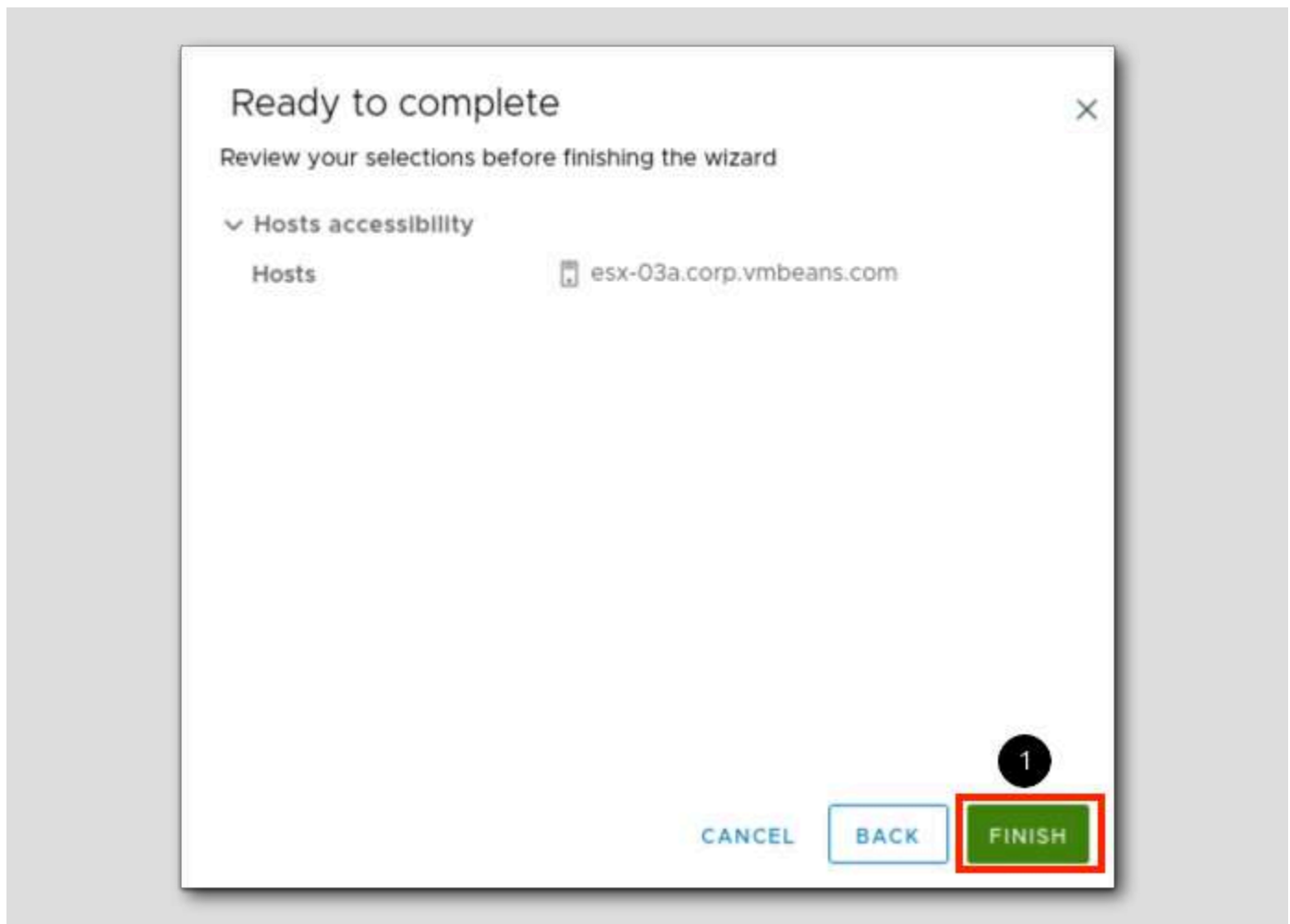
[463]



1. Click the checkbox next to esx-03a.corp.vmbeans.com
2. Click Next.

Review to Complete

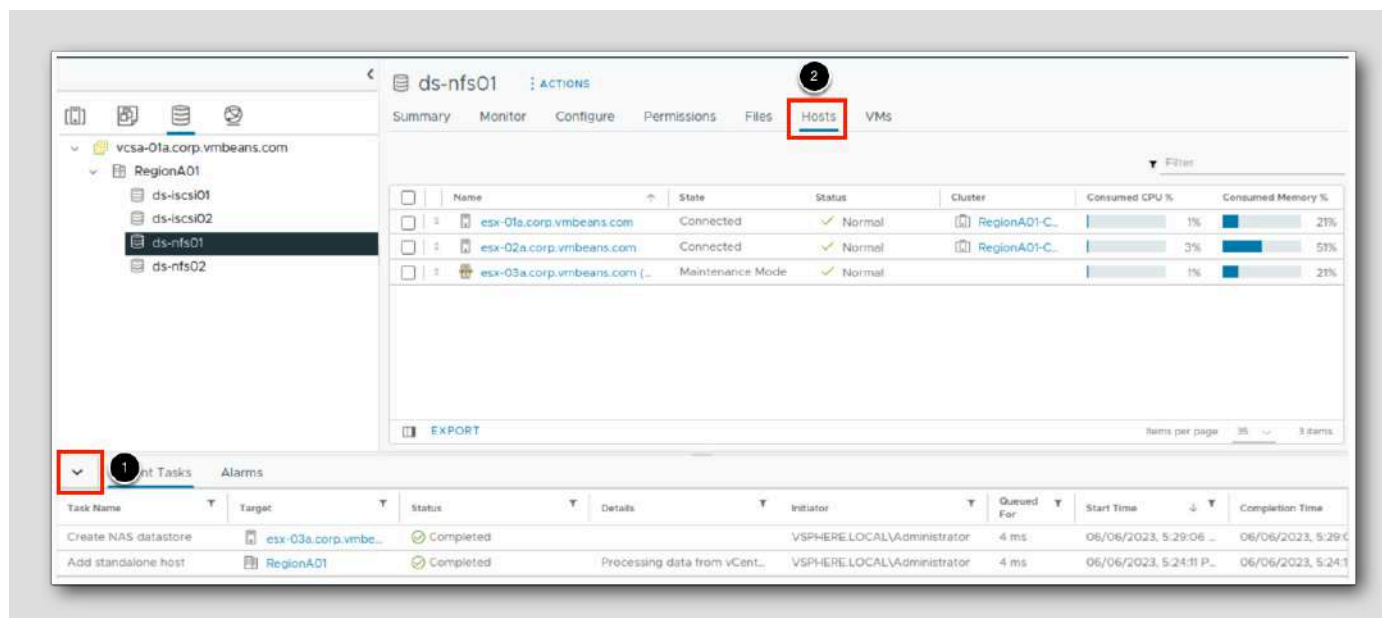
[464]



1. Review the information and click Finish to complete the mounting process

Mount NFS Datastore - Monitor Task

[465]



1. The mount task can be monitored in **Recent Tasks**.
2. Once the mount completes, it can be verified by clicking on the **Hosts** tab.

This will show all hosts in the inventory that have mounted this datastore.

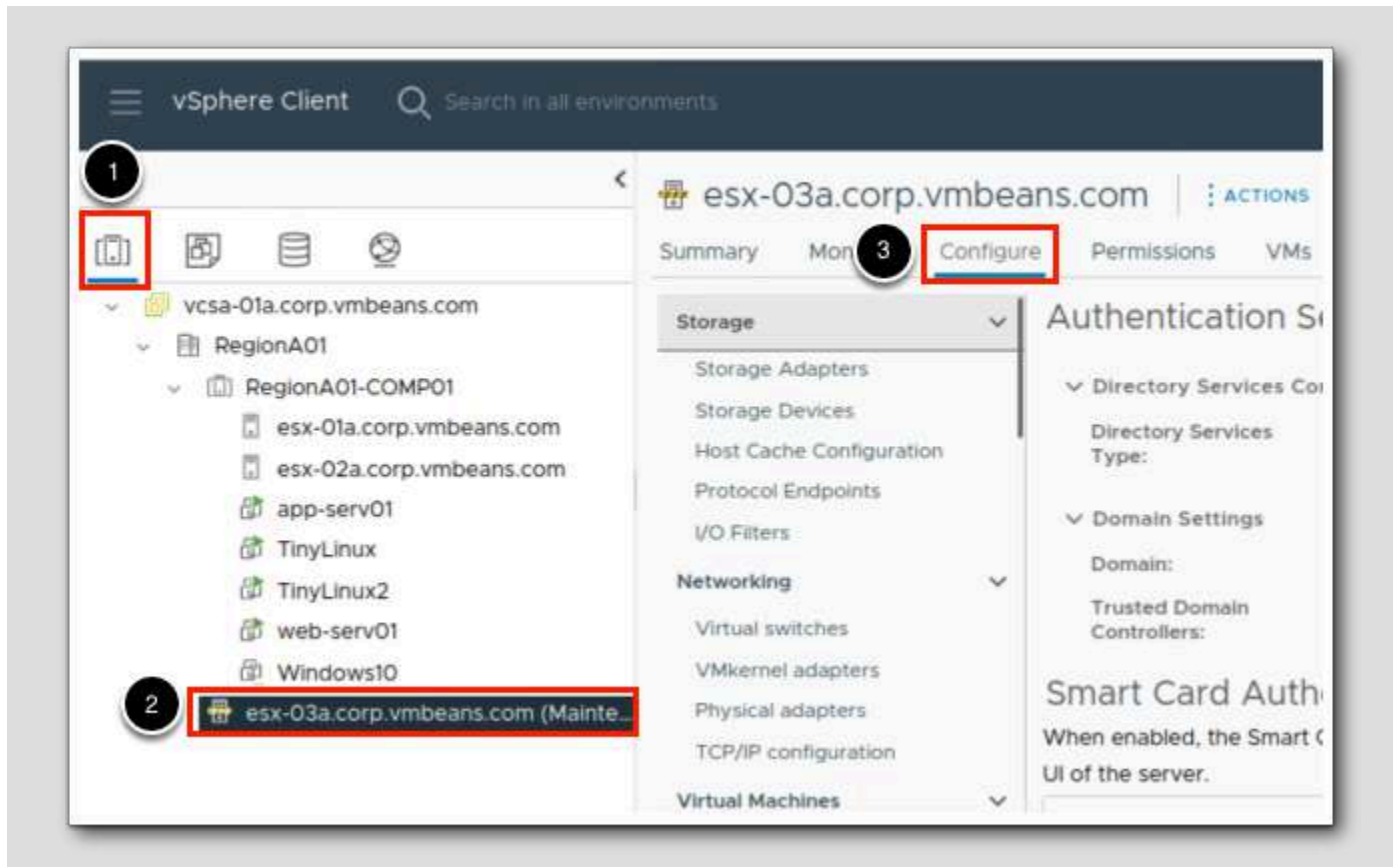
For additional practice, perform the same steps to mount the other NFS datastore, **ds-nfs02** to the **esx-03a.corp.vmbeans.com** host.

Add iSCSI Target to an ESXi host

[466]

iSCSI devices are presented via an iSCSI Target. Think of this as the host for the iSCSI devices. The ESXi host needs to know where to look for the devices, so this section will go through the process of pointing the ESXi host at the iSCSI target and discovering which LUNs are available.

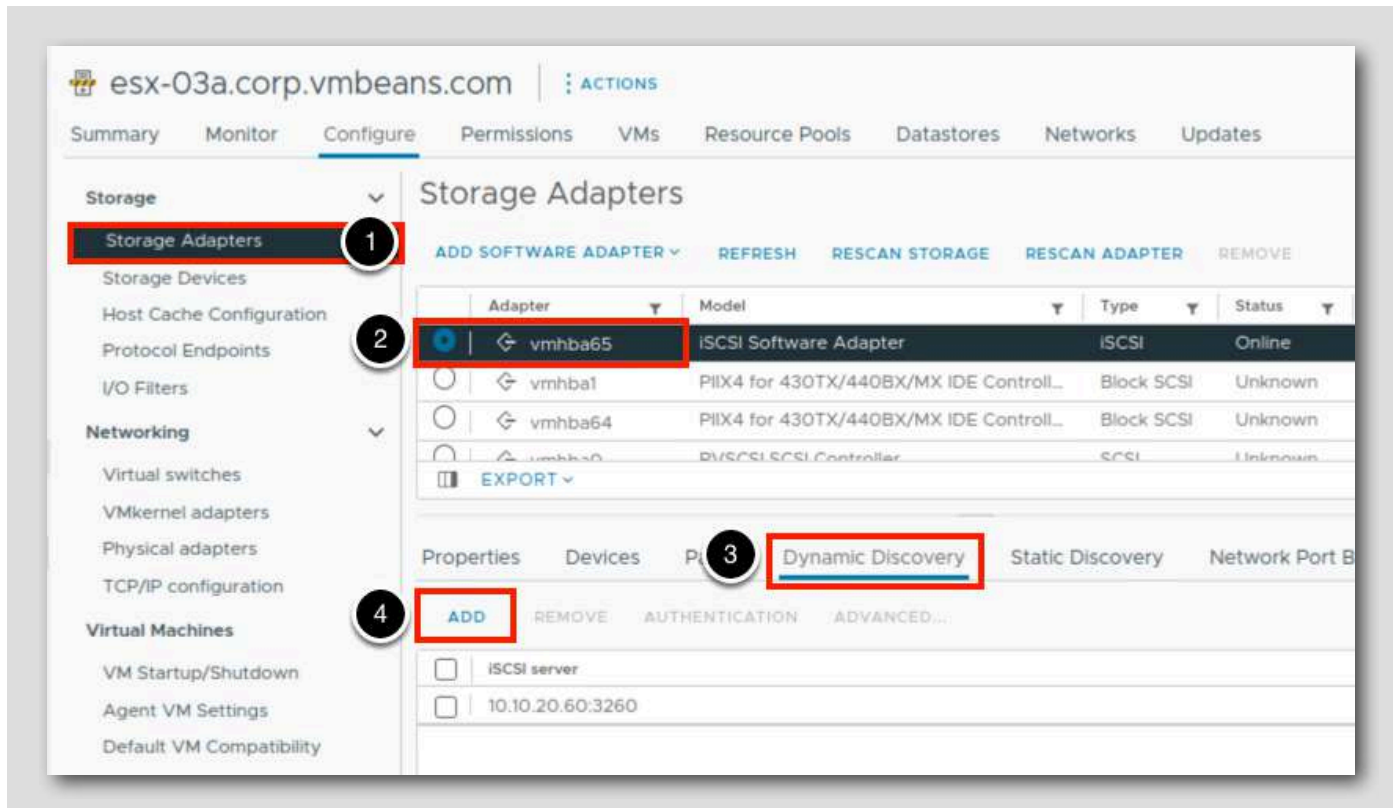
Select Hosts and Clusters



1. Select the Hosts and Clusters icon.
2. Click on esx-03a.corp.vmbeans.com (Maintenance Mode).
3. Click the Configure tab.

Perform Dynamic Discovery

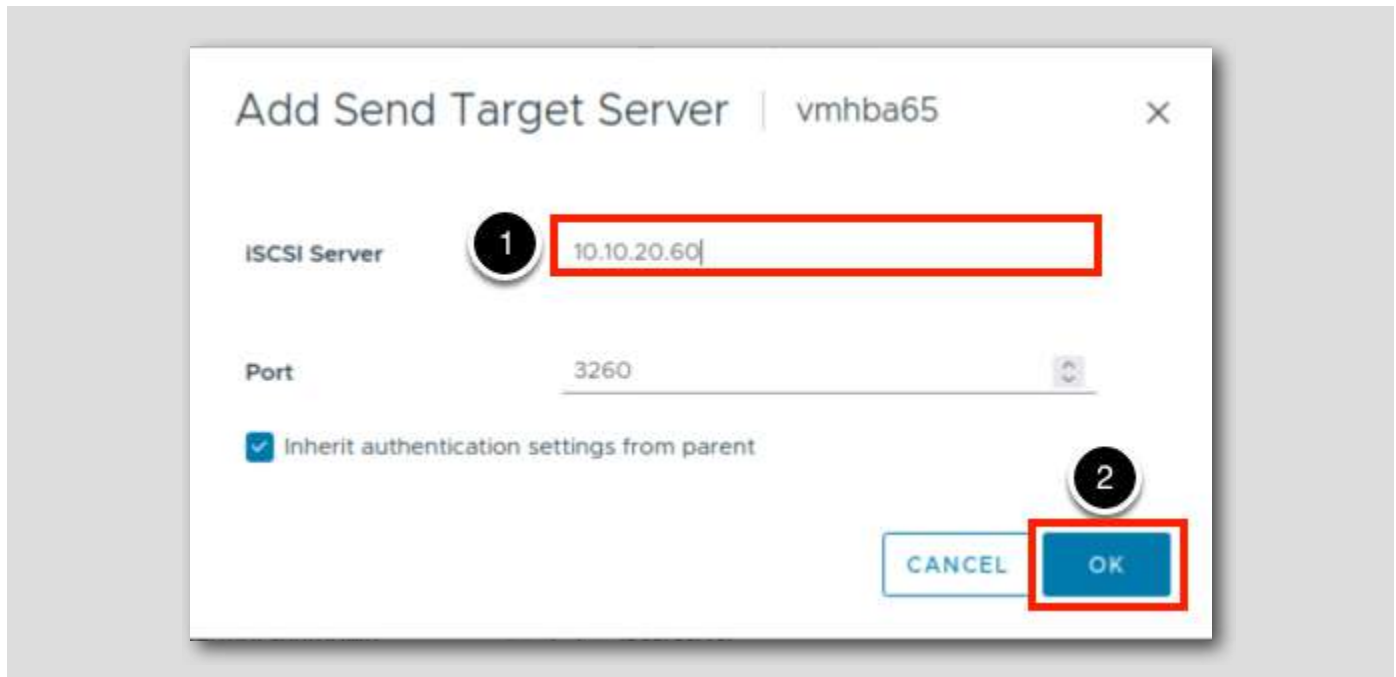
[468]



1. Select "Storage Adapters"
2. Select the "vmhba65" adapter in the iSCSI Software Adapter section.
3. Click on "Dynamic Discovery" - notice that the list of iSCSI Servers is currently empty.
4. Click "Add"

Add Send Target Server

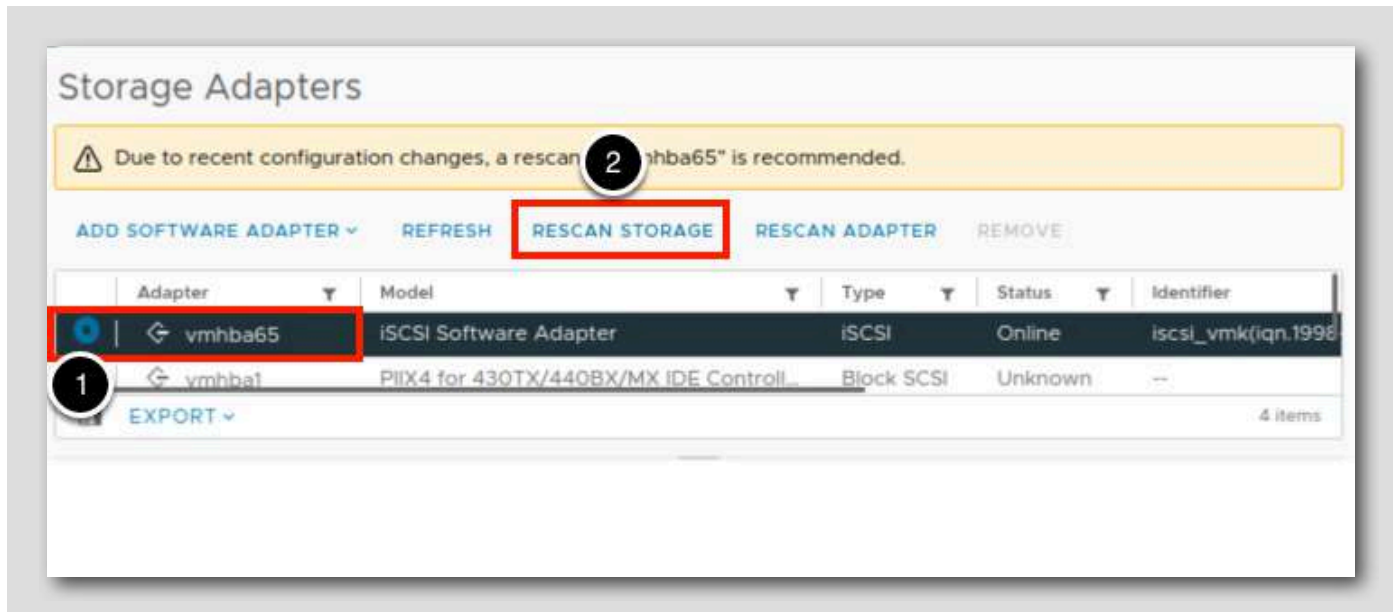
[469]



1. Enter the iSCSI Server Address: 10.10.20.60
2. Select OK.

Rescan the iSCSI storage adapter

[470]



Once the new Target has been added, a message will appear in yellow to remind you of the need to tell the adapter to reach out and query the iSCSI Target.

1. Click on the **vmhba65** iSCSI adapter to select it.
2. Click the **Rescan Storage...** icon to rescan.

Rescan Storage

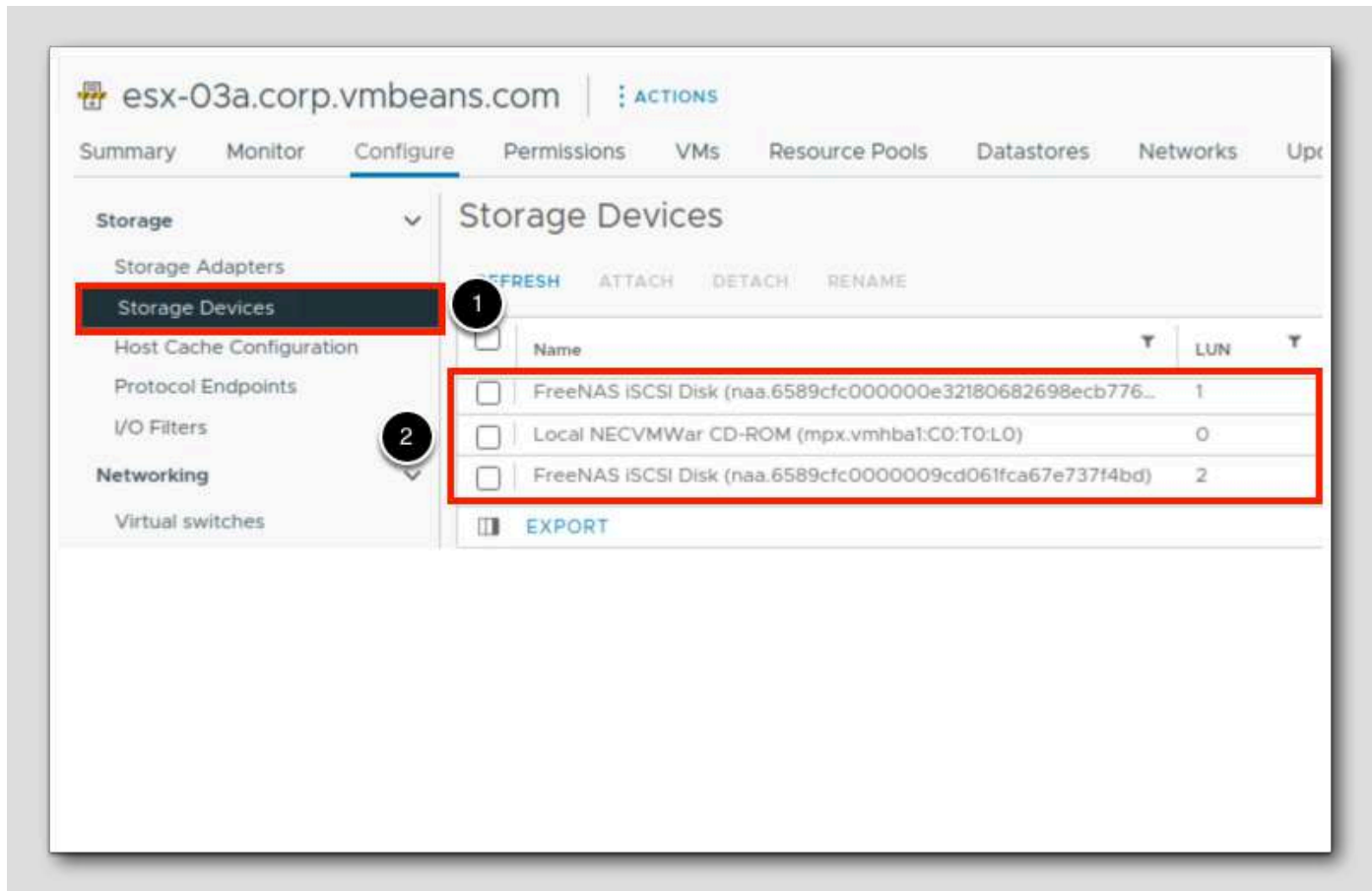
[471]



1. Leave the default options selected and click OK.

Verify iSCSI Devices are Visible

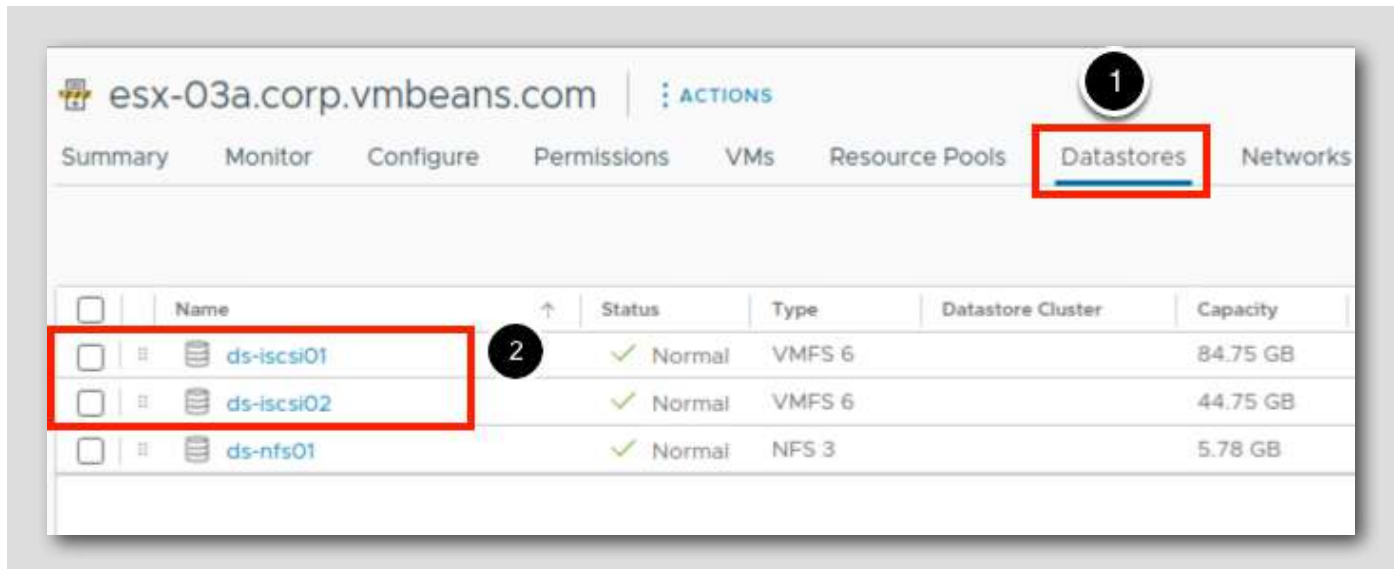
[472]



1. Once the rescan is complete, Click on **Storage Devices**.
2. You should now see two iSCSI disks connected, both with 45GB of capacity.

Verify iSCSI Datastore Availability

[473]

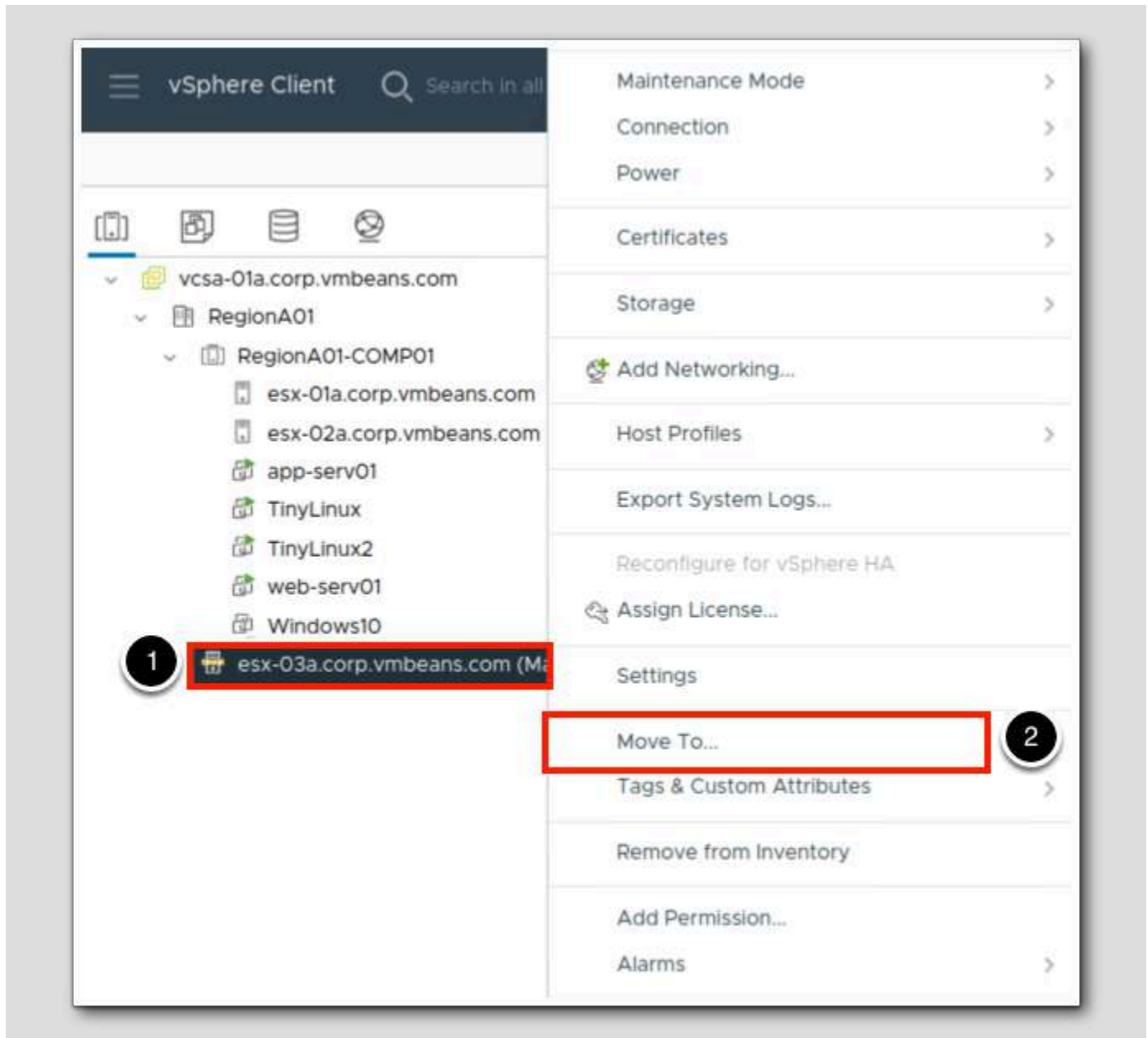


1. Click on the Datastores tab.

Notice that the two iSCSI datastores are now visible to the esx-03a.corp.vmbeans.com host.

Move into the Cluster

[474]

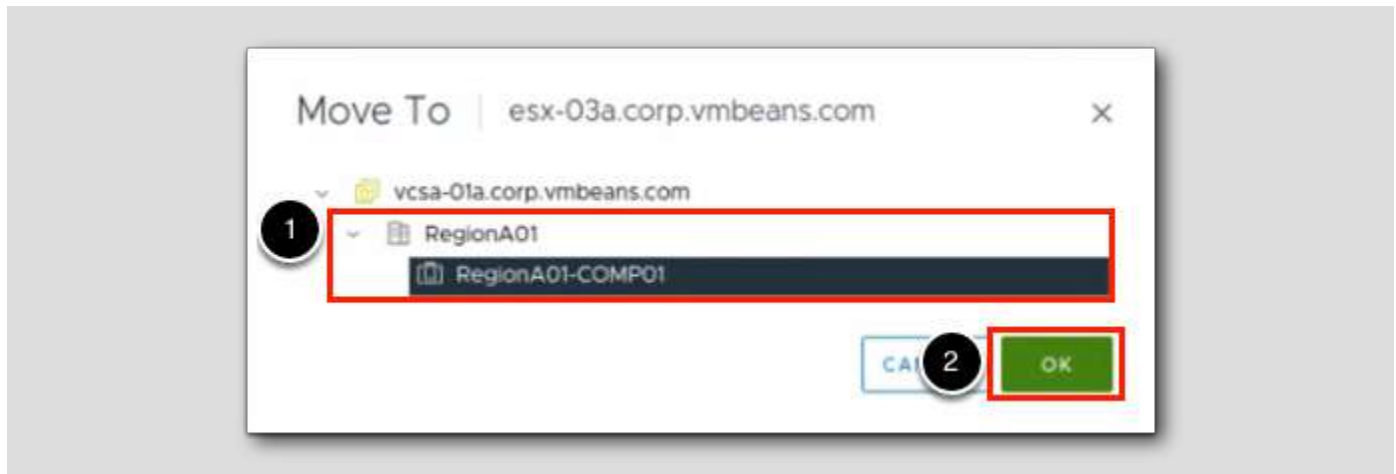


Now that we have the storage configured, move the esx-03a-corp.vmbeans.com into RegionA01-COMP01.

1. Right-click on esx-03a.corp.vmbeans.com
2. Select Move To...

Move To

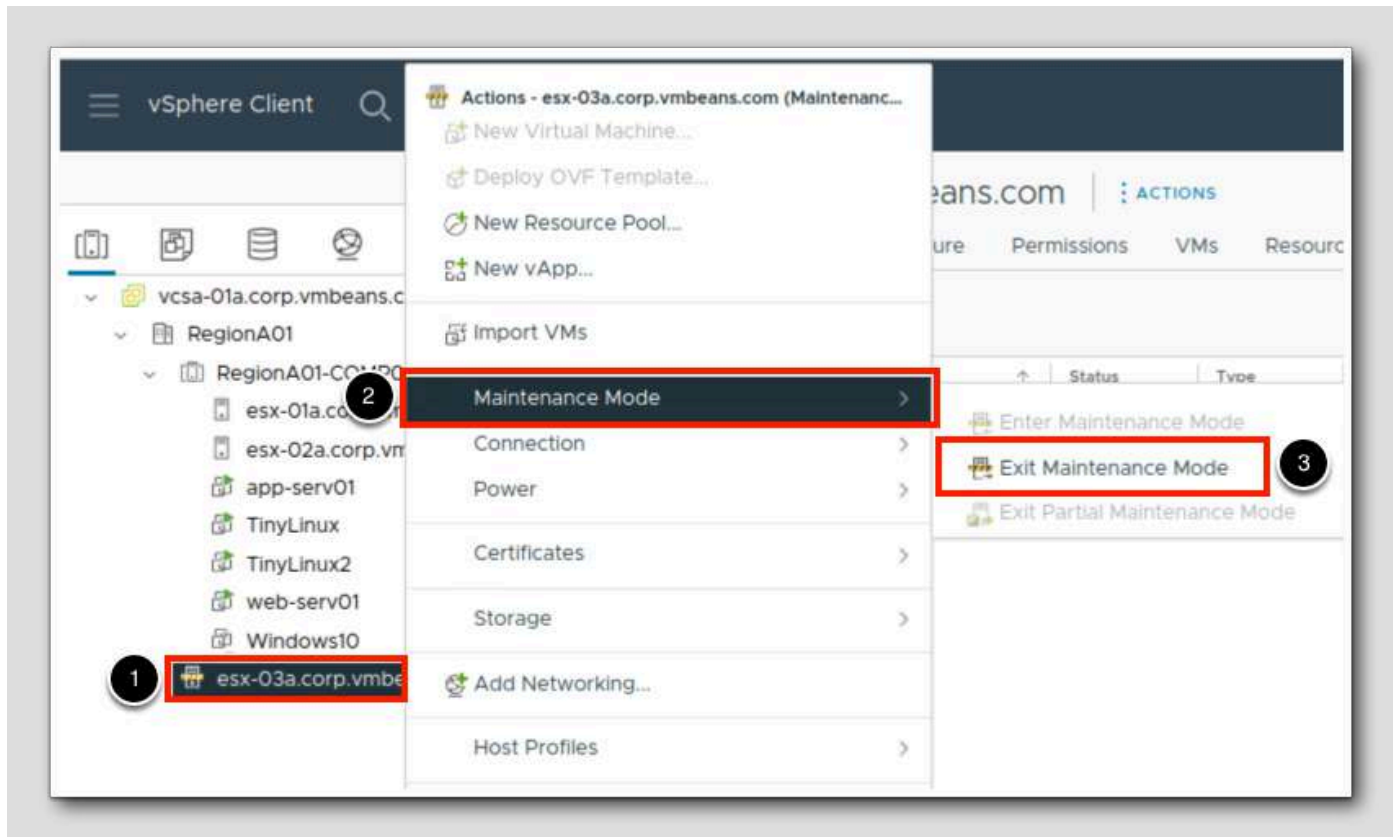
[475]



1. Expand **RegionA01** and select **RegionA01-COMP01**.
2. Click **OK**.

Exit Maintenance Mode

[476]

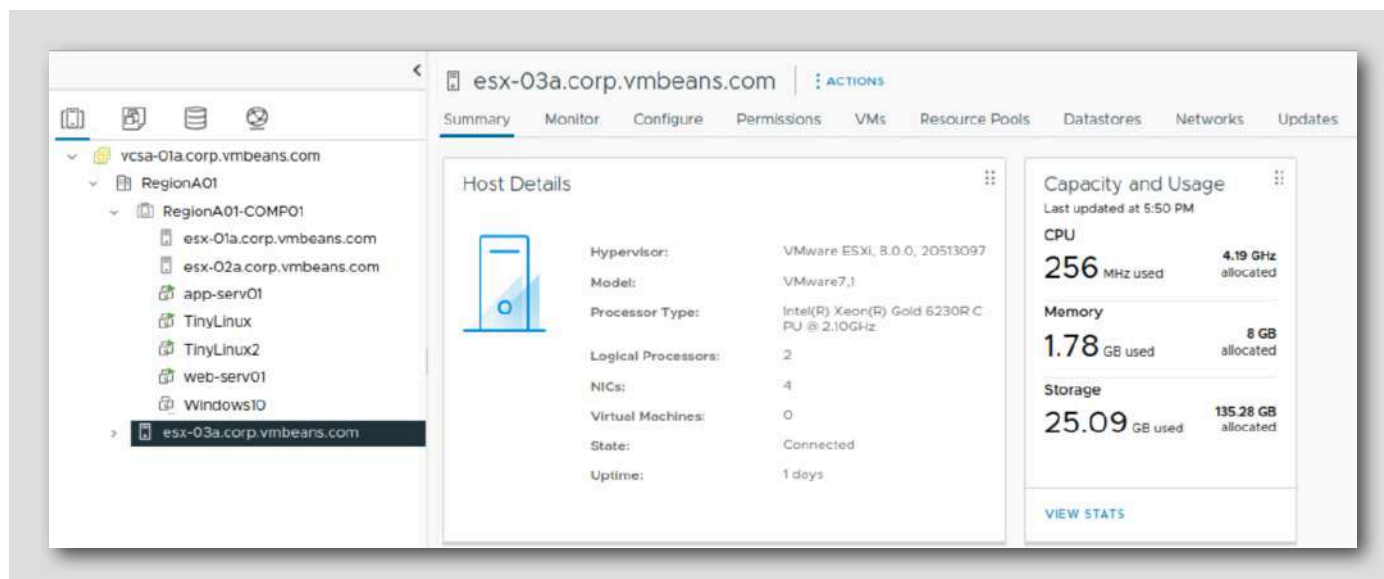


The host has been added to the cluster. Now it can exit Maintenance Mode and participate in the cluster.

1. Right-click on esx-03a.corp.vmbeans.com
2. Select Maintenance Mode.
3. Click Exit Maintenance Mode.

Ready to Go

[477]



After a minute or two, the host will exit Maintenance Mode. If you enabled vSphere HA on the cluster, the HA agent will be configured and started before the host shows a Status of Normal. The process occurs fairly quickly, so a refresh of the Web Client may be required to show the current state.

Note that basic networking for virtual machines, vMotion, and IP Storage have been preconfigured on this host for the purpose of this lab exercise. Adding the new host to a distributed switch would typically be done prior to taking the host out of Maintenance Mode, but is not required for this exercise. Feel free to migrate this switch to the vDS if you would like the practice.

This host is now able to handle workloads for the cluster.

Storage vMotion

[478]

Planned downtime typically accounts for over 80% of datacenter downtime. Hardware maintenance, server migration, and firmware updates all require downtime for physical servers. To minimize the impact of this downtime, organizations are forced to delay maintenance until inconvenient and difficult-to-schedule downtime windows.

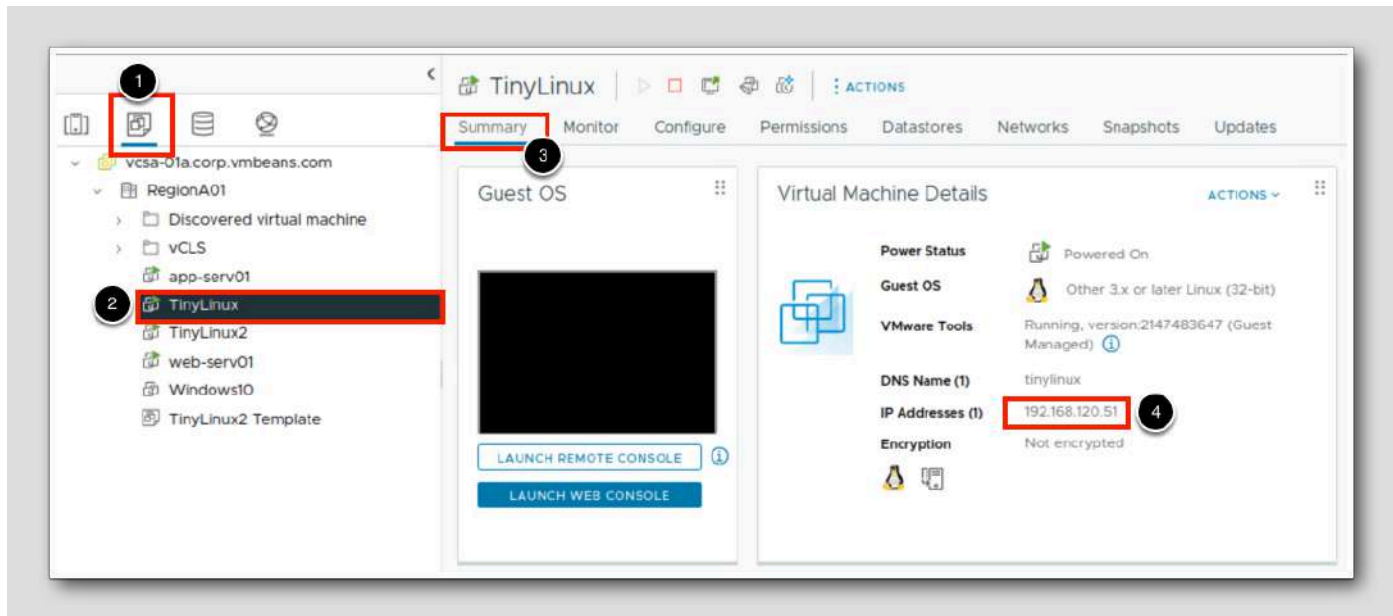
The vMotion and Storage vMotion functionality in vSphere makes it possible for organizations to reduce planned downtime because workloads in a VMware environment can be dynamically moved to different physical servers or to different underlying storage without service interruption. Administrators can perform faster and completely transparent maintenance operations, without being forced to schedule inconvenient maintenance windows. With vSphere vMotion and Storage vMotion, organizations can:

- Eliminate downtime for common maintenance operations.
- Eliminate planned maintenance windows.
- Perform maintenance at any time without disrupting users and services.

In this lesson, you will learn how to work with vMotion and move virtual machines to different hosts within the cluster.

Navigate to Virtual Machines and Templates

[479]



Before the Storage vMotion, we'll verify there is no downtime for the virtual machine by constantly ping it. To ping it, we will need the IP address of the virtual machine, TinyLinux.

1. Click the VMs and Templates tab.
2. Select TinyLinux.
3. Ensure you are on the Summary tab.
4. Note the IP Address of TinyLinux, 192.168.120.51

Open a Command Prompt

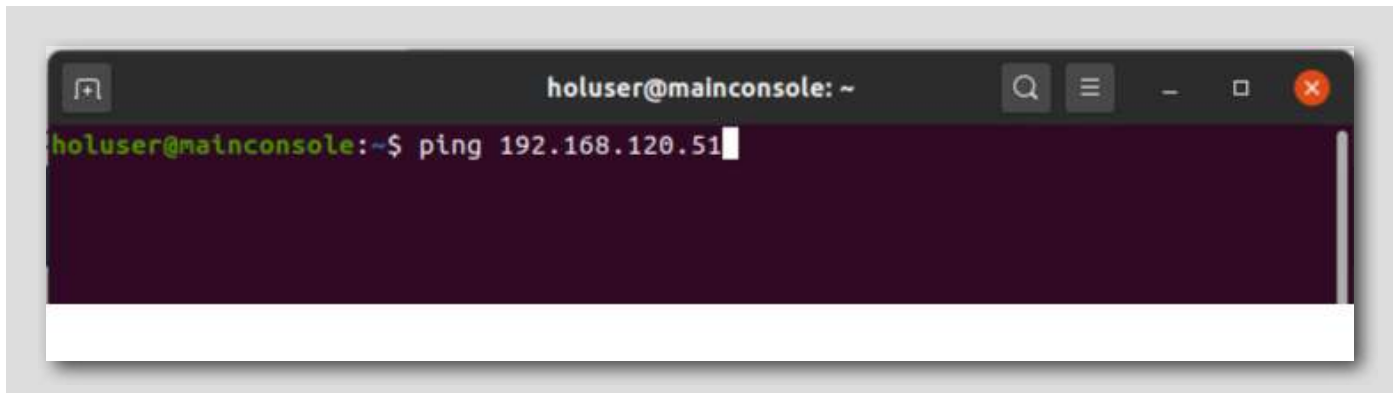
[480]



1. Click on the icon to open a command prompt from the LinuxTask Bar.

Ping TinyLinux-01

[481]

A terminal window titled 'holuser@mainconsole: ~' with search, menu, and window control icons. The command 'ping 192.168.120.51' is entered at the prompt 'holuser@mainconsole:~\$'.

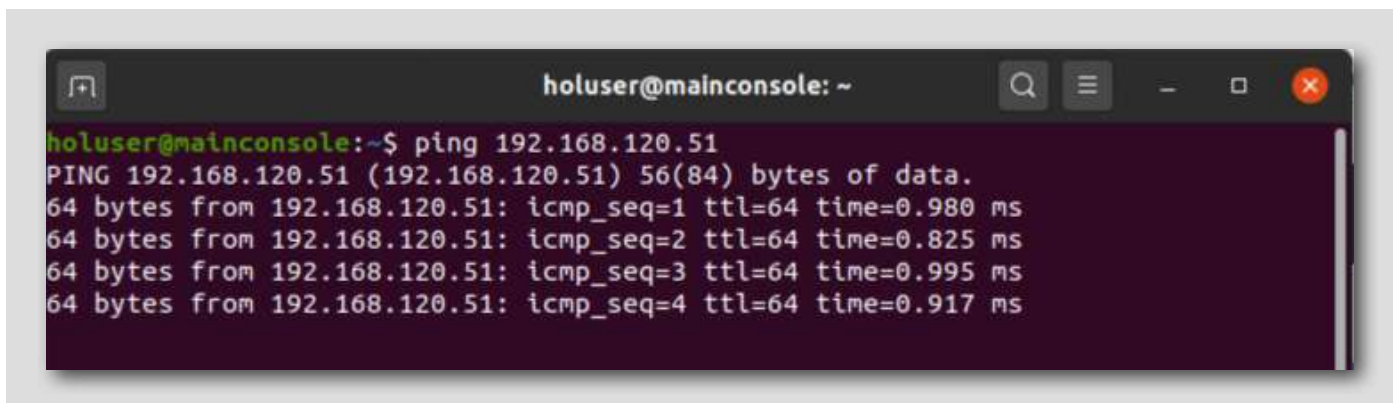
```
holuser@mainconsole:~$ ping 192.168.120.51
```

Issue the following in the command prompt and press the Enter key:

```
ping 192.168.120.51
```

Ping Results

[482]

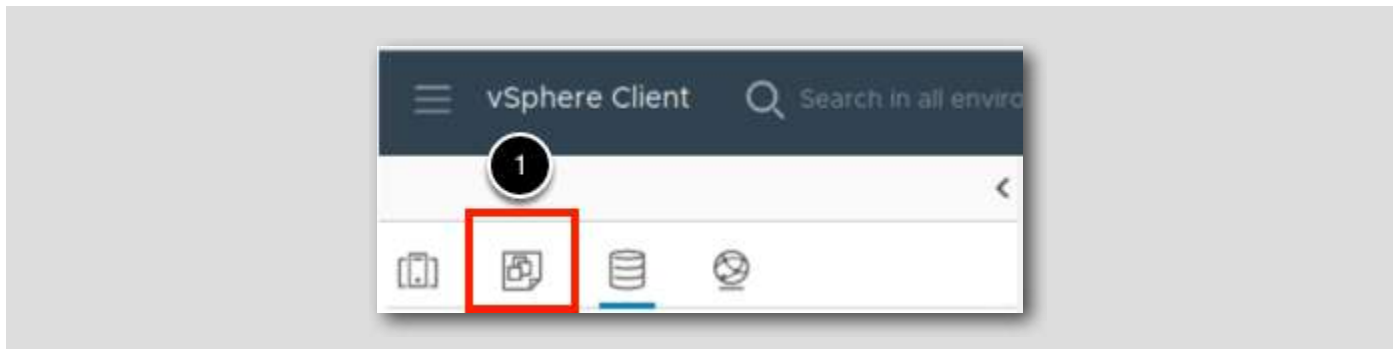
A terminal window titled 'holuser@mainconsole: ~' showing the output of the 'ping 192.168.120.51' command. The output shows four successful ping requests with varying times.

```
holuser@mainconsole:~$ ping 192.168.120.51
PING 192.168.120.51 (192.168.120.51) 56(84) bytes of data:
64 bytes from 192.168.120.51: icmp_seq=1 ttl=64 time=0.980 ms
64 bytes from 192.168.120.51: icmp_seq=2 ttl=64 time=0.825 ms
64 bytes from 192.168.120.51: icmp_seq=3 ttl=64 time=0.995 ms
64 bytes from 192.168.120.51: icmp_seq=4 ttl=64 time=0.917 ms
```

You should now see a continuous ping to TinyLinux.

VMs and Templates

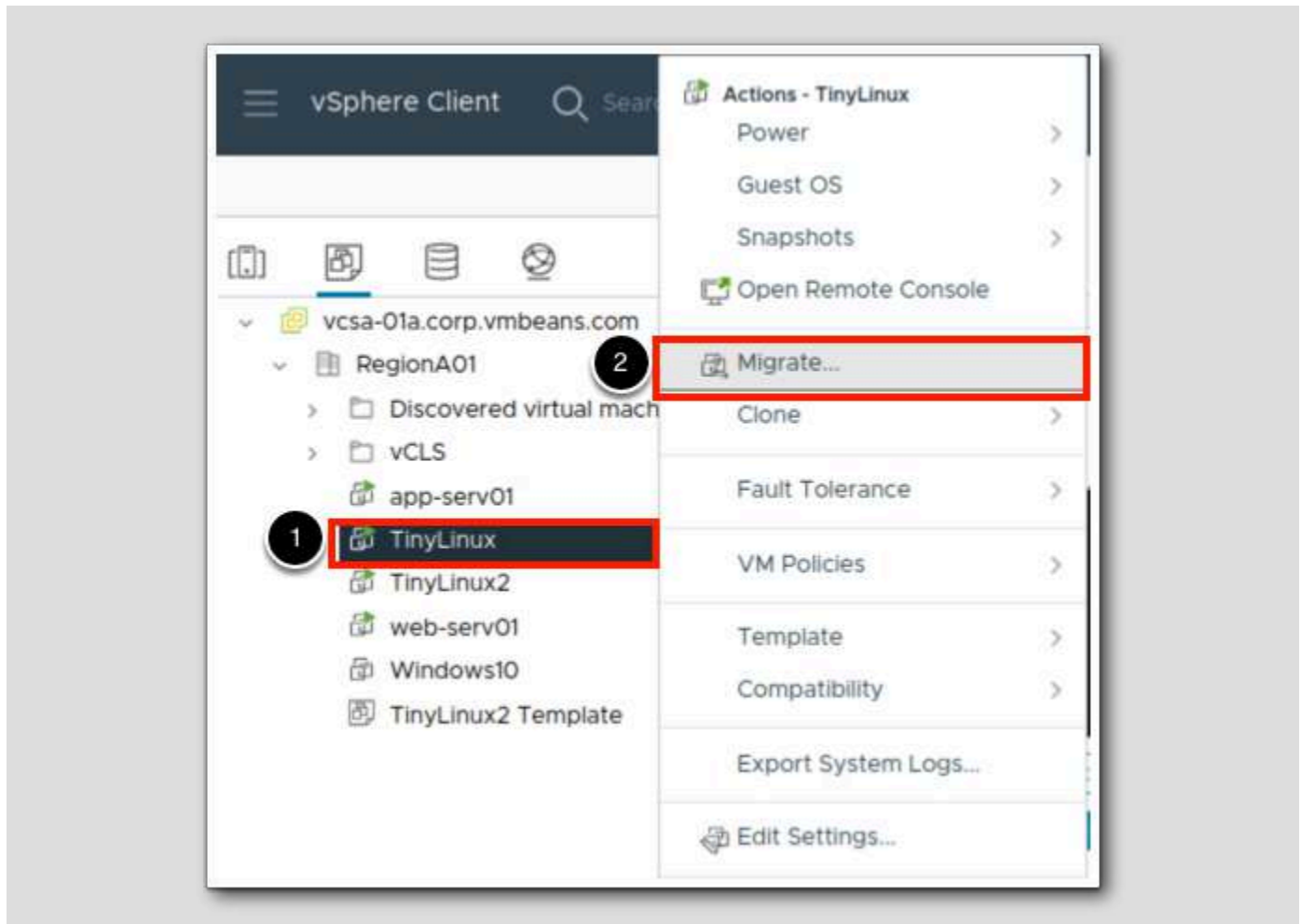
[483]



1. Click the VMs and Templates icon.

List Virtual Machines on a Specified Datastore

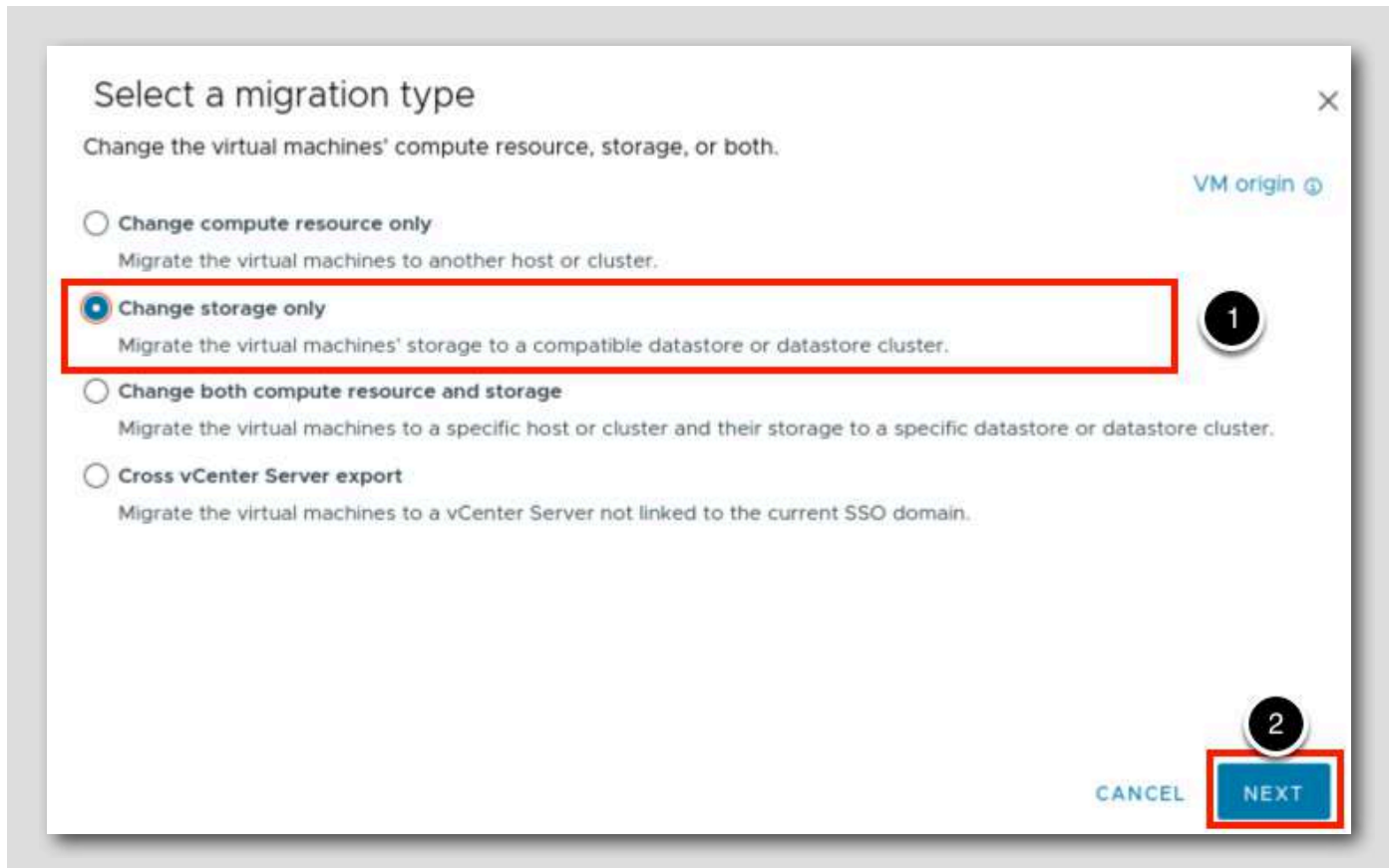
[484]



1. Right-click on TinyLinux
2. Select Migrate

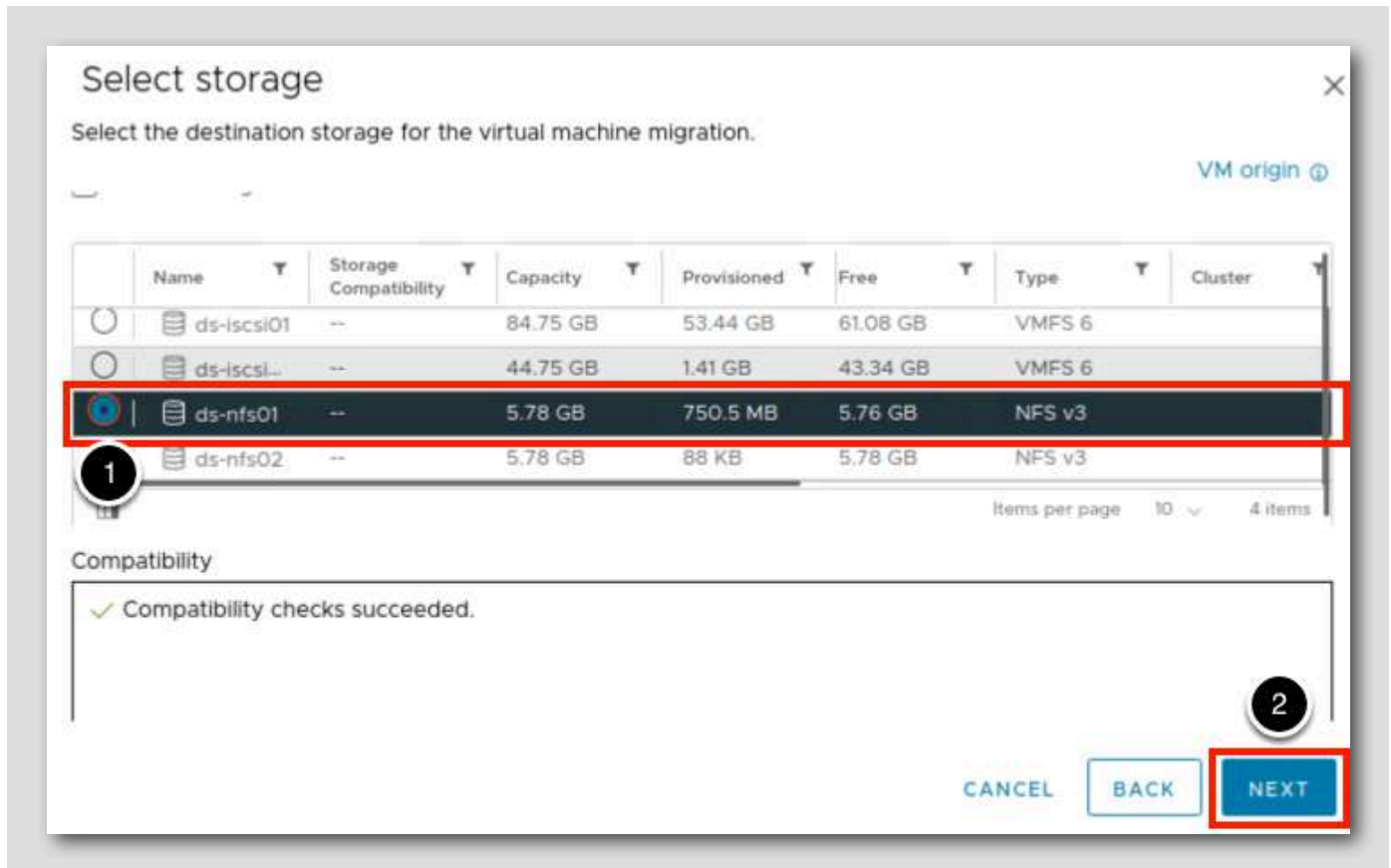
Note: depending on which lessons you have completed, the available datastores and virtual machines may be different than the images.

Migrate Datastore



1. Select the radio button to **Change storage only**. Note that as of vSphere 6.5 (and higher) we do have the ability to change compute, network, and storage in the same vMotion operation.
2. Click **Next**.

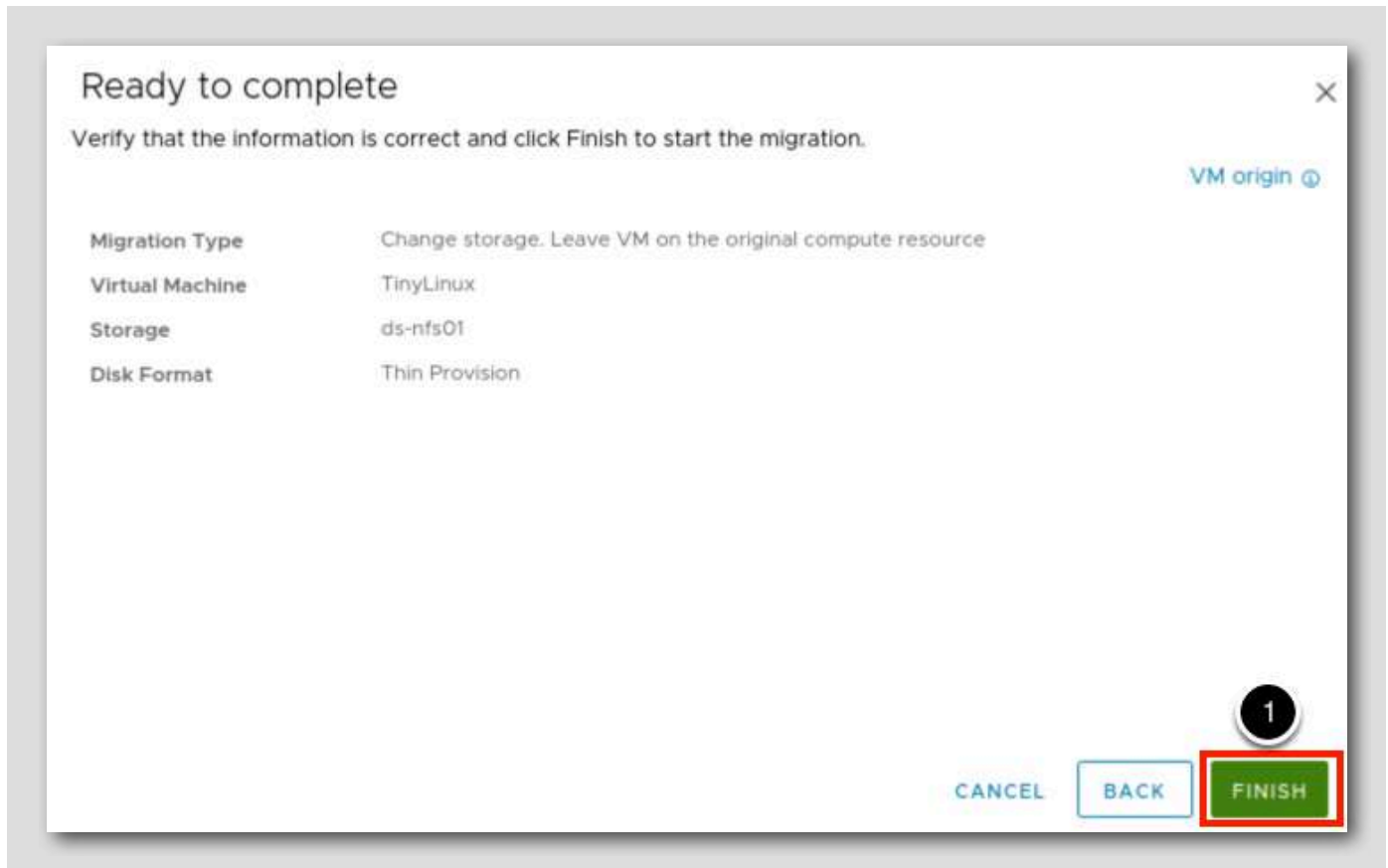
Storage Policy



1. Select the **ds-nfs01** datastore
2. Click **Next** to accept the settings for the storage move.

Ready to Complete

[487]

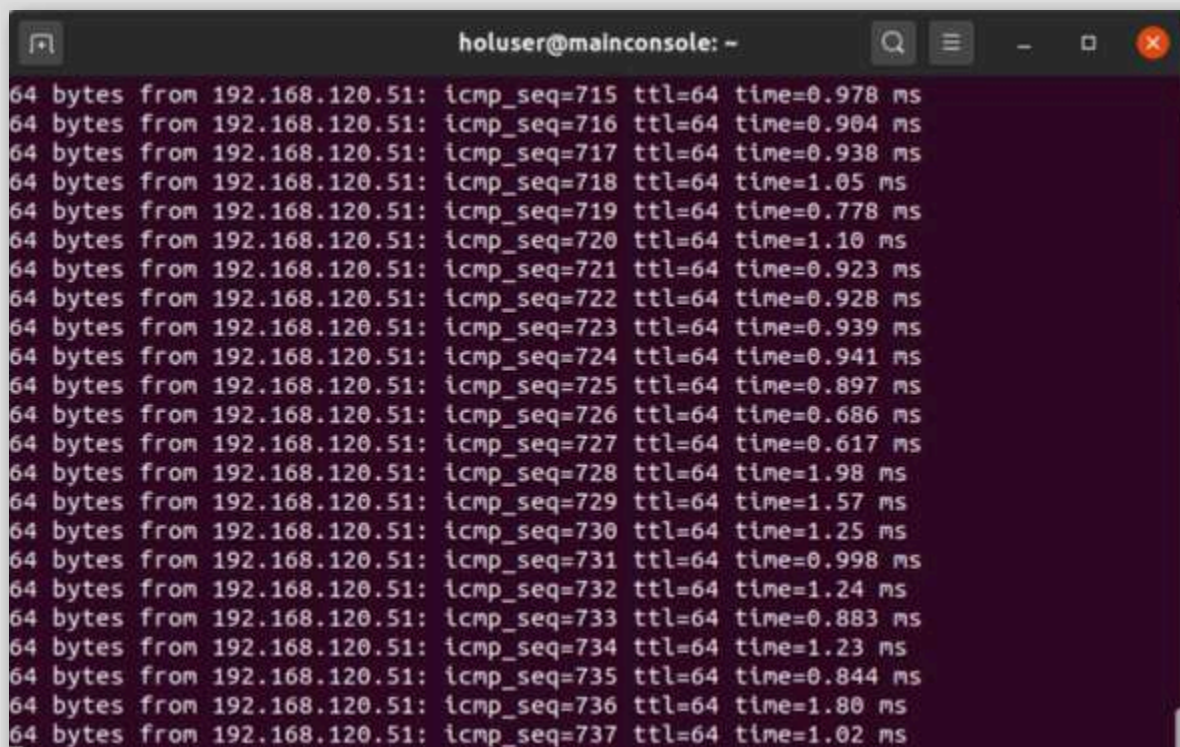


1. Verify your selections on the Ready to complete screen and click **Finish** to start the migration.

Feel free to monitor the operation within the Recent Tasks pane or move on to the next step.

Confirm No Packets Were Dropped

[488]



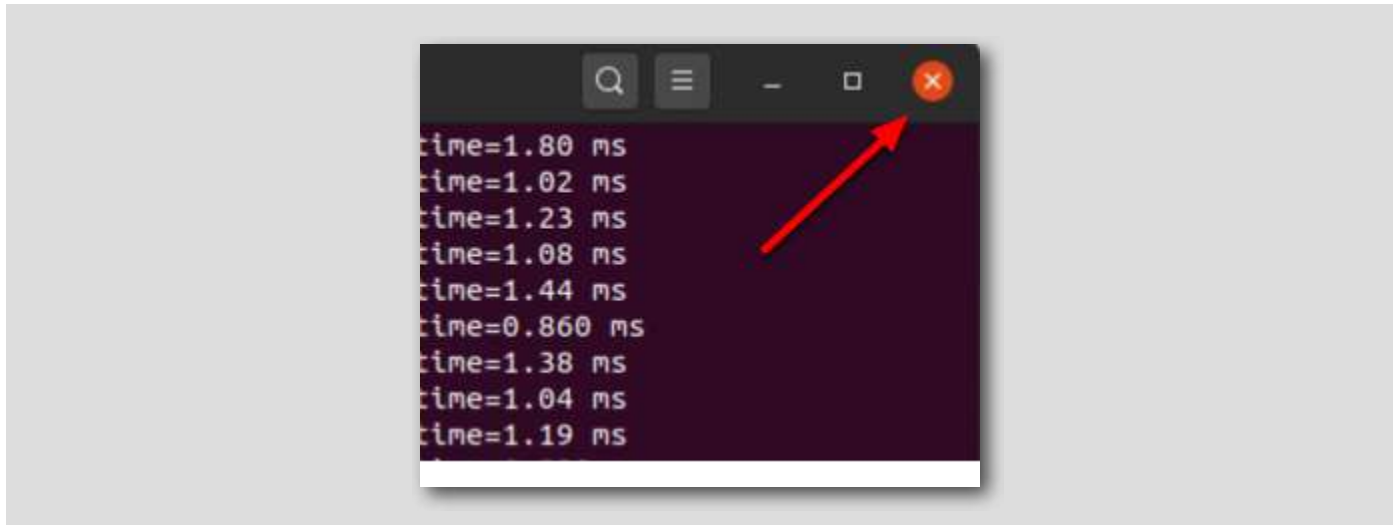
```
holuser@mainconsole: ~
64 bytes from 192.168.120.51: icmp_seq=715 ttl=64 time=0.978 ms
64 bytes from 192.168.120.51: icmp_seq=716 ttl=64 time=0.904 ms
64 bytes from 192.168.120.51: icmp_seq=717 ttl=64 time=0.938 ms
64 bytes from 192.168.120.51: icmp_seq=718 ttl=64 time=1.05 ms
64 bytes from 192.168.120.51: icmp_seq=719 ttl=64 time=0.778 ms
64 bytes from 192.168.120.51: icmp_seq=720 ttl=64 time=1.10 ms
64 bytes from 192.168.120.51: icmp_seq=721 ttl=64 time=0.923 ms
64 bytes from 192.168.120.51: icmp_seq=722 ttl=64 time=0.928 ms
64 bytes from 192.168.120.51: icmp_seq=723 ttl=64 time=0.939 ms
64 bytes from 192.168.120.51: icmp_seq=724 ttl=64 time=0.941 ms
64 bytes from 192.168.120.51: icmp_seq=725 ttl=64 time=0.897 ms
64 bytes from 192.168.120.51: icmp_seq=726 ttl=64 time=0.686 ms
64 bytes from 192.168.120.51: icmp_seq=727 ttl=64 time=0.617 ms
64 bytes from 192.168.120.51: icmp_seq=728 ttl=64 time=1.98 ms
64 bytes from 192.168.120.51: icmp_seq=729 ttl=64 time=1.57 ms
64 bytes from 192.168.120.51: icmp_seq=730 ttl=64 time=1.25 ms
64 bytes from 192.168.120.51: icmp_seq=731 ttl=64 time=0.998 ms
64 bytes from 192.168.120.51: icmp_seq=732 ttl=64 time=1.24 ms
64 bytes from 192.168.120.51: icmp_seq=733 ttl=64 time=0.883 ms
64 bytes from 192.168.120.51: icmp_seq=734 ttl=64 time=1.23 ms
64 bytes from 192.168.120.51: icmp_seq=735 ttl=64 time=0.844 ms
64 bytes from 192.168.120.51: icmp_seq=736 ttl=64 time=1.80 ms
64 bytes from 192.168.120.51: icmp_seq=737 ttl=64 time=1.02 ms
```

Go back to the command prompt and review the results of the ping. You can use the scroll bar to see if there were any dropped packets.

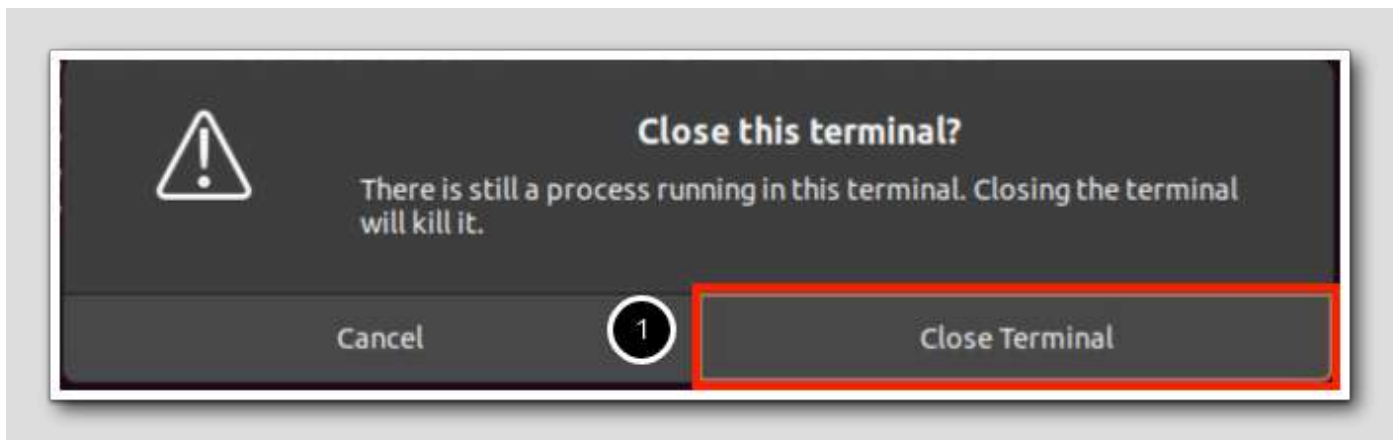
You may see instances where the time field increases to 2ms, but otherwise no packets should have dropped.

Stop the Ping

[489]



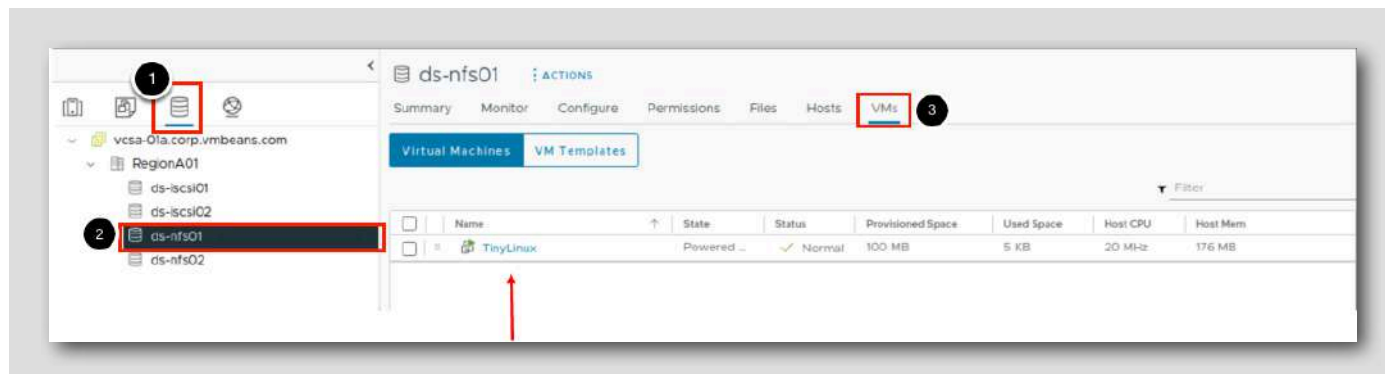
Click the 'X' to stop the ping and close the command window.



1. Click Close Terminal

Confirm Storage vMotion

[490]



The Storage vMotion progress can be monitored in the Recent Tasks panel.

1. Once complete, click on **Datastores**
2. Select the **ds-nfs01** datastore and notice that the **TinyLinux** virtual machine is listed.
3. Click **VMs**

The virtual machine's storage has been migrated from iSCSI to NFS storage without the need to take the virtual machine offline.

Managing Virtual Machine Disks

[491]

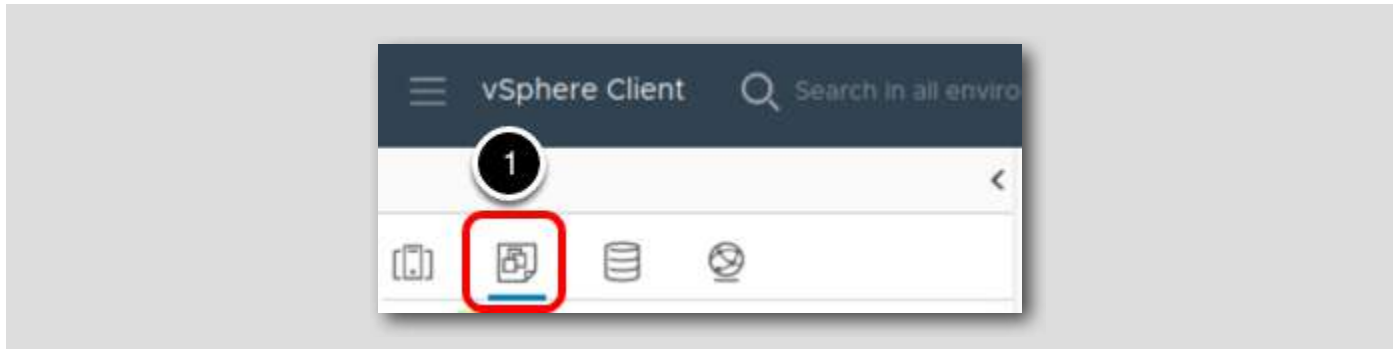
When working with Virtual Machines, you can create a virtual disk or use an existing virtual disk. A virtual disk comprises one or more files on the file system that appear as a single hard disk to the guest operating system. These disks are portable among hosts.

You use the "Create Virtual Machine" wizard to add virtual disks during virtual machine creation. However, in this lesson you will work with an existing Virtual Machine in the inventory.

This lesson will walk you through the process of adding a new virtual disk to an existing Virtual Machine. Additionally, you will extend the Virtual Machine's original disk to a larger capacity.

Navigate to the VMs and Templates management pane

[492]

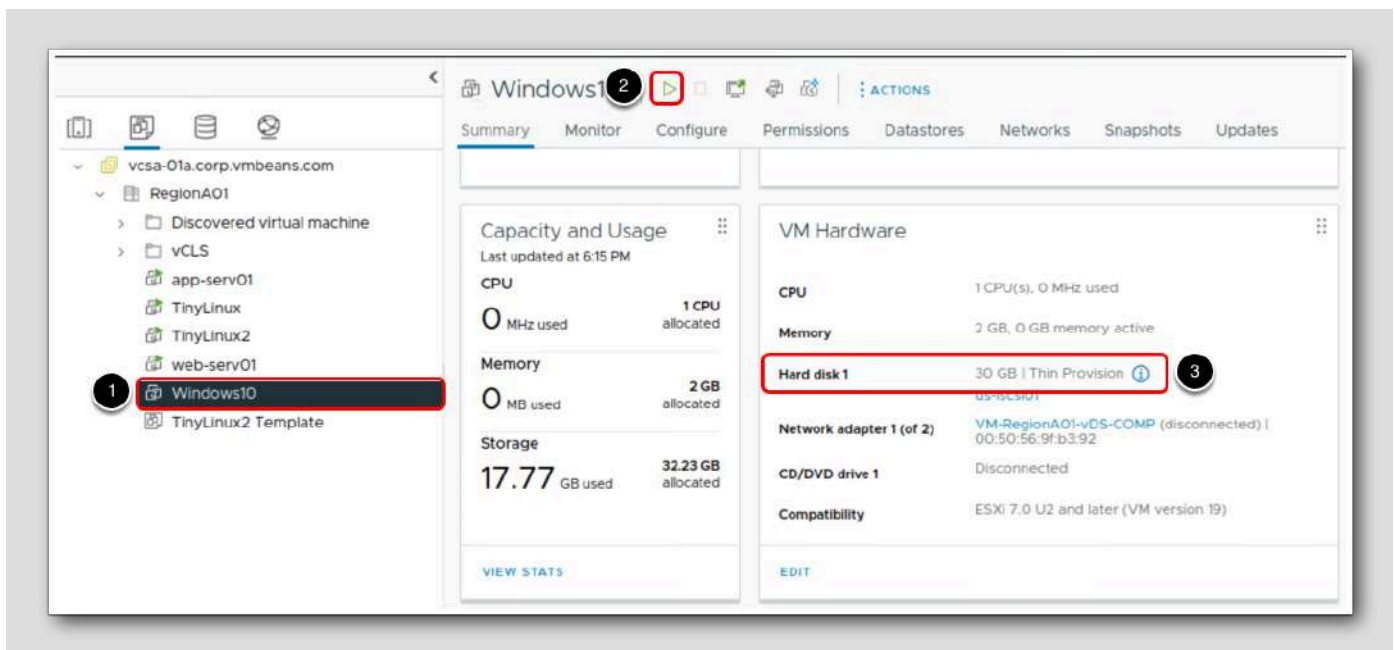


1. Select VMs and Templates.

From this view, we can see that there are several existing Virtual Machines in our vSphere environment. In the next step, we will add a new virtual disk to the **Windows10** Virtual Machine.

Verify windows10 Storage

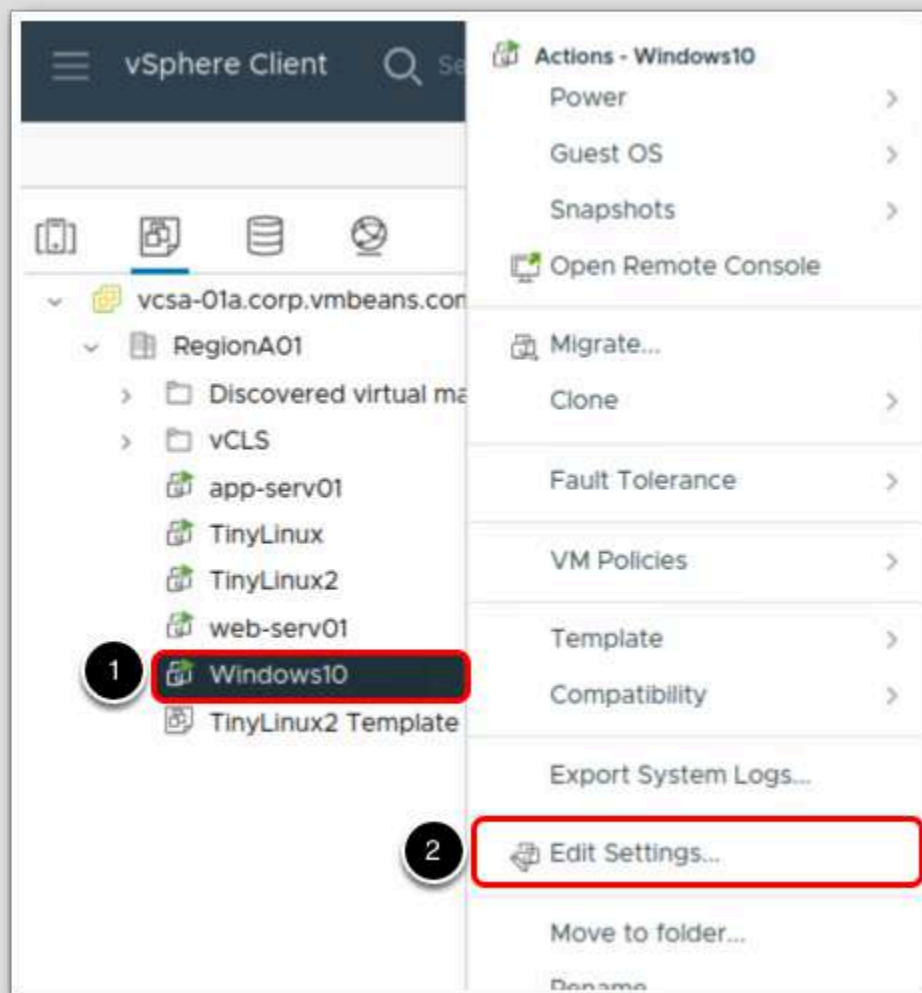
[493]



1. Select Virtual Machine **Windows10** and click the **Summary** tab.
2. If Windows10 VM is not powered on, click the **power on (play)** button.
3. In the VM Hardware pane, note the original disk configuration - single hard disk with a capacity of 30.00 GB.

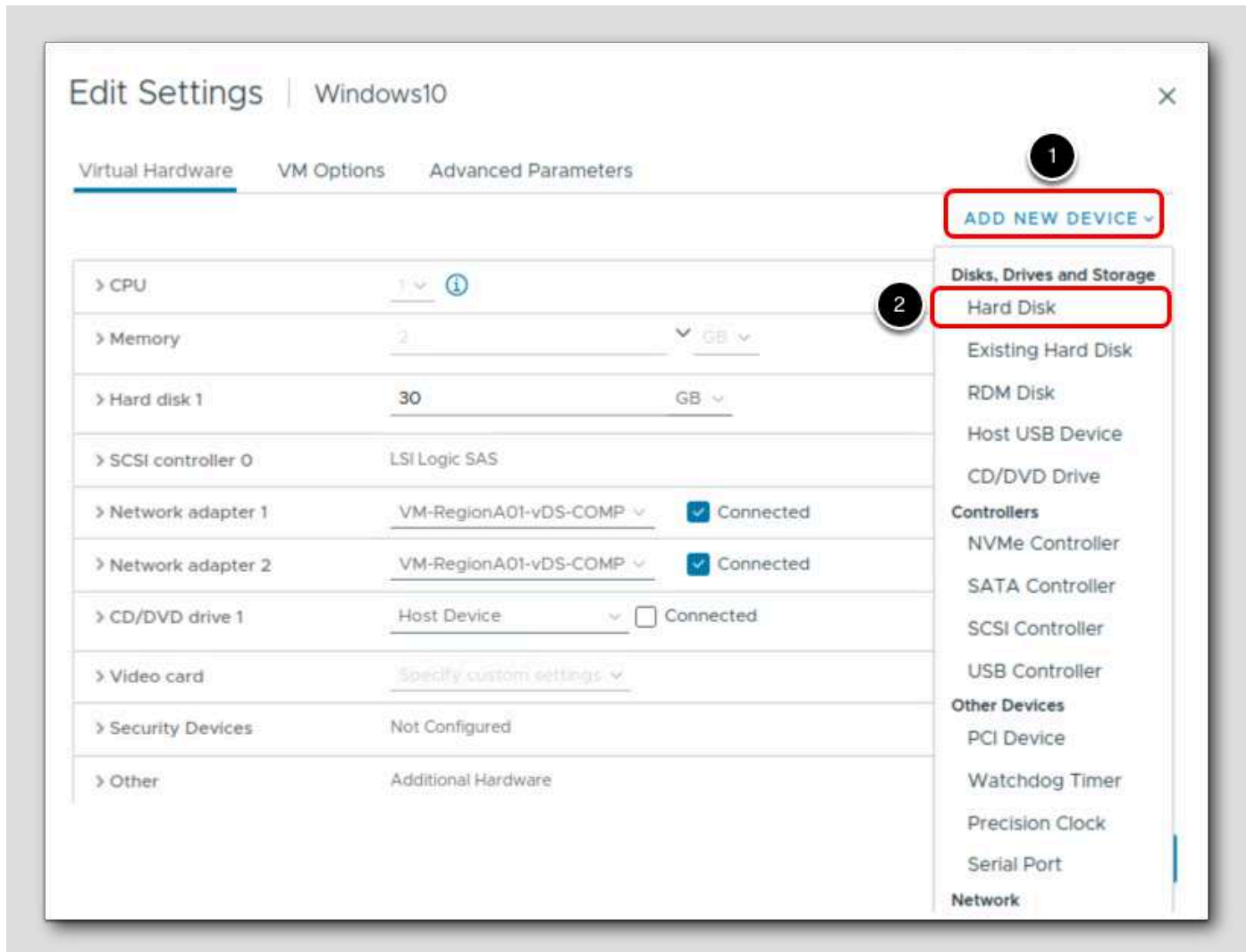
Edit VM Settings

[494]



1. Right-click on **Windows10**
2. Select **Edit Settings**

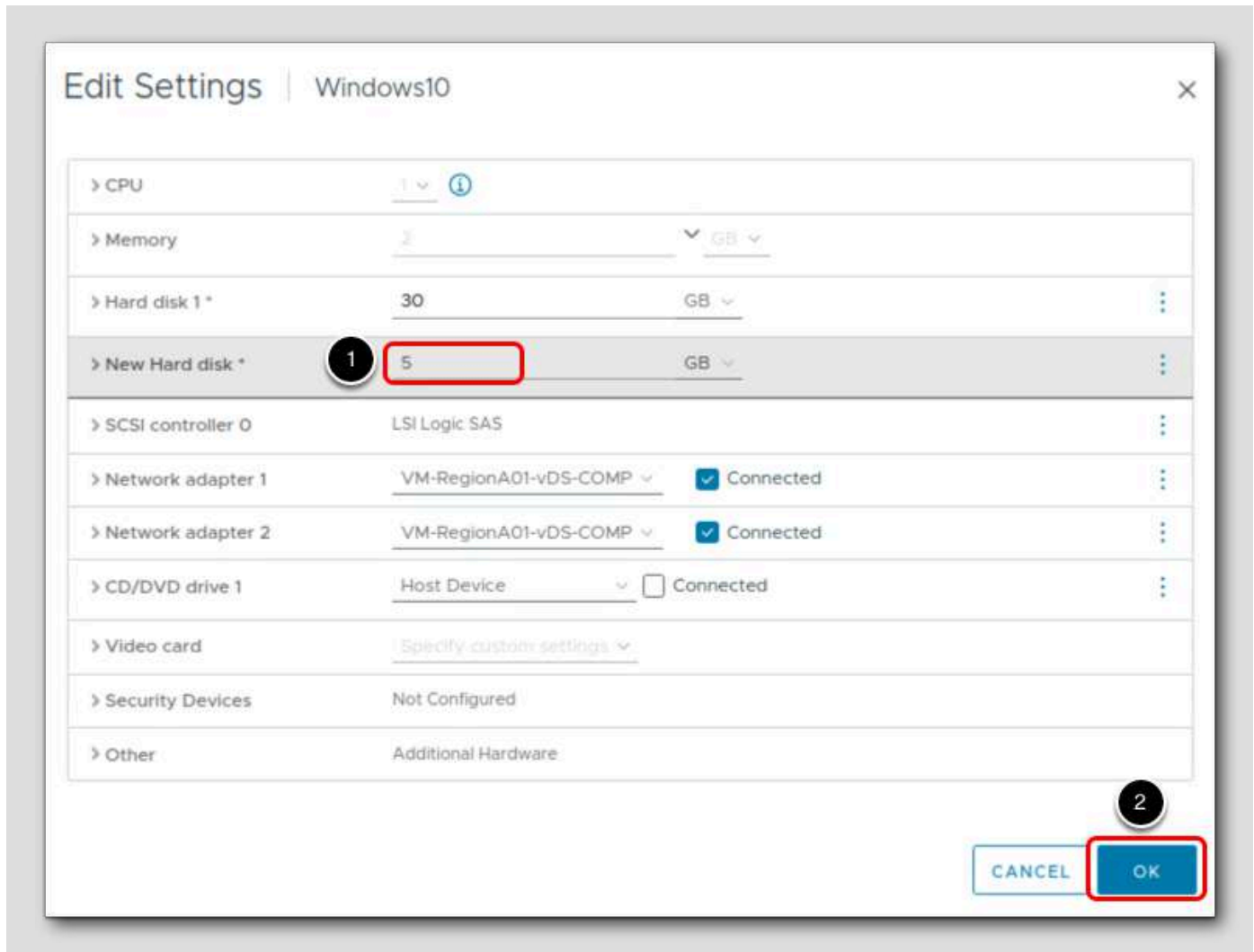
Add New Device



1. Click the **Add New Device** button.

2. Click **Hard Disk**.

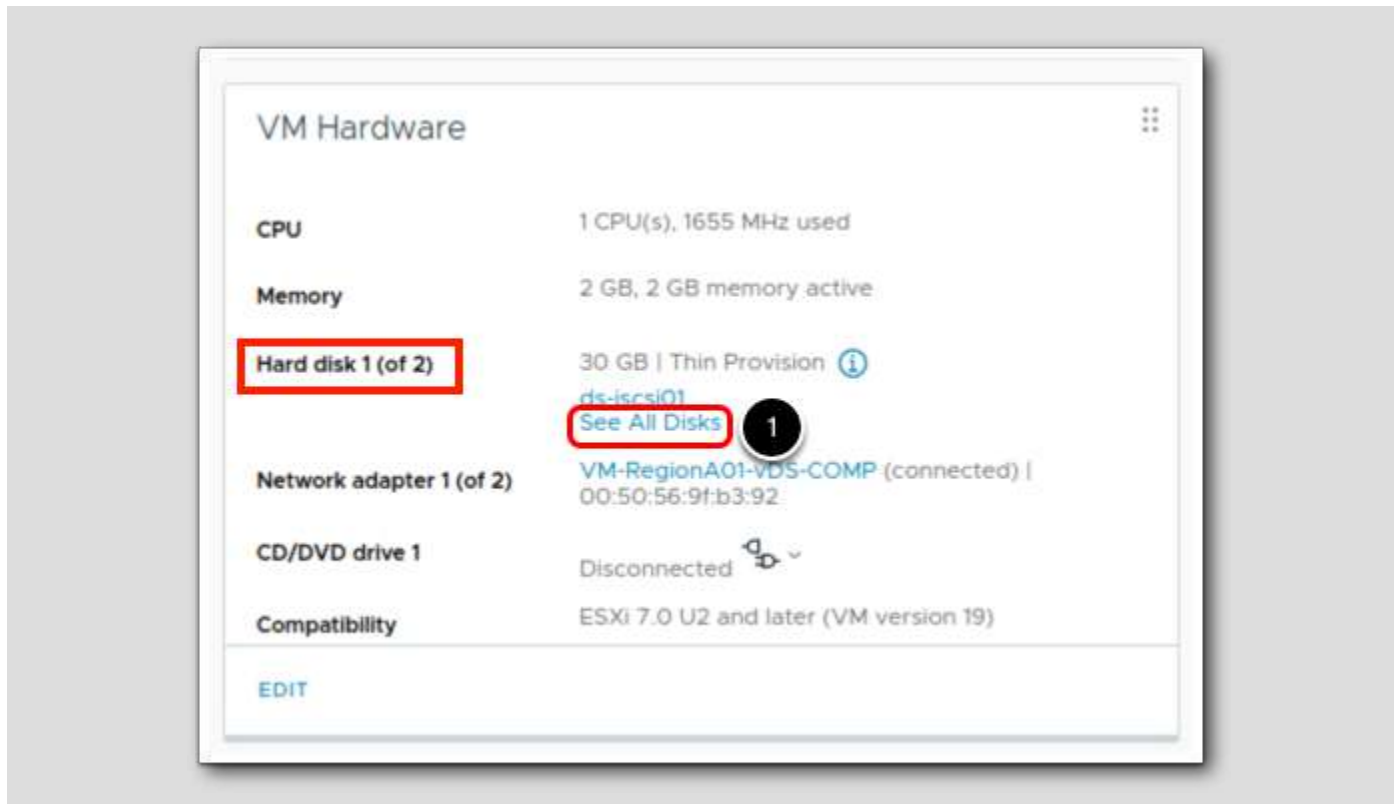
Configure Size and Provisioning Settings



1. Decrease the size to 5 GB.
2. Click OK to create the new virtual disk.

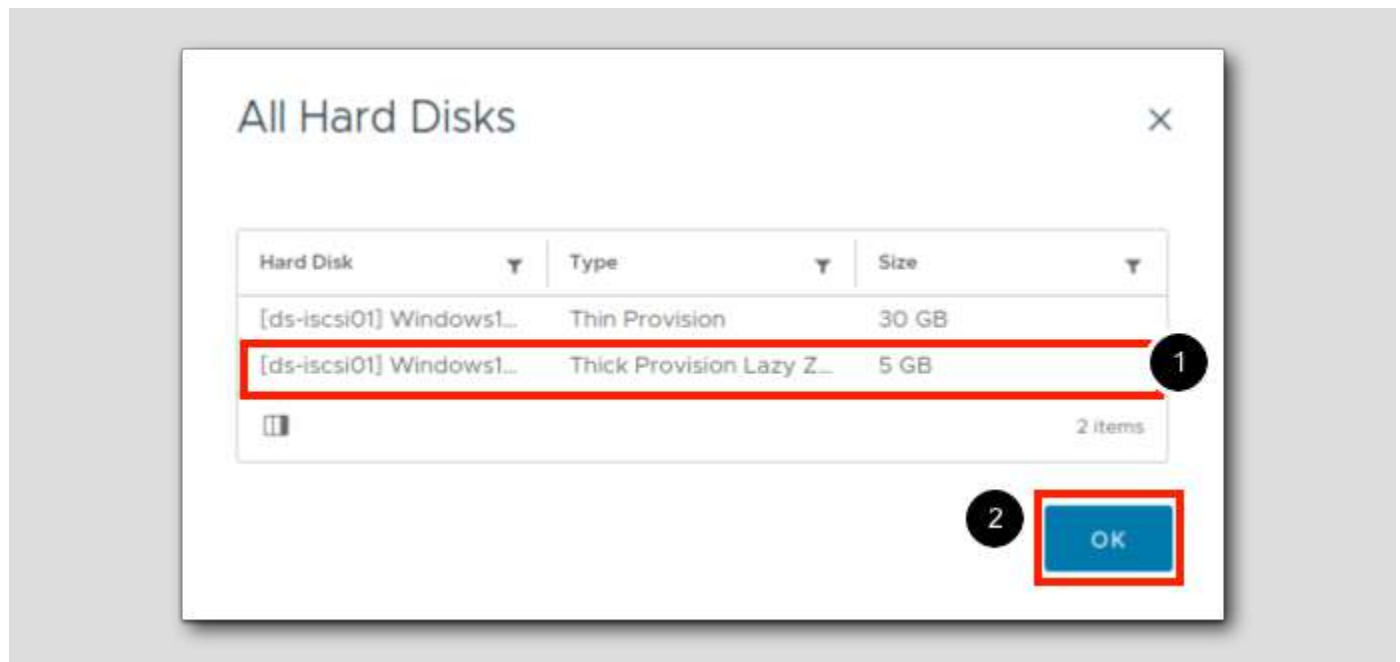
New Hard Disk

[497]



You can follow the progress in the Recent Tasks pane

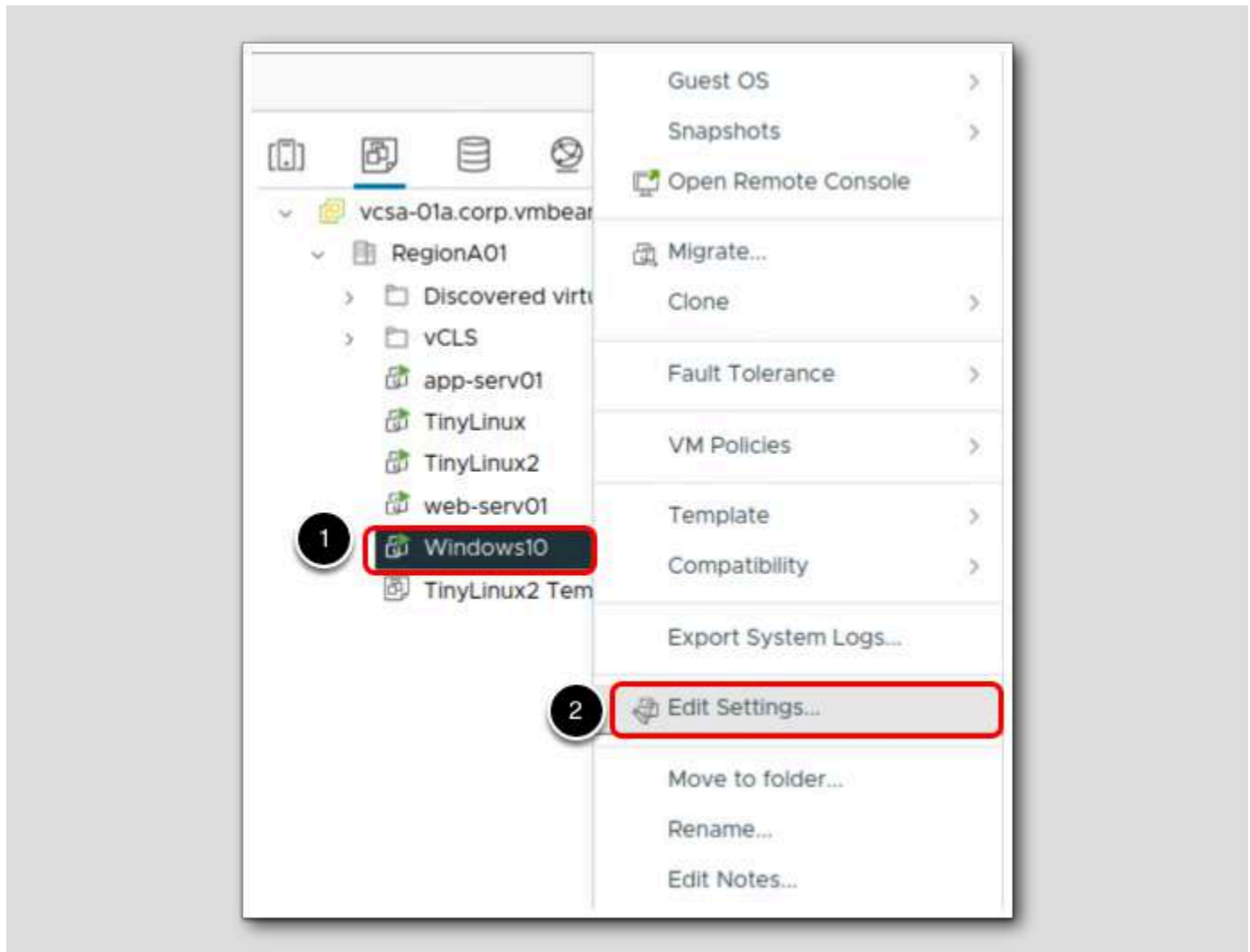
1. You should now see an message that says "Hard disk 1 (of 2)". Click on "See All Disks"



1. Hard disk 2 with a capacity of 5 GB is now available to the Windows10 VM.
2. Click Ok

Extend an Existing Virtual Disk

[498]

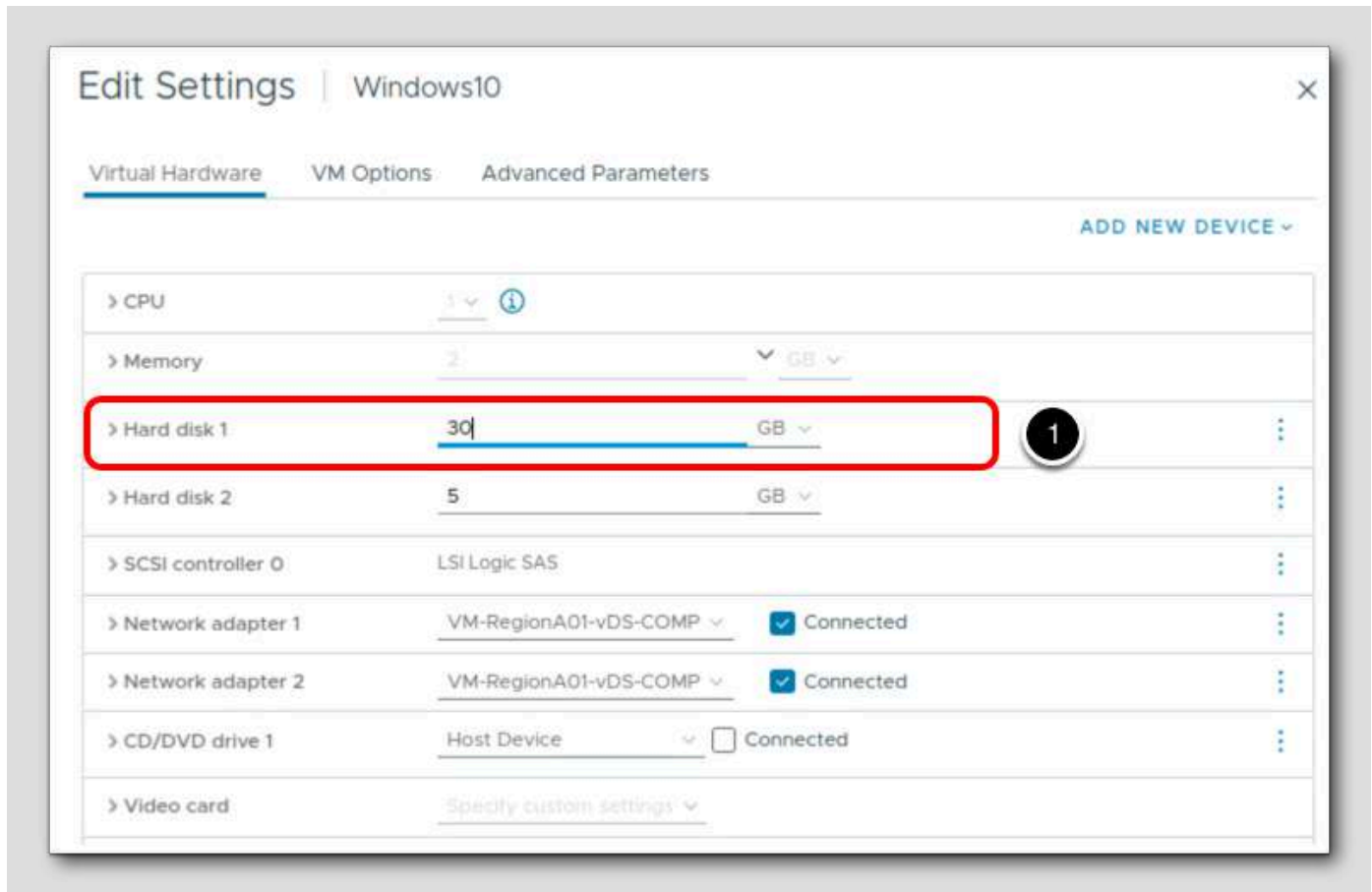


In this section, you will extend an existing Virtual Disk for a Virtual Machine.

1. Right-click the Virtual Machine **Windows10**.
2. Select **Edit Settings**.

Hard disk 1 Settings

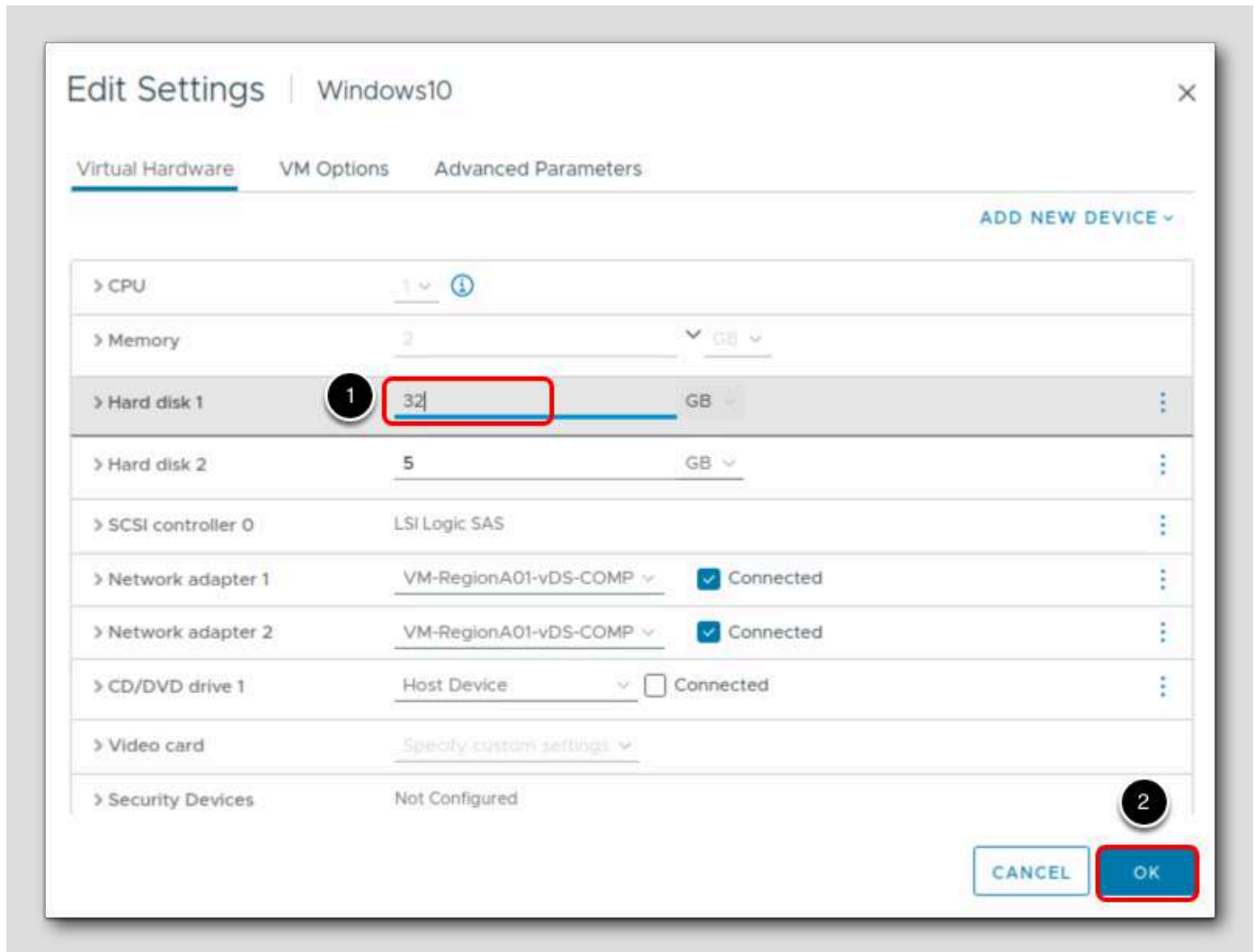
[499]



1. In the Edit Settings wizard, note the capacity for Hard disk 1 is 30 GB.

Extend Hard disk 1

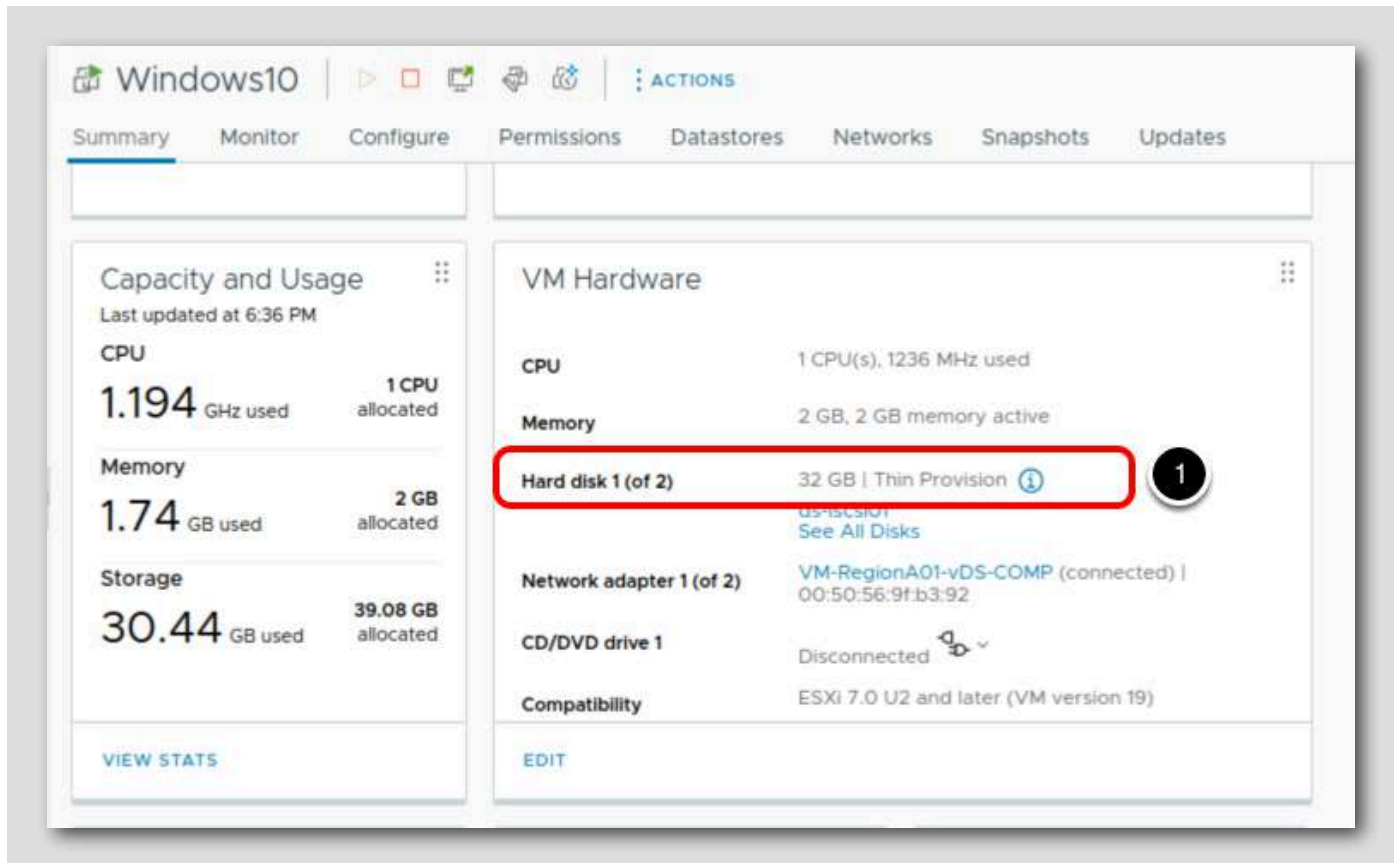
[500]



1. Type 32 Hard disk 1 capacity field.
2. Click OK.

Monitor Task Progress

[501]

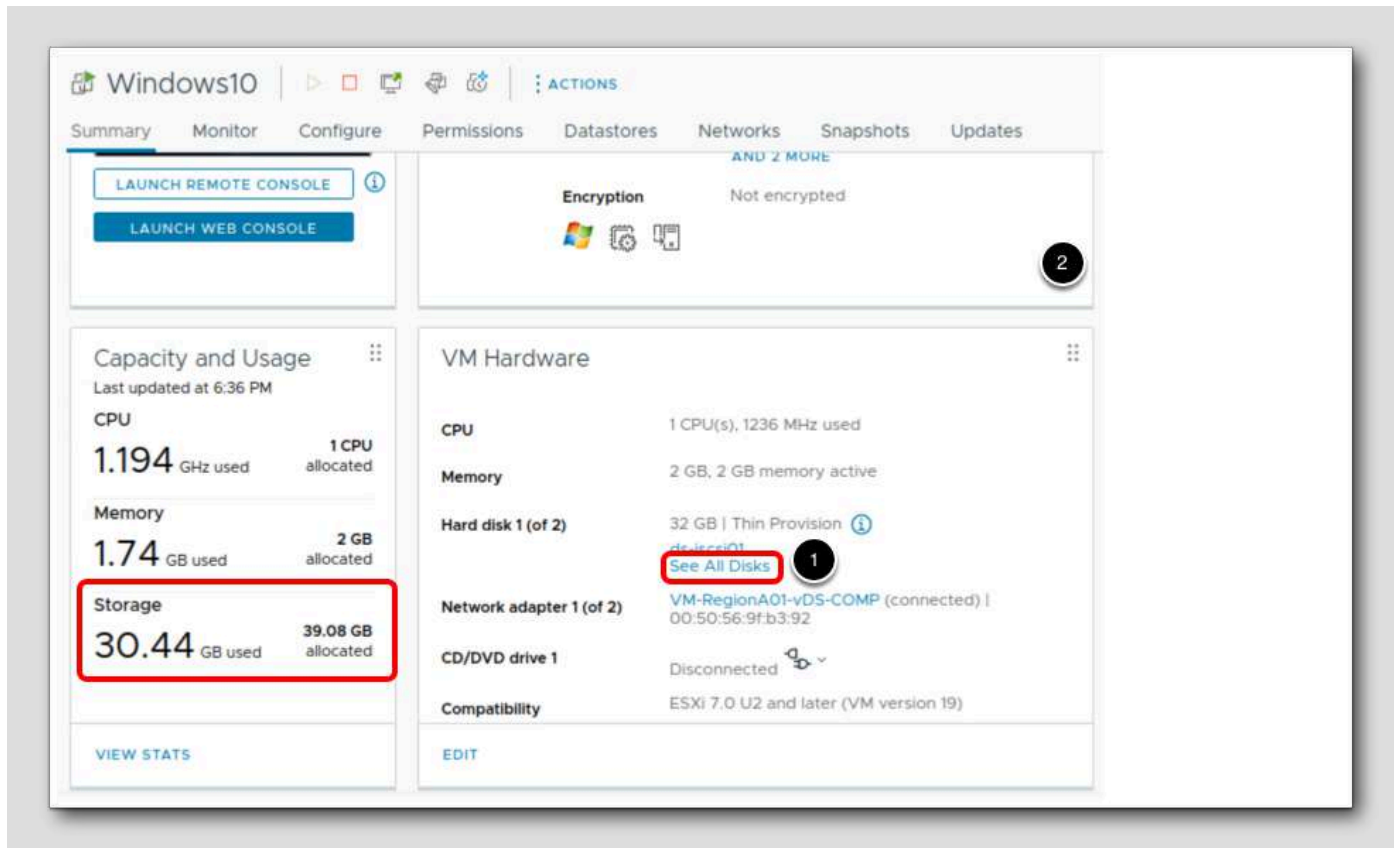


You can follow the progress in the Recent Tasks pane.

1. You should now see **Hard disk 1** with a capacity of 32 GB available to the **windows10** VM.

Review the Virtual Disk Configuration

[502]



1. You can See each of the configured virtual disks and associated capacity within the **See All Disks** option
2. Note that due to Thin Provisioning, the total consumed storage for the virtual disks is less than 32GB!

Working with Virtual Machine Snapshots

[503]

Snapshots preserve the state and data of a virtual machine at the time you take the snapshot. Snapshots are useful when you must revert repeatedly to the same virtual machine state, but you do not want to create multiple virtual machines. You can also take multiple snapshots of a virtual machine to create restoration positions in a linear process. With multiple snapshots, you can save many positions to accommodate many kinds of work processes. The Snapshot Manager in the vSphere Web Client provides several operations for creating and managing virtual machine snapshots and snapshot trees. These operations let you create snapshots, restore any snapshot in the snapshot hierarchy, delete snapshots, and more.

A Virtual Machine snapshot preserves the following information:

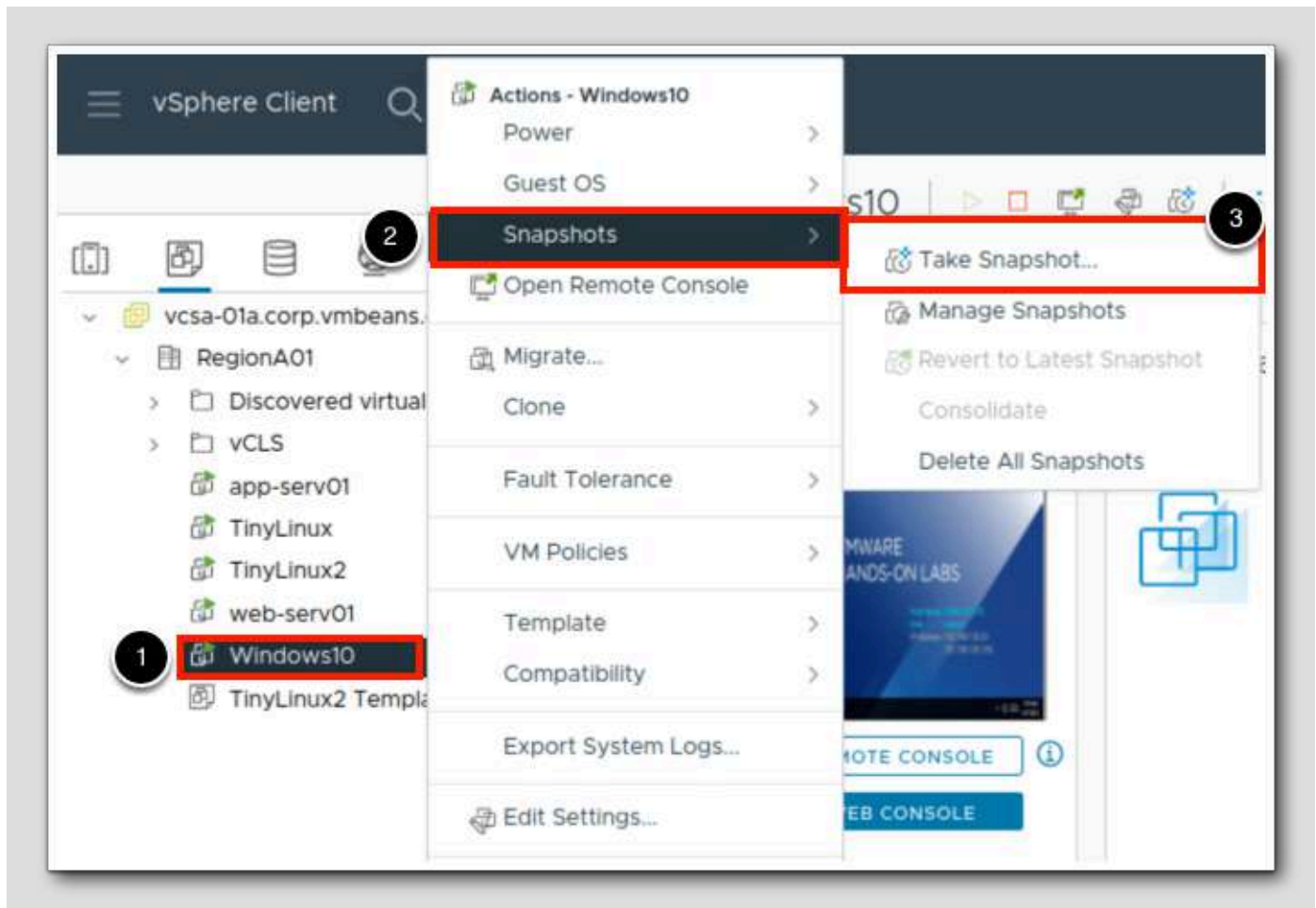
- **Virtual machine settings** - The virtual machine directory, which includes disks that were added or changed after you took the snapshot.

- **Power state** - The virtual machine can be powered on, powered off, or suspended.
- **Disk state** - State of all the virtual machine's virtual disks.
- **Memory state** (optional) - The contents of the virtual machine's memory.

In this section, you will create a Virtual Machine snapshot, make changes to the Virtual Machine's hardware and configuration state, and then revert back to the original state of the Virtual Machine by leveraging the vSphere Web Client Snapshot Manager.

Take a Virtual Machine Snapshot

[504]



In this step, you'll take a Snapshot of a Virtual Machine.

1. Right-click **Windows10**.
2. Select **Snapshots**.
3. Click **Take Snapshot**.

Enter a Name and Description for the VM Snapshot

[505]

Take snapshot

Name **1** Snapshot #1

Description **2** Snapshot taken prior to VM settings change

☐ Include virtual machine's memory **3**

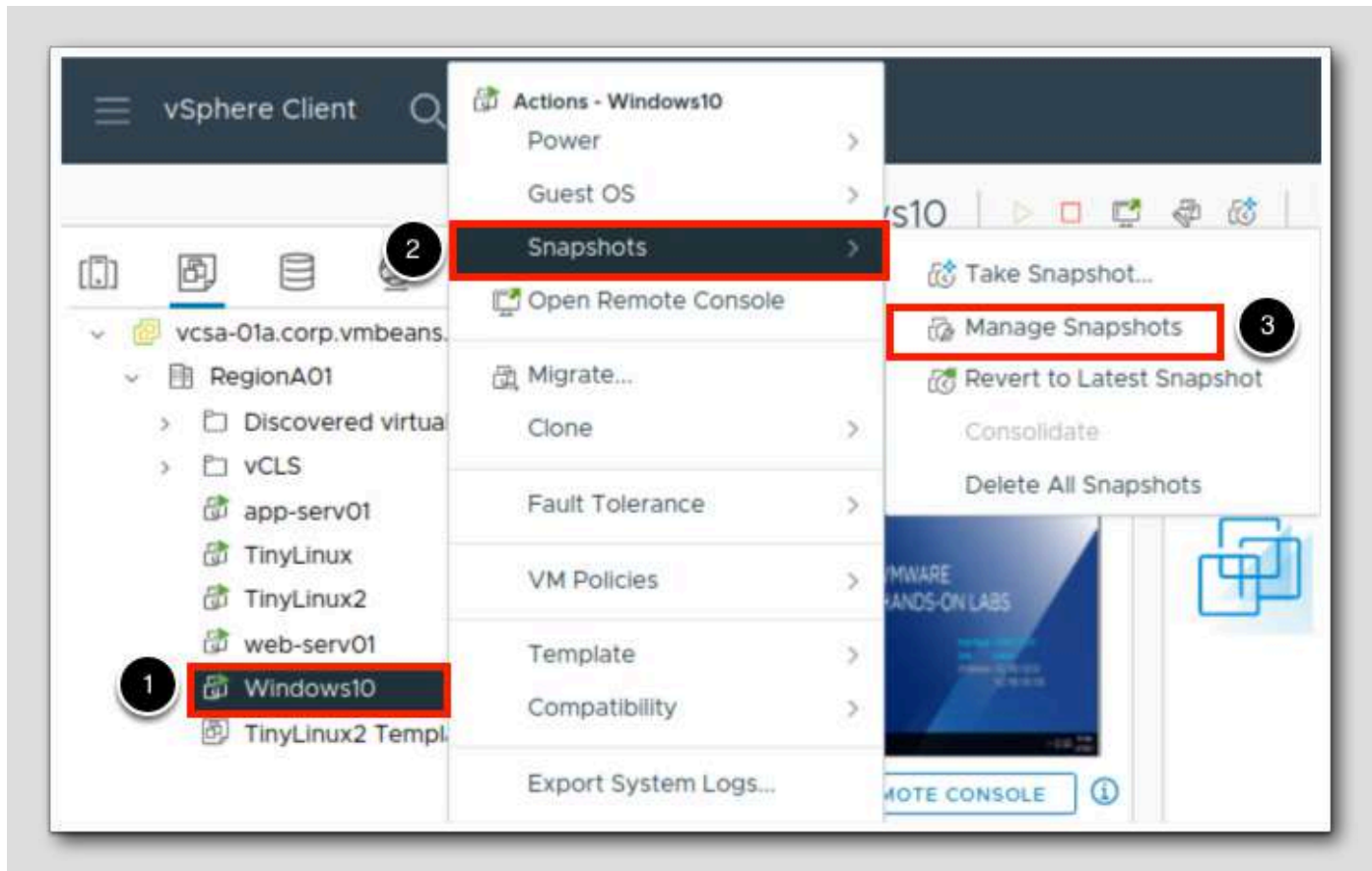
☐ Quiesce guest file system (requires VM tools)

4 CANCEL CREATE

1. In the Take Snapshot window, provide a name for the Snapshot point - **Snapshot #1**
2. Provide a description for the Snapshot point - **Snapshot taken prior to VM settings change**
3. Uncheck the **Snapshot the virtual machine's memory** box.
4. Click **CREATE**

Note: When you take a snapshot of a powered-on virtual machine, you are given the option to capture the running VMs memory state. In our case, since we are in a lab environment, this will generate unneeded I/O.

Open the Snapshots Tab

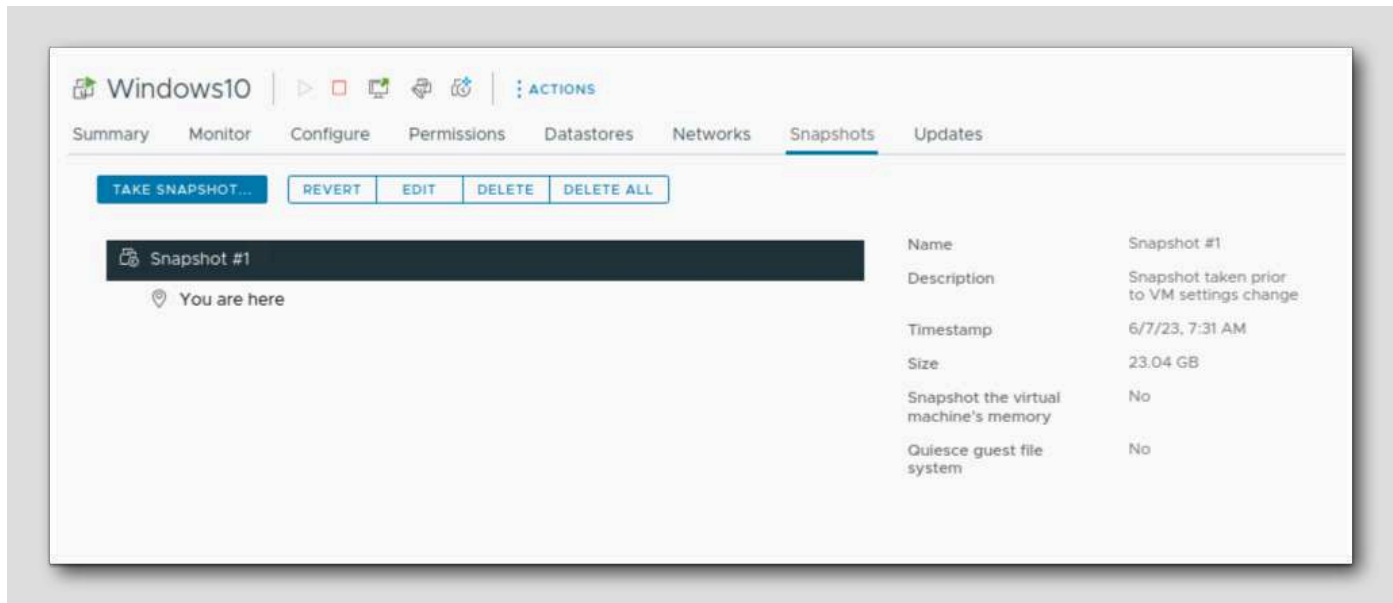


Note the progress in the Recent Tasks pane. Once the snapshot task is complete:

1. Right-click **Windows10**.
2. Select **Snapshots**.
3. Click **Manage Snapshots**.

Snapshot Details

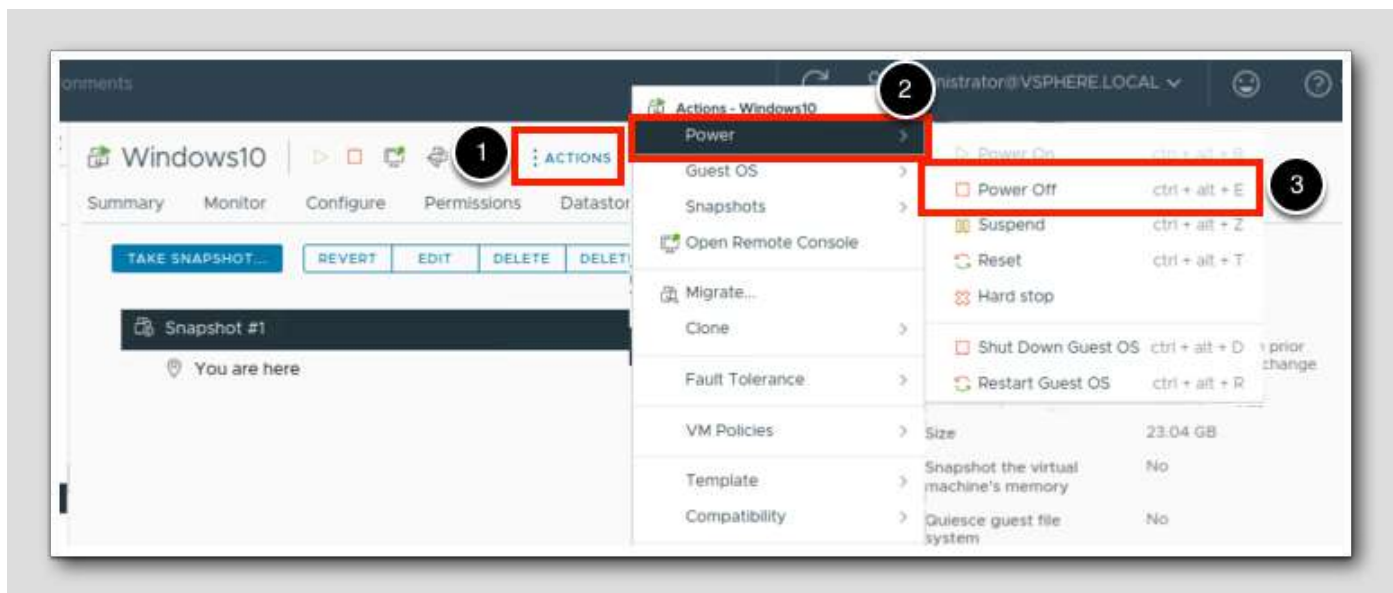
[507]



Here you can view the details of the snapshot and verify it was taken.

Change the Virtual Machine Settings

[508]

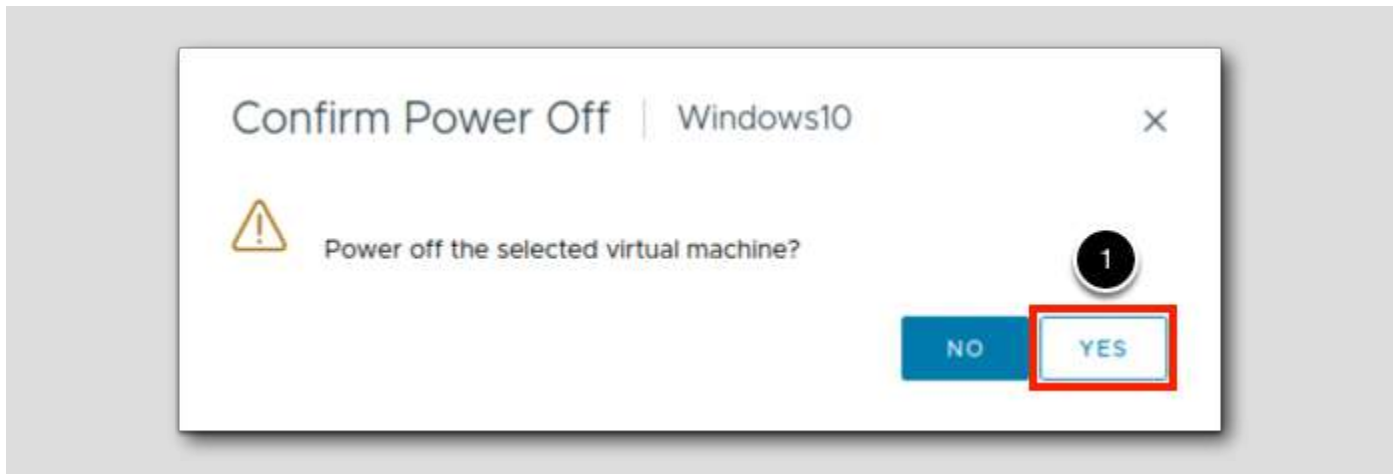


In this section, you will change the memory configuration for the Virtual Machine.

To change the memory configuration for **Windows10**, we will need to shut it down.

1. Click the **Actions** menu.
2. Select **Power**.
3. Click on **Power Off**.

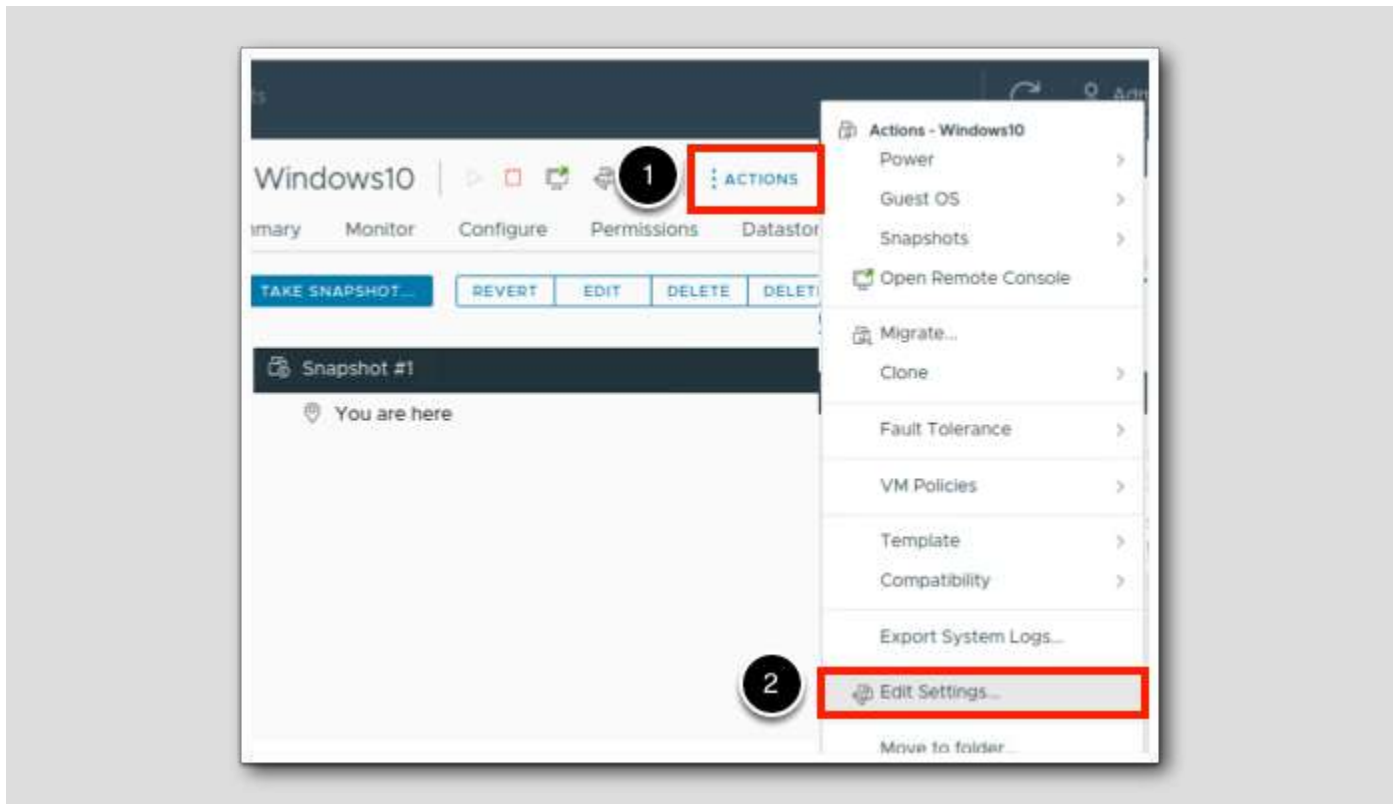
NOTE: This is not the proper way to shut the VM down gracefully, but for our lab environment, it provides a quick way to power off a machine.



1. Click the **Yes** button to power off the virtual machine.

Launch the Edit Settings Wizard

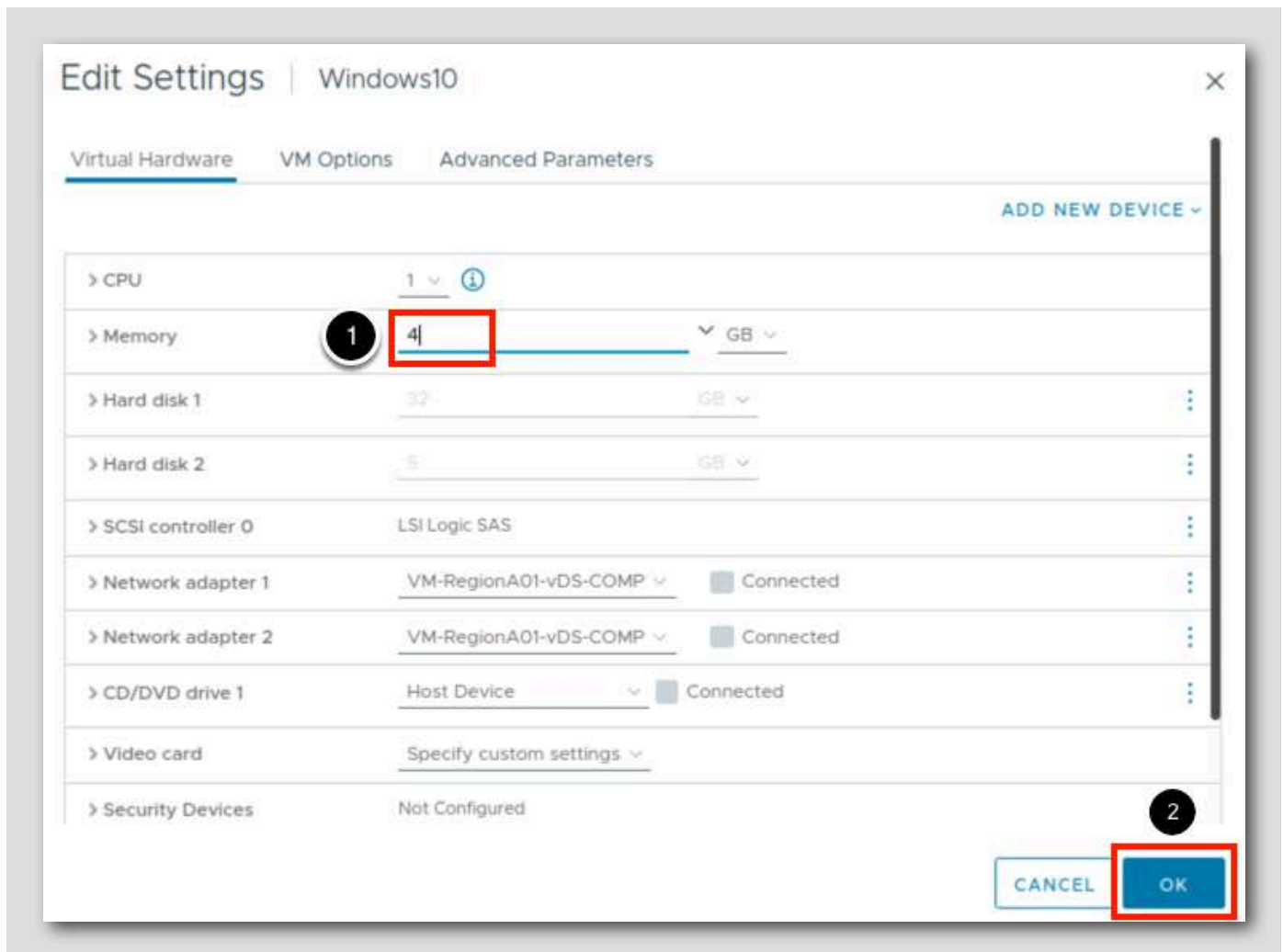
[509]



1. Click the "Actions" drop-down menu.
2. Select "Edit Settings..."

Change the Virtual Machine Settings

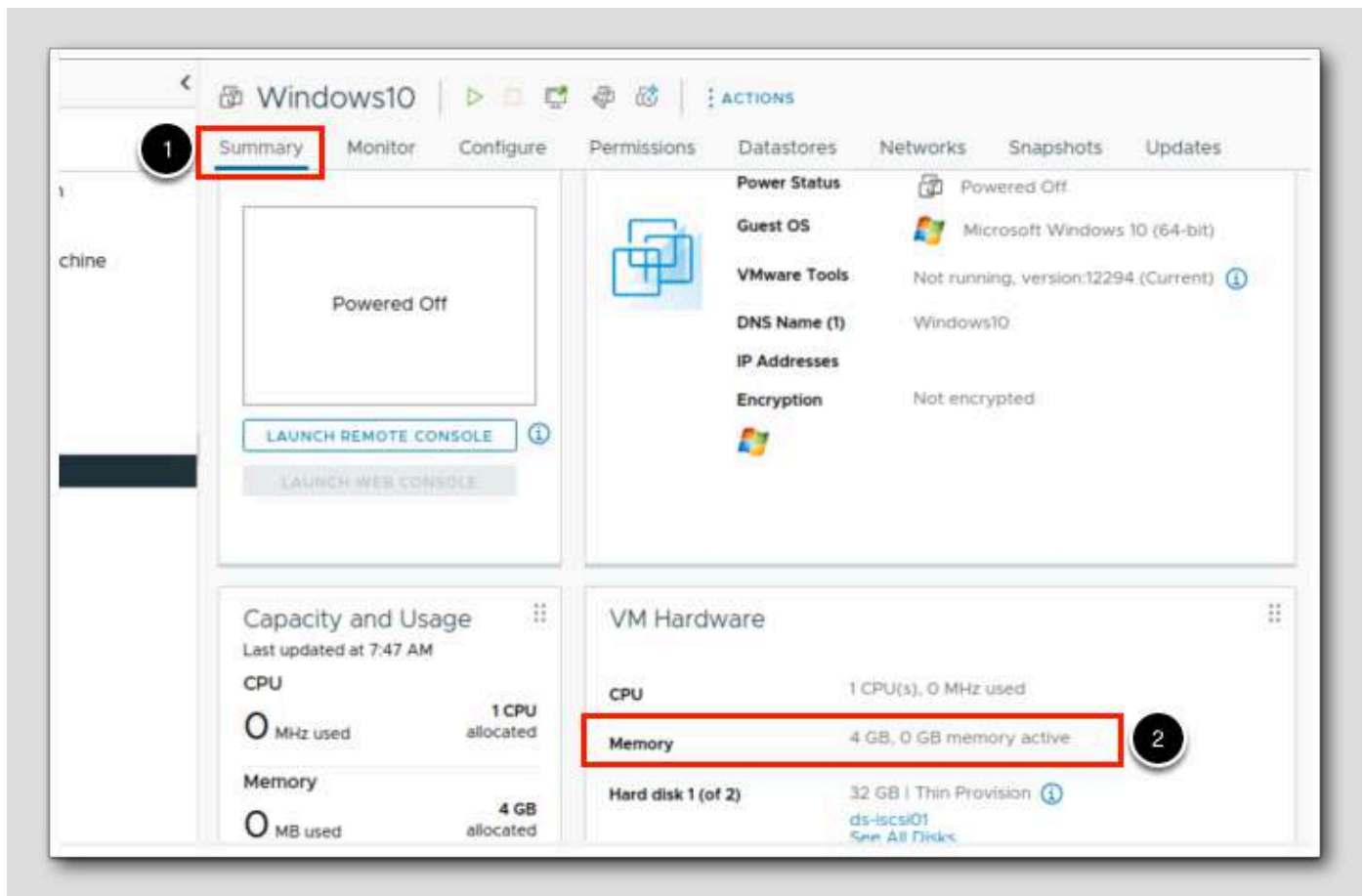
[510]



1. In the Memory field, change this setting to "4".
2. Click Ok to continue

Summary Tab

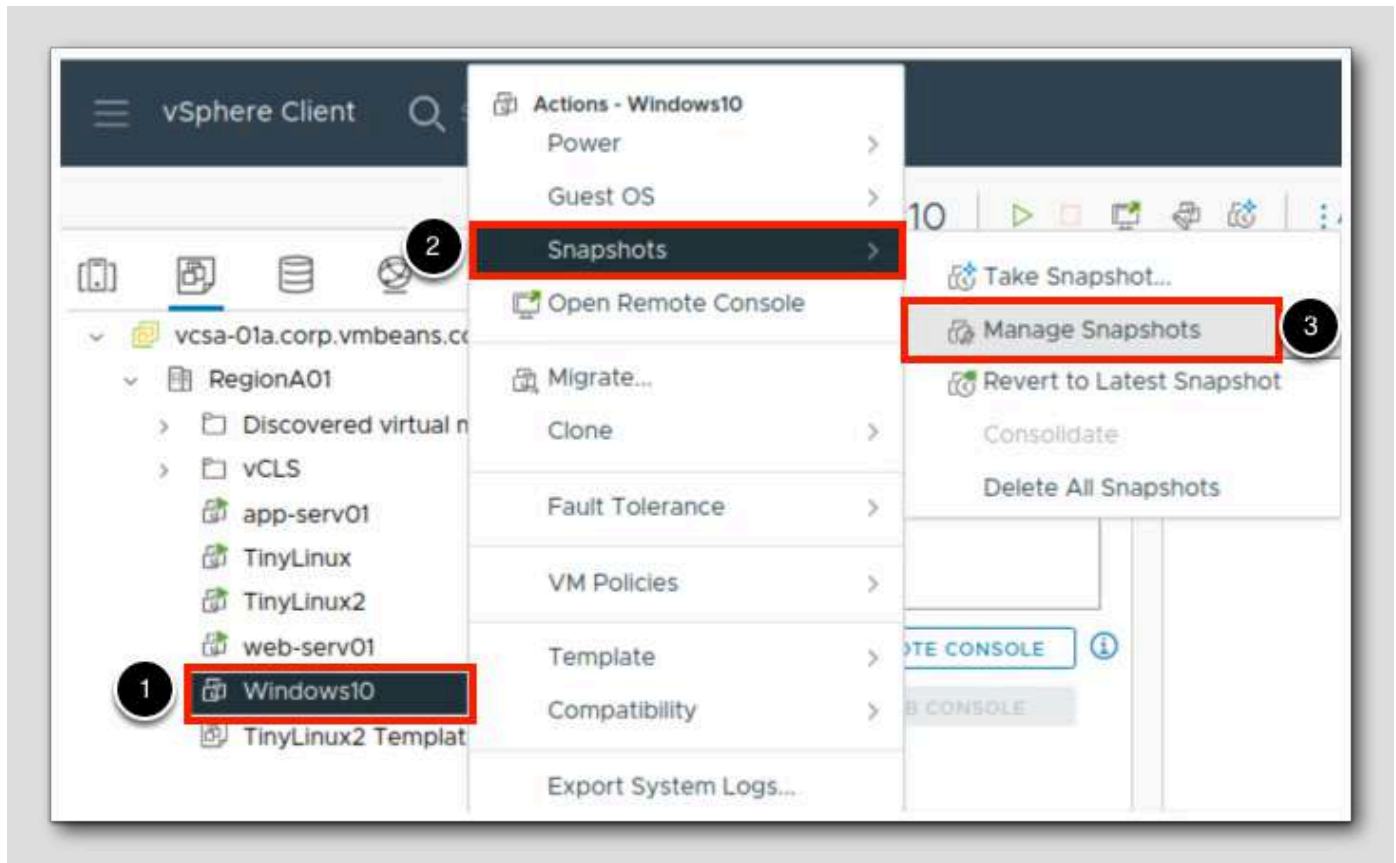
[511]



1. Make sure you are on the Summary tab for Windows10.
2. Verify the memory has been updated.

Revert Virtual Machine settings using the Snapshot Manager

[512]

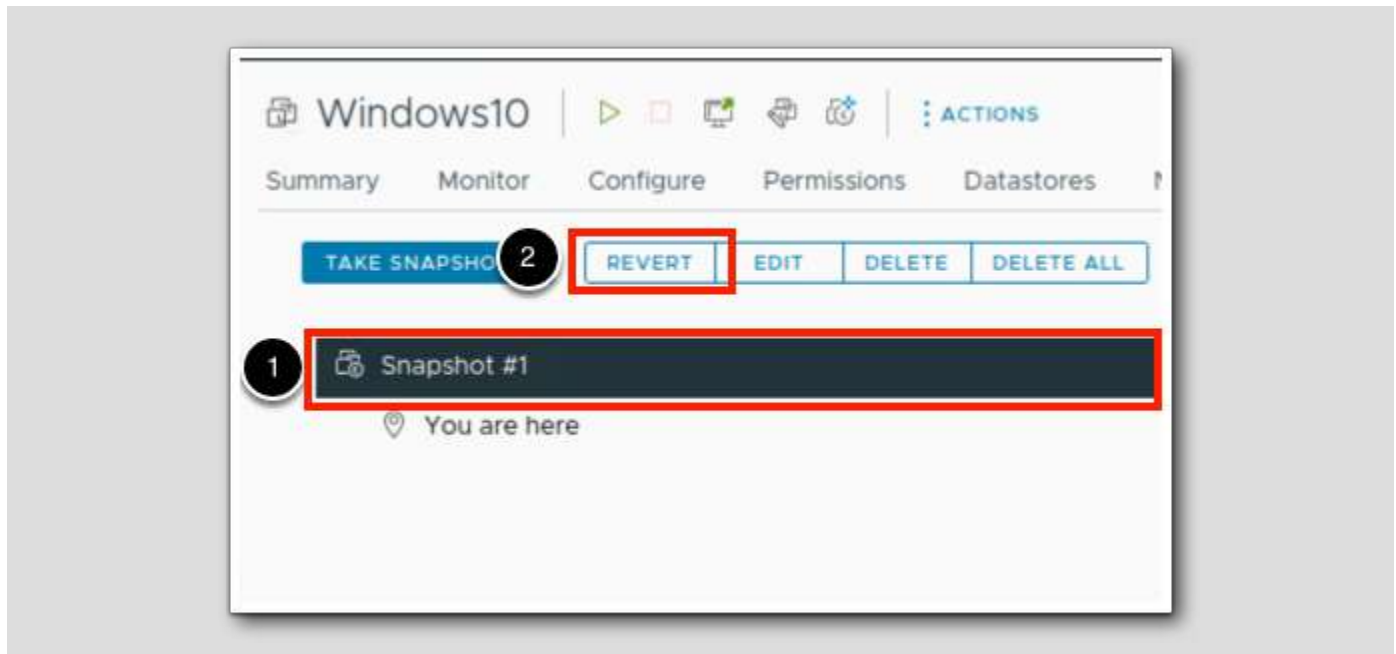


In this section, you revert the Virtual Machine's configuration back to the original state using the Snapshot Manager.

1. Right-click **Windows10**.
2. Select **Snapshots**.
3. Click **Manage Snapshots**.

Select the VM Snapshot to Revert to

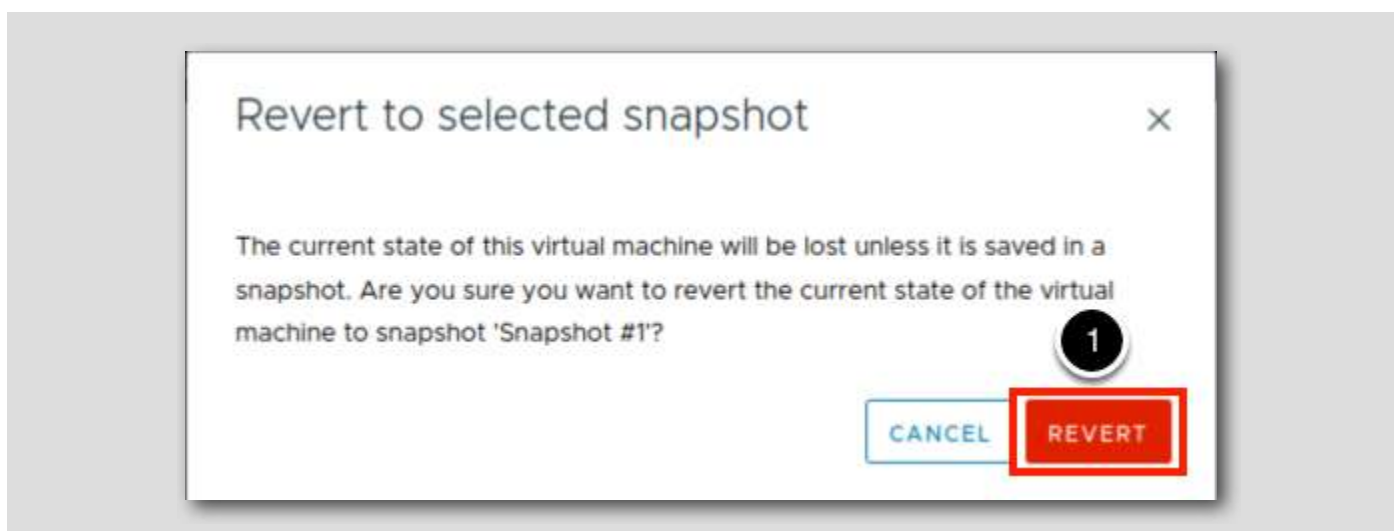
[513]



1. Make sure Snapshot #1 is selected.
2. Click the Revert button.

Confirm Revert to Snapshot

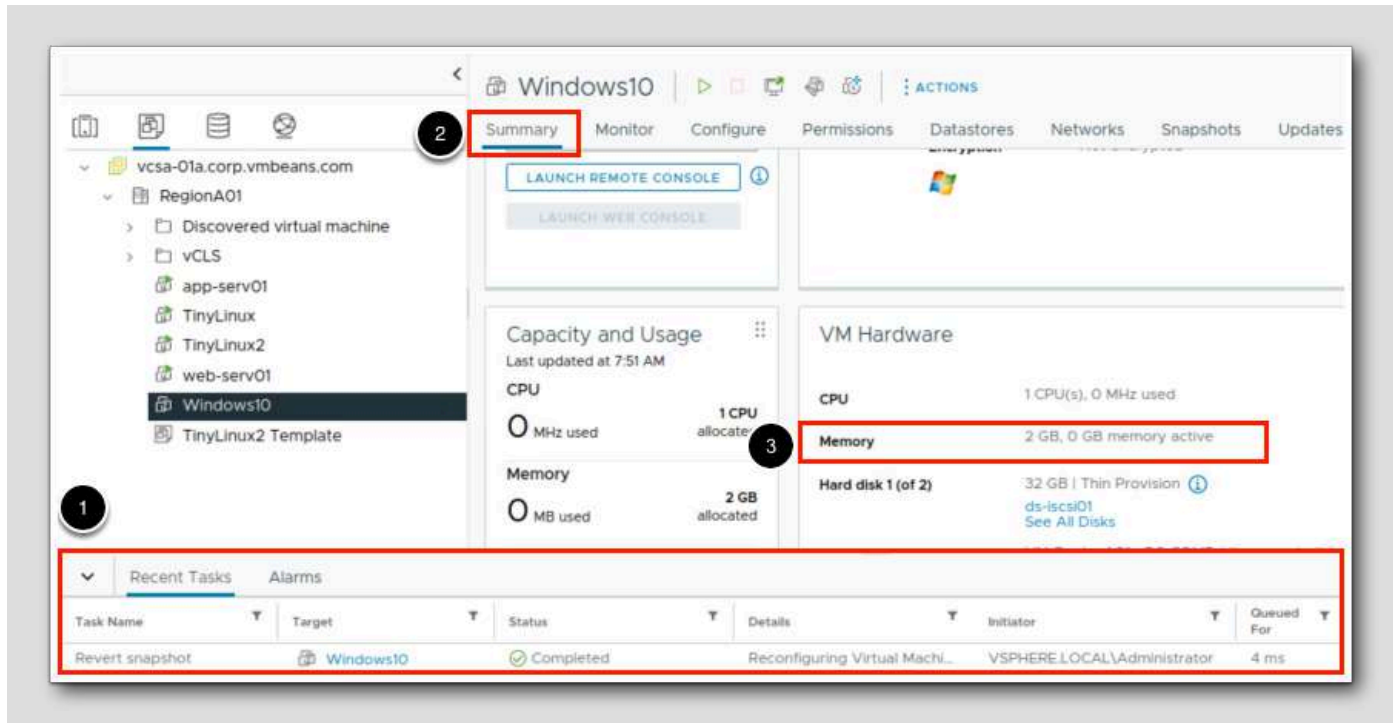
[514]



1. Click **Revert** to confirm action.

Monitor Task Progress

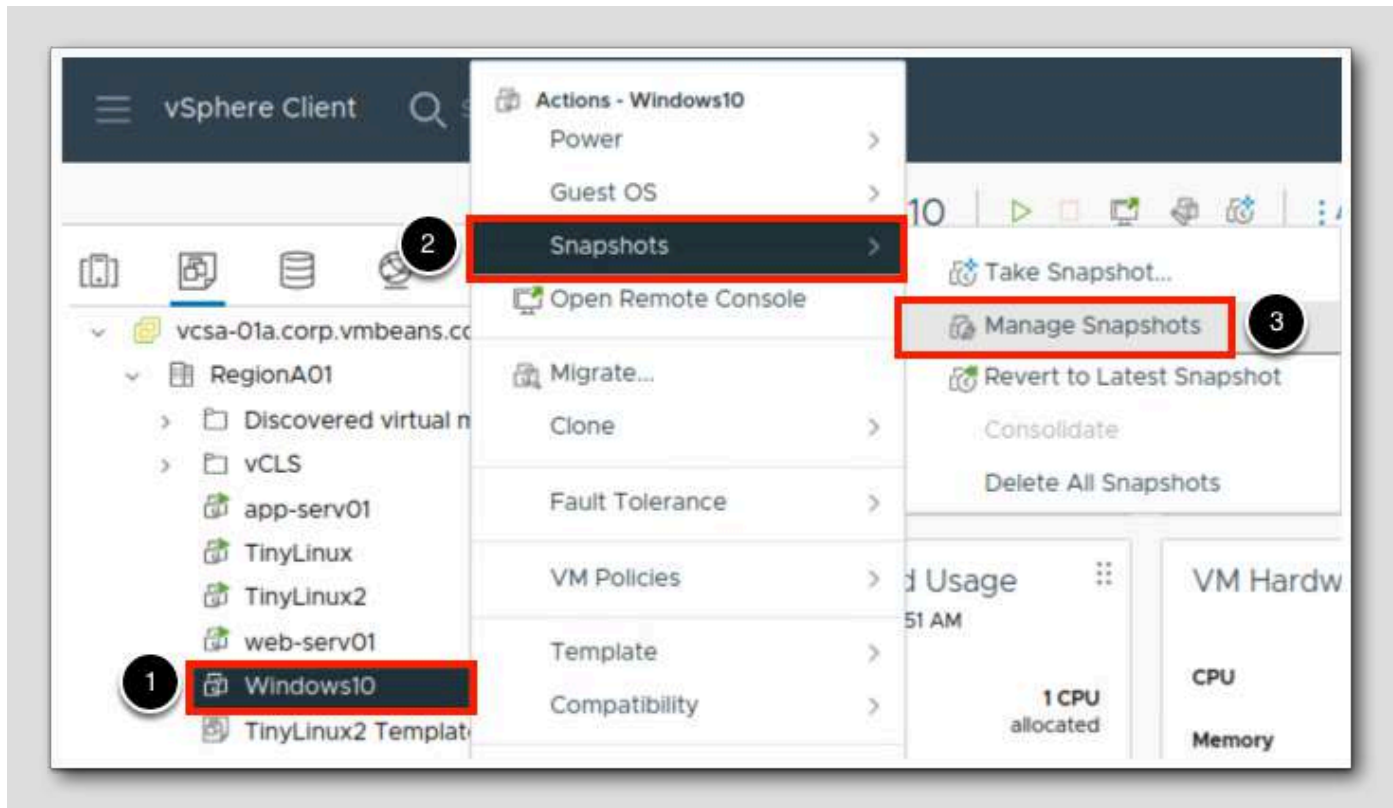
[515]



1. Note the progress in the **Recent Tasks** pane.
2. Click on the **Summary** Tab
3. Note the **Memory** configuration has reverted back to 2 GB.

Delete Snapshot #1

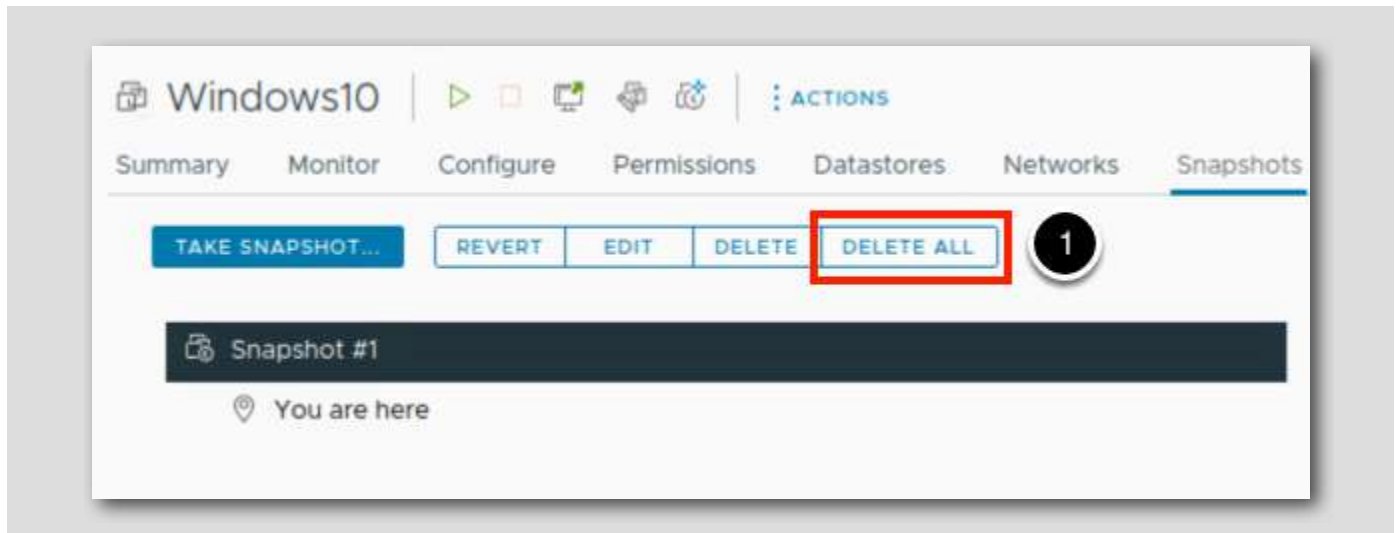
[516]



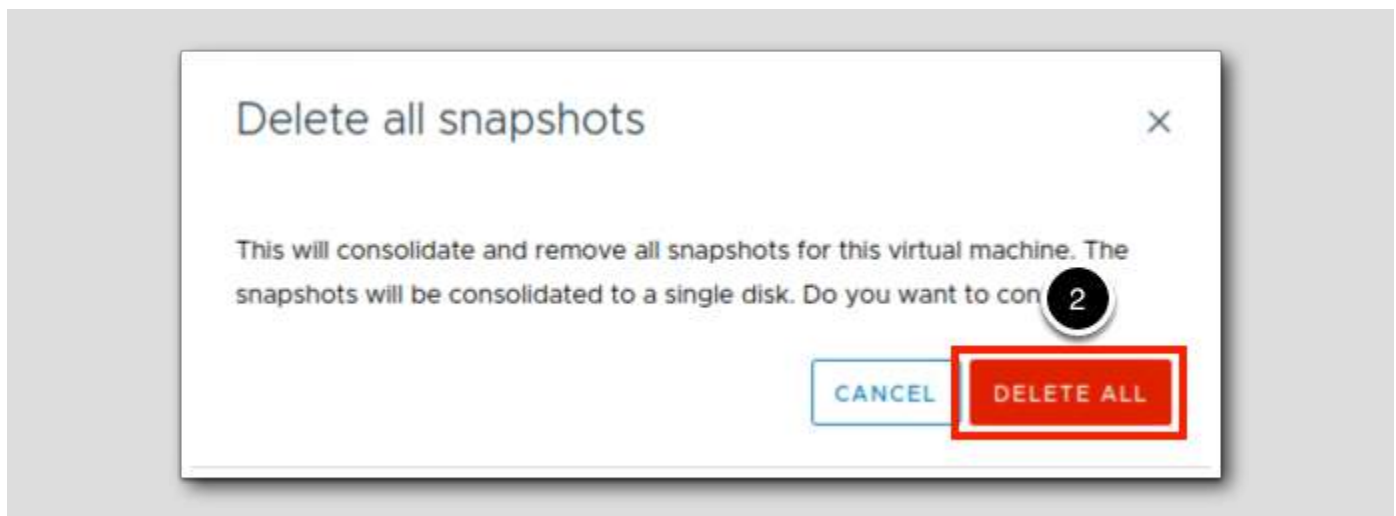
Here you can go and delete the taken snapshot.

1. Right-click **Windows10**.
2. Select **Snapshots**.
3. Click **Manage Snapshots**.

Select the VM Snapshot to Delete All



1. Click the **Delete All** button to remove the snapshot.

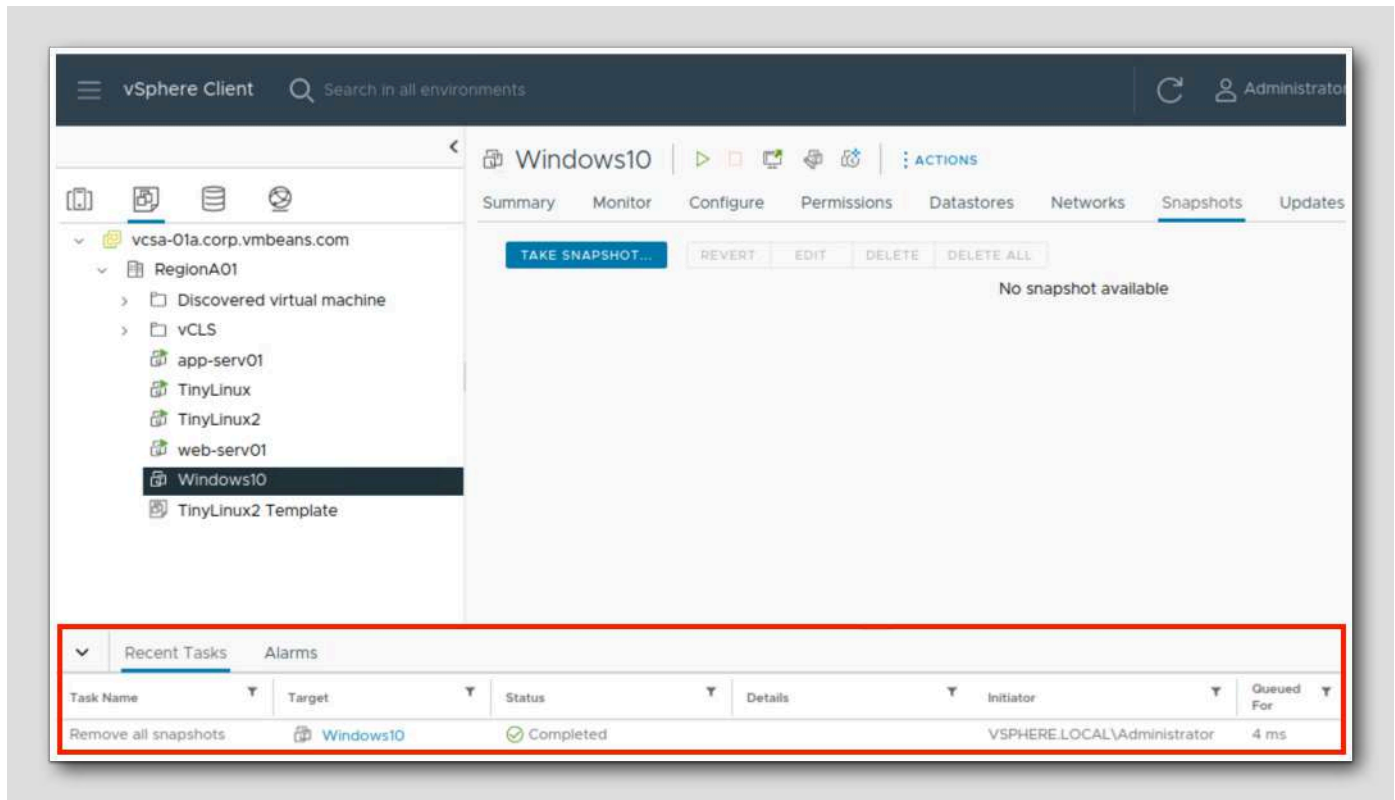


2. Click **Delete All** to confirm the deletion of all the snapshots.

It is a best practice to delete virtual machine snapshots when they are no longer needed. Over time the snapshot delta can grow to be quite large which could result in issues consolidating the virtual machine files and lead to performance issues.

Snapshot Removed

[518]

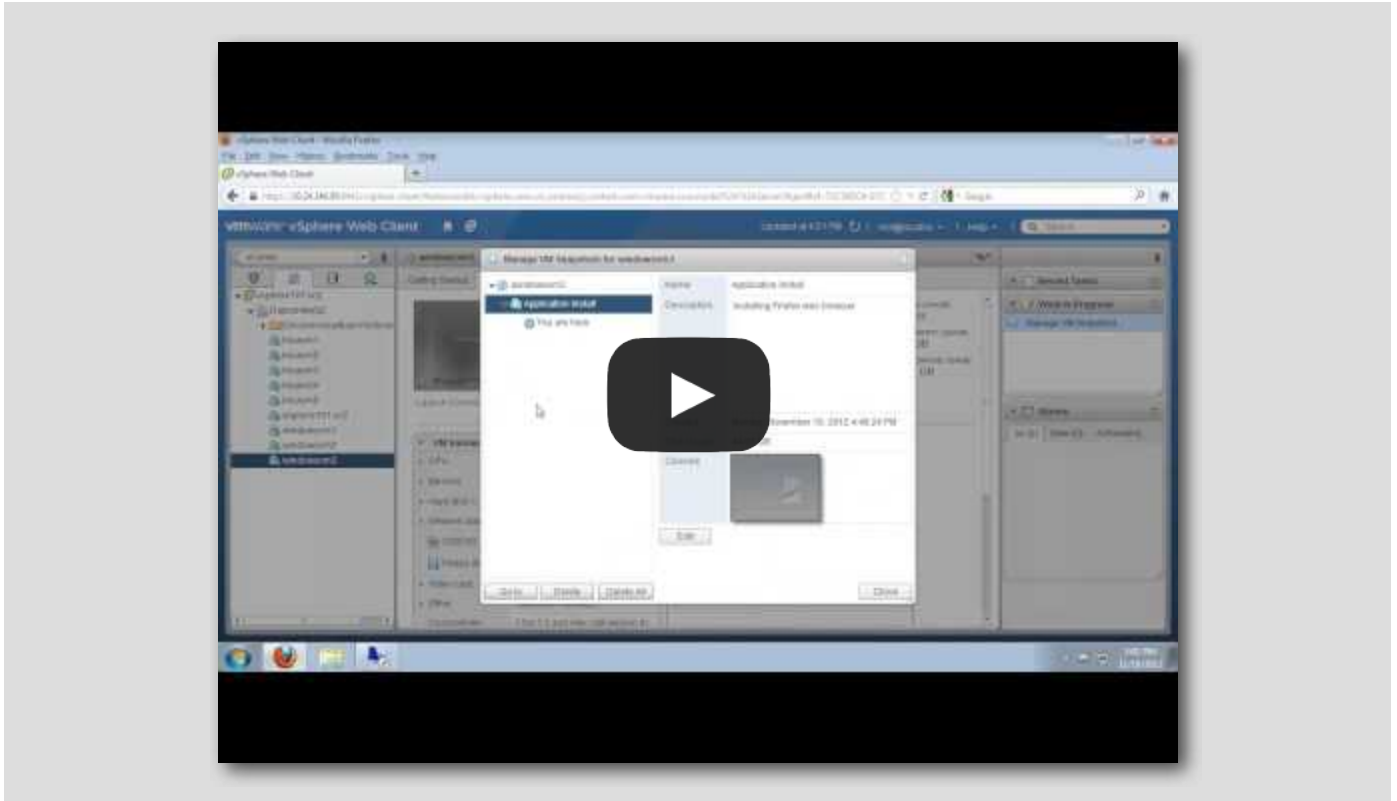


You can watch the progress of the snapshot being deleted in the Recent Tasks window.

Video: More on Virtual Machine Snapshots (2:33)

[519]

<https://www.youtube.com/watch?v=7AVIWifTEMM>



For more information on vSphere Virtual Machine Snapshots, be sure to check out this video.

vSphere Datastore Cluster

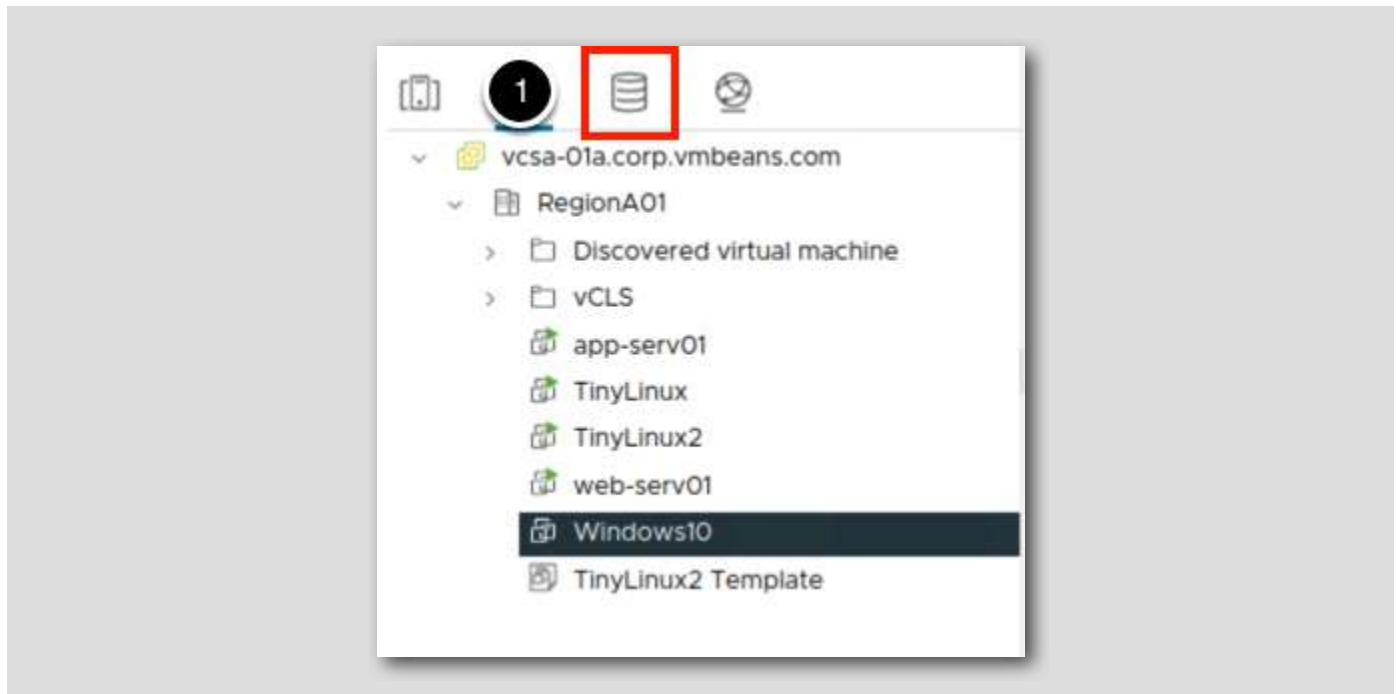
[520]

A vSphere Datastore Cluster balances I/O and storage capacity across a group of vSphere datastores. Depending on the level of automation desired, Storage Dynamic Resource Scheduler will place and migrate virtual machines in order to balance out datastore utilization across the Datastore Cluster.

In this section, you will create a vSphere Datastore Cluster using two iSCSI datastores.

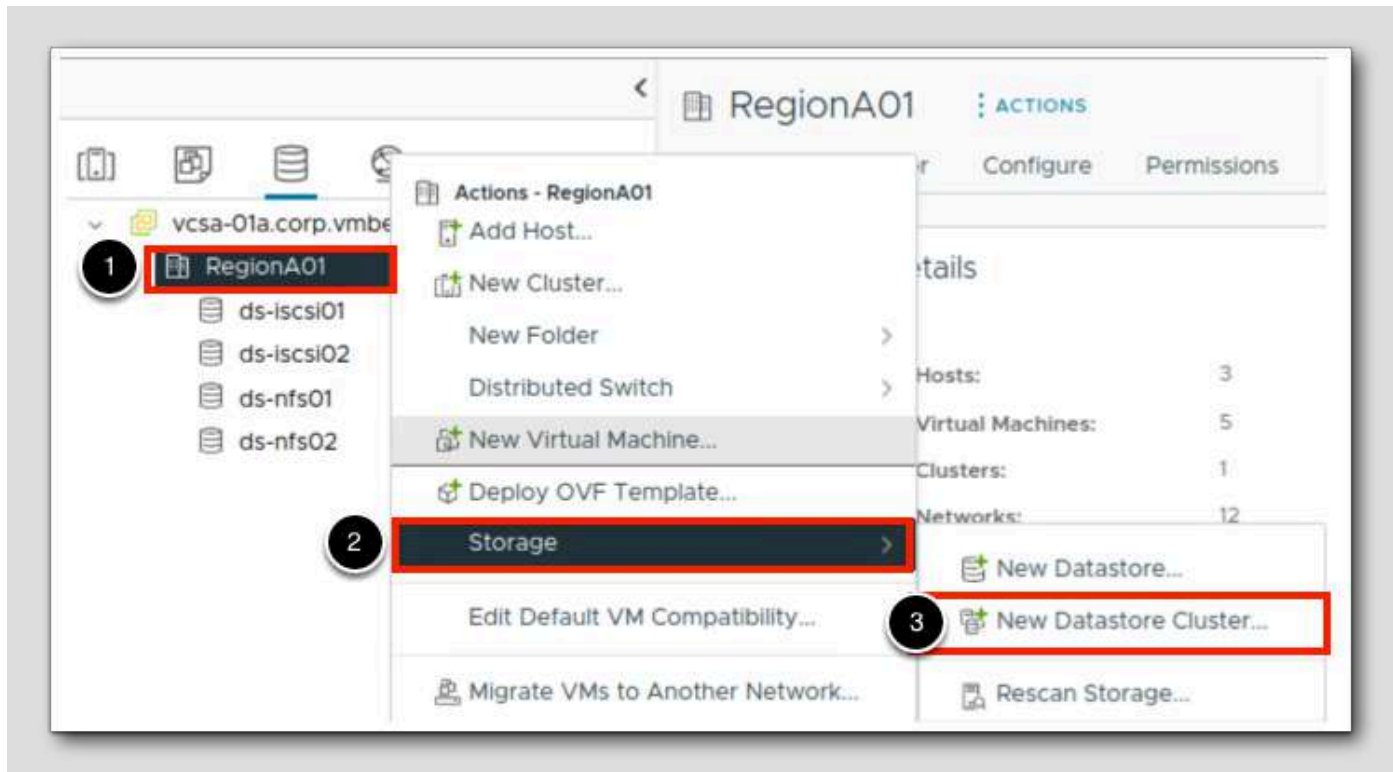
Navigate to Storage

[521]



1. Click on the **Storage** icon

New Datastore Cluster



1. Right Click on RegionA01
2. Select Storage
3. Click New Datastore Cluster...

New Datastore Cluster - Name and Location

[523]

Name and Location

Datastore cluster name: DatastoreCluster-01 1

Location: RegionA01

Datastore type: NFS 3 2

☒ Turn ON Storage DRS

vSphere Storage DRS enables vCenter Server to manage datastores as an aggregate pool of storage resources.

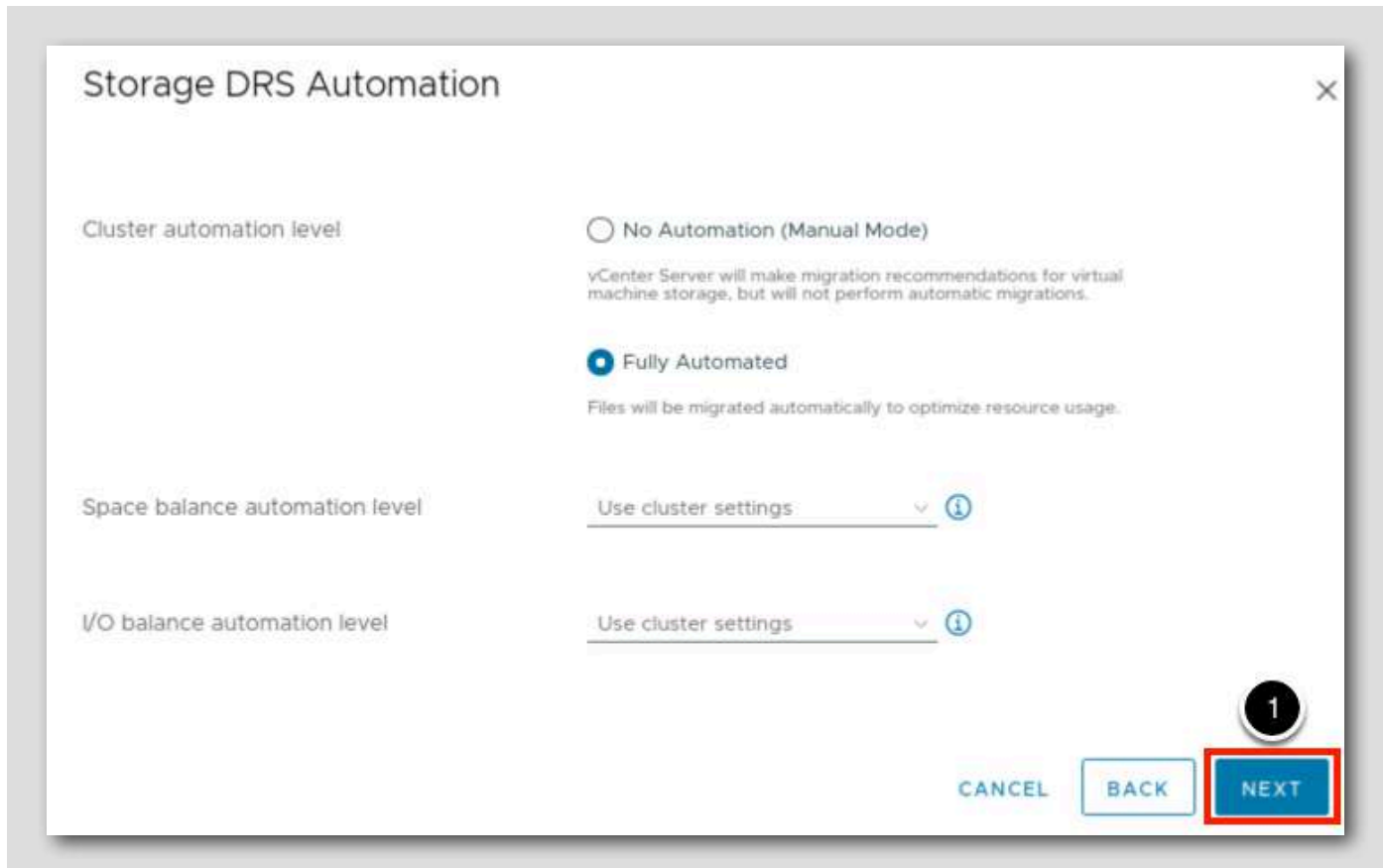
vSphere Storage DRS also enables vCenter Server to manage the assignment of virtual machines to datastores, suggesting placement when virtual machines are created, migrated or cloned, and migrating running virtual machines to balance load and enforce placement rules.

CANCEL NEXT 3

1. Enter DatastoreCluster-01 for the name
2. Select NFS 3 as the Datastore Type
3. Select Next

New Datastore Cluster - Storage DRS Automation

[524]



The image shows a 'Storage DRS Automation' configuration window. It has a title bar with a close button (X) in the top right corner. The window is divided into three sections for configuration:

- Cluster automation level:** This section has two radio button options. The first is 'No Automation (Manual Mode)' with a description: 'vCenter Server will make migration recommendations for virtual machine storage, but will not perform automatic migrations.' The second option, 'Fully Automated', is selected with a blue dot and has a description: 'Files will be migrated automatically to optimize resource usage.'
- Space balance automation level:** This section has a dropdown menu set to 'Use cluster settings' and an information icon (i) to its right.
- I/O balance automation level:** This section also has a dropdown menu set to 'Use cluster settings' and an information icon (i) to its right.

At the bottom right of the window, there are three buttons: 'CANCEL', 'BACK', and 'NEXT'. The 'NEXT' button is highlighted with a red rectangular border. Above the 'NEXT' button, there is a small black circle containing the number '1', indicating a step in a sequence.

1. Leave the defaults settings and select Next

New Datastore Cluster - Storage DRS Runtime Settings

[525]

Storage DRS Runtime Settings [X]

I/O Metric inclusion ☒ Enable I/O metric for SDRS recommendations ⓘ
Select this option if you want I/O metrics considered as a part of any SDRS recommendations or automated migrations in this data store cluster.

I/O latency threshold Dictates the minimum I/O latency for each datastore below which I/O load balancing moves are not considered.
5 ms 100 ms **15** ms

Space threshold Runtime thresholds govern when Storage DRS performs or recommends migrations (based on the selected automation level).
☒ Utilized space
50 % 100 % **80** %
Dictates the minimum level of consumed space for each datastore that is the threshold for act.

1

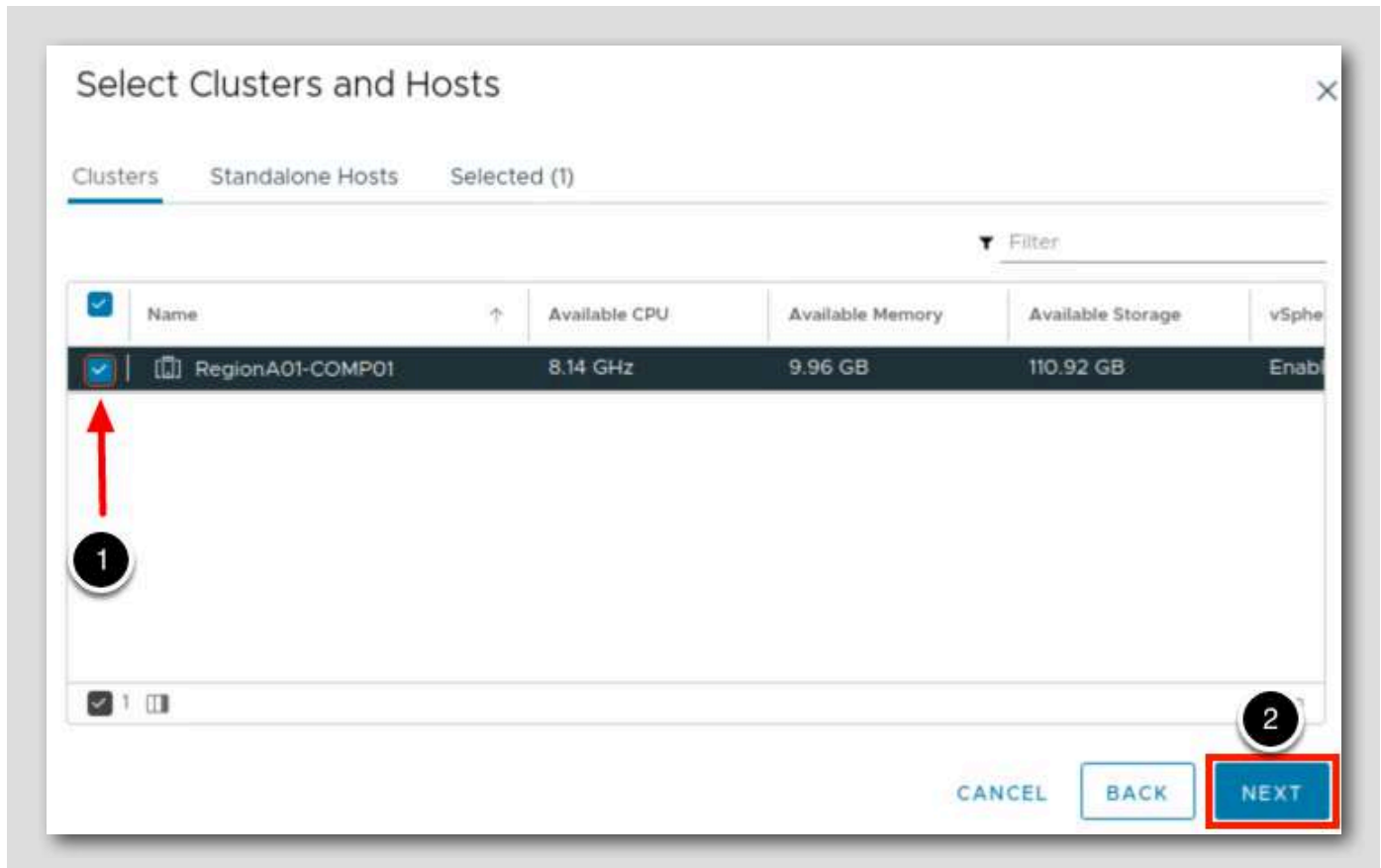
CANCEL BACK **NEXT**

Storage DRS provides multiple options for tuning the sensitivity of storage cluster balancing.

1. Leave the defaults for now and select **Next**

New Datastore Cluster - Select Clusters and Hosts

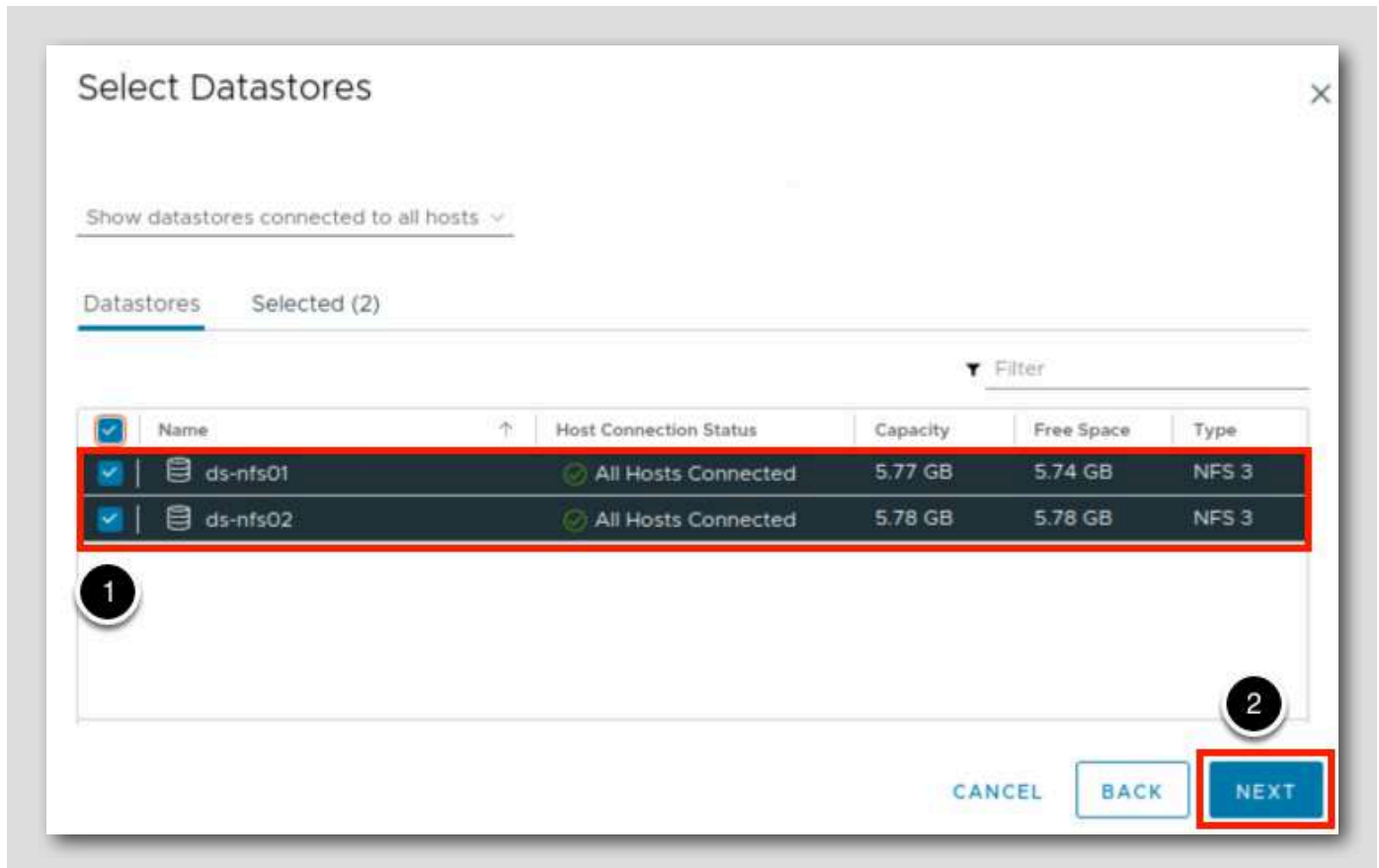
[526]



1. Because there are no standalone hosts, please select **RegionA01-COMP01**
2. Click the **Next** button

New Datastore Cluster - Select Datastores

[527]



1. Select the `ds-iscsi01` and `ds-iscsi02` datastores for the new Datastore Cluster
2. Click **Next**

New Datastore Cluster- Ready to Complete

[528]

Ready to Complete [X]

▼ Name and Location

Datastore cluster name:	DatastoreCluster-01
Datastore type:	NFS 3
Storage DRS:	Enabled

▼ Storage DRS Automation

Cluster automation level:	Fully Automated
Space balance automation level:	Use cluster settings
I/O balance automation level:	Use cluster settings
Rule enforcement automation level:	Use cluster settings

CANCEL BACK **FINISH**

1

1. Review the Storage DRS settings and click the **Finish** button

New Datastore Cluster- Summary

[529]

RegionA01 ACTIONS

Summary Monitor Configure Permissions Hosts & Clusters VMs Datastores Networks Updates

Datacenter Details

Hosts: 3
Virtual Machines: 5
Clusters: 1
Networks: 12
Datastores: 4

Capacity and Usage
Last updated at 7:59 AM

CPU
253 MHz used 12.57 GHz allocated

Memory
7.57 GB used 23.99 GB allocated

Storage
30.13 GB used 141.06 GB allocated

Recent Tasks

Task Name	Target	Status	Details	Initiator	Queued For	Start Time
Move datastores into a d...	DatastoreCluster-01	Completed		VSPHERE.LOCAL\Administrator	2 ms	06/07/2023, 8:05:36 ...
Configure Storage DRS	DatastoreCluster-01	Completed		VSPHERE.LOCAL\Administrator	3 ms	06/07/2023, 8:05:36 ...
Create a datastore cluster	RegionA01	Completed		VSPHERE.LOCAL\Administrator	4 ms	06/07/2023, 8:05:36 ...

1. View the **Recent Tasks** to check the progress of the operation.

Conclusion

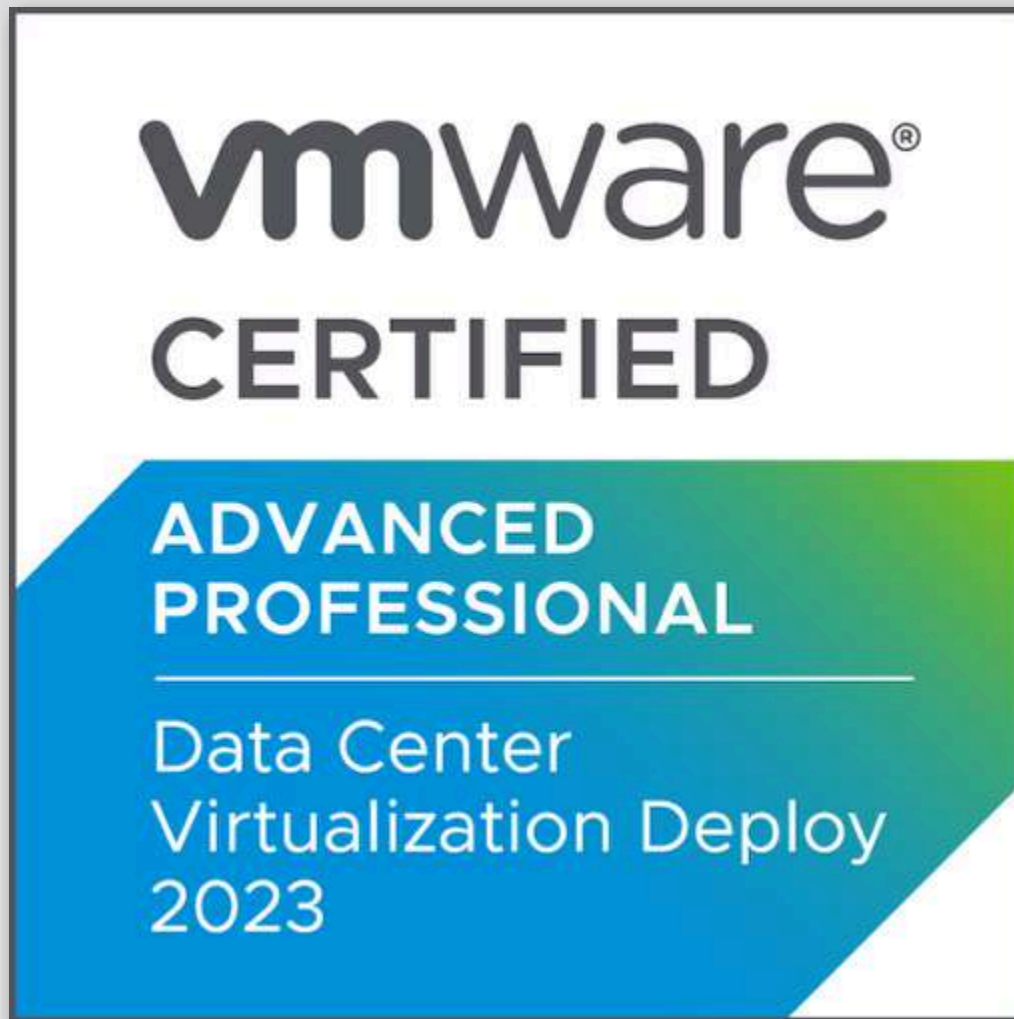
[530]

Leveraging vSphere Datastore Clusters in your vSphere environment can help to ensure datastores are filled evenly and I/O is spread out across the group of datastores in the cluster. Storage DRS can automate the initial placement of new virtual machines and adjust virtual machine placement to maintain an even distribution of I/O across the datastore cluster.

Certification Path

[531]

Learn and practice with Hands-On Labs to help prepare for several VMware Certifications.



This lab can help you study for the industry-recognized VCAP-DCV Deploy 2023 certification which validates that you know how to deploy and optimize VMware vSphere infrastructures.

Learn more here: <https://www.vmware.com/learning/certification/vcap-dcv-deploy.html>

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Data Center
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2023

Additional Information

Additional Resources

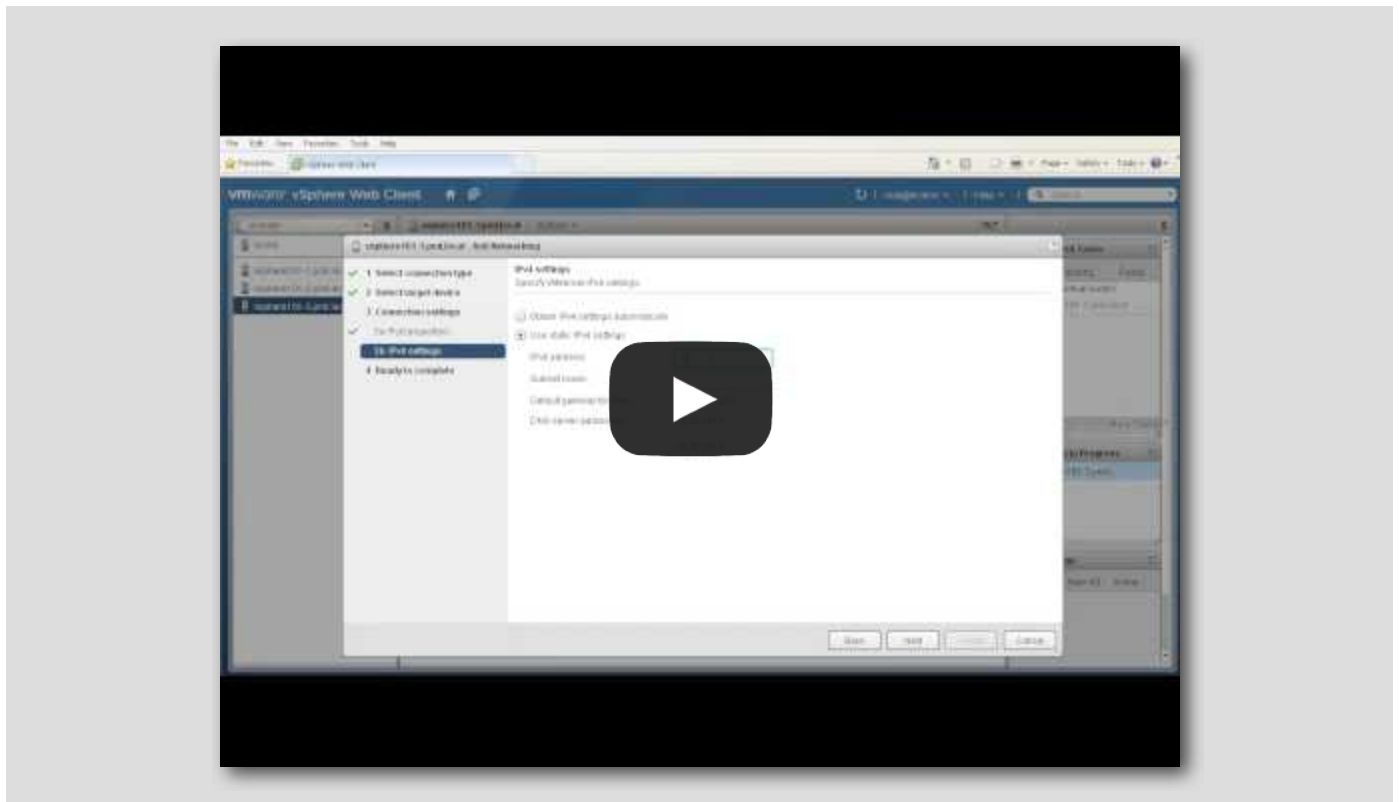
[533]

This section provides supplementary documentation and videos.

Video: How to Configure a vSphere Standard Switch (VSS) (4:22)

[534]

<https://www.youtube.com/watch?v=XpXuhK0c-f4>

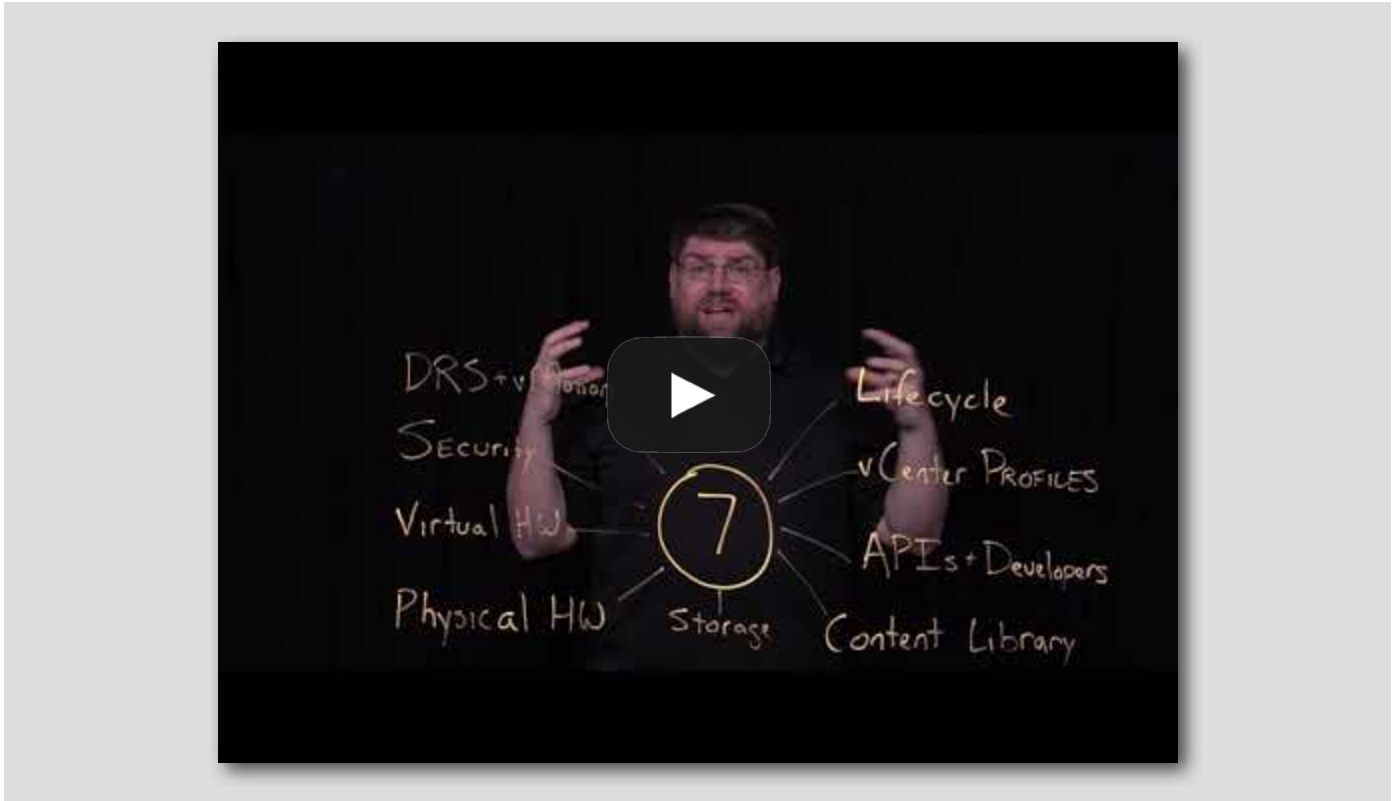


This video shows how to use the VMware vSphere web client to configure basic networking for your vSphere hosts using the vSphere Standard Switch (VSS).

Video: vSphere 8 - Overview (8:18)

[535]

<https://www.youtube.com/watch?v=XkP6QCutw9k>



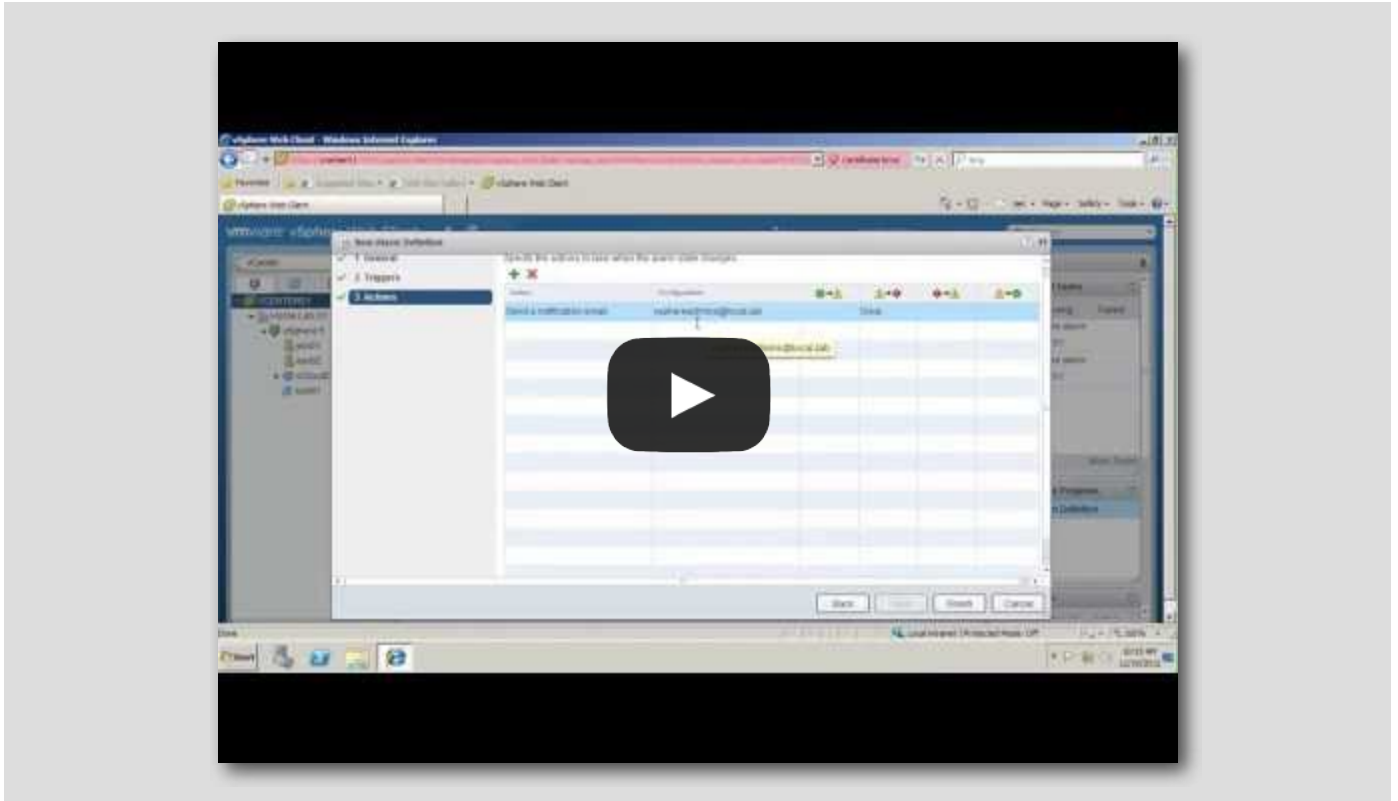
Light board illustration of the vCenter Server High Availability options for each deployment type.

Video: Configure Alarms and Notification for VMware vSphere (5:20)

[536]

This video shows how to use the VMware vSphere web client to configure vCenter Server alarms and alerts and how to enable email notification.

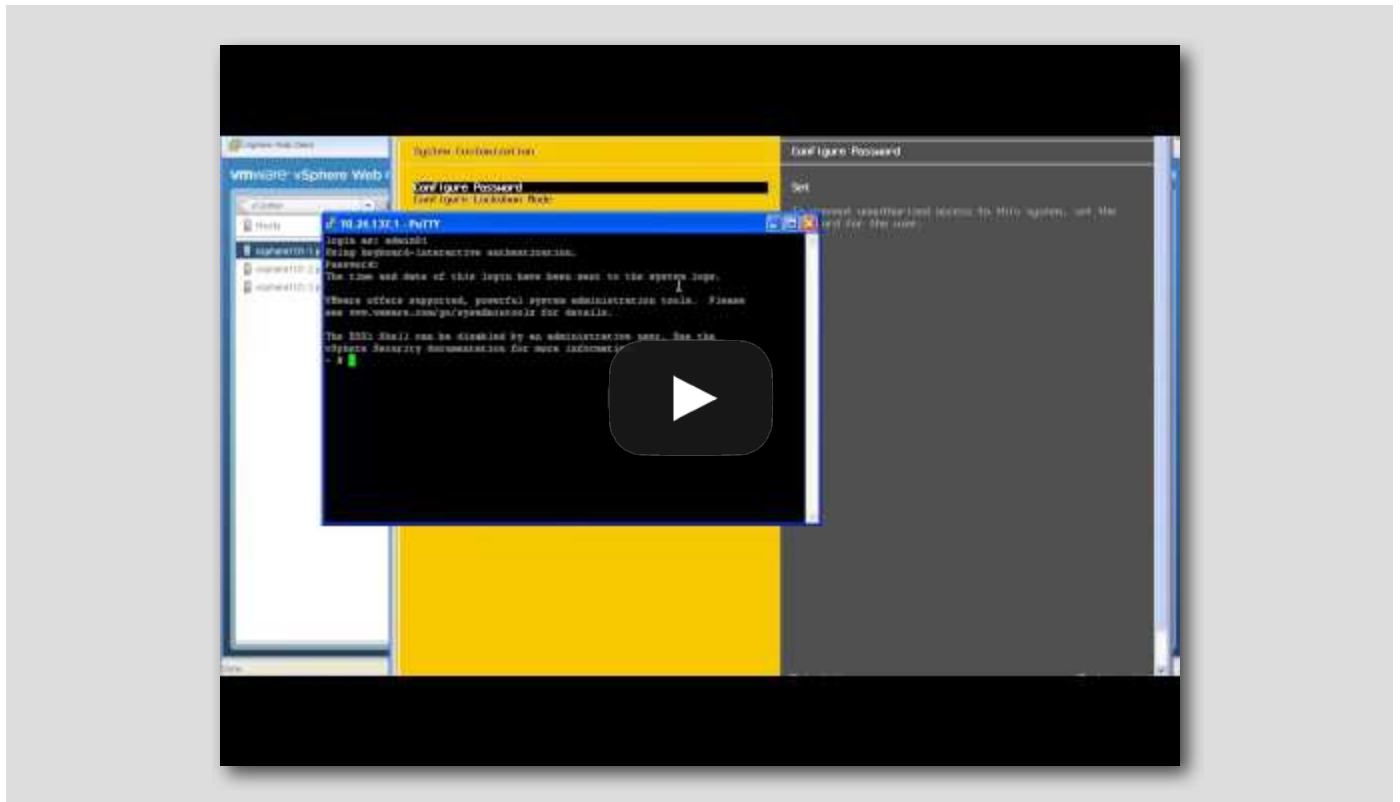
<https://www.youtube.com/watch?v=8vWNVBDPcu4>



Video: Enable vSphere Host Lockdown Mode for VMware vSphere (4:48)

This video shows how to secure VMware vSphere hosts with Lockdown Mode in order to limit direct access to the host console and to require administrators manage hosts through vCenter Server.

<https://www.youtube.com/watch?v=gWib2HHu3bE>

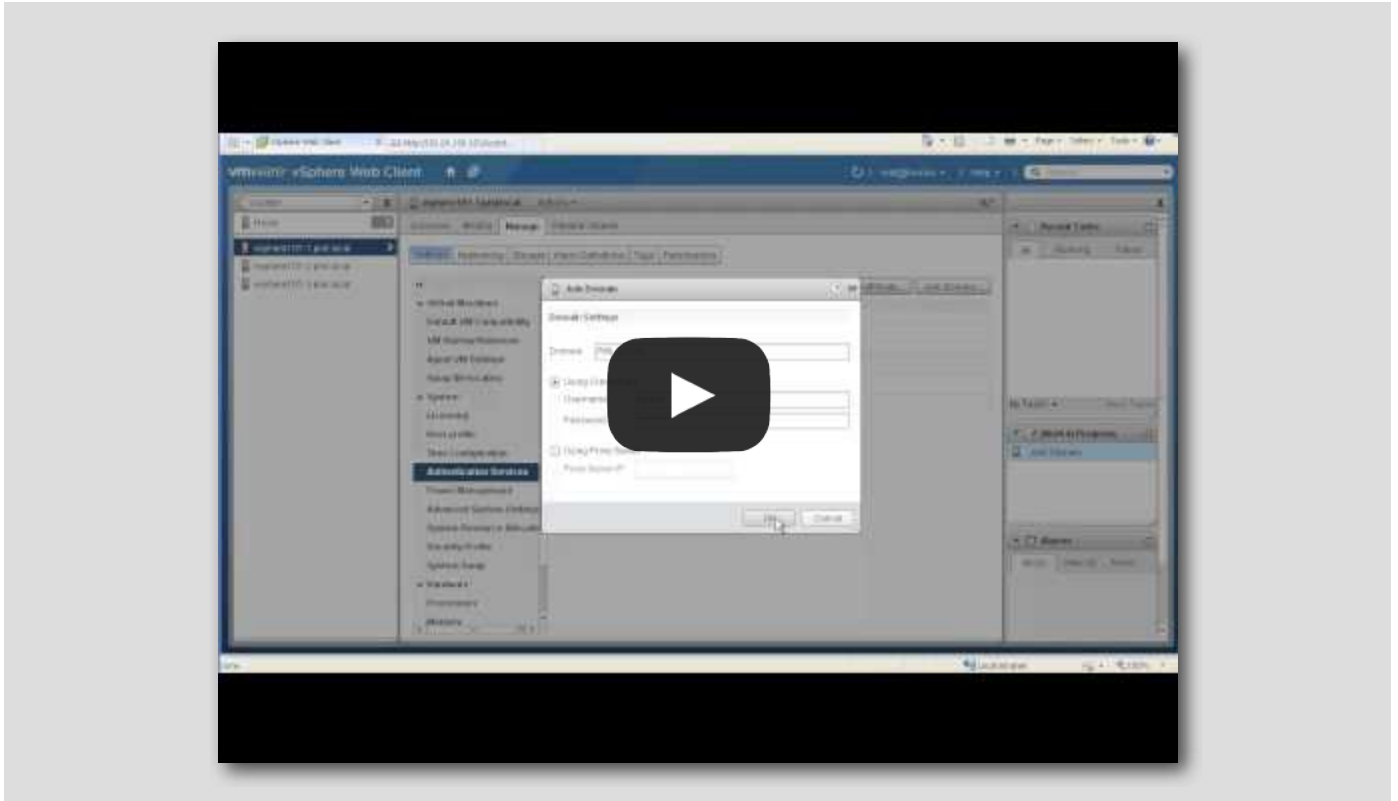


(Optional) Video: Add VMware vSphere Hosts to Active Directory (3:40)

[538]

This video shows how to join a VMware vSphere host to a Microsoft Active Directory (AD) domain in order to allow administrators to use their Active Directory credentials to access and manage hosts.

https://www.youtube.com/watch?v=H74M__Eshtw

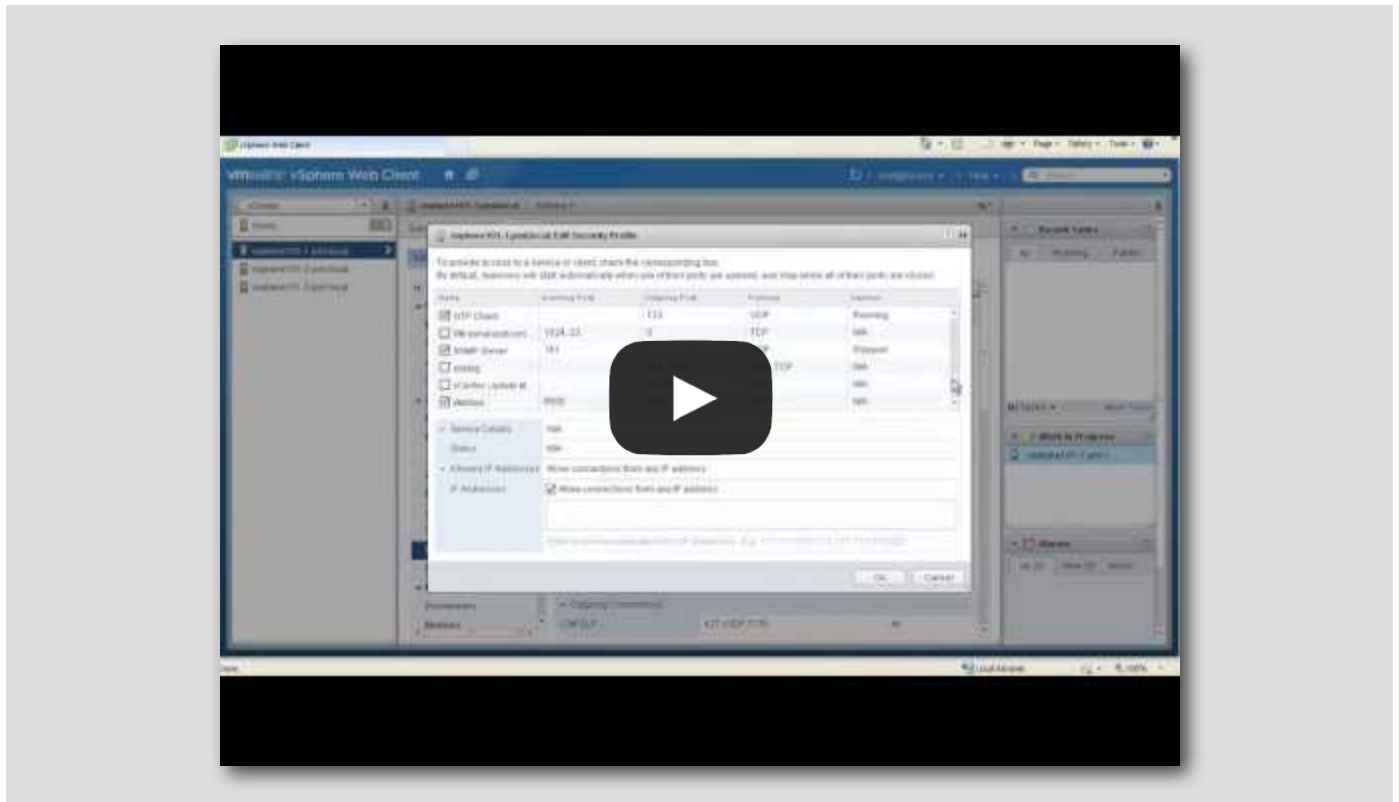


Video: Configure vSphere Host Firewall for VMware vSphere (4:34)

[539]

This video shows how to use the VMware ESXi Firewall on the vSphere host to block incoming and outgoing communication and to manage the services running on the host.

<https://www.youtube.com/watch?v=bzjsjQdnTuk>



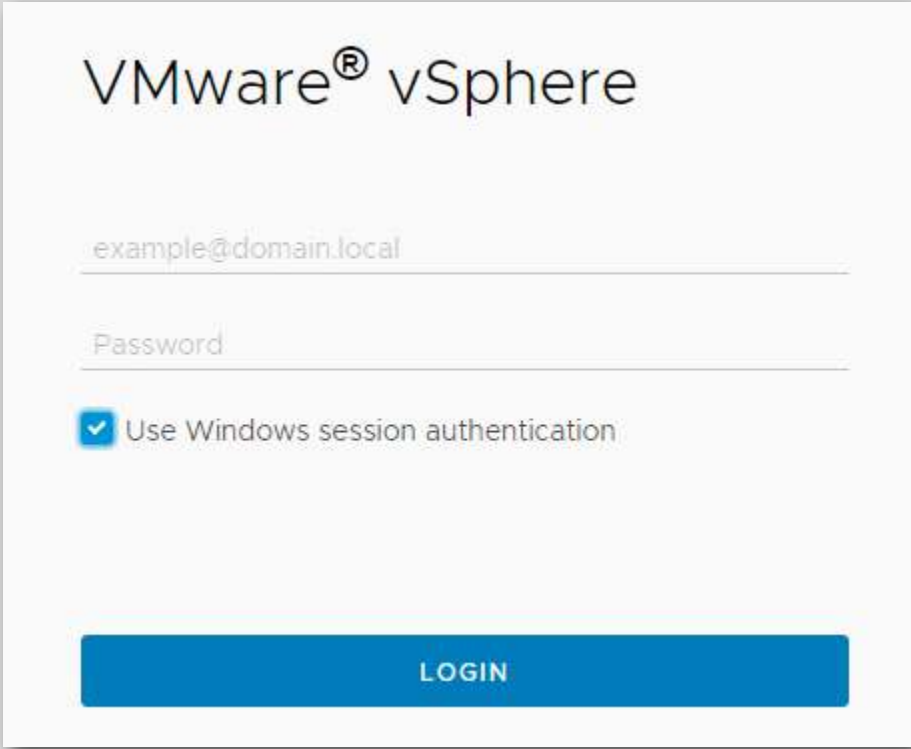
REFERENCE - Unlock vCenter Single Sign On Users in the vSphere Web Client

A vCenter Single Sign On user account might be locked when a user exceeds the allowed number of failed login attempts. After a user account is locked, the user cannot log in to the Single Sign On system until the account is unlocked, either manually or after a certain amount of time has elapsed.

You specify the conditions under which a user account is locked in the Single Sign On Lockout Policy. Locked user accounts appear on the Users and Groups administration page. Users with appropriate privileges can manually unlock Single Sign On user accounts before the specified amount of time has elapsed. You must be a member of the Single Sign On Administrators group to unlock a Single Sign On user.

Locked Out User

[541]

A screenshot of the VMware vSphere login interface. The form is white with a blue border and is centered on a gray background. At the top, it says "VMware® vSphere". Below that are two input fields: the first contains "example@domain.local" and the second is labeled "Password". Below the password field is a checkbox that is checked, with the text "Use Windows session authentication". At the bottom of the form is a large blue button with the word "LOGIN" in white capital letters.

VMware® vSphere

example@domain.local

Password

☒ Use Windows session authentication

LOGIN

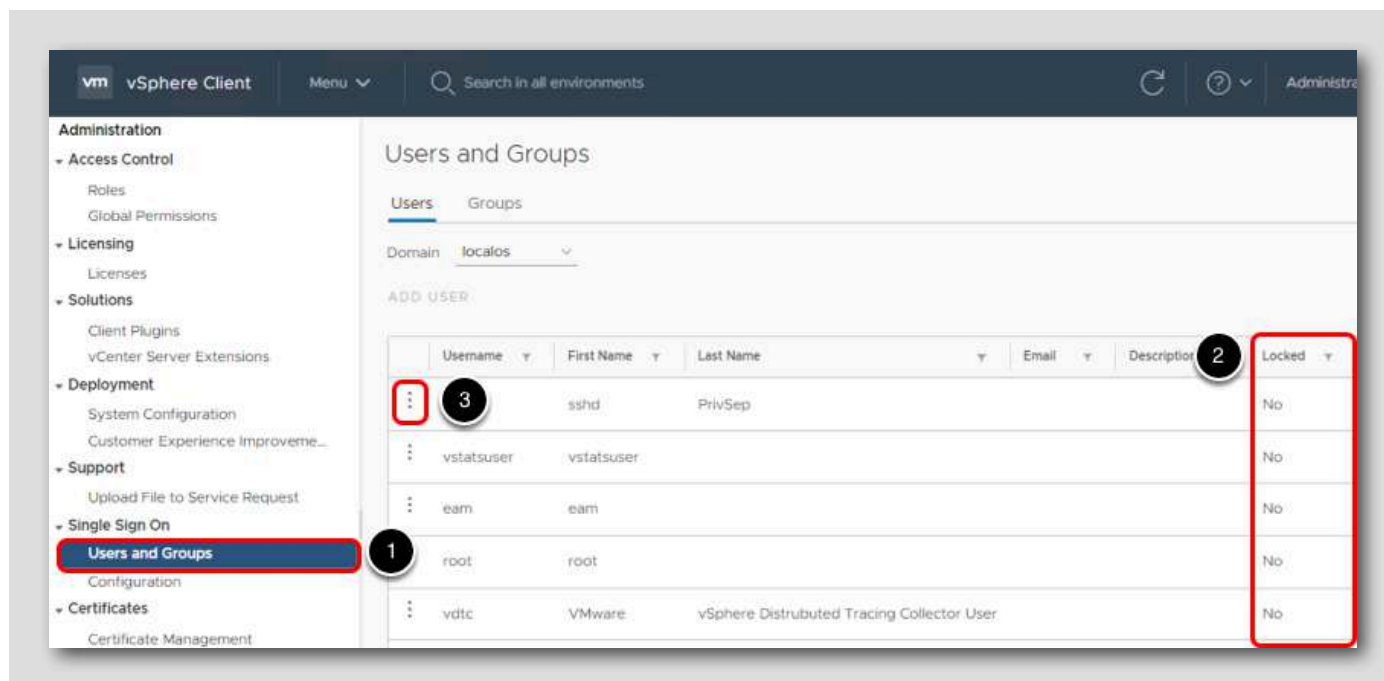
By default, after three failed login attempts, the Users' account is locked.

In the lab, this policy has been disabled in order to prevent login issues that frequently occur with non-US keyboards.

This section has been included for reference purposes only.

Unlocking a User

[542]



Login to the vSphere Web Client as a user with SSO Admin privileges and navigate Menu --> Administration.

1. Click on Users and Groups
2. Locate the locked user account -- it will show as "Yes" in the "Locked" column if the user is locked
3. Click the **tree dots** on the left and select the **unlock** option. Alternatively, if the kebabs menu is not in screen, you can click on **More**.

Log out of the Web Client.

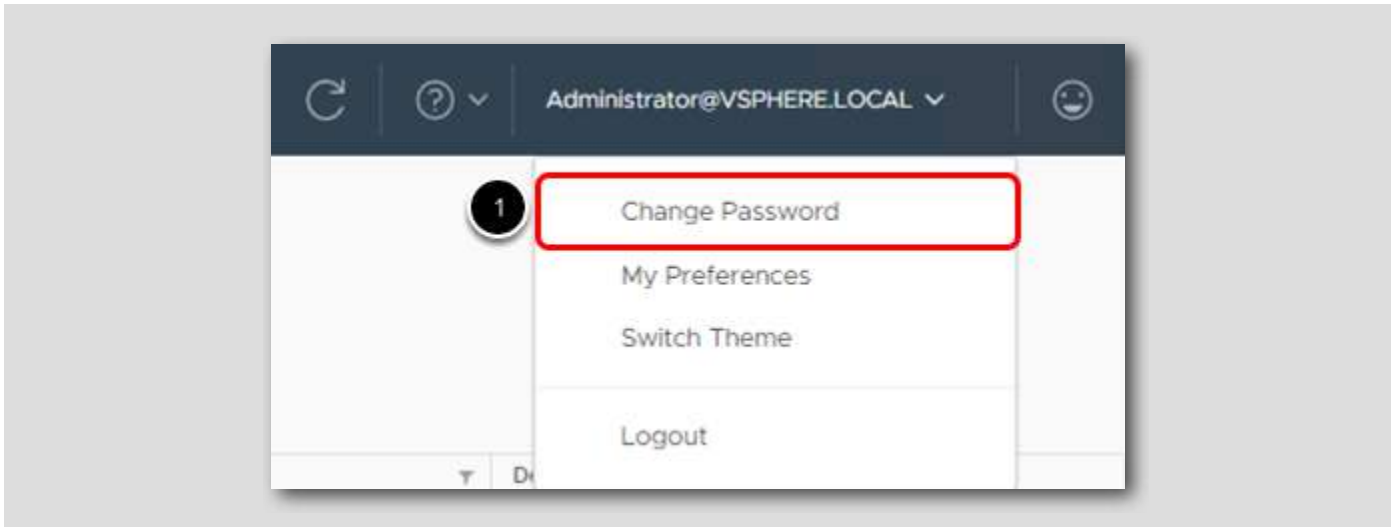
Change Your Password in the vSphere Web Client

[543]

Depending on your vCenter Single Sign On privileges, you might not be able to view or edit your Single Sign On user profile. However, all users can change their Single Sign On passwords in the vSphere Web Client. The password policy defined in the vCenter Single Sign-On configuration tool determines when your password expires. **By default, Single Sign-On passwords expire after 90 days in vSphere 6**, but your system administrator might change this depending on the policy of your organization. If you choose to keep the defaults, remember to change the password for the administrator@vsphere.local account password every 90 days or it will lock out on day 91.

Change Password

[544]

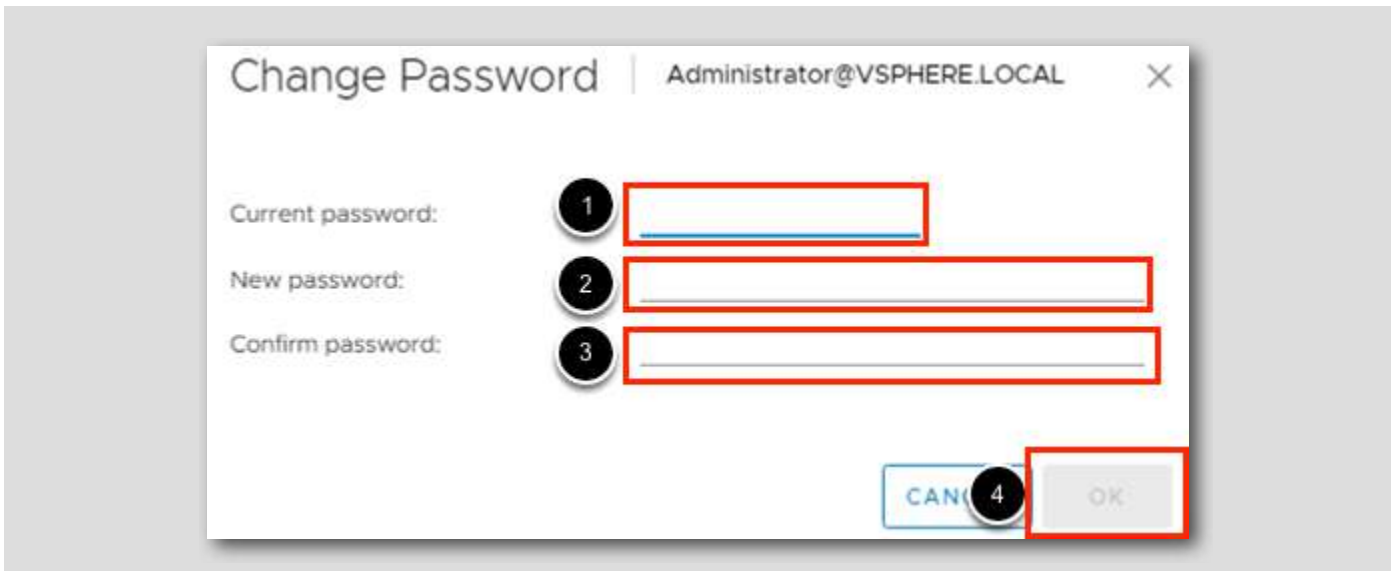


In the upper navigation pane, click your user name to pull down the menu.

1. Click on Change Password

Change Password Dialog

[545]

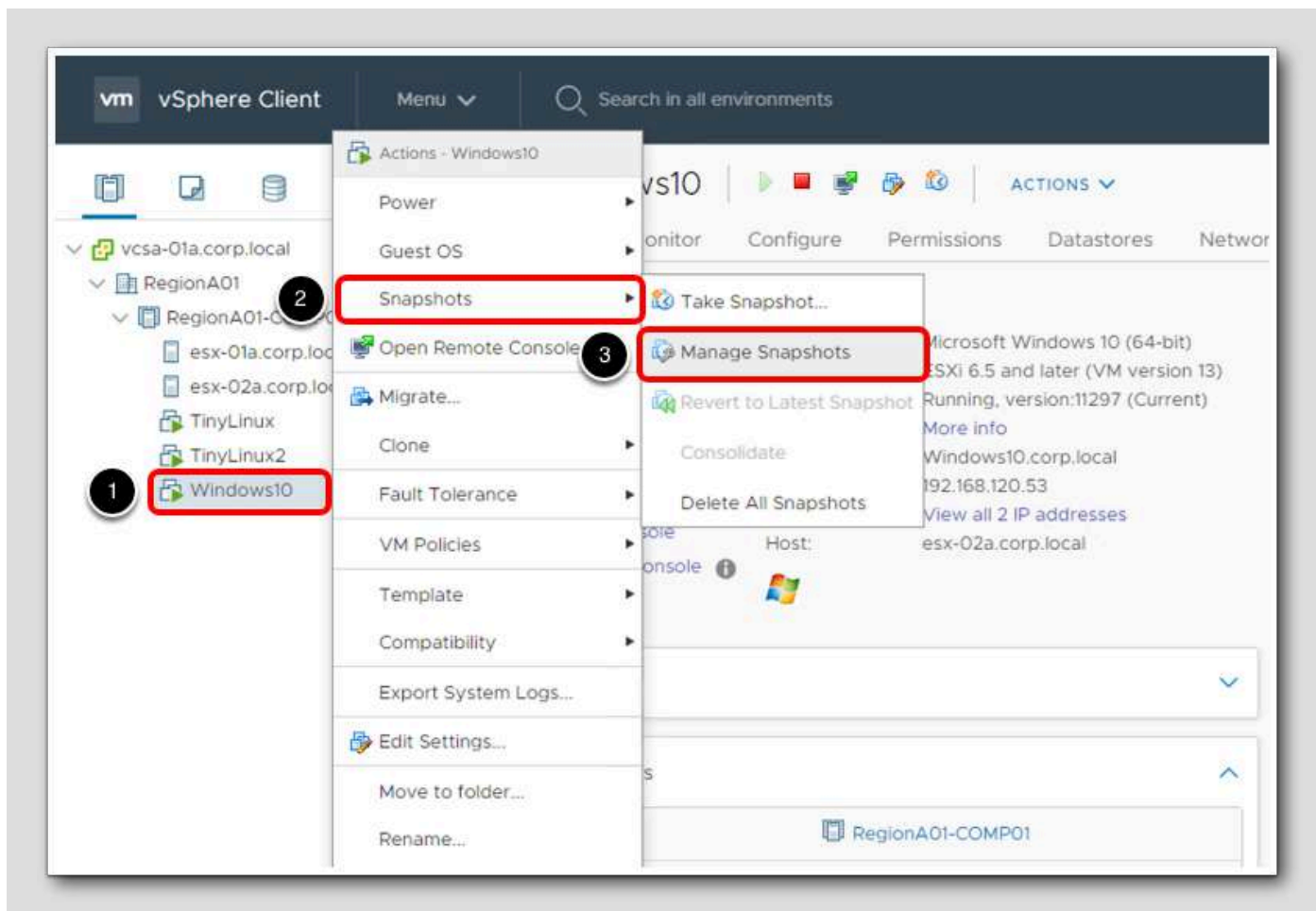


1. Select Change Password and type your current password.
2. Select New password and Type a new password
3. Confirm your new password
4. Click the OK button to make the change.

NOTE: If you do change the password, please make sure to remember it for other activities in the lab.

Snapshot Manager

[546]



In this section, you revert the Virtual Machine's configuration back to the original state using the Snapshot Manager.

1. Right-click **Windows10**
2. Select **Snapshots**
3. Click **Manage Snapshots**

Video: What is vSphere Storage DRS? (5:08)

[547]

<https://www.youtube.com/watch?v=z77xmaxoNec>



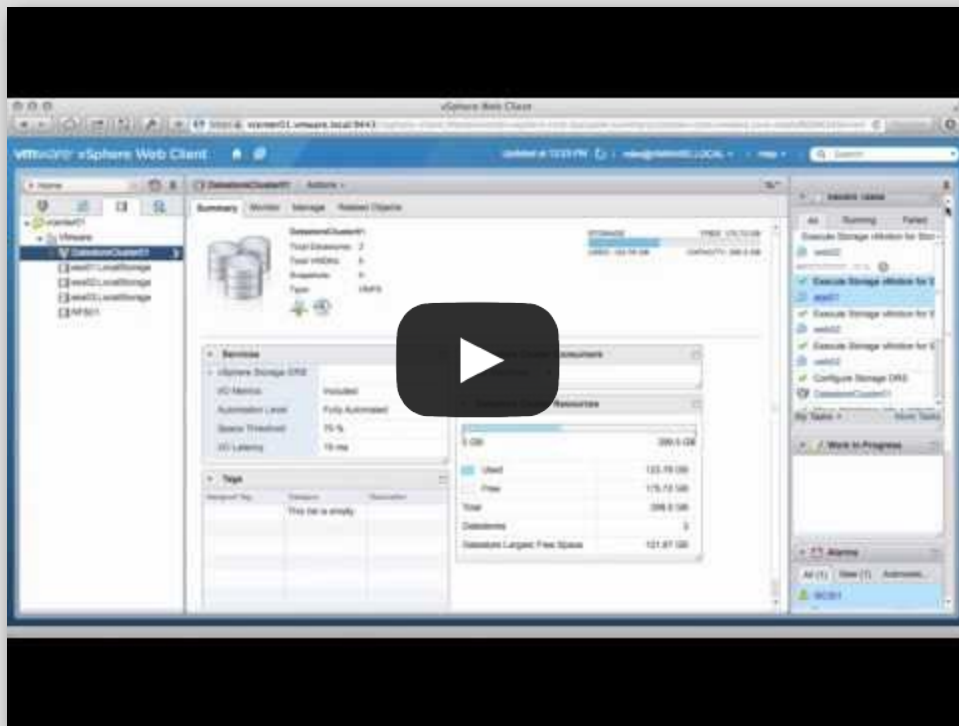
This animated video shows how VMware Storage DRS reduces the time and complexity of provisioning virtual machines by aggregating data stores into a single pool, called a datastore cluster, enabling rapid placement of virtual machines and virtual machine disks.

Video: Creating a Datastore Cluster with Storage DRS (3:23)

[548]

This video reviews the process of creating and managing a datastore cluster in a vSphere environment.

<https://www.youtube.com/watch?v=gATLj6pUxnk>



Appendix

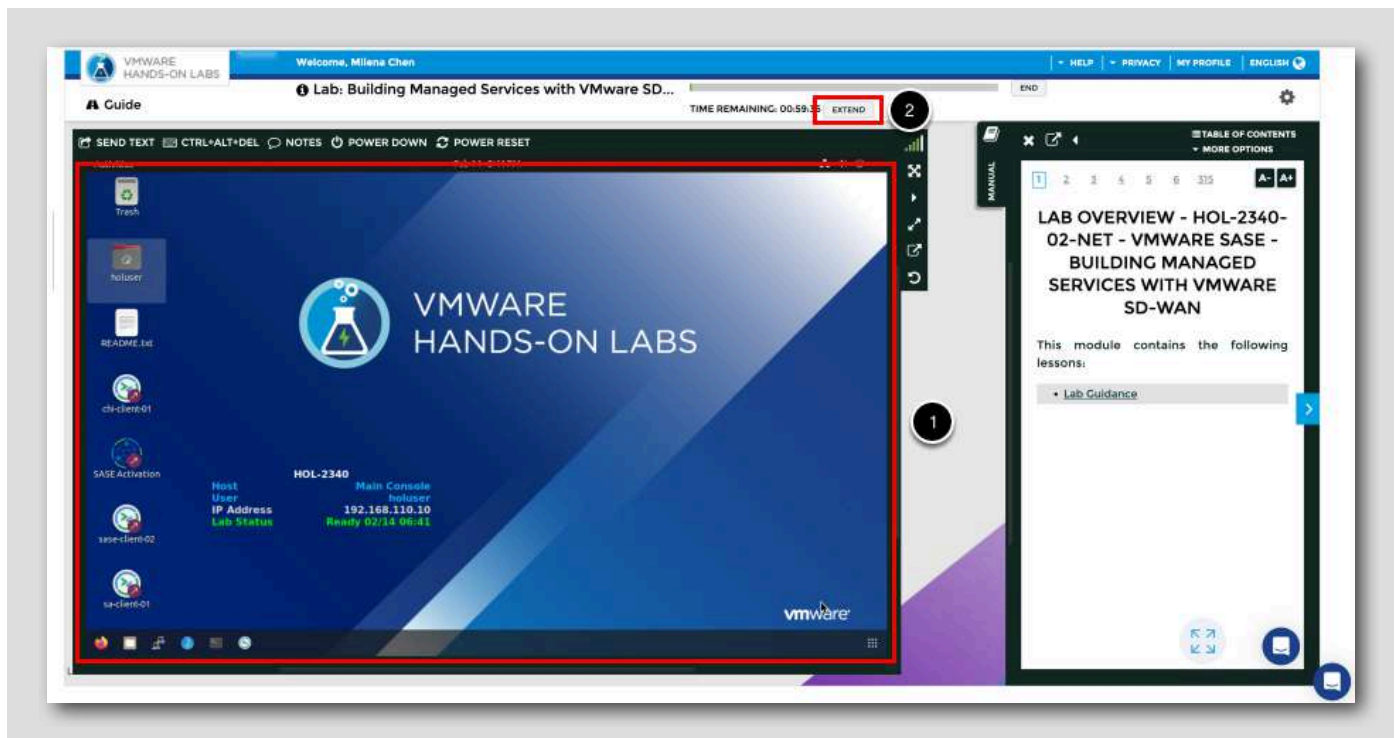
Hands-on Labs Interface

[550]

Welcome to Hands-on Labs! This overview of the interface and features will help you to get started quickly. Click next in the manual to explore the Main Console or use the Table of Contents to return to the Lab Overview page or another module.

Location of the Main Console

[551]



1. The area in the large RED box contains the Main Console. The Lab Manual is on the tab to the right of the Main Console.
2. Your lab starts with a the timer. The lab cannot be saved and will end when the timer expires. Click the **EXTEND** button to increase the time allowed. The amount of time you can extend will depend on the lab.

Alternate Methods of Keyboard Data Entry

[552]

In this lab you will input text into the Main Console. Besides directly typing in the console, there are two very helpful methods of entering data which make it easier to enter complex data.

Click and Drag Lab Manual Content Into Console Active Window

[553]

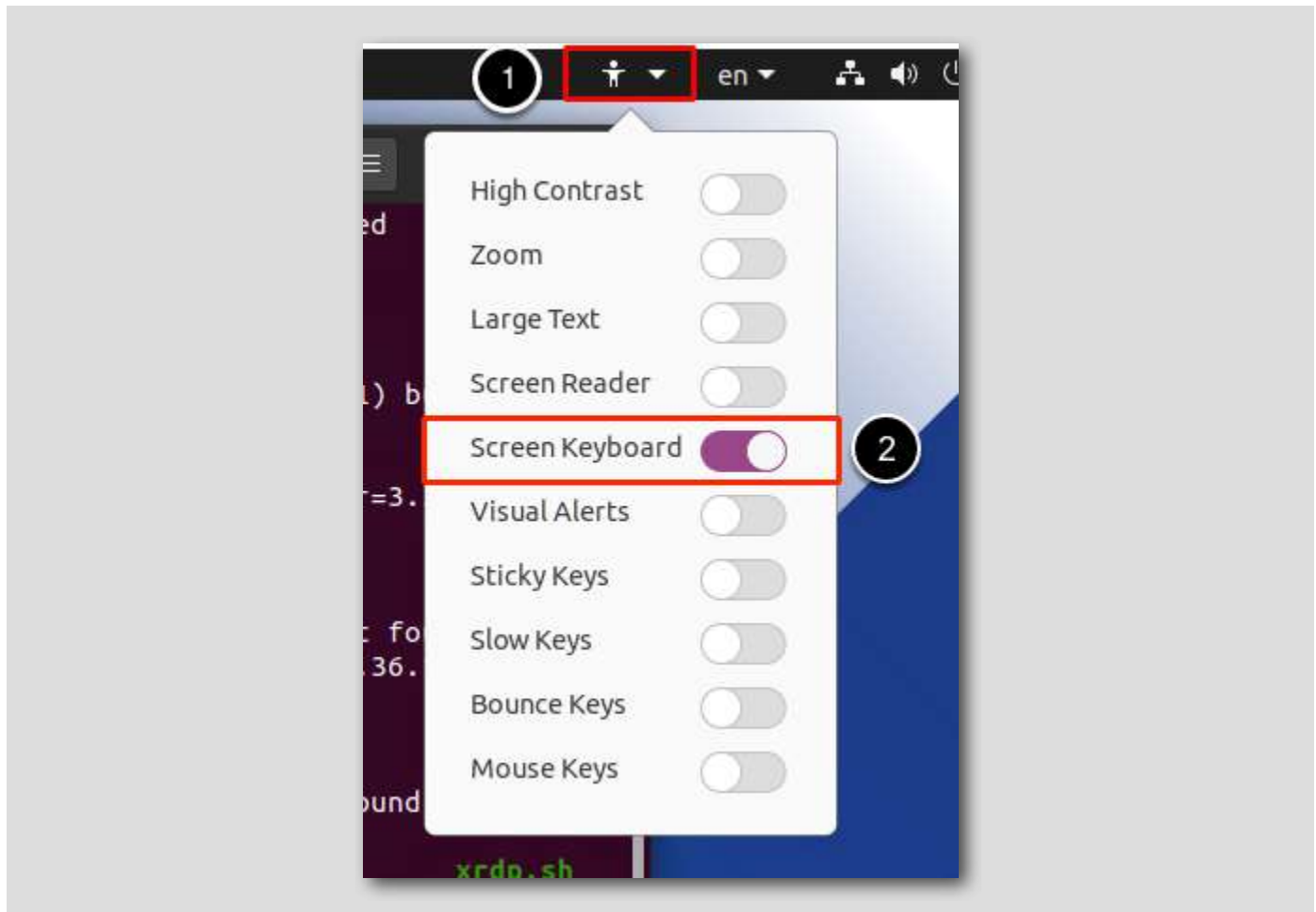
<https://www.youtube.com/watch?v=xS07n6GzGuo>



You can also click and drag text and Command Line Interface (CLI) commands directly from the Lab Manual into the active window in the Main Console.

Accessing the Online International Keyboard

[554]



You can also use the Online International Keyboard found in the Main Console.

1. Click on the human icon (Universal Access) on the top taskbar
2. Enable Screen Keyboard

The Keyboard Is Now Enabled

[555]



The keyboard will now be enabled and will autohide and appear when needed; e.g., when you click in a text field or terminal.

Return to Lab Guidance

[556]

Use the Table of Contents to return to the Lab Overview page or another module.

